Contents

Morse Theory, verified: the machine-checked edition **1**

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A companion to "Morse Theory, Rewritten", the expanded edition of J. Milnor's Morse Theory *(Princeton, 1963). That edition is written to be read; this one is written to be checked. It records, chapter by chapter, exactly which statements of the book have been verified by the Lean 4 proof assistant against the Mathlib library and the formalization of Milnor's* Topology from the Differentiable Viewpoint *(the companion project), what each verified statement says when read back in ordinary language, which hypotheses the book left silent, where the formal statement or proof departs from the book's, and what remains unverified and why. Nothing here needs to be read in Lean; every checked statement is restated in prose, and the Lean names are given only so that the claims can be audited.*

Where to start. The project has three parts:

- *Morse Theory, Rewritten*, the book for reading;
- *Morse Theory, Checked*, the Lean formalization and the reports on it;
- *Morse Theory, the Shortest Path*, the main theorems and exactly what their formal proofs use.

This edition is the way into the second part. Its four reports each answer one question:

- **What is proved?** This edition: every checked result in plain language, with all its hypotheses, and the list of statements to sign off.
- **How is it proved?** *Morse Theory, proof by proof* (`Exposition/ProofByProof.md`): the formal proofs retold as mathematics, with a trace tag on each step naming the Lean declaration that carries it.
- **How does it differ from Milnor?** *Morse Theory beside Lean*, the comparison (`Exposition/morse-beside-lean`) each result of the book beside the checked statement, with a verdict (the same, restated, hypotheses made explicit, stronger, narrower) and the differences.
- **Are the comparison's verdicts right?** The round-trip audit (`RoundTrip/REPORT.md`): Milnor's statements are formalized blind, without sight of the project's Lean, and proved equivalent, or not, to the checked ones; the definitions under the main theorems are checked against blind textbook definitions in the same way.

A result has the same number, V2.8 say, in this edition, the comparison, the rewritten book's notes and the shortest path.

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0. Summary

Chapter	Milnor	Content	Status	Lean lines
Chapter 1	§§1–2	critical points, Hessian and index, the Morse lemma, flows (Lemma 2.4)	**partly verified**	4,572

Chapter	Milnor	Content	Status	Lean lines
Chapter 2	§3	Theorems 3.1, 3.2, 3.5, Lemmas 3.6–3.7, Remarks 3.3–3.4	**partly verified**	6,359
Chapter 3	§§4–5	Reeb's theorem, the torus, the Morse inequalities in singular homology, the Euler characteristic	**partly verified**	13,353
Chapter 4	§6	focal points, existence and density of Morse functions, compact abstract manifolds	**partly verified**	3,720
Chapter 5	§7	the index bound behind the Lefschetz theorem	**partly verified**	2,804
Chapter 6	§§8–9	connections, parallel transport and curvature, for submanifolds of $\mathbb{R}^K$	**partly verified**	3,346
Chapter 7	§10	geodesics, the exponential map, the Gauss lemma, the Hopf–Rinow theorem	**partly verified**	5,962
Chapter 8	§§11–13	the path space, the energy, the first and second variation formulas	**partly verified**	4,064
Chapter 9	§§14–15	Jacobi fields and the index theorem	**partly verified**	5,288
Chapter 10	§§16–17	the finite-dimensional approximation, the full path space	**partly verified**	13,564

Chapter	Milnor	Content	Status	Lean lines
Chapter 11	§§18–19	non-conjugate points, topology and curvature	**partly verified**	4,322
Chapter 12	§§20–21	bi-invariant metrics, matrix groups, their geodesics and curvature	**partly verified**	3,922
Chapter 13	§22	manifolds of minimal geodesics, Freudenthal's theorem	**partly verified**	3,048
Chapter 14	§23	Bott periodicity for the unitary group	**partly verified**	2,106
Chapter 15	§24	Bott periodicity for the orthogonal group	**partly verified**	9,682
Appendix	Appendix	the homotopy type of a monotone union	**partly verified**	2,720

Total: 88,832 lines of Lean in 216 files under `Morse/` (`wc -l`, excluding unfinished `WIP_` files), on top of the companion project. Toolchain: Lean 4.33.1, Mathlib 4.33.1. To reproduce: `lake build`, then `lake env lean -DmaxSynthPendingDepth=3 Morse/Test.lean`, which prints the axioms of every main theorem. Edition of 5 November 2026.

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1. How to read a verified statement

The statements are the contract. A machine-checked proof guarantees that the *stated* theorem follows from Lean's axioms. It guarantees nothing about whether the statement is the one Milnor meant. The reader's job, and the only mathematical judgement this document asks for, is to read each prose restatement and confirm that it says what the book means. Where the restatement is weaker or stronger than the book, or uses a different but equivalent definition, this is flagged as a **Deviation**.

Three settings. Mathlib's manifolds are abstract spaces with atlases. The results of §2 are proved for abstract manifolds. Most of Part I is proved for submanifolds of a Euclidean space in the sense of the companion project (subsets with slice charts), and carried to abstract *compact* manifolds through Mathlib's Whitney embedding theorem; non-compact abstract manifolds would need a Whitney embedding that Mathlib does not have. Part II, and Part III as far as it is verified (§§11–19), are verified for submanifolds of $\mathbb{R}^K$ with the induced metric (every Riemannian manifold is such a submanifold by Nash's theorem, which is quoted, not proved); Part IV (§§20–24) for compact groups

of matrices and the spaces $\Omega_k(n)$ of §24, as submanifolds of a space of matrices, and in its algebraic cores; the Appendix for arbitrary topological spaces. Homology is Mathlib's singular homology, extended here by relative homology, the exact sequences of pairs and triples, and excision (Chapter 3).

What "smooth" means. Smooth means C^∞. Hausdorffness and second countability, which Milnor uses silently, are explicit hypotheses wherever they are needed; submanifolds of a Euclidean space have both automatically.

Axioms. Every verified theorem depends only on the three standard axioms of Lean's Mathlib (`propext`, `Classical.choice`, `Quot.sound`), as `Morse/Test.lean` prints; there is no `sorry`.

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2. Chapter 1 (Milnor §§1–2): partly verified

Milnor's §1 introduces the subject through the height function on a torus standing on a plane; §2 defines critical points, the Hessian, the index and the nullity, proves Hadamard's lemma (Lemma 2.1) and the Morse lemma (Lemma 2.2), states that non-degenerate critical points are isolated (Corollary 2.3), lists examples of degenerate critical points, and proves that a compactly supported vector field generates a one-parameter group of diffeomorphisms (Lemma 2.4). All of §2 is verified except the remark of p. 4 that regular sublevel sets are manifolds with boundary, the vector-field form of the Hessian, example (b), Lemma 2.4 on a manifold that is not a vector space, and the closing remark of p. 11. The definitions, Corollary 2.3 and the Morse lemma are verified in two settings: on a real normed space E (finite-dimensional wherever dimensions are counted), which is Milnor's situation after a choice of coordinates, and on an abstract smooth manifold without boundary modelled on such an E, with no Hausdorff or countability hypothesis. For a function $F|N$ on a slice submanifold N of a Euclidean space, in the sense of the companion project, they are computed in slice parametrizations (V1.6); this is the setting of most of Chapter 2. Lemma 2.4 is verified on a complete real normed space, together with the smooth dependence of solutions on the initial point, which Milnor quotes. Of §1, the critical points of the torus, their indices, and Milnor's cases (1) and (5) are verified for an explicit torus of revolution; the identification of the intermediate sublevel sets and the cell attachments are not.

V1.1 (the torus of §1; `Morse.torus`, `Morse.Torus.isCriticalPointM_height_iff`, `Morse.Torus.pointP_spec`, `Morse.Torus.pointQ_spec`, `Morse.Torus.pointR_spec`, `Morse.Torus.pointS_spec`, `Morse.Torus.sublevel_he` `Morse.Torus.sublevel_height_eq_univ`, `Morse.theorem_3_1_torus`). Let $0 < \rho < R$ and let $T \subseteq \mathbb{R}^3$ be the torus $(\sqrt{x_1^2 + x_2^2} - R)^2 + x_3^2 = \rho^2$, whose axis of revolution is the x_3-axis; it is a compact slice submanifold of dimension 2 (V1.6). Let $f(x) = x_1 + R + \rho$, the height above the plane $x_1 = -(R + \rho)$, which touches T at $p = (-(R + \rho), 0, 0)$. Then $f|T$ has exactly four critical points, p, $q = (-(R - \rho), 0, 0)$, $r = (R - \rho, 0, 0)$ and $s = (R + \rho, 0, 0)$, at the heights 0, 2ρ, $2R$ and $2R + 2\rho$; all four are non-degenerate (V1.4), of indices 0, 1, 1, 2; $T^a = \emptyset$ for $a < 0$ and $T^a = T$ for $a \geq 2R + 2\rho$; and if $a \leq b$ and $[a, b]$ contains none of the four heights, a C^∞ diffeomorphism of T carries T^a onto T^b, T^a is a strong deformation retract of T^b (V2.1), and the inclusion $T^a \to T^b$ is a homotopy equivalence. *Book:* pp. 1–3: for the torus of Diagram 1, tangent to the plane V, with f the height above V, the sublevel set M^a is successively empty, a 2-cell, a cylinder, a compact genus-one surface bounded by a circle, and all of M as a passes the values of f at p, q, r, s; these four points are the critical points, with normal forms $x^2 + y^2$ at p, const $+x^2 - y^2$ at q and r, const $-x^2 - y^2$ at s. *Hidden hypotheses:* the torus must stand on the plane with its axis parallel to it, and $\rho < R$; lying flat, its height function has two circles of degenerate critical points.

Deviation (value differs): case (1) reads $a < 0 < f(p)$ in the scan, whereas $f(p) = 0$; the hypothesis is $a < 0$. *Deviation (narrower):* cases (2)–(4) and the cell attachments are not verified; between critical heights only Theorem 3.1 (V2.3) is applied. The Lean file is `Morse/Chapter3/Torus.lean`, since its proofs use Theorem 3.1 on submanifolds. The details, with their own sign-off items, are V3.22–V3.25 in Chapter 3.

V1.2 (attaching a cell; `Morse.AttachCell`, `Morse.cellDesc`). For a topological space X, an integer $n \geq 0$ and a continuous map $\varphi : S^{n-1} \to X$, the space $X \cup_\varphi e^n$ is the pushout, in the category of topological spaces, of φ and the inclusion of the unit sphere S^{n-1} in the closed unit ball D^n of $\mathbb{R}^n$: the quotient of the disjoint union $X \sqcup D^n$ by the identification of each $u \in S^{n-1}$ with $\varphi(u)$, so that a map out of it is a pair of maps on X and on D^n which agree along φ. For $n = 0$ the sphere is empty and D^0 is a point. The disk and the sphere are Mathlib's, those of its CW complexes. *Book:* pp. 2–3, the same definition (the topological sum of Y and e^k with each $x \in S^{k-1}$ identified with $g(x)$), with the same convention for $k = 0$. *Hidden hypotheses:* none.

V1.3 (critical points, Hessian, index and nullity in coordinates; `Morse.IsCriticalPoint`, `Morse.hessian`, `Morse.hessian_symm`, `Morse.nullSpace`, `Morse.nullity`, `Morse.index`, `Morse.index_isGreatest`, `Morse.IsNondegenerate`, `Morse.isNondegenerate_iff_nullity_eq_zero`, `Morse.IsNondegenerateCriticalPoint`, `Morse.isNondegenerate_hessian_iff_det_ne_zero`). Let E be a real normed space and $f : E \to \mathbb{R}$. A point p is a *critical point* of f if f is differentiable at p with derivative 0. The *Hessian* of f at p is the second derivative $d^2 f_p$, a continuous bilinear form on E, symmetric when f is C^2 at p; it is defined at every point but only used at critical points. For a continuous bilinear form H on E, the *null space* is $\{v : H(v, w) = 0 \text{ for all } w\}$, the *nullity* is its dimension, H is *non-degenerate* if its null space is 0, and the *index* of H is the negative index of inertia of the quadratic form $v \mapsto H(v, v)$, which for finite-dimensional E is the largest dimension of a subspace on which $H(v, v) < 0$ for all $v \neq 0$. A *non-degenerate critical point* is a critical point at which $d^2 f_p$ is non-degenerate. For finite-dimensional E, non-degenerate means nullity 0; for $E = \mathbb{R}^n$ and f of class C^2 at p, it means that the matrix $(\partial^2 f / \partial x^i \partial x^j(p))$ is non-singular. *Book:* pp. 4–5: critical points ($f_* = 0$, restated in local coordinates; the display is lost in the scan), non-degeneracy (the matrix of second partial derivatives is non-singular), the index and nullity of a bilinear functional, and the remark that a critical point is non-degenerate exactly when the nullity is 0. *Hidden hypotheses:* none. *Deviation (restated):* the index is defined as Mathlib's negative index of inertia (`sigNeg`), and Milnor's definition as a maximum is a theorem (`Morse.index_isGreatest`); non-degeneracy is defined by a zero null space, and Milnor's matrix definition is a theorem.

V1.4 (the same on a manifold, and Morse functions; `Morse.IsCriticalPointM`, `Morse.hessianM`, `Morse.indexM`, `Morse.nullityM`, `Morse.IsNondegenerateCriticalPointM`, `Morse.IsMorseFunction`, `Morse.isCriticalPointM_iff_milnor`). Let M be a smooth manifold without boundary modelled on a real normed space E, with no Hausdorff or countability hypothesis, $f : M \to \mathbb{R}$ and $p \in M$. The point p is a *critical point* of f if the derivative $df_p : T_pM \to \mathbb{R}$ is 0; equivalently, df_p is not surjective, which is the companion project's definition. The *Hessian* f_{**} at p is the second derivative at $\kappa(p)$ of $f \circ \kappa^{-1}$, where κ is the preferred chart at p: a bilinear form on E, which Mathlib identifies with T_pM through the same chart. The index and the nullity of f at p are those of f_{**}, and p is a non-degenerate critical point if it is critical and f_{**} is non-degenerate (V1.3). A *Morse function* is a C^∞ function on M all of whose critical points are non-degenerate. *Book:* pp. 4–5; the Morse functions are the differentiable functions with no degenerate critical points of Theorem 3.5 (p. 20). *Hidden hypotheses:* none. *Deviation (restated):* Milnor defines $f_{**}(v, w) = \tilde{v}_p(\tilde{w}(f))$ for vector fields $\tilde{v}$, $\tilde{w}$ extending v, w, and checks that this is symmetric and independent of the

extensions. That formula is not formalized: the formal definition is his coordinate expression (the matrix of second partial derivatives, p. 5, which is his formula for extensions with constant coefficients), and its independence of the extensions is replaced by its independence of the chart (V1.5). Mathlib's derivative is 0 at a point where f is not differentiable, so such a point counts as critical; for smooth f this never matters.

V1.5 (independence of the coordinate system; `Morse.hessian_comp`, `Morse.IsSmoothCoord`, `Morse.isSmoothCoord_iff_mem_maximalAtlas`, `Morse.IsSmoothCoord.isCriticalPointM_iff`, `Morse.IsSmoothCoord.isCriticalPointM_iff_forall_partial`, `Morse.IsSmoothCoord.hessian_comp_symm`, `Morse.IsSmoothCoord.isNondegenerate_iff`, `Morse.IsSmoothCoord.index_eq`, `Morse.IsSmoothCoord.nullit` `Morse.IsSmoothCoord.isNondegenerate_iff_det_ne_zero`). On normed spaces: if ψ is C^2 at q, f is C^2 at $\psi(q)$ and $df_{\psi(q)} = 0$, then $d^2(f \circ \psi)_q(v, w) = d^2 f_{\psi(q)}(d\psi_q v, d\psi_q w)$. On M as in V1.4: a *smooth coordinate system* is a homeomorphism y of an open subset of M onto an open subset of a real normed space E' which is C^∞ with C^∞ inverse (for $E' = E$ these are exactly the charts of the maximal atlas). Let p lie in the domain of y. If f is differentiable at p, then p is critical if and only if $y(p)$ is a critical point of $f \circ y^{-1}$, that is, for $E' = \mathbb{R}^n$, if and only if all the partial derivatives $\partial f/\partial y^i(p)$ vanish. If f is C^2 at p and p is critical, the Hessian of $f \circ y^{-1}$ at $y(p)$ is f_{**} composed in both arguments with the derivative of y^{-1} at $y(p)$ (the matrix $(\partial^2 f/\partial y^i \partial y^j(p))$ represents f_{**} in the basis $\partial/\partial y^i|_p$); hence non-degeneracy, index and nullity are those of $f \circ y^{-1}$ at $y(p)$, and for $E' = \mathbb{R}^n$, p is non-degenerate if and only if $\det(\partial^2 f/\partial y^i \partial y^j(p)) \neq 0$. *Book:* p. 4, non-degeneracy "can be checked directly" not to depend on the coordinate system; p. 5, the matrix of second partial derivatives represents f_{**}. *Hidden hypotheses:* none. *Deviation (stronger):* f need only be C^2 at p, and the coordinates may take values in any normed space. Criticality is used exactly once, to kill the first-order term $df(d^2\psi)$ of the chain rule.

V1.6 (functions on a slice submanifold; `Milnor.IsSliceSubmanifold`, `Morse.sliceParam`, `Morse.tangentSubspace`, `Morse.tangentSubspace_eq_range_fderiv_sliceParam`, `Morse.isCriticalPointM_c` `Morse.isCriticalPointM_comp_val_iff_forall_tangent`, `Morse.isCriticalPointM_comp_val_iff_gradient` `Morse.hessian_comp_sliceParam`, `Morse.index_hessian_comp_sliceParam`, `Morse.nullity_hessian_comp_sl` `Morse.isNondegenerateCriticalPointM_comp_val_iff_sliceParam`, `Morse.isCriticalPointM_normSq_sub_i` `Morse.hessian_normSq_sub_comp`, `Morse.indexM_normSq_sub_eq`, `Morse.nullityM_normSq_sub_eq`). This is the setting of every later statement about a submanifold of Euclidean space. V is a real normed space (Milnor's $\mathbb{R}^K$) and $N \subseteq V$ is a slice submanifold modelled on Z in the sense of the companion project: every point of N lies in the domain of a slice chart, a C^∞ diffeomorphism σ of an open subset of V onto an open subset of $Z \times W$ which carries the points of N in its domain exactly onto $Z \times \{0\}$. N need not be closed; it is a smooth manifold modelled on Z, with the charts $q \mapsto \sigma(q)_1$ and the parametrizations $x(z) = \sigma^{-1}(z, 0)$. Let $T_q N \subseteq V$ be the range of the derivative of the inclusion $N \to V$ at q; it is the range of dx. Let $F : V \to \mathbb{R}$ and let $q \in N$ lie in the domain of σ. If F is differentiable at q, then q is a critical point of $F|N$ if and only if $\sigma(q)_1$ is a critical point of $F \circ x$, if and only if dF_q vanishes on $T_q N$, and, when V is a complete inner product space, if and only if $\nabla F(q) \perp T_q N$. If F is C^2 at q and q is critical for $F|N$, the Hessian of $F \circ x$ at $\sigma(q)_1$ is f_{**} composed with the derivative of the parametrization, so non-degeneracy, index and nullity of $F|N$ at q are those of $F \circ x$. For V an inner product space and $L_p(y) = \|y - p\|^2$: q is a critical point of $L_p|N$ if and only if $q - p \perp T_q N$; for any x of class C^2 at u, the second derivative of $L_p \circ x$ at u is $2(\langle \partial x/\partial v, \partial x/\partial w\rangle + \langle \partial^2 x/\partial v\, \partial w, x(u) - p\rangle)$; and at a critical point q the index and the nullity of $L_p|N$ are those of this form at $u = \sigma(q)_1$. *Book:* §2 is written for abstract manifolds, and the reduction to a parametrization is used without comment later; the statements on L_p are Milnor's criterion of p. 35 and his formula of p. 36 (§6),

proved in this chapter's files. *Hidden hypotheses:* none. *Deviation (not in the book):* the identifications themselves. The Hessian of $F \circ x$ is not the restriction of the ambient Hessian of F: the chain rule adds $dF_q(\partial^2 x/\partial v\,\partial w)$, which does not vanish at a critical point of $F|N$ and which, for L_p, is the second-fundamental-form term of Milnor's formula.

V1.7 (Lemma 2.1, Hadamard's lemma; `Morse.exists_hadamard_coord`, `Morse.exists_hadamard_on`, `Morse.exists_hadamard_two_on`, `Morse.contDiffOn_intervalIntegral`). Let $V \subseteq \mathbb{R}^n$ be open and convex with $0 \in V$, and f smooth on V with $f(0) = 0$. Then $f(x) = \sum_i x_i g_i(x)$ on V for some functions g_i smooth on V with $g_i(0) = \partial f/\partial x_i(0)$. More generally, let V be open and star-shaped about 0 in a real normed space E and f smooth on V, with values in a complete normed space and $f(0) = 0$. Then $f(x) = L(x)\,x$ on V for a family of linear maps L smooth on V with $L(0) = df_0$; and if moreover $df_0 = 0$, then $f(x) = B(x)(x,x)$ on V for a family of bilinear maps B smooth on V, with $B(x)$ symmetric for each $x \in V$ and $B(0) = \frac{1}{2}d^2 f_0$. (The proofs take $L(x) = \int_0^1 df_{tx}\,dt$ and $B(x) = \int_0^1 (1-t)\,d^2 f_{tx}\,dt$.) *Book:* p. 5, Lemma 2.1, for f smooth in a convex neighbourhood V of 0 in $\mathbb{R}^n$; on p. 7 Milnor applies it twice, symmetrizes, and asserts without proof that the coefficients at 0 are $\frac{1}{2}\,\partial^2 f/\partial x^i \partial x^j(0)$, which is the second-order form. *Hidden hypotheses:* V open; and the smoothness of the g_i, which the book does not argue, is differentiation under the integral sign on an open set (`Morse.contDiffOn_intervalIntegral`: if h is C^n on an open $U \subseteq E \times \mathbb{R}$ containing $V \times [0,1]$, V open, then $x \mapsto \int_0^1 h(x,t)\,dt$ is C^n on V). *Deviation (stronger):* star-shaped instead of convex, any normed source and any complete target. *Proof:* the second-order form is Taylor's formula with integral remainder rather than two applications of Lemma 2.1, which makes the value $B(0) = \frac{1}{2}d^2 f_0$ immediate.

V1.8 (Lemma 2.2, the Morse lemma; `Morse.morse_lemma`, `Morse.exists_morse_chart`, `Morse.morse_lemmaM`, `Morse.exists_morse_chartM`). Let E be a finite-dimensional real normed space, $n = \dim E$, f smooth on an open set $U \ni p$, and p a critical point at which $d^2 f_p$ is non-degenerate, of index λ. Then there is a homeomorphism y of an open neighbourhood $U' \subseteq U$ of p onto an open subset of $\mathbb{R}^n$, C^∞ with C^∞ inverse, such that $y(p) = 0$ and $f = f(p) - (y^1)^2 - \cdots - (y^\lambda)^2 + (y^{\lambda+1})^2 + \cdots + (y^n)^2$ throughout U'. In coordinate-free form, there is such a φ onto an open subset of E with $\varphi(p) = 0$, $d\varphi_p = \mathrm{id}$ and $f = f(p) + \frac{1}{2}d^2 f_p(\varphi,\varphi)$ on its domain. On a smooth manifold M without boundary modelled on E, with no Hausdorff hypothesis, for f smooth on an open $U \ni p$ and p a non-degenerate critical point of index λ (V1.4): there is a smooth coordinate system y of M (V1.5) with values in $\mathbb{R}^n$, domain an open neighbourhood of p inside U, $y(p) = 0$, and the same normal form on its domain; and there is a chart φ of the maximal atlas, with domain an open neighbourhood of p inside U, $\varphi(p) = 0$ and $f = f(p) + \frac{1}{2}f_{**}(\varphi,\varphi)$ on its domain. *Book:* p. 6, Lemma 2.2. *Hidden hypotheses:* none. *Deviation (stronger):* f need only be smooth near p. *Proof:* not Milnor's induction, which completes one square at a time and does not justify the linear change of the remaining coordinates making $H_{rr}(0) \neq 0$, but one application of the inverse function theorem: with $f(p+x) - f(p) = B(x)(x,x)$ and $B_0 = \frac{1}{2}d^2 f_p$ (V1.7), the map $T \mapsto B_0(T\cdot, T\cdot) + B_0(T\cdot,\cdot) - B_0(\cdot, T\cdot)$ on linear maps of E has invertible derivative $H \mapsto 2B_0(H\cdot,\cdot)$ at the identity, which yields linear maps $C(x)$, smooth in x, with $C(0) = \mathrm{id}$ and $B(x)(v,v) = B_0(C(x)v, C(x)v)$ (`Morse.exists_sqrt_family`), so that $\varphi(p+x) = C(x)x$. Sylvester's law of inertia, from Mathlib, then diagonalizes B_0 with the negative squares first (`Morse.exists_linearEquiv_diagForm`).

V1.9 (the index is forced by the normal form; `Morse.index_eq_of_morse_chart`, `Morse.isNondegenerateCritic` `Morse.indexM_eq_of_morse_chart`). On a real normed space E, $n = \dim E$: if f is C^2 at a critical point p and z is a homeomorphism of an open neighbourhood of p onto an open subset of $\mathbb{R}^n$, C^∞

with C^∞ inverse, with $z(p) = 0$ and $f = f(p) - \sum_{i\le k}(z^i)^2 + \sum_{i>k}(z^i)^2$ on its domain, $k \le n$, then k is the index of f at p. On M as in V1.4, with no hypothesis on f: if z is a smooth coordinate system around p with values in $\mathbb{R}^n$, $z(p) = 0$, $k \le n$, and f has this form on its domain, then p is a non-degenerate critical point of f of index k. *Book:* p. 6, the first paragraph of the proof of Lemma 2.2. *Hidden hypotheses:* none. *Deviation (stronger):* on M, smoothness, criticality and non-degeneracy are conclusions, not hypotheses. The proof is Milnor's, with the independence of coordinates (V1.5) that he uses without comment.

V1.10 (Corollary 2.3; `Morse.IsNondegenerateCriticalPoint.exists_nhds_isCriticalPoint_eq`, `Morse.IsNondegenerateCriticalPointM.exists_nhds_isCriticalPointM_eq`). Let E be finite-dimensional. If $f : E \to \mathbb{R}$ is C^2 at p and p is a non-degenerate critical point, then p has a neighbourhood in which it is the only critical point of f. The same holds on a manifold M without boundary modelled on E, for f of class C^2 at p. *Book:* p. 8, Corollary 2.3, stated after Lemma 2.2 without proof. *Hidden hypotheses:* none. *Deviation (stronger):* only C^2 at p. *Proof:* not through the Morse lemma: $x \mapsto df_x$ has the injective derivative d^2f_p at p, so it does not vanish at the points $x \ne p$ near p.

V1.11 (finitely many critical points; `Morse.isClosed_setOf_isCriticalPointM`, `Morse.isDiscrete_inter_setO` `Morse.finite_inter_setOf_isCriticalPointM`, `Morse.IsMorseFunction.finite_setOf_isCriticalPointM`). Let M be as in V1.4. The critical set of a C^1 function on M is closed. If E is finite-dimensional and f is C^2 with all its critical points non-degenerate, the critical points of f form a discrete set and every compact set contains only finitely many of them; in particular a Morse function on a compact M has finitely many critical points. *Book:* not stated; Milnor uses it on p. 23, where the critical values have no cluster point because each M^a is compact. *Hidden hypotheses:* none; Hausdorffness is not needed. *Deviation (not in the book):* a consequence of V1.10 that the book uses silently.

V1.12 (degenerate critical points, pp. 8–9; `Morse.Examples.cube_degenerate`, `Morse.Examples.isCriticalPoin` `Morse.Examples.hessian_cube`, `Morse.Examples.monkey_degenerate`, `Morse.Examples.isCriticalPoint_mon` `Morse.Examples.hessian_monkey`, `Morse.Examples.isCriticalPoint_sqFst_iff`, `Morse.Examples.not_isNon` `Morse.Examples.isCriticalPoint_sqMul_iff`, `Morse.Examples.not_isNondegenerate_sqMul`, `Morse.not_isNondegenerate_of_frequently`). (a) The only critical point of x^3 on $\mathbb{R}$ is 0, where the Hessian is 0, so it is degenerate. (c) The only critical point of $x^3 - 3xy^2$ on $\mathbb{R}^2$ is the origin, where the Hessian is 0. (d) The critical points of x^2 on $\mathbb{R}^2$ are the points of the line $x = 0$, and all are degenerate. (e) The critical points of x^2y^2 are the points of the two coordinate axes, and all are degenerate. In (d) and (e) degeneracy is the contrapositive of V1.10: a critical point that is a limit of other critical points, of a function C^2 there, is degenerate. *Book:* pp. 8–9, examples (a), (c), (d), (e). *Hidden hypotheses:* none. *Deviation (stronger):* in (a) and (c) the origin is also the only critical point. *Deviation (value differs):* in (d) the book says the critical set is the x axis; since $\partial f/\partial x = 2x$ and $\partial f/\partial y = 0$, it is the y axis. The formalization noticed this.

V1.13 (one-parameter groups; `Morse.IsOneParameterGroup`, `Morse.Generates`, `Morse.IsOneParameterGroup.z` `Morse.IsOneParameterGroup.isIntegralCurve`). On a real normed space E, a *one-parameter group of diffeomorphisms* is a map $\varphi : \mathbb{R} \times E \to E$, C^∞ in (t, x), such that each φ_t is a C^∞ diffeomorphism of E and $\varphi_{t+s} = \varphi_t \circ \varphi_s$ for all t, s. A vector field $X : E \to E$ *generates* φ if for every x the curve $h \mapsto \varphi_h(x)$ has derivative $X(x)$ at $h = 0$. Then $\varphi_0 = \mathrm{id}$, and if φ is generated by X, every orbit $t \mapsto \varphi_t(x)$ is an integral curve of X. *Book:* pp. 9–10, on a manifold; the orbit computation is on p. 10. *Hidden hypotheses:* none. *Deviation (narrower):* only for $M = E$, where a tangent vector is a vector and Milnor's $X_q(f) = \lim_{h\to 0}(f(\varphi_h(q)) - f(q))/h$ becomes

the derivative of $h \mapsto \varphi_h(q)$. That $\varphi_0 = \mathrm{id}$, used implicitly on p. 10, follows from the group law because φ_0 is injective.

V1.14 (global flows, and smooth dependence on the initial point; `Morse.IsBddLipschitz`, `Morse.flow`, `Morse.IsBddLipschitz.isIntegralCurve_flow`, `Morse.IsBddLipschitz.eq_flow`, `Morse.IsBddLipschitz.flow_add`, `Morse.IsBddLipschitz.continuous_flow`, `Morse.IsBddLipschitz.flowHom` `Morse.IsBddLipschitz.of_contDiff`, `Morse.contDiff_flow`, `Morse.contDiff_flow_uncurry`, `Morse.flowDiffeomorph`). Let E be a complete real normed space and $X : E \to E$ bounded and globally Lipschitz, for instance C^1 with compact support. Through every point there is an integral curve of X defined on all of $\mathbb{R}$, every integral curve is a flow line, and the flow satisfies $\varphi_0 = \mathrm{id}$ and $\varphi_{s+t} = \varphi_s \circ \varphi_t$, is jointly continuous in (t, x), and each φ_t is a homeomorphism with inverse φ_{-t}. If X is C^n, $1 \le n \le \infty$, with compact support, then $(t, x) \mapsto \varphi_t(x)$ is C^n on $\mathbb{R} \times E$ and each φ_t is a C^n diffeomorphism. *Book:* p. 10, quoted as well known (citing Graves): locally the equation has a unique solution depending smoothly on the initial conditions; p. 11, "not difficult to verify" that the φ_t obtained by iteration are well defined, smooth, and form a group. *Hidden hypotheses:* none. *Deviation (restated):* Milnor quotes the local theory and iterates it; the formal statements are global, for bounded Lipschitz or compactly supported fields on a vector space, and the smooth dependence, which Mathlib lacks, is proved rather than quoted. *Proof:* on $[0, 1]$ the flow line through x solves $\Psi(\gamma) = x$ with $\Psi(\gamma) = \gamma - \int_0^{\cdot} X(\gamma)$ in the Banach space $C([0, 1], E)$; for a Lipschitz constant below 1, Ψ is a homeomorphism with everywhere invertible derivative (Neumann series) and $\gamma \mapsto X \circ \gamma$ is as smooth as X, so the time-one map is C^n (`Morse.Flow.voltHomeomorph`, `Morse.Flow.contDiff_voltHomeomorph_symm`, `Morse.Flow.contDiff_nem`). General times come from $\varphi_t = (\varphi_{t/N})^N$ and joint smoothness from the flow of $(x, \tau) \mapsto (\rho(\tau)X(x), 0)$ on $E \times \mathbb{R}$; Milnor's iteration of $\varphi_{\epsilon_0/2}$ is replaced by Picard–Lindelöf on intervals of any length, which a bounded field allows.

V1.15 (Lemma 2.4; `Morse.lemma_2_4`). Let E be a complete real normed space and $X : E \to E$ a C^∞ vector field which vanishes outside a compact set. Then there is exactly one one-parameter group of diffeomorphisms of E (V1.13) generated by X, namely its flow (V1.14). *Book:* p. 10, Lemma 2.4, for a manifold M. *Hidden hypotheses:* none on E; on a manifold, the uniqueness of integral curves needs M Hausdorff. *Deviation (narrower):* only for $M = E$; if E is infinite-dimensional the hypothesis forces $X = 0$, so the content is the case $E = \mathbb{R}^n$. The uniqueness, implicit in the book, holds among all one-parameter groups generated by X, because uniqueness of integral curves is global in time.

What is not verified in this chapter

- §1: the identification of the sublevel sets of the torus in cases (2)–(4) (a 2-cell, a cylinder, a genus-one surface bounded by a circle) and the cell attachments of p. 2. Theorem 3.2 on submanifolds (V2.8) has not been applied to the torus. The normal forms at p, q, r, s follow from V1.1 and V1.8 but are not restated.
- p. 4: that M^a is a smooth manifold with boundary $f^{-1}(a)$ when a is not a critical value. Not restated; it is the companion project's Lemma 3 (`Milnor.isHalfSliceSubmanifold_of_regularValue`) applied to $a - f$.
- pp. 4–5: Milnor's intrinsic definition of the Hessian through extending vector fields, and its independence of the extensions; his coordinate formula is used instead (V1.4).
- p. 8, example (b), $e^{-1/x^2} \sin^2(1/x)$: its smoothness at 0 needs the flatness of e^{-1/x^2} against the growth of the derivatives of $\sin^2(1/x)$; not done. Nor are the (non-)submanifold claims

of (d) and (e).

- p. 10: Lemma 2.4 on a manifold other than a vector space. The smooth dependence of V1.14 is proved on vector spaces only (Mathlib has no smooth dependence on initial conditions); for submanifolds of $\mathbb{R}^K$, Chapter 2 uses instead the ambient flow of a field tangent to N (V2.3).
- p. 11: the Remark that compact support cannot be dropped (d/dt on $(0,1)$); not formalized.

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3. Chapter 2 (Milnor §3): partly verified

Milnor's §3 proves that the homotopy type of M^a changes only at critical values (Theorem 3.1); that passing a non-degenerate critical point of index λ attaches a λ-cell (Theorem 3.2, with Remark 3.3 for several critical points on one level and Remark 3.4 for the critical level itself); two lemmas on attaching cells (Lemmas 3.6 and 3.7); and that a manifold carrying a function with only non-degenerate critical points and compact sublevel sets has the homotopy type of a CW-complex with one λ-cell for each critical point of index λ (Theorem 3.5). Theorem 3.1 is verified in three settings: on a finite-dimensional real inner product space (Milnor's $\mathbb{R}^n$ with a Riemannian metric); for $f = F|N$ on a slice submanifold $N \subseteq \mathbb{R}^K$ in the sense of the companion project (V1.6), closed or not, with F smooth on $\mathbb{R}^K$; and on an abstract compact Hausdorff manifold, through Mathlib's Whitney embedding. The unnumbered Remark after it is verified on an explicit example. Theorem 3.2 is verified on $\mathbb{R}^n$ and on slice submanifolds, both as a strong deformation retraction onto $M^{c-\epsilon}$ with an embedded cell and as a homotopy equivalence with Milnor's adjunction space; Remark 3.3 on slice submanifolds; Remark 3.4 on $\mathbb{R}^n$; Lemmas 3.6 and 3.7 for arbitrary spaces. Theorem 3.5 is verified for compact slice submanifolds, with "CW-complex" replaced by an iterated adjunction space with exactly C_λ cells of dimension λ, because Mathlib has no cellular approximation theorem. Not verified: Theorems 3.2 and 3.5 and Remarks 3.3 and 3.4 on abstract manifolds, the non-compact case of Theorem 3.5, and the CW structure.

V2.1 (sublevel sets and strong deformation retracts; `Morse.sublevel`, `Morse.IsStrongDeformationRetract`, `Morse.IsStrongDeformationRetract.exists_homotopyEquiv`). For a real function f on a set and $c \in \mathbb{R}$, $M^c = \{x : f(x) \le c\}$. A subset A of a topological space X is a *strong deformation retract* of X if there are a continuous map $r : X \to X$ with values in A and a homotopy $(t, x) \mapsto r_t(x)$, $0 \le t \le 1$, from $r_0 = \mathrm{id}_X$ to $r_1 = r$ with $r_t(a) = a$ for all t and all $a \in A$. If $A \subseteq B$ are subsets of a space and A is a strong deformation retract of B, the inclusion $A \to B$ is a homotopy equivalence (the proof takes r_1 as its homotopy inverse). *Book:* p. 12 (M^a); p. 13, the deformation r_t in the proof of Theorem 3.1, and the passage from deformation retract to homotopy equivalence. *Hidden hypotheses:* none. *Deviation (stronger):* Milnor uses "deformation retract" without a definition; every deformation he constructs fixes the smaller set at all times, and every formal statement asserts this strong form, which implies the weaker ones.

V2.2 (Theorem 3.1 on a Euclidean space; `Morse.theorem_3_1`, `Morse.exists_diffeomorph_image_sublevel_eq`). Let E be a finite-dimensional real inner product space, $f : E \to \mathbb{R}$ of class C^∞, $a \le b$, and suppose that $f^{-1}[a,b]$ is compact and contains no critical point of f. Then there is a C^∞ diffeomorphism Φ of E with $\Phi(M^a) = M^b$; M^a is a strong deformation retract of M^b; the inclusion $M^a \to M^b$ is a homotopy equivalence; and M^a and M^b are homeomorphic. *Book:* p. 12, Theorem 3.1, for a smooth function on a manifold and $a < b$. *Hidden hypotheses:* none; the inner product is Milnor's Riemannian metric. *Deviation (restated):* "M^a is diffeomorphic to M^b" is formalized as what the proof gives, a diffeomorphism of the whole space carrying M^a onto M^b (Milnor's φ_{b-a}); $a = b$ is allowed. *Proof:* Milnor's. The cut-off ρ, smooth with compact support and equal

to $1/\|\nabla f\|^2$ on $f^{-1}[a,b]$, is constructed (`Morse.exists_cutoff`), and the flow of $\rho\nabla f$ is V1.14; everything after $\frac{d}{dt}f(\varphi_t(q)) = 1$ is proved for any continuous flow on a topological space along which a continuous function has derivative 1 while its value lies in $[a,b]$ (`Morse.IsSlabFlow`, `Morse.IsSlabFlow.image_sublevel`, `Morse.IsSlabFlow.isStrongDeformationRetract`): each trajectory crosses the slab once, at unit speed, which is what Milnor's "clearly" needs, and his two-case r_t is the single continuous formula $r_t(q) = \varphi_{t\min(0,\,a-f(q))}(q)$.

V2.3 (Theorem 3.1 on a submanifold of Euclidean space; `Morse.theorem_3_1_submanifold`, `Morse.theorem_3_1_submanifold_of_isCompact`). Let V be a finite-dimensional real normed space, $N \subseteq V$ a slice submanifold modelled on a finite-dimensional Z (V1.6; N need not be closed), $F : V \to \mathbb{R}$ of class C^∞ and $f = F|N$. Let $a \le b$, and suppose that $f^{-1}[a,b] \subseteq N$ is compact and contains no critical point of f, as a function on the manifold N. Then there is a C^∞ diffeomorphism Φ of N with $\Phi(N^a) = N^b$, N^a is a strong deformation retract of N^b, and the inclusion $N^a \to N^b$ is a homotopy equivalence. When N is compact, the compactness of $f^{-1}[a,b]$ need not be assumed. *Book:* p. 12, Theorem 3.1. *Hidden hypotheses:* none; N is Hausdorff and second countable as a subset of V. *Deviation (narrower):* f must be the restriction of a smooth function on V. *Proof:* no Riemannian metric is used. Slice charts and bump functions give a smooth field Y on V, tangent to N along N, with $dF(Y) > 0$ on $f^{-1}[a,b]$; the field ρY, with $\rho\, dF(Y) = 1$ there, replaces $\rho\,\mathrm{grad}\, f$, its flow on V preserves N (`Morse.flow_mem_of_tangent`), and the slab-flow argument of V2.2 runs on N.

V2.4 (Theorem 3.1 on a compact manifold; `Morse.theorem_3_1_of_compactSpace`). Let M be a compact Hausdorff smooth manifold without boundary modelled on a finite-dimensional real normed space, $f : M \to \mathbb{R}$ of class C^∞, $a \le b$, and suppose that $f^{-1}[a,b]$ contains no critical point of f. Then a C^∞ diffeomorphism of M carries M^a onto M^b, M^a is a strong deformation retract of M^b, and the inclusion $M^a \to M^b$ is a homotopy equivalence. *Book:* p. 12, Theorem 3.1. *Hidden hypotheses:* Hausdorff. *Deviation (narrower):* M compact. *Proof:* Mathlib's Whitney embedding for compact manifolds makes M diffeomorphic to a compact slice submanifold $N \subseteq \mathbb{R}^{K+1}$ (`Morse.exists_diffeomorph_sliceSubmanifold`); f read on N is the restriction of a smooth function on $\mathbb{R}^{K+1}$ (`Morse.exists_contDiff_comp_val_eq`), so V2.3 applies, and its conclusions are carried back to M.

V2.5 (the Remark on p. 13: compactness cannot be omitted; `Morse.RemarkP13.remark_p13`, `Morse.RemarkP13.remark_p13_not_isStrongDeformationRetract`, `Morse.RemarkP13.remark_p13_not_homeom` `Morse.RemarkP13.remark_p13_not_isCompact`, `Morse.RemarkP13.remark_p13_not_isCriticalPoint`, `Morse.RemarkP13.remark_p13_plane`). Let $M = \mathbb{C} \setminus \{i\}$, $f(z) = |z|^2$ on M, $a = \frac{1}{4}$ and $b = 4$. No $z \in \mathbb{C}$ with $\frac{1}{4} \le |z|^2 \le 4$ is a critical point of $|z|^2$, but $f^{-1}[\frac{1}{4}, 4] \subseteq M$ is not compact; $M^{1/4}$ (the closed disc of radius $\frac{1}{2}$) and M^4 (the closed disc of radius 2 without i) are not homotopy equivalent, so $M^{1/4}$ is not a strong deformation retract of M^4, and the two are not homeomorphic. With i put back, Theorem 3.1 (V2.2) applies: in $\mathbb{C}$ the disc of radius $\frac{1}{2}$ is a strong deformation retract of the disc of radius 2. *Book:* p. 13, the Remark after Theorem 3.1, with Diagram 3 (p. 14): a surface from which a point p between the levels a and b has been removed, for which M^a is "clearly" not a deformation retract of M^b. The Remark is unnumbered (Milnor's Remark 3.3 is the statement of p. 19, V2.9). *Hidden hypotheses:* none. *Deviation (restated):* the pictured surface is replaced by the punctured plane, and the conclusion is the stronger one that the two sublevel sets are not even homotopy equivalent; M is open in $\mathbb{C}$, so the critical points of f are those of $|z|^2$. *Proof:* the "clearly" is the non-contractibility of the punctured disc, proved by lifting a null-homotopy of $z \mapsto z - i$ through the covering $\exp : \mathbb{C} \to \mathbb{C} \setminus \{0\}$ (`Morse.RemarkP13.not_contractibleSpace`).

V2.6 (Theorem 3.2 on a Euclidean space; `Morse.theorem_3_2`, `Morse.theorem_3_2_homotopyEquiv`, `Morse.theorem_3_2_attach`). Let E be a finite-dimensional real inner product space, $f : E \to \mathbb{R}$ of class C^∞, p a critical point at which $d^2 f_p$ is non-degenerate, λ its index (V1.3) and $c = f(p)$, and suppose that for some $\epsilon_0 > 0$ the set $f^{-1}[c - \epsilon_0, c + \epsilon_0]$ is compact and contains no critical point other than p. Then there is $\epsilon_1 > 0$ such that for every ϵ with $0 < \epsilon \le \epsilon_1$ there is a map χ from the closed unit ball D^λ of $\mathbb{R}^\lambda$ to E, continuous and injective on D^λ, with $\chi(z) \in M^{c-\epsilon}$ exactly when $|z| = 1$ and $\chi(D^\lambda) \subseteq M^{c+\epsilon}$, such that $M^{c-\epsilon} \cup \chi(D^\lambda)$ is a strong deformation retract of $M^{c+\epsilon}$. Hence the inclusion of $M^{c-\epsilon} \cup \chi(D^\lambda)$ in $M^{c+\epsilon}$ is a homotopy equivalence, and there is an injective continuous $\varphi : S^{\lambda-1} \to M^{c-\epsilon}$ such that $M^{c+\epsilon}$ is homotopy equivalent to $M^{c-\epsilon} \cup_\varphi e^\lambda$ (V1.2). *Book:* p. 14, Theorem 3.2: for all sufficiently small ϵ, $M^{c+\epsilon}$ has the homotopy type of $M^{c-\epsilon}$ with a λ-cell attached (the scan prints "a k-cell" and renders the index as "A"; both are read as λ, as the proof requires). *Hidden hypotheses:* none on E (on a manifold, Hausdorff: V2.8). *Deviation (stronger):* the conclusion is a strong deformation retraction onto $M^{c-\epsilon}$ with an embedded cell, which is what the proof produces; "for all sufficiently small ϵ" is read as "for all ϵ in some $(0, \epsilon_1]$". *Proof:* Milnor's, through his four Assertions (`Morse.sublevel_milnorF_add`, `Morse.isCriticalPoint_milnorF_iff`, `Morse.isStrongDeformationRetract_milnorF`, `Morse.isStrongDeformationRetract_handle`), with these additions. His μ (C^∞, $\mu(0) > \epsilon$, $\mu = 0$ on $[2\epsilon, \infty)$, $-1 < \mu' \le 0$), whose existence he does not show, is $\mu(r) = \frac{3}{4}\int_r^{2\epsilon} \beta$ for a smooth plateau β (`Morse.milnorMu`, `Morse.isMilnorMu_milnorMu`); Assertion 2 uses $\partial F/\partial \eta \ge 1$, where the scan shows > 1, which fails where $\mu' = 0$; and the three cases of r_t in Assertion 4 are one formula $(x, y) \mapsto (x, \sigma_t(x, y)\, y)$, continuous on the whole patch, which pastes with the identity because it moves only points of the compact ellipsoid $\xi + 2\eta \le 2\epsilon$ and keeps them there (`Morse.CellModel.retractModel`, `Morse.CellModel.continuous_retractModel`).

V2.7 (a cell meeting a closed set in its boundary sphere is attached to it; `Morse.isPushout_of_cell`, `Morse.exists_isPushout_union_cell`, `Morse.exists_isPushout_union_cell_of_t2`, `Morse.exists_homotopy`). Let A be a closed subset of a Hausdorff space M and $\chi : \mathbb{R}^n \to M$ continuous and injective on D^n, with $\chi(z) \in A$ exactly when $|z| = 1$, for $z \in D^n$. Then the subspace $A \cup \chi(D^n)$ of M is the pushout of $\varphi = \chi|S^{n-1} : S^{n-1} \to A$ and $S^{n-1} \to D^n$, that is, $A \cup_\varphi e^n$ (V1.2), and φ is injective. Consequently, if $f : M \to \mathbb{R}$ is continuous, $a \le b$, χ is as above for $A = M^a$, $\chi(D^n) \subseteq M^b$, and $M^a \cup \chi(D^n)$ is a strong deformation retract of M^b, then M^b is homotopy equivalent to $M^a \cup_\varphi e^n$. *Book:* p. 16: $e^\lambda \cap M^{c-\epsilon}$ is the boundary $\dot{e}^\lambda$, "so that" e^λ is attached to $M^{c-\epsilon}$ as required. *Hidden hypotheses:* M Hausdorff and A closed, so that the cell is closed and χ is a homeomorphism onto it. *Deviation (not in the book):* the identification of the subspace with the adjunction space is a lemma that Milnor does not state (Lemma D of the exposition). *Proof:* the universal property of the pushout is checked directly, by the pasting lemma for the closed cover of $A \cup \chi(D^n)$ by A and $\chi(D^n)$.

V2.8 (Theorem 3.2 in a Morse patch, and on a submanifold; `Morse.IsMorsePatch`, `Morse.IsMilnorMu`, `Morse.patch_theorem_3_2`, `Morse.theorem_3_2_submanifold`, `Morse.theorem_3_2_attach_submanifold`). (i) The topological half, in any Hausdorff space M. Let X, Y be finite-dimensional real inner product spaces with $\dim X = \lambda$, $f : M \to \mathbb{R}$, $p \in M$, and φ a homeomorphism of an open neighbourhood of p onto an open subset of $X \times Y$ with $\varphi(p) = 0$, $f = f(p) - |x|^2 + |y|^2$ on its domain, and image containing the ball $|x|^2 + |y|^2 \le 2\epsilon$, $\epsilon > 0$. Let μ satisfy Milnor's four conditions (V2.6), and let $F = f - \mu(|x|^2 + 2|y|^2)$ on the domain of φ and $F = f$ elsewhere. If $F^{-1}(-\infty, f(p) - \epsilon]$ is a strong deformation retract of $M^{f(p)+\epsilon}$ (Assertion 3), then the conclusion of V2.6 about the cell holds in M for this ϵ. (ii) Let V, Z be finite-dimensional, $N \subseteq V$ a slice submanifold modelled on

Z, $F : V \to \mathbb{R}$ of class C^∞, $f = F|N$, $p \in N$ a non-degenerate critical point of f of index λ (V1.4), $c = f(p)$, and suppose that for some $\epsilon_0 > 0$ the set $f^{-1}[c-\epsilon_0, c+\epsilon_0]$ is compact and contains no critical point of f other than p. Then the conclusions of V2.6 hold in N: there is $\epsilon_1 > 0$ such that for every $\epsilon \in (0, \epsilon_1]$ there is a λ-cell $\chi(D^\lambda) \subseteq N^{c+\epsilon}$, χ continuous and injective on D^λ, meeting $N^{c-\epsilon}$ exactly in $\chi(S^{\lambda-1})$, such that $N^{c-\epsilon} \cup \chi(D^\lambda)$ is a strong deformation retract of $N^{c+\epsilon}$; and $N^{c+\epsilon}$ is homotopy equivalent to $N^{c-\epsilon} \cup_\varphi e^\lambda$ for an injective φ. *Book:* p. 14, Theorem 3.2. *Hidden hypotheses:* Hausdorff, used for the ellipsoid $\xi + 2\eta \le 2\epsilon$ to be closed in M (for the smoothness of F and the continuity of r_t) and in V2.7; automatic for $N \subseteq V$. *Deviation (narrower):* in (ii), f is the restriction of an ambient smooth function. *Proof:* in (ii) the Morse coordinates are V1.8 on the manifold N, taken inside one slice chart; Assertion 3 is V2.3 applied to F, which requires extending F to a smooth function on V (`Morse.exists_contDiff_comp_val_eq_of_chart`), a step that Milnor's intrinsic Theorem 3.1 does not need.

V2.9 (Remark 3.3, several critical points on one level; `Morse.remark_3_3_submanifold`). Let V, Z, N, F and $f = F|N$ be as in V2.8 (ii), S a finite set of non-degenerate critical points of f, all with value c, and suppose that for some $\epsilon_0 > 0$ the set $f^{-1}[c-\epsilon_0, c+\epsilon_0]$ is compact and all its critical points lie in S. Then there is $\epsilon_1 > 0$ such that for every $\epsilon \in (0, \epsilon_1]$ there are cells $e_p = \chi_p(D^{\lambda_p})$, $p \in S$, λ_p the index of p, each χ_p continuous and injective on D^{λ_p}, meeting $N^{c-\epsilon}$ exactly in $\chi_p(S^{\lambda_p - 1})$ and contained in $N^{c+\epsilon}$, the cells pairwise disjoint, such that $N^{c-\epsilon} \cup \bigcup_{p \in S} e_p$ is a strong deformation retract of $N^{c+\epsilon}$. *Book:* p. 19, Remark 3.3: for k non-degenerate critical points in $f^{-1}(c)$, "a similar proof" shows that $M^{c+\epsilon}$ has the homotopy type of $M^{c-\epsilon} \cup e^{\lambda_1} \cup \cdots \cup e^{\lambda_k}$. *Hidden hypotheses:* as for Theorem 3.2, a compact $f^{-1}[c-\epsilon_0, c+\epsilon_0]$ with no other critical points; and Morse charts with pairwise disjoint domains, which needs Hausdorffness. *Deviation (narrower):* only on slice submanifolds. That the union is $N^{c-\epsilon}$ with the cells attached one at a time is part of V2.12 (`Morse.isFiniteCellComplexType_union_cells`).

V2.10 (Remark 3.4, the critical level; `Morse.remark_3_4`, `Morse.IsStrongDeformationRetract.of_retract`). Under the hypotheses of V2.6 (E a finite-dimensional real inner product space), there is $\epsilon_1 > 0$ such that for every $\epsilon \in (0, \epsilon_1]$: M^c is a strong deformation retract of $M^{c+\epsilon}$; and there is a map χ from D^λ to E, continuous and injective on D^λ, with $\chi(z) \in M^{c-\epsilon}$ exactly when $|z| = 1$, such that the part of $M^{c-\epsilon} \cup \chi(D^\lambda)$ lying in M^c is a strong deformation retract of M^c. (The whole cell lies in M^c, since $f = c - \epsilon|z|^2$ on it; the proof uses this, but the formal statement does not record it.) *Book:* p. 20, the remark printed "Remark 2.4" in the scan and cited as Remark 3.4 on p. 24: a simple modification of the proof of 3.2 shows that M^c is a deformation retract of $M^{c+\epsilon}$, and combining this with 3.2 one sees "easily" that $M^{c-\epsilon} \cup e^\lambda$ is one of M^c. *Hidden hypotheses:* none. *Deviation (narrower):* only on E, not on submanifolds. *Proof:* the second half uses a lemma that Milnor does not state: if $A \subseteq B \subseteq C$ and both A and B are strong deformation retracts of C, then A is a strong deformation retract of B (only the retraction of C onto B is used).

V2.11 (iterated adjunction spaces, in place of CW-complexes; `Morse.IsCellAttachment`, `Morse.IsFiniteCellComplexType`). A space K is an *iterated adjunction space with cells* $(\lambda_1, \dots, \lambda_k)$ if it is obtained from the empty space by attaching a λ_1-cell, then a λ_2-cell, and so on, each along an arbitrary continuous map of its boundary sphere into the space built so far (V1.2). A space X *has the homotopy type of a finite cell complex with cells* $(\lambda_1, \dots, \lambda_k)$ if it is homotopy equivalent to such a K. *Book:* p. 20, the "CW-complex" of Theorem 3.5, for which Milnor refers to J. H. C. Whitehead. *Hidden hypotheses:* none. *Deviation (narrower):* this is weaker than a CW-complex with these cells: the attaching maps need not be cellular, and the cells are attached in the listed order rather than by dimension. Milnor makes the attaching maps cellular by the cellular approximation theorem (p. 23), which Mathlib does not have; with it and Lemma 3.6 such a space is homotopy equivalent

to a CW-complex with the same number of cells in each dimension, a step that is not formalized.

V2.12 (Theorem 3.5 for a compact submanifold of Euclidean space; `Morse.theorem_3_5'`, `Morse.theorem_3_5_count'`, `Morse.theorem_3_5`, `Morse.theorem_3_5_count`). Let V be a finite-dimensional real normed space, $N \subseteq V$ a compact slice submanifold modelled on a finite-dimensional Z, $F : V \to \mathbb{R}$ of class C^∞, and suppose that every critical point of $f = F|N$ is non-degenerate. Then the critical points of f can be listed without repetition as $p_1, \dots, p_k$, with $f(p_1) \leq \dots \leq f(p_k)$, so that N has the homotopy type of a finite cell complex with cells $(\lambda(p_1), \dots, \lambda(p_k))$ in the sense of V2.11, $\lambda(p)$ the index of p. In particular N has the homotopy type of such a complex whose number of cells of dimension λ is, for every λ, the number C_λ of critical points of index λ. If the critical values are distinct, the list is strictly increasing (`Morse.theorem_3_5`, `Morse.theorem_3_5_count`). *Book:* p. 20, Theorem 3.5: if f has no degenerate critical points and every M^a is compact, M has the homotopy type of a CW-complex with one cell of dimension λ for each critical point of index λ; proof pp. 23–24. *Hidden hypotheses:* that M^a is empty below the first critical value (p. 23) needs a global minimum to be a critical point (`Morse.isCriticalPointM_of_forall_le`); that the critical values do not accumulate is V1.11. *Deviation (narrower):* N compact and f the restriction of an ambient function; the non-compact case (infinitely many critical points, Whitehead's theorem) is not done; and the conclusion is an iterated adjunction space, not a CW-complex (V2.11). *Proof:* Milnor's induction over regular values: Theorem 3.1 between critical levels (V2.3), Theorem 3.2 or Remark 3.3 at each level (V2.8, V2.9), V2.7 to attach disjoint cells one at a time (`Morse.isFiniteCellComplexType_union_cells`), and Lemma 3.7 (V2.14) to carry the cells along homotopy equivalences; Lemma 3.6 is not needed, since the attaching maps are not made cellular.

V2.13 (Lemma 3.6; `Morse.lemma_3_6`). Let X be a topological space, $n \geq 0$, and $\varphi_0, \varphi_1 : S^{n-1} \to X$ homotopic maps. Then there is a homotopy equivalence $k : X \cup_{\varphi_0} e^n \to X \cup_{\varphi_1} e^n$ which is the identity on X. *Book:* p. 20, Lemma 3.6 (Whitehead). *Hidden hypotheses:* none. *Proof:* Milnor's formulas for k and ℓ; his "not difficult to verify" that $k\ell$ and ℓk are homotopic to the identities is an explicit homotopy (`Morse.lemma36Homotopy`). Homotopies on an adjunction space are built through the exponential law for the locally compact $[0,1]$ (`Morse.cellHomotopy`), which is the fact, used silently by Milnor, that $[0,1]\times$ (a quotient map) is a quotient map.

V2.14 (Lemma 3.7, and the Assertion of p. 22; `Morse.lemma_3_7`, `Morse.homotopic_of_leftInv_rightInv`). Let X, Y be topological spaces, $\varphi : S^{n-1} \to X$ continuous and $f : X \to Y$ a homotopy equivalence. Then the map $F : X \cup_\varphi e^n \to Y \cup_{f\varphi} e^n$ equal to f on X and to the identity on the cell is a homotopy equivalence. And for maps F, L, R: if $LF \simeq \mathrm{id}$ and $FR \simeq \mathrm{id}$, then $RF \simeq \mathrm{id}$. *Book:* p. 21, Lemma 3.7 (after Hilton): any homotopy equivalence f extends to a homotopy equivalence F; p. 22, the Assertion that a map with a left and a right homotopy inverse is a homotopy equivalence, with R (or L) a two-sided homotopy inverse. *Hidden hypotheses:* none. *Deviation (stronger):* the extension is named, as in Milnor's proof. *Proof:* Milnor's; it uses the formula for k from the proof of Lemma 3.6, not only its statement.

V2.15 (the Remark on p. 24, at regular values; `Morse.isFiniteCellComplexType_sublevel'`, `Morse.isFiniteCellComplexType_sublevel`). Let V, Z, N, F and $f = F|N$ be as in V2.12, and let a be a real number which is not a critical value of f. Then the critical points p with $f(p) < a$ can be listed without repetition as $p_1, \dots, p_j$, with $f(p_1) \leq \dots \leq f(p_j)$, so that N^a has the homotopy type of a finite cell complex with cells $(\lambda(p_1), \dots, \lambda(p_j))$ (V2.11). *Book:* p. 24, the unnumbered Remark at the end of §3: each M^a has the homotopy type of a finite CW-complex with one cell of dimension λ for each critical point of index λ in M^a, even when a is a critical value. *Hidden*

hypotheses: none. *Deviation (narrower):* only for a not a critical value (a critical value would need Remark 3.4 on submanifolds), and with the iterated adjunction space of V2.11.

What is not verified in this chapter

- Theorem 3.1 on a non-compact abstract manifold (with $f^{-1}[a,b]$ compact): the transfer through Whitney's theorem needs a closed embedding of a non-compact manifold into Euclidean space, which Mathlib does not have. On submanifolds of $\mathbb{R}^K$, closed or not, it is V2.3.
- Theorems 3.2 and 3.5 and Remarks 3.3 and 3.4 on abstract manifolds, compact or not: not transferred through the Whitney embedding, as Theorem 3.1 is (V2.4).
- Remark 3.4 on submanifolds, and with it the Remark of p. 24 at a critical value.
- The CW structure in Theorem 3.5 and in the Remark of p. 24: Mathlib has no cellular approximation theorem (V2.11).
- Theorem 3.5 for non-compact M with compact sublevel sets: with finitely many critical points it needs a deformation of M onto some M^a obtained by patching flows; with infinitely many it needs Whitehead's theorem and the fact that a manifold has the homotopy type of a CW-complex, neither of which is in Mathlib.
- The Remark of p. 17 (Smale's handles: $M^{c-\epsilon} \cup H$ is diffeomorphic to $M^{c+\epsilon}$): it is Theorem 3.1 applied to F, but the diffeomorphism is not recorded.

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4. Chapter 3 (Milnor §§4–5): partly verified

Milnor's §4 proves Reeb's theorem (Theorem 4.1, p. 25) from Theorem 3.1 and the lemma of Morse, adds two remarks, and sketches two examples: a manifold carrying a function with three critical points, and $\mathbb{C}P^n$ (pp. 26–27). His §5 (pp. 28–31) defines subadditive functions and proves Lemma 5.1, the weak Morse inequalities (Theorem 5.2), Lemma 5.3, the strong inequalities (4_λ) and Corollary 5.4, for a compact manifold and singular homology over a field. Verified: Reeb's theorem for a compact submanifold of a finite-dimensional real normed space, with every step of Milnor's short proof made explicit (V3.1–V3.6; for an abstract compact manifold it is V4.24); the whole algebra of §5, for an abstract "homology of a filtration" that records exactly the finiteness and the exact sequences of triples that Milnor uses (V3.7–V3.15); formula (3) of Theorem 5.2 by the route of §6, Application 2 (p. 36), without homology: on a compact submanifold N of $\mathbb{R}^{K+1}$, $K \geq 1$, of positive dimension, $\sum_\lambda (-1)^\lambda C_\lambda$ is the same integer for every Morse function, namely the integer $\chi(N)$ of the Poincaré–Hopf theorem of the Milnor project (V3.16–V3.21); and **§5 in singular homology, with no hypothesis left** (V3.26–V3.40). For the last, the homology Milnor quotes is built first, Mathlib having only absolute singular homology: relative singular homology with the exact sequences of pairs and triples, homotopy invariance for pairs, excision, the Mayer–Vietoris sequence, and the homology of disks, spheres and attached cells (V3.26–V3.31). Singular homology over any field is then an instance of the abstract structure (V3.32); the homology of each slab between regular values is computed from Theorems 3.1 and 3.2 (V3.33–V3.34); and Theorem 5.2, (4_λ), Corollary 5.4 and the lacunary principle hold for the Betti numbers over every field of every compact submanifold of a Euclidean space and, through the Whitney embedding, of every compact Hausdorff manifold (V3.35–V3.37, V3.39). The Poincaré–Hopf integer $\chi(N)$, and the Euler number of an abstract compact manifold (V4.18), **are identified with the homological Euler characteristic** $\sum_i (-1)^i \dim H_i(N;F)$, the same for every field F (V3.38). The

setting is that of slice submanifolds (the Milnor project's `Milnor.IsSliceSubmanifold`: subsets covered by slice charts, with the manifold structure those charts give), which are automatically Hausdorff and second countable. Also verified is the example of $\mathbb{C}P^n$ (V3.41–V3.43): it is the submanifold of rank-one Hermitian projections, Milnor's coordinates are slice coordinates on it, his function has the $n+1$ critical points and the indices he states, and the cells and the homology over every field follow (`Morse/Chapter3/ProjectiveSpace.lean`, 450 lines; `ProjectiveMorse.lean`, 431). Not verified: Milnor's two remarks on Reeb's theorem, the example with three critical points, integral homology, and non-compact manifolds (the list at the end). The torus of §1 (pp. 1–3) belongs to the book pages of Chapter 1, but its Lean files (`Morse/Chapter3/TorusSurface.lean`, `Morse/Chapter3/Torus.lean`) live with this chapter and use V3.20, so it is verified here (V3.22–V3.25, with its homology in V3.40). Lean: `Morse/Chapter3/` and `Morse/Homology/`.

V3.1 (extrema of a function; `Morse.isCriticalPointM_of_isLocalMin`, `Morse.isCriticalPointM_of_isLocalM` `Morse.indexM_eq_zero_of_isLocalMin`, `Morse.indexM_eq_finrank_of_isLocalMax`). Let M be a manifold modelled on a real normed space E and $f : M \to \mathbb{R}$. A local minimum or local maximum p of f is a critical point (for every f: in the formal definition of Chapter 1 a point where f is not differentiable counts as critical). If moreover M is a smooth manifold without boundary, E is finite-dimensional with $n = \dim E$, f is smooth on an open set containing p, and p is a non-degenerate critical point, then the index of f at p is 0 if p is a local minimum and n if p is a local maximum. *Book:* used without comment in the proof of Theorem 4.1 (p. 25), where the small sublevel sets at the minimum and the maximum are said to be closed n-cells "by 2.2". *Hidden hypotheses:* none. *Deviation (not in the book):* Milnor never states it. *Proof:* in a Morse chart (`Morse.morse_lemmaM`), f decreases along an axis carrying a minus sign, so a minimum has none, and increases along an axis carrying a plus sign, so a maximum has only minus signs.

V3.2 (Morse charts at the extrema; small sublevel sets are closed balls; `Morse.exists_morse_chart_of_isLocall` `Morse.exists_morse_chart_of_isLocalMax`, `Morse.exists_image_closedBall_eq_sublevel`). (i) In the setting of the second half of V3.1, if p is a non-degenerate local minimum (maximum) of f, f smooth on an open $U \ni p$, there are smooth coordinates y on an open subset of U containing p, with values in $\mathbb{R}^n$ with its Euclidean norm, such that $y(p) = 0$ and $f = f(p) + \|y\|^2$ (resp. $f = f(p) - \|y\|^2$) on the domain of y. (ii) Let X be a compact topological space, $f : X \to \mathbb{R}$ continuous with a strict global minimum at p ($f(p) < f(x)$ for all $x \neq p$), and y a homeomorphism from an open neighbourhood of p onto an open subset of a real normed space F, with $y(p) = 0$ and $f = f(p) + \|y\|^2$ on its domain. Then there is $\delta > 0$ such that for every $0 < \varepsilon < \delta$ the map $v \mapsto y^{-1}(\sqrt{\varepsilon}\, v)$ is continuous and injective on the closed unit ball of F, maps it onto $\{f \le f(p) + \varepsilon\}$, and maps the unit sphere onto the level set $\{f = f(p) + \varepsilon\}$. (Applied to $-f$, the same holds at a strict maximum.) *Book:* p. 25: if ε is small enough, $f^{-1}[0, \varepsilon]$ and $f^{-1}[1-\varepsilon, 1]$ are closed n-cells by Lemma 2.2. *Hidden hypotheses:* the lemma of Morse describes f only near p; that the whole sublevel set lies in the Morse chart needs compactness and the uniqueness of the minimum (off a small coordinate ball $f \ge f(p) + \eta$ for some $\eta > 0$); the gluing at the end of the proof needs the boundary sphere of the cell to be carried onto the level set, which "closed n-cell" does not say. *Deviation (restated):* the closed cell is produced as the image of the closed unit ball under a continuous injective map, with its boundary sphere going onto the level set.

V3.3 (the two critical points are the minimum and the maximum; `Morse.exists_min_max_of_ncard_eq_two`). Let M be a compact smooth manifold without boundary modelled on a finite-dimensional real normed space E with $n = \dim E \ge 1$, and $f : M \to \mathbb{R}$ smooth with exactly two critical points, both non-degenerate. Then there are points p, q with $f(p) < f(q)$ such that the critical points are exactly p and q, $f(p) < f(x)$ for every $x \neq p$, $f(x) < f(q)$ for every $x \neq q$, the index of f is 0 at

p and n at q. *Book:* p. 25: the two critical points must be the minimum and the maximum. *Hidden hypotheses:* $n \geq 1$ (in dimension 0 every point is critical and f may be constant); non-degeneracy, to see that f is not constant (a constant f would make its minimum point also a maximum, of index $n \neq 0$); that no other point takes an extreme value, which V3.2(ii) needs. Hausdorffness is not needed.

V3.4 (Theorem 3.1 also carries the boundary level; `Morse.exists_diffeomorph_image_sublevel_level`, `Morse.IsSlabFlow.image_level`). Let N be a slice submanifold of a finite-dimensional real normed space V, modelled on a finite-dimensional space Z, let $F : V \to \mathbb{R}$ be smooth and $f = F|_N$, and let $a \leq b$ with $f^{-1}[a,b]$ compact and containing no critical point of f. Then there is a smooth diffeomorphism Φ of N with $\Phi(N^a) = N^b$ and $\Phi(f^{-1}(a)) = f^{-1}(b)$, where $N^c = \{f \leq c\}$. *Book:* p. 25: M^ε is homeomorphic to $M^{1-\varepsilon}$ by Theorem 3.1. *Hidden hypotheses:* the gluing needs a homeomorphism that also carries the level $f^{-1}(\varepsilon)$ onto $f^{-1}(1-\varepsilon)$; the statement of Theorem 3.1 does not give this, its proof does (the flow moves the levels of the slab at unit speed). *Deviation (stronger):* the second conclusion is added; Φ is the time-$(b-a)$ map of the flow used for Theorem 3.1 on submanifolds (`Morse.theorem_3_1_submanifold`).

V3.5 (two closed balls glued along their boundary spheres form a sphere; `Morse.alexanderConeHomeomorph`, `Morse.nonempty_homeomorph_sphere_of_union_closedBall`). (i) Alexander's trick: if g is a homeomorphism of the unit sphere of a real normed space E, the cone $x \mapsto \|x\|\, g(x/\|x\|)$, $0 \mapsto 0$, is a homeomorphism of E that preserves norms and extends g. (ii) Let X be a Hausdorff space, $n \geq 0$, D^n and S^{n-1} the closed unit ball and unit sphere of $\mathbb{R}^n$, and $h_1, h_2 : D^n \to X$ continuous and injective, with $h_1(D^n) \cup h_2(D^n) = X$, $h_1(D^n) \cap h_2(D^n) = h_1(S^{n-1})$ and $h_2(S^{n-1}) = h_1(S^{n-1})$. Then X is homeomorphic to the unit sphere $S^n \subseteq \mathbb{R}^{n+1}$. *Book:* p. 25: M is the union of two closed n-cells matched along their common boundary, and a homeomorphism with S^n is "now easy to construct"; none is constructed. *Hidden hypotheses:* the identification $h_2^{-1} \circ h_1$ of the boundary spheres is in general not the identity and must be extended over a ball (Alexander's trick); the inverse is continuous because a continuous bijection from a compact space onto a Hausdorff space is a homeomorphism. *Deviation (not in the book):* Milnor's "easy" step, made explicit. The cone is in general not differentiable at 0, which is where smoothness is lost (compare Milnor's Remark 2).

V3.6 (Theorem 4.1, Reeb's theorem; `Morse.reeb`, `Morse.reeb_of_finrank_pos`, `Morse.reeb_of_finrank_eq_zer` `Morse.reeb_of_contMDiff`). Let N be a compact slice submanifold of a finite-dimensional real normed space V, modelled on a finite-dimensional space of dimension n, and $f = F|_N$ with $F : V \to \mathbb{R}$ smooth. If f has exactly two critical points, both non-degenerate, then N is homeomorphic to the unit sphere $S^n \subseteq \mathbb{R}^{n+1}$. For a compact $N \subseteq \mathbb{R}^{K+1}$ modelled on $\mathbb{R}^n$, $n \geq 1$, the same holds for every smooth $f : N \to \mathbb{R}$ (`Morse.reeb_of_contMDiff`; such an f is the restriction of a smooth function on $\mathbb{R}^{K+1}$, V3.20). *Book:* Theorem 4.1 (p. 25): a compact manifold carrying a differentiable function with only two critical points, both non-degenerate, is homeomorphic to a sphere. *Hidden hypotheses:* "only two" is read as exactly two (on a nonempty compact manifold of positive dimension the extrema already give two); the sphere is S^n with $n = \dim N$; N has no boundary. In dimension 0 Milnor's proof does not apply (f may be constant), but every point is critical, so N has two points (`Morse.reeb_of_finrank_eq_zero`). *Deviation (narrower):* N is a submanifold of a vector space; the abstract compact (Hausdorff) case is V4.24. The conclusion is a homeomorphism, as in Milnor. *Proof:* Milnor's, made complete: V3.3 gives the minimum p and the maximum q; V3.2 gives the closed cells $\{f \leq f(p)+\varepsilon\}$ and $\{f \geq f(q)-\varepsilon\}$ for small ε; V3.4 on $[f(p)+\varepsilon, f(q)-\varepsilon]$ carries the first onto $\{f \leq f(q)-\varepsilon\}$ and its boundary onto $\{f = f(q)-\varepsilon\}$; V3.5 glues the two cells. The normalization $f(p) = 0$, $f(q) = 1$ is not needed.

V3.7 (subadditive and additive functions; `Morse.IsSubadditive`, `Morse.IsAdditive`). Spaces are replaced by the elements of a preordered set ι, $a \le b$ standing for $X_a \subseteq X_b$. A function on pairs is an integer-valued function $S(a,b)$ of two indices, used only for $a \le b$, where $S(a,b)$ stands for Milnor's $S(X_b, X_a)$ (the smaller space is written first). S is subadditive if $S(a,c) \le S(a,b) + S(b,c)$ whenever $a \le b \le c$, and additive if equality always holds. *Book:* p. 28: S is an integer-valued function on certain pairs of spaces, subadditive if $S(X,Z) \le S(X,Y) + S(Y,Z)$ whenever $X \supset Y \supset Z$, additive if equality holds. *Hidden hypotheses:* the class of pairs on which S is defined must be closed under the passage from $(X,Y), (Y,Z)$ to (X,Z); Milnor does not say which pairs. Here every pair $a \le b$ of the index set is allowed. *Deviation (restated):* indices of a preorder in place of spaces, and the reversed argument order.

V3.8 (Lemma 5.1; `Morse.IsSubadditive.le_sum`, `Morse.IsAdditive.eq_sum`, `Morse.le_sum_of_chain`). If S is subadditive and $X_0 \le X_1 \le \ldots \le X_n$ with $n \ge 1$, then $S(X_0, X_n) \le \sum_{k=0}^{n-1} S(X_k, X_{k+1})$, that is, in Milnor's notation, $S(X_n, X_0) \le \sum_{i=1}^{n} S(X_i, X_{i-1})$; if S is additive, equality holds. `Morse.le_sum_of_chain` assumes subadditivity only on the triples $X_0 \le X_m \le X_{m+1}$, which is all the induction uses. *Book:* Lemma 5.1 (p. 28), by induction on n; with $X_0 = \emptyset$ it gives inequality (1). *Hidden hypotheses:* $n \ge 1$: for $n = 0$ the statement would be $S(X_0, X_0) \le 0$, which subadditivity does not give (it gives $S(X,X) \ge 0$).

V3.9 (the homology of a filtration and its Morse numbers; `Morse.LongExactSeq`, `Morse.FiltrationHomology`, `Morse.morseNumber`, `Morse.altSum`, `Morse.FiltrationHomology.ofSteps`). This is the structure in which §5 is verified. Fix a field F and a preordered set ι. A *homology of a filtration* assigns to every degree $i \ge 0$ and indices a, b a vector space $H_i(a,b)$ over F, to be read as $H_i(X_b, X_a; F)$ (only $a \le b$ is ever used), such that $H_i(a,b)$ is finite-dimensional for $a \le b$, and assigns to every triple $a \le b \le c$ a long exact sequence of F-linear maps

$$\cdots \to H_i(a,b) \to H_i(a,c) \to H_i(b,c) \xrightarrow{\partial} H_{i-1}(a,b) \to \cdots \to H_0(a,b) \to H_0(a,c) \to H_0(b,c) \to 0,$$

exact at every term, the last condition being the surjectivity of $H_0(a,c) \to H_0(b,c)$. Nothing else is assumed: no functoriality, no compatibility between the sequences of different triples, no vanishing of $H_*(a,a)$, no excision. For a chain $X_0 \le X_1 \le \ldots \le X_n$ the Betti numbers are $R_i = \dim H_i(X_0, X_n)$ (Milnor's $R_i(M)$ when $X_0 = \emptyset$, $X_n = M$), and the Morse numbers are $C_i = \sum_{k<n} \dim H_i(X_k, X_{k+1})$. The truncated alternating sum of a sequence d is $S_\lambda(d) = d_\lambda - d_{\lambda-1} + \cdots \pm d_0$. On a chain indexed by $\mathbb{N}$, `Morse.FiltrationHomology.ofSteps` builds the structure from the exact sequences and the finiteness of the slabs $H_i(k, k+1)$ and of the $H_i(k,k)$ alone, so that the finiteness of $H_*(M;F)$ is a consequence rather than an assumption. *Book:* pp. 28–30: singular homology with coefficients in a field F, the exact sequence of a triple $X \supset Y \supset Z$, and $R_\lambda(X,Y) = \dim H_\lambda(X,Y;F)$ for the pairs where it is finite. *Hidden hypotheses:* that the sequence of a triple ends with $\to 0$ (Milnor's exact sequence "$\ldots \to D \to 0$", p. 29) is the surjectivity of $H_0(X,Z) \to H_0(X,Y)$. *Deviation (restated):* this is an abstract structure, not singular homology. Every statement V3.10–V3.15 is a theorem about every such structure; singular homology over a field is an instance (V3.32), and the slab computation of p. 29 that applies it to a manifold is V3.33–V3.34.

V3.10 (R_λ is subadditive and χ is additive; `Morse.LongExactSeq.finrank_le`, `Morse.LongExactSeq.finiteDim`, `Morse.FiltrationHomology.isSubadditive_finrank`, `Morse.altSum_finrank_eq_zero_of_exact`, `Morse.LongExactSeq.altSum_eq_of_finrank_eq_zero`, `Morse.FiltrationHomology.isAdditive_eulerChar`). For a homology of a filtration: (i) for every λ, $(a,b) \mapsto \dim H_\lambda(a,b)$ is subadditive; and in the segment $H_\lambda(a,b) \to H_\lambda(a,c) \to H_\lambda(b,c)$ of a long exact sequence, the middle term is finite-

dimensional when the outer two are. (ii) The alternating sum of the dimensions in a finite exact sequence $0 \to V_n \to \dots \to V_0 \to 0$ of finite-dimensional spaces is 0. (iii) If $H_{N+1}(a,b) = 0$ for all $a \le b$, the truncated Euler characteristic $\chi_N(a,b) = \sum_{i \le N} (-1)^i \dim H_i(a,b)$ is additive. *Book:* p. 28: R_λ is subadditive, "as is easily seen" from the exact segment $H_\lambda(Y,Z) \to H_\lambda(X,Z) \to H_\lambda(X,Y)$; the Euler characteristic $\chi(X,Y) = \sum (-1)^\lambda R_\lambda(X,Y)$ is additive, asserted without argument. *Hidden hypotheses:* the closedness of the class of pairs with finite-dimensional homology under triples (the second half of (i)); additivity of χ needs the total homology to be finite, here replaced by truncation at N with vanishing in degree $N+1$. *Deviation (restated):* χ is truncated at a degree N; it is Milnor's χ when the homology vanishes above N.

V3.11 (Theorem 5.2, the weak Morse inequalities; `Morse.FiltrationHomology.finrank_le_morseNumber`, `Morse.FiltrationHomology.sum_neg_one_pow_finrank_eq`, `Morse.morseNumber_eq_card`, `Morse.morseNumber_eq_card_of_single`, `Morse.FiltrationHomology.weak_morse_inequality`). For a homology of a filtration and a chain $X_0 \le \dots \le X_n$ with $n \ge 1$: (2) $R_\lambda \le C_\lambda$ for every λ; (3) if no slab $H_{N+1}(X_k, X_{k+1})$ is nonzero, then $\sum_{i \le N} (-1)^i R_i = \sum_{i \le N} (-1)^i C_i$. In terms of critical points: let P be a finite set (the critical points), each $p \in P$ carrying an index $\mathrm{ind}(p)$ and a step $\mathrm{step}(p) < n$ (the slab containing it), and assume the *cell hypothesis* $\dim H_i(X_k, X_{k+1}) = \#\{p \in P : \mathrm{step}(p) = k,\ \mathrm{ind}(p) = i\}$ for all $k < n$ and all i. Then $C_i = \#\{p \in P : \mathrm{ind}(p) = i\}$ and (2) reads $R_\lambda \le \#\{p \in P : \mathrm{ind}(p) = \lambda\}$; with one critical point per slab this is Milnor's count. *Book:* Theorem 5.2 (p. 29): on the compact manifold M, $R_\lambda(M) \le C_\lambda$ and $\sum (-1)^\lambda R_\lambda(M) = \sum (-1)^\lambda C_\lambda$, C_λ the number of critical points of index λ; just before, $H_*(M^{a_i}, M^{a_{i-1}})$ is computed as the coefficient group in dimension λ_i and 0 otherwise, "by excision". *Hidden hypotheses:* Milnor's choice of $a_1 < \dots < a_k$ with exactly i critical points in M^{a_i} needs distinct critical values (the cell hypothesis allows any number of critical points per slab); his chain starts at $\emptyset = M^{a_0}$ with a_0 never introduced (here X_0 is arbitrary and R_i is the homology of the pair); the slab identification is not literally an excision (one thickens first). *Deviation (narrower):* the topological input, the slab computation of p. 29 (Theorems 3.1 and 3.2, homotopy invariance, excision and $H_*(D^\lambda, S^{\lambda-1})$), is the cell hypothesis, not a theorem; and R_λ is a dimension in the abstract structure of V3.9, not a Betti number of a manifold. Both are supplied for singular homology in V3.32–V3.35.

V3.12 (Lemma 5.3; `Morse.finrank_range_eq_altSum`, `Morse.LongExactSeq.altSum_le`, `Morse.FiltrationHomology.isSubadditive_altSum`). (i) If $V_{n+1} \xrightarrow{h} V_n \to \dots \to V_0 \to 0$ is exact at $V_0, \dots, V_n$ and all V_k are finite-dimensional, then $\mathrm{rank}\, h = \dim V_n - \dim V_{n-1} + \dots \pm \dim V_0$. (ii) For the sequence of a triple $a \le b \le c$, $S_\lambda(a,c) \le S_\lambda(a,b) + S_\lambda(b,c)$, where $S_\lambda = R_\lambda - R_{\lambda-1} + \dots \pm R_0$ of the pair; the proof is Milnor's, through the identity $\mathrm{rank}(\partial : H_{\lambda+1}(b,c) \to H_\lambda(a,b)) = S_\lambda(a,b) - S_\lambda(a,c) + S_\lambda(b,c)$, which is proved as a lemma of its own. (iii) Hence S_λ is subadditive. *Book:* Lemma 5.3 (pp. 29–30), with Milnor's computation of $\mathrm{rank}\, h$ and of $\mathrm{rank}\, \partial$. *Hidden hypotheses:* finite dimensions, for rank–nullity.

V3.13 (the Morse inequalities (4_λ); `Morse.FiltrationHomology.altSum_le_morseNumber`, `Morse.FiltrationHomology.morse_inequality`). For a homology of a filtration and a chain with $n \ge 1$: $R_\lambda - R_{\lambda-1} + \dots \pm R_0 \le C_\lambda - C_{\lambda-1} + \dots \pm C_0$ for every λ; under the cell hypothesis of V3.11, with $C_i = \#\{p \in P : \mathrm{ind}(p) = i\}$. *Book:* p. 30, inequality (4_λ), from Lemma 5.1 for S_λ on $\emptyset \subset M^{a_1} \subset \dots \subset M^{a_k}$. *Hidden hypotheses:* as in V3.11. *Deviation (narrower):* as in V3.11.

V3.14 (consequences of (4_λ); `Morse.le_of_altSum_le`, `Morse.altSum_eq_of_altSum_le`, `Morse.sum_neg_one_pow_eq_of_altSum_le`, `Morse.FiltrationHomology.finrank_eq_zero_of_forall_ne`, `Morse.FiltrationHomology.euler_characteristic_eq`). Let b, c be integer sequences with

(4_λ), $S_\lambda(b) \le S_\lambda(c)$, for every λ. Then $b_\lambda \le c_\lambda$ for every λ; if $c_{\lambda+1} = 0$ and $b_{\lambda+1} \ge 0$, then $S_\lambda(b) = S_\lambda(c)$; if $b \ge 0$ and $c_{N+1} = 0$, then $\sum_{i\le N}(-1)^i b_i = \sum_{i\le N}(-1)^i c_i$. For a filtration with the cell hypothesis, $R_l = 0$ if no critical point has index l, and, if every index is at most N, $\sum_{i\le N}(-1)^i R_i = \sum_{p\in P}(-1)^{\mathrm{ind}(p)}$, the left side being the whole alternating sum. *Book:* p. 30: adding (4_λ) and $(4_{\lambda-1})$ gives (2_λ); comparing (4_λ) with $(4_{\lambda-1})$ for $\lambda > n$ gives (3); if $C_{\lambda+1} = 0$ then $R_{\lambda+1} = 0$ and (4_λ) is an equality. *Hidden hypotheses:* only $\lambda = n+1$ matters in the comparison for $\lambda > n$, and it uses $C_{n+1} = 0$ and $R_{n+1} \ge 0$; that the left side of (3) is the whole alternating sum uses $R_i = 0$ for $i > n$, which is (2). Milnor's intermediate step $R_{\lambda+1} = 0$ is not needed: $R_{\lambda+1} \ge 0$ suffices.

V3.15 (Corollary 5.4 and the lacunary principle; `Morse.eq_of_altSum_le_succ`, `Morse.eq_of_altSum_le_zero`, `Morse.FiltrationHomology.finrank_eq_morseNumber_of_succ`, `Morse.FiltrationHomology.finrank_eq_mor` `Morse.eq_of_altSum_le_of_lacunary`, `Morse.lacunary_of_odd`, `Morse.FiltrationHomology.finrank_eq_mor`
For sequences $b \ge 0$ and c satisfying every (4_λ), and so for the Betti and Morse numbers of a chain: if $\lambda \ge 1$ and $c_{\lambda+1} = c_{\lambda-1} = 0$, then $b_\lambda = c_\lambda$ and $b_{\lambda+1} = b_{\lambda-1} = 0$; if $c_1 = 0$, then $b_0 = c_0$ and $b_1 = 0$; and if no two consecutive c_i are nonzero (for instance, if every odd c_i vanishes), then $b_i = c_i$ for every i. *Book:* Corollary 5.4 (p. 31): if $C_{\lambda+1} = C_{\lambda-1} = 0$ then $R_\lambda = C_\lambda$ and $R_{\lambda+1} = R_{\lambda-1} = 0$; Milnor adds that this gives the homology of complex projective space (§4) without Theorem 3.5. *Hidden hypotheses:* for $\lambda = 0$ the condition $C_{\lambda-1} = 0$ is empty. *Deviation (not in the book):* the lacunary form. Corollary 5.4 yields Betti numbers over a field, not integral homology groups. For the Betti numbers of a compact manifold it is V3.37 and V3.39; $\mathbb{C}P^n$ is not constructed (see the list at the end).

V3.16 (the numbers C_λ of a function; `Morse.critCount`, `Morse.finsum_neg_one_pow_indexM_eq_sum_critCoun`
For $f : M \to \mathbb{R}$ on a manifold modelled on a real normed space E, $C_\lambda(f)$ is the number of critical points of f of index λ, set to 0 if there are infinitely many. If E is finite-dimensional and f has finitely many critical points, $\sum_{p \text{ critical}}(-1)^{\mathrm{index}_p f} = \sum_{\lambda=0}^{\dim E}(-1)^\lambda C_\lambda(f)$. *Book:* p. 29: C_λ is the number of critical points of index λ. *Hidden hypotheses:* finiteness, which holds for a Morse function on a compact manifold (non-degenerate critical points are isolated, `Morse.IsMorseFunction.finite_setOf_isCriticalPointM`); $C_\lambda = 0$ for $\lambda > \dim M$, which is why Milnor's sum stops at $\pm C_n$ with $n = \dim M$. *Deviation (restated):* the convention for infinitely many critical points never applies in the theorems below.

V3.17 (the Poincaré–Hopf integer; `Morse.IsEulerNumber`, `Morse.exists_isEulerNumber`). Let $N \subseteq \mathbb{R}^{K+1}$ be a compact nonempty slice submanifold modelled on $\mathbb{R}^{d+1}$. An integer χ is an *Euler number* of N if for every smooth tangent vector field v on N whose zeros are finitely many distinct points $z_1, \dots, z_j$, $\sum_i \mathrm{index}_{z_i}(v) = \chi$, the index being that of the Milnor project (`Milnor.Tub.indexN`, its V6.8: the index at z_i of the extension $x \mapsto v(r(x)) + (x - r(x))$ of v to a tubular neighbourhood, r the nearest-point retraction). For $K \ge 1$ an Euler number exists: this is the Poincaré–Hopf theorem of the Milnor project (`Milnor.Tub.poincare_hopf`, its V6.9), where it is the degree of the Gauss map of the boundary of a tubular neighbourhood. It is unique: by V3.20 every Euler number equals the alternating count of the critical points of any Morse function, and N carries one (V4.13). Below it is written $\chi(N)$. *Book:* Milnor's χ is the Euler characteristic $\sum(-1)^\lambda R_\lambda$ (p. 28); in Application 2 (pp. 36–37) he uses that the index sum of a vector field is a topological invariant (citing Steenrod, §39.7). *Hidden hypotheses:* N compact and nonempty; $K \ge 1$, i.e. ambient dimension at least 2; the zeros indexed without repetition. *Deviation (restated):* $\chi(N)$ is defined through vector fields; its identification with $\sum_i (-1)^i \dim H_i(N; F)$ is V3.38.

V3.18 ($\mathrm{sgn}\det = (-1)^{\mathrm{index}}$; `Morse.index_eq_ncard_eigenvalues_neg`, `Morse.isNondegenerate_iff_bijective`

`Morse.sgn_det_eq_neg_one_pow_index`, `Morse.sgn_det_eq_neg_one_pow_index_of_symm`). Let H be a symmetric bilinear form on a finite-dimensional real inner product space and A the operator with $H(v,w) = \langle Av, w\rangle$. The index of H (the largest dimension of a subspace on which H is negative definite) is the number of negative eigenvalues of A; H is non-degenerate iff A is bijective; and if H is non-degenerate, $\operatorname{sgn}\det A = (-1)^{\operatorname{index} H}$. The right side does not involve the inner product, so the sign is the same for the operators representing H with respect to any two inner products. *Book:* used without comment in Application 2 (p. 36), in the claim that $(-1)^\lambda$ is the index of grad f. *Hidden hypotheses:* none. *Deviation (not in the book). Proof:* the spectral theorem and Sylvester's law of inertia, from Mathlib.

V3.19 (the index of the gradient field is $(-1)^\lambda$; `Morse.tangentialGradient`, `Morse.inner_tangentialGradient`, `Morse.tangentialGradient_eq_zero_iff`, `Morse.dret_eq_starProjection`, `Morse.contMDiff_tangentialGra` `Morse.index_hessT`, `Morse.isNondegenerate_hessT_iff`, `Morse.bijective_dvT_tangentialGradient`, `Morse.indexN_tangentialGradient`). Let $N \subseteq \mathbb{R}^{K+1}$, $K \geq 1$, be a compact nonempty slice submanifold modelled on $\mathbb{R}^{d+1}$, and $F : \mathbb{R}^{K+1} \to \mathbb{R}$ smooth, $f = F|_N$. The tangential gradient $v(q) = P_q \nabla F(q)$, P_q the orthogonal projection onto T_qN, is the gradient of f for the metric induced from $\mathbb{R}^{K+1}$ ($\langle v(q), t\rangle = dF_q(t)$ for $t \in T_qN$); its zeros are exactly the critical points of f; it is a smooth tangent field. If q is a non-degenerate critical point of f of index λ, the derivative of v at q is an invertible endomorphism of T_qN whose determinant has sign $(-1)^\lambda$, and the index of v at q is $(-1)^\lambda$. *Book:* Application 2 (p. 36): $(-1)^\lambda$ is the index of grad f at a point where f has index λ, asserted without proof. *Hidden hypotheses:* a metric must be chosen (here the induced one); the Hessian of $F|_N$ is not the restriction of the Hessian of F, since at a critical point $\nabla F(q)$ is normal to N, not zero; the derivative of v at q is an endomorphism of T_qN only because $v(q) = 0$ (proved in the Milnor project, its V6.13; `Milnor.Tub.dvT`). *Deviation (not in the book):* Milnor gives no proof. *Proof:* F is replaced by $F \circ r$, r the nearest-point retraction, whose derivative at a point of N is P_q: $F \circ r$ agrees with F on N and is critical at q as a function on $\mathbb{R}^{K+1}$, $v = \nabla(F \circ r)$ on N, and the Hessian of f in a slice parametrization is the Hessian of $F \circ r$ restricted to T_qN; then V3.18, and the index of a non-degenerate zero (`Milnor.Tub.indexN_eq_sgn_det`, the Milnor project's V6.14).

V3.20 (formula (3) without homology; `Morse.IsEulerNumber.finsum_eq_of_ambient`, `Morse.exists_contDiff_comp_val_eq`, `Morse.IsEulerNumber.finsum_eq`, `Morse.IsEulerNumber.sum_critCou` `Morse.exists_eulerNumber_eq_sum_critCount`, `Morse.sum_critCount_eq_of_isMorseFunction`, `Morse.IsEulerNumber.eq_sum_critCount`). Let $N \subseteq \mathbb{R}^{K+1}$, $K \geq 1$, be a compact nonempty slice submanifold of dimension $d+1 \geq 1$. There is an integer $\chi(N)$, an Euler number of N in the sense of V3.17, such that every Morse function f on N (smooth, with only non-degenerate critical points) satisfies

$$\sum_{\lambda=0}^{d+1} (-1)^\lambda C_\lambda(f) = \sum_{q \text{ critical}} (-1)^{\operatorname{index}_q f} = \chi(N).$$

Consequently the alternating count is the same for all Morse functions on N, and every Euler number of N equals the alternating count of any one Morse function. Every smooth $f : N \to \mathbb{R}$ is the restriction of a smooth function on $\mathbb{R}^{K+1}$ (`Morse.exists_contDiff_comp_val_eq`). *Book:* p. 29: for a compact M and a function with isolated, non-degenerate critical points, $\chi(M) = C_0 - C_1 + C_2 - \cdots \pm C_n$, recorded as formula (3) of Theorem 5.2; Application 2 (pp. 36–37). *Hidden hypotheses:* "isolated" is redundant for non-degenerate critical points; $\pm C_n$ takes $n = \dim M$; distinct critical values are not needed; in Application 2, "any differentiable function" must be one with only non-degenerate critical points, and "the critical points of X" are the zeros of X; $\dim N \geq 1$ and $K \geq 1$

(the Milnor project computes the index of a non-degenerate zero only in positive dimension, and its Poincaré–Hopf theorem needs $K \geq 1$). *Deviation (restated):* $\chi(N)$ is the Poincaré–Hopf integer; what is verified here is the topological invariance of $\sum(-1)^\lambda C_\lambda$. Its equality with $\sum(-1)^i R_i(N)$, formula (3) as Milnor states it, is V3.35, and the identification of $\chi(N)$ with the homological Euler characteristic is V3.38. N is a submanifold of Euclidean space; for an abstract compact manifold see V4.19. *Proof:* Application 2 run in the direction that needs no homology: the gradient field of f (V3.19) has index sum $\sum(-1)^\lambda C_\lambda(f)$, and every tangent field has the index sum $\chi(N)$.

V3.21 (the sphere; `Morse.isEulerNumber_sph`, `Morse.sum_critCount_sph`, `Morse.heightFun`, `Morse.isCriticalPointM_heightFun_iff`, `Morse.hessT_heightFun`, `Morse.indexM_heightFun_pos`, `Morse.indexM_heightFun_neg`, `Morse.isMorseFunction_heightFun`, `Morse.critCount_heightFun`, `Morse.sum_critCount_heightFun`, `Morse.IsEulerNumber.eq_sph`). Let $S^n \subseteq \mathbb{R}^{n+1}$, $n \geq 1$, be the unit sphere, as the slice submanifold `Milnor.Sphere.sphSub` of the Milnor project. Its Euler number is $1 + (-1)^n$ (from `Milnor.Sphere.indexSum_sph`), so every Morse function on S^n has $\sum_\lambda(-1)^\lambda C_\lambda = 1 + (-1)^n$. For a unit vector e, the height function $h(x) = \langle e, x\rangle$ on S^n has exactly the critical points $\pm e$; its Hessian on $T_q S^n = q^\perp$ is $-\langle e, q\rangle\langle\cdot,\cdot\rangle$; it is a Morse function of index n at e and 0 at $-e$, so $C_0 = C_n = 1$ and all other $C_\lambda = 0$; and every Euler number of S^n equals $1 + (-1)^n$, recomputed from h by V3.20. *Book:* not in the book. *Hidden hypotheses:* $n \geq 1$. *Deviation (not in the book):* a check of V3.20 on S^n against the Milnor project's direct computation of the index sum; h also satisfies the hypotheses of Reeb's theorem (V3.6).

The next four items are Milnor's opening example (§1, pp. 1–3), from the book pages of Chapter 1. They are verified here because their Lean files live with this chapter and use V3.20.

V3.22 (the torus of Diagram 1 is a compact surface; `Morse.torus`, `Morse.Torus.poly`, `Morse.Torus.mem_torus_iff`, `Morse.Torus.dpoly_ne_zero`, `Morse.Torus.isRegularValue_poly1`, `Morse.isSliceSubmanifold_torus`, `Morse.isCompact_torus`, `Morse.nonempty_torus`, `Morse.Torus.mem_tange` `Morse.Torus.finrank_tangentSubspace_torus`). For $0 < \rho < R$ let $T = \{x \in \mathbb{R}^3 : (\sqrt{x_1^2 + x_2^2} - R)^2 + x_3^2 = \rho^2\}$, the torus of revolution about the x_3-axis with core radius R and tube radius ρ. T is the zero set of $g(x) = (\|x\|^2 + R^2 - \rho^2)^2 - 4R^2(x_1^2 + x_2^2)$, and dg does not vanish on T; hence T is a 2-dimensional slice submanifold of $\mathbb{R}^3$, compact and nonempty, and its tangent plane at q is $\ker dg_q$, of dimension 2. *Book:* p. 1: a torus M tangent to a plane V, drawn standing upright (Diagram 1); no equations are given, and nothing is said about why M is a smooth compact surface. *Hidden hypotheses:* $\rho < R$ (a ring torus, which is smooth; it also makes the four critical values of V3.23 distinct, which (3) and (4) of p. 1 presuppose). *Deviation (restated):* the surface is fixed as the torus of revolution.

V3.23 (the height function: four critical points, of indices $0, 1, 1, 2$; `Morse.Torus.height`, `Morse.Torus.isCriticalPointM_height_iff`, `Morse.Torus.criticalValues_height`, `Morse.Torus.onAxis_sp` `Morse.Torus.pointP_spec`, `Morse.Torus.pointQ_spec`, `Morse.Torus.pointR_spec`, `Morse.Torus.pointS_spec` `Morse.Torus.isMorseFunction_height`, `Morse.Torus.critCount_height`, `Morse.hessian_comp_sliceParam_c` `Morse.isNondegenerate_and_index_of_diag`). Let $f = x_1 + R + \rho$ on T, the height above the plane $V = \{x_1 = -(R + \rho)\}$, which touches T at $p = (-(R + \rho), 0, 0)$; the axis of revolution is parallel to V, so the torus stands upright. The critical points of f are exactly p, $q = (-(R-\rho), 0, 0)$, $r = (R - \rho, 0, 0)$ and $s = (R + \rho, 0, 0)$, with critical values $0, 2\rho, 2R, 2R + 2\rho$. All four are non-degenerate, of indices $0, 1, 1, 2$; f is a Morse function on T, with $C_0 = 1$, $C_1 = 2$, $C_2 = 1$ and $C_\lambda = 0$ for $\lambda \geq 3$. *Book:* p. 3: the points p, q, r, s at which the homotopy type of M^a changes are the critical points of f; in suitable coordinates $f = x^2 + y^2$ at p, $f = \text{const} + x^2 - y^2$ at q and r, $f = \text{const} - x^2 - y^2$ at s. *Hidden hypotheses:* the upright position (lying flat, the height function

has degenerate critical circles); the critical points and the normal forms are asserted without verification. *Deviation (restated):* the normal forms are not restated for the torus; they follow from the lemma of Morse (`Morse.morse_lemmaM`) with the indices computed here. *Proof:* Lagrange's rule. At a point $(a,0,0)$ of T, $dg = \kappa\, dx_1$ with $\kappa \neq 0$, so $G = f - \kappa^{-1} g$ agrees with f on T and is critical at the point as a function on $\mathbb{R}^3$; the Hessian of $f|_T$ is then d^2G restricted to the tangent plane, a diagonal form whose entries are, up to sign, the principal curvatures.

V3.24 (Milnor's (1) and (5), and Theorem 3.1 on the torus; `Morse.Torus.sublevel_height_eq_empty`, `Morse.Torus.sublevel_height_eq_univ`, `Morse.theorem_3_1_torus`). $T^a = \emptyset$ for $a < 0$, and $T^a = T$ for $a \geq 2R + 2\rho = f(s)$. If $a \leq b$ and $[a,b]$ contains none of the critical values $0, 2\rho, 2R, 2R + 2\rho$, then a smooth diffeomorphism of T carries T^a onto T^b, T^a is a strong deformation retract of T^b, and the inclusion $T^a \to T^b$ is a homotopy equivalence; in particular the T^a with a in any one of $(0, 2\rho)$, $(2\rho, 2R)$, $(2R, 2R + 2\rho)$ are diffeomorphic to one another. *Book:* p. 1: (1) M^a is vacuous if $a < 0 < f(p)$ (as printed in the scan); (5) M^a is the full torus if $f(s) < a$; (2)–(4) M^a is a 2-cell, a cylinder, a compact genus-one surface bounded by a circle. *Hidden hypotheses:* $f(p) = 0$ because V touches M at p, so the hypothesis of (1) must be read as $a < 0 = f(p)$; (5) holds for $a = f(s)$ as well. *Deviation (narrower):* of (2)–(4) only what Theorem 3.1 gives is verified; the homeomorphism types are not.

V3.25 (the Euler number of the torus is 0; `Morse.isEulerNumber_torus`, `Morse.IsEulerNumber.eq_torus`, `Morse.sum_critCount_torus`, `Morse.IsEulerNumber.eq_torus_of_rotation`). 0 is an Euler number of T (V3.17), and every Euler number of T is 0: by V3.20 from $C_0 - C_1 + C_2 = 1 - 2 + 1$, and independently of Morse theory, because the rotation field $(-x_2, x_1, 0)$ is a smooth tangent field on T without zeros. Hence every Morse function on T has $C_0 - C_1 + C_2 = 0$. *Book:* not in the book; p. 2 describes the passage from (1) to (5) as attaching a 0-cell, two 1-cells and a 2-cell, which gives the same count. *Hidden hypotheses:* none. *Deviation (not in the book):* the value is the Poincaré–Hopf integer; by V3.38 it is also the homological Euler characteristic of T (V3.40).

The next six items build the homology that Milnor quotes in §5 (Mathlib, v4.33.1, has the singular chain complex, absolute singular homology and its homotopy invariance, but no relative homology). Below, $\Bbbk$ denotes a field (Milnor's F; the letter F is kept for ambient functions), and the spaces whose homology is taken lie in the universe of the coefficients, a formality of the type theory.

V3.26 (relative singular homology and the exact sequence of a pair; `Morse.Homology.relChains`, `Morse.Homology.relHomology`, `Morse.Homology.relShortComplex_shortExact`, `Morse.Homology.rel`, `Morse.Homology.relHomology_exact`, `Morse.Homology.relHomology_exact`, `Morse.Homology.relHomology_e` `Morse.Homology.epi_homologyMap_rel _zero`, `Morse.Homology.rel _naturality`, `Morse.Homology.relHomolo` `Morse.Homology.isZero_relHomology_diag`, `Morse.Homology.isIso_relHomology_map`, `Morse.Homology.relHomology_map_eq_of_homotopy`, `Morse.Homology.isIso_relHomology_map_of_isStrongD` `Morse.Homology.isZero_relHomology_of_isStrongDeformationRetract`, `Morse.Homology.singularHomology` Coefficients are an object R of an abelian category with coproducts (for instance a field, or any commutative ring); pairs (X, A) are embeddings $A \hookrightarrow X$ (Mathlib's `TopPair`). The relative chain complex $C(X, A)$ is the cokernel of the chain map $C(A) \to C(X)$ of singular chains, which is injective, so that $0 \to C(A) \to C(X) \to C(X, A) \to 0$ is exact; $H_n(X, A)$ is its homology, and both are functorial in maps of pairs. The sequence

$$\cdots \to H_{n+1}(X, A) \xrightarrow{\partial} H_n(A) \to H_n(X) \to H_n(X, A) \xrightarrow{\partial} H_{n-1}(A) \to \cdots \to H_0(X) \to H_0(X, A) \to 0$$

is exact at every place, and ∂ is natural in maps of pairs. $H_n(X, \emptyset) \cong H_n(X)$ naturally, and

$H_n(X, X) = 0$. A map of pairs inducing isomorphisms in the homology of both spaces induces isomorphisms in relative homology (the five lemma). Maps of pairs that are homotopic through maps of pairs induce the same maps in relative homology. If $A \subseteq B \subseteq X$ are subsets of a space and B is a strong deformation retract of X, then $H_n(B, A) \to H_n(X, A)$ is an isomorphism and $H_n(X, B) = 0$. With these data singular homology is a homotopy-invariant homology pretheory in the sense of Mathlib's formulation of the Eilenberg–Steenrod axioms. *Book:* p. 28: homology of pairs with coefficients in a field and its exact sequences; p. 29: pairs replaced by homotopy equivalent ones; all quoted as standard. *Hidden hypotheses:* the first equality of p. 29 needs homotopy equivalences of pairs, here strong deformation retracts. *Deviation (not in the book):* built here, since Mathlib has no relative singular homology. *Proof:* homotopy invariance for pairs from the absolute one, through the cylinder $(X \times I, A \times I)$ and the five lemma.

V3.27 (the exact sequence of a triple; `Morse.Homology.TopTriple`, `Morse.Homology.TopTriple.ofSubsets`, `Morse.Homology.cokernel_comp_shortExact`, `Morse.Homology.TopTriple.shortComplex_shortExact`, `Morse.Homology.TopTriple.` , `Morse.Homology.TopTriple.exact` , `Morse.Homology.TopTriple.exact` , `Morse.Homology.TopTriple.exact` , `Morse.Homology.TopTriple.epi_map _zero`, `Morse.Homology.TopTriple.` For a triple $Z \subseteq Y \subseteq X$ (two composable embeddings, for instance subsets $A \subseteq B \subseteq D$ of a space), the inclusions of pairs give a short exact sequence $0 \to C(Y, Z) \to C(X, Z) \to C(X, Y) \to 0$, hence the sequence

$$\cdots \to H_{n+1}(X, Y) \xrightarrow{\partial} H_n(Y, Z) \to H_n(X, Z) \to H_n(X, Y) \xrightarrow{\partial} H_{n-1}(Y, Z) \to \cdots \to H_0(X, Z) \to H_0(X, Y$$

exact at every place, whose ∂ is the boundary of the pair (X, Y) followed by $H_n(Y) \to H_n(Y, Z)$. *Book:* p. 28, the segment $H_\lambda(Y, Z) \to H_\lambda(X, Z) \to H_\lambda(X, Y)$; p. 30, "the homology exact sequence of a triple $X \supset Y \supset Z$", used with its final "$\to 0$" (p. 29). *Hidden hypotheses:* none. *Deviation (not in the book):* quoted by Milnor, built here. *Proof:* the third isomorphism theorem $(C(X)/C(Z))/(C(Y)/C(Z)) \cong C(X)/C(Y)$, in any abelian category.

V3.28 (excision; `Morse.Homology.smallChainsHomotopyEquiv`, `Morse.Homology.excision_of_union`, `Morse.Homology.excision`, `Morse.Homology.excision_of_closure_subset_interior`**).** (i) If the interiors of a family of subsets of a space X cover X, the inclusion of the chains made of simplices each lying in one member of the family into all singular chains is a chain homotopy equivalence (any ring of coefficients). With coefficients in a commutative ring R, for subsets of a space E, interiors and closures taken in X: (ii) if $A, B \subseteq X$ have interiors covering X, then $(B, A \cap B) \to (X, A)$ induces isomorphisms $H_n(B, A \cap B) \cong H_n(X, A)$ for all n; (iii) if $Z \subseteq A \subseteq X$ and the closure of Z lies in the interior of A, then $(X \setminus Z, A \setminus Z) \to (X, A)$ induces isomorphisms in all degrees; the same holds with closure and interior taken in E. *Book:* p. 29, "by excision", quoted. *Hidden hypotheses:* the step Milnor calls excision is not literally an instance of (iii) (V3.31). *Deviation (not in the book):* proved here (Hatcher, *Algebraic Topology*, Proposition 2.21 and Theorem 2.20), by barycentric subdivision with the diameter bound $(n/(n+1))^k$.

V3.29 (the Mayer–Vietoris sequence; `Morse.Homology.mvShortComplex_shortExact`, `Morse.Homology.mvMap` , `Morse.Homology.mvMap` , `Morse.Homology.mv` , `Morse.Homology.mv_exact` , `Morse.Homology.mv_exact` , `Morse.Homology.mv_exact` **).** With coefficients in a commutative ring R: if $A, B \subseteq X$ have interiors covering X, then

$$\cdots \to H_{n+1}(X) \xrightarrow{\partial} H_n(A \cap B) \xrightarrow{(i_*, -i'_*)} H_n(A) \oplus H_n(B) \xrightarrow{j_* + j'_*} H_n(X) \xrightarrow{\partial} H_{n-1}(A \cap B) \to \cdots,$$

i, i', j, j' the inclusions, is exact at every place. *Book:* not used in §5. *Hidden hypotheses:* none.

Deviation (not in the book): part of the infrastructure; the naturality of ∂ is not proved. *Proof:* the short exact sequence $0 \to C(A \cap B) \to C(A) \oplus C(B) \to C^{\{A,B\}}(X) \to 0$ and (i) of V3.28.

V3.30 (the homology of disks, spheres and cells; `Morse.Homology.relHomologyDiskPairIso`, `Morse.Homology.isZero_relHomology_diskPair`, `Morse.Homology.finrank_relHomology_diskPair`, `Morse.Homology.sphereHomologyZeroIso`, `Morse.Homology.sphereHomologyTopIso`, `Morse.Homology.sphereH` `Morse.Homology.isZero_sphereHomology`, `Morse.Homology.finrank_sphereHomology`, `Morse.Homology.cellPair`, `Morse.Homology.cellPairIso`, `Morse.Homology.isZero_relHomology_cellPair`, `Morse.Homology.finrank_homology_imageSphere`, `Morse.Homology.relHomologyTopDiskPairIso`**).** Coefficients in a commutative ring R; D^n and S^{n-1} are the closed unit ball and the unit sphere of $\mathbb{R}^n$ ($S^{-1} = \emptyset$). $H_i(D^n, S^{n-1}) \cong R$ for $i = n$, and $H_i(D^n, S^{n-1}) = 0$ for $i \neq n$. For $n \geq 1$, $H_0(S^n) \cong R$, $H_n(S^n) \cong R$ and $H_i(S^n) = 0$ for $i \neq 0, n$; $H_0(S^0) \cong R \oplus R$. Over a field, $\dim H_i(D^n, S^{n-1}) = [i = n]$ and $\dim H_i(S^n) = [i = 0] + [i = n]$. More generally, if E is Hausdorff and $\chi : \mathbb{R}^k \to E$ is continuous and injective on D^k, the pair $(\chi(D^k), \chi(S^{k-1}))$ has the homology of (D^k, S^{k-1}); an image of S^n under a map continuous and injective on it has the homology of S^n; and so has Mathlib's disk pair, in every universe. *Book:* p. 29: $H_*(e^\lambda, \dot{e}^\lambda)$ is the coefficient group in dimension λ and 0 otherwise, quoted. *Hidden hypotheses:* none. *Deviation (not in the book):* computed here. *Proof:* induction on k, through the boundary of the triple $\{q\} \subseteq \chi(S^k) \subseteq \chi(D^{k+1})$, q a point onto which the cell contracts, and excision for the upper hemisphere of S^k attached to the lower one (V3.31).

V3.31 (attaching cells to a closed set; `Morse.Homology.isIso_relHomology_map_cellPair`, `Morse.Homology.relHomology_cellAttach_iso`, `Morse.Homology.relHomologyCellAttachTopIso`, `Morse.Homology.isZero_relHomology_cellAttach`, `Morse.Homology.finrank_relHomology_cellAttach`, `Morse.Homology.finrank_relHomology_cellsAttach`, `Morse.Homology.relHomology_cellsAttach_iso`, `Morse.Homology.isZero_relHomology_cellsAttach`**).** Let E be a Hausdorff space, $Y \subseteq X \subseteq E$ with Y closed, and $\chi_j : \mathbb{R}^{k_j} \to E$, $j = 1, \dots, m$, maps continuous and injective on the closed unit balls D^{k_j}, whose cells $e_j = \chi_j(D^{k_j})$ are pairwise disjoint, such that $\chi_j(z) \in Y$ exactly when $\|z\| = 1$, and $X = Y \cup e_1 \cup \cdots \cup e_m$. Coefficients in a commutative ring R. For one cell of dimension k, the inclusion $(e_1, \chi_1(S^{k-1})) \to (X, Y)$ and the characteristic map $(D^k, S^{k-1}) \to (X, Y)$ induce isomorphisms in all degrees: $H_k(X, Y) \cong R$ and $H_i(X, Y) = 0$ for $i \neq k$ (the first statement holds for the unit ball of any finite-dimensional normed space). For several cells, $H_i(X, Y) \cong R^{c_i}$, c_i the number of cells of dimension i, so $H_i(X, Y) = 0$ when no cell has dimension i, and over a field $\dim H_i(X, Y) = c_i$. *Book:* p. 29: $H_*(M^{a_{i-1}} \cup e^{\lambda_i}, M^{a_{i-1}}) = H_*(e^{\lambda_i}, \dot{e}^{\lambda_i})$ "by excision". *Hidden hypotheses:* this is not literally excision: the closure of the set to be excised is not inside the interior of $M^{a_{i-1}}$. One first deforms X minus the centre of the cell onto Y (pushing the punctured cell radially onto its boundary), then excises; Y must be closed. With several cells on one level (Remark 3.3) the exact sequence of a triple gives only an inequality of dimensions; additivity uses the disjointness of the cells. *Deviation (stronger):* any number of disjoint cells, and any commutative ring of coefficients; this is the exposition's Lemma E.

The remaining items are Milnor's statements, in singular homology over a field.

V3.32 (the Morse inequalities for any filtration of a space; `Morse.Homology.TopTriple.longExactSeq`, `Morse.Homology.subsetHomology`, `Morse.Homology.filtrationHomology`, `Morse.Homology.singularMorseNum` `Morse.Homology.finiteDimensional_subsetHomology_of_steps`, `Morse.Homology.singular_weak_morse_ine` `Morse.Homology.singular_morse_inequality`, `Morse.Homology.singular_euler_characteristic`, `Morse.Homology.singularMorseNumber_eq_card`, `Morse.Homology.singular_weak_morse_inequality_of_cel` `Morse.Homology.singular_morse_inequality_of_cells`, `Morse.Homology.singular_euler_characteristic_`

Let $X_0 \subseteq X_1 \subseteq \cdots$ be subsets of a space E. With $H_i(a,b) = H_i(X_b, X_a; \Bbbk)$ for $a \leq b$ and the exact sequences of triples of V3.27, singular homology is a homology of a filtration in the sense of V3.9, provided every step (X_{k+1}, X_k) has finite-dimensional homology. Consequently, for $n \geq 1$, if the steps $k < n$ have finite-dimensional homology, so has (X_n, X_0), and with $b_i = \dim H_i(X_n, X_0; \Bbbk)$ and $C_i = \sum_{k<n} \dim H_i(X_{k+1}, X_k; \Bbbk)$: $b_l \leq C_l$; $b_l - b_{l-1} + \cdots \pm b_0 \leq C_l - C_{l-1} + \cdots \pm C_0$ for every l; and $\sum_{i \leq N} (-1)^i b_i = \sum_{i \leq N} (-1)^i C_i$ if no step has homology in degree $N+1$. Under the cell hypothesis of V3.11 ($\dim H_i(X_{k+1}, X_k; \Bbbk)$ is the number of points of a finite set P of critical points that lie in step k and have index i), C_i is the number of points of P of index i, the three statements hold with these C_i, and the alternating sum is $\sum_{p \in P} (-1)^{\operatorname{ind} p}$ when every index is at most N. *Book:* pp. 28–30: Lemma 5.1 and Lemma 5.3 applied to $\emptyset = M^{a_0} \subset \cdots \subset M^{a_k} = M$, giving Theorem 5.2 and (4_λ). *Hidden hypotheses:* as in V3.10–V3.11; finiteness of $H_*(X_n, X_0)$ is a consequence, not an assumption. *Deviation (restated):* the instance of V3.9 that V3.9–V3.15 lacked; it applies to any filtration of any space.

V3.33 (Betti numbers, and the homology of a slab in the abstract; `Morse.Homology.spaceHomology`, `Morse.Homology.bettiNumber`, `Morse.Homology.bettiNumber_eq_of_homeomorph`, `Morse.Homology.finrank_s` `Morse.Homology.subsetHomologyEquivOfIsSDRBottom`, `Morse.Homology.CellAttachHomology`, `Morse.Homology.cellAttachHomology`, `Morse.Homology.finrank_subsetHomology_of_cells`). For a space X, $b_i(X) = \dim_{\Bbbk} H_i(X; \Bbbk)$ (Mathlib's singular homology), Milnor's $R_i(X)$; it is 0 by convention if $H_i(X; \Bbbk)$ is infinite-dimensional, which never happens for the spaces below. Homeomorphic spaces have the same Betti numbers, and $H_i(E, \emptyset) = H_i(E)$. If $A \subseteq B \subseteq D$ and A is a strong deformation retract of B, then $H_i(D, A) \cong H_i(D, B)$, by the sequence of the triple, in which $H_*(B, A) = 0$ (the deformation of B onto A need not extend to D). The cell computation of V3.31 (several disjoint cells attached to a closed subset of a Hausdorff space give relative homology of dimension the number of cells in each degree) holds over every field. The slab lemma: let $A \subseteq B \subseteq X' \subseteq X$ be subsets of a Hausdorff space, B closed, and $e_1, \ldots, e_m$ pairwise disjoint cells attached to B as in V3.31, with $B \cup e_1 \cup \cdots \cup e_m \subseteq X'$; if A is a strong deformation retract of B, $B \cup e_1 \cup \cdots \cup e_m$ one of X', and X' one of X, then $H_i(X, A; \Bbbk)$ is finite-dimensional, of dimension the number of cells of dimension i. *Book:* p. 28, R_λ; p. 29: $H_*(M^{a_i}, M^{a_{i-1}}) = H_*(M^{a_{i-1}} \cup e^{\lambda_i}, M^{a_{i-1}}) = H_*(e^{\lambda_i}, \dot{e}^{\lambda_i})$. *Hidden hypotheses:* the first equality is about pairs, not spaces: the top of a pair moves along deformations that fix the bottom, but the bottom moves only through the sequence of a triple; this needs strong deformation retracts, whereas Milnor states Theorem 3.2 only up to homotopy type. *Deviation (restated):* for $A = M^{a_{i-1}}$, $B = M^{c-\varepsilon}$, $X' = M^{c+\varepsilon}$, $X = M^{a_i}$, c the critical value, this is the slab computation; V3.34 supplies the hypotheses.

V3.34 (the Morse filtration of a compact submanifold; `Morse.exists_separating_seq`, `Morse.isCriticalPointM_of_forall_ge`, `Morse.Homology.sublevel_step_critical`, `Morse.Homology.sublev` `Morse.Homology.exists_morse_filtration`, `Morse.Homology.exists_morse_filtration'`, `Morse.Homology.exists_cells_filtration`, `Morse.Homology.critCount_eq_card`, `Morse.Homology.finiteDi` Let N be a compact slice submanifold of a finite-dimensional real normed space V, modelled on a finite-dimensional space, and $F : V \to \mathbb{R}$ smooth such that every critical point of $f = F|_N$ is non-degenerate. (i) If $a' \leq a$ are not critical values and all critical values in (a', a) are equal, then $H_i(N^a, N^{a'}; \Bbbk)$ is finite-dimensional, of dimension the number of critical points of index i in $f^{-1}(a', a)$. (ii) There are $n \geq 1$ and non-critical values $a_0 \leq a_1 \leq \ldots$ with $N^{a_0} = \emptyset$, $N^{a_n} = N$ and at most one critical value in each (a_k, a_{k+1}); by (i), each step $(N^{a_{k+1}}, N^{a_k})$ has finite-dimensional homology over every field, of dimension the number of critical points of index i in $f^{-1}(a_k, a_{k+1})$. (iii) Hence $H_i(N; \Bbbk)$ is finite-dimensional for every i. These statements are made for every space

homeomorphic to N, homology being carried along the homeomorphism (for use in V3.39), and for N itself. *Book:* p. 29: for compact M and f with isolated, non-degenerate critical points, $a_1 < \cdots < a_k$ are chosen so that M^{a_i} contains exactly i critical points and $M^{a_k} = M$, and $H_*(M^{a_i}, M^{a_{i-1}})$ is the coefficient group in dimension λ_i and 0 otherwise. *Hidden hypotheses:* Milnor's choice of the a_i needs distinct critical values; here each slab contains at most one critical value, with any number of critical points on it (Remark 3.3); $M^{a_0} = \emptyset$ and $M^{a_k} = M$ rest on the minimum and the maximum of f being critical values; the finiteness of $R_\lambda(M)$ is a consequence; "isolated" is redundant (V3.16). *Proof:* in (i), with no critical value in (a', a), $N^{a'}$ is a strong deformation retract of N^a (`Morse.theorem_3_1_submanifold`); otherwise the slab lemma of V3.33, with Remark 3.3 at the critical value c (`Morse.remark_3_3_submanifold`) and Theorem 3.1 on $[a', c-\varepsilon]$ and $[c+\varepsilon, a]$.

V3.35 (Theorem 5.2 in singular homology; `Morse.Homology.bettiNumber_le_critCount`, `Morse.Homology.bettiNumber_le_critCount'`,`Morse.Homology.sum_bettiNumber_eq_sum_critCount`, `Morse.Homology.sum_bettiNumber_eq_sum_critCount'`,`Morse.Homology.bettiNumber_eq_zero_of_finrank_` `Morse.Homology.subsingleton_spaceHomology_of_finrank_lt`). In the setting of V3.34, with $n = \dim N$, $b_i = \dim H_i(N;\Bbbk)$ for any field $\Bbbk$, and $C_\lambda = C_\lambda(f)$ (V3.16): (2) $b_\lambda \le C_\lambda$ for every λ; (3) $\sum_{i=0}^{n}(-1)^i b_i = \sum_{\lambda=0}^{n}(-1)^\lambda C_\lambda$; and $H_i(N;\Bbbk) = 0$ for $i > n$, so that the left side of (3) is the whole alternating sum. The same holds for every space homeomorphic to N. *Book:* p. 29, Theorem 5.2 (2) and (3), on a compact manifold M, C_λ the number of critical points of index λ. *Hidden hypotheses:* as in V3.34; $R_\lambda(M) = 0$ for $\lambda > n$, which makes $\sum(-1)^\lambda R_\lambda(M)$ a finite sum, follows from (2). *Deviation (narrower):* M is a compact submanifold of a Euclidean space and f the restriction of a smooth function on the ambient space; abstract compact manifolds are V3.39. *Proof:* V3.32 applied to the filtration of V3.34.

V3.36 (the inequalities (4_λ) in singular homology; `Morse.Homology.altSum_bettiNumber_le`, `Morse.Homology.altSum_bettiNumber_le'`). In the setting of V3.35, $b_\lambda - b_{\lambda-1} + \cdots \pm b_0 \le C_\lambda - C_{\lambda-1} + \cdots \pm C_0$ for every λ. *Book:* p. 30, (4_λ). *Hidden hypotheses:* as in V3.34. *Deviation (narrower):* as in V3.35.

V3.37 (Corollary 5.4 and the lacunary principle in singular homology; `Morse.Homology.bettiNumber_eq_critC` `Morse.Homology.bettiNumber_eq_critCount_of_zero`,`Morse.Homology.bettiNumber_eq_critCount_of_lacu` In the setting of V3.35: if $\lambda \ge 1$ and $C_{\lambda+1} = C_{\lambda-1} = 0$, then $b_\lambda = C_\lambda$ and $b_{\lambda+1} = b_{\lambda-1} = 0$; if $C_1 = 0$, then $b_0 = C_0$ and $b_1 = 0$; and if no two consecutive C_i are nonzero, then $b_i = C_i$ for every i. *Book:* p. 31, Corollary 5.4, with the remark that it gives the homology groups of $\mathbb{C}P^n$ (§4) without Theorem 3.5. *Hidden hypotheses:* for $\lambda = 0$ the condition $C_{\lambda-1} = 0$ is empty. *Deviation (not in the book):* the lacunary form. Over every field; integral homology groups are not derived, and $\mathbb{C}P^n$ is not constructed.

V3.38 (χ is the homological Euler characteristic; `Morse.IsEulerNumber.eq_sum_bettiNumber`, `Morse.Homology.eulerNumberM_eq_sum_bettiNumber`). (i) Let $N \subseteq \mathbb{R}^{K+1}$, $K \ge 1$, be a compact nonempty slice submanifold of dimension $d+1 \ge 1$. Every Euler number χ of N (V3.17, the Poincaré–Hopf integer) satisfies $\chi = \sum_{i=0}^{d+1}(-1)^i \dim H_i(N;\Bbbk)$ for every field $\Bbbk$; the same holds for every space homeomorphic to N. (ii) For a compact Hausdorff smooth manifold M without boundary, modelled on a finite-dimensional real normed space, the integer $\chi(M)$ of V4.18 ($\sum_\lambda(-1)^\lambda C_\lambda$ for any Morse function) equals $\sum_{i=0}^{\dim M}(-1)^i \dim H_i(M;\Bbbk)$ for every field $\Bbbk$. In particular this alternating sum does not depend on the field. *Book:* p. 28, $\chi(X,Y) = \sum(-1)^\lambda R_\lambda(X,Y)$; p. 29, formula (3), $\chi(M) = C_0 - C_1 + \cdots \pm C_n$; Application 2 (pp. 36–37): $\chi(M)$ is the index sum of every vector field. *Hidden hypotheses:* that $\chi(M)$ is the

same over every field, never stated; passing from M to an embedded copy uses the invariance of homology under homeomorphisms. *Deviation (restated):* the three descriptions of χ (by homology, by critical points, by index sums of vector fields) are identified, which closes the circle left open in V3.20 and V4.19. *Proof:* (3) of V3.35 for a Morse function $L_p = \|\cdot - p\|^2$ (Theorem 6.6, V4.12), compared with V3.20, resp. V4.19, through a Whitney embedding (V4.16).

V3.39 (the Morse inequalities on a compact manifold; `Morse.Homology.finiteDimensional_spaceHomology_of` `Morse.Homology.bettiNumber_eq_zero_of_isCompact`, `Morse.Homology.exists_whitney_normSq`, `Morse.Homology.exists_ambient_of_isMorseFunction`, `Morse.Homology.finiteDimensional_spaceHomology` `Morse.Homology.bettiNumber_eq_zero_of_compactSpace`, `Morse.Homology.bettiNumber_le_critCount_of_` `Morse.Homology.altSum_bettiNumber_le_of_compactSpace`, `Morse.Homology.sum_bettiNumber_eq_sum_crit` `Morse.Homology.bettiNumber_eq_critCount_of_lacunary_of_compactSpace`). The homology over any field of a compact k-dimensional slice submanifold of $\mathbb{R}^n$, or of a space homeomorphic to one, is finite-dimensional and vanishes above degree k. Let M be a compact Hausdorff smooth manifold without boundary, modelled on a finite-dimensional real normed space, of dimension n. Then $H_i(M;\Bbbk)$ is finite-dimensional and is 0 for $i > n$, and for every Morse function g on M, with $b_i = \dim H_i(M;\Bbbk)$ and $C_\lambda = C_\lambda(g)$: $b_\lambda \le C_\lambda$; (4_λ); $\sum_{i\le n}(-1)^i b_i = \sum_{\lambda\le n}(-1)^\lambda C_\lambda$; and $b_i = C_i$ for every i if no two consecutive C_i are nonzero. *Book:* §5 is stated for compact manifolds; p. 36 quotes Whitney's embedding theorem. *Hidden hypotheses:* Hausdorff, for the embedding; homology is invariant under homeomorphisms, which carries the results from the embedded copy. *Deviation (restated):* proved through a Whitney embedding $M \cong N \subseteq \mathbb{R}^{K+1}$ (V4.16), on which g becomes the restriction of a smooth ambient function (V3.20) with the same numbers C_λ (V4.15); in dimension 0 directly from (3).

V3.40 (examples: two critical points, the torus, the spheres; `Morse.Homology.bettiNumber_of_ncard_eq_two`, `Morse.Homology.torus_bettiNumber`, `Morse.Homology.sphere_sum_bettiNumber`, `Morse.Homology.sphere_be` `Morse.Homology.circle_bettiNumber`). Over any field: (i) if a compact Hausdorff manifold M of dimension $n \ge 2$ carries a smooth function with exactly two critical points, both non-degenerate, then $b_0 = b_n = 1$ and $b_i = 0$ otherwise; (ii) the torus of V3.22 has $b_0 \le 1$, $b_1 \le 2$, $b_2 \le 1$, $b_i = 0$ for $i \ge 3$, and $b_0 - b_1 + b_2 = 0$; (iii) $\sum_i (-1)^i b_i(S^n) = 1 + (-1)^n$ for $n \ge 1$, and for $n \ge 2$, $b_0(S^n) = b_n(S^n) = 1$ and $b_i(S^n) = 0$ otherwise; (iv) the circle has $b_0 = b_1 \le 1$. *Book:* Reeb's situation (p. 25) and the torus of §1 (pp. 1–3), with §5; the spheres are not discussed in §5. *Hidden hypotheses:* $n \ge 2$ in (i) and (iii), where the lacunary principle applies ($C_0 = C_n = 1$); for the circle it does not ($C_0 = C_1 = 1$). *Deviation (not in the book):* checks of V3.35–V3.39; (iii) agrees with the direct computation of V3.30, which also gives $b_0 = b_1 = 1$ for the circle. For the torus only the bounds and the Euler characteristic follow; its Betti numbers $1, 2, 1$ are not derived.

V3.41 ($\mathbb{C}P^n$ as a compact submanifold, with Milnor's coordinates; `Morse.CP.exists_vec_of_mem`, `Morse.CP.Model`, `Morse.CP.param`, `Morse.CP.coord`, `Morse.CP.coord_param`, `Morse.CP.nsq_coord_lt_and_par` `Morse.CP.exists_re_diag_pos`, `Morse.CP.isSliceSubmanifold`, `Morse.CP.exists_sliceChart`, `Morse.CP.isCompact`). Here $\mathbb{C}P^n$ is the set of Hermitian projections of trace 1 in the space $M_{n+1}(\mathbb{C})$ of complex matrices (the Grassmannian of V14.16 with $m = 1$). Every point of it is $P = zz^*$, that is $P_{ab} = z_a\bar{z}_b$, for a unit vector $z \in \mathbb{C}^{n+1}$. For $0 \le k \le n$ let U_k be the set of P with $\operatorname{Re} P_{kk} > 0$; every point of $\mathbb{C}P^n$ lies in some U_k. On $\mathbb{R}^{2n} = \mathbb{C}^n$, with coordinates $z_j = x_j + iy_j$ ($j \ne k$), put $\operatorname{param}_k(z) = vv^*$, where $v_k = \sqrt{1 - |z|^2}$ and $v_j = z_j$ for $j \ne k$; and for a matrix P put $\operatorname{coord}_k(P) = (P_{jk}/\sqrt{\operatorname{Re} P_{kk}})_{j\ne k}$. Then $\operatorname{coord}_k(\operatorname{param}_k(z)) = z$ for $|z| < 1$, and for $P \in \mathbb{C}P^n \cap U_k$, $|\operatorname{coord}_k(P)| < 1$ and $\operatorname{param}_k(\operatorname{coord}_k(P)) = P$. $\mathbb{C}P^n$ is a compact slice submanifold of $M_{n+1}(\mathbb{C})$ of dimension $2n$, and at every $x \in \mathbb{C}P^n \cap U_k$ it has a slice chart whose

parametrization is $z \mapsto \mathrm{param}_k(\mathrm{coord}_k(x) + z)$ near 0. *Book:* p. 26: $\mathbb{C}P_n$ as the equivalence classes of $(n+1)$-tuples $(z_0, \dots, z_n)$ with $\sum |z_j|^2 = 1$; on the set U_0 where $z_0 \neq 0$, $|z_0|\, z_j/z_0 = x_j + iy_j$, and $x_1, y_1, \dots, x_n, y_n$ "are the required coordinate functions, mapping U_0 diffeomorphically onto the open unit ball in $\mathbb{R}^{2n}$". *Hidden hypotheses:* "diffeomorphically" presupposes a smooth structure on $\mathbb{C}P_n$. *Deviation (restated):* $\mathbb{C}P^n$ is the submanifold of rank-one projections, with the smooth structure induced from the space of matrices; Milnor's coordinates are then entries of P, visibly independent of the representative z, and they are the coordinates of slice charts.

V3.42 (Milnor's function on $\mathbb{C}P^n$, its critical points and their indices; `Morse.CP.height`, `Morse.CP.height_param`, `Morse.CP.isCriticalPointM_height_iff`, `Morse.CP.isNondegenerate_indexM_heig` `Morse.CP.isNondegenerate_of_isCriticalPointM`, `Morse.CP.bijective_rank`, `Morse.CP.critCount_height`). For real numbers $c_0, \dots, c_n$ let $f(P) = \sum_j c_j \operatorname{Re} P_{jj}$ on $\mathbb{C}P^n$, so that $f(zz^*) = \sum_j c_j |z_j|^2$. In the coordinates of V3.41, $f(\mathrm{param}_k(z)) = c_k + \sum_{j \neq k} (c_j - c_k)(x_j^2 + y_j^2)$ for $|z| \leq 1$. Suppose the c_j are distinct. Then the critical points of f are exactly the $n+1$ points $p_k = \mathrm{param}_k(0)$ (the projections onto the coordinate axes); p_k is non-degenerate, of index $2\,\#\{j : c_j < c_k\}$; so f is a Morse function. The map $k \mapsto \#\{j : c_j < c_k\}$ is a bijection of $\{0, \dots, n\}$, and so the number C_λ of critical points of index λ is 1 for $\lambda = 0, 2, \dots, 2n$ and 0 for every other λ. *Book:* pp. 26–27: $f(z_0 : z_1 : \cdots : z_n) = \sum c_j |z_j|^2$ with distinct real constants c_j; $f = c_0 + \sum_{j=1}^n (c_j - c_0)(x_j^2 + y_j^2)$ on U_0, so the only critical point of f within U_0 is the centre p_0, non-degenerate, of index twice the number of j with $c_j < c_0$; similarly at $p_1, \dots, p_n$; "every possible even index between 0 and $2n$ occurs exactly once". *Hidden hypotheses:* the scan of p. 26 prints the definition as $f = c_j |z_j|$; the display that follows shows that $\sum c_j |z_j|^2$ is meant. That every even index occurs once is the bijectivity of $k \mapsto \#\{j : c_j < c_k\}$, Milnor's tacit renumbering $c_0 < \cdots < c_n$. *Deviation:* none in substance; the critical points, their non-degeneracy and their indices are read in the slice parametrizations of V3.41 (V1.6).

V3.43 (the cells and the homology of $\mathbb{C}P^n$; `Morse.CP.exists_cells`, `Morse.CP.bettiNumber_eq`, `Morse.bettiNumber_complexProjectiveSpace`). $\mathbb{C}P^n$ has the homotopy type of a space built from the empty set by attaching cells one after another (V2.12), exactly one of each dimension $0, 2, \dots, 2n$. For every field $\Bbbk$, $\dim H_\lambda(\mathbb{C}P^n; \Bbbk) = 1$ if λ is even and $\lambda \leq 2n$, and 0 otherwise. *Book:* p. 27: by Theorem 3.5, $\mathbb{C}P_n$ has the homotopy type of a CW-complex of the form $e^0 \cup e^2 \cup e^4 \cup \cdots \cup e^{2n}$; it follows that the integral homology groups are $\mathbb{Z}$ for $i = 0, 2, 4, \dots, 2n$ and 0 for other values of i. *Hidden hypotheses:* none. *Deviation (narrower):* an iterated adjunction space rather than a CW-complex (V2.12); the homology over fields, by the lacunary principle (V3.37, V3.39), not the integral groups.

What is not verified in this chapter

- **Remark 1** (p. 25, Reeb's theorem with degenerate critical points): Milnor cites two papers. A proof needs M. Brown's monotone union theorem, which is not in Mathlib (a proof by that route is in `Exposition/Chapter3.md`, §1).
- **Remark 2** (pp. 25–26, a non-standard smooth 7-sphere): out of scope (characteristic classes, the signature theorem). V3.5 shows where smoothness is lost.
- **Three critical points** (p. 26: indices $0, n/2, n$ by Poincaré duality, and the homotopy type of an $n/2$-sphere with an n-cell attached): not formalized. Singular homology with its exact sequences is now available (V3.26–V3.31), but Poincaré duality (mod 2), which the argument needs, is not.
- $\mathbb{C}P^n$ (pp. 26–27): verified in V3.41–V3.43, except for the integral homology (next item) and

for the CW-complex in the strict sense, which is an iterated adjunction space here (V2.12).

- **Integral homology**: §5 works over a field, and so do V3.32–V3.40. The integral groups $H_i(\mathbb{C}P^n;\mathbb{Z})$ of p. 27, and more generally the exposition's Proposition G (in the lacunary case $H_\lambda(M;\mathbb{Z})$ is free of rank C_λ), are not derived, although relative homology and the cell computations are available with integer coefficients (V3.26–V3.31).
- **Non-compact manifolds**: everything above is for compact N or M, as in Milnor's §5; the inequalities for the compact sublevel sets of a function on a non-compact manifold are not stated, and abstract non-compact manifolds would need a Whitney embedding that Mathlib does not have.
- **The torus, (2)–(4) of p. 1 and the cell attachments of p. 2**: the sublevel sets are not shown to be a disk, a cylinder and a torus with an open disk removed, and the attachments are not derived; Theorem 3.2 on submanifolds is proved in Chapter 2 but has not been applied to the torus, and the homeomorphism types would need explicit homeomorphisms. Its Betti numbers $1, 2, 1$ are not derived either: Morse theory with the height function gives only $b_0 \leq 1$, $b_1 \leq 2$, $b_2 \leq 1$ and $b_0 - b_1 + b_2 = 0$ (V3.40).

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5. Chapter 4 (Milnor §6): partly verified

Milnor's §6 (pp. 32–38) takes a manifold M of dimension k in $\mathbb{R}^n$, its normal bundle N and the endpoint map $E(q,v) = q + v$, and defines focal points (p. 32). He quotes Sard's theorem (Theorem 6.1) to get Corollary 6.2 (almost no point is focal), introduces the fundamental forms and the principal curvatures, locates the focal points on a normal line (Lemma 6.3, through Assertion 6.4), relates them to the squared distance functions $L_p(q) = \|q-p\|^2$ (Lemma 6.5), and proves that L_p is a Morse function for almost every p (Theorem 6.6). Corollary 6.7 (every manifold has a Morse function with compact sublevel sets, via Whitney's embedding), Applications 1 and 2, Corollary 6.8 (approximation by Morse functions) and Lemma 6.9 (the index of L_p) follow. The numbering, checked against the scan of pp. 32–38, is: 6.1 Sard's theorem, 6.2 the corollary on focal points, 6.3 the lemma on focal points along a normal line (principal curvatures), 6.4 an assertion inside its proof, 6.5 the lemma on degenerate critical points of L_p, 6.6 the theorem, 6.7 and 6.8 corollaries, 6.9 the index lemma. Verified: every numbered statement of pp. 32–38 for a slice submanifold $M \subseteq \mathbb{R}^n$ modelled on $\mathbb{R}^k$ with complement $\mathbb{R}^w$ (the Milnor project's `Milnor.IsSliceSubmanifold`; M need not be closed or compact, except that Corollary 6.7 needs M closed, and $k = n$ is allowed), with Sard's theorem for smooth maps into a vector space (V4.1–V4.13, V4.20, V4.21, V4.23; Application 2 for a compact submanifold of $\mathbb{R}^{K+1}$ is V3.20, its Lean files being those of Chapter 3); and, through Mathlib's Whitney embedding of compact manifolds, Corollary 6.7, Application 2 and Corollary 6.8 for an abstract compact Hausdorff manifold without boundary modelled on a finite-dimensional real normed space (V4.14–V4.19, V4.22), together with two results of earlier chapters whose Lean files live here, Reeb's theorem and Theorem 3.1 for such manifolds (V4.24, V4.25). Corollary 6.8 is proved **without Milnor's hypothesis that f be bounded**. In Application 2 and throughout, χ is the Poincaré–Hopf integer of the Milnor project (V3.17); its identification with the homological Euler characteristic is in Chapter 3 (V3.38). Not verified: Corollary 6.7 for non-compact abstract manifolds, Application 1, and the items listed at the end.

V4.1 (the normal bundle and the endpoint map; `Milnor.Problems.normalBundleSet`, `Milnor.Problems.isSliceSubmanifold_normalBundle`, `Milnor.Problems.normalSum`, `Morse.mem_normalBundl` `Morse.slice_dim_add_eq_of_nonempty`, `Morse.normalEnd`, `Morse.normalEnd_id`). Throughout, $M \subseteq \mathbb{R}^n$ is a slice submanifold modelled on $\mathbb{R}^k$ with complement $\mathbb{R}^w$; if M is nonempty,

$k + w = n$. The normal bundle is $N = \{(q, v) \in M \times \mathbb{R}^n : v \perp T_qM\}$; when $k + w = n$ it is a slice submanifold of $\mathbb{R}^n \times \mathbb{R}^n$ of dimension n (the Milnor project's V8.11). The endpoint map is $E(q, v) = q + v$; more generally, for a map $\Gamma : \mathbb{R}^n \to \mathbb{R}^n$, $E_\Gamma(q, v) = \Gamma(q) + v$, smooth over every open set on which Γ is smooth, with $E_{\mathrm{id}} = E$. *Book:* p. 32: $M \subset \mathbb{R}^n$ a manifold of dimension $k < n$, differentiably embedded; N is an n-dimensional manifold ("not difficult to show"); $E(q, v) = q + v$. *Hidden hypotheses:* "differentiably embedded" must mean a submanifold (an injectively immersed manifold need not be one, V4.14); for slice charts, $k + w = n$ needs $M \neq \emptyset$. *Deviation (stronger):* $k = n$ is allowed and M need not be closed or compact; E_Γ is not in the book and serves V4.20.

V4.2 (focal points and their multiplicity; `Morse.endpointNullity`, `Morse.IsFocalPointMult`, `Morse.IsFocalPoint`, `Morse.focalSet`, `Morse.endpointNullity_pos_iff`, `Morse.isFocalPoint_iff`, `Morse.focalSet_eq_image_critPoints`). Assume $k + w = n$. The nullity of E at $x \in N$ is the dimension of the kernel of dE_x, the tangent space of N at x being identified with $\mathbb{R}^n$ by the chart of N at x. A point $e \in \mathbb{R}^n$ is a focal point of (M, q) with multiplicity μ if $(q, e - q) \in N$ and the nullity of E at $(q, e - q)$ is $\mu > 0$; a focal point of (M, q) if this holds for some μ; a focal point of M if it is a focal point of (M, q) for some $q \in M$. The nullity is positive exactly at the critical points of E (the points where dE is not onto); so e is a focal point of (M, q) iff $(q, e - q) \in N$ is a critical point of E, and the set of focal points of M is the set of critical values of E. *Book:* p. 32 (the definition, with $e = q + v$ and $(q, v) \in N$); p. 33 (the proof of Corollary 6.2). *Hidden hypotheses:* the nullity does not depend on the chart (two charts differ by an isomorphism of $\mathbb{R}^n$).

V4.3 (Theorem 6.1, Sard's theorem on a manifold; `Morse.measure_image_inter_critPoints_eq_zero`, `Morse.measure_image_critPoints_eq_zero`). Let M be a second countable smooth manifold without boundary modelled on a finite-dimensional real normed space E, F a finite-dimensional real normed space with a Haar measure μ, $U \subseteq M$ open and $f : M \to F$ smooth on U. Then the image under f of the points of U where df is not onto is μ-null; in particular, for f smooth on M, the set of critical values is null. E and F are required to lie in the same universe of Lean's type theory, a formality with no mathematical content (as in the Milnor project). *Book:* Theorem 6.1 (p. 33): for a C^1 map between manifolds of the same dimension with countable bases, the image of the set of critical points (where the Jacobian is singular) has measure 0; Milnor refers to de Rham and gives no proof. *Hidden hypotheses:* "measure 0" in the target needs a measure class; only $\mathbb{R}^n$ is used, where all Haar measures have the same null sets. *Deviation (narrower):* f is smooth, not merely C^1, and the target is a vector space; on the other hand the dimensions are arbitrary. Every use in §6 is for a smooth map into $\mathbb{R}^n$. *Proof:* countably many charts and the Euclidean Sard theorem of the Milnor project (`Milnor.sard`).

V4.4 (Corollary 6.2; `Morse.measure_focalSet_eq_zero`). If $k + w = n$, the set of focal points of M is null for every Haar measure on $\mathbb{R}^n$. *Book:* Corollary 6.2 (p. 33): for almost all $x \in \mathbb{R}^n$, x is not a focal point of M. *Hidden hypotheses:* N has a countable basis, automatic for a subset of $\mathbb{R}^{2n}$.

V4.5 (the fundamental forms; `Morse.xParam`, `Morse.uParam`, `Morse.dParam`, `Morse.d2Param`, `Morse.firstFF`, `Morse.secondFF`, `Morse.normalForm`). For $q \in M$, Milnor's coordinates $u \mapsto x(u)$ are the inverse of the slice chart of M chosen at q, with $x(u_q) = q$; dx and d^2x are its first and second derivatives at u_q (Milnor's $\partial x/\partial u^i$ and $\partial^2 x/\partial u^i \partial u^j$). The first fundamental form is the bilinear form $g(\xi, \eta) = \langle dx\,\xi, dx\,\eta\rangle$ on $\mathbb{R}^k$; the second fundamental form in a direction $v \in \mathbb{R}^n$ is $\ell_v(\xi, \eta) = \langle v, d^2x(\xi, \eta)\rangle$; and for a linear map T of $\mathbb{R}^n$ the form $(\xi, \eta) \mapsto \langle T\,dx\,\xi, dx\,\eta\rangle - \ell_v(\xi, \eta)$ is the one attached to E_Γ with $T = d\Gamma_q$ (V4.8), equal to $g - \ell_v$ for $T = \mathrm{id}$. *Book:* pp. 33–34: $g_{ij} = \partial x/\partial u^i \cdot \partial x/\partial u^j$; ℓ_{ij} is the normal component of $\partial^2 x/\partial u^i \partial u^j$,

and for a unit normal v the matrix $(v \cdot \ell_{ij})$ is the second fundamental form in the direction v. *Hidden hypotheses:* none. *Deviation (restated):* ℓ_v is defined for every v by the inner product with d^2x; for v normal at q it is Milnor's $(v \cdot \ell_{ij})$, the tangential part of d^2x being orthogonal to v. The coordinates are those of the chosen slice chart; Milnor allows any coordinates.

V4.6 (principal curvatures; `Morse.tanSpace`, `Morse.shapeOp`, `Morse.isSymmetric_shapeOp`, `Morse.principalCurvatures`, `Morse.finrank_eigenspace_shapeOp`). The shape operator S_v of M at q in the direction v is the endomorphism of the tangent space $T_qM = \operatorname{range} dx$ with $\langle S_v a, b\rangle = \ell_v(dx^{-1}a, dx^{-1}b)$; it is self-adjoint. The principal curvatures of M at q in the direction v are its k eigenvalues, with multiplicity, in decreasing order; the multiplicity of K is the dimension of the eigenspace of K. *Book:* p. 34: with coordinates chosen so that (g_{ij}) is the identity at q, the eigenvalues of $(v \cdot \ell_{ij})$ are the principal curvatures, and the reciprocals of the nonzero ones the principal radii of curvature. *Hidden hypotheses:* that coordinates with $g_{ij}(q) = \delta_{ij}$ exist, and that the eigenvalues do not depend on them. *Deviation (restated):* no normalization: the eigenvalues of S_v are those of $g^{-1}(v \cdot \ell)$, which are Milnor's K_i when g is the identity. That S_v does not depend on the parametrization is not proved, and nothing uses it. The scan lists the curvatures as $K_1, \dots, K_n$; there are k of them, and Lemma 6.3 itself uses $1 \le i \le k$.

V4.7 (Lemma 6.3; `Morse.focalMult`, `Morse.focalMult_add_smul`, `Morse.isFocalPoint_add_smul_iff`, `Morse.focalMult_add_inv_smul`, `Morse.sum_focalMult_add_smul_le`). Assume $k + w = n$, and let $\operatorname{mult}_q(e)$ be the multiplicity of e as a focal point of (M, q), 0 if it is not one. For $q \in M$, $v \perp T_qM$ and $t \in \mathbb{R}$: $\operatorname{mult}_q(q + tv) = 0$ if $t = 0$, and otherwise it is the multiplicity of $1/t$ as an eigenvalue of S_v. Hence $q + tv$ is a focal point of (M, q) iff $t = K^{-1}$ for a nonzero principal curvature K in the direction v, the multiplicity of $q + K^{-1}v$ is that of K, and for any finitely many values of t the multiplicities of the points $q + tv$ add up to at most k. *Book:* Lemma 6.3 (p. 34), for a unit normal v: the focal points of (M, q) on the normal line $\{q + tv\}$ are exactly the points $q + K_i^{-1}v$ with $K_i \neq 0$, so there are at most k of them, each counted with its multiplicity. *Hidden hypotheses:* $t = 0$ must be set aside in "$1/t$ is an eigenvalue" (q is never a focal point of (M, q), V4.9); the multiplicity of an eigenvalue of a symmetric matrix as a root of the characteristic polynomial equals the dimension of its eigenspace, which is what the proof produces. *Deviation (stronger):* v is any normal vector, not necessarily a unit vector. In place of Milnor's normalized coordinates, for $t \neq 0$ the map dx carries the null space of $g - t\ell_v$ onto the eigenspace of S_v for $1/t$ (`Morse.map_nullSpace_firstFF_sub_eq_eigenspace`).

V4.8 (the Jacobian of the endpoint map; `Morse.mem_range_mfd_val_normalBundle_iff`, `Morse.hasMFDerivAt_normalEnd`, `Morse.finrank_ker_mfd_normalEnd`, `Morse.endpointNullity_eq_nullity`). Assume $k + w = n$ and let $(q, v) \in N$. A pair $(a, b) \in \mathbb{R}^n \times \mathbb{R}^n$ is tangent to N at (q, v) iff $a = dx\,\xi$ for some $\xi \in \mathbb{R}^k$ and $\langle b, dx\,\eta\rangle + \langle v, d^2x(\xi, \eta)\rangle = 0$ for every $\eta \in \mathbb{R}^k$. For Γ differentiable at q, the derivative of E_Γ at (q, v) is $(a, b) \mapsto d\Gamma_q(a) + b$ on this tangent space, and its nullity equals the nullity of the form $(\xi, \eta) \mapsto \langle d\Gamma_q\,dx\,\xi, dx\,\eta\rangle - \langle v, d^2x(\xi, \eta)\rangle$ on $\mathbb{R}^k$. For $\Gamma = \mathrm{id}$: the nullity of E at (q, v) is the nullity of $g - \ell_v$. *Book:* pp. 34–35, in the proof of Lemma 6.3: coordinates (u, t) on N built from orthonormal normal fields w_α, and an $n \times n$ matrix whose upper left block is $(g_{ij} - \sum_\alpha t^\alpha w_\alpha \cdot \ell_{ij})$. *Hidden hypotheses:* the fields w_α exist only locally, and Milnor does not say why they exist; his sentence that the rank of the matrix equals the rank of the upper left block (p. 35) does not hold as written, the two ranks differing by $n - k$; what holds, and is needed, is that the nullities agree. *Deviation (restated):* no normal frame is used. *Proof:* the tangent space of N is the kernel of the derivative of the local defining function of N used by the Milnor project; its second condition is the derivative of Milnor's identity $w_\alpha \cdot \partial x/\partial u^j = 0$.

V4.9 (Assertion 6.4; `Morse.isFocalPointMult_add_smul_iff`, `Morse.not_isFocalPoint_self`). Assume $k+w=n$. For $q \in M$, $v \perp T_qM$, $t \in \mathbb{R}$ and $\mu \in \mathbb{N}$: $q+tv$ is a focal point of (M,q) with multiplicity μ iff the form $g - t\ell_v$ (in the coordinates of V4.5) has nullity $\mu > 0$. In particular q is never a focal point of (M,q), g being non-degenerate. *Book:* Assertion 6.4 (p. 35), for a unit normal v: $q+tv$ is a focal point of (M,q) with multiplicity μ iff the matrix $(g_{ij} - t\, v \cdot \ell_{ij})$ is singular with nullity μ. *Hidden hypotheses:* the case $t=0$, left implicit. *Deviation (stronger):* v need not be a unit vector.

V4.10 (the critical points and the Hessian of L_p; `Morse.isCriticalPointM_normSq_sub_iff_mem_normalBundle`, `Morse.hessian_normSq_sub_xParam`). For $p \in \mathbb{R}^n$ let $L_p(q) = \|q-p\|^2$ on M. Then q is a critical point of L_p iff $p - q \perp T_qM$, that is, iff $(q, p-q) \in N$; and at every $q \in M$ the Hessian of $L_p \circ x$ at u_q is $2(g - \ell_{p-q})$. *Book:* p. 35: L_p has a critical point at q iff $q-p$ is normal to M at q; p. 36: at a critical point the second derivatives are $2(\partial x/\partial u^i \cdot \partial x/\partial u^j + \partial^2 x/\partial u^i \partial u^j \cdot (x-p))$, which on setting $p = x + tv$ becomes $2(g_{ij} - t\, v \cdot \ell_{ij})$ (the second display is cut off in the scan; the value is the one used on p. 38). *Hidden hypotheses:* setting $p = q + tv$ with v a unit normal presupposes that $p-q$ is normal (i.e. that q is critical) and that $p \neq q$. *Deviation (stronger):* the formula holds at every point, in the direction $p-q$ itself, so no unit vector and no case $p=q$ are needed.

V4.11 (Lemma 6.5; `Morse.nullityM_normSq_sub_eq_endpointNullity`, `Morse.isFocalPointMult_iff`, `Morse.isDegenerate_normSq_sub_iff_isFocalPoint`). Assume $k+w=n$, and let $p \in \mathbb{R}^n$, $q \in M$. If q is a critical point of L_p, the nullity of L_p at q (the nullity of its Hessian) equals the nullity of E at $(q, p-q)$. Hence p is a focal point of (M,q) with multiplicity μ iff q is a critical point of L_p of nullity $\mu > 0$, and q is a degenerate critical point of L_p iff p is a focal point of (M,q). *Book:* Lemma 6.5 (p. 36): q is a degenerate critical point of L_p iff p is a focal point of (M,q), and the nullity of q is the multiplicity of p. *Hidden hypotheses:* the case of a non-critical q (then $p-q$ is not normal at q and p is not a focal point of (M,q)) is left implicit.

V4.12 (Theorem 6.6; `Morse.mem_focalSet_of_not_isMorseFunction`, `Morse.ae_isMorseFunction_normSq_sub`, `Morse.ae_volume_isMorseFunction_normSq_sub`, `Morse.exists_mem_isMorseFunction_normSq_sub`, `Morse.exists_isMorseFunction_normSq_sub`). For every slice submanifold $M \subseteq \mathbb{R}^n$ and every Haar measure μ on $\mathbb{R}^n$ (in particular Lebesgue measure), for μ-almost every p the function L_p is a Morse function on M (smooth, with only non-degenerate critical points). If L_p is not a Morse function, p is a focal point of M. Every nonempty open subset of $\mathbb{R}^n$ contains a p for which L_p is a Morse function. *Book:* Theorem 6.6 (p. 36): for almost all $p \in \mathbb{R}^n$, L_p has no degenerate critical points. *Hidden hypotheses:* none; $M = \emptyset$ is trivial, and is the only case in which $k+w=n$ may fail. *Deviation (stronger):* M need not be closed or compact, $k=n$ is allowed, and any Haar measure may be used.

V4.13 (Corollary 6.7 for submanifolds of $\mathbb{R}^n$; `Morse.isCompact_sublevel_normSq_sub`, `Morse.ae_isMorseFunction_normSq_sub_and_isCompact_sublevel`, `Morse.exists_isMorseFunction_isCompa`, `Morse.exists_isMorseFunction_finite_critPoints_sublevel`, `Morse.ae_isMorseFunction_normSq_sub_fi`, `Morse.exists_isMorseFunction_finite_critPoints`). If the slice submanifold $M \subseteq \mathbb{R}^n$ is a closed subset, then for every p and every a the sublevel set $\{q \in M : L_p(q) \le a\}$ is compact; so for almost every p, L_p is a Morse function on M with compact sublevel sets, and M carries a Morse function f each of whose sublevel sets M^a is compact and contains only finitely many critical points. If M is compact, for almost every p, L_p is a Morse function with finitely many critical points. *Book:* Corollary 6.7 (p. 36): on any manifold M there is a differentiable function with no degenerate critical points for which each M^a is compact; the proof combines Theorem 6.6 with an embedding of M as a closed subset of $\mathbb{R}^{2n+1}$ (Whitney). *Hidden hypotheses:* the embedding

must be closed (for the open interval in $\mathbb{R}$, every L_p has a non-compact sublevel set); "any manifold" must be read as Hausdorff with a countable basis (on the long line every continuous real function is eventually constant, so some sublevel set is not compact). *Deviation (narrower):* closed submanifolds of $\mathbb{R}^n$; abstract compact manifolds are V4.17; non-compact abstract manifolds are not covered.

V4.14 (the image of an embedding is a submanifold; `Morse.exists_continuousLinearEquiv_coprod`, `Morse.exists_sliceChart_range`, `Morse.isSliceSubmanifold_range`, `Morse.contMDiff_ofInjective_symm`, `Morse.diffeomorphRange`, `Morse.exists_diffeomorph_range`**).** Let M be a smooth manifold without boundary modelled on a finite-dimensional real normed space E, F a finite-dimensional real normed space, and $e : M \to F$ smooth, a topological embedding (a homeomorphism onto its image), with injective derivative at every point. For finite-dimensional Z, W with $\dim Z = \dim E$ and $\dim Z + \dim W = \dim F$, the image $e(M)$ is a slice submanifold of F modelled on Z with complement W, and $e : M \to e(M)$ is a smooth diffeomorphism. *Book:* not stated; taken for granted in the proof of Corollary 6.7 (p. 36), where Theorem 6.6 is applied to the image of Whitney's embedding. *Hidden hypotheses:* e must be a homeomorphism onto its image (the figure eight, an injectively immersed line, is not a submanifold); the inverse of e is smooth because of the slice charts, not merely because e is a smooth bijection ($x \mapsto x^3$ on $\mathbb{R}$). *Deviation (not in the book). Proof:* the local immersion theorem, from the inverse function theorem on manifolds of the Milnor project (`Milnor.isLocalDiffeomorphAt_of_isInvertible_mfderiv`) applied to $(z, w) \mapsto e(\varphi^{-1}(Tz)) + Lw$, φ a chart of M, $T : Z \cong E$, and L completing de_x to an isomorphism.

V4.15 (the Morse notions are invariant under diffeomorphisms; `Morse.isCriticalPointM_comp_symm_iff`, `Morse.setOf_isCriticalPointM_comp_symm`, `Morse.hessianM_comp_symm_apply`, `Morse.isNondegenerate_hes` `Morse.indexM_comp_symm`, `Morse.nullityM_comp_symm`, `Morse.IsMorseFunction.comp_diffeomorph_symm`, `Morse.isMorseFunction_comp_symm_iff`, `Morse.critCount_comp_symm`, `Morse.ncard_setOf_isCriticalPoint` Let $g : M \to M'$ be a smooth diffeomorphism of smooth manifolds without boundary modelled on real normed spaces E, E', and $f : M \to \mathbb{R}$. For every f, q is a critical point of $f \circ g^{-1}$ iff $g^{-1}(q)$ is a critical point of f, and the numbers of critical points agree. If f is C^2 at a critical point $g^{-1}(q)$, the Hessian of $f \circ g^{-1}$ at q is the Hessian of f at $g^{-1}(q)$ pulled back by $d(g^{-1})_{q'}$ so non-degeneracy, index and nullity agree there. f is a Morse function iff $f \circ g^{-1}$ is, and for f of class C^2 the numbers C_λ (V3.16) of f and $f \circ g^{-1}$ agree. *Book:* not stated; used without comment wherever a manifold is replaced by its image under an embedding (p. 36). *Hidden hypotheses:* none. *Deviation (not in the book).*

V4.16 (every compact manifold is a compact submanifold of Euclidean space; `Morse.exists_isSliceSubmanifo` `Morse.exists_embedding_euclidean`, `Morse.exists_diffeomorph_sliceSubmanifold`**).** Let M be a compact Hausdorff smooth manifold without boundary modelled on a finite-dimensional real normed space E, $n = \dim E$. Then M has a smooth closed embedding with injective derivative into some $\mathbb{R}^m$ (Mathlib's Whitney embedding, `exists_embedding_euclidean_of_compact`), whose image is a slice submanifold modelled on E; such an embedding exists into $\mathbb{R}^{K+1}$ with $K \geq 1$ and $K + 1$ as large as desired; and M is diffeomorphic to a compact, closed slice submanifold $N \subseteq \mathbb{R}^{K+1}$, $K \geq 1$, modelled on $\mathbb{R}^n$. *Book:* p. 36, in the proof of Corollary 6.7: an n-dimensional manifold can be embedded differentiably as a closed subset of $\mathbb{R}^{2n+1}$ (Whitney, *Geometric Integration Theory*, p. 113). *Hidden hypotheses:* Hausdorff. *Deviation (narrower):* compact manifolds only; the ambient dimension is not $2n + 1$, which plays no role ($K \geq 1$ is arranged by two extra zero coordinates, for the Poincaré–Hopf theorem).

V4.17 (Corollary 6.7 for a compact manifold; `Morse.exists_isMorseFunction_of_compactSpace`). Every compact Hausdorff smooth manifold without boundary, modelled on a finite-dimensional real normed space, carries a Morse function, and it has finitely many critical points. *Book:* Theorem 6.6 and Corollary 6.7 (p. 36). *Hidden hypotheses:* Hausdorff, for the embedding; for compact M every M^a is compact. *Deviation (narrower):* compact M only; the function is L_p on an embedded copy (V4.16, V4.13), pulled back by the diffeomorphism (V4.15).

V4.18 (the Euler number of an abstract manifold; `Morse.eulerNumberM`). For a manifold M modelled on E, $\chi(M) = \sum_{\lambda=0}^{\dim E} (-1)^\lambda C_\lambda(f)$ for one Morse function f on M chosen once and for all, and $\chi(M) = 0$ if M has no Morse function. *Book:* p. 29: $\chi(M) = C_0 - C_1 + C_2 - \cdots \pm C_n$, where χ is the Euler characteristic of p. 28. *Hidden hypotheses:* the value must not depend on the choice; that is V4.19, for compact Hausdorff M. *Deviation (restated):* defined through critical points, not homology; for compact Hausdorff M it equals $\sum_i (-1)^i \dim H_i(M; F)$ for every field F (V3.38).

V4.19 (formula (3) and Application 2 on a compact manifold; `Morse.sum_critCount_eq_of_isMorseFunction_o` `Morse.sum_critCount_eq_eulerNumberM`, `Morse.exists_eulerNumber_of_compactSpace`, `Morse.sum_critCount_of_finrank_eq_zero`, `Morse.IsEulerNumber.eq_eulerNumberM`, `Morse.eulerNumberM_eq_of_diffeomorph`, `Morse.exists_isEulerNumber_eulerNumberM`). Let M be a compact Hausdorff smooth manifold without boundary modelled on a finite-dimensional real normed space. (i) Every Morse function f on M has $\sum_{\lambda=0}^{\dim M} (-1)^\lambda C_\lambda(f) = \chi(M)$; in dimension 0 both sides are the number of points of M. (ii) If $\dim M = d+1 \geq 1$ and $N \subseteq \mathbb{R}^{K+1}$, $K \geq 1$, is a compact nonempty slice submanifold modelled on $\mathbb{R}^{d+1}$ and diffeomorphic to M (for instance the image of any embedding, V4.14), every Euler number of N (V3.17) equals $\chi(M)$; so the Poincaré–Hopf integer does not depend on the embedding. (iii) χ is invariant under diffeomorphisms between compact Hausdorff manifolds. (iv) If M is nonempty of dimension $d+1$, it is diffeomorphic to a compact $N \subseteq \mathbb{R}^{K+1}$ on which every smooth tangent field with finitely many zeros has index sum $\chi(M)$. *Book:* p. 29, formula (3); Application 2 (pp. 36–37): on a compact M the index sum of grad f is $\chi(M)$, and the index sum of any vector field is $\chi(M)$, the index sum being a topological invariant (Steenrod, §39.7). *Hidden hypotheses:* Hausdorff; in Application 2 "any differentiable function" must be a Morse function. *Deviation (restated):* $\chi(M)$ is the integer of V4.18, which equals the Poincaré–Hopf integer of every embedded copy and, by V3.38, the homological Euler characteristic; vector fields and their indices are defined in the Milnor project only for submanifolds of Euclidean space, so Application 2 is stated on an embedded copy; the topological invariance that Milnor quotes from Steenrod is replaced by the Poincaré–Hopf theorem and invariance under diffeomorphisms.

V4.20 (the generic perturbation behind Corollary 6.8; `Morse.negGrad`, `Morse.exists_local_extension_of_con` `Morse.mem_image_critPoints_of_not_isNondegenerate`, `Morse.ae_isMorseFunction_add_inner`, `Morse.exists_mem_isMorseFunction_add_inner`). For every slice submanifold $M \subseteq \mathbb{R}^n$, every smooth $f : M \to \mathbb{R}$ and every Haar measure on $\mathbb{R}^n$: for almost every $a \in \mathbb{R}^n$, $f + \langle a, \cdot \rangle$ is a Morse function on M, and such a exist in every nonempty open set. *Book:* not in this form; Milnor's proof of Corollary 6.8 (p. 37) uses L_p on an embedding whose first coordinate is f. For $f = \frac{1}{2}\|x\|^2$ the statement is Theorem 6.6, since $\frac{1}{2}\|x\|^2 + \langle a, x \rangle = \frac{1}{2} L_{-a}(x) - \frac{1}{2}\|a\|^2$. *Hidden hypotheses:* none; no boundedness, compactness or closedness. *Deviation (not in the book):* the standard linear perturbation, used in place of Milnor's construction. *Proof:* that of Theorem 6.6 with L_p replaced by $f + \langle a, \cdot \rangle$ and E by E_Γ, $\Gamma = -\nabla F$ for a local extension F of f (constant along the slices of a slice chart): a degenerate critical point q makes $(q, a + \nabla F(q))$ a critical point of E_Γ with value a (V4.8), and Sard's theorem (V4.3) on countably many pieces of N finishes.

V4.21 (Corollary 6.8 on a submanifold of $\mathbb{R}^n$; `Morse.exists_isMorseFunction_approx`, `Morse.ballDiffeo`, `Morse.isSliceSubmanifold_ballImage`, `Morse.isMorseFunction_of_ballImage`, `Morse.ae_isMorseFunction_add_inner_ballDiffeo`, `Morse.exists_isMorseFunction_approx_uniform`). Let $M \subseteq \mathbb{R}^n$ be any slice submanifold, $f : M \to \mathbb{R}$ smooth (bounded or not), $\varepsilon > 0$, $r \in \mathbb{N}$, and finitely many smooth coordinate systems y_j on open subsets of M, with values in $\mathbb{R}^k$, with compact sets K_j inside their ranges. There is a Morse function g on M with $|g - f| < \varepsilon$ on all of M and $\|D^i((g-f)\circ y_j^{-1})(u)\| < \varepsilon$ for every j, every $i \le r$ and every $u \in K_j$. One can take $g = f + \langle a, \beta(\cdot)\rangle$ with a small, where $\beta(y) = y/\sqrt{1+\|y\|^2}$ is the diffeomorphism of $\mathbb{R}^n$ onto its open unit ball; for compact M also $g = f + \langle a, \cdot\rangle$ (`Morse.exists_isMorseFunction_approx`). *Book:* Corollary 6.8 (p. 37): any bounded smooth $f : M \to \mathbb{R}$ can be uniformly approximated by a smooth g without degenerate critical points, and g can be chosen so that its i-th derivatives on a given compact $K \subset M$ uniformly approximate those of f, for $i \le k$. *Hidden hypotheses:* derivatives on a manifold exist only in coordinates, so the approximation on K must be read in fixed coordinate systems covering K (finitely many suffice); Milnor's embedding h with bounded image and first coordinate f is not constructed (one can take $h = (f, \beta \circ h')$ for any embedding h'). *Deviation (stronger):* **no boundedness hypothesis on** f: the perturbation $\langle a, \beta(\cdot)\rangle$ is bounded by $\|a\|$; and M need not be compact. M is a submanifold of $\mathbb{R}^n$; abstract compact M is V4.22. *Proof:* V4.20 applied to the slice submanifold $\beta(M)$ and $f \circ \beta^{-1}$, and the Morse property carried back along β.

V4.22 (Corollary 6.8 on a compact manifold; `Morse.ae_isMorseFunction_add_inner_comp`, `Morse.exists_isMorseFunction_approx_of_compactSpace`). (i) Let M be a smooth manifold without boundary modelled on a finite-dimensional real normed space, $e : M \to \mathbb{R}^n$ smooth, a topological embedding with injective derivative, and $f : M \to \mathbb{R}$ smooth. Then $f + \langle a, e(\cdot)\rangle$ is a Morse function for almost every a (any Haar measure). (ii) If M is compact and Hausdorff, then for every smooth f, $\varepsilon > 0$, $r \in \mathbb{N}$, and finitely many smooth coordinate systems y_j (with values in a real normed space) with compact sets K_j inside their ranges, there is a Morse function g with $|g - f| < \varepsilon$ on M and every derivative of order $\le r$ of $(g-f)\circ y_j^{-1}$ of norm $< \varepsilon$ on K_j; one can take $g = f + \langle a, e(\cdot)\rangle$, e Mathlib's Whitney embedding. *Book:* Corollary 6.8 (p. 37). *Hidden hypotheses:* Hausdorff; Milnor's boundedness of f is automatic for compact M. *Deviation (narrower):* (ii) for compact M only; the perturbation is not Milnor's.

V4.23 (Lemma 6.9, the index of L_p; `Morse.index_firstFF_sub_secondFF`, `Morse.indexM_normSq_sub_eq_card`, `Morse.indexM_normSq_sub_eq_finsum_focalMult`). (i) The index of $g - \ell_v$ is the number of principal curvatures in the direction v, counted with multiplicity, that are greater than 1. (ii) At a critical point q of L_p, the index of L_p is the number of principal curvatures of M at q in the direction $p - q$ that are greater than 1. (iii) If $k + w = n$ and q is a non-degenerate critical point of L_p, the index of L_p at q is $\sum_{t\in[0,1]} \mathrm{mult}_q(q + t(p-q))$ (all but finitely many terms vanish): the number of focal points of (M, q) on the segment from q to p, each counted with its multiplicity. *Book:* Lemma 6.9 (p. 37), proved on p. 38: the index of $2(g_{ij} - t\, v\cdot \ell_{ij})$ is its number of negative eigenvalues, which for (g_{ij}) the identity is the number of eigenvalues of $(v \cdot \ell_{ij})$ that are $\ge 1/t$; compare with Lemma 6.3. *Hidden hypotheses:* the eigenvalues to count are those $> 1/t$, not $\ge 1/t$: equality would make p a focal point of (M, q), i.e. q degenerate (V4.11), and this is where non-degeneracy enters; the case $p = q$ (index 0, and the segment $\{q\}$ holds no focal point). *Deviation (restated):* the direction $p - q$ is not normalized: if $p - q = tv$ with $\|v\| = 1$, the curvatures in the direction $p - q$ are t times those in the direction v, so "greater than 1" is Milnor's "greater than $1/t$". *Proof:* Sylvester's law of inertia in an orthonormal eigenbasis of S_v, in which $g - \ell_v$ is $\sum_i (1 - \kappa_i)c_i^2$.

V4.24 (Reeb's theorem on a compact manifold; `Morse.reeb_of_compactSpace`). Let M be a compact Hausdorff smooth manifold without boundary modelled on a real normed space of finite dimension n, and $f : M \to \mathbb{R}$ smooth with exactly two critical points, both non-degenerate. Then M is homeomorphic to the unit sphere $S^n \subseteq \mathbb{R}^{n+1}$. This is Theorem 4.1 of Chapter 3, placed here because its Lean file is `Morse/Chapter4/Whitney.lean`. *Book:* Theorem 4.1 (p. 25), in Milnor's own setting of an abstract compact manifold. *Hidden hypotheses:* Hausdorff; the others as in V3.6. *Proof:* V3.6 on an embedded copy (V4.16), the critical points and their non-degeneracy carried over by V4.15.

V4.25 (Theorem 3.1 on a compact manifold). This is V2.4 in Chapter 2, where it belongs in book order; it is proved through the Whitney embedding of this chapter (V4.14–V4.16).

What is not verified in this chapter

- **Theorem 6.1 in Milnor's generality** (C^1 maps between manifolds of the same dimension): only smooth maps into a vector space are treated, which is every use Milnor makes of it; the Milnor project proves Sard's theorem for smooth maps only.
- **Corollary 6.7 for a non-compact abstract manifold**, and with it the abstract forms of V4.17 and V4.22 for non-compact M: Mathlib's Whitney embedding is for compact manifolds only, so a non-compact manifold has no closed embedding into Euclidean space here. The submanifold forms (V4.13, V4.21) need the embedding as input.
- **Application 1** (p. 36: a differentiable manifold has the homotopy type of a CW-complex): Theorem 3.5 is proved only for compact submanifolds of Euclidean space and only as an iterated adjunction space, since Mathlib has no cellular approximation (Chapter 2); Theorems 3.2 and 3.5 are not transferred to abstract manifolds; the non-compact case needs Whitehead's theorem.
- **Application 2 as Milnor argues it**: χ enters as the Poincaré–Hopf integer (V3.17, V4.18), identified with $\sum_i (-1)^i \dim H_i(M; F)$ only afterwards (V3.38); the topological invariance quoted from Steenrod is not formalized; vector fields on an abstract manifold are not treated, only on an embedded copy.
- **Independence of the principal curvatures from the parametrization**: not proved, and not needed (V4.6).
- **Milnor's own proof of Corollary 6.8**, through L_p on an embedding with first coordinate f: not formalized; replaced by V4.20–V4.22.

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6. Chapter 5 (Milnor §7): partly verified

Milnor's §7 (pp. 39–42) applies Part I to complex submanifolds of $\mathbb{C}^n$. A closed complex submanifold of complex dimension k has the homotopy type of a CW-complex of dimension at most k (Theorem 7.2), so a non-singular affine variety of real dimension $2k$ has no homology above dimension k (Theorem 7.1), and Lefschetz's theorems on hyperplane sections follow (Corollary 7.3, Theorem 7.4). The only new mathematics in the section is algebraic: at a critical point q of $L_p = \| \cdot - p\|^2$ on a complex submanifold, the Hessian has index at most k. That algebraic core is verified, in a stronger form than Milnor's (at every point of a holomorphic chart, critical or not, degenerate or not), together with Assertions 1 and 2, read as statements about quadratic forms and about Hessians in a holomorphic chart. The setting is linear algebra: a finite-dimensional complex vector space V (Milnor's $\mathbb{C}^k$, the domain of a holomorphic chart), a complex inner product space F

(Milnor's $\mathbb{C}^n$), and a map $w : V \to F$ holomorphic near a point (Milnor's inclusion $M \to \mathbb{C}^n$ read in holomorphic coordinates). The topological statements of §7 are verified as well (V5.13–V5.19), for submanifolds of $\mathbb{R}^{2n}$ whose tangent spaces are complex subspaces: the index bound for L_p on such a submanifold, at every critical point; Theorem 7.2, as a countable cell complex with cells of dimension at most k; Theorem 7.1, over a field; and, under the hypothesis that a neighbourhood of the hyperplane section deforms into it (which Milnor draws from the triangulability of algebraic varieties), Corollary 7.3 over a field and Theorem 7.4, for a closed subset of $\mathbb{C}P^n$ whose affine part is such a submanifold. Focal points of complex submanifolds are never used. Not verified: integral coefficients, the passage from algebraic varieties to these hypotheses, and the deformation hypothesis itself (the list at the end). Lean: `Morse/Chapter5/IndexBound.lean` (545 lines), on the definitions of §2 in `Morse/Chapter1/Nondegenerate.lean`, for V5.1–V5.12; `ComplexIndex.lean` (230), `Lefschetz.lean` (351), `LefschetzSlab.lean` (364), `HyperplaneSection.lean` (423), `ProjectiveSection.lean` (310) and `LefschetzHomotopy.lean` (581) for V5.13–V5.19.

Conventions. A complex vector space is also regarded as a real one (restriction of scalars), and "bilinear form" means a continuous real-bilinear form. Mathlib's Hermitian product $\langle a, b\rangle$ is conjugate-linear in the first slot, Milnor's $\sum_\alpha a_\alpha \bar{b}_\alpha$ in the second; their real parts agree, and only real parts occur below. A *complex structure* on a real vector space is a real-linear J with $J^2 = -1$; on a complex vector space, $J = i\cdot$.

V5.1 (Hessian, index, nullity; `Morse.hessian`, `Morse.index`, `Morse.index_isGreatest`, `Morse.nullSpace`, `Morse.nullity`). For $f : E \to \mathbb{R}$ on a real normed space E and $q \in E$, the Hessian $d^2 f_q$ is the second Fréchet derivative, the bilinear form $(v, w) \mapsto d(x \mapsto df_x(w))_q(v)$; it is defined at every point (it is 0 at a point where $x \mapsto df_x$ is not differentiable). For a bilinear form H, the null space is $\{v : H(v, w) = 0 \text{ for all } w\}$ and the nullity is its dimension; the index is Mathlib's negative index of inertia (`sigNeg`) of $v \mapsto H(v, v)$, and on a finite-dimensional space `Morse.index_isGreatest` proves it is the largest dimension of a subspace on which $H(v, v) < 0$ for $v \neq 0$. *Book:* p. 5: the index of a bilinear functional is the maximal dimension of a subspace on which it is negative definite, the nullity is the dimension of its null space; the Hessian f_{**} is defined at a critical point (p. 4) and is the matrix of second partial derivatives in local coordinates (p. 5). *Hidden hypotheses:* none. *Deviation (restated):* the formal Hessian exists at every point; at a critical point it is the matrix of f_{**} in the given linear coordinates. The nullity is a Mathlib `finrank`, which is 0 for an infinite-dimensional null space; the statements below that compare nullities hold as stated in any dimension and are meaningful in finite dimension.

V5.2 (Milnor's form Q' and the orthogonal map J; `Morse.reBilin`, `Morse.reBilin_apply`, `Morse.reBilin_I_smul`, `Morse.reInner`, `Morse.reInner_I_smul`, `Morse.realMulI`, `Morse.complexStructureEq`). For a complex vector space V and a continuous complex-bilinear $B : V \times V \to \mathbb{C}$, $\operatorname{Re} B$ is a real bilinear form; for $B(z, z') = \sum b_{hj} z^h z'^j$ on $\mathbb{C}^k$ its quadratic form is Milnor's Q'. Multiplication by i is a real-linear isomorphism J with $J^2 = -1$ (inverse $-J$), and $\operatorname{Re} B(iv, iw) = -\operatorname{Re} B(v, w)$ for all v, w. On a complex inner product space the real inner product $\operatorname{Re}\langle v, w\rangle$ (the Euclidean inner product of $\mathbb{R}^{2k}$ when $V = \mathbb{C}^k$) satisfies $\operatorname{Re}\langle iv, iw\rangle = \operatorname{Re}\langle v, w\rangle$: J is orthogonal. *Book:* p. 39: substituting $x^h + iy^h$ for z^h in $Q(z) = \sum b_{hj} z^h z^j$ and taking real parts gives a real quadratic form Q' in $2k$ variables; the identity $Q(iz) = -Q(z)$ shows that an orthogonal change of variables carries Q' into $-Q'$. *Hidden hypotheses:* none. *Deviation (stronger):* the sign change is proved for the bilinear form, not only for the quadratic form.

V5.3 (Assertion 1; `Morse.nullity_reBilin_sub_smul_reInner_eq`, `Morse.eigenspace_eq_nullSpace_sub_smul`, `Morse.finrank_eigenspace_neg_eq`). Let V be a complex inner product space, B a continuous

complex-bilinear form on V and $e \in \mathbb{R}$. Then $\operatorname{Re} B - e \operatorname{Re}\langle\cdot,\cdot\rangle$ and $\operatorname{Re} B + e \operatorname{Re}\langle\cdot,\cdot\rangle$ have the same nullity. If A is a real-linear operator on V with $\operatorname{Re}\langle Au, w\rangle = \operatorname{Re} B(u, w)$ for all u, w (the matrix of Q' in orthonormal real coordinates), then the e-eigenspace of A is the null space of $\operatorname{Re} B - e \operatorname{Re}\langle\cdot,\cdot\rangle$, and the e- and $(-e)$-eigenspaces of A have the same dimension. *Book:* p. 39, Assertion 1: if e is an eigenvalue of Q' with multiplicity μ, then so is $-e$; after the orthogonal change of variables of V5.2, Assertion 1 "clearly follows". *Hidden hypotheses:* none. *Proof:* as in the book; the step left implicit is that J, orthogonal with $\operatorname{Re} B(Ju, Jw) = -\operatorname{Re} B(u, w)$, carries $\operatorname{Re} B - e \operatorname{Re}\langle\cdot,\cdot\rangle$ into $-(\operatorname{Re} B + e \operatorname{Re}\langle\cdot,\cdot\rangle)$, and a linear change of variables does not change the nullity.

V5.4 (Assertion 1 relative to an invariant form; `Morse.nullity_sub_smul_eq_nullity_add_smul`, `Morse.nullity_sub_smul_eq_nullity_add_smul'`). Let E be a real normed space, J a continuous real-linear map with $J^2 = -1$, G a bilinear form with $G(Jv, Jw) = G(v, w)$ and L a bilinear form with $L(Jv, Jw) = -L(v, w)$, for all v, w. Then for every real t the forms $G - tL$ and $G + tL$ have the same nullity, and for every real e the forms $L - eG$ and $L + eG$ have the same nullity. *Book:* not stated. On p. 40 Milnor passes from Assertion 1 (eigenvalues for the coordinate inner product of $\mathbb{R}^{2k}$) to focal points, which are governed by the second fundamental form relative to the first fundamental form (g_{ij}) (Assertion 6.4, p. 35). *Deviation (not in the book):* this is the form of Assertion 1 that Assertion 2 needs (Assertion 1a of the exposition). With $G = (g_{ij})$, which is J-invariant by the Cauchy–Riemann equations, and L the second fundamental form in a normal direction v, the two nullities $G \mp tL$ are the multiplicities of $q \pm tv$ as focal points; V5.3 gives the same conclusion only when the real coordinates are orthonormal at q, an assumption Milnor makes in §6 (p. 34) but not on p. 40.

V5.5 (real orthogonality to a complex subspace; `Morse.inner_eq_zero_of_re_inner_eq_zero`). Let T be a complex linear subspace of a complex inner product space and v a vector with $\operatorname{Re}\langle a, v\rangle = 0$ for every $a \in T$. Then $\langle a, v\rangle = 0$ for every $a \in T$. *Book:* p. 40, the parenthesis that the linear terms of $\sum_\alpha w_\alpha(z)\bar{v}_\alpha$ vanish since v is orthogonal to M. *Hidden hypotheses:* the tangent space TM_q is a complex subspace, and orthogonality is meant in the real sense; the lemma is what turns real orthogonality into the vanishing of the Hermitian products. *Deviation (not in the book):* stated and proved, but not used by V5.6–V5.11, which avoid power series.

V5.6 (the Hessian of L_p in holomorphic coordinates; `Morse.hessian_normSq_sub_holomorphic`). Let V be a complex normed space, F a complex inner product space, $w : V \to F$ complex C^2 on a neighbourhood of q (in particular holomorphic there), and $p \in F$. Then the real Hessian at q of $z \mapsto \|w(z) - p\|^2$ is

$$(\xi, \xi') \longmapsto 2\Big(\operatorname{Re}\langle dw_q\xi,\ dw_q\xi'\rangle + \operatorname{Re}\,\big\langle w(q) - p,\ d^2w_q(\xi, \xi')\big\rangle \Big),$$

where dw_q and d^2w_q are the complex first and second derivatives. The first term is unchanged by $(\xi, \xi') \mapsto (i\xi, i\xi')$ (it is the first fundamental form of w), the second changes sign (it is the real part of a complex bilinear form). *Book:* p. 40: in holomorphic coordinates the real inner product $w \cdot v$ is a constant plus $Q'(x, y)$ plus higher terms, and "clearly" the quadratic terms Q' determine the second fundamental form in the normal direction v. *Hidden hypotheses:* none. *Deviation (restated):* Milnor's remark is replaced by an explicit formula for the whole Hessian of L_p in the chart. For $p = w(q) + tv$, v a unit normal, the second inner product is $-2t\,Q'(\xi)$ on the diagonal, Q being Milnor's quadratic part of $\sum_\alpha w_\alpha\bar{v}_\alpha$.

V5.7 (Assertion 2, in a holomorphic chart; `Morse.nullity_hessian_normSq_sub_add_eq_sub`). With V, F, w, q as in V5.6, for every $n \in F$ and every real t, the Hessians at q of $\|w(\cdot) - (w(q) +$

$tn)\|^2$ and of $\|w(\cdot) - (w(q) - tn)\|^2$ have the same nullity. *Book:* p. 40, Assertion 2: the focal points of (M, q) along any normal line occur in pairs symmetric about q; if $q + tv$ is a focal point, then $q - tv$ is a focal point with the same multiplicity. *Hidden hypotheses:* none for the formal statement (n need be neither normal nor a unit vector, and q need not be critical). *Deviation (restated):* the statement concerns Hessians in a holomorphic chart, not focal points. For a unit normal n these nullities are the multiplicities of $w(q) \pm tn$ as focal points, by Assertion 6.4 and Milnor's p. 36 formula for the Hessian of L_p; those are formalized for real slice submanifolds of $\mathbb{R}^n$ (Chapter 4, `Morse.isFocalPointMult_add_smul_iff`), but a holomorphic chart of a complex submanifold is not identified with a chart of that real submanifold, so the passage to focal points is not verified. *Proof:* $J = i\cdot$ carries one Hessian into the other (V5.6), as in V5.4. Milnor's route through the eigenvalues of Q' (V5.3) is not used.

V5.8 (the Andreotti–Frankel identity; `Morse.apply_J_apply_J_eq`, `Morse.apply_I_smul_apply_I_smul_eq`, `Morse.apply_I_smul_apply_I_smul_eq_two_mul_norm_sq_sub`). Let G, L, H be bilinear forms on a real normed space and J any map of it such that, for all v, $H(v, v) = G(v, v) - L(v, v)$, $G(Jv, Jv) = G(v, v)$ and $L(Jv, Jv) = -L(v, v)$. Then $H(Jv, Jv) = 2G(v, v) - H(v, v)$. On a complex vector space with $J = i\cdot$ and $L = \operatorname{Re} B$, B complex bilinear, the condition on L is automatic; on a complex inner product space with $G(v, v) = \|v\|^2$ this reads $H(iv, iv) = 2\|v\|^2 - H(v, v)$. *Book:* not in Milnor's text; it replaces the count of focal points on p. 41 (Milnor attributes the proof of §7 to Andreotti and Frankel, p. 39). *Deviation (not in the book).*

V5.9 (the index bound; `Morse.two_mul_index_le_finrank`, `Morse.index_le_finrank_complex`). Let E be a finite-dimensional real normed space, H a bilinear form and J an injective real-linear map of E with $H(v, v) + H(Jv, Jv) \geq 0$ for all v. Then $2 \cdot \operatorname{index} H \leq \dim_{\mathbb{R}} E$. On a finite-dimensional complex vector space V, if $H(v, v) + H(iv, iv) \geq 0$ for all v, then $\operatorname{index} H \leq \dim_{\mathbb{C}} V$. *Book:* not stated; it is the abstract form of the conclusion on p. 41 that the index of L_p at q is at most k. *Deviation (not in the book). Proof:* if H is negative definite on W, then $W \cap JW = 0$, since a non-zero $Jw \in W$ would make both $H(w, w)$ and $H(Jw, Jw)$ negative; so $2 \dim W \leq \dim E$.

V5.10 (the index bound for forms of Milnor's shape; `Morse.index_le_finrank_of_eq_sub_re`, `Morse.index_le_finrank_of_hermitian`, `Morse.index_hessian_le_finrank_complex`). Let V be a complex vector space of finite dimension k, G a bilinear form with $G(v, v) \geq 0$ and $G(iv, iv) = G(v, v)$, B a continuous complex-bilinear form, and H a bilinear form with $H(v, v) = G(v, v) - \operatorname{Re} B(v, v)$ for all v. Then $\operatorname{index} H \leq k$. In particular: on a complex inner product space, every form with quadratic form $\|v\|^2 - \operatorname{Re} B(v, v)$ has index at most k; and for any $f : V \to \mathbb{R}$ and $q \in V$ whose Hessian (V5.1) has, on the diagonal, the form $G - \operatorname{Re} B$, the Hessian has index at most k. *Book:* p. 41: the index of L_p at a critical point q is at most k. *Hidden hypotheses:* none. *Deviation (stronger):* neither criticality, nor non-degeneracy, nor any smoothness of f is assumed; only the shape of the Hessian matters. *Proof:* V5.8 and V5.9: if $H(v, v) < 0$ then $H(iv, iv) = 2G(v, v) - H(v, v) > 0$. Milnor's count of focal points (Lemmas 6.3 and 6.9 with Assertion 2) is not used.

V5.11 (the index of L_p is at most k, in a holomorphic chart; `Morse.index_hessian_normSq_sub_holomorphic_le`). Let V be a complex vector space of finite dimension k, F a complex inner product space, $w : V \to F$ complex C^2 on a neighbourhood of q, and $p \in F$. Then the Hessian at q of $z \mapsto \|w(z) - p\|^2$ has index at most k. *Book:* p. 41: the index of L_p at a critical point q of $L_p : M \to \mathbb{R}$ is at most k, M a complex submanifold of complex dimension k and L_p without degenerate critical points. *Hidden hypotheses:* none. *Deviation (stronger):* q need not be a critical point, the Hessian need not be non-degenerate, and dw_q need not be injective. *Deviation (restated):* the statement is about the Hessian in a holomorphic chart. When w parametrizes M near a critical

point $w(q)$ of $L_p|M$, this Hessian is that of $L_p|M$ in the real coordinates (x, y), and its index is the index of L_p at $w(q)$ by the chart independence of §2; that identification needs M as a real slice submanifold with w^{-1} among its charts, which is not formalized. *Proof:* by V5.6 the Hessian H satisfies $H(\xi,\xi) + H(i\xi, i\xi) = 4\|dw_q\xi\|^2 \geq 0$; then V5.9.

V5.12 (no negative semidefinite Hessian; `Morse.exists_pos_apply_of_eq_sub_re`). If, in V5.10, G is moreover positive definite ($G(v,v) > 0$ for $v \neq 0$) and $V \neq 0$, then $H(v,v) > 0$ for some v. *Book:* not in Milnor's text. *Deviation (not in the book):* with $G(\xi,\xi) = 2\|dw_q\xi\|^2$ (V5.6) it says that, where dw_q is injective, the Hessian of L_p in a holomorphic chart of positive dimension is never negative semidefinite; the consequence drawn in the exposition (Chapter 5, §4), that L_p has no local maximum on a complex submanifold of positive dimension, is not formalized.

V5.13 (complex submanifolds, and the index bound at every critical point of L_p; `Morse.IsOrthComplexStructur` `Morse.IsComplexSubmanifold`, `Morse.two_mul_indexM_normSq_sub_le`, `Morse.indexM_normSq_sub_le`). Let V be a finite-dimensional real inner product space and J an orthogonal linear map of V with $J^2 = -1$ (for $V = \mathbb{C}^m = \mathbb{R}^{2m}$: multiplication by i). A slice submanifold $N \subseteq V$ is called a *complex submanifold* (for J) if J maps every tangent space T_xN into itself. For such an N, every $p \in V$ and every critical point q of $L_p = \|\cdot - p\|^2$ on N, degenerate or not, twice the index of L_p at q is at most $\dim_{\mathbb{R}} N$; so if $\dim_{\mathbb{R}} N = 2k$, the index is at most k. *Book:* pp. 39–41: for a complex manifold M of complex dimension k bianalytically embedded in $\mathbb{C}^n$, the index of L_p at a non-degenerate critical point is $\leq k$, by the focal points (Assertions 1 and 2, Lemmas 6.3 and 6.9). *Hidden hypotheses:* none. *Deviation (restated):* "complex" is the invariance of the tangent spaces under J, which a bianalytically embedded complex manifold has; nothing else about it is used. This is the bridge that V5.7 and V5.11 lacked: the bound is for the Hessian of L_p on the real submanifold, computed in a slice chart, where the complex structure read in the chart satisfies the identity of V5.8. *Deviation (stronger):* every critical point, without non-degeneracy; no focal points.

V5.14 (the homology of a slab without critical points of a given index; `Morse.critValuesIn`, `Morse.subsingleton_subsetHomology_sublevel`, `Morse.subsingleton_subsetHomology_slab`, `Morse.exists_regular_levels`). Let N be a slice submanifold of a finite-dimensional real inner product space, F a smooth function on the ambient space, $f = F|N$, and $\Bbbk$ a field. Let $a \leq b \leq a_0$ be such that $f^{-1}(-\infty, a_0]$ is compact (or only the slab $f^{-1}[\ell, a_0]$, for some $\ell \leq a$), every critical point of f in it is non-degenerate, and a and b are not critical values. If no critical point q with $a < f(q) < b$ has index j, then $H_j(N^b, N^a; \Bbbk) = 0$, where $N^c = f^{-1}(-\infty, c]$. A Morse function on N which is bounded below by 0 and has compact sublevel sets has regular values $a_0 < 0 \leq a_1 \leq a_2 \leq \ldots$ tending to infinity. *Book:* §3 and §5 (Theorems 3.1, 3.2, Remark 3.3; the exact sequences), as §7 uses them for L_p (p. 39) and for $1/L_p$ (p. 42). *Hidden hypotheses:* for $-L_p$ and $1/L_p$ the sublevel sets are not compact; Theorems 3.1 and 3.2 only need the compactness of the slab $f^{-1}[a, b]$. *Deviation (not in the book as a statement):* the form in which Chapter 3's computation (V3.33–V3.34) is used in §7, by induction on the number of critical values in the slab.

V5.15 (Theorem 7.2; `Morse.theorem_7_2`). Let $M \subseteq \mathbb{R}^n$ be a slice submanifold of dimension $2k$ which is closed in $\mathbb{R}^n$ and complex for an orthogonal J with $J^2 = -1$ (V5.13). Then M has the homotopy type of a countable cell complex (VA.18) all of whose cells have dimension at most k. *Book:* p. 39, Theorem 7.2: a complex manifold of complex dimension k, bianalytically embedded as a closed subset of $\mathbb{C}^n$, has the homotopy type of a k-dimensional CW-complex. *Hidden hypotheses:* "k-dimensional" means of dimension at most k ($\mathbb{C}$ is contractible). *Deviation (restated):* a closed submanifold with complex tangent spaces, of pure dimension $2k$; a countable cell complex in the sense of VA.18, with cells attached along arbitrary maps. *Proof:* for almost every p, L_p is a Morse

function on M (V4.12), with compact sublevel sets because M is closed; its critical points have index $\leq k$ (V5.13); the sublevel sets between regular values are obtained by attaching cells, and the tower of these attachments has the homotopy type of M (V10.20, VA.19).

V5.16 (Theorem 7.1, over a field; `Morse.theorem_7_1`). Under the hypotheses of V5.15, $H_i(M;\Bbbk) = 0$ for every field $\Bbbk$ and every $i > k$. *Book:* p. 39, Theorem 7.1 (Andreotti and Frankel): a non-singular affine algebraic variety $M \subset \mathbb{C}^n$ of real dimension $2k$ has $H_i(M;\mathbb{Z}) = 0$ for $i > k$. *Hidden hypotheses:* that such a variety is a closed complex submanifold of pure dimension. *Deviation (narrower):* coefficients in a field; "non-singular affine algebraic variety" is replaced by the hypotheses of V5.15. *Proof:* from the filtration by sublevel sets (V5.14, assembled as in V10.25), not from cellular chains.

V5.17 (hyperplane sections, for an abstract pair; `Morse.HypothesisN`, `Morse.le_indexM_neg_normSq_sub`, `Morse.subsingleton_subsetHomology_univ_of_section`, `Morse.corollary_7_3_of_section`). Let X be a compact space, $A \subseteq X$ closed, M as in V5.15, and e a homeomorphism of M onto $X \setminus A$ such that $\|e^{-1}(y)\| \to \infty$ as $y \to A$. *Hypothesis (N)* for A in X: there are an open $U \supseteq A$ and a homotopy $H : [0,1] \times U \to X$ with H_0 the inclusion, $H_1(U) \subseteq A$, and $H_t(a) = a$ for all t and all $a \in A$. Under Hypothesis (N), for every field $\Bbbk$: $H_r(X, A;\Bbbk) = 0$ for $r < k$; hence the map $H_r(A;\Bbbk) \to H_r(X;\Bbbk)$ induced by the inclusion is bijective for $r + 1 < k$ and surjective for $r < k$. On the way: at a non-degenerate critical point of L_p on M, the function $-L_p$ has index at least k. *Book:* pp. 41–42: for Corollary 7.3 the exact sequence of the pair and Lefschetz duality, with Theorem 7.1 for $V - (V \cap P)$; for Theorem 7.4 "the hypothesis that some neighborhood U of $V \cap P$ can be deformed into $V \cap P$ within V". *Hidden hypotheses:* the deformation must keep $V \cap P$ pointwise fixed; Lefschetz duality has hypotheses that are not stated. *Deviation (restated):* an abstract pair (X, A) with $X \setminus A$ a closed complex submanifold of $\mathbb{C}^n$ going to infinity at A; no duality: with $V_R = A \cup e\{L_p \geq R\}$ inside U, the map $H_r(V_R, A) \to H_r(X, A)$ is zero by (N), and $H_r(X, V_R) \cong H_r(M, \{L_p \geq R\})$ vanishes for $r < k$ by excision and V5.14 for $-L_p$ on the compact slab $\{L_p \leq R\}$; then the exact sequence of the triple. *Deviation (narrower):* field coefficients; (N) is a hypothesis.

V5.18 (Corollary 7.3, Lefschetz, over a field; `Morse.CP.affinePart`, `Morse.CP.hyperplaneSection`, `Morse.CP.affineHomeomorph`, `Morse.CP.tendsto_norm_affineHomeomorph_symm`, `Morse.corollary_7_3`). Let $V \subseteq \mathbb{C}P^n$ (V3.41) be closed and P the hyperplane $z_0 = 0$, the set of projections Q with $Q_{00} = 0$. The *affine part* of V is the set of $u \in \mathbb{C}^n = \mathbb{R}^{2n}$ for which the projection onto the line of $(1, u)$ belongs to V; it is homeomorphic to $V \setminus P$, and its points go to infinity as their images approach $V \cap P$. Suppose the affine part is a slice submanifold of $\mathbb{R}^{2n}$ of dimension $2k$, complex for multiplication by i, and suppose Hypothesis (N) for $V \cap P$ in V (V5.17). Then for every field $\Bbbk$ the map $H_r(V \cap P;\Bbbk) \to H_r(V;\Bbbk)$ induced by the inclusion is bijective for $r < k - 1$ and surjective for $r = k - 1$. *Book:* p. 41, Corollary 7.3 (Lefschetz): for an algebraic variety V of complex dimension k in $\mathbb{C}P_n$ and a hyperplane P containing the singular points of V, the inclusion $V \cap P \to V$ induces isomorphisms of homology groups in dimensions less than $k - 1$, and is onto in dimension $k - 1$. *Hidden hypotheses:* V must have pure dimension k (the exposition gives a counterexample otherwise); and (N). *Deviation (narrower):* field coefficients; (N) assumed; the hyperplane $z_0 = 0$ only (another hyperplane is brought there by a unitary change of coordinates, which is not formalized); "algebraic variety with its singular points in P" is replaced by the hypothesis on the affine part.

V5.19 (Theorem 7.4, Lefschetz; `Morse.theorem_7_4_of_section`, `Morse.theorem_7_4`, `Morse.theorem_7_4_of_embedding`, `Morse.theorem_7_4_bijective`, `Morse.theorem_7_4_surjective`). Under the hypotheses of V5.18: for $r < k$, every continuous map $h : I^r \to V$ is homotopic,

relative to $h^{-1}(V \cap P)$, to a map into $V \cap P$; the same holds for maps from any compact space that admits a continuous injection into a real vector space of dimension $< k$, and for the abstract pair (X, A) of V5.17. Consequently, for every $x \in V \cap P$, the map $\pi_i(V \cap P, x) \to \pi_i(V, x)$ induced by the inclusion is bijective for $i + 1 < k$ and surjective for $i < k$. *Book:* pp. 41–42, Theorem 7.4 (Lefschetz): under the same hypotheses the relative homotopy group $\pi_r(V, V \cap P)$ is zero for $r < k$. The proof uses the function f equal to 0 on $V \cap P$ and to $1/L_p$ elsewhere, whose critical points have index $\geq 2k - k = k$, and the hypothesis that a neighbourhood U of $V \cap P$ can be deformed into $V \cap P$ within V, which "can be proved, for example, using the theorem that algebraic varieties can be triangulated". *Hidden hypotheses:* as in V5.17; and $1/L_p$ is infinite at p when $p \in V$. *Deviation (restated):* relative homotopy groups are not in Mathlib; the statement is the criterion for their vanishing, in the relative form above, with $r = 0$ included. Milnor's $1/L_p$ is replaced by $1/(1 + L_p)$. *Proof:* the pushing lemma of Chapter 13 (V13.13) for $-L_p$ on a compact slab pushes every map into a small neighbourhood V_R of A, after localizing to the part of the map that lies outside it; (N) pushes V_R into A.

What is not verified in this chapter

- **Integral coefficients** in Theorem 7.1 and Corollary 7.3 (pp. 39, 41): the homology of the sublevel filtrations is computed over a field (V5.16–V5.18).
- **Algebraic varieties:** that a non-singular affine variety, or the affine part of a projective variety of pure dimension whose singular points lie in P, is a closed submanifold with complex tangent spaces, and that a bianalytically embedded complex manifold is one. These are the hypotheses of V5.13–V5.19; and Theorem 7.2 gives a countable cell complex with cells attached along arbitrary maps (VA.18), not a CW-complex in the strict sense.
- **Assertion 2 as a statement about focal points** (p. 40), and **Milnor's count on p. 41** (at most $2k$ focal points on the normal line, symmetric about q, hence at most k between p and q, via Lemmas 6.3 and 6.9): replaced by V5.8–V5.11 and V5.13, which need neither.
- **Hypothesis (N)** (p. 42): the deformation of a neighbourhood of $V \cap P$ into $V \cap P$, which Milnor obtains from the triangulability of algebraic varieties (quoted). It is a hypothesis of V5.17–V5.19, in the form that keeps $V \cap P$ fixed. **Lefschetz duality**, which Milnor's proof of Corollary 7.3 quotes, is neither formalized nor needed (V5.17).
- **Relative homotopy groups** as objects (pp. 41–42): Theorem 7.4 is stated as the criterion for the vanishing of $\pi_r(V, V \cap P)$ (V5.19). **Hyperplanes other than** $z_0 = 0$ need a unitary change of coordinates, which is not formalized.

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7. Chapter 6 (Milnor §§8–9): partly verified

Milnor's §§8–9 (pp. 43–54) are a rapid course on connections for an abstract Riemannian manifold: affine connections and Christoffel symbols, covariant derivatives along curves (Lemma 8.1), parallel transport (Lemma 8.2), compatibility with the metric (Lemma 8.3, Corollary 8.4), symmetric connections and the fundamental Lemma 8.6, parametrized surfaces (Lemma 8.7), the curvature tensor and its tensor property (Lemma 9.1), Lemma 9.2, and the four symmetries of Lemma 9.3; the sectional curvature is defined later, on p. 100. Mathlib (as used here) has covariant derivatives on vector bundles, their torsion and metric compatibility, but no curvature tensor, no Christoffel symbols, no covariant derivative along curves, no parallel transport and no Levi-Civita connection. The chapter is therefore verified in three layers. (1) On an abstract manifold, with Mathlib's `CovariantDerivative`: Milnor's rules, his symmetric and compatible connections identified with

Mathlib's torsion-free and metric ones, the sign of his curvature (the negative of the Kobayashi–Nomizu R), and the uniqueness half of Lemma 8.6. (2) For a submanifold N of a Euclidean space with the induced metric: the substance of §§8–9 — the Levi-Civita connection with rules (1)–(4), Lemma 8.6 (existence and uniqueness), Lemmas 8.1–8.3 and 8.7, Corollary 8.4, parallel transport as a linear isometry, the curvature tensor, Lemmas 9.1–9.3 — plus the Gauss equation and the curvature of the round sphere. Every Riemannian manifold is isometric to such an N by Nash's embedding theorem, which is quoted in the Lean files but neither proved nor used, so these results are statements about embedded manifolds with the induced metric. (3) At one point, for any trilinear map on an inner product space satisfying Lemma 9.3 (1)–(3): Lemma 9.3 (4) by Milnor's octahedron, and the theory of sectional curvature. What is not verified is listed at the end. Lean: `Morse/Chapter6/` — `Connection.lean` (248 lines), `CurvatureAlgebra.lean` (499), `Projection.lean` (756), `Submanifold.lean` (780), `LinearODE.lean` (232), `Parallel.lean` (461), `Surface.lean` (239), `SphereCurvature.lean` (131).

The embedded setting (layer 2). V is a finite-dimensional real inner product space (Milnor's $\mathbb{R}^K$) and $N \subseteq V$ a slice submanifold in the sense of the companion project (`Milnor.IsSliceSubmanifold`): every point of N lies in the domain of a diffeomorphism σ from an open subset of V onto an open subset of $Z \times W$ that carries N onto $Z \times \{0\}$, and the maps $q \mapsto (\sigma q)_1$ make N a smooth manifold modelled on Z. Z is assumed finite-dimensional throughout, and W in the results that use the symmetry of the second fundamental form; when N is non-empty both hold automatically, $Z \times W$ being linearly isomorphic to V. N carries the induced metric: tangent vectors at q are vectors of V, with the inner product of V. A map $F : N \to E$ into a normed space is smooth, or differentiable at q, as a map of manifolds; a curve $c : \mathbb{R} \to N$ or a surface $s : \mathbb{R}^2 \to N$ is smooth when it is smooth as a map into V. No closedness or compactness of N is assumed anywhere.

V6.1 (tangent spaces, the projection P_q, derivatives along N, vector fields; `Morse.tangentSubspace`, `Morse.tanProj`, `Morse.dN`, `Morse.dN_eq_fderiv`, `Morse.IsTangentField`). $T_qN \subseteq V$ is the range of the derivative at q of the inclusion $N \to V$, and P_q is the orthogonal projection of V onto T_qN. For $F : N \to E$, the linear map $dN\,F\,q : V \to E$ is the derivative of F at q read on ambient vectors: on $v \in T_qN$ it is $dF_q(v)$ (its values off T_qN depend on a chart and are never used). If $F = G$ on N near q for some $G : V \to E$ differentiable at q, then $dN\,F\,q\,v = dG_q(v)$ for $v \in T_qN$. A vector field is a map $X : N \to V$ with $X(q) \in T_qN$ for every q; it is smooth if it is smooth as a map $N \to V$. *Book:* Milnor works on an abstract manifold with tangent spaces TM_p, vector fields and directional derivatives X_pf (pp. 43–44). *Hidden hypotheses:* none. *Deviation (narrower):* the embedded setting.

V6.2 (the projection is smooth, and tangent vectors extend to fields; `Morse.contMDiff_tanProj`, `Morse.tanProj_eq_gramProj`, `Morse.contDiffAt_gramProj`, `Morse.projField`, `Morse.contMDiff_projField`, `Morse.projField_apply_of_mem`). The map $q \mapsto P_q$ from N to the operators on V is smooth. Hence for each $u \in V$ the field $q \mapsto P_qu$ is a smooth vector field on N, equal to u at every q with $u \in T_qN$: every tangent vector extends to a smooth vector field on all of N. *Book:* not stated; Milnor checks the smoothness of what he builds chart by chart (condition (1), p. 44). *Deviation (not in the book). Proof:* in a slice chart with parametrization x, $P_{x(u)} = A(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}$ with $A = dx_u$, which is injective, and this expression is smooth in A near an injective operator.

V6.3 (affine connections on an abstract manifold; `Morse.cov_smul_left`, `Morse.cov_add_right`, `Morse.cov_smul_right`). Let M be a smooth manifold in Mathlib's sense (modelled on a real normed space) and ∇ a Mathlib covariant derivative on its tangent bundle, $X_p \vdash Y$ being the value at X_p of the linear map $\nabla Y(p) : T_pM \to T_pM$. Then Milnor's (3), $(fX) \vdash Y = f\,(X \vdash Y)$, holds at every point; (2), additivity in Y, holds at every point where both summands are

differentiable; and (4), $X \vdash (fY) = (Xf)\,Y + f\,(X \vdash Y)$, holds at every point where f and Y are differentiable. Dependence on X_p only, and linearity in X_p, are part of the definition. *Book:* pp. 43–44: an affine connection at p assigns to X_p and Y a vector $X_p \vdash Y$, bilinear, with the Leibniz rule; a global one satisfies the smoothness condition (1), and conditions (1)–(4) "can be taken as the definition". *Hidden hypotheses:* differentiability of f and Y at the point, in (2) and (4). *Deviation (restated):* Milnor's connection is Mathlib's `CovariantDerivative`, his condition (1) the separate class `ContMDiffCovariantDerivative`; Mathlib's object is defined on all sections and is unconstrained on sections not differentiable at the point.

V6.4 (the connection of the induced metric and Milnor's rules (1)–(4); `Morse.covAt`, `Morse.leviCivita`, `Morse.contMDiff_leviCivita`, `Morse.leviCivita_add_left`, `Morse.leviCivita_add_righ` `Morse.leviCivita_smul_left`, `Morse.leviCivita_smul_right`). In the embedded setting put $v \vdash Y = P_q(dY_q(v))$ for $v \in T_qN$ and $Y : N \to V$, and $(X \vdash Y)(q) = P_q(dY_q(X(q)))$. Then $X \vdash Y$ is tangent to N at every point, and: (1) $X \vdash Y$ is smooth when X is a smooth vector field and Y is smooth; (2) it is additive in X, and additive in Y at every point where both summands are differentiable; (3) $(fX) \vdash Y = f\,(X \vdash Y)$ for every function f; (4) at q, $X \vdash (fY) = (Xf)\,Y + f\,(X \vdash Y)$ with $Xf = df_q(X(q))$, when f and Y are differentiable at q and $X(q)$, $Y(q)$ are tangent. *Book:* p. 44, conditions (1)–(4). *Hidden hypotheses:* differentiability in (2) and (4). *Deviation (narrower):* the embedded setting; the connection is a global formula, with no Christoffel symbols.

V6.5 (covariant derivative along a curve; `Morse.velocity`, `Morse.IsFieldAlong`, `Morse.covDeriv`, `Morse.deriv_mem_tangentSubspace`, `Morse.covDeriv_add`, `Morse.covDeriv_smul`, `Morse.covDeriv_comp`). For a curve $c : \mathbb{R} \to N$, a vector field along c is a map $X : \mathbb{R} \to V$ with $X(t) \in T_{c(t)}N$ for all t; the velocity $c'(t)$ is tangent when c is smooth; and $DX/dt\,(t) = P_{c(t)}(X'(t))$. Milnor's axioms hold: (a) $D(X+Y)/dt = DX/dt + DY/dt$ at t when X, Y are differentiable at t; (b) $D(fX)/dt = f'X + f\,DX/dt$ at t when f, X are differentiable at t and $X(t)$ is tangent; (c) if c is smooth and $Y : N \to V$ is differentiable at $c(t)$, then $D(Y \circ c)/dt\,(t) = c'(t) \vdash Y$. *Book:* pp. 45–46: parametrized curves, vector fields along a curve, the velocity field, and the axioms (a), (b), (c). *Hidden hypotheses:* differentiability at t (Milnor's fields along curves are smooth by definition). *Deviation (narrower):* the embedded setting.

V6.6 (Lemma 8.1; `Morse.IsCovDerivAlong`, `Morse.isCovDerivAlong_covDeriv`, `Morse.IsCovDerivAlong.eq_co` Let c be a smooth curve in N. Call an operation D on maps $\mathbb{R} \to V$ admissible if it satisfies (a) for smooth fields along c, (b) for smooth f and smooth fields along c, and (c) for smooth vector fields Y on N. Then $X \mapsto P(X')$ of V6.5 is admissible, and every admissible D agrees with it on every smooth field along c. *Book:* p. 46, Lemma 8.1 (the scan prints it as "Lemma 0.1"): there is one and only one operation $V \mapsto DV/dt$ satisfying (a), (b), (c). *Hidden hypotheses:* none beyond the smoothness conventions. *Deviation (narrower):* the embedded setting. *Proof:* instead of a chart, a field along c is expanded as $X = \sum_i \langle e_i, X\rangle\,(E_i \circ c)$ in the smooth fields $E_i(q) = P_q e_i$ of V6.2 (e_i an orthonormal basis of V), and (a)–(c) determine DX/dt. This avoids what Milnor's coordinate proof leaves implicit: locality in t and the extension of the coordinate fields to all of M.

V6.7 (global solutions of linear equations; `Morse.linearODE_eqOn`, `Morse.linearODE_eq`, `Morse.exists_skewLinearODE`). Let $A : \mathbb{R} \to \mathrm{End}(E)$ be continuous. (i) On a real normed space E, two solutions of $x' = A(t)x$ on an open interval (or on $\mathbb{R}$) that agree at one time agree everywhere. (ii) If E is a complete real inner product space and $\langle A(t)x, x\rangle = 0$ for all t and x, then for every t_0 and x_0 there is a solution on all of $\mathbb{R}$ with $x(t_0) = x_0$. *Book:* p. 47, proof of Lemma 8.2: the equations are linear, so the solutions are determined by their initial values and "can be defined

for all relevant values of t" (Milnor cites Graves). *Hidden hypotheses:* a global existence theorem for linear equations; Mathlib has local existence (Picard–Lindelöf) and Grönwall uniqueness only. *Deviation (narrower):* global existence is proved only for skew-adjoint coefficients, the case needed in V6.8. *Proof:* solutions of the skew equation have constant norm, so the equation may be truncated outside a ball and solved by Picard–Lindelöf on any compact interval; solutions on nested intervals are glued by (i).

V6.8 (Lemma 8.2; `Morse.IsParallel`, `Morse.exists_unique_isParallel`, `Morse.IsParallel.contDiff`). A field X along c is parallel if it is differentiable and $DX/dt = 0$ at every t. For a smooth curve c, every $t_0 \in \mathbb{R}$ and every $v \in T_{c(t_0)}N$ there is exactly one parallel field X along c with $X(t_0) = v$; it is defined on all of $\mathbb{R}$, and it is smooth. *Book:* p. 47, Lemma 8.2, with initial time 0 and smooth fields. *Hidden hypotheses:* none. *Deviation (stronger):* any initial time, and uniqueness among differentiable fields, not only smooth ones. *Deviation (narrower):* the embedded setting. *Proof:* a tangent field is parallel iff $X' = P'X$, where $P(t) = P_{c(t)}$, and uniqueness is V6.7 (i) for this equation. It has no a priori bound, so existence solves instead $X' = (P'P - PP')X$, whose coefficient is skew-adjoint (V6.7 (ii)); since $PP'P = 0$, the normal part $U = X - PX$ satisfies $U' = -P'U$, so a solution that starts tangent stays tangent (V6.7 (i) again) and is then parallel. On an abstract manifold Milnor's coordinate equations would also have to be patched from chart to chart along the curve; here there is one equation in V.

V6.9 (parallel translation is a linear isometry; `Morse.IsParallel.inner_eq`, `Morse.parallelTransport`, `Morse.inner_parallelTransport`, `Morse.parallelTransport_self`, `Morse.parallelTransport_trans`). For parallel fields X, Y along any curve c in N, $\langle X(t), Y(t)\rangle$ does not depend on t: the induced connection is compatible with the metric in Milnor's sense. For a smooth curve c and times t_0, t_1, the map sending $v \in T_{c(t_0)}N$ to the value at t_1 of the parallel field through v at t_0 is a linear isometry of $T_{c(t_0)}N$ onto $T_{c(t_1)}N$; it is the identity when $t_1 = t_0$, and translating from t_0 to t_1 and then to t_2 is translating from t_0 to t_2. *Book:* p. 47: V_t is obtained from V_0 by parallel translation; a connection is compatible with the metric if parallel translation preserves inner products. *Hidden hypotheses:* none. *Deviation (stronger):* linearity, surjectivity and transitivity of parallel translation are stated; Milnor uses them without comment. *Deviation (narrower):* compatibility is proved for the induced connection, not assumed.

V6.10 (Lemma 8.3; `Morse.hasDerivAt_inner_covDeriv`). For any curve c in N and fields X, Y along c differentiable at t, the derivative of $\langle X, Y\rangle$ at t is $\langle DX/dt, Y\rangle + \langle X, DY/dt\rangle$. *Book:* p. 47, Lemma 8.3, for a connection compatible with the metric; the proof expands V and W in a parallel orthonormal frame. *Hidden hypotheses:* none in the statement; Milnor's proof needs Lemma 8.2 for n initial vectors and the smoothness of the coefficients $\langle V, P_i\rangle$. *Deviation (narrower):* the induced connection. *Proof:* the product rule in V and $\langle PX', Y\rangle = \langle X', Y\rangle$ for tangent Y; no parallel transport is used, so here Lemma 8.3 comes before Lemma 8.2.

V6.11 (Corollary 8.4; `Morse.dN_inner_eq_covAt`, `Morse.IsCompatible`, `Morse.isCompatible_iff_isMetricCom` *Embedded:* if Y, Y' are differentiable at q with $Y(q)$, $Y'(q)$ tangent, and $v \in T_qN$, then $v\langle Y, Y'\rangle = \langle v \vdash Y, Y'(q)\rangle + \langle Y(q), v \vdash Y'\rangle$. *Abstract:* on a manifold whose tangent bundle carries a Riemannian metric (Mathlib's `RiemannianBundle`), a covariant derivative is called compatible (`Morse.IsCompatible`) if Milnor's identity holds at every point for all vector fields differentiable there; for a finite-dimensional model space and a C^1 metric this is Mathlib's `IsMetricCompatible`. *Book:* p. 48, Corollary 8.4, deduced from Lemma 8.3 with a curve whose velocity at 0 is X_p. *Hidden hypotheses:* differentiability at the point. *Deviation (restated, abstract):* Milnor's definition of compatibility (parallel translation preserves inner products, p. 47) cannot

be stated for Mathlib's connections, which have no parallel transport; Corollary 8.4 is taken as the definition instead.

V6.12 (the second fundamental form; `Morse.secondFundForm`, `Morse.secondFundForm_eq_sub_tanProj`, `Morse.sub_tanProj_dN_eq_of_eq`, `Morse.secondFundForm_mem_orthogonal`, `Morse.dN_tanProj_mem_of_mem_c` `Morse.inner_secondFundForm_comm`, `Morse.secondFundForm_comm`). For $v \in T_qN$ let D_v be the derivative of $q \mapsto P_q$ in the direction v, and $\mathrm{II}_q(v,w) = D_v w$. For every vector field W differentiable at q, $\mathrm{II}_q(v, W(q)) = dW_q(v) - P_q(dW_q(v))$, the normal part of $dW_q(v)$, which therefore depends only on v and $W(q)$. For $v, w \in T_qN$, $\mathrm{II}_q(v,w)$ is normal to N and $\mathrm{II}_q(v,w) = \mathrm{II}_q(w,v)$; and $\langle \mathrm{II}_q(v,x), y\rangle = \langle x, \mathrm{II}_q(v,y)\rangle$ for all $x, y \in V$ (D_v is self-adjoint; it maps T_qN to normal vectors and normal vectors to T_qN). *Book:* not in §§8–9; Milnor meets the second fundamental form in §6 (p. 34). *Deviation (not in the book):* it is the tool for symmetry and curvature below. *Proof of the symmetry:* in a slice chart σ, T_xN is the kernel of $d(\sigma_2)_x$ for $x \in N$, so $d(\sigma_2)_x \circ P_x = 0$ on N; differentiating along N gives $d(\sigma_2)_q(\mathrm{II}(v,w)) = -d^2(\sigma_2)_q(v,w)$, which is symmetric in v, w, so $\mathrm{II}(v,w) - \mathrm{II}(w,v)$ is tangent as well as normal.

V6.13 (Definition 8.5, symmetric connections; `Morse.IsSymmetric`, `Morse.isSymmetric_iff_torsion_eq_zero`, `Morse.bracket`, `Morse.bracket_mem`, `Morse.dN_bracket`, `Morse.leviCivita_sub_leviCivita`). *Abstract:* a covariant derivative is symmetric (`Morse.IsSymmetric`) if $X \vdash Y - Y \vdash X = [X,Y]$ at every point where X and Y are differentiable, $[X,Y]$ being Mathlib's Lie bracket of vector fields (which has Milnor's sign, $[X,Y]f = X(Yf) - Y(Xf)$); for a finite-dimensional model space this is "the torsion is zero". *Embedded:* the bracket of two maps $X, Y : N \to V$ is $[X,Y](q) = dY_q(X(q)) - dX_q(Y(q))$. For vector fields differentiable at q it is tangent to N at q; for smooth f and smooth vector fields, $X(Yf) - Y(Xf) = [X,Y]f$, where $Xf = df(X)$; and the induced connection is symmetric: $X \vdash Y - Y \vdash X = [X,Y]$ at q for vector fields differentiable at q. *Book:* p. 48, Definition 8.5, with $[X,Y]$ the Poisson bracket $[X,Y]f = X(Yf) - Y(Xf)$. *Hidden hypotheses:* differentiability at the point; in the embedded setting, that the ambient formula for the bracket is tangent and acts on functions as Milnor's bracket, neither of which is automatic. *Proof:* tangency of the bracket is the symmetry of II (V6.12): the normal part of $[X,Y]$ is $\mathrm{II}(X,Y) - \mathrm{II}(Y,X)$; the action on functions uses the symmetry of second derivatives.

V6.14 (Lemma 8.6, the fundamental lemma; `Morse.IsSymmCompat`, `Morse.isSymmCompat_leviCivita`, `Morse.IsSymmCompat.two_mul_inner`, `Morse.IsSymmCompat.eq_leviCivita`, `Morse.koszul`, `Morse.two_mul_inner_cov_eq_koszul`, `Morse.cov_eq_of_isSymmetric_of_isCompatible`, `Morse.cov_eq_of_torsion_eq_zero_of_isMetricCompatible`). *Embedded:* call an operation D on maps $N \to V$ symmetric and compatible if, on smooth vector fields, $D(X,Y)$ is tangent, $D(X,Y) - D(Y,X) = [X,Y]$, and $X\langle Y, Y'\rangle = \langle D(X,Y), Y'\rangle + \langle Y, D(X,Y')\rangle$. The induced connection $P(dY(X))$ is such an operation (existence), every such operation satisfies the Koszul formula

$$2\langle D(X,Y), Z\rangle = X\langle Y,Z\rangle + Y\langle X,Z\rangle - Z\langle X,Y\rangle + \langle [X,Y],Z\rangle - \langle [X,Z],Y\rangle - \langle [Y,Z],X\rangle$$

on smooth vector fields, and every such operation agrees with $P(dY(X))$ on smooth vector fields at every point (uniqueness). *Abstract:* for a manifold with a Riemannian metric on its tangent bundle, two covariant derivatives that are both symmetric and compatible (V6.11, V6.13) satisfy the Koszul formula and agree on every vector field differentiable at the point; with Mathlib's predicates, two torsion-free metric connections agree on such fields (finite-dimensional model space, C^1 metric). *Book:* p. 48, Lemma 8.6: a Riemannian manifold possesses one and only one symmetric connection compatible with its metric; uniqueness by the Christoffel identities (p. 49), and "one can verify" the converse. *Hidden hypotheses:* the existence half needs a converse of Lemma 8.3, which

Milnor does not state, and a patching of the local connections. *Deviation (stronger):* the embedded uniqueness assumes no linearity or tensoriality of D, only the three properties on smooth fields. *Deviation (narrower):* existence is proved only in the embedded setting; on an abstract manifold only uniqueness, and only on sections differentiable at the point (a Mathlib covariant derivative is unconstrained elsewhere). *Proof:* the Koszul formula, the coordinate-free form of Milnor's first Christoffel identity, replaces his linear system on p. 49 (whose list of three unknowns contains $\langle \partial_i \vdash \partial_j, \partial_k \rangle$ twice, by the symmetry he quotes, a misprint); uniqueness applies it with Z the field $x \mapsto P_x(D(X,Y)(q) - (X \vdash Y)(q))$ of V6.2.

V6.15 (Lemma 8.7; `Morse.velX`, `Morse.velY`, `Morse.covDerivX`, `Morse.covDerivY`, `Morse.covDerivX_eq`, `Morse.covDerivX_velY`). For a smooth surface $s : \mathbb{R}^2 \to N$ the velocity fields $\partial s/\partial x$, $\partial s/\partial y$ are tangent, and $DW/\partial x$ is defined, as in Milnor, as the covariant derivative of $x \mapsto W(x, y_0)$ along the curve $x \mapsto s(x, y_0)$ (similarly $DW/\partial y$); at a point p where W is differentiable it equals $P_{s(p)}(\partial W/\partial x\,(p))$. Then $\frac{D}{\partial x}\frac{\partial s}{\partial y} = \frac{D}{\partial y}\frac{\partial s}{\partial x}$ at every point. *Book:* p. 50, Lemma 8.7, for a symmetric connection; the proof is to express both sides in local coordinates and compute. *Hidden hypotheses:* joint smoothness of $DW/\partial x$ in (x, y), which Milnor defines one curve at a time; here it is visible from the formula. *Deviation (narrower):* the induced connection, which is symmetric (V6.13). *Proof:* both sides are $P(\partial^2 s/\partial x \partial y)$.

V6.16 (the curvature tensor and its sign; `Morse.milnorCurvature`, `Morse.nomizuCurvature`, `Morse.milnorCurvature_eq_neg`, `Morse.milnorCurvature_swap`, `Morse.curvature`). For a covariant derivative on an abstract manifold, Milnor's curvature of vector fields is $R(X,Y)Z = -X \vdash (Y \vdash Z) + Y \vdash (X \vdash Z) + [X,Y] \vdash Z$; it is the negative of Nomizu's (and Kobayashi–Nomizu's) $\nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z$, and $R(Y,X)Z = -R(X,Y)Z$. In the embedded setting the same formula, with the induced connection and the bracket of V6.13, defines $R(X,Y)Z$ for maps $N \to V$. *Book:* p. 51, the definition, with the footnote that Nomizu gives R the opposite sign. *Hidden hypotheses:* none. Mathlib defines no curvature, so there is no Mathlib sign to compare with.

V6.17 (Lemma 9.1, made explicit; `Morse.curvOp`, `Morse.curvature_eq`, `Morse.curvTensor`, `Morse.curvature_eq_curvTensor`, `Morse.curvature_congr`). In the embedded setting, for smooth vector fields X, Y, Z and every q,

$$R(X,Y)Z\,(q) = \mathrm{II}_q(Y_q, \mathrm{II}_q(X_q, Z_q)) - \mathrm{II}_q(X_q, \mathrm{II}_q(Y_q, Z_q)),$$

that is, $R(X,Y) = D_Y D_X - D_X D_Y$ on $T_q N$. Consequently $R(X,Y)Z$ at q depends only on X_q, Y_q, Z_q, and it is the value of a trilinear map $T_qN \times T_qN \times T_qN \to T_qN$, the curvature tensor of N at q. *Book:* p. 51, Lemma 9.1: the value of $R(X,Y)Z$ at p depends only on X_p, Y_p, Z_p, and the correspondence is trilinear. *Hidden hypotheses:* none. *Deviation (stronger):* an explicit formula. *Deviation (narrower):* the embedded setting. *Proof:* with a smooth ambient representative G of Z, $Y \vdash (X \vdash Z)$ at q is $D_Y D_X Z$ plus terms that cancel by the symmetry of d^2G and against $[X,Y] \vdash Z$. Milnor treats X explicitly and says the identities for Y and Z "are easily obtained by similar computations"; for Z the computation is a different one.

V6.18 (the Gauss equation; `Morse.inner_curvOp`, `Morse.gauss_equation`). For all $x, y, z, w \in V$,

$$\langle D_y D_x z - D_x D_y z, w\rangle = \langle \mathrm{II}(x,z), \mathrm{II}(y,w)\rangle - \langle \mathrm{II}(x,w), \mathrm{II}(y,z)\rangle$$

at q; hence, for smooth vector fields X, Y, Z and any $W : N \to V$,

$$\langle R(X,Y)Z, W\rangle = \langle \mathrm{II}(X,Z), \mathrm{II}(Y,W)\rangle - \langle \mathrm{II}(X,W), \mathrm{II}(Y,Z)\rangle$$

at every point, with Milnor's sign of R. With $Z = X$, $W = Y$ and the symmetry of II (V6.12) this gives $\langle R(X,Y)X, Y\rangle = \langle \mathrm{II}(X,X), \mathrm{II}(Y,Y)\rangle - \|\mathrm{II}(X,Y)\|^2$, Gauss's curvature for a surface in $\mathbb{R}^3$. *Book:* not in Milnor's text. *Deviation (not in the book).*

V6.19 (Lemma 9.2; `Morse.covDerivY_covDerivX_sub`). For a smooth surface $s : \mathbb{R}^2 \to N$ and a smooth map $W : \mathbb{R}^2 \to V$ with $W(p) \in T_{s(p)}N$ for all p,

$$\frac{D}{\partial y}\frac{D}{\partial x}W - \frac{D}{\partial x}\frac{D}{\partial y}W = R\Big(\frac{\partial s}{\partial x}, \frac{\partial s}{\partial y}\Big)W$$

at every point, with $R(x,y) = D_y D_x - D_x D_y$ the curvature of V6.17. *Book:* p. 52, Lemma 9.2, again proved by a computation in local coordinates. *Hidden hypotheses:* joint smoothness of $DW/\partial x$ (as in V6.15). *Deviation (narrower):* the embedded setting. *Proof:* from $W = PW$, $PD_vP = 0$, and the fact that D_v maps normal vectors to tangent ones; no symmetry of the connection is used.

V6.20 (Lemma 9.3 (1)–(3); `Morse.IsCurvatureTensor`, `Morse.inner_add_inner_eq_zero_iff`, `Morse.isCurvatureTensor_curvTensor`, `Morse.curvature_add_swap`, `Morse.curvature_bianchi`, `Morse.inner_curvature_add_inner`). A trilinear map R on a real inner product space is called a curvature tensor (`Morse.IsCurvatureTensor`) if it satisfies (1) $R(X,Y)Z + R(Y,X)Z = 0$, (2) $R(X,Y)Z + R(Y,Z)X + R(Z,X)Y = 0$, (3) $\langle R(X,Y)Z, W\rangle + \langle R(X,Y)W, Z\rangle = 0$. For a linear map T, $\langle TZ, W\rangle$ is skew in Z, W iff $\langle TZ, Z\rangle = 0$ for all Z. The curvature tensor of N at each point (V6.17) satisfies (1)–(3). For maps X, Y, Z of N into V, (1) holds with no hypothesis; for smooth vector fields, (2) and (3) hold at every point. *Book:* p. 53, Lemma 9.3 (1)–(3), and "clearly equivalent" for the reduction of (3) to $\langle R(X,Y)Z, Z\rangle = 0$. *Hidden hypotheses:* Milnor proves (2) and (3) for extensions with vanishing brackets, which must be constructed (coordinate fields cut off by a bump function). *Deviation (narrower):* the embedded setting. *Proof:* no extensions are needed; (2) is the symmetry of II and (3) the self-adjointness of D_v (V6.12).

V6.21 (Lemma 9.3 (4), the octahedron; `Morse.octahedron`, `Morse.IsCurvatureTensor.inner_comm_pairs`, `Morse.inner_curvature_comm_pairs`). Let $T(X,Y,Z,W)$ be any real function of four arguments that is skew in (X,Y), skew in (Z,W) and satisfies $T(X,Y,Z,W) + T(Y,Z,X,W) + T(Z,X,Y,W) = 0$. Then $T(X,Y,Z,W) = T(Z,W,X,Y)$. Hence every curvature tensor (V6.20) satisfies (4) $\langle R(X,Y)Z, W\rangle = \langle R(Z,W)X, Y\rangle$, and so does R on smooth vector fields of N. *Book:* p. 54: (4) from (1), (2), (3) by the octahedron, the four shaded faces being instances of (2), "making use of (1) and (3)". *Hidden hypotheses:* none. *Deviation (stronger):* no linearity of T is used. *Proof:* Milnor's; the four face identities, whose signs he leaves to the reader, are written out.

V6.22 (sectional curvature; `Morse.sectionalCurvature`, `Morse.IsCurvatureTensor.sectionalCurvature_smul`, `Morse.IsCurvatureTensor.neg`, `Morse.IsCurvatureTensor.inner_neg_apply_swap`). For a trilinear map R and vectors X, Y, put $K(X,Y) = \langle R(X,Y)X, Y\rangle / (\|X\|^2\|Y\|^2 - \langle X, Y\rangle^2)$ (Lean's convention gives 0 when X, Y are dependent); for an orthonormal pair this is Milnor's $\langle R(A,B)A, B\rangle$. If R satisfies (1)–(3), K depends only on the plane: $K(aX + bY, cX + dY) = K(X,Y)$ whenever $ad - bc \neq 0$. With the standard sign, $-R$ also satisfies (1)–(3), and $\langle (-R)(X,Y)Y, X\rangle = \langle R(X,Y)X, Y\rangle$: the sectional curvature is the same number in both conventions. *Book:* p. 100, remark after Lemma 19.1: for orthogonal unit vectors A, B at p, $\langle R(A,B)A, B\rangle$ is the sectional curvature determined by A and B. *Hidden hypotheses:* that this number depends only on the plane spanned by A and B is not stated in the book. *Deviation (stronger):* defined for any independent pair, and shown to depend only on the plane.

V6.23 (the sectional curvatures determine R; `Morse.IsCurvatureTensor.polarization`, `Morse.IsCurvatureTensor.ext`, `Morse.IsCurvatureTensor.ext_of_sectionalCurvature`,

`Morse.IsCurvatureTensor.ext_of_orthonormal`**).** For a curvature tensor R, $6\langle R(X,Y)Z,W\rangle$ is an explicit combination of the values $k(A,B)=\langle R(A,B)A,B\rangle$ at sums of X,Y,Z,W (polarization). Hence two curvature tensors are equal as soon as they have the same $\langle R(X,Y)X,Y\rangle$ for all X,Y, or the same sectional curvature on all independent pairs, or the same $\langle R(A,B)A,B\rangle$ on all orthonormal pairs (Milnor's definition). *Book:* not stated. *Deviation (not in the book):* Lemma 7.2 of the exposition.

V6.24 (constant curvature; `Morse.curvatureOne, Morse.isCurvatureTensor_curvatureOne, Morse.sectionalCurvature_smul_curvatureOne, Morse.IsCurvatureTensor.eq_smul_curvatureOne_of_orth Morse.IsCurvatureTensor.apply_eq_of_sectionalCurvature_eq, Morse.neg_smul_curvatureOne_apply`**).** $R_1(X,Y)Z=\langle X,Z\rangle Y-\langle Y,Z\rangle X$ is a curvature tensor, and κR_1 has sectional curvature κ on every independent pair. Conversely, if a curvature tensor R has $\langle R(A,B)A,B\rangle=\kappa$ for all orthonormal A,B, or $K(X,Y)=\kappa$ for all independent X,Y, then $R(X,Y)Z=\kappa(\langle X,Z\rangle Y-\langle Y,Z\rangle X)$; with the standard sign this is $\kappa(\langle Y,Z\rangle X-\langle X,Z\rangle Y)$. *Book:* not stated in §§8–9 or on p. 100. *Deviation (not in the book):* Corollary 7.3 of the exposition.

V6.25 (the round sphere; `Milnor.Sphere.Sph, Milnor.Sphere.sphSub, Morse.tanProj_sph, Morse.secondFundForm_sph, Morse.curvOp_sph, Morse.curvTensor_sph, Morse.sectionalCurvature_sph, Morse.inner_curvTensor_sph_orthonormal`**).** For the unit sphere $S^n\subseteq\mathbb{R}^{n+1}$ (a slice submanifold, from the companion project) with the induced metric: $P_q v=v-\langle q,v\rangle q$; $\mathrm{II}_q(v,w)=-\langle v,w\rangle q$ for tangent v,w; the curvature tensor is $R(x,y)z=\langle x,z\rangle y-\langle y,z\rangle x$ with Milnor's sign; so S^n has sectional curvature 1, both as $K(x,y)=1$ for independent tangent x,y and as $\langle R(A,B)A,B\rangle=1$ for orthonormal A,B. *Book:* not in §§8–9; the computation is the application in §8 of the exposition. *Deviation (not in the book). Proof:* coordinate-free, from the Gauss equation (V6.18).

What is not verified in this chapter

- **§§8–9 on an abstract Riemannian manifold**, beyond V6.3, V6.11, V6.13, V6.14 (uniqueness) and V6.16: the existence of the Levi-Civita connection, Christoffel symbols and formulas (5), (6), the coordinate formula for DV/dt, Lemmas 8.1–8.3 and 8.7, the tensor property of R (Lemma 9.1), Lemma 9.2, and (2)–(4) of Lemma 9.3 for vector fields. Mathlib has no curvature, Christoffel symbols, covariant derivative along curves, parallel transport or Levi-Civita connection; the embedded results reach an arbitrary Riemannian manifold only through Nash's embedding theorem, which is neither in Mathlib nor proved here.
- **The bridge between the two settings:** the induced connection of N as a Mathlib `CovariantDerivative` on the tangent bundle of N, with the induced metric as a Riemannian bundle, is not constructed. The embedded existence (V6.14) and the abstract uniqueness (V6.14) are parallel results, not one theorem.
- **Milnor's side remarks:** the converse "if $\Gamma^k_{ij}=\Gamma^k_{ji}$ then the connection is symmetric" (p. 48), the footnote on the covariant second derivative (p. 48), and the construction of parallel fields along a surface when $R=0$ (pp. 52–53).
- **The geometric meaning of sectional curvature** (p. 100: the Gaussian curvature of the surface swept out by geodesics tangent to the plane): it needs geodesics and the exponential map (§10, Chapter 7 of this edition), not used here.

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8. Chapter 7 (Milnor §10): partly verified

Milnor's §10 (pp. 55–66) defines geodesics on a connected Riemannian manifold and notes that their speed is constant (p. 55), quotes the existence and uniqueness theorem for ordinary differential equations (Theorem 10.1), obtains geodesics on a fixed time interval for small initial velocities (Lemma 10.2), defines the exponential map and geodesic completeness (pp. 57–58), builds uniformly normal neighbourhoods (Lemma 10.3), and proves that short geodesics are shortest (Theorem 10.4, from the Gauss lemma 10.5 and Lemma 10.6) and that shortest paths parametrized by arc length are geodesics (Corollary 10.7). He then defines the distance ρ, proves Corollary 10.8, the Hopf–Rinow theorem (Theorem 10.9) and Corollary 10.10, states its converse, and ends with examples (pp. 65–66). Verified: every numbered statement from Theorem 10.1 through Corollary 10.10 (Lemma 10.6 in a one-sided form), and the converse of Corollary 10.10, with the definitions they need (geodesics, the exponential map, piecewise smooth paths and their lengths, the distance, minimal geodesics, metric and geodesic completeness), for a submanifold N of a Euclidean space with the induced metric, in the embedded setting of Chapter 6; Theorem 10.1 itself for a finite-dimensional normed space. Every Riemannian manifold is isometric to such an N by Nash's embedding theorem, which is quoted in the Lean files but neither proved nor used, so these results are statements about embedded manifolds with the induced metric. In this setting the Hausdorff property that Milnor uses silently is automatic, and his standing assumption that M is connected is made only where it is needed (finiteness of ρ, Theorem 10.9 for all pairs of points, Corollary 10.10, the Hopf–Rinow equivalence); elsewhere ρ may take the value ∞. Proved more sharply than in the book: **the equality case of Theorem 10.4** (a path of the same length is a monotone reparametrization of the geodesic, V7.16), which Corollary 10.7 needs; the uniqueness of geodesics in the large (V7.5), the smooth dependence in Lemma 10.2 (V7.6) and the Gauss lemma in every direction (V7.12); and **Theorem 10.9 with completeness at a single point** (V7.26), which yields the four-way Hopf–Rinow equivalence (V7.29). Supplied where the book is silent or refers elsewhere: the continuity of ρ (V7.21), the continuity of exp on its whole domain (V7.23), the converse of Corollary 10.10 (V7.28); and, not in the book, the completeness of closed submanifolds, among them the sphere (V7.30). The smoothness of exp on its whole domain, and the extension of a geodesic on $[0,1]$ past its ends, are proved later (Chapters 9 and 11; V7.7 and the list at the end). **Not verified:** maximal geodesics, the convexity remark after Lemma 10.3, the remarks on minimal geodesics of p. 62 and the examples of pp. 65–66 (the list at the end). Lean: `Morse/Chapter7/` — `ODE.lean` (235 lines), `Geodesic.lean` (368), `GeodesicLocal.lean` (447), `Exp.lean` (337), `NormalNbhd.lean` (764), `GaussLemma.lean` (419), `Path.lean` (394), `Shortest.lean` (542), `Minimal.lean` (329), `Distance.lean` (386), `Metric.lean` (309), `GeodesicFlow.lean` (766), `HopfRinow.lean` (664).

The embedded setting. As in Chapter 6: V is a finite-dimensional real inner product space (Milnor's $\mathbb{R}^K$) and $N \subseteq V$ a slice submanifold modelled on Z with complement W (V1.6), Z and W finite-dimensional (automatic when N is non-empty), with the induced metric. $T_qN \subseteq V$ is the tangent space at q, P_q the orthogonal projection onto it (V6.1), II the second fundamental form (V6.12) and $D/dt = P \circ d/dt$ the covariant derivative along curves (V6.5). Curves are maps $\mathbb{R} \to V$, of which only the values on the interval considered matter. A map defined on a subset of V, $V \times V$ or $V \times V \times \mathbb{R}$ is called smooth there if it is the restriction of a map that is smooth on an open set containing the subset (or, where so stated, near each of its points); this is Milnor's notion of smoothness for subsets of Euclidean space. N need not be closed, compact or connected.

V7.1 (geodesics; `Morse.IsGeodesicOn`, `Morse.hasDerivAt_mem_tangentSubspace`, `Morse.IsGeodesicOn.deriv_`
`Morse.IsGeodesicOn.tanProj_deriv_deriv`, `Morse.isGeodesicOn_univ_iff_covDeriv`,

`Morse.isGeodesicOn_const`). Let $s \subseteq \mathbb{R}$ be open, in practice an open interval. A curve γ is a geodesic on s if it is C^∞ on s, $\gamma(t) \in N$ for $t \in s$, and the acceleration $\gamma''(t)$ is orthogonal to $T_{\gamma(t)}N$ for every $t \in s$. The velocity at t of a curve that lies in N near t and is differentiable at t is tangent to N; so for a geodesic $\gamma'(t) \in T_{\gamma(t)}N$, and the condition on γ'' reads $P_{\gamma(t)}\gamma''(t) = 0$, Milnor's $\frac{D}{dt}\frac{d\gamma}{dt} = 0$ with D/dt of V6.5. Formally: a smooth curve $c : \mathbb{R} \to N$ is a geodesic on $\mathbb{R}$ iff $\frac{D}{dt}\frac{dc}{dt}(t) = 0$ for all t (V6.5 defines D/dt for curves on all of $\mathbb{R}$). Constant curves at points of N are geodesics. *Book:* p. 55: on a connected Riemannian manifold M, a parametrized path $\gamma : I \to M$, I any interval, is a geodesic if its acceleration $\frac{D}{dt}\frac{d\gamma}{dt}$ vanishes identically, i.e. its velocity field is parallel along γ. *Hidden hypotheses:* none. *Deviation (narrower):* the embedded setting, where a geodesic is a curve whose acceleration in V is normal to N; geodesics live on open sets, so a geodesic on a closed interval is represented by one on a slightly larger open interval (V7.7).

V7.2 (constant speed and homogeneity; `Morse.IsGeodesicOn.hasDerivAt_norm_sq`, `Morse.IsGeodesicOn.norm` `Morse.IsGeodesicOn.pathLength_eq`, `Morse.IsGeodesicOn.comp_mul_add`). For a geodesic γ on s, $\frac{d}{dt}\|\gamma'\|^2 = 2\langle\gamma'', \gamma'\rangle = 0$ on s. If s is an open interval (more generally, preconnected), the speed $\|\gamma'\|$ is constant on s, and the length of γ on any $[a, b] \subseteq s$ (V7.11) is $\|\gamma'(t_0)\|\,(b - a)$ for every $t_0 \in s$. For all real a, b the curve $t \mapsto \gamma(at + b)$ is a geodesic on $\{t : at + b \in s\}$. *Book:* p. 55: $\frac{d}{dt}\langle\frac{d\gamma}{dt}, \frac{d\gamma}{dt}\rangle = 2\langle\frac{D}{dt}\frac{d\gamma}{dt}, \frac{d\gamma}{dt}\rangle = 0$, so the speed is constant and the parameter is a linear function of arc length; p. 57 (homogeneity): if $t \mapsto \gamma(t)$ is a geodesic, so is $t \mapsto \gamma(ct)$ for every constant c. *Hidden hypotheses:* none. *Deviation (stronger):* all affine changes of parameter $t \mapsto at + b$ ($a = 0$ gives a constant curve).

V7.3 (the geodesic equation; `Morse.IsGeodesicOn.deriv_deriv_eq`, `Morse.isGeodesicOn_of_deriv_deriv_eq`, `Morse.geodesicSpray`, `Morse.IsGeodesicOn.mem_tangentBundleSet`, `Morse.isGeodesicOn_iff_hasDerivAt_g` `Morse.pinvL`, `Morse.chartAcc`, `Morse.chartSpray`, `Morse.contDiffOn_chartSpray`, `Morse.isGeodesicOn_char` Let γ be C^∞ on an open set s with $\gamma(s) \subseteq N$. Then γ is a geodesic on s iff $\gamma'' = \mathrm{II}_\gamma(\gamma', \gamma')$ on s, iff the lift $t \mapsto (\gamma(t), \gamma'(t))$, which then lies in $TN = \{(x, v) : x \in N,\ v \in T_xN\}$, is an integral curve on s of the geodesic spray $(x, v) \mapsto (v, \mathrm{II}_x(v, v))$. In a slice chart with parametrization x (V1.6) on the open set $T \subseteq Z$, put $A^+ = (A^\mathsf{T}A)^{-1}A^\mathsf{T}$ for injective A and $\Gamma(u)(w, w) = (dx_u)^+(d^2x_u(w, w))$. The field $(u, w) \mapsto (w, -\Gamma(u)(w, w))$ is smooth on $T \times Z$, and if (u, w) is a C^∞ solution of $u' = w$, $w' = -\Gamma(u)(w, w)$ on an open set s with $u(s) \subseteq T$, then $x \circ u$ is a geodesic on s. *Book:* p. 55: in local coordinates the geodesic condition becomes the system $\frac{d^2u^k}{dt^2} + \sum_{i,j}\Gamma^k_{ij}\frac{du^i}{dt}\frac{du^j}{dt} = 0$, so the existence of geodesics rests on a system of second-order equations (the scan shows only the Γ-term of this display, a defect of the scan, not a misprint). *Hidden hypotheses:* none. *Deviation (restated):* $(dx_u)^+d^2x_u(w, w)$ is the coordinate vector, in the basis $\partial x/\partial u^k$, of the tangential part of $d^2x_u(w, w)$, that is $\big(\sum_{i,j}\Gamma^k_{ij}w^iw^j\big)_k$ for the Christoffel symbols of the induced metric; the Lean files define no Christoffel symbols, so this identification is a remark, not a theorem. In the chart only the direction "solutions are geodesics" is stated, which is what existence needs (V7.6); uniqueness uses the global form (V7.5).

V7.4 (Theorem 10.1; `Morse.exists_localFlow`, `Morse.theorem_10_1`, `Morse.ode_eventuallyEq`, `Morse.ode_eqOn`, `Morse.eqOn_of_eventuallyEq`, `Morse.theorem_10_1_unique`). *Existence and smooth dependence.* Let E be a finite-dimensional real normed space (Milnor's $\mathbb{R}^n$), $U \subseteq E \times E$ open, $F : E \times E \to E$ smooth on U, and $z_1 \in U$. There are a neighbourhood W of z_1, an $\varepsilon > 0$ and a map $(z, t) \mapsto u_z(t)$, smooth on $W \times (-\varepsilon, \varepsilon)$, such that for every $z = (u_0, v_0) \in W$: $u_z(0) = u_0$, $u_z'(0) = v_0$, and $(u_z(t), u_z'(t)) \in U$ and $u_z''(t) = F(u_z(t), u_z'(t))$ for $|t| < \varepsilon$. In first-order form: for G smooth on an open set $U \ni x_1$ of E there are a smooth $\varphi : \mathbb{R} \times E \to E$ and $r, \tau > 0$ such that for $\|x - x_1\| < r$ the curve $t \mapsto \varphi(t, x)$ solves $x' = G(x)$ in U for $|t| < \tau$,

with $\varphi(0,x)=x$. *Uniqueness.* Let E be any real normed space and G of class C^1 on an open set $U\subseteq E$. Two solutions of $x'=G(x)$ with values in U that agree at one time agree near it, and, if both are defined on the same open interval (more generally, open preconnected set), on all of it. In particular, if F is C^1 on $U\subseteq E\times E$, two pairs (u_i,v_i) with $u_i'=v_i$, $v_i'=F(u_i,v_i)$ and $(u_i,v_i)\in U$ on an open interval s, with the same position and velocity at one time of s, have $u_1=u_2$ on s. *Book:* p. 56, Theorem 10.1, for an n-tuple F of C^∞ functions near $(u_1,v_1)\in\mathbb{R}^{2n}$: a neighbourhood W and an $\varepsilon>0$ such that for $(u_0,v_0)\in W$ the equation has a unique solution for $|t|<\varepsilon$ with the given initial conditions, and $(u_0,v_0,t)\mapsto u(t)$ is C^∞ on $W\times(-\varepsilon,\varepsilon)$; the proof passes to the first-order system $du/dt=v$, $dv/dt=F(u,v)$ and quotes Graves (the scan shows only part of the displayed equation). *Hidden hypotheses:* none. *Deviation (stronger):* uniqueness on every common interval of definition, for C^1 equations on any normed space. *Proof:* proved, not quoted: G times a bump function equal to 1 near x_1 is a compactly supported smooth field whose global flow is smooth in (t,x) (V1.14), and its bounded speed keeps the flow lines from points near x_1, for a uniform time, where the bump function is 1; uniqueness is Grönwall's inequality (Mathlib) near each time, spread over the interval by connectedness.

V7.5 (uniqueness of geodesics; `Morse.projRep`, `Morse.IsGeodesicOn.eventuallyEq`, `Morse.IsGeodesicOn.eqO`**)** Two geodesics on open sets s_1, s_2 containing t_0, with the same position and the same velocity at t_0, agree near t_0. Two geodesics on the same open interval (more generally, open preconnected set) s, with the same position and velocity at one time of s, agree on all of s. *Book:* Theorem 10.1 gives uniqueness for small times in one coordinate system (p. 56); Lemmas 10.2 and 10.3 assert uniqueness of geodesics on whole intervals, and p. 58 says that $\exp_q(v)$, if defined at all, is uniquely defined. *Hidden hypotheses:* uniqueness in the large, by a connectedness argument; on an abstract manifold it also needs the Hausdorff property, automatic here. *Deviation (not in the book):* the statement, which is the exposition's Lemma A. *Proof:* near t_0 both lifts (γ_i,γ_i') solve $(x,v)'=(v,dQ_x(v)\,v)$, where Q is the smooth extension of $x\mapsto P_x$ to an open set of V given by a slice chart at $\gamma_1(t_0)$ (V6.2); this field is smooth, so V7.4 applies.

V7.6 (Lemma 10.2; `Morse.LocalSpray`, `Morse.nonempty_localSpray`, `Morse.LocalSpray.geo`, `Morse.LocalSpray.isGeodesicOn_geo`, `Morse.LocalSpray.contDiffOn_geo`, `Morse.lemma_10_2`**).** For every $q_0\in N$ there are an open set $U\ni q_0$ of V, an $\varepsilon>0$ and a map $\Gamma: V\times V\times\mathbb{R}\to V$, smooth on $U\times\{v:\|v\|<\varepsilon\}\times(-2,2)$, such that for every $p\in N\cap U$ and every $v\in T_pN$ with $\|v\|<\varepsilon$, the curve $\gamma_v=\Gamma(p,v,\cdot)$ is a geodesic on $(-2,2)$ with $\gamma_v(0)=p$ and $\gamma_v'(0)=v$, and every geodesic on $(-2,2)$ with this position and velocity at 0 equals γ_v on $(-2,2)$. *Book:* p. 56, Lemma 10.2: for p in a neighbourhood U of p_0 and $v\in TM_p$ of length $<\varepsilon$ there is a unique geodesic $\gamma_v:(-2,2)\to M$ with $\gamma_v(0)=p$ and $\frac{d\gamma_v}{dt}(0)=v$; the proof (pp. 56–57) gets ε_1, ε_2 from Theorem 10.1 in a coordinate system and puts $\gamma_v(t)=\gamma_{v/\varepsilon_2}(\varepsilon_2 t)$ for $\|v\|<\varepsilon<\varepsilon_1\varepsilon_2$. *Hidden hypotheses:* the step from the coordinate neighbourhood W of Theorem 10.1 to the condition "$p\in U$, $\|v\|<\varepsilon_1$" needs a bound for the coordinates of v by $\|v\|$, uniform for p near p_0; uniqueness on $(-2,2)$ is V7.5. *Deviation (stronger):* $\gamma_v(t)$ depends smoothly on (p,v,t), which Milnor does not state here but uses on p. 58 (V7.8). *Proof:* Milnor's, with Theorem 10.1 (V7.4) applied to the chart equation of V7.3 near $(u_0,0)$ and the rescaling by half the time of existence; the coordinates of v are $(d\sigma_p v)_1$, σ the slice chart, and $\|d\sigma_p v\|\le\|d\sigma_p\|\,\|v\|$ with $\|d\sigma_p\|$ bounded near q_0.

V7.7 (the exponential map; `Morse.IsGeodesicFrom`, `Morse.expDom`, `Morse.exp`, `Morse.IsGeodesicFrom.mem`, `Morse.isGeodesicFrom_of_isGeodesicOn`, `Morse.IsGeodesicFrom.eqOn`, `Morse.IsGeodesicFrom.exp_eq`, `Morse.exp_zero`, `Morse.zero_mem_expDom`, `Morse.IsGeodesicFrom.exp_smul`, `Morse.IsGeodesicFrom.isGeode`**)** For $q,v\in V$, a geodesic from q with initial velocity v is a geodesic γ on an open interval $(-\delta,1+\delta)$, $\delta>0$, with $\gamma(0)=q$ and $\gamma'(0)=v$ (then $q\in N$, and $v\in T_qN$ by V7.1); a geodesic on an open

interval containing 0 and 1 is one, from $\gamma(0)$ with initial velocity $\gamma'(0)$. The domain of exp is the set of pairs (q, v) for which such a geodesic exists, and $\exp_q(v) = \gamma(1)$ (off the domain $\exp_q(v)$ is defined to be q, a placeholder). Two geodesics from q with initial velocity v agree on an open interval containing $[0, 1]$, so $\exp_q(v) = \gamma(1)$ for every such γ. For $q \in N$, $(q, 0)$ is in the domain and $\exp_q(0) = q$. If γ is a geodesic from q with initial velocity v, there is $\delta > 0$ such that γ is a geodesic on $(-\delta, 1 + \delta)$ and, for each t in this interval, $s \mapsto \gamma(ts)$ is a geodesic from q with initial velocity tv and $\exp_q(tv) = \gamma(t)$; so $t \mapsto \exp_q(tv)$ is a geodesic near $[0, 1]$, equal to γ. *Book:* p. 57: for $v \in TM_q$ and a geodesic $\gamma : [0, 1] \to M$ with $\gamma(0) = q$ and $\frac{d\gamma}{dt}(0) = v$, $\exp_q(v) = \gamma(1)$, and $\gamma(t) = \exp_q(tv)$; p. 58: $\exp_q(v)$ is defined for small $\|v\|$, in general not for large v, but uniquely defined whenever defined. *Hidden hypotheses:* well-definedness is the uniqueness in the large of V7.5; $\gamma(t) = \exp_q(tv)$ uses homogeneity (V7.2) together with V7.5. *Deviation (restated):* Milnor's geodesic on the closed interval $[0, 1]$ (with one-sided derivatives at the ends) is replaced by one on an open interval containing $[0, 1]$. Restricting to $[0, 1]$ gives a geodesic in Milnor's sense with the same value $\gamma(1)$, and both domains contain every $(q, v) \in TN$ with v small enough (V7.8). Conversely, a geodesic on $[0, 1]$ in Milnor's sense (a curve in N, C^∞ on $[0, 1]$ with one-sided derivatives at the ends, and a geodesic on $(0, 1)$) agrees on $[0, 1]$ with $t \mapsto \exp_{\gamma(0)}(t\gamma'(0))$, which is a geodesic on an open interval containing $[0, 1]$ (V9.9 (ii), `Morse.exists_initial_ray`), so that $(\gamma(0), \gamma'(0))$ is in the formal domain (V11.1 (iii), `Morse.mem_expDom_of_isGeodesicOn`): the two domains coincide. This is proved in Chapters 9 and 11, not in `Morse/Chapter7/`.

V7.8 (the tangent manifold, exp **near the zero section, and** $d(\exp_q)_0$**;** `Milnor.Problems.tangentBundleSet`, `Milnor.Problems.isSliceSubmanifold_tangentBundle`, `Morse.exists_exp_eq`, `Morse.exists_isOpen_exp`, `Morse.contMDiffOn_exp`, `Morse.hasFDerivAt_exp_zero`**).** $TN = \{(q, v) \in V \times V : q \in N,\ v \in T_qN\}$ is a slice submanifold of $V \times V$ modelled on $Z \times Z$ (the companion project). (i) For every $q_0 \in N$ there are an open set $U \ni q_0$ of V, an $\varepsilon > 0$ and a map $F : V \times V \to V$, smooth on $U \times \{v : \|v\| < \varepsilon\}$, such that for $p \in N \cap U$ and $v \in T_pN$ with $\|v\| < \varepsilon$, (p, v) is in the domain of exp and $\exp_p(v) = F(p, v)$. (ii) There is an open set $O \subseteq V \times V$ containing $(q, 0)$ for every $q \in N$ such that every point of $TN \cap O$ is in the domain of exp, and exp is smooth on $TN \cap O$: near each point of $TN \cap O$ it is the restriction of a smooth map on an open set of $V \times V$, and $(q, v) \mapsto \exp_q(v)$ is smooth on the open subset $TN \cap O$ of the manifold TN. (iii) For each $q \in N$, $\exp_q$, as a map on the vector space T_qN, has derivative at 0 the inclusion $T_qN \to V$. *Book:* p. 58: TM is given a C^∞ structure by the coordinates $(u^1, \dots, u^n, t^1, \dots, t^n)$; by Lemma 10.2, $(q, v) \mapsto \exp_q(v)$ is defined and differentiable throughout a neighbourhood of $(p, 0)$ in TM; in the Jacobian of $(q, v) \mapsto (q, \exp_q(v))$ at $(p, 0)$, the formula $F_*(\partial/\partial t^j) = \partial/\partial u_2^j$ amounts to $d(\exp_p)_0 = \mathrm{id}$. *Hidden hypotheses:* the smooth dependence in Lemma 10.2 (V7.6). *Deviation (restated):* Milnor's tangent manifold is the embedded $TN \subseteq V \times V$. *Deviation (stronger):* one open neighbourhood of the whole zero section.

V7.9 (normal radius; `Morse.NormalCoord`, `Morse.IsNormalRadius`, `Morse.expToN`, `Morse.NormalCoord.expDiff`**).** For $q \in N$ and $\varepsilon > 0$ let $B_\varepsilon(q) = \{v \in T_qN : \|v\| < \varepsilon\}$. ε is a normal radius at q if there are open sets $O, U \subseteq V$ and maps F smooth on O and G smooth on U such that: $B_\varepsilon(q) \subseteq O$, $B_\varepsilon(q)$ lies in the domain of $\exp_q$, and $\exp_q = F$ on $B_\varepsilon(q)$; $\exp_q$ maps $B_\varepsilon(q)$ into U, with $G(\exp_q v) = v$ for $v \in B_\varepsilon(q)$; and G maps $U_q = N \cap U$ into $B_\varepsilon(q)$, with $\exp_q(G(y)) = y$ for $y \in U_q$. Thus $\exp_q$ is a bijection of $B_\varepsilon(q)$ onto the open subset U_q of N, and it and its inverse are restrictions of smooth maps on open sets of V. A normal radius yields a diffeomorphism in Mathlib's sense:

$\exp_q$, as a map from the vector space T_qN to the manifold N, restricts to a `PartialDiffeomorph` of the open ε-ball onto U_q. *Book:* p. 59, Lemma 10.3 (3): $\exp_q$ maps the open ε-ball in TM_q diffeomorphically onto an open set U_q. *Hidden hypotheses:* none. *Deviation (restated):* the name is the exposition's; "diffeomorphically" is made precise in Milnor's sense for subsets of Euclidean space, and Mathlib's sense follows.

V7.10 (Lemma 10.3; `Morse.LocalSpray.chartExp`, `Morse.LocalSpray.exp_eq_chartExp`, `Morse.LocalSpray.chartF`, `Morse.LocalSpray.jacobian`, `Morse.LocalSpray.hasFDerivAt_chartF`, `Morse.lemma_10_3`, `Morse.lemma_10_3_geodesic`, `Morse.exists_isNormalRadius`, `Morse.exists_expDiffeo`). For each $p \in N$ there are an open set $O \ni p$ of V and an $\varepsilon > 0$ such that, with $W = N \cap O$: (1) for all $q_1, q_2 \in W$ there is exactly one $v \in T_{q_1}N$ with $\|v\| < \varepsilon$ and $\exp_{q_1}(v) = q_2$; and if γ is a geodesic from q_1 (V7.7) with initial velocity v, $\|v\| < \varepsilon$, and $\gamma(1) = q_2$, then v is this vector, so $\gamma(t) = \exp_{q_1}(tv)$ on $[0,1]$; (2) there are an open set $O_2 \subseteq V \times V$ containing $W \times W$ and a map G smooth on O_2 such that $G(q_1, q_2)$ is the vector of (1) for all $q_1, q_2 \in W$; (3) ε is a normal radius (V7.9) at every $q \in W$, and $U_q \supseteq W$ by (1). In particular every point of N has a normal radius, and $\exp_q$ restricts to a `PartialDiffeomorph` of an open ball of T_qN onto an open neighbourhood of q in N. *Book:* p. 59, Lemma 10.3: a neighbourhood W of p and an $\varepsilon > 0$ such that (1) any two points of W are joined by a unique geodesic of length $< \varepsilon$, (2) it depends smoothly on them, in that $(q_1, v) \in TM$ depends differentiably on (q_1, q_2), and (3) for $q \in W$, $\exp_q$ maps the open ε-ball diffeomorphically onto an open set $U_q \supseteq W$; proved on pp. 58–59 from the nonsingular Jacobian of $F(q, v) = (q, \exp_q v)$ at $(p, 0)$ by the "implicit function theorem". *Hidden hypotheses:* the theorem used is the inverse function theorem; taking the neighbourhood V' of $(p, 0)$ of the form $\{(q, v) : q \in U', \|v\| < \varepsilon\}$ needs such sets to form a neighbourhood base of $(p, 0)$ in TM, that is a comparison between $\|v\|$ and the coordinates of v, uniform near p; uniqueness in (1) among geodesics of length $< \varepsilon$ uses that the initial vector of such a geodesic has norm equal to its length (V7.2). *Deviation (restated):* in (1), "unique geodesic of length $< \varepsilon$" becomes the unique initial vector v, $\|v\| < \varepsilon$, the geodesic $t \mapsto \exp_{q_1}(tv)$ on $[0,1]$ having length $\|v\|$ (V7.15). *Proof:* Milnor's, in the slice chart at p: $E(u, w) = \sigma_1(\exp(x(u), dx_u w))$, x the parametrization, is given by the flow of the chart equation of V7.3, hence smooth (V7.4), and $(u, w) \mapsto (u, E(u, w))$ has derivative $(a, b) \mapsto (a, a + b)$ at $(u_0, 0)$ (Milnor's matrix $\begin{pmatrix} I & I \\ 0 & I \end{pmatrix}$ has the images of $\partial/\partial u^i$ and $\partial/\partial t^j$ as rows); the companion project's inverse function theorem (`Milnor.ContMDiff.isLocalDiffeomorphAt_of_isInvertible_mfderiv`) gives the local inverse from which W, ε, G and the inverses in (3) are read off.

V7.11 (piecewise smooth paths and their lengths; `Morse.pathLength`, `Morse.IsPiecewiseSmooth`, `Morse.IsAdmissible`, `Morse.IsPiecewiseSmooth.isAdmissible`, `Morse.IsAdmissible.integral_deriv`, `Morse.IsAdmissible.norm_sub_le_pathLength`, `Morse.IsAdmissible.pathLength_add`, `Morse.IsAdmissible.eq_of_pathLength_eq_zero`, `Morse.pathLength_eq_of_norm_deriv`, `Morse.IsPiecewiseSmooth.mono`, `Morse.IsPiecewiseSmooth.concat`, `Morse.pathLength_concat`, `Morse.IsPiecewiseSmooth.reverse`, `Morse.pathLength_reverse`, `Morse.IsPiecewiseSmooth.translate`, `Morse.pathLength_translate`). Let c be a curve in a real normed space (complete, for the statements involving $\int c'$) and $a \le b$. The length of c on $[a, b]$ is $L(c|[a, b]) = \int_a^b \|c'(t)\|\, dt$ (c' is taken to be 0 where c is not differentiable, which for the paths below happens only on a null set). c is piecewise smooth on $[a, b]$ if it is continuous on $[a, b]$ and there is a finite set B of break points such that c is C^∞, with one-sided derivatives at the ends, on every $[s, t] \subseteq [a, b]$, $s < t$, with no point of B in (s, t). c is admissible on $[a, b]$ if it is continuous on $[a, b]$, differentiable at

all but countably many points of (a,b), and $\|c'\|$ is integrable on $[a,b]$. A path in N is one with $c([a,b]) \subseteq N$. Piecewise smooth paths are admissible. For admissible paths $\int_a^b c' = c(b) - c(a)$, so $\|c(b) - c(a)\| \le L(c|[a,b])$; length is additive under subdivision; a path of length 0 is constant; a path with $\|c'\| = k$ wherever it is differentiable has length $k(b-a)$. Restriction to a subinterval, concatenation of two paths with a common endpoint, reversal $t \mapsto c(a+b-t)$ and translation of the parameter preserve piecewise smoothness; under concatenation lengths add, and under reversal and translation they are unchanged. *Book:* p. 67 (§11): a piecewise smooth path is a map $\omega : [0,1] \to M$ with a subdivision $0 = t_0 < \cdots < t_k = 1$ such that each $\omega|[t_{i-1}, t_i]$ is C^∞; from Theorem 10.4 on (p. 59) such paths on intervals $[a,b]$ and their lengths $\int \|d\omega/dt\|\, dt$ are used without comment. *Hidden hypotheses:* integrability of $\|d\omega/dt\|$; additivity and invariance of length under subdivision, concatenation, reversal and translation. *Deviation (restated):* the subdivision is given as a finite set of break points. *Deviation (stronger):* the lower bounds for lengths (V7.13–V7.16) are proved for admissible paths, a class that contains the piecewise smooth paths and is closed under restriction.

**V7.12 (Lemma 10.5, the Gauss lemma; `Morse.NormalCoord.norm_fderiv_F_apply`, `Morse.NormalCoord.inner_`
`Morse.NormalCoord.fderiv_G_fderiv_F`, `Morse.NormalCoord.injective_fderiv_F`, `Morse.gauss_lemma`, `Morse.gauss_lemma_orthogonal`).** Let ε be a normal radius at $q \in N$ and regard $\exp_q$ as a map on the vector space T_qN. For $w, x \in T_qN$ with $\|w\| < \varepsilon$,

$$\langle d(\exp_q)_w(w),\ d(\exp_q)_w(x)\rangle = \langle w, x\rangle;$$

in particular, if $x \perp w$, the radial velocity $d(\exp_q)_w(w)$ is orthogonal to $d(\exp_q)_w(x)$. Moreover $d(\exp_q)_w$ is injective on T_qN, and $\|d(\exp_q)_{rw}(w)\| = \|w\|$ for $0 \le r \le 1$ (these two are stated for the smooth extension F of V7.9, whose derivative on T_qN is that of $\exp_q$). *Book:* p. 60, Lemma 10.5: in U_q the geodesics through q are the orthogonal trajectories of the hypersurfaces $\{\exp_q(v) : \|v\| = \text{constant}\}$; proof with the surface $f(r,t) = \exp_q(rv(t))$, $\|v(t)\| = 1$. *Hidden hypotheses:* that these sets are hypersurfaces whose tangent vectors are the $\frac{d}{dt}\exp_q(r_0 v(t))$ uses that $\exp_q$ is a diffeomorphism on the ball (V7.9). *Deviation (stronger):* the identity holds for every $x \in T_qN$, not only for $x \perp w$, and needs no curve $v(t)$; Milnor's orthogonality of trajectories is its case $x \perp w$, the vectors $x \perp w$ forming the tangent space at w of the sphere $\{\|v\| = \|w\|\}$. *Proof:* Milnor's, with $f(r,s) = \exp_q(r(w+sx))$ and $g(r) = \langle \partial f/\partial r, \partial f/\partial s\rangle$ at $s = 0$: in V, $\langle \partial^2 f/\partial r^2, \partial f/\partial s\rangle = 0$ because the acceleration of the radial geodesic is normal and $\partial f/\partial s$ is tangent, and $\langle \partial f/\partial r, \partial^2 f/\partial s\,\partial r\rangle = \frac{1}{2}\frac{\partial}{\partial s}\|w+sx\|^2 = \langle w, x\rangle$ by constant speed, the symmetry of second derivatives in V replacing Lemma 8.7; so $g' = \langle w, x\rangle$, $g(0) = 0$, and $g(1) = \langle w, x\rangle$ by the mean value theorem.

**V7.13 (Lemma 10.6; `Morse.NormalCoord.hasDerivAt_radius`, `Morse.NormalCoord.radius_sub_le_pathLength`
`Morse.NormalCoord.direction_eq_of_pathLength_eq`).** Let ε be a normal radius at q, with U_q and $G = \exp_q^{-1}$ as in V7.9. Let c be admissible on $[a,b]$ with $c(t) \in U_q$ and $c(t) \ne q$ for $t \in [a,b]$, and write $c(t) = \exp_q(r(t)v(t))$ with $r(t) = \|G(c(t))\| \in (0,\varepsilon)$ and $v(t) = G(c(t))/r(t)$, a unit vector. Then $L(c|[a,b]) \ge r(b) - r(a)$, and if equality holds, v is constant on $[a,b]$. Pointwise: at each $t \in (a,b)$ where c is differentiable, r is differentiable with $|r'(t)| \le \|c'(t)\|$, and $r'(t) = \|c'(t)\|$ implies $v'(t) = 0$. *Book:* p. 60, Lemma 10.6, for a piecewise smooth $\omega : [a,b] \to U_q - \{q\}$: $\int_a^b \|d\omega/dt\|\, dt \ge |r(b) - r(a)|$, with equality only if r is monotone and v constant; the proof (p. 61) splits $d\omega/dt = \frac{\partial f}{\partial r}r'(t) + \frac{\partial f}{\partial t}$ into orthogonal parts. *Hidden*

hypotheses: the step from $\partial f/\partial t = 0$ to $dv/dt = 0$ needs $d(\exp_q)_{rv}$ injective (V7.12) and $r > 0$. *Deviation (narrower):* the bound is one-sided, $r(b) - r(a)$ rather than $|r(b) - r(a)|$, which is the case Theorem 10.4 uses, and the equality clause gives only that v is constant; the monotone parametrization appears in V7.16. *Deviation (stronger):* admissible paths. *Proof:* without the orthogonal splitting: with $w = G \circ c$, the Gauss lemma gives $\langle d(\exp_q)_w(w), c'\rangle = \langle w, w'\rangle = rr'$, and the Cauchy–Schwarz inequality with $\|d(\exp_q)_w(w)\| = \|w\|$ gives $|r'| \le \|c'\|$, equality forcing w' to be a multiple of w; the integrated forms use the fundamental theorem of calculus off a countable set.

V7.14 (paths leaving a normal ball; `Morse.NormalCoord.ballSet`, `Morse.NormalCoord.closedBallSet`, `Morse.NormalCoord.isCompact_closedBallSet`, `Morse.NormalCoord.mem_closedBallSet`, `Morse.NormalCoord.exists_crossing`). Let ε be a normal radius at q and $0 < s < \varepsilon$. The open normal ball $\exp_q\{v \in T_qN : \|v\| < s\}$ is the intersection of N with an open set of V; the closed normal ball $\exp_q\{v \in T_qN : \|v\| \le s\}$ is compact, hence closed in V, and consists of the points $y \in U_q$ with $\|G(y)\| \le s$. If c is an admissible path in N on $[a, b]$ with $c(a) = q$ and $c(b)$ outside the open normal ball of radius s, there are $a \le t_1 < t_2 \le b$ with $c(t_1) = q$, $c(t) \in U_q \setminus \{q\}$ for $t_1 < t \le t_2$, $c(t)$ in the open normal ball for $t_1 \le t < t_2$, $\|G(c(t_2))\| = s$, and $L(c|[t_1, t_2]) \ge s$. (t_2 is the first exit from the open ball and t_1 the last visit of q before it; after t_2 the path may leave U_q.) *Book:* p. 61, proof of Theorem 10.4: for every $\delta > 0$ a path from q to a point of the shell of radius r contains a segment that joins the shells of radii δ and r and lies between them, of length $\ge r - \delta$; then $\delta \to 0$. *Hidden hypotheses:* a competing path may leave U_q and come back; the segment exists because closed normal balls are closed in M (compact in a Hausdorff space), which puts the first exit point on the shell. *Deviation (restated):* the shells of radius $\delta \to 0$ are replaced by the last visit of q, the limit being taken inside the proof; this is the exposition's Lemma C for paths starting at q.

V7.15 (Theorem 10.4; `Morse.NormalCoord.radial`, `Morse.NormalCoord.pathLength_exp_smul`, `Morse.NormalCoord.norm_le_pathLength`, `Morse.theorem_10_4`). (i) If ε is a normal radius at q and $v \in T_qN$, $\|v\| < \varepsilon$, then $t \mapsto \exp_q(tv)$ is a geodesic near $[0, 1]$ with initial velocity v and length $\|v\|$ on $[0, 1]$, and every admissible path in N, on any $[a, b]$, from q to $\exp_q(v)$ has length $\ge \|v\|$, whether or not it stays in U_q. (ii) For each $p \in N$ there are an open set $O \ni p$ of V and an $\varepsilon > 0$ such that for any $q_1, q_2 \in N \cap O$ there is $v \in T_{q_1}N$ with $\|v\| < \varepsilon$ and $\exp_{q_1}(v) = q_2$, the geodesic $\gamma(t) = \exp_{q_1}(tv)$, $0 \le t \le 1$, has length $\|v\|$, and every piecewise smooth path c in N on any $[a, b]$ from q_1 to q_2 satisfies $L(c|[a, b]) \ge \|v\|$, with equality only if $c([a, b]) = \gamma([0, 1])$. *Book:* p. 59, Theorem 10.4: for W, ε as in Lemma 10.3, $\gamma : [0, 1] \to M$ the geodesic of length $< \varepsilon$ joining two points of W, and any other piecewise smooth path $\omega : [0, 1] \to M$ joining them, $\int_0^1 \|d\gamma/dt\|\, dt \le \int_0^1 \|d\omega/dt\|\, dt$, with equality only if the point sets $\omega([0, 1])$ and $\gamma([0, 1])$ coincide; proof on p. 61. *Hidden hypotheses:* the shell-crossing step (V7.14). *Deviation (stronger):* paths on any parameter interval, and in (i) admissible paths. *Proof:* (ii) takes O and ε from Lemma 10.3 (V7.10), so v is the vector of its item (1) and ε a normal radius at q_1; the inequality is (i), which is V7.14 with $s = \|v\|$; the equality clause is V7.16 (for $v = 0$ the path has length 0 and is constant).

V7.16 (the equality case of Theorem 10.4, sharp form; `Morse.NormalCoord.eq_exp_of_pathLength_eq`, `Morse.IsAdmissible.continuousOn_pathLength`, `Morse.IsAdmissible.monotoneOn_pathLength`, `Morse.NormalCoord.image_eq_of_pathLength_eq`). Let ε be a normal radius at q, $v \in T_qN$ with

$0 < \|v\| < \varepsilon$, and c an admissible path in N on $[a,b]$ from q to $\exp_q(v)$ of length exactly $\|v\|$. Then

$$c(t) = \exp_q(\rho(t)v), \qquad \rho(t) = \frac{L(c|[a,t])}{\|v\|}, \qquad a \le t \le b,$$

where ρ is continuous and nondecreasing with $\rho(a) = 0$ and $\rho(b) = 1$: $c = \gamma \circ \rho$ is a monotone reparametrization of the geodesic $\gamma(t) = \exp_q(tv)$, $0 \le t \le 1$. Milnor's conclusion $c([a,b]) = \gamma([0,1])$ follows. *Book:* p. 59, the equality clause of Theorem 10.4 (the point sets coincide); p. 61: if the images differ, a strict inequality is "easily" obtained. *Hidden hypotheses:* the equality analysis, left to the reader. *Deviation (stronger):* a monotone reparametrization instead of equal images. Equal images are too weak for Corollary 10.7 (V7.17), whose proof needs this form. *Proof:* with t_1, t_2 from V7.14, the lengths on $[a,t_1]$ and $[t_2,b]$ vanish, so c is constant there; for $t_1 < t \le t_2$, (i) of V7.15 on $[t_1,t]$ and Lemma 10.6 on $[t,t_2]$ force $L(c|[t_1,t]) = r(t)$ and equality in Lemma 10.6 on $[t,t_2]$, so the direction of $G(c(t))$ is that of v.

V7.17 (Corollary 10.7; `Morse.isGeodesicOn_of_locally`, `Morse.minimal_segment`, `Morse.corollary_10_7`, `Morse.corollary_10_7_extend`). Let c be a piecewise smooth path in N on $[0,\ell]$, parametrized by arc length ($\|c'(t)\| = 1$ at every $t \in (0,\ell)$ where c is differentiable), whose length is at most that of every piecewise smooth path in N, on any parameter interval, from $c(0)$ to $c(\ell)$. Then c is a geodesic on $(0,\ell)$, in particular smooth there, also at its break points; and c agrees on $[0,\ell]$ with a geodesic defined on an open interval $(-\delta, \ell+\delta)$. On the way: being a geodesic is a local property, and if c maps $[\alpha,\beta]$ into a set W as in Lemma 10.3 (V7.10), it agrees on $[\alpha,\beta]$ with a geodesic defined on an open set containing $[\alpha,\beta]$. *Book:* p. 61, Corollary 10.7, for a path $\omega : [0,\ell] \to M$ parametrized by arc length whose length is at most that of any other path from $\omega(0)$ to $\omega(\ell)$; the proof applies Theorem 10.4 to each segment of ω that lies in a set W of Lemma 10.3 and has length $< \varepsilon$. *Hidden hypotheses:* the equality clause of Theorem 10.4 as stated (equal images) does not make such a segment a geodesic, a path running along γ at varying speed having the same image; the proof needs V7.16 together with the unit speed, which makes ρ affine. Also needed: each segment is minimal between its endpoints (replacing it by the geodesic of Lemma 10.3 gives a competitor, by concatenation, V7.11), the segments must contain the break points in their interiors, and being a geodesic is local. *Deviation (stronger):* the extension to an open interval containing $[0,\ell]$.

V7.18 (the distance; `Morse.pathLengths`, `Morse.riemDist`, `Morse.riemDist_le_pathLength`, `Morse.riemDist_self`, `Morse.riemDist_eq_top_of_notMem`, `Morse.riemDist_comm`, `Morse.riemDist_triangle`, `Morse.NormalCoord.ofReal_le_riemDist`, `Morse.eq_of_riemDist_eq_zero`). For $p,q \in V$, $\rho(p,q) \in [0,\infty]$ is the infimum of the lengths of the piecewise smooth paths c in N, on any $[a,b]$, with $c(a) = p$ and $c(b) = q$ (∞ if there is none, as when $q \notin N$). Then $\rho(p,q) \le L(c|[a,b])$ for every such path; $\rho(p,p) = 0$ for $p \in N$; $\rho(p,q) = \rho(q,p)$; $\rho(p,r) \le \rho(p,q) + \rho(q,r)$; if ε is a normal radius at p and $0 < s < \varepsilon$, every point outside the open normal ball of radius s about p is at distance $\ge s$ from p; and for $p,q \in N$, $\rho(p,q) = 0$ implies $p = q$. So ρ is a metric on N with values in $[0,\infty]$. *Book:* p. 62: $\rho(p,q)$ is the greatest lower bound of the arc-lengths of the piecewise smooth paths joining p and q, and this "clearly" makes M a metric space. *Hidden hypotheses:* finiteness needs M connected (Milnor's standing assumption, p. 55); positivity is not clear, since a path from p to q may leave every normal neighbourhood of p: the lower bound needs the shell-crossing lemma (V7.14). *Deviation (restated):* ρ is ∞ between points that no path joins, and N need not be connected.

V7.19 (metric balls and the topology; `Morse.NormalCoord.riemDist_exp_le`, `Morse.NormalCoord.riemDist_ex`, `Morse.NormalCoord.riemDist_lt_iff`, `Morse.hasBasis_nhdsWithin_riemDist`). Let ε be a normal radius at $p \in N$. For $v \in T_pN$ with $\|v\| < \varepsilon$, $\rho(p, \exp_p v) = \|v\|$. For $0 < r \le \varepsilon$, $\rho(p,y) < r$

iff y lies in the normal ball $\exp_p\{v \in T_pN : \|v\| < r\}$. Consequently, for every $p \in N$, the metric balls $\{y : \rho(p,y) < r\}$, $r > 0$, form a basis of the neighbourhoods of p in N (with the topology of a subspace of V): the topology defined by ρ is the topology of N. *Book:* p. 62: compatibility of the metric with the topology of M follows "easily" from Theorem 10.4. *Hidden hypotheses:* that small metric balls are normal balls (V7.14 again), and that normal balls of small radius lie in any given neighbourhood of p (continuity of $\exp_p$ at 0). *Deviation (restated):* compatibility is stated as the neighbourhood-basis property at each point; the metric-space structure on N that it yields is V7.21.

V7.20 (Corollary 10.8; `Morse.IsShortJoin`, `Morse.exists_isShortJoin`, `Morse.corollary_10_8`). Let $K \subseteq N$ be compact. There is $\delta > 0$ such that for all $y, z \in K$ with $\rho(y,z) < \delta$: there is $v \in T_yN$ with $\|v\| < \delta$ and $\exp_y(v) = z$, the initial vector of the geodesic $t \mapsto \exp_y(tv)$, $0 \le t \le 1$, of length $\|v\|$, from y to z; this geodesic is minimal, $\rho(y,z) = \|v\|$; v is the only $w \in T_yN$ with $\|w\| < \delta$ and $\exp_y(w) = z$; and v depends smoothly on the endpoints: there are an open set $O_2 \ni (y,z)$ of $V \times V$ and a map G smooth on O_2 with $G(y,z) = v$ such that for all $y', z' \in N$ with $(y',z') \in O_2$ and $\rho(y',z') < \delta$, $G(y',z') \in T_{y'}N$, $\|G(y',z')\| < \delta$ and $\exp_{y'}(G(y',z')) = z'$. *Book:* p. 62, Corollary 10.8: any two points of K at distance less than δ are joined by a unique geodesic of length less than δ, which is minimal and depends differentiably on its endpoints; proof: cover K by sets W as in Lemma 10.3 and take δ so small that any two points of K at distance $< \delta$ lie in a common W. *Hidden hypotheses:* the choice of δ is a Lebesgue-number argument with a finite subcover; uniqueness among geodesics of length $< \delta$ needs δ at most the ε of each set of the cover, else Lemma 10.3 (1) does not apply; minimality is Theorem 10.4, through $\rho(y,z) = \|v\|$ (V7.19). *Deviation (restated):* uniqueness is stated for initial vectors (a geodesic from y with initial velocity w, defined near $[0,1]$ and ending at z, has length $\|w\|$ and $\exp_y(w) = z$, by V7.2 and V7.7); "depends differentiably" means that near each pair the initial vector is the restriction of a smooth map on an open set of $V \times V$, on the pairs at distance $< \delta$. *Proof:* each $x \in K$ is the centre of a normal ball of radius s_x, with $2s_x$ at most the ε of Lemma 10.3 at x and the radius of a metric ball about x inside its set W; finitely many of these balls cover K, δ is the least of their s_x, and the triangle inequality puts y and z in one W.

V7.21 (continuity and finiteness of ρ; the metric space (N,ρ); `Morse.ofReal_norm_sub_le_riemDist`, `Morse.eventually_riemDist_lt`, `Morse.continuousOn_riemDist`, `Morse.continuousOn_riemDist_left`, `Morse.continuous_riemDist`, `Morse.riemDist_ne_top`, `Morse.riemEMetricSpace`, `Morse.riemMetricSpace`, `Morse.riemMetricSpace_dist`, `Morse.riemMetricSpace_topology`). For all $p, q \in V$, $\|q - p\| \le \rho(p,q)$. For $p \in N$ and every $r > 0$, all points of N near p are at distance $< r$ from p. The function $(x,y) \mapsto \rho(x,y)$, with values in $[0,\infty]$, is continuous on $N \times N$ (with the topology of a subspace of $V \times V$); in particular $y \mapsto \rho(y,q)$ is continuous on N for every q. If N is connected, $\rho(p,q) < \infty$ for all $p, q \in N$. Hence ρ makes N an extended metric space whose topology is the topology of N (by V7.19), and, for connected N, a metric space with distance ρ and the topology of N. *Book:* p. 62: ρ "clearly" makes M a metric space, compatible with its topology by Theorem 10.4; finiteness rests on the standing assumption that M is connected (p. 55); the continuity of ρ is used without comment in the proof of Theorem 10.9 (p. 63: a point of the compact shell S at minimum distance from q, and the equality at the supremum t_0 "by continuity"). *Hidden hypotheses:* connectedness, for finiteness; the continuity of ρ, which the book never states. *Deviation (restated):* the two metric spaces are structures on N given by name, not instances, since N, a subset of V, already carries the metric of V. "Connected" is preconnected in the formal statements here and below, which also admits $N = \emptyset$. *Proof:* the points at finite

distance from p form a subset of N that is open and closed, metric balls being neighbourhoods (V7.19); continuity follows from $|\rho(x,y)-\rho(x',y')| \le \rho(x,x')+\rho(y,y')$ and V7.19.

V7.22 (metric and geodesic completeness; `Morse.IsRiemComplete`, `Morse.isRiemComplete_iff_completeSpace` `Morse.IsRiemBounded`, `Morse.IsGeodesicallyComplete`, `Morse.isGeodesicOn_univ_exp_smul`, `Morse.isGeodesicallyComplete_iff`). (N,ρ) is *complete* if every sequence of points $x_n \in N$ that is Cauchy for ρ (for every $\varepsilon>0$, $\rho(x_m,x_n)<\varepsilon$ for all large m,n) converges for ρ to some $y \in N$, i.e. $\rho(x_n,y)\to 0$; for connected N this is the completeness of the metric space of V7.21. A set $X \subseteq V$ is *bounded* if $\rho(x,y)\le d$ for all $x,y\in X$ and some real d (a nonempty bounded set lies in N, since $\rho(x,x)=\infty$ for $x\notin N$). N is *geodesically complete* if $\exp_q(v)$ is defined (V7.7) for every $q\in N$ and every $v\in T_qN$. If $\exp_q$ is defined on all of T_qN, then $t\mapsto \exp_q(tv)$ is a geodesic on $\mathbb{R}$ for every $v\in T_qN$; and N is geodesically complete iff every geodesic on an open interval is the restriction of a geodesic on $\mathbb{R}$. *Book:* p. 58: the definition, "clearly equivalent" to the requirement that every geodesic segment $\gamma_0:[a,b]\to M$ extends to a geodesic $\gamma:\mathbb{R}\to M$, the form recalled on p. 62; p. 64: bounded subsets, and M complete as a metric space (every Cauchy sequence converges). *Hidden hypotheses:* the equivalence glues geodesics, which needs their uniqueness in the large (V7.5). *Deviation (restated):* Milnor's segments on closed intervals are replaced by geodesics on open intervals, as in V7.7.

V7.23 (the geodesic flow, and the continuity of exp; `Morse.IsGeodesicOn.exp_smul_eq`, `Morse.geoFlow`, `Morse.exists_localFlow_exp`, `Morse.isGeodesicOn_exp_smul_union`, `Morse.eventually_mem_expDom_and_continuousWithinAt_exp`, `Morse.continuousOn_exp`). (i) If γ is a geodesic on an open interval containing 0, then $\gamma(t)=\exp_{\gamma(0)}(t\,\gamma'(0))$ for every t in the interval. So the curves $t\mapsto \exp_y(tw)$ are all the geodesics, and the geodesic flow is read off from exp: it sends (y,w) and t to the position and velocity at time t of $s\mapsto\exp_y(sw)$. (ii) For $x_0\in N$ and $R>0$ there are an open set $U\ni x_0$ of V, a time $\tau>0$ and a map Φ, continuous on $U\times\{w:\|w\|<R\}\times(-\tau,\tau)$, such that for $y\in N\cap U$ and $w\in T_yN$ with $\|w\|<R$ the curve $t\mapsto\exp_y(tw)$ is a geodesic on $(-\tau,\tau)$ with velocity w at 0, whose position and velocity at time t are $\Phi(y,w,t)$. (iii) If $t\mapsto\exp_y(tw)$ is a geodesic on an open interval $I\ni 0$, $s\in I$, (y',w') is its position and velocity at time s, and $t\mapsto\exp_{y'}(tw')$ is a geodesic on an open interval $J\ni 0$, then $t\mapsto\exp_y(tw)$ is a geodesic on $I\cup(s+J)$, and its position and velocity at time $s+t$, $t\in J$, are those of $t\mapsto\exp_{y'}(tw')$ at time t. (iv) The domain of exp is open in TN (V7.8) and exp is continuous on it: if (p,v) is in the domain, so is every $(q,w)\in TN$ near (p,v), and exp, as a function on TN, is continuous at (p,v). In particular, if $\exp_p$ is defined on all of T_pN, it is continuous there. *Book:* p. 57, $\gamma(t)=\exp_q(tv)$; the extension of geodesics (pp. 58, 62); Corollary 10.10 (p. 64) uses that $\exp_p$ maps the disk of radius d in TM_p onto a compact set, which needs (iv). *Hidden hypotheses:* (iv), the global dependence of geodesics on their initial data: Lemma 10.2 concerns small vectors only. *Deviation (narrower):* continuity of exp on its domain, not the smoothness asserted in the exposition's Lemma B; maximal geodesics are not constructed, geodesics being glued by (iii) instead. *Proof:* (ii) is Lemma 10.2 with smooth dependence (V7.6), rescaled by homogeneity; for (iv) the compact arc $\gamma([0,1])$ is covered by finitely many sets U of (ii) with a uniform time $2\tau_0$, and the geodesics with initial data near (p,v) are continued by (iii) in steps of length τ_0, their position and velocity being at each step a composition of continuous maps.

V7.24 (minimal geodesics; `Morse.IsMinimalGeodesic`, `Morse.exp_mem_of_mem_expDom`, `Morse.isMinimalGeodesic_exp_smul`). A curve γ is a minimal geodesic on $[a,b]$, $a\le b$, if

it is a geodesic on some open set containing $[a,b]$ and $L(\gamma|[a,b]) \le L(c|[a',b'])$ for every piecewise smooth path c in N, on any $[a',b']$, with $c(a') = \gamma(a)$ and $c(b') = \gamma(b)$. Where $\exp_q(v)$ is defined, it lies in N. If $v \in T_pN$, $t \mapsto \exp_p(tv)$ is a geodesic on $\mathbb{R}$ and $\rho(p, \exp_p v) = \|v\|$, then $t \mapsto \exp_p(tv)$ is a minimal geodesic on $[0,1]$, of length $\|v\|$. *Book:* pp. 61–62: a geodesic $\gamma : [a,b] \to M$ is minimal if its length is at most that of any other piecewise smooth path joining its endpoints. *Hidden hypotheses:* none. *Deviation (restated):* a geodesic on an open set containing $[a,b]$ in place of one on $[a,b]$, as in V7.7.

V7.25 (the two steps of Milnor's proof of Theorem 10.9; `Morse.NormalCoord.exists_riemDist_eq_add`, `Morse.broken_geodesic_eq`). (i) Let ε be a normal radius at $p \in N$ (V7.9), $0 < \delta < \varepsilon$, and q with $\delta < \rho(p,q) < \infty$. There is $w \in T_pN$ with $\|w\| = \delta$ and $\rho(p,q) = \delta + \rho(\exp_p w, q)$: the point $p_0 = \exp_p w$ of the sphere of radius δ about p is a point of that sphere nearest to q. (ii) Let γ be a geodesic on $\mathbb{R}$ with $\|\gamma'(0)\| = 1$ (so of unit speed), $0 < t_0 < t_1$, and σ a geodesic on an open interval containing $[t_0, t_1]$ with $\|\sigma'(t_0)\| = 1$ and $\sigma(t_0) = \gamma(t_0)$. If $\rho(\gamma(0), \sigma(t_1)) \ge t_1$, that is, if the broken path (γ on $[0,t_0]$, then σ), of length t_1, is minimal, then $\sigma(t_1) = \gamma(t_1)$. *Book:* p. 63: since every path from p to q passes through the shell S, $\rho(p,q) = \min_{s\in S}(\rho(p,s)+\rho(s,q)) = \delta + \rho(p_0,q)$; pp. 63–64: the broken geodesic has minimal length, so it is an unbroken geodesic by Corollary 10.7, "and hence coincides with γ". *Hidden hypotheses:* in (i), $\delta < \rho(p,q)$, not only $\delta < \varepsilon$ (otherwise q lies inside the shell and paths from p to q need not meet it); that every path crosses the shell is V7.14; the minimum exists because the shell is compact and nonempty and $\rho(\cdot, q)$ is continuous (V7.21). In (ii), unit speed on both pieces, for Corollary 10.7 (V7.17), and the uniqueness of geodesics in the large (V7.5) from a time before the break, carried to t_1 by continuity.

V7.26 (Theorem 10.9, Hopf–Rinow; `Morse.theorem_10_9`, `Morse.theorem_10_9'`). (i) Let $p \in N$ be such that $\exp_p$ is defined on all of T_pN. Every $q \in N$ with $\rho(p,q) < \infty$ is $\exp_p(v)$ for some $v \in T_pN$ with $\|v\| = \rho(p,q)$, and $t \mapsto \exp_p(tv)$ is a minimal geodesic on $[0,1]$ (V7.24) from p to q. (ii) If N is connected and geodesically complete, any two points $p, q \in N$ are joined by a minimal geodesic γ on $[0,1]$, $\gamma(0) = p$, $\gamma(1) = q$, whose length is $\rho(p,q)$. *Book:* p. 62, Theorem 10.9: if M is geodesically complete, any two points can be joined by a minimal geodesic; proof pp. 62–64, with Diagram 10. *Hidden hypotheses:* those of V7.25, and $\delta' < r - t_0$ in the inductive step (otherwise the new time $t_0 + \delta'$ overshoots $r = \rho(p,q)$); the supremum argument is a continuous induction on the set of $t \in [\delta, r]$ with $\rho(\gamma(t), q) = r - t$, closed by the continuity of ρ (V7.21); connectedness, for $\rho(p,q) < \infty$. *Deviation (stronger):* (i) uses completeness only at p, as the exposition remarks after the proof. *Proof:* Milnor's, with V7.25 (i) at p and at $\gamma(t_0)$, and V7.25 (ii) for the broken geodesic.

V7.27 (Corollary 10.10; `Morse.isCompact_closure_of_isRiemBounded`, `Morse.isRiemComplete_of_forall_isC` `Morse.corollary_10_10`). (i) If N is connected and $\exp_p$ is defined on all of T_pN for some $p \in N$, every bounded set $X \subseteq V$ (V7.22) has compact closure, contained in N. (ii) If every bounded set has compact closure contained in N, then (N, ρ) is complete. (iii) Hence, if N is connected and geodesically complete, every bounded set has compact closure in N, and (N, ρ) is complete. *Book:* p. 64, Corollary 10.10: if M is geodesically complete, every bounded subset of M has compact closure; consequently M is complete as a metric space; proof: for X of diameter d and $p \in X$, $\exp_p$ maps the disk of radius d in TM_p onto a compact set which contains X, by Theorem 10.9. *Hidden hypotheses:* the continuity of $\exp_p$ on the whole disk (V7.23 (iv)); connectedness, so that distances are finite and Theorem 10.9 applies to all of X; for completeness, that a Cauchy sequence is bounded and converges when a subsequence does. *Deviation (stronger):* (i) needs completeness at one point only. The closure is taken in V; as it lies in N, it is also the closure in N.

V7.28 (the converse of Corollary 10.10; `Morse.isGeodesicallyComplete_of_isRiemComplete`). If (N, ρ) is complete, then N is geodesically complete. *Book:* p. 64: conversely, if M is complete as a metric space, it is "not difficult, using Lemma 10.3" to prove that M is geodesically complete; the details are left to Hopf and Rinow. *Hidden hypotheses:* none; connectedness is not needed. *Deviation (not in the book):* the proof, the exposition's, which uses the uniform local existence of Lemma 10.2 (V7.23 (ii)) rather than Lemma 10.3. *Proof:* if the geodesic $\gamma(t) = \exp_q(tv)$ existed only up to a time T_0, then $\rho(\gamma(t), \gamma(t')) \le \|v\|\,|t - t'|$ would make $\gamma(t_n)$, $t_n \uparrow T_0$, a Cauchy sequence; it converges to some $x \in N$, near which all geodesics of speed $\|v\|$ exist for a uniform time, so γ continues past T_0 (V7.23 (iii)).

V7.29 (the Hopf–Rinow theorem; `Morse.hopf_rinow`, `Morse.hopf_rinow_of_isRiemComplete`, `Morse.isCompact_of_isClosed_of_isRiemBounded`). For a connected nonempty N the following are equivalent: (1) (N, ρ) is complete; (2) N is geodesically complete; (3) $\exp_p$ is defined on all of T_pN for some $p \in N$; (4) every bounded set has compact closure, contained in N. If N is connected and (N, ρ) is complete, then N is geodesically complete, every geodesic on an open interval is the restriction of a geodesic on $\mathbb{R}$, any two points of N are joined by a minimal geodesic on $[0, 1]$ whose length is their distance, and every bounded subset of N that is closed in N is compact (which follows from (4) alone). *Book:* p. 64: after Corollary 10.10 and its converse, geodesic and metric completeness are no longer distinguished, and M is called a complete Riemannian manifold. *Hidden hypotheses:* connectedness. *Deviation (stronger):* condition (3), the exposition's remark that completeness at one point suffices, and the compactness of closed bounded sets are not in the book. *Proof:* (2) $\Rightarrow$ (3) as $N \neq \emptyset$, (3) $\Rightarrow$ (4) is V7.27 (i), (4) $\Rightarrow$ (1) is V7.27 (ii), (1) $\Rightarrow$ (2) is V7.28; the second statement adds V7.22 and V7.26.

V7.30 (closed and compact submanifolds; the sphere; `Morse.tendsto_riemDist_of_tendsto`, `Morse.isRiemComplete_of_isClosed`, `Morse.hopf_rinow_of_isClosed`, `Morse.hopf_rinow_of_isCompact`, `Morse.sphere_hopf_rinow`). Convergence in N implies convergence for ρ. If N is closed in V, then (N, ρ) is complete. Hence a connected N that is closed in V, in particular a connected compact one, has all the properties listed in the second statement of V7.29. The unit sphere $S^n \subseteq \mathbb{R}^{n+1}$, $n \ge 1$ (the companion project's slice submanifold `Milnor.Sphere.sphSub`), is geodesically complete, and any two of its points are joined by a minimal geodesic on $[0, 1]$ whose length is their distance. *Book:* not stated; p. 65 shows that the geodesics of S^n are the great circles and discusses which arcs are minimal; the exposition (Chapter 7, §5) notes that compact Riemannian manifolds are complete. *Hidden hypotheses:* none. *Deviation (not in the book):* in the embedded setting this is the source of complete manifolds, among them compact matrix groups. The great circles are not identified as the geodesics of S^n. *Proof:* a ρ-Cauchy sequence is Cauchy in V, by $\|x - y\| \le \rho(x, y)$ (V7.21), so it converges in V; its limit lies in the closed set N, and convergence in N gives convergence for ρ.

What is not verified in this chapter

- **Maximal geodesics** (the exposition's Lemma B): the domain of exp is shown to be open in TN and exp continuous on it (V7.23), which is all that Corollary 10.10 needs; the maximal geodesic through each point is not formalized, geodesics being glued instead. The smoothness of exp on its whole domain, not only near the zero section (V7.8), is proved in Chapter 11 (V11.1: `Morse.exists_contDiffOn_exp`, `Morse.contMDiffOn_exp_expDom`), and the extension of a geodesic on $[0, 1]$ to an open interval, which makes Milnor's domain of exp equal to the formal one, in Chapters 9 and 11 (V7.7).

- **The remark after Lemma 10.3** (p. 59): W can be chosen so that the geodesic joining any two of its points lies in W (after J. H. C. Whitehead). Not formalized in this chapter; it is proved in Chapter 10, in the form the modified proof of Theorem 17.1 needs (V10.17: `Morse.IsGoodBall`, `Morse.exists_isGoodBall_ball`): the part of N in a small ball of the ambient space contains the short geodesic joining any two of its points.
- **The remarks on minimal geodesics** (p. 62): sufficiently small segments of a geodesic are minimal, long geodesics need not be, and minimal geodesics need not be unique. Minimal geodesics are defined (V7.24) and produced by Theorem 10.9 (V7.26), but these remarks are not formalized.
- **The examples** of pp. 65–66 (straight lines in $\mathbb{R}^n$ and inscribed polygons, great circles on S^n and the geodesics joining two points of S^n, meridians of surfaces of revolution, the cylinder) and the footnote on $U(n)$ (p. 57): not formalized in this chapter, except that S^n is complete and has minimal geodesics between any two points (V7.30). Later chapters prove that the geodesics of S^n are the great circles, with the list of those joining two points (V10.23), that the geodesics of an open subset of a Euclidean space are the straight segments (V10.27), and that the exponential map of a group of matrices such as $U(n)$ is the matrix exponential (V12.15). Inscribed polygons, the continuum of minimal geodesics between antipodal points, surfaces of revolution and the cylinder are not formalized.
- **Lemma 10.6 as stated:** the two-sided bound $|r(b) - r(a)|$ and the monotonicity of r in the equality case (V7.13); Theorem 10.4 uses only the one-sided form, and its equality case is V7.16.
- **§10 on an abstract Riemannian manifold:** Mathlib has no Levi-Civita connection, geodesics or exponential map, and the Christoffel-symbol form of the geodesic equation is not formalized (V7.3); the embedded results reach an arbitrary Riemannian manifold only through Nash's embedding theorem, which is neither in Mathlib nor proved here.

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9. Chapter 8 (Milnor §§11–13): partly verified

Milnor's §11 (pp. 67–69) introduces the space $\Omega = \Omega(M; p, q)$ of piecewise smooth paths from p to q, its "tangent spaces" $T\Omega_\omega$ (an analogy: Ω is given no topology before §16), variations and their variation fields, and critical paths of a function on Ω. §12 (pp. 70–73) defines the energy E, compares it with the length by Schwarz's inequality, proves Lemma 12.1 (on a complete manifold E takes its minimum d^2 exactly at the minimal geodesics), the first variation formula (Theorem 12.2) and Corollary 12.3 (the critical paths of E are the geodesics). §13 (pp. 74–76) defines the Hessian E_{**} at a geodesic through 2-parameter variations and proves the second variation formula (Theorem 13.1), Corollary 13.4 (E_{**} is a well-defined symmetric bilinear form), Remark 13.5 and Lemma 13.6 (at a minimal geodesic E_{**} is positive semi-definite). Verified, for a submanifold N of a Euclidean space with the induced metric, in the embedded setting of Chapters 6–7: every numbered statement of §§11–13 except Lemma 12.1, of which only the inequality $E(\gamma) \le E(\omega)$ at a minimal geodesic is checked (V8.6); the definitions they need; and the exposition's Lemma 11.A (every finite family of fields along a path is the family of variation fields of one variation, without completeness), Lemma 12.A, formula (12.2′) and Proposition 13.A. Every Riemannian manifold is such an N by Nash's embedding theorem, which is quoted, not proved. As in the book, Ω is given no manifold structure; how $T\Omega_\omega$ and E_{**} are formalized is explained below. Proved more sharply than in the book: the first variation at every value of the parameter (V8.9); Theorem 13.1 with the existence of the second derivative as part of the conclusion (V8.13); the symmetry of

E_{**} without assuming $E \circ \bar{\alpha}$ of class C^2 (V8.16); and Lemma 13.6 from minimality alone (V8.18). Theorem 14.1 (the null space of E_{**} consists of the Jacobi fields vanishing at the ends) is also proved, in `Morse/Chapter8/Jacobi.lean`, and is reported with Chapter 9 (§14). **Not verified:** Lemma 12.1 in full, Milnor's motivating constructions F_* and f_{**} and the example of p. 69, the index of E_{**} as a number, and the exposition's Lemma 13.B (the list at the end). Lean: `Morse/Chapter8/` — `Piecewise.lean` (448 lines), `Energy.lean` (576), `FirstVariation.lean` (572), `Critical.lean` (590), `SecondVariation.lean` (575), `Hessian.lean` (595); the directory also holds `Jacobi.lean` (704), which belongs to Chapter 9.

The embedded setting. As in Chapters 6 and 7: V is a finite-dimensional real inner product space (Milnor's $\mathbb{R}^K$) and $N \subseteq V$ a slice submanifold modelled on Z with complement W, both finite-dimensional, with the induced metric. $T_qN \subseteq V$ is the tangent space and P_q the orthogonal projection onto it (V6.1), extended by $P_x = 0$ for $x \notin N$, so that $P_{\omega(t)}$ makes sense for every t; $D/dt = P \circ d/dt$ along curves (V6.5); R is Milnor's curvature tensor, with his sign (V6.16–V6.17). Paths, fields and variations are maps defined on all of $\mathbb{R}$ (in the time variable), of which only the values for $t \in [0,1]$ matter. $c'(t)$ is the derivative of c at t, taken to be 0 where c is not differentiable, which for the paths below happens at finitely many points only. Piecewise smooth is V7.11's notion: continuous on the interval, and C^∞, with one-sided derivatives at the ends, on every closed piece between consecutive points of a finite set of break points. N need not be closed, compact or connected.

How Ω, $T\Omega_\omega$ and E_{**} are formalized. As in the book, Ω carries no topology or manifold structure. $T\Omega_\omega$ is *defined* as a vector space of fields along ω (V8.1), related to Ω only through variations: the variation field of every variation lies in it (V8.2), and every finite family of its elements is the family of variation fields of one variation (V8.3). Every derivative "on Ω" is a derivative of a function $u \mapsto E(\bar{\alpha}(u))$ of finitely many real variables. E_{**} is *defined* as the bilinear form on $T\Omega_\gamma$ given by the right side of Milnor's second variation formula (V8.14); Theorem 13.1 (V8.13) and Lemma 11.A (V8.3) then show that Milnor's definition through 2-parameter variations is well defined and agrees with it.

V8.1 (the path space and its "tangent spaces"; `Morse.pathSpace`, `Morse.prj`, `Morse.prj_eq_self_iff`, `Morse.pathTangent`, `Morse.mem_pathTangent`, `Morse.mem_pathTangent_congr`). For $p, q \in V$, $\Omega(N; p, q)$ is the set of maps ω that are piecewise smooth on $[0,1]$, with $\omega(t) \in N$ for $t \in [0,1]$, $\omega(0) = p$ and $\omega(1) = q$. For a map ω, $T\Omega_\omega$ is the real vector space of maps W that are piecewise smooth on $[0,1]$ and tangent along ω ($P_{\omega(t)}W(t) = W(t)$ for $t \in [0,1]$, that is $W(t) \in T_{\omega(t)}N$ when $\omega(t) \in N$), with $W(0) = 0$ and $W(1) = 0$; it depends only on ω on $[0,1]$. *Book:* p. 67: $\Omega(M; p, q)$ is the set of piecewise smooth paths $\omega : [0,1] \to M$ from p to q (each $\omega|[t_{i-1}, t_i]$ of class C^∞ for some subdivision), thought of as "something like an infinite dimensional manifold", to be topologized in §16; $T\Omega_\omega$ is the vector space of piecewise smooth vector fields W along ω with $W(0) = 0$ and $W(1) = 0$. *Hidden hypotheses:* smoothness on a closed piece means one-sided derivatives at its ends. *Deviation (restated):* paths and fields are maps on $\mathbb{R}$ considered on $[0,1]$, and the subdivision is a finite set of break points (V7.11); a field's break points need not be those of ω.

V8.2 (variations and variation fields; `Morse.IsVariation`, `Morse.IsVariation.isPiecewiseSmooth`, `Morse.IsVariation.mem_pathSpace`, `Morse.IsVariation.comp`, `Morse.varField`, `Morse.hasFDerivWithinAt_m` `Morse.IsVariation.prj_varDeriv`, `Morse.IsVariation.varField_mem`). Let P be a real normed space (Milnor's $\mathbb{R}$ or $\mathbb{R}^n$) and $U \ni 0$ an open subset of P. A map $\alpha(u,t)$, $u \in P$, $t \in \mathbb{R}$, is a variation with break points B (a finite set) if $\alpha(u,t) \in N$ for $u \in U$ and $t \in [0,1]$; $\alpha(u,0)$ and

$\alpha(u,1)$ do not depend on $u \in U$; and α is C^∞ on every closed strip $U \times [s,t]$, $0 \le s < t \le 1$, with no point of B in (s,t). It is a variation of $\omega = \alpha(0,\cdot)$, and every $\alpha(u,\cdot)$, $u \in U$, lies in $\Omega(N;\omega(0),\omega(1))$; composed with a smooth map into P that sends 0 into U it is again a variation (for instance a slice $u_2 \mapsto \alpha(u_1,u_2,\cdot)$). For $P = \mathbb{R}$ the variation field is $W_t = \frac{\partial\alpha}{\partial u}(0,t)$; every $\frac{\partial\alpha}{\partial u}(u,t)$ is tangent to N at $\alpha(u,t)$, and $W \in T\Omega_\omega$. Derivatives within a set (for instance one-sided derivatives along the edge of a strip) of maps into N are tangent to N. *Book:* p. 68: a variation of ω keeping the end points fixed is a map $\bar\alpha : (-\varepsilon,\varepsilon) \to \Omega$ with $\bar\alpha(0) = \omega$ and a subdivision such that $\alpha(u,t) = \bar\alpha(u)(t)$ is C^∞ on each strip $(-\varepsilon,\varepsilon) \times [t_{i-1},t_i]$; an n-parameter variation replaces $(-\varepsilon,\varepsilon)$ by a neighbourhood of 0 in $\mathbb{R}^n$; $W_t = \frac{\partial\alpha}{\partial u}(0,t)$, and "clearly" $W \in T\Omega_\omega$. *Hidden hypotheses:* behind "clearly": W is continuous at a break point t_i because both adjacent strips compute the velocity of the same curve $u \mapsto \alpha(u,t_i)$, tangent because that curve lies in N, and zero at the ends because they are fixed. *Deviation (stronger):* any open parameter set containing 0, in any normed space.

V8.3 (Lemma 11.A: every family of tangent fields comes from one variation; `Morse.exists_variation`, `Morse.exists_variation_one`, `Morse.exists_variation_two`). Let $\omega \in \Omega(N;p,q)$, P a real normed space, and Λ a family, piecewise smooth on $[0,1]$, of continuous linear maps $\Lambda(t) : P \to V$ with values in $T_{\omega(t)}N$ and $\Lambda(0) = \Lambda(1) = 0$. There are $\varepsilon > 0$ and a finite set B such that $\alpha(u,t) = \exp_{\omega(t)}(\Lambda(t)u)$ is a variation with break points B on the ball $\|u\| < \varepsilon$ (V8.2), with $\alpha(0,t) = \omega(t)$ and derivative $\Lambda(t)$ in u at $u = 0$, for $t \in [0,1]$. In particular every $W \in T\Omega_\omega$ is the variation field of $\alpha(u,t) = \exp_{\omega(t)}(uW_t)$, $|u| < \varepsilon$; and for $W_1, W_2 \in T\Omega_\omega$, $\alpha(u_1,u_2,t) = \exp_{\omega(t)}(u_1W_1(t) + u_2W_2(t))$ is a 2-parameter variation of ω with $\frac{\partial\alpha}{\partial u_i}(0,0,t) = W_i(t)$. *Book:* p. 68: given $W \in T\Omega_\omega$ there is a variation with $\bar\alpha(0) = \omega$ and $\frac{d\bar\alpha}{du}(0) = W$; "in fact one can set" $\bar\alpha(u)(t) = \exp_{\omega(t)}(uW_t)$; p. 74: 2-parameter variations with given fields are chosen without comment. *Hidden hypotheses:* exp is defined only near the zero section (Lemma 10.2), so ε must be chosen uniformly along ω, by the compactness of $\omega([0,1])$ and a bound for $\|\Lambda\|$; completeness of N is not needed. *Deviation (stronger):* any number of parameters (the exposition's Lemma 11.A). *Proof:* exp is defined and smooth on an open neighbourhood of the zero section of TN (V7.8), and the derivative in u at 0 is $\Lambda(t)$ because $d(\exp_q)_0$ is the inclusion of T_qN.

V8.4 (critical paths of the energy; `Morse.IsCriticalPath`). A map ω is a critical path of the energy E (V8.5) if for every 1-parameter variation α with $\alpha(0,t) = \omega(t)$ for $t \in [0,1]$, the function $u \mapsto E(\alpha(u,\cdot))$ is differentiable at 0 with derivative 0. *Book:* p. 69: ω is a critical path of $F : \Omega \to \mathbb{R}$ iff $\frac{dF(\bar\alpha(u))}{du}\Big|_{u=0}$ is zero for every variation $\bar\alpha$ of ω. *Hidden hypotheses:* that the derivative exists (for E it always does, V8.9). *Deviation (narrower):* only for $F = E$, the case of Corollary 12.3.

V8.5 (energy and length, Lemma 12.A; `Morse.energy`, `Morse.pathLength_sq_le_energy`, `Morse.pathLength_sq_eq_energy_iff`, `Morse.energy_eq_of_norm_deriv`). For a map c into a real normed space and $a \le b$, $E_a^b(c) = \int_a^b \|c'(t)\|^2\,dt$, $E = E_0^1$, and $L_a^b(c) = \int_a^b \|c'(t)\|\,dt$ is the length of V7.11. If c is piecewise smooth on $[a,b]$, then $(L_a^b(c))^2 \le (b-a)E_a^b(c)$, with equality iff $\|c'\|$ takes one and the same value at all points of (a,b) where c is differentiable. An admissible path (V7.11) of constant speed k there has $E_a^b = k^2(b-a)$. *Book:* p. 70: $E_a^b(\omega) = \int_a^b \|d\omega/dt\|^2\,dt$, $E = E_0^1$, L_a^b, and by Schwarz's inequality $(L_a^b)^2 \le (b-a)E_a^b$, with equality iff $\|d\omega/dt\|$ is constant, i.e. iff the parameter is proportional to arc length. *Hidden hypotheses:* for a speed that is only piecewise continuous, "constant" can only mean constant off the break points. *Deviation*

(stronger): any $a \le b$ (Milnor: $0 \le a < b \le 1$) and any normed target. *Proof:* the variance identity $(b-a)\int_a^b (g-\bar g)^2 = (b-a)E_a^b - (L_a^b)^2$, $g = \|c'\|$ with mean $\bar g$ (the exposition's Lemma 12.A).

V8.6 (the inequality of Lemma 12.1; `Morse.IsMinimalPath`, `Morse.energy_le_of_isMinimalPath`). A path $\gamma \in \Omega(N; p, q)$ is minimal if $L(\gamma) \le L(\omega)$ for every $\omega \in \Omega(N; p, q)$. If γ is minimal and a geodesic (V8.10), then $E(\gamma) \le E(\omega)$ for every $\omega \in \Omega(N; p, q)$. *Book:* p. 70, Lemma 12.1: if M is complete and p, q are at distance d, then $E : \Omega(M; p, q) \to \mathbb{R}$ takes its minimum d^2 precisely on the set of minimal geodesics from p to q; the proof is the chain $E(\gamma) = L(\gamma)^2 \le L(\omega)^2 \le E(\omega)$. *Hidden hypotheses:* completeness serves only to make a minimal geodesic exist (V7.26); the inequality holds on any N. *Deviation (narrower):* only the inequality; neither the value d^2 of the minimum nor the converse (a path of energy d^2 is a minimal geodesic, which needs the equality cases of V8.5 and Corollary 10.7) is formalized. Minimality is among the paths of Ω, parametrized on $[0, 1]$; by affine reparametrization this is Milnor's notion (V7.24), an equivalence not stated formally. *Proof:* Milnor's chain, with the constant speed of a geodesic (V7.2) and V8.5.

V8.7 (velocity jumps, acceleration and the first variation; `Morse.jump`, `Morse.jump_eq_zero`, `Morse.covD`, `Morse.covD_eq_covDeriv`, `Morse.accel`, `Morse.firstVariation`, `Morse.finsum_jump_eq_sum`, `Morse.firstVariation_congr_field`). For a map f into V, the jump of its derivative at t is $\Delta_t f' = f'(t^+) - f'(t^-)$, the difference of the right and the left derivative of f at t; it vanishes where f is differentiable. Along a path ω, $\frac{DX}{dt}(t) = P_{\omega(t)}X'(t)$ (for ω in N, D/dt of V6.5), $V = \omega'$ and $A = \frac{D}{dt}V$. The first variation of ω in the direction W is

$$-\sum_{0<t<1} \langle W_t, \Delta_t V\rangle - \int_0^1 \langle W_t, A_t\rangle \, dt,$$

the sum being finite: for ω differentiable off a finite set B, it is the sum over the points of B in $(0, 1)$. It depends only on W on $[0, 1]$. *Book:* p. 71: $V_t = d\omega/dt$, $A_t = \frac{D}{dt}\frac{d\omega}{dt}$, $\Delta_t V = V_{t^+} - V_{t^-}$, which is 0 for all but finitely many $t \in (0, 1)$. *Hidden hypotheses:* none. *Deviation (restated):* the sum is written over all of $(0, 1)$, its terms vanishing off the break points.

V8.8 (calculus on closed strips; `Morse.pdU`, `Morse.pdT`, `Morse.hasDerivWithinAt_pdT`, `Morse.pdU_pdT`, `Morse.hasDerivAt_integral_strip`, `Morse.piecewise_induction`, `Morse.integral_eq_of_pi` `Morse.hasDerivAt_integral_of_piecewise`). For F smooth on a closed strip $U \times [a, b]$, $U \subseteq \mathbb{R}$ open, $a < b$: the partial derivatives $\partial F/\partial u$ and $\partial F/\partial t$ computed within the strip are smooth on it, are the one-sided t-derivatives on the edges $t = a$, $t = b$, and the mixed partials agree on the whole closed strip, edges included. If F and F' are continuous on the strip and F' is the u-derivative of F at its interior points, then $u \mapsto \int_a^b F(u, t)\, dt$ has derivative $\int_a^b F'(u_0, t)\, dt$ at every $u_0 \in U$; the same holds for an integrand that on each piece of a finite subdivision agrees, inside the piece, with a function of class C^1 on the closed strip. If $\int_s^t g = F^-(t) - F^+(s)$ on every piece $[s, t]$ of $[a, b]$ with break points B, then $\int_a^b g = F^-(b) - F^+(a) - \sum_{x \in B \cap (a,b)} (F^+(x) - F^-(x))$; and properties of intervals that hold on pieces and glue at break points hold on every interval. *Book:* used without comment in the proofs of Theorems 12.2 and 13.1 (pp. 71–72, 74–76): differentiation under the integral sign, integration by parts on each $[t_{i-1}, t_i]$, and adding up. *Hidden hypotheses:* smoothness on the closed strips; compactness, for the domination bound; the symmetry of second derivatives on the edges of the strips, where the t-derivatives are one-sided. *Deviation (not in the book).*

V8.9 (Theorem 12.2 and formula (12.2′); `Morse.IsVariation.hasDerivAt_energy`, `Morse.first_variation`).

Let α be a 1-parameter variation (V8.2), with parameter set $U \subseteq \mathbb{R}$. For every $u_0 \in U$, $u \mapsto E(\alpha(u,\cdot))$ is differentiable at u_0, and

$$\frac{1}{2}\frac{d}{du}E(\alpha(u,\cdot))\Big|_{u=u_0} = -\sum_t \Big\langle \frac{\partial\alpha}{\partial u}, \Delta_t \frac{\partial\alpha}{\partial t}\Big\rangle - \int_0^1 \Big\langle \frac{\partial\alpha}{\partial u}, \frac{D}{\partial t}\frac{\partial\alpha}{\partial t}\Big\rangle\, dt,$$

everything at (u_0,t): the first variation (V8.7) of $\alpha(u_0,\cdot)$ in the direction $\frac{\partial\alpha}{\partial u}(u_0,\cdot)$. In particular, if $\alpha(0,t)=\omega(t)$ on $[0,1]$ and W is the variation field, $\frac{1}{2}\frac{dE(\bar\alpha(u))}{du}(0) = -\sum_t\langle W_t, \Delta_t V\rangle - \int_0^1\langle W_t, A_t\rangle\, dt$. *Book:* p. 71, Theorem 12.2, at $u=0$ (the scan loses the leading minus sign of the displayed statement; the end of the proof, p. 72, has it); the proof differentiates under the integral sign, uses Lemma 8.3 and "Lemma 8.5" to replace $\frac{D}{\partial u}\frac{\partial\alpha}{\partial t}$ by $\frac{D}{\partial t}\frac{\partial\alpha}{\partial u}$, integrates by parts on each $[t_{i-1},t_i]$ and adds up. *Hidden hypotheses:* the reference should be to Lemma 8.7 (8.5 is the definition of a symmetric connection); differentiation under the integral sign and the symmetry of mixed partials on the closed strips (V8.8); the boundary terms telescope into the jump terms because $\partial\alpha/\partial u$ is continuous in t. *Deviation (stronger):* the formula at every u, formula (12.2′) of the exposition, which §13 differentiates.

V8.10 (geodesics of Ω; criticality through the first variation; `Morse.IsUnbrokenGeodesic`, `Morse.isCriticalPath_iff`, `Morse.IsUnbrokenGeodesic.firstVariation_eq_zero`). A map ω is an (unbroken) geodesic if it is C^∞ on $[0,1]$, with one-sided derivatives at the ends, and a geodesic on $(0,1)$ in the sense of V7.1, so that $A=0$ there. For $\omega\in\Omega(N;p,q)$: ω is a critical path (V8.4) iff its first variation (V8.7) vanishes in every direction $W\in T\Omega_\omega$. For a geodesic, the first variation vanishes in every direction. *Book:* p. 73: the path $\omega\in\Omega$ is a geodesic iff ω is C^∞ on the whole interval $[0,1]$ and its acceleration $\frac{D}{dt}\frac{d\omega}{dt}$ is identically zero; "clearly a geodesic is a critical point". *Hidden hypotheses:* the converse direction of the "iff" needs Lemma 11.A (V8.3). *Deviation (restated):* the acceleration is required to vanish on $(0,1)$; at the ends it then vanishes by continuity. *Proof:* Theorem 12.2 (V8.9) and Lemma 11.A (V8.3).

V8.11 (Corollary 12.3; `Morse.accel_eq_zero_of_critical`, `Morse.prj_jump`, `Morse.jump_eq_zero_of_critical`, `Morse.exists_isGeodesicOn_through`, `Morse.isGeodesicOn_of_glue`, `Morse.corollary_12_3`). A path $\omega\in\Omega(N;p,q)$ is a critical path of E iff it is a geodesic (V8.10). The steps, for ω critical and smooth on the pieces of a finite set B: the acceleration vanishes at every $t\in(0,1)\setminus B$; the one-sided velocities, hence the jumps $\Delta_t V$, are tangent to N, and every jump $\Delta_t V$, $0<t<1$, vanishes; through every $q\in N$ with any velocity $v\in T_qN$ at any time x there is a geodesic on an interval around x; and a path which is a geodesic on both sides of x, smooth up to x from each side, and differentiable at x, is a geodesic near x. *Book:* p. 73, Corollary 12.3; proof: a variation with $W(t)=f(t)A(t)$, f positive except at the t_i, gives $A=0$ on each piece; one with $W(t_i)=\Delta_{t_i}V$ gives $\Delta_{t_i}V=0$, so ω is C^1; then ω is C^∞ "from the uniqueness theorem for differential equations". *Hidden hypotheses:* the test variations exist only by Lemma 11.A (V8.3); $W=fA$ must be continuous at the t_i, where the one-sided accelerations differ; a field with the prescribed values $\Delta_{t_i}V$ must be constructed; the last step needs the lift (ω,ω') to solve the geodesic equation at the former break point itself (a derivative-extension argument) before uniqueness applies. *Proof:* test fields $W=\varphi A$ and $W=\varphi P(\Delta_x V)$, with $\varphi\ge 0$ a bump function at one point x, supported where ω is smooth, give $A(x)=0$ and $\|\Delta_x V\|^2=0$; then uniqueness for the first-order geodesic system (V7.3, V7.4) identifies ω near x with the geodesic through $(\omega(x),\omega'(x))$.

V8.12 (the curvature identity (13.3), and jumps under differentiation; `Morse.curvR`, `Morse.curvR_of_mem`, `Morse.curvature_identity`, `Morse.IsVariation.hasDerivAt_jump`). $R(a,b)c$ at $x\in N$ is Mil-

nor's curvature tensor of N at x (V6.17, with his sign), and 0 for $x \notin N$. (i) For a map $F(u,t)$, smooth on an open subset O of $\mathbb{R}^2$ with values in N, at every point of O

$$\frac{D}{\partial u}\frac{D}{\partial t}\frac{\partial F}{\partial t} = \frac{D}{\partial t}\frac{D}{\partial t}\frac{\partial F}{\partial u} + R\Big(\frac{\partial F}{\partial t}, \frac{\partial F}{\partial u}\Big)\frac{\partial F}{\partial t},$$

with $\frac{D}{\partial u}X = P(\partial X/\partial u)$ and $\frac{D}{\partial t}X = P(\partial X/\partial t)$. (ii) For a 1-parameter variation α, $u_0 \in U$ and $0 < x < 1$, the jump $\Delta_x \frac{\partial \alpha}{\partial t}(u,\cdot)$ is differentiable in u at u_0, with derivative $\Delta_x \frac{\partial}{\partial t}\frac{\partial \alpha}{\partial u}(u_0,\cdot)$. *Book:* pp. 75–76: the curvature formula $\frac{D}{\partial u_1}\frac{D}{\partial t}V - \frac{D}{\partial t}\frac{D}{\partial u_1}V = R(V,W_1)V$ (Lemma 9.2) and $\frac{D}{\partial u_1}V = \frac{D}{\partial t}W_1$ (Lemma 8.7) give (13.3), $\frac{D}{\partial u_1}\frac{D}{\partial t}V = \frac{D^2W_1}{dt^2} + R(V,W_1)V$; on p. 75, "rearranging the second term" of the differentiated first variation formula. *Hidden hypotheses:* Lemmas 8.7 and 9.2 are applied to a surface defined only on the strips, near an interior point (Chapter 6 states them for surfaces on all of $\mathbb{R}^2$, V6.15, V6.19); differentiating a one-sided t-derivative in u needs the symmetry of mixed partials on the edges of the strips (V8.8). *Deviation (restated):* (i) is (13.3) for a surface, with $x = t$, $y = u$ in Lemma 9.2.

V8.13 (Theorem 13.1, the second variation formula; `Morse.secondVariation`, `Morse.second_variation`, `Morse.second_variation_eq`). For a path γ with velocity $V = \gamma'$ and fields W_1, W_2 along it, let

$$\Phi(W_1,W_2) = -\sum_{0<t<1}\Big\langle W_2(t), \Delta_t\frac{DW_1}{dt}\Big\rangle - \int_0^1 \Big\langle W_2, \frac{D^2W_1}{dt^2} + R(V,W_1)V\Big\rangle\, dt,$$

where $\Delta_t\frac{DW_1}{dt} = P_{\gamma(t)}(\Delta_t W_1)$ is the jump of $\frac{DW_1}{dt} = PW_1'$. Let γ be a geodesic (V8.10) and $\alpha(u_1,u_2,t)$ a 2-parameter variation of γ (parameter set $U \subseteq \mathbb{R}^2$, $\alpha(0,0,t) = \gamma(t)$ on $[0,1]$) with variation fields $W_i = \frac{\partial \alpha}{\partial u_i}(0,0,\cdot)$. Then $u_1 \mapsto \frac{\partial}{\partial u_2}E(\alpha(u_1,u_2,\cdot))\big|_{u_2=0}$ is differentiable at $u_1 = 0$, and its derivative, Milnor's $\frac{\partial^2 E}{\partial u_1 \partial u_2}(0,0)$, equals $2\Phi(W_1,W_2)$. Two 2-parameter variations of γ whose variation fields agree on $[0,1]$ have the same $\frac{\partial^2 E}{\partial u_1 \partial u_2}(0,0)$. *Book:* p. 75, Theorem 13.1: $\frac{1}{2}\frac{\partial^2 E}{\partial u_1 \partial u_2}(0,0) = \Phi(W_1,W_2)$ (the scan drops the $-\sum_t$ in front of the first term of the displayed statement; (13.2) on the same page has it), introduced on p. 74 to show that E_{**} is well defined; proof pp. 75–76, from 12.2, the vanishing of $\Delta_t V$ and $\frac{D}{dt}V$ along a geodesic, (13.2) and (13.3). *Hidden hypotheses:* (12.2′) at every $(u_1,0)$ (V8.9); differentiating the jump and integral terms in u_1 (V8.8, V8.12); the curvature identity on the strips (V8.12). *Deviation (restated):* the mixed second derivative is the iterated derivative $\partial/\partial u_1$ of $\partial E/\partial u_2$ along $u_2 = 0$, which is what Milnor's proof computes; its existence is part of the conclusion, and $E \circ \bar{\alpha}$ is not assumed to be of class C^2.

V8.14 (the Hessian E_{**}; `Morse.energyHessian`, `Morse.energyHessian_apply`, `Morse.IsVariation.partial _me` `Morse.IsVariation.partial _mem`, `Morse.hasDerivAt_energyHessian`, `Morse.secondVariation_congr`). For $\gamma \in \Omega(N;p,q)$, $E_{**}(W_1,W_2) = 2\Phi(W_1,W_2)$ (V8.13) is a bilinear form on the vector space $T\Omega_\gamma$ (V8.1; bilinearity is V8.15– V8.16), depending only on the values of the fields on $[0,1]$. The variation fields of a 2-parameter variation of γ lie in $T\Omega_\gamma$. If γ is a geodesic, then for every 2-parameter variation of γ with variation fields W_1, W_2, Milnor's $\frac{\partial^2 E}{\partial u_1 \partial u_2}(0,0)$ (as in V8.13) exists and equals $E_{**}(W_1,W_2)$; and by Lemma 11.A (V8.3) every pair $W_1, W_2 \in T\Omega_\gamma$ is the pair of variation fields of such a variation. So Milnor's definition is well defined and agrees with E_{**}. *Book:* p. 74: for a geodesic γ, $E_{**} : T\Omega_\gamma \times T\Omega_\gamma \to \mathbb{R}$, the Hessian of E at γ, is defined by analogy with the Hessian f_{**} of a function on a manifold: choose a 2-parameter variation with fields W_1, W_2 and put $E_{**}(W_1,W_2) = \frac{\partial^2 E(\bar{\alpha}(u_1,u_2))}{\partial u_1 \partial u_2}(0,0)$. *Hidden hypotheses:* that such variations exist

(Lemma 11.A); well-definedness is Theorem 13.1. *Deviation (restated):* Ω has no manifold structure, here as in the book; E_{**} is defined by Milnor's formula and identified with his definition by the statements above. It is defined for every $\gamma \in \Omega$; only at a geodesic is it the second derivative of E.

V8.15 (Proposition 13.A, the Hessian as an integral; `Morse.hessForm`, `Morse.secondVariation_eq_hessForm`, `Morse.energyHessian_eq_integral`). For $\gamma \in \Omega(N; p, q)$ and $W_1, W_2 \in T\Omega_\gamma$,

$$\tfrac{1}{2} E_{**}(W_1, W_2) = \int_0^1 \Big(\Big\langle \frac{DW_1}{dt}, \frac{DW_2}{dt} \Big\rangle - \langle R(V, W_1)V, W_2 \rangle \Big)\, dt.$$

Book: not in the book. *Hidden hypotheses:* none; γ need not be a geodesic. *Deviation (not in the book):* the exposition's Proposition 13.A. *Proof:* one integration by parts on each piece (Lemma 8.3, V6.10), the boundary terms telescoping into the jump terms, W_2 vanishing at the ends.

V8.16 (Corollary 13.4; `Morse.inner_curvR_comm`, `Morse.hessForm_comm`, `Morse.hessForm_add_right`, `Morse.hessForm_smul_right`, `Morse.energyHessian_isSymm`). For $\gamma \in \Omega(N; p, q)$, E_{**} is a symmetric bilinear form on $T\Omega_\gamma$. For tangent vectors v, w_1, w_2 at $x \in N$, $\langle R(v, w_1)v, w_2 \rangle = \langle R(v, w_2)v, w_1 \rangle$. *Book:* p. 76, Corollary 13.4: $E_{**}(W_1, W_2) = \frac{\partial^2 E}{\partial u_1 \partial u_2}(0, 0)$ is a well defined, symmetric, bilinear function of W_1, W_2; well defined and bilinear by the second variation formula, symmetric by $\frac{\partial^2 E}{\partial u_1 \partial u_2} = \frac{\partial^2 E}{\partial u_2 \partial u_1}$. *Hidden hypotheses:* that symmetry of mixed partials needs $E \circ \bar{\alpha}$ of class C^2 near $(0, 0)$, which the book does not address; bilinearity in W_1 needs the linearity of $W \mapsto \Delta_t \frac{DW}{dt}$ and $W \mapsto \frac{D^2 W}{dt^2}$ on piecewise smooth fields. *Deviation (restated):* well-definedness is V8.13–V8.14; symmetry and bilinearity come from the integral form (V8.15) and the symmetry of the curvature term (Lemma 9.3 (4), here from the Gauss equation, V6.18), not from second derivatives.

V8.17 (Remark 13.5; `Morse.remark_13_5`, `Morse.hasDerivAt_deriv_energy`). Let $\gamma \in \Omega(N; p, q)$ be a geodesic and α a 1-parameter variation of γ with variation field W. Then $u \mapsto \frac{d}{du} E(\alpha(u, \cdot))$ is differentiable at 0, and $\frac{d^2 E(\bar{\alpha}(u))}{du^2}(0) = E_{**}(W, W)$. *Book:* p. 76, Remark 13.5, through the 2-parameter variation $\beta(u_1, u_2) = \bar{\alpha}(u_1 + u_2)$. *Hidden hypotheses:* that β is a 2-parameter variation with both variation fields equal to W. *Proof:* Milnor's, V8.13 for β.

V8.18 (Lemma 13.6; `Morse.lemma_13_6`). If $\gamma \in \Omega(N; p, q)$ is a geodesic and is minimal in the sense of V8.6, then $E_{**}(W, W) \geq 0$ for every $W \in T\Omega_\gamma$. *Book:* p. 76, Lemma 13.6: if γ is a minimal geodesic from p to q, then E_{**} is positive semi-definite, hence its index λ is zero; proof: $E(\bar{\alpha}(u)) \geq E(\gamma) = E(\bar{\alpha}(0))$ forces $\frac{d^2 E(\bar{\alpha}(u))}{du^2}(0) \geq 0$. *Hidden hypotheses:* only minimality is used, not completeness; the energy inequality is V8.6; that a minimum forces a nonnegative second derivative needs an argument when only the second derivative at 0 is known to exist (if it were negative, 0 would also be a local maximum, so the function would be locally constant and its second derivative 0). *Deviation (narrower):* the index of E_{**} (the largest dimension of a subspace on which it is negative definite) is not defined formally; index 0 is the positive semi-definiteness proved. *Proof:* for $W \in T\Omega_\gamma$, Lemma 11.A gives a variation with field W; $u \mapsto E(\bar{\alpha}(u))$ has a minimum at 0 (V8.6) and second derivative $E_{**}(W, W)$ there (V8.17).

What is not verified in this chapter

- **Lemma 12.1 in full** (p. 70): that the minimum of E on Ω is d^2 and is attained exactly at the minimal geodesics. Only $E(\gamma) \leq E(\omega)$ at a minimal geodesic is formalized (V8.6); the

existence of a minimal geodesic on a complete N is V7.26, but the equality cases (V8.5 and Corollary 10.7) are not assembled into the converse.

- **Milnor's motivating constructions** (pp. 67–69, 74): the induced map F_* of a function on Ω, the Hessian f_{**} of a function on a finite-dimensional manifold and its independence of the chosen map, and the example of p. 69 (a minimum of F at which all the derivatives $dF(\bar{\alpha}(u))/du$ exist is a critical path). Critical paths are defined for E only (V8.4).
- **The index of E_{**}** (p. 76: "hence the index λ of E_{**} is zero"): not defined (V8.18); it belongs with the index theorem (§15).
- **A topology or manifold structure on Ω**: none, as in §§11–15 of the book; Palais's Hilbert manifold of H^1 paths is not formalized.
- **The exposition's Lemma 13.B** (rescaling the Hessian of a geodesic on $[0,\tau]$, used in Chapter 9): not formalized.
- **§§11–13 on an abstract Riemannian manifold**: as in Chapters 6–7, only for submanifolds of a Euclidean space with the induced metric; Nash's embedding theorem is quoted, not proved.

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10. Chapter 9 (Milnor §§14–15): partly verified

Milnor's §14 (pp. 77–82) defines Jacobi fields along a geodesic, conjugate points and their multiplicity, and the null space and nullity of E_{**}. It proves that the null space consists of the Jacobi fields vanishing at both ends (Theorem 14.1) and that the nullity is less than n (Remark 14.2), treats flat manifolds and the antipodal points of S^n (Examples 1–2), shows that the Jacobi fields are exactly the variation fields of variations through geodesics (Lemmas 14.3–14.4), and adds two remarks on short segments (Remarks 14.5–14.6). §15 (pp. 83–87) proves the Morse index theorem (Theorem 15.1, with Corollary 15.2): the index of E_{**} is the number of interior conjugate points, counted with multiplicity. The proof splits $T\Omega_\gamma$ into the broken Jacobi fields of a fine subdivision and the fields vanishing at its points (Lemma 15.3), reduces the index to the finite-dimensional first summand (Lemma 15.4), and follows the index $\lambda(\tau)$ of $\gamma|[0,\tau]$ through Assertions (1)–(4). Verified, for a submanifold N of a Euclidean space with the induced metric, in the embedded setting of Chapters 6–8: every numbered statement of §§14–15 except Examples 1 and 2 and part of Remark 14.6 (negative parameters, and the absence of cluster points along an entire geodesic), Remark 14.5 being restated for the neighbourhoods of Lemma 10.3; the definitions they need (Jacobi fields, conjugate parameters and their multiplicity, the null space and nullity of E_{**}, the index of a bilinear form, fine subdivisions, broken Jacobi fields, $\lambda(\tau)$); Milnor's parenthetical remark on p. 81 (V9.10); and the exposition's Lemma 14.A, Corollary 14.B and Lemmas 15.A–15.C. Every Riemannian manifold is such an N by Nash's embedding theorem, which is quoted, not proved. Proved more sharply than in the book: the solution theory of the Jacobi equation, Remark 14.6 and the index identity along any smooth curve in N, not only along geodesics (V9.2, V9.12, V9.24); Remark 14.2 also for the constant geodesic (V9.7); and the lower bound of Assertion (4) for every $\tau' > \tau$ (V9.23). The proofs depart from Milnor's in three places. **Lemma 14.4 is proved from initial data**, with the variation $\alpha(u,t) = \exp_{c(u)}(t\,w(u))$, so that neither minimal geodesics nor a short initial segment are needed (V9.9). **Positivity on T' comes from a Poincaré estimate**, which also gives the two properties of a fine subdivision and Assertion (2), without minimal geodesics (V9.16, V9.18, V9.20). **$\lambda(\tau)$ is the index of E_{**} on the fields supported in** $[0,\tau]$, so that no Hessian along $\gamma|[0,\tau]$ on its own parameter interval is needed (V9.17, V9.20). **Not verified:** Examples 1 and 2, the rest of Remark 14.6, and the exposition's Lemma 14.C and §5 (the list at the end). Lean: `Morse/Chapter8/Jacobi.lean` (704 lines; Theorem 14.1) and `Morse/Chapter9/` — `FormIndex.lean` (379), `JacobiODE.lean` (256),

`JacobiSystem.lean` (570), `JacobiSpace.lean` (507), `IndexForm.lean` (458), `BrokenJacobi.lean` (466), `IndexReduction.lean` (437), `IndexContinuity.lean` (450), `IndexAssertions.lean` (419), `IndexTheorem.lean` (286), `JacobiVariation.lean` (386), `GeodesicVariation.lean` (674).

The embedded setting. As in Chapter 8: V is a finite-dimensional real inner product space and $N \subseteq V$ a slice submanifold modelled on Z with complement W, with the induced metric; every tangent space T_xN, $x \in N$, has dimension $n = \dim Z$ (`Morse.finrank_tanAt`); P_x is the orthogonal projection onto T_xN ($P_x = 0$ for $x \notin N$), $D/dt = P \circ d/dt$, and R is Milnor's curvature tensor, with his sign (V6.16–V6.17, V8.12). $\Omega = \Omega(N; p, q)$, $T\Omega_\gamma$, the geodesics of Ω, the form Φ of the second variation formula and $E_{**} = 2\Phi$ are those of V8.1, V8.10, V8.13 and V8.14. A *smooth curve in N on $[0,1]$* (`Morse.IsSmoothCurveIn`) is a map $\gamma : \mathbb{R} \to V$ that is C^∞ on $[0,1]$, with one-sided derivatives at the ends, and has values in N there; every geodesic of Ω is one (`Morse.IsUnbrokenGeodesic.isSmoothCurveIn`), and every such curve lies in $\Omega(N; \gamma(0), \gamma(1))$ (`Morse.IsSmoothCurveIn.mem_pathSpace`). Along it the velocity $V(t)$, the derivative $P'(t)$ of $t \mapsto P(t) = P_{\gamma(t)}$, and $\frac{DX}{dt}(t) = P(t)X'(t)$ are computed within $[0,1]$, so one-sided at the ends (`Morse.velAlong`, `Morse.dpAlong`, `Morse.covAlong`). Curves and fields are maps on $\mathbb{R}$ of which only the values on $[0,1]$ matter; where a space of fields is to be finite-dimensional, its elements are normalized to be constant off $[0,1]$. N need not be closed, compact or connected.

How the index, the nullity and $\lambda(\tau)$ are formalized. $T\Omega_\gamma$ is infinite-dimensional, and its elements (V8.1) are maps on $\mathbb{R}$ whose values off $[0,1]$ play no role. Hence three conventions. The index of a bilinear form is a supremum in $\{0, 1, 2, \dots\} \cup \{\infty\}$ (V9.13); that it is finite for E_{**} is part of Theorem 15.1, as in the book. The null space of E_{**} is read on $[0,1]$ (V9.6), since the literal kernel of the formal E_{**} contains every field vanishing on $[0,1]$. And Milnor's $\lambda(\tau)$, the index of the Hessian $(E_0^\tau)_{**}$ of $\gamma_\tau = \gamma|[0,\tau]$, is the index of E_{**} (of γ on $[0,1]$) restricted to the fields of $T\Omega_\gamma$ vanishing outside $[0,\tau]$ (V9.17, V9.20). These are Milnor's fields along γ_τ vanishing at 0 and τ, extended by zero as in his proof of Assertion (1), and on them $\frac{1}{2}E_{**}(W_1, W_2) = \int_0^\tau (\langle \frac{DW_1}{dt}, \frac{DW_2}{dt} \rangle - \langle R(V, W_1)V, W_2 \rangle)\, dt$ (V8.15), the integral that represents $\frac{1}{2}(E_0^\tau)_{**}$ (Proposition 13.A on $[0,\tau]$). The Hessian on $[0,\tau]$ itself is not formalized, and the proof does not need it.

V9.1 (linear equations: global solutions, the exposition's Lemma 14.A; `Morse.exists_linearODE`, `Morse.linSol`, `Morse.linSol_add`, `Morse.linSol_smul`, `Morse.eqOn_Icc_linSol`, `Morse.contDiffOn_Icc_of_h`
Let E be a complete real normed space and $t \mapsto A(t)$ a continuous map from $\mathbb{R}$ to the continuous linear maps $E \to E$. (i) For every $t_0 \in \mathbb{R}$ and $x_0 \in E$ there is a solution of $x' = A(t)x$ on all of $\mathbb{R}$ with $x(t_0) = x_0$; the solution with value ξ at s depends linearly on ξ. (ii) If $a < b$ and f is continuous on $[a,b]$ and solves the equation on (a,b), then for every $s \in [a,b]$, end points included, f agrees on $[a,b]$ with the solution whose value at s is $f(s)$. (iii) If A is C^∞ on $[a,b]$ (one-sided derivatives at the ends) and $f'(t) = A(t)f(t)$ at every $t \in [a,b]$, then f is C^∞ on $[a,b]$. *Book:* p. 77: the Jacobi equation, a linear second-order equation, has $2n$ linearly independent solutions, each "defined throughout γ", all of class C^∞, and a Jacobi field is determined by $J(0)$ and $\frac{DJ}{dt}(0)$; no proof. *Hidden hypotheses:* a global existence theorem for linear equations; the local theorem gives solutions on intervals of length about $1/\sup\|A\|$ only. *Deviation (stronger):* any continuous coefficients on any complete space, extending V6.7 (ii), where they had to be skew-adjoint. *Proof:* the equation is truncated radially outside a ball and solved on compact intervals by Picard–Lindelöf; Grönwall's inequality, forward and backward in time, bounds the solution so that the truncation is never active (the exposition uses Picard iteration on the whole interval); (iii) is a bootstrap.

V9.2 (Jacobi fields and the Jacobi system; `Morse.jacobiExpr`, `Morse.IsJacobiField`,

`Morse.curvAlong`, `Morse.jacS`, `Morse.jacSys`, `Morse.IsSmoothCurveIn.contDiffOn_jacSys`, `Morse.IsSmoothCurveIn.continuous_jacA`, `Morse.jacSol`, `Morse.jacSol_mem_tanAt`, `Morse.isJacobiField_j` `Morse.IsJacobiField.eqOn_jacSol`, `Morse.IsJacobiField.eqOn_zero`). For maps γ and J the Jacobi operator is $\frac{D^2J}{dt^2} + R(V,J)V$, $V = \gamma'$, with D/dt of V8.7. J is a *Jacobi field along* γ if it is C^∞ on $[0,1]$ (one-sided derivatives at the ends), tangent along γ ($J(t) \in T_{\gamma(t)}N$ for $t \in [0,1]$), and

$$\frac{D^2J}{dt^2} + R(V,J)V = 0 \quad \text{on } (0,1).$$

Let γ be a smooth curve in N on $[0,1]$, $S = P'P - PP'$ and $K(t)v = R(V,P(t)v)V$. The Jacobi system is the first-order linear system $J' = SJ + Y$, $Y' = SY - KJ$ for (J,Y) in $V \times V$; its coefficients are smooth on $[0,1]$, and extended to $\mathbb{R}$ as constants they are continuous, so its solution with any value ζ at any time s exists on $\mathbb{R}$ (V9.1). If $s \in [0,1]$ and $\zeta = (x,y)$ with $x, y \in T_{\gamma(s)}N$, both components of this solution stay tangent on $[0,1]$, the first is a Jacobi field and the second is its covariant derivative there. Conversely, for every Jacobi field J and every $s \in [0,1]$, $(J, \frac{DJ}{dt})$ agrees on $[0,1]$ with the solution whose value at s is $(J(s), \frac{DJ}{dt}(s))$; in particular a Jacobi field with $J(s) = 0$ and $\frac{DJ}{dt}(s) = 0$ vanishes on $[0,1]$. *Book:* p. 77: a vector field J along a geodesic γ is a Jacobi field if it satisfies the Jacobi equation; in orthonormal parallel fields $P_1, \dots, P_n$ along γ, $J = \sum f^i P_i$, it becomes a linear system with coefficients $\langle R(V,P_j)V, P_i\rangle$; a Jacobi field is completely determined by its initial conditions $J(0)$, $\frac{DJ}{dt}(0)$. *Hidden hypotheses:* the global solution theory (V9.1); parallel frames along γ (Lemma 8.2), which V6.8 provides only for curves defined on all of $\mathbb{R}$. *Deviation (restated):* no frame: the unknown is $(J, \frac{DJ}{dt})$ in $V \times V$, and the term S (the device of V6.8) keeps solutions with tangent initial values tangent, their normal parts $(1-P)J$, $(1-P)Y$ solving a homogeneous linear system and vanishing at s. *Deviation (stronger):* along any smooth curve in N, not only along a geodesic, and with initial values at any $s \in [0,1]$.

V9.3 (the space of Jacobi fields, the exposition's Corollary 14.B; `Morse.jacobiSpace`, `Morse.IsJacobiField.comp_projIcc_mem`, `Morse.jacobiSpace_eq_zero`, `Morse.jacEval`, `Morse.jacEval_bijective`, `Morse.jacEquiv`, `Morse.finrank_jacobiSpace`, `Morse.exists_jacobi`**).** Let γ be a smooth curve in N on $[0,1]$. $\mathcal{J}(\gamma)$ is the vector space of the Jacobi fields along γ that are constant off $[0,1]$ ($J(t) = J(0)$ for $t < 0$ and $J(t) = J(1)$ for $t > 1$); every Jacobi field agrees on $[0,1]$ with one of them, and an element of $\mathcal{J}(\gamma)$ vanishing on $[0,1]$ is 0. For every $s \in [0,1]$, $J \mapsto (J(s), \frac{DJ}{dt}(s))$ is a linear isomorphism $\mathcal{J}(\gamma) \to T_{\gamma(s)}N \times T_{\gamma(s)}N$: every pair of tangent vectors at $\gamma(s)$ is the value and covariant derivative at s of exactly one field of $\mathcal{J}(\gamma)$, and $\dim \mathcal{J}(\gamma) = 2n$. *Book:* p. 77: "$2n$ linearly independent solutions", determined by $J(0)$ and $\frac{DJ}{dt}(0)$. *Hidden hypotheses:* that every pair of initial values occurs (global existence, V9.1). *Deviation (stronger):* the isomorphism at every $s \in [0,1]$, along any smooth curve in N.

V9.4 (conjugate parameters and their multiplicity; `Morse.IsConjugate`, `Morse.conjSpace`, `Morse.conjMultiplicity`, `Morse.isConjugate_iff`**).** Let γ be a smooth curve in N on $[0,1]$. Parameters a and b are *conjugate along* γ if $a, b \in [0,1]$, $a \neq b$, and some Jacobi field along γ vanishes at a and at b without vanishing identically on $[0,1]$. Their *multiplicity* is the dimension of the space of fields of $\mathcal{J}(\gamma)$ vanishing at a and b (finite, by V9.3); for $a \neq b$ in $[0,1]$ they are conjugate iff the multiplicity is positive. Below, $\nu(t)$ is the multiplicity of 0 and t, which is 0 for $t \in (0,1]$ not conjugate to 0. *Book:* p. 77: for $a \neq b$, $p = \gamma(a)$ and $q = \gamma(b)$ are conjugate along γ if a non-zero Jacobi field vanishes for $t = a$ and $t = b$, and the multiplicity is the dimension of the vector space of all such fields; footnote: if γ has self-intersections, one should say that the parameter values a and b are conjugate. *Hidden hypotheses:* none. *Deviation (restated):* conjugacy is a relation between parameter values, as the footnote advises; in §15 "the conjugate points $\gamma(t)$"

are read as the conjugate parameters t.

V9.5 (Theorem 14.1: the null vectors of E_{**} are the Jacobi fields; `Morse.IsJacobiField.mem_pathTangent`, `Morse.IsJacobiField.secondVariation_eq_zero`, `Morse.jacobiExpr_eq_zero_of_null`, `Morse.jump_eq_prj_jump`, `Morse.jump_eq_zero_of_null`, `Morse.deriv_deriv_eq_of_jacobiExpr`, `Morse.smooth_near_of_null`, `Morse.theorem_14_1`). Let $\gamma \in \Omega(N; p, q)$ be a geodesic (V8.10) and $W_1 \in T\Omega_\gamma$. Then $E_{**}(W_1, W_2) = 0$ for every $W_2 \in T\Omega_\gamma$ iff W_1 is a Jacobi field along γ; in particular such a W_1 is C^∞ on all of $[0, 1]$. The steps: a Jacobi field vanishing at 0 and 1 lies in $T\Omega_\gamma$, and for a Jacobi field J (along any curve) $\Phi(J, Y) = 0$ for every field Y (V8.13); for a null vector W_1, the Jacobi operator of W_1 vanishes off its break points, and at every $t \in (0, 1)$ the jump of W_1' is tangent and vanishes; for a smooth tangent field the Jacobi equation gives the second-order equation $W'' = -P(P'W') - R(V, W)V + (1 - P)(P''W + 2P'W')$ in V; and W_1 is C^∞ near every point of $(0, 1)$. *Book:* p. 78, Theorem 14.1 (first sentence); proof pp. 78–79: a Jacobi field vanishing at the ends lies in the null space by the second variation formula; conversely, testing against $W_2 = f(t)(\frac{D^2W_1}{dt^2} + R(V, W_1)V)$, f vanishing exactly at the break points, makes W_1 a Jacobi field on each piece, testing against a field W_2' with $W_2'(t_i) = \Delta_{t_i} \frac{DW_1}{dt}$ shows that $\frac{DW_1}{dt}$ has no jumps, and the pieces "fit together" since a solution is determined by $W_1(t_i)$ and $\frac{DW_1}{dt}(t_i)$. *Hidden hypotheses:* the field W_2' with prescribed values at the break points has to be constructed; fitting the pieces together needs Jacobi fields along all of γ with given initial values (V9.2), stated in the book without proof. *Deviation (restated):* in V, the absence of jumps of $\frac{DW_1}{dt} = PW_1'$ makes W_1 of class C^1 only because the normal parts of the two one-sided derivatives agree (both are $\mathrm{II}(V, W_1)$). *Proof:* the test fields are bump functions, supported inside one piece or centred at one break point, times $\frac{D^2W_1}{dt^2} + R(V, W_1)V$ or times the tangent projection of the jump; the pieces are joined by a local argument instead of the uniqueness of Jacobi fields: the second-order equation above holds on both sides of a former break point x, with a right side continuous across x, so W_1' is differentiable at x, and a solution of a smooth second-order equation is C^∞ (a bootstrap).

V9.6 (Theorem 14.1: the null space and the nullity; `Morse.restrictIcc`, `Morse.hessNullSpace`, `Morse.hessNullity`, `Morse.hessNullSpace_eq`, `Morse.theorem_14_1_nullity`, `Morse.hessNullity_pos_iff`). For $\gamma \in \Omega(N; p, q)$, the *null space* of E_{**} is the space of the restrictions to $[0, 1]$, extended as constants, of the $W_1 \in T\Omega_\gamma$ with $E_{**}(W_1, W_2) = 0$ for all $W_2 \in T\Omega_\gamma$, and the *nullity* ν is its dimension. If γ is a geodesic, the null space is the space of fields of $\mathcal{J}(\gamma)$ vanishing at 0 and 1; so ν is the multiplicity of 0 and 1 (V9.4), in particular finite, and $\nu > 0$ (E_{**} is degenerate) iff 0 and 1 are conjugate along γ. *Book:* pp. 77–78: the null space of E_{**} is the space of the $W_1 \in T\Omega_\gamma$ with $E_{**}(W_1, W_2) = 0$ for all W_2, the nullity ν is its dimension, and E_{**} is degenerate if $\nu > 0$; p. 78, Theorem 14.1: hence E_{**} is degenerate iff p and q are conjugate along γ, and the nullity is their multiplicity; p. 79: the nullity is always finite. *Hidden hypotheses:* finiteness needs the identification with a subspace of the $2n$-dimensional $\mathcal{J}(\gamma)$ (V9.3). *Deviation (restated):* the fields of $T\Omega_\gamma$ (V8.1) are maps on $\mathbb{R}$ whose values off $[0, 1]$ play no role, so the literal kernel of the formal E_{**} contains every field vanishing on $[0, 1]$ and is infinite-dimensional; the null space is therefore taken modulo the values off $[0, 1]$. "p and q conjugate" reads "0 and 1 conjugate" (V9.4).

V9.7 (Remark 14.2; `Morse.finrank_jacobiSpace_zero`, `Morse.isJacobiField_smul_vel`, `Morse.eq_const_of_velAlong_one_eq_zero`, `Morse.isJacobiField_const_curve`, `Morse.remark_14_2`). Along a smooth curve γ in N on $[0, 1]$, the fields of $\mathcal{J}(\gamma)$ vanishing at 0 form a space of dimension n. If γ is moreover a geodesic on $(0, 1)$, then $t \mapsto tV(t)$ is a Jacobi field, and γ is constant on $[0, 1]$ if $V(1) = 0$; along a constant curve, $t \mapsto ty$ is a Jacobi field for every tangent vector y. Hence, for

a geodesic $\gamma \in \Omega(N; p, q)$ and $n \geq 1$, the nullity of E_{**} (V9.6) satisfies $\nu < n$. *Book:* p. 79, Remark 14.2: $0 \leq \nu < n$; the Jacobi fields vanishing at $t = 0$ form a space of dimension precisely n, and $J_t = tV_t$ is a Jacobi field vanishing at 0 but not at 1. *Hidden hypotheses:* $J_1 = V_1 \neq 0$ requires γ to be non-constant, and $\nu < n$ requires $n \geq 1$ (for $n = 0$, $\nu = 0 = n$). *Deviation (stronger):* the constant geodesic ($p = q$) is included, with ty, $y \neq 0$, in place of tV.

V9.8 (Lemma 14.3; `Morse.IsJacobiField.congr_curve`, `Morse.lemma_14_3`). Let $\varepsilon > 0$ and let $\alpha(u, t)$ be C^∞ on $(-\varepsilon, \varepsilon) \times [0, 1]$ (one-sided derivatives along the edges $t = 0, 1$), with values in N there, such that each $t \mapsto \alpha(u, t)$, $|u| < \varepsilon$, is a geodesic on $(0, 1)$ (V7.1). Then $W(t) = \frac{\partial \alpha}{\partial u}(0, t)$ is a Jacobi field along $\alpha(0, \cdot)$. Being a Jacobi field along γ depends only on γ on $[0, 1]$. *Book:* p. 79: a variation of γ through geodesics, not necessarily keeping the end points fixed, is a C^∞ map $\alpha : (-\varepsilon, \varepsilon) \times [0, 1] \to M$ with $\alpha(0, t) = \gamma(t)$ and each $\bar{\alpha}(u)$ a geodesic; p. 80, Lemma 14.3: its variation field is a Jacobi field; proof: apply $\frac{D}{\partial u}$ to $\frac{D}{\partial t}\frac{\partial \alpha}{\partial t} = 0$ and use Lemmas 9.2 and 8.7. *Hidden hypotheses:* Lemmas 9.2 and 8.7 are applied to a map defined on a strip and smooth up to its edges (V8.12). *Proof:* Milnor's, with the identity (13.3) of V8.12 on the open strip $(-\varepsilon, \varepsilon) \times (0, 1)$; the smoothness of W up to $t = 0, 1$ comes from that of α on the closed strip.

V9.9 (Lemma 14.4; `Morse.exists_smooth_ray`, `Morse.exists_initial_ray`, `Morse.RaysGood`, `Morse.raysGood_of_isGeodesicOn`, `Morse.lemma_14_4`). (i) For $x_1 \in N$ and $R > 0$ there are an open set $U \ni x_1$ of V, an $h > 0$ and a map $\Phi(y, w, t)$, C^∞ on $U \times \{\|w\| < R\} \times (-h, h)$, such that for $y \in N \cap U$ and $w \in T_y N$ with $\|w\| < R$, $t \mapsto \exp_y(tw)$ is a geodesic on $(-h, h)$, equal to $\Phi(y, w, t)$ there. (ii) A smooth curve γ in N on $[0, 1]$ that is a geodesic on $(0, 1)$ agrees on $[0, 1]$ with $t \mapsto \exp_{\gamma(0)}(tV(0))$, which is a geodesic on an open interval containing $[0, 1]$; this is the extension of a geodesic segment past its end points that V7.7 leaves unformalized. (iii) Let c, w be C^∞ on an open set $U_0 \ni 0$ of $\mathbb{R}$, with $c(u) \in N$ and $w(u) \in T_{c(u)}N$. If $t \mapsto \exp_{c(0)}(tw(0))$ is a geodesic on $(-\delta, B + \delta)$, $B \geq 0$, $\delta > 0$, there are $\varepsilon, \eta > 0$ such that every $t \mapsto \exp_{c(u)}(tw(u))$, $|u| < \varepsilon$, is a geodesic on $(-\eta, B+\eta)$ and $(u, t) \mapsto \exp_{c(u)}(tw(u))$ is C^∞ on $(-\varepsilon, \varepsilon) \times (-\eta, B+\eta)$. (iv) **Lemma 14.4:** let γ be a smooth curve in N on $[0, 1]$ that is a geodesic on $(0, 1)$, and J a Jacobi field along γ. There are $\varepsilon > 0$ and a map α satisfying the hypotheses of Lemma 14.3 (V9.8) on $(-\varepsilon, \varepsilon) \times [0, 1]$, with $\alpha(0, t) = \gamma(t)$ and $\frac{\partial \alpha}{\partial u}(0, t) = J(t)$ for $t \in [0, 1]$. *Book:* p. 81, Lemma 14.4: every Jacobi field along a geodesic $\gamma : [0, 1] \to M$ may be obtained by a variation of γ through geodesics. Proof: take a neighbourhood U of $\gamma(0)$ in which any two points are joined by a unique minimal geodesic depending differentiably on the end points, and δ with $\gamma(t) \in U$ for $0 \leq t \leq \delta$; minimal geodesics between the points of curves $a(u)$, $b(u)$ give Jacobi fields along $\gamma|[0, \delta]$ with arbitrary values at 0 and δ, so $J \mapsto (J(0), J(\delta))$ is onto, hence an isomorphism (both spaces have dimension $2n$); "using the compactness of $[0, 1]$", the geodesics extend over $[0, 1]$ for small u. *Hidden hypotheses:* the construction returns $\bar{\alpha}(0) = \gamma|[0, \delta]$ only if $\gamma|[0, \delta]$ is *the* minimal geodesic from $\gamma(0)$ to $\gamma(\delta)$, which $\gamma([0, \delta]) \subseteq U$ does not say; for Milnor's U it is true, by an argument the book omits (the exposition supplies it after its Lemma 14.4, and ensures it directly by a bound on the length of $\gamma|[0, \delta]$); the compactness step is the smooth dependence of geodesics on their initial data over a compact interval, (iii), with the openness of the domain of exp, neither among the statements of §10; and that the extended variation still has J as variation field needs the uniqueness of Jacobi fields (V9.2). *Proof:* departs from Milnor's. The variation is built from initial data: $\alpha(u, t) = \exp_{c(u)}(t\, w(u))$ with $c(u) = \exp_{\gamma(0)}(uJ(0))$ and $w(u) = P_{c(u)}(V(0) + u\frac{DJ}{dt}(0))$. It is defined and smooth on $(-\varepsilon, \varepsilon) \times [0, 1]$ by (i)–(iii); its variation field is a Jacobi field (V9.8) with value $J(0)$ and covariant derivative $\frac{DJ}{dt}(0)$ at $t = 0$ (by the symmetry of the mixed partials and $PP'P = 0$), hence equals J (V9.2). No minimal geodesic, restriction to $[0, \delta]$ or dimension

count is used: the formal statement has no minimality hypothesis, and the minimality of $\gamma|[0,\delta]$ is neither assumed nor proved.

V9.10 (Jacobi fields at two non-conjugate parameters; `Morse.jacEvalEnds`, `Morse.jacEvalEnds_bijective_of_`
Let γ be a smooth curve in N on $[0,1]$, and $a \neq b$ in $[0,1]$ parameters that are not conjugate along γ (V9.4). Then $J \mapsto (J(a), J(b))$ is a linear isomorphism $\mathcal{J}(\gamma) \to T_{\gamma(a)}N \times T_{\gamma(b)}N$: a Jacobi field is determined by its values at a and b, and these values can be prescribed. *Book:* p. 81, in the proof of Lemma 14.4: more generally a Jacobi field is determined by its values at any two non-conjugate points (in parentheses, without proof). *Hidden hypotheses:* none; injectivity is the definition of non-conjugacy. *Deviation (stronger):* the values can be prescribed, by the dimension count (V9.3).

V9.11 (Remark 14.5; `Morse.IsNormalRadius.isSmoothCurveIn_ray`, `Morse.remark_14_5_of_isNormalRadius`, `Morse.remark_14_5_bijective`, `Morse.remark_14_5`). (i) Let ε be a normal radius at $q \in N$ (V7.9), $v \in T_qN$ and $\|v\| < \varepsilon$. Then $\sigma(t) = \exp_q(tv)$ is a smooth curve in N on $[0,1]$; a Jacobi field along σ vanishing at $t = 0$ and $t = 1$ vanishes on $[0,1]$; and $J \mapsto (J(0), J(1))$ is a linear isomorphism $\mathcal{J}(\sigma) \to T_qN \times T_{\sigma(1)}N$. (ii) Every $p \in N$ has an open neighbourhood O in V, with an $\varepsilon > 0$, such that for all $q_1, q_2 \in N \cap O$ there is $v \in T_{q_1}N$ with $\|v\| < \varepsilon$ and $\exp_{q_1}(v) = q_2$, ε is a normal radius at q_1, and a Jacobi field along $t \mapsto \exp_{q_1}(tv)$ vanishing at both ends vanishes on $[0,1]$. *Book:* pp. 81–82, Remark 14.5: the argument of Lemma 14.4 shows that in any such neighbourhood U the Jacobi fields along a geodesic segment in U are uniquely determined by their values at the end points. *Hidden hypotheses:* that argument applies to a segment that is the minimal geodesic between its end points (V9.9). *Deviation (restated):* Milnor's U is replaced by the neighbourhoods $N \cap O$ of Lemma 10.3 (V7.10), and "a geodesic segment in U" by $t \mapsto \exp_{q_1}(tv)$, $\|v\| < \varepsilon$, the geodesic of V7.15 (ii), which is shortest among all paths from q_1 to q_2. *Deviation (stronger):* the end values can be prescribed. *Proof:* injectivity from the local inverse of $\exp_q$: the Jacobi field with $J(0) = 0$ and $\frac{DJ}{dt}(0) = w$ is the variation field $t \mapsto d(\exp_q)_{tv}(tw)$ of the variation $\exp_q(t(v + uw))$ through geodesics (V9.8 and V9.2), and $d(\exp_q)_v$ is injective on T_qN because $\exp_q$ has a smooth inverse near $\exp_q(v)$; surjectivity by the dimension count (V9.3). Milnor goes the other way: surjectivity from the end-point construction of Lemma 14.4, then the dimension count.

V9.12 (Remark 14.6; `Morse.remark_14_6`). For every smooth curve γ in N on $[0,1]$ there is $\delta > 0$ such that no $t \in (0,\delta)$ is conjugate to 0 along γ. *Book:* p. 82, Remark 14.6: the proof also shows that there is a neighbourhood $(-\delta,\delta)$ of 0 such that $\gamma(t)$ is not conjugate to $\gamma(0)$ for $t \in (-\delta,\delta)$; and (by 15.2) the conjugate points of $\gamma(0)$ along the entire geodesic have no cluster points. *Hidden hypotheses:* in Milnor's argument, as in V9.9, that $\gamma|[0,t]$ is the minimal geodesic between its end points. *Deviation (narrower):* only $t > 0$, γ being given on $[0,1]$; the last sentence is not formalized. *Deviation (stronger):* along any smooth curve in N. *Proof:* $\delta = \min(h_0, 1)$, since no two parameters at distance at most h_0 are conjugate (V9.16 (ii)); Milnor's neighbourhood U is not used.

V9.13 (the index and the nullity of a bilinear form; `Morse.IsNegDefOn`, `Morse.IsPosDefOn`, `Morse.formIndex`, `Morse.le_formIndex`, `Morse.exists_isNegDefOn_of_formIndex_eq`, `Morse.formIndex_le_of_map`, `Morse.formIndex_eq_of_equiv`, `Morse.formIndex_eq_sigNeg`). Let B be a bilinear form on a real vector space X. B is negative (positive) definite on a subspace S if $B(x,x) < 0$ (> 0) for every $x \neq 0$ in S. The *index* of B is the supremum, in $\{0,1,2,\dots\} \cup \{\infty\}$, of the dimensions of the finite-dimensional subspaces on which B is negative definite; a finite index is the dimension of such a subspace. A linear map L with $B'(Lx,Ly) = B(x,y)$ does not decrease the index (it is injective on every negative definite subspace), so a linear isomorphism

carrying B to B' preserves it; on a finite-dimensional space the index is Mathlib's negative index of inertia (`sigNeg`) of $x \mapsto B(x,x)$, as in V1.3. The *nullity* of B is the dimension of its null space $\{x : B(x,y) = 0 \text{ for all } y\}$; for E_{**} the null space is read on $[0,1]$ (V9.6). *Book:* p. 83: the index λ of E_{**} is the maximum dimension of a subspace of $T\Omega_\gamma$ on which E_{**} is negative definite; p. 5: the index and the nullity of a bilinear functional on a vector space. *Hidden hypotheses:* that the maximum exists, i.e. is finite; for E_{**} this is part of Theorem 15.1 (V9.19, V9.24). *Deviation (restated):* a supremum over finite-dimensional subspaces, possibly ∞; a negative definite subspace of infinite dimension contains finite-dimensional ones of every dimension, and so also gives ∞.

V9.14 (linear algebra, the exposition's Lemmas 15.A and 15.B; `Morse.finrank_add_finrank_le`, `Morse.exists_isPosDefOn_add`, `Morse.eventually_isNegDefOn`, `Morse.eventually_isPosDefOn`, `Morse.formIndex_eq_restrict_of_isCompl`, `Morse.ker_eq_map_ker_restrict`, `Morse.le_formIndex_of_comp`
Let B be a bilinear form on a real vector space X. (1) If $\dim X = m < \infty$, a negative definite and a positive definite subspace have dimensions adding up to at most m. (2) (Sylvester) If $\dim X = m < \infty$ and B is symmetric, with index λ and nullity ν, then B is positive definite on a subspace of dimension $m - \lambda - \nu$. (3) If (B_τ) is a family of bilinear forms on X indexed by a topological space, $S \subseteq X$ is a finite-dimensional subspace, $\tau \mapsto B_\tau(x,y)$ is continuous at τ_0 for all $x, y \in S$, and B_{τ_0} is negative (positive) definite on S, then so is B_τ for all τ near τ_0. (4) (Lemma 15.B) If $X = A \oplus C$, $B(a,c) = B(c,a) = 0$ for all $a \in A$, $c \in C$, and B is positive definite on C, then B and its restriction to A have the same index, and the null space of B is that of its restriction to A. (5) (Milnor's compound matrix) If B is symmetric, $U : \mathbb{R}^\lambda \to X$ is linear with $B(Ua, Ua) < 0$ for $a \neq 0$, and $J, Y : \mathbb{R}^\nu \to X$ are linear with $B(Jb, Ua) = 0$, $B(Jb, Jb') = 0$ and $B(Jb, Yb') = 2\langle b, b'\rangle$ for all a, b, b', then the index of B is at least $\lambda + \nu$. *Book:* used without proof on pp. 85–87: Lemma 15.4 ("the proof is straightforward"), the persistence of definiteness for $H_{\tau'}$ near H_τ in Assertions (3) and (4), a positive definite subspace of dimension $ni - \lambda(\tau) - \nu$, and the compound matrix, "certainly negative definite" for small c. *Hidden hypotheses:* in (5), that a negative definite Gram matrix makes the $\lambda + \nu$ fields linearly independent. *Deviation (not in the book):* stated and proved as lemmas. *Proof:* (3) by the compactness of the unit sphere of S; (5) by (3): for small $c \neq 0$, B is negative definite on the image of $(a,b) \mapsto Ua + c^{-1}Jb - cYb$, whose Gram matrix is a perturbation of the block-diagonal one with blocks $(B(Ua_i, Ua_j))$ and $-4I$.

V9.15 (the second variation formula for piecewise Jacobi fields; `Morse.jacobiExpr_congr`, `Morse.isJacobiField_zero`, `Morse.energyHessian_eq_jumps`). Let $\gamma \in \Omega(N; p, q)$, $X, Y \in T\Omega_\gamma$, and B a finite set such that near every $t \in (0,1) \setminus B$ the field X agrees with some Jacobi field along γ (the zero field is one). Then

$$E_{**}(X,Y) = -2 \sum_{b \in B \cap (0,1)} \left\langle Y(b), \Delta_b \frac{DX}{dt} \right\rangle,$$

$\Delta_b \frac{DX}{dt} = P_{\gamma(b)}(\Delta_b X')$ being the jump of $\frac{DX}{dt}$ (V8.13). γ need not be a geodesic. *Book:* p. 84, in the proof of Lemma 15.3: for a broken Jacobi field W_1 and $W_2 \in T'$ the second variation formula reduces to its jump terms; p. 87: by the second variation formula one sees "easily" the values of $E_{**}(J_h, \cdot)$. *Hidden hypotheses:* none. *Proof:* the formula of V8.13, whose integrand vanishes off the finite set $B \cup \{1\}$.

V9.16 (positivity near zeros, and fine subdivisions; `Morse.jacBound`, `Morse.fineMesh`, `Morse.fineMesh_pos`, `Morse.integrand_ge`, `Morse.integral_norm_sq_le`, `Morse.hessForm_ge`, `Morse.eqOn_zero_of_hessForm`, `Morse.eq_zero_of_short`, `Morse.jacEvalEnds_bijective`, `Morse.IsFine`, `Morse.exists_isFine`). Let γ be a smooth curve in N on $[0,1]$, $C \geq 0$ a bound

for $\|P'(t)\|^2 + \|K(t)\|$ on $[0,1]$ (K as in V9.2), and $h_0 = 1/(C+1) > 0$. (i) *A Poincaré estimate.* If $X \in T\Omega_\gamma$ and every $t \in [0,1]$ has a zero $s \in [0,t]$ of X with $t - s \le h_0$, then

$$\int_0^1 \|X\|^2\,dt \le \int_0^1 \left(\left\| \frac{DX}{dt} \right\|^2 - \langle R(V,X)V, X\rangle \right) dt = \tfrac{1}{2} E_{**}(X,X)$$

(the equality is V8.15); so $E_{**}(X,X) \le 0$ forces $X = 0$ on $[0,1]$. (ii) If $0 \le a < b \le 1$ and $b - a \le h_0$, a field of $\mathcal{J}(\gamma)$ vanishing at a and b is 0, and $J \mapsto (J(a), J(b))$ is a linear isomorphism $\mathcal{J}(\gamma) \to T_{\gamma(a)}N \times T_{\gamma(b)}N$; in particular no two parameters at distance at most h_0 are conjugate. (iii) For $\tau > 0$, a finite set $B \subseteq (0,\tau)$ of division points is *fine for* $[0,\tau]$ if consecutive points of $\{0\} \cup B \cup \{\tau\}$ are at most h_0 apart; for every $\tau > 0$ there is one whose last point is within $h_0/2$ of τ. *Book:* p. 83: each $\gamma(t)$ lies in an open set U in which any two points are joined by a unique minimal geodesic depending differentiably on the end points (§10), and the subdivision is chosen "sufficiently fine" that each $\gamma|[t_{i-1}, t_i]$ lies in such a U and is minimal. Milnor uses two consequences: along each segment the Jacobi fields are determined by their end values, which can be prescribed (Remark 14.5); and each segment has smaller energy than any other path between its end points (p. 84). *Hidden hypotheses:* that such subdivisions exist (a Lebesgue number of a finite cover, and a length bound for minimality, Theorem 10.4). *Deviation (restated):* "sufficiently fine" becomes the explicit mesh h_0, which depends only on bounds for P' and the curvature along γ; γ need not be a geodesic, and neither exp nor minimality enters. *Proof:* on a piece $[s,t]$ starting at a zero, $\|X(t)\|^2 \le L_s^t(X)^2 \le (t-s)E_s^t(X)$ (V7.11, V8.5), whence $\int_0^1 \|X\|^2 \le h_0 \int_0^1 \|X'\|^2$; and the integrand is at least $\|X'\|^2 - C\|X\|^2$, since $\|\frac{DX}{dt}\|^2 \ge \|X'\|^2 - \|P'\|^2\|X\|^2$ for a tangent field and $\langle R(V,X)V,X\rangle \le \|K\|\,\|X\|^2$. For (ii), the field equal to J on $[a,b]$ and to 0 elsewhere has $E_{**} = 0$ by V9.15 (its only break points a, b are zeros), so it vanishes by (i); J then vanishes near the midpoint of $[a,b]$, hence everywhere (V9.2).

V9.17 (fields along $\gamma|[0,\tau]$, broken Jacobi fields, and T'; `Morse.fieldsUpTo`, `Morse.fieldsUpTo_apply_self`, `Morse.IsBrokenJacobi`, `Morse.brokenSpace`, `Morse.nodeZero`, `Morse.exists_isBrokenJacobi`, `Morse.IsBrokenJacobi.eqOn_zero`, `Morse.exists_brokenSpace_eq`, `Morse.nodeSpace`, `Morse.nodeEval_bijective`). Let γ be a smooth curve in N on $[0,1]$. For $\tau \in \mathbb{R}$, the *fields along* $\gamma|[0,\tau]$ are the fields of $T\Omega_\gamma$ vanishing outside $[0,\tau]$; for $0 \le \tau \le 1$ they vanish at τ. Let $0 < \tau \le 1$ and B be fine for $[0,\tau]$ (V9.16). A map is a *broken Jacobi field* on $[0,\tau]$ for B if on each piece between consecutive points of $\{0\} \cup B \cup \{\tau\}$ it agrees with a field of $\mathcal{J}(\gamma)$. Milnor's $T\Omega_\gamma(t_0, \dots, t_k)$ is the space of fields along $\gamma|[0,\tau]$ that are broken Jacobi fields, and his T' the space of fields along $\gamma|[0,\tau]$ vanishing at the points of B. Broken Jacobi fields with arbitrary tangent values at 0, τ and the points of B exist, and one vanishing at all these points vanishes on $[0,\tau]$; so $W \mapsto (W(b))_{b\in B}$ is a linear isomorphism from the broken Jacobi fields along $\gamma|[0,\tau]$ onto $\Sigma = \bigoplus_{b\in B} T_{\gamma(b)}N$. *Book:* p. 83: $T\Omega_\gamma(t_0, \dots, t_k)$, the fields W along γ vanishing at 0 and 1 with $W|[t_{i-1}, t_i]$ a Jacobi field along $\gamma|[t_{i-1}, t_i]$ for each i, a finite-dimensional space of broken Jacobi fields; p. 84: T', the fields with $W(t_0) = W(t_1) = \cdots = W(t_k) = 0$, and "it follows from §14.5" that the broken Jacobi field with given values at the t_i exists and is unique; p. 86: along γ_τ, for $0 < t_1 < \cdots < t_i < \tau$, the broken Jacobi fields form a space isomorphic to Σ, since a broken Jacobi field is determined by its values at the break points. *Hidden hypotheses:* Remark 14.5 on each segment, and the gluing of the pieces. *Deviation (restated):* everything is along $\gamma|[0,\tau]$ for every $\tau \in (0,1]$, as fields on $[0,1]$ vanishing outside $[0,\tau]$ (Milnor uses $\tau = 1$ here and γ_τ on p. 86); a piece is the restriction of a Jacobi field along all of γ, which is equivalent to being a Jacobi field along $\gamma|[t_{i-1}, t_i]$ by V9.1–V9.2, an equivalence not stated formally. *Proof:* existence by induction

over the division points, gluing segment by segment with V9.16 (ii); uniqueness from V9.16 (ii) on each piece.

V9.18 (Lemma 15.3; `Morse.energyHessian_broken_eq`, `Morse.isCompl_broken_nodeZero`, `Morse.energyHessian_broken_nodeZero`, `Morse.energyHessian_nodeZero_pos`). Let γ be a smooth curve in N on $[0,1]$, $0 < \tau \leq 1$ and B fine for $[0,\tau]$. (i) The fields along $\gamma|[0,\tau]$ are the direct sum of the broken Jacobi fields and T' (V9.17). (ii) For a broken Jacobi field W_1 along $\gamma|[0,\tau]$ and every field Y along $\gamma|[0,\tau]$, $E_{**}(W_1, Y) = -2\sum_{b\in B}\langle Y(b), \Delta_b \frac{DW_1}{dt}\rangle$; in particular $E_{**}(W_1, W_2) = 0$ for $W_2 \in T'$. (iii) $E_{**}(W, W) > 0$ for every $W \neq 0$ in T'. *Book:* p. 84, Lemma 15.3: $T\Omega_\gamma = T\Omega_\gamma(t_0, \dots, t_k) \oplus T'$, the two subspaces are mutually perpendicular with respect to E_{**}, and E_{**} restricted to T' is positive definite; proof pp. 84–85: $W = W_1 + (W - W_1)$; perpendicularity from the second variation formula; $E_{**}(W, W) \geq 0$ on T' because each $\gamma|[t_{i-1}, t_i]$ is minimal, and $E_{**}(W, W) = 0$ puts W in the null space, so W is a Jacobi field, and T' contains no Jacobi field other than 0. *Hidden hypotheses:* the existence and uniqueness of W_1 (V9.17); that T' contains no non-zero Jacobi field needs the uniqueness of Jacobi fields on short segments. *Deviation (restated):* along $\gamma|[0,\tau]$ for every $\tau \in (0,1]$ (Milnor's lemma is $\tau = 1$), for fine subdivisions in the sense of V9.16. *Proof:* (ii) is V9.15; (iii) departs from Milnor's: a field of T' has a zero within h_0 before every point, so V9.16 (i) gives $E_{**}(W, W) > 0$ directly, without minimal geodesics, variations or Theorem 14.1.

V9.19 (Lemma 15.4; `Morse.lemma_15_4`, `Morse.lemma_15_4_ker`, `Morse.eq_jacobi_of_jumps_zero`, `Morse.finrank_ker_broken`, `Morse.finiteDimensional_brokenSpace`, `Morse.indexUpTo_le_finrank`, `Morse.indexUpTo_lt_top`). Let γ, τ and B be as in V9.18. The index (V9.13) of E_{**} on the fields along $\gamma|[0,\tau]$ equals its index on the broken Jacobi fields; the null space of E_{**} on the fields along $\gamma|[0,\tau]$ is the null space of its restriction to the broken Jacobi fields, and the latter has dimension $\nu(\tau)$ (V9.4), because a broken Jacobi field whose jumps $\Delta_b \frac{DW}{dt}$ all vanish agrees on $[0,\tau]$ with a single field of $\mathcal{J}(\gamma)$. In particular the index is at most the (finite) dimension of the broken Jacobi fields, so it is finite for every $\tau \in (0,1]$. *Book:* p. 85, Lemma 15.4: the index (or the nullity) of E_{**} equals the index (or the nullity) of E_{**} restricted to the broken Jacobi fields; in particular the index λ is always finite; "the proof is straightforward". *Hidden hypotheses:* the proof is the abstract Lemma 15.B (V9.14 (4)), applied to Lemma 15.3. *Deviation (restated):* along $\gamma|[0,\tau]$ (Milnor states $\tau = 1$ and applies it to γ_τ on p. 86), and for subdivisions fine in the sense of V9.16 rather than Milnor's; the theorem needs only one fine subdivision. *Deviation (stronger):* the nullity on the broken Jacobi fields is identified with $\nu(\tau)$ directly; Milnor gets this on p. 86 from Theorem 14.1 for γ_τ.

V9.20 ($\lambda(\tau)$, Assertions (1) and (2); `Morse.indexUpTo`, `Morse.indexUpTo_one`, `Morse.indexUpTo_mono`, `Morse.indexUpTo_eq_zero`). For a smooth curve γ in N on $[0,1]$ and $\tau \in \mathbb{R}$, $\lambda(\tau)$ is the index of E_{**} restricted to the fields along $\gamma|[0,\tau]$ (V9.17); $\lambda(1)$ is the index of E_{**} on all of $T\Omega_\gamma$. (1) $\lambda(\tau) \leq \lambda(\tau')$ for $\tau \leq \tau'$. (2) $\lambda(\tau) = 0$ for $0 < \tau \leq h_0$ (V9.16). *Book:* p. 85: $\gamma_\tau = \gamma|[0,\tau]$, a geodesic from $\gamma(0)$ to $\gamma(\tau)$, and $\lambda(\tau)$ the index of the associated Hessian $(E_0^\tau)_{**}$, so that $\lambda(1)$ is the index sought; Assertion (1): λ is monotone, because a field along γ_τ vanishing at both ends extends by zero to one along $\gamma_{\tau'}$; Assertion (2): $\lambda(\tau) = 0$ for small τ, because γ_τ is then a minimal geodesic (Lemma 13.6). *Hidden hypotheses:* (1) needs the extension by zero to preserve the Hessian (the integral form, V8.15); (2) needs short geodesics to be minimal (Theorem 10.4). *Deviation (restated):* $\lambda(\tau)$ is the index of a restriction of E_{**}, not of a Hessian along γ_τ on its own interval (see "How the index, the nullity and $\lambda(\tau)$ are formalized" above); with this definition (1) is immediate, and no rescaling of γ_τ to $[0,1]$ (the exposition's Lemma 13.B) is needed. *Proof:* (2) from V9.16 (i): a field along $\gamma|[0,\tau]$ with $\tau \leq h_0$ has a zero within h_0 before every point, so E_{**} is

positive definite on these fields; minimality and Lemma 13.6 are not used.

V9.21 (the forms H_τ and their continuity, the exposition's Lemma 15.C; `Morse.validFor`, `Morse.Hform`, `Morse.Hform_isSymm`, `Morse.formIndex_Hform`, `Morse.finrank_ker_Hform`, `Morse.brokenOf_eqOn`, `Morse.continuous_linFlow`, `Morse.continuousWithinAt_of_injective`, `Morse.continuousWithinAt_Hform`, `Morse.mem_validFor_of_near`). Let γ be a smooth curve in N on $[0,1]$ and B a finite set; τ is *valid* for B if $0<\tau\le 1$ and B is fine for $[0,\tau]$. For valid τ, H_τ is the symmetric bilinear form on $\Sigma=\bigoplus_{b\in B}T_{\gamma(b)}N$ given by E_{**} on the broken Jacobi fields along $\gamma|[0,\tau]$ with these values at the points of B (V9.17); $H_\tau=0$ for other τ. For valid τ, the index of H_τ is $\lambda(\tau)$ and its nullity is $\nu(\tau)$. For $x,y\in\Sigma$, $\tau\mapsto H_\tau(x,y)$ is continuous on the set of valid τ. If B is fine for $[0,\tau]$ and the last point m of $\{0\}\cup B$ is within $h_0/2$ of τ, every $\tau'\in(m,\tau+h_0/2)$ with $\tau'\le 1$ is valid. *Book:* p. 86: with the subdivision chosen so that $t_i<\tau<t_{i+1}$, $\lambda(\tau)$ is the index of a quadratic form H_τ on the broken Jacobi fields along γ_τ for $0<t_1<\cdots<t_i<\tau$, a space isomorphic to $\Sigma=TM_{\gamma(t_1)}\oplus\cdots\oplus TM_{\gamma(t_i)}$, which is independent of τ; "evidently" H_τ varies continuously with τ. *Hidden hypotheses:* that the subdivision stays fine for τ' near τ; and, for the continuity, that t_i and τ are not conjugate (so that the Jacobi field with value x_i at t_i and 0 at τ exists), the linear dependence of Jacobi fields on their initial values, and the continuity of the solution of an injective linear system. *Proof:* by V9.18 (ii), $H_\tau(x,y)=-2\sum_b\langle y_b,\Delta_b\frac{DW_x}{dt}\rangle$; the broken Jacobi fields for different valid τ agree up to the last division point m, so only $\frac{DY^\tau}{dt}(m)$ depends on τ, where Y^τ is the Jacobi field with $Y^\tau(m)=x_m$ and $Y^\tau(\tau)=0$; its initial vector solves $L(\tau)z=c(\tau)$, with L and c continuous (propagators of V9.1) and $L(\tau_0)$ injective because m and τ_0 are closer than h_0 (V9.16 (ii)), and the solution of such a system is continuous at τ_0. This replaces Cramer's rule in the exposition's proof.

V9.22 (Assertion (3); `Morse.assertion_3`). For a smooth curve γ in N on $[0,1]$ and $0<\tau\le 1$: $\lambda(\tau')=\lambda(\tau)$ for all $\tau'<\tau$ close enough to τ. *Book:* p. 85, Assertion (3): $\lambda(\tau-\varepsilon)=\lambda(\tau)$ for all sufficiently small $\varepsilon>0$; proof p. 86: H_τ is negative definite on a subspace of Σ of dimension $\lambda(\tau)$, hence so is $H_{\tau'}$ for τ' near τ, giving $\lambda(\tau')\ge\lambda(\tau)$, and Assertion (1) gives the reverse inequality for $\tau'<\tau$. *Hidden hypotheses:* that $\lambda(\tau)$ is finite and attained (V9.13, V9.19). *Proof:* Milnor's, with V9.21 and V9.14 (3).

V9.23 (Assertion (4); `Morse.assertion_4_le`, `Morse.energyHessian_cut`, `Morse.exists_dual`, `Morse.assertion_4_ge`, `Morse.assertion_4`). Let γ be a smooth curve in N on $[0,1]$ and $0<\tau<1$. (i) $\lambda(\tau')\le\lambda(\tau)+\nu(\tau)$ for all $\tau'>\tau$ close enough to τ. (ii) If $J\in\mathcal{J}(\gamma)$ vanishes at 0 and τ, and $\hat J$ equals J on $[0,\tau]$ and 0 elsewhere, then $E_{**}(\hat J,Z)=2\langle Z(\tau),\frac{DJ}{dt}(\tau)\rangle$ for every $Z\in T\Omega_\gamma$. (iii) Linearly independent tangent vectors $d_1,\dots,d_\nu$ at a point of N have tangent vectors $x_1,\dots,x_\nu$ there with $\langle d_h,x_k\rangle=\delta_{hk}$. (iv) $\lambda(\tau')\ge\lambda(\tau)+\nu(\tau)$ for every $\tau'\in(\tau,1]$. Hence $\lambda(\tau')=\lambda(\tau)+\nu(\tau)$ for all $\tau'>\tau$ close enough to τ. *Book:* p. 86, Assertion (4): if ν is the nullity of $(E_0^\tau)_{**}$, then $\lambda(\tau+\varepsilon)=\lambda(\tau)+\nu$ for all sufficiently small $\varepsilon>0$. Proof of $\le$ (p. 86): H_τ is positive definite on a subspace of dimension $ni-\lambda(\tau)-\nu$, hence so is $H_{\tau'}$ for τ' near τ. Proof of $\ge$ (pp. 86–87): with $W_1,\dots,W_\lambda$ spanning a space on which $(E_0^\tau)_{**}$ is negative definite, $J_1,\dots,J_\nu$ independent Jacobi fields vanishing at 0 and τ, and fields X_k with $\langle\frac{DJ_h}{dt}(\tau),X_k(\tau)\rangle=\delta_{hk}$, all extended by 0 beyond τ, the fields W_i, $c^{-1}J_h-cX_h$ span a $(\lambda+\nu)$-dimensional space on which $(E_0^{\tau+\varepsilon})_{**}$ is negative definite, for small c. *Hidden hypotheses:* for $\le$, Sylvester's law of inertia and the fact that the nullity of H_τ is $\nu(\tau)$ (Lemma 15.4 and Theorem 14.1 for γ_τ); for $\ge$, the linear independence of the $\frac{DJ_h}{dt}(\tau)$ (uniqueness, V9.2), the construction of the X_k, and the linear independence of the $\lambda+\nu$ fields (V9.14 (5)). *Deviation (restated):* ν is $\nu(\tau)$, the multiplicity of 0 and τ (V9.4), which is the nullity of E_{**} along $\gamma|[0,\tau]$ by V9.19 and V9.21. *Deviation (stronger):* (iv) holds for every

$\tau' \in (\tau, 1]$, not only for τ' near τ. *Proof:* Milnor's, with V9.14 (1)–(3) and (5); (ii) is V9.15, and the fields X_k are a bump function times the vectors of (iii).

V9.24 (Theorem 15.1, the Morse index theorem; `Morse.eq_finsum_of_jumps`, `Morse.indexUpTo_one_eq`, `Morse.theorem_15_1`). (i) Let $\ell, \nu : \mathbb{R} \to \{0, 1, 2, \dots\}$ and $h_0 > 0$, with ℓ non-decreasing on $(0, 1]$, $\ell = 0$ on $(0, h_0]$, $\ell(\tau') = \ell(\tau)$ for $\tau' < \tau$ near τ (for $0 < \tau \le 1$), and $\ell(\tau') = \ell(\tau) + \nu(\tau)$ for $\tau' > \tau$ near τ (for $0 < \tau < 1$). Then $\nu(t) \neq 0$ for only finitely many $t \in (0, 1)$, and $\ell(1) = \sum_{0<t<1} \nu(t)$. (ii) For every smooth curve γ in N on $[0, 1]$, $\nu(t) \neq 0$ for only finitely many $t \in (0, 1)$, and $\lambda(1) = \sum_{0<t<1} \nu(t)$. (iii) **Theorem 15.1:** let $\gamma \in \Omega(N; p, q)$ be a geodesic (V8.10). The parameters $t \in (0, 1)$ conjugate to 0 along γ form a finite set, and the index of E_{**} on $T\Omega_\gamma$ (V9.13) is

$$\lambda = \sum_{0<t<1} \nu(t),$$

the number of these parameters, each counted with its multiplicity (V9.4); in particular λ is finite. *Book:* p. 83, Theorem 15.1 (Morse): the index λ of E_{**} is equal to the number of points $\gamma(t)$, $0 < t < 1$, conjugate to $\gamma(0)$ along γ, each counted with its multiplicity, and is always finite; p. 87: the theorem "clearly follows" from Assertions (2), (3) and (4). *Hidden hypotheses:* the finiteness of the set of conjugate parameters has to be proved first (from Assertions (1) and (4) and the finiteness of $\lambda(1)$), before the jumps can be added up; then $\lambda(\tau) - \sum_{t<\tau} \nu(t)$ is locally constant on $(0, 1]$, hence constant, and it vanishes for small τ, where Assertion (4) also excludes conjugate parameters. *Deviation (restated):* parameters for points (V9.4). *Deviation (stronger):* (ii) holds along any smooth curve in N; when γ is not a geodesic, E_{**} is the form of V8.14, not the Hessian of the energy. *Proof:* (i) applied to $\ell = \lambda$ and ν, by Assertions (1)–(4) (V9.20, V9.22, V9.23); $\lambda(1)$ is the index on all of $T\Omega_\gamma$ (V9.20).

V9.25 (Corollary 15.2; `Morse.corollary_15_2`). For a geodesic $\gamma \in \Omega(N; p, q)$, the set of parameters conjugate to 0 along γ is finite. *Book:* p. 83, Corollary 15.2, "an immediate consequence": a geodesic segment $\gamma : [0, 1] \to M$ contains only finitely many points conjugate to $\gamma(0)$ along γ. *Hidden hypotheses:* none. *Deviation (restated):* parameters for points (V9.4). *Proof:* V9.24, plus the parameter $t = 1$.

What is not verified in this chapter

- **Examples 1 and 2** (pp. 79–80): along a flat manifold no two points are conjugate; on S^n, antipodal points are conjugate along a great semicircle with multiplicity $n - 1$, an upper bound that needs Remark 14.2. Not formalized in this chapter. Chapter 10 proves Example 2, for every great circle and every parameter (V10.23: `Morse.isJacobiField_sin_smul`, `Morse.conjMultiplicity_greatCircle`), and Example 1 for the open subsets of a Euclidean space (V10.27: `Morse.not_isConjugate_of_top`); a flat manifold embedded with positive codimension is not covered.
- **Remark 14.6 for $t < 0$, and its last sentence** (p. 82): the conjugate points of $\gamma(0)$ along an entire geodesic have no cluster points. The geodesics of Ω are given on $[0, 1]$, and the reparametrization of a geodesic to $[0, 1]$ (the exposition's Lemma 14.C) is not formalized.
- **Milnor's subdivisions** (p. 83): Lemmas 15.3–15.4 are proved for subdivisions of mesh at most h_0 (V9.16), not for his subdivisions into minimal segments inside neighbourhoods U; the theorem needs only one fine subdivision.
- **$\lambda(\tau)$ as the index of the Hessian of γ_τ on $[0, \tau]$**: no Hessian on $[0, \tau]$ is formalized, nor the exposition's Lemma 13.B; $\lambda(\tau)$ is defined by restriction (V9.20).

- **The index of E_{**} at a minimal geodesic** (p. 76, Lemma 13.6, "hence the index λ of E_{**} is zero"): positive semi-definiteness is V8.18, and the index is now defined (V9.13), but the vanishing of the index is not stated formally.
- **The exposition's §5** (the index of every great-circle route on S^n, the explicit matrix on S^2, the optics remark) and its interactive figure: not formalized in this chapter; the index $k(n-1)$ of the geodesics joining two non-antipodal points of S^n is V10.23.
- **§§14–15 on an abstract Riemannian manifold**: only for submanifolds of a Euclidean space with the induced metric; Nash's embedding theorem is quoted, not proved.

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11. Chapter 10 (Milnor §§16–17): partly verified

Milnor's §16 (pp. 88–92) topologizes the path space $\Omega = \Omega(M; p, q)$ of a connected Riemannian manifold by a metric d for which the energy E is continuous and, for complete M, replaces the paths of energy less than c by the finite-dimensional manifold B of broken geodesics with fixed break times (Lemma 16.1). The energy $E' = E|B$ is smooth, its sublevel sets B^a, $a < c$, are compact deformation retracts of the sets Ω^a, and its critical points are the geodesics, with the index and nullity of E_{**} (Theorem 16.2); so Ω^a has the homotopy type of a finite CW-complex with one λ-cell for each geodesic in Ω^a of index λ (Theorem 16.3), and a minimum of E' is a minimal geodesic (the Remark on p. 92). §17 (pp. 93–97) shows that the inclusion of Ω into the space Ω^* of continuous paths is a homotopy equivalence (Theorem 17.1), so that Ω has the homotopy type of a CW-complex (Corollary 17.2); proves the fundamental theorem (Theorem 17.3: for complete M and p, q not conjugate along any geodesic, Ω has the homotopy type of a countable CW-complex with one λ-cell for each geodesic from p to q of index λ); finds the cells of the loop space of S^n (Corollary 17.4) and its homology for $n > 2$; and concludes that two non-conjugate points of a manifold of the homotopy type of S^n, $n > 2$, are joined by infinitely many geodesics (Corollary 17.5), with a Remark citing Serre. Verified, for a submanifold N of a Euclidean space with the induced metric, in the embedded setting of Chapters 6–9: Milnor's metric, a pseudometric on paths given as maps on $\mathbb{R}$, with the exposition's Lemmas 16.A and 16.B; Lemma 16.1; all of Theorem 16.2, with the continuity of Milnor's deformation r_u (the exposition's Lemma 16.C); Theorem 16.3, for a level a that is not the energy of a geodesic; the Remark on p. 92 for the energy; Theorem 17.1, for every N; Theorem 17.3 for Ω, with N closed in the Euclidean space; Corollary 17.4 for the path space $\Omega(S^n; p, q)$, $q \neq \pm p$, $n \geq 2$, and its homology over every field for $n \geq 3$; and the example $\mathbb{R}^{n+1} \setminus \{0\}$, which shows that Corollary 17.5 needs the completeness Milnor omits. Every Riemannian manifold is such an N by Nash's embedding theorem, which is quoted, not proved. As in Part I and the Appendix, a "CW-complex" is an iterated adjunction space, or a mapping telescope of such spaces, whose attaching maps need not be cellular (V2.11, VA.18). Proved more sharply than in the book: Lemma 16.1 without Milnor's hypothesis $\Omega^c \neq \emptyset$ (V10.4). **One step of the printed proof of Theorem 17.1 does not hold as we read it:** Milnor's "Clearly h is continuous" (p. 94) fails for the metric d of Ω; the theorem is true, and it is verified with the exposition's modified proof (V10.19). The proofs depart from Milnor's in three further places. **Lemma 15.4 is proved anew for the subdivision of §16**, which need not be fine in the sense of §15 (V10.10). **Theorem 16.3 applies Theorem 3.5 to a compact sublevel set** of a smooth extension of E' (V10.12). **Theorem 17.3 takes Milnor's second route**, Theorem A of the Appendix, with a tower that commutes only up to homotopy, so that neither Corollary 17.2, Whitehead's theorem nor nested subdivisions are needed (V10.20, V10.22). Corollary 17.2, for N closed in V, and the loop-space form of Corollary 17.4, with the homology

of the loop space of S^n for $n \geq 3$, are obtained by moving the end point to a non-conjugate one (V10.28–V10.30). **Not verified:** Milnor's own route to Corollary 17.2, Theorem 17.3 for Ω^*, the homology of the path spaces of S^2 and integral homology, Corollary 17.5 itself, and Serre's remark (the list at the end). Lean: `Morse/Chapter10/`, 30 files of 13,564 lines in all — for §16 `Product.lean` (220 lines), `Broken.lean` (806), `BrokenPaths.lean` (248), `BrokenEnergy.lean` (246), `BrokenVariation.lean` (357), `BrokenSplitting.lean` (707), `BrokenIndex.lean` (424), `SublevelCW.lean` (319), `Finite.lean` (340), `PathMetric.lean` (400), `Retraction.lean` (528), `RetractionCont.lean` (320), `RetractionConv.lean` (510), `Deformation.lean` (302), `Omega.lean` (228); for §17 `FullPathSpace.lean` (471), `GoodBalls.lean` (630), `FullPathDeform.lean` (502), `GoodJoin.lean` (194), `FullPathOmega.lean` (879), `FullPathTheorem.lean` (113), `RelativeCells.lean` (583), `FundamentalTheorem.lean` (561), `SublevelHomology.lean` (534), `PathHomology.lean` (553), `SphereGeodesics.lean` (881), `SpherePaths.lean` (107), `Incomplete.lean` (591), `CWType.lean` (199).

The embedded setting. As in Chapters 8 and 9: V is a finite-dimensional real inner product space and $N \subseteq V$ a slice submanifold modelled on Z with complement W (V1.6), with the induced metric and its distance ρ (V7.18), which satisfies $\|x - y\| \leq \rho(x, y)$ (V7.21); $p, q \in V$. $\Omega = \Omega(N; p, q)$ is the set of maps that are piecewise smooth on $[0, 1]$, with values in N there, from p to q (V8.1); $E(\omega) = E_0^1(\omega)$ is the energy (V8.5), $\Omega^a = \{\omega \in \Omega : E(\omega) \leq a\}$ and $\operatorname{Int}\Omega^c = \{\omega \in \Omega : E(\omega) < c\}$. A *geodesic* of Ω is an unbroken geodesic (V8.10); it has constant speed, so $E = L^2$ for it, and Milnor's "length $\leq \sqrt{a}$" reads "energy $\leq a$". E_{**}, its index and its nullity are those of V8.14, V9.13 and V9.6; at a geodesic the nullity is 0 iff 0 and 1 are not conjugate along it (Theorem 14.1, V9.6). Milnor's completeness of M enters only through the compactness of the balls $\{x \in N : \rho(p, x) \leq r\}$, which holds when N is closed in V (then N is complete, V7.30), or when N is connected and complete (V7.22). N need not be compact or connected unless this is stated.

How the path space is formalized. The elements of Ω are maps $\mathbb{R} \to V$ whose values off $[0, 1]$ play no role, and Milnor's metric d sees only $[0, 1]$, so it is a pseudometric: distinct maps that agree on $[0, 1]$ are at distance 0 (V10.1). Three conventions follow. Identities between paths hold on $[0, 1]$, unless a map is completed off $[0, 1]$ to make them exact (V10.7). Milnor's homeomorphisms between sets of paths and sets of break points become homotopy equivalences whose composites are the identity up to paths at distance 0 (V10.5). And geodesics are counted by their values on $[0, 1]$: "one cell for each geodesic" means one cell for each class of geodesics that agree on $[0, 1]$. The manifold B of broken geodesics is represented by its set U of break points, an open subset of $N \times \cdots \times N$ (V10.4).

V10.1 (Milnor's metric on Ω; `Morse.PathSp`, `Morse.PathSp.speed`, `Morse.PathSp.supDist`, `Morse.PathSp.l2Dist`, `Morse.PathSp.instPseudoEMetricSpace`, `Morse.PathSp.edist_def`, `Morse.PathSp.edist_eq_zero_of_eqOn`). For $\omega, \eta \in \Omega$, with speeds $\dot{s}_\omega(t) = \|\frac{d\omega}{dt}(t)\|$ and $\dot{s}_\eta(t)$,

$$d(\omega, \eta) = \sup_{0 \leq t \leq 1} \rho(\omega(t), \eta(t)) + \Big(\int_0^1 (\dot{s}_\omega - \dot{s}_\eta)^2 \, dt \Big)^{1/2}.$$

This is an extended pseudometric on Ω (formally with values in $[0, \infty]$; symmetric, with the triangle inequality), and two paths that agree on $[0, 1]$ are at distance 0. From now on Ω and its subsets carry the topology of d. *Book:* p. 88: Ω can be topologized by the distance $d(\omega, \omega') = \max_t \rho(\omega(t), \omega'(t)) + [\int_0^1 (\frac{ds}{dt} - \frac{ds'}{dt})^2 dt]^{1/2}$, with s, s' the arc lengths. *Hidden hypotheses:* the triangle inequality for the second term is Minkowski's inequality in L^2; $\frac{ds}{dt}$ exists except at finitely

many points. *Deviation (restated):* on maps $\mathbb{R} \to V$, d is a pseudometric, not a metric; on paths $[0,1] \to N$ it is Milnor's metric.

V10.2 (Lemmas 16.A and 16.B of the exposition: the energy is continuous; `Morse.PathSp.abs_sqrt_energy_sub`, `Morse.PathSp.lipschitzWith_sqrt_energy`, `Morse.PathSp.continuous_energy`, `Morse.PathSp.isClosed_sub`, `Morse.PathSp.isOpen_sublevel_lt`, `Morse.PathSp.exists_energy_lt_of_tendsto`). (i) For $\omega, \eta \in \Omega$ and $0 \le a \le b \le 1$, $|\sqrt{E_a^b(\omega)} - \sqrt{E_a^b(\eta)}| \le (\int_0^1 (\dot{s}_\omega - \dot{s}_\eta)^2 dt)^{1/2}$; hence $\sqrt{E_a^b}$ is 1-Lipschitz for d, E_a^b is continuous on Ω, Ω^c is closed and $\operatorname{Int}\Omega^c$ is open. (ii) (Lemma 16.B) If $\omega_n \to \omega$ in Ω, then for every $\epsilon > 0$ there is $\delta > 0$ with $E_a^b(\omega_n) < \epsilon$ for all n and all $0 \le a \le b \le 1$ with $b - a < \delta$. *Book:* p. 88: the last term of d is added so that $E_a^b(\omega) = \int_a^b (\frac{ds}{dt})^2 dt$ is a continuous function on Ω; $\Omega^c = E^{-1}([0,c])$ is closed and $\operatorname{Int}\Omega^c = E^{-1}([0,c))$ is open. *Hidden hypotheses:* $\sqrt{E_a^b} = \|\dot{s}\|_{L^2[a,b]}$ and Minkowski's inequality on $[a,b]$; with the uniform term alone the energy is not continuous. *Deviation (not in the book):* (ii), used for the continuity of r_u (V10.7).

V10.3 (the ball S, subdivisions, and Milnor's choices; `Morse.closedRiemBall`, `Morse.mem_closedRiemBall_of_`, `Morse.isCompact_closedRiemBall_of_isClosed`, `Morse.isCompact_closedRiemBall_of_isRiemComplete`, `Morse.corollary_10_8'`, `Morse.Subdivision`, `Morse.BrokenSetup`, `Morse.exists_brokenSetup`, `Morse.exists_brokenSetup_uniform`, `Morse.BrokenSetup.IsBroken`, `Morse.BrokenSetup.brokenPaths`). Let $c > 0$ and $S = \{x \in N : \rho(p,x) \le \sqrt{c}\}$. (i) Every $\omega \in \Omega$ with $E(\omega) \le c$ maps $[0,1]$ into S. (ii) S is compact if N is closed in V, or if N is connected and complete. (iii) For every compact $K \subseteq N$ there is $\delta > 0$ such that any $y, z \in K$ with $\rho(y,z) < \delta$ are joined as in Corollary 10.8 (V7.20) at the scale δ, and δ is at most a normal radius (V7.9) at every point of K. (iv) A *subdivision* is $0 = t_0 < t_1 < \cdots < t_{m+1} = 1$, with m interior points (Milnor's $k = m+1$). *Milnor's data* consist of $p, q \in N$, $c > 0$ with S compact, a δ as in (iii) for $K = S$ (Milnor's ε), and a subdivision with $c\,(t_{i+1} - t_i) < \delta^2$ for every i. If $p, q \in N$ and S is compact, there is $\eta > 0$ such that every subdivision with all $t_{i+1} - t_i < \eta$ is part of such data, and some subdivision into equal parts is. (v) For Milnor's data, $B = \operatorname{Int}\Omega(t_0, \dots, t_k)^c$ is the set of $\omega \in \Omega$ with $E(\omega) < c$ that are geodesics on each open interval (t_i, t_{i+1}), and $B^a = \{\omega \in B : E(\omega) \le a\}$. *Book:* p. 88: $\Omega(t_0, \dots, t_k)$, the paths ω with $\omega|[t_{i-1}, t_i]$ a geodesic for each i, and $\operatorname{Int}\Omega(t_0, \dots, t_k)^c = (\operatorname{Int}\Omega^c) \cap \Omega(t_0, \dots, t_k)$; p. 89: every $\omega \in \Omega^c$ lies in the ball S, by $L^2 \le E \le c$; S is compact since M is complete; by 10.8 there is $\varepsilon > 0$ such that two points of S at distance less than ε are joined by a unique geodesic of length less than ε, depending differentiably on them; and each $t_i - t_{i-1} < \varepsilon^2/c$. *Hidden hypotheses:* the compactness of S is the Hopf–Rinow theorem (Corollary 10.10, V7.27). *Deviation (restated):* completeness enters only as the compactness of S; δ is also asked to be at most a normal radius on S, which (iii) provides at no cost and which keeps the ends of each segment non-conjugate (V10.10).

V10.4 (Lemma 16.1; `Morse.BrokenSetup.energy'`, `Morse.BrokenSetup.nodes`, `Morse.BrokenSetup.joinVec`, `Morse.BrokenSetup.path`, `Morse.BrokenSetup.nodeVal`, `Morse.contDiffOn_Icc_of_isGeodesicOn`, `Morse.BrokenSetup.exists_segVec`, `Morse.BrokenSetup.energy_path`, `Morse.BrokenSetup.lemma_16_1`, `Morse.BrokenSetup.exists_isOpen_nodes`, `Morse.isSliceSubmanifold_pi_inter`, `Morse.tangentSubspace_p`. Fix Milnor's data. For $y = (y_1, \dots, y_m) \in V^m$ put $x_0 = p$, $x_j = y_j$ $(1 \le j \le m)$, $x_{m+1} = q$, and

$$E'(y) = \sum_{i=0}^{m} \frac{\rho(x_i, x_{i+1})^2}{t_{i+1} - t_i};$$

let U be the set of $y \in N^m$ with every $\rho(x_i, x_{i+1})$ finite and $E'(y) < c$. (i) $U = N^m \cap O$ for an open $O \subseteq V^m$; for every open $O \subseteq V^m$, $N^m \cap O$ is a slice submanifold of V^m modelled on

Z^m, with tangent space $T_{y_1}N \times \cdots \times T_{y_m}N$ at y. (ii) For $y \in U$, the broken geodesic ω_y, equal on $[t_i, t_{i+1}]$ to $t \mapsto \exp_{x_i}(\frac{t-t_i}{t_{i+1}-t_i}\, v(x_i, x_{i+1}))$, where $v(x, x')$ is the initial vector of the short geodesic from x to x' (V7.20), lies in B, has break points $(\omega_y(t_1), \dots, \omega_y(t_m)) = y$, and $E(\omega_y) = E'(y)$ (the exposition's formula $(*)$). (iii) For $\omega \in B$ and each i there is $w \in T_{\omega(t_i)}N$ with $\|w\| < \delta$, $\omega(t) = \exp_{\omega(t_i)}(\frac{t-t_i}{t_{i+1}-t_i}\, w)$ for $t_i \le t \le t_{i+1}$, and $E_{t_i}^{t_{i+1}}(\omega) = \|w\|^2/(t_{i+1} - t_i)$; the break points y of ω lie in U, and ω_y agrees with ω on $[0, 1]$. So $y \mapsto \omega_y$ and $\omega \mapsto (\omega(t_1), \dots, \omega(t_m))$ are inverse bijections between U and B, paths being compared on $[0, 1]$. *Book:* p. 88, Lemma 16.1: for M complete and $c > 0$ with $\Omega^c \neq \emptyset$, for all sufficiently fine subdivisions $\operatorname{Int}\Omega(t_0, \dots, t_k)^c$ can be given the structure of a smooth finite dimensional manifold "in a natural way"; proof, p. 89: for a broken geodesic, $(L_{t_{i-1}}^{t_i}\omega)^2 = (t_i - t_{i-1})E_{t_{i-1}}^{t_i}\omega < \varepsilon^2$, so each segment is determined by its end points, and ω by $(\omega(t_1), \dots, \omega(t_{k-1})) \in M \times \cdots \times M$; this correspondence is a homeomorphism onto an open subset of the product, whose differentiable structure is taken over. *Hidden hypotheses:* that the image is the open set U: the vertices of a point of U lie in S by the Cauchy–Schwarz inequality, $\rho(p, x_i) \le \sum_j \rho(x_j, x_{j+1}) \le (\sum_j (t_{j+1} - t_j))^{1/2} E'(y)^{1/2} < \sqrt{c}$, and consecutive ones are at distance less than δ; that a geodesic segment is the radial geodesic of its initial vector (V9.9 (ii)) and, that vector being shorter than δ, the short geodesic (uniqueness in Corollary 10.8); that a broken geodesic is smooth up to the ends of its segments. *Deviation (stronger):* $\Omega^c \neq \emptyset$ is not needed. *Deviation (restated):* the manifold is U, an open subset of N^m, rather than a structure transported to B; the topologies are compared in V10.5.

V10.5 (the correspondence is a homeomorphism up to paths at distance 0; `Morse.BrokenSetup.continuous_pat` `Morse.BrokenSetup.continuous_nodeVal`, `Morse.BrokenSetup.homotopyEquivNodes`). For Milnor's data, $y \mapsto \omega_y$ is continuous from U (with the topology of V^m) to Ω, and $\omega \mapsto (\omega(t_1), \dots, \omega(t_m))$ is continuous from Ω to V^m. For every $a \in \mathbb{R}$ they restrict to continuous maps between $B^a \subseteq \Omega$ and $\{y \in U : E'(y) \le a\}$, inverse to each other up to paths at distance 0; so the two sets are homotopy equivalent (for $a \ge c$, B and U). *Book:* p. 89: "Evidently" $\omega \mapsto (\omega(t_1), \dots, \omega(t_{k-1}))$ defines a homeomorphism between $\operatorname{Int}\Omega(t_0, \dots, t_k)^c$ and an open subset of $M \times \cdots \times M$. *Hidden hypotheses:* the continuity of $\omega \mapsto \omega(t_i)$ uses $\|x - y\| \le \rho(x, y)$; that of $y \mapsto \omega_y$ for d needs, for the uniform term, the continuity of the short geodesics in their end points (Corollary 10.8 and the continuity of exp, V7.23), and, for the L^2 term, that of the speeds $\rho(x_i, x_{i+1})/(t_{i+1} - t_i)$. *Deviation (restated):* a homotopy equivalence, indeed a homeomorphism up to the identification of paths at distance 0, in place of a homeomorphism, since paths are maps on $\mathbb{R}$.

V10.6 (Theorem 16.2: E' is smooth and B^a is compact; `Morse.contMDiffAt_of_locally_eq`, `Morse.BrokenSetup.exists_smooth_energy'`, `Morse.BrokenSetup.contMDiff_energy'`, `Morse.BrokenSetup.continuousOn_energy'`, `Morse.BrokenSetup.isCompact_sublevel`). For Milnor's data: (i) every $y_0 \in U$ has an open neighbourhood O in V^m with a function smooth on O that equals E' on $U \cap O$; hence E' is continuous on U, and smooth on the manifold $N^m \cap O$ for every open $O \subseteq V^m$ with $N^m \cap O \subseteq U$ (a function on a slice submanifold which near a point is the restriction of a smooth function on an open set of the ambient space is smooth there). (ii) For $a < c$, $\{y \in U : E'(y) \le a\}$ is compact. *Book:* p. 90, Theorem 16.2: $E' : B \to \mathbb{R}$ is smooth, and for each $a < c$ the set $B^a = (E')^{-1}[0, a]$ is compact; p. 90: E' depends smoothly on the break points, by the explicit formula $E'(\omega) = \sum_i \rho(\omega(t_{i-1}), \omega(t_i))^2/(t_i - t_{i-1})$; p. 91: B^a is homeomorphic to a set of $(k-1)$-tuples in $S \times \cdots \times S$, closed in a compact set. *Hidden hypotheses:* ρ is not smooth on the diagonal; E' is smooth because $\rho(x, x')^2 = \|v(x, x')\|^2$ with v smooth (Corollary 10.8); the set of tuples is closed only because the distances are finite. *Deviation (restated):* as reproduced in

the scan, the condition defining the tuples reads $\rho(p_{i-1},p_i)^c/(t_i-t_{i-1}) \le a$, with exponent c and no summation sign; the formal condition is $E'(p_1,\dots,p_{k-1}) \le a$, the sum with exponent 2 (see `Exposition/Findings.md`). Compactness is stated for the set of break points.

V10.7 (Milnor's deformation r_u, and the exposition's Lemma 16.C; `Morse.BrokenSetup.retr`, `Morse.BrokenSetup.retr_mem_pathSpace`, `Morse.BrokenSetup.energy_retr_le`, `Morse.BrokenSetup.retr_ze` `Morse.BrokenSetup.retr_one`, `Morse.BrokenSetup.retr_of_mem_brokenPaths`, `Morse.BrokenSetup.retrSeg_` `Morse.BrokenSetup.deform`, `Morse.BrokenSetup.continuousOn_deform`). For Milnor's data, $\omega \in \Omega$ with $E(\omega) < c$ and $t_i \le u \le t_{i+1}$, $r_u(\omega)$ is the broken geodesic ω_y of the break points y of ω on $[0,t_i]$, the short geodesic from $\omega(t_i)$ to $\omega(u)$, run at constant speed, on $[t_i,u]$, and ω on $[u,1]$; at $u = t_{i+1}$ the formulas of the two neighbouring segments agree. On $[0,1]$: $r_u(\omega) \in \Omega$, $E(r_u(\omega)) \le E(\omega)$, $r_0(\omega) = \omega$, $r_1(\omega) = \omega_y$, and $r_u(\omega) = \omega$ for $\omega \in B$. Completed by ω off $[0,1]$, and set equal to ω when $u \notin [0,1]$ or $E(\omega) \ge c$, $(u,\omega) \mapsto r_u(\omega)$ is a map $\mathbb{R} \times \Omega \to \Omega$, and (Lemma 16.C) it is continuous on $[0,1] \times \operatorname{Int}\Omega^c$. *Book:* p. 91: the retraction $r : \operatorname{Int}\Omega^c \to B$, $r(\omega)$ being on each $[t_{i-1},t_i]$ the geodesic of length less than ε from $\omega(t_{i-1})$ to $\omega(t_i)$; "Clearly" $E(r(\omega)) \le E(\omega) < c$; for $t_{i-1} \le u \le t_i$, $r_u(\omega)$ is $r(\omega)$ on $[0,t_{i-1}]$, the minimal geodesic from $\omega(t_{i-1})$ to $\omega(u)$ on $[t_{i-1},u]$, and ω on $[u,1]$; r_0 is the identity, $r_1 = r$, and "it is easily verified" that $r_u(\omega)$ is continuous in both variables. *Hidden hypotheses:* the short geodesic exists since $\rho(\omega(t_i),\omega(u))^2 \le (u-t_i)E(\omega) < \delta^2$; the formulas must agree where neighbouring intervals meet; r_u fixes B because a sub-arc of a short geodesic is the short geodesic between its ends (uniqueness in Corollary 10.8); $E(r_u(\omega)) \le E(\omega)$ holds segment by segment, by Schwarz's inequality $\rho^2 \le (b-a)E_a^b$; for the continuity, the uniform term needs the continuity of short geodesics in their end points and the equicontinuity of the paths of energy less than c, and the L^2 term, for u near t_i, the bound $\rho(\omega(t_i),\omega(u))^2/(u-t_i) \le E_{t_i}^u(\omega)$ with Lemma 16.B (V10.2). *Deviation (restated):* Milnor's minimal geodesic is the short geodesic of V7.20, which is minimal.

V10.8 (Theorem 16.2: B^a is a strong deformation retract of Ω^a; `Morse.BrokenSetup.deform_zero`, `Morse.BrokenSetup.energy_deform_le`, `Morse.BrokenSetup.deform_one_mem_brokenPaths`, `Morse.BrokenSetup.deform_of_mem_brokenPaths`, `Morse.BrokenSetup.theorem_16_2_deformationRetract`, `Morse.BrokenSetup.homotopyEquivBroken`). For Milnor's data and $a < c$, the deformation of V10.7 is continuous on $[0,1] \times \Omega^a$, does not increase the energy (so every r_u maps Ω^a into itself), starts at the identity, ends in B^a ($r_1(\omega) \in B$ for every $\omega \in \operatorname{Int}\Omega^c$), and fixes every point of B for $0 \le u \le 1$. So B^a is a strong deformation retract of Ω^a, and the inclusion $B^a \subseteq \Omega^a$ is a homotopy equivalence; by V10.7, B is likewise a strong deformation retract of $\operatorname{Int}\Omega^c$. *Book:* p. 90, Theorem 16.2: B^a is a deformation retract of Ω^a, with the footnote that B itself is similarly a deformation retract of $\operatorname{Int}\Omega^c$; p. 91: since $E(r_u(\omega)) \le E(\omega)$, "it is clear" that each B^a is a deformation retract of Ω^a. *Hidden hypotheses:* those of V10.7. *Deviation (restated):* the strong deformation retraction is stated through these properties of $(u,\omega) \mapsto r_u(\omega)$, which hold exactly, not only on $[0,1]$, thanks to the completion of V10.7.

V10.9 (Theorem 16.2: the critical points of E'; `Morse.exists_curve_of_mem_tangentSubspace`, `Morse.BrokenSetup.exists_variation_path`, `Morse.BrokenSetup.varField_path_node`, `Morse.BrokenSetup.firstVariation_path`, `Morse.BrokenSetup.fderiv_eq_jumps`, `Morse.BrokenSetup.isCri` `Morse.BrokenSetup.eqOn_path_nodeVal_of_isUnbrokenGeodesic`). Fix Milnor's data. (i) Every tangent vector of a slice submanifold at a point is the velocity at 0 of a smooth curve in it. If $u \mapsto y(u)$ is a smooth curve in U near 0, then $(u,t) \mapsto \omega_{y(u)}(t)$ is, for $|u| < \varepsilon$, a variation (V8.2) through broken geodesics, smooth on each strip $(-\varepsilon,\varepsilon) \times [t_i,t_{i+1}]$, whose variation field takes the value $y'(0)_j$ at t_j. (ii) At $\gamma = \omega_y$, $y \in U$, the first variation (V8.7) in the direction X is $-\sum_j \langle X(t_j), \Delta_{t_j}\gamma' \rangle$. (iii) Let $O \subseteq V^m$ be open with $N^m \cap O \subseteq U$, and F smooth on V^m with

$F = E'$ on $N^m \cap O$. For $y \in N^m \cap O$ and $w \in T_{y_1}N \times \cdots \times T_{y_m}N$, $dF_y(w) = -2\sum_j \langle w_j, \Delta_{t_j}\omega'_y\rangle$; and y is a critical point (V1.4) of $E' = F|(N^m \cap O)$ iff ω_y is an unbroken geodesic. (iv) For every geodesic $\gamma \in \Omega$ with $E(\gamma) < c$, the break points y of γ lie in U, ω_y agrees with γ on $[0,1]$ and is an unbroken geodesic, and $E'(y) = E(\gamma)$. *Book:* p. 90, Theorem 16.2: the critical points of E' are precisely the critical points of E in $\operatorname{Int}\Omega^c$, namely the unbroken geodesics from p to q of length less than $\sqrt{c}$; p. 91: every geodesic is a broken geodesic, so every critical point of E in $\operatorname{Int}\Omega^c$ lies in B, and with the first variation formula (§12.2) "it is clear" that the critical points of E' are the unbroken geodesics. *Hidden hypotheses:* that every tangent vector of B at γ is the variation field of a variation through broken geodesics, smooth on the strips, which needs the smooth dependence of a geodesic segment on its end points over the whole segment (V9.9 (iii)); testing one break point at a time, with the tangent vector $\Delta_{t_j}\gamma'$; a broken geodesic without velocity jumps is unbroken (Corollary 12.3, V8.11). *Deviation (restated):* the critical points of E are the unbroken geodesics by Corollary 12.3; E' is read through a smooth ambient extension F, as functions on submanifolds are in Part I (V1.6), and the statement holds for every such F and O.

V10.10 (broken Jacobi fields for the subdivision of §16, and Lemma 15.4 for it; `Morse.IsJacobiField.comp_affi` `Morse.nonneg_of_isLocalMin_of_hasDerivAt`, `Morse.BrokenSetup.energy'_nodeVal_le`, `Morse.BrokenSetup.not_isConjugate_seg`, `Morse.BrokenSetup.exists_brokenSpace_node`, `Morse.BrokenSetup.brokenSpace_eq_zero`, `Morse.BrokenSetup.brokenNodeMap_bijective`, `Morse.BrokenSetup.energyHessian_nodeZero_nonneg`, `Morse.BrokenSetup.energyHessian_nodeZero_pos`, `Morse.BrokenSetup.isCompl_brokenSpace_nodeZero`, `Morse.BrokenSetup.formIndex_brokenSpace`, `Morse.BrokenSetup.finrank_ker_brokenSpace`**).** Fix Milnor's data and $y \in U$ with $\gamma = \omega_y$ an unbroken geodesic. (i) A Jacobi field along γ, restricted to a segment and reparametrized affinely, is a Jacobi field along the reparametrized segment. (ii) For every i, t_i and t_{i+1} are not conjugate along γ (V9.4). (iii) The broken Jacobi fields along γ for the break points $t_1, \dots, t_m$ (V9.17, with $\tau = 1$) with arbitrary tangent values at the t_j exist and are unique: $W \mapsto (W(t_1), \dots, W(t_m))$ is a linear isomorphism onto $T_{y_1}N \times \cdots \times T_{y_m}N$. (iv) For every $\omega \in \Omega$, $E'(\omega(t_1), \dots, \omega(t_m)) \le E(\omega)$; hence $E_{**}(W,W) \ge 0$ for every $W \in T\Omega_\gamma$ vanishing at the t_j, and $E_{**}(W,W) > 0$ if moreover $W \neq 0$ and W vanishes off $[0,1]$. (v) The fields of $T\Omega_\gamma$ vanishing off $[0,1]$ are the direct sum of the broken Jacobi fields and of those vanishing at the t_j; the index of E_{**} is the index of its restriction to the broken Jacobi fields, and the nullity of this restriction is the nullity of E_{**}. *Book:* pp. 91–92: TB_γ is identified with the space $T\Omega_\gamma(t_0, \dots, t_k)$ of broken Jacobi fields of §15; p. 92: the equality of the indices and nullities "is an immediate consequence of Lemma 15.4". *Hidden hypotheses:* Lemma 15.4 (V9.19) is proved for a subdivision chosen in §15, fine in the sense of V9.16, which the subdivision of §16 need not be; its proof needs (a) Jacobi fields on each segment with arbitrary end values, and unique, and (b) E_{**} positive definite on the fields vanishing at the break points. For §16, (a) holds because each segment is the radial geodesic of a vector shorter than a normal radius (V9.11 (i)), and (b) because the broken geodesic minimizes the energy among the paths through its break points, (a) excluding null vectors. *Deviation (not in the book):* (iv)–(v) are Lemma 15.4 proved anew for this subdivision. *Proof:* (iv) from Schwarz's inequality on each segment, Remark 13.5 (V8.17) and the fact that a minimum has a nonnegative second derivative; a null vector vanishing at the t_j would be a Jacobi field on each segment vanishing at both ends, hence 0 by (ii).

V10.11 (Theorem 16.2: the index and the nullity of E'; `Morse.hasDerivAt_deriv_line`, `Morse.BrokenSetup.chartBrokenEquiv`, `Morse.BrokenSetup.hessian_apply_self`, `Morse.BrokenSetup.hessi` `Morse.BrokenSetup.theorem_16_2_index`**).** Let O and F be as in V10.9 (iii), and $y \in N^m \cap O$ with $\gamma = \omega_y$ an unbroken geodesic; let κ be the slice chart of $N^m \cap O$ at y, x its parametrization

and $z_0 = \kappa(y)$. For z in the model space Z^m let Λz be the broken Jacobi field (V10.10 (iii)) with the values $dx_{z_0}(z)$ at the break points; Λ is a linear isomorphism onto the broken Jacobi fields. (i) For G of class C^2 near z_0, the second derivative at 0 of $u \mapsto G(z_0 + uz)$ is $G_{**}(z, z)$. (ii) The Hessian of $F \circ x$ at z_0 is E_{**} on the broken Jacobi fields: $(F \circ x)_{**}(z, z') = E_{**}(\Lambda z, \Lambda z')$. (iii) **Theorem 16.2:** the index of E' at y (V1.4) equals the index of E_{**} at γ (V9.13), and the nullity of E' at y equals the nullity of E_{**} at γ (V9.6). *Book:* p. 90, Theorem 16.2: the index [or the nullity] of the Hessian E'_{**} at each critical point γ is equal to the index [or the nullity] of E_{**} at γ; p. 92: for a variation $\bar{\alpha} : (-\varepsilon, \varepsilon) \to B$ of γ through broken geodesics the variation field is "clearly a broken Jacobi field" (compare §14.3); then Lemma 15.4. *Hidden hypotheses:* the variation field of a variation through broken geodesics is, on each segment, a Jacobi field along the segment reparametrized to $[0, 1]$ (Lemma 14.3, V9.8, with V10.10 (i)); that it is the broken Jacobi field with the same values at the break points needs their uniqueness (V10.10 (ii)); the Hessians are compared first as quadratic forms, then polarized. *Proof:* $F(x(z_0 + uz)) = E(\omega_{x(z_0+uz)})$ for a variation through broken geodesics (V10.9 (i)), so by Remark 13.5 (V8.17) the second derivative is $E_{**}(W, W)$, W its variation field, which is Λz; then V10.10 (v).

V10.12 (Theorem 16.3 in the model; `Morse.exists_contDiff_eq_of_locally`, `Morse.isFiniteCellComplexType` `Morse.BrokenSetup.exists_contDiff_eq_energy'`, `Morse.BrokenSetup.geodIndex`, `Morse.BrokenSetup.theo` `Morse.BrokenSetup.theorem_16_3_count`, `Morse.BrokenSetup.theorem_16_3_nonconj`**).** (i) If $K \subseteq S'$ are subsets of a finite-dimensional normed space, K compact, and f agrees on S' near each point of K with a function smooth on an open set, then f agrees on K with a function smooth everywhere. (ii) (Theorem 3.5 for a compact sublevel set) Let N be a slice submanifold, F smooth on V, $f = F|N$, and a_0 a real number that is not a critical value of f, with $N^{a_0} = f^{-1}(-\infty, a_0]$ compact and all critical points of f in N^{a_0} non-degenerate. Then the critical points of f in N^{a_0}, listed by non-decreasing value, are finitely many, and N^{a_0} has the homotopy type of an iterated adjunction space (V2.11) with one cell of dimension the index for each of them. (iii) **Theorem 16.3 in the model:** for Milnor's data and $a < c$, suppose $E'(y) \neq a$ for every $y \in U$ with ω_y a geodesic, and E_{**} has nullity 0 at every such ω_y with $E'(y) \leq a$ (equivalently, 0 and 1 are not conjugate along it). Then the $y \in U$ with ω_y a geodesic and $E'(y) \leq a$ are finitely many, and $\{y \in U : E'(y) \leq a\}$ has the homotopy type of an iterated adjunction space with one cell of dimension the index of E_{**} at ω_y for each of them, attached in order of non-decreasing energy; the number of cells of dimension λ is the number of these geodesics of index λ. *Book:* p. 90, Theorem 16.3 (V10.13); proof: "This follows from 16.2 together with §3.5". *Hidden hypotheses:* Theorem 3.5 (V2.12) concerns a function on a manifold with compact sublevel sets and no degenerate critical point, while B is not compact and only the critical points in B^a are assumed non-degenerate; and a must not be a critical value of E' (Milnor's hypothesis does not exclude a geodesic of energy exactly a, and then B^a is a critical sublevel set). *Deviation (narrower):* a is not the energy of a geodesic; a cell complex in the sense of V2.11. *Proof:* departs from the exposition's, which reparametrizes E' on $\{E' < a'\}$ for a level $a' > a$ below the next critical value: E' is extended by (i) from the compact $\{E' \leq (a + c)/2\}$ to a smooth function on V^m, and (ii) is applied on the manifold $\{y \in U : E'(y) < (a + c)/2\}$, with V10.9 for the critical points and V10.11 for their indices.

V10.13 (Theorem 16.3 for Ω^a; `Morse.BrokenSetup.theorem_16_3_Omega`, `Morse.BrokenSetup.theorem_16_3_Om` Let Milnor's data be given with $c > a$ (by V10.3 they exist for every $c > 0$ when N is closed in V, or connected and complete), and suppose that no geodesic $\gamma \in \Omega$ has $E(\gamma) = a$ and that E_{**} has nullity 0 at every geodesic $\gamma \in \Omega$ with $E(\gamma) \leq a$. Then the geodesics in Ω^a are finitely many, and Ω^a, with the metric d, has the homotopy type of an iterated adjunction space with one cell

of dimension λ for each geodesic in Ω^a of index λ, attached in order of non-decreasing energy. *Book:* p. 90, Theorem 16.3: if M is complete and p, q are not conjugate along any geodesic of length $\leq \sqrt{a}$, then Ω^a has the homotopy type of a finite CW-complex with one cell of dimension λ for each geodesic in Ω^a at which E_{**} has index λ; in particular Ω^a contains only finitely many geodesics. *Hidden hypotheses:* as in V10.12; and the homotopy equivalences $\Omega^a \simeq B^a$ (Theorem 16.2) and between B^a and its break points (V10.5). *Deviation (narrower):* a is not the energy of a geodesic; non-conjugacy is stated as nullity 0, equivalent by Theorem 14.1 (V9.6); the CW-complex is a cell complex in the sense of V2.11. *Proof:* V10.12 (iii), transported along V10.5 and V10.8; every geodesic of Ω^a is some ω_y on $[0, 1]$ (V10.9 (iv)).

V10.14 (the Remark on p. 92: a minimum of E'; `Morse.BrokenSetup.nodeVal_mem_nodes_of_energy_lt`, `Morse.BrokenSetup.exists_energy_minimizing`). For Milnor's data, every $\omega \in \Omega$ with $E(\omega) < c$ has its break points y in U, with $E'(y) \leq E(\omega)$; and if some $\omega_0 \in \Omega$ has $E(\omega_0) < c$, there is $y \in U$ such that ω_y is an unbroken geodesic and $E(\omega_y) \leq E(\omega)$ for every $\omega \in \Omega$. *Book:* p. 92, Remark: an alternative proof of the existence of a minimal geodesic joining two points of a complete manifold: if Ω^a is non-vacuous, B^a is compact and non-vacuous, E' takes its minimum on B^a at some γ, and this γ "will be the required minimal geodesic". *Hidden hypotheses:* that γ minimizes E on all of Ω needs $E(\omega) \geq E'(\omega(t_1), \dots, \omega(t_m))$ for every ω (Schwarz's inequality); that it is a geodesic, that a minimum is a critical point (V10.9); that it is shortest, a reparametrization of paths by arc length. *Deviation (narrower):* minimality is proved for the energy; that ω_y is a minimal geodesic in the sense of V7.24 is not stated.

V10.15 (Ω^* and the inclusion i; `Morse.pathCSet`, `Morse.PathC`, `Morse.PathSp.toPathC`, `Morse.PathSp.edist_toPathC_le`, `Morse.PathSp.continuous_toPathC`, `Morse.pathIncl`). $\Omega^* = \Omega^*(N; p, q)$ is the set of continuous maps $\omega : [0, 1] \to V$ with values in N, $\omega(0) = p$ and $\omega(1) = q$, as a subspace of the space of continuous maps $[0, 1] \to V$ with the uniform distance $d^*(\omega, \eta) = \max_t \|\omega(t) - \eta(t)\|$, whose topology is the compact-open topology. The map $i : \Omega \to \Omega^*$ restricts a path to $[0, 1]$; $d^*(i\omega, i\eta) \leq d(\omega, \eta)$, so i is continuous. *Book:* p. 93: Ω^*, the space of all continuous paths $\omega : [0, 1] \to M$ from p to q with the compact-open topology, which is also induced by $d^*(\omega, \omega') = \max_t \rho(\omega(t), \omega'(t))$; since $d \geq d^*$, the natural map $i : \Omega \to \Omega^*$ is continuous. *Hidden hypotheses:* none. *Deviation (restated):* d^* is measured with the norm of V rather than with ρ; on N the two distances define the same topology (V7.21), so both give the compact-open topology, and $\|x - y\| \leq \rho(x, y)$ gives the inequality.

V10.16 (convergence in Ω controls the velocities; `Morse.PathSp.tendsto_integral_norm_deriv_sub_sq`, `Morse.PathSp.tendsto_of_tendsto_apply_of_integral`, `Morse.tendsto_integral_deriv_comp_sub_sq`). (i) If $\omega_j \to \omega$ in Ω (along any filter), then $\int_0^1 \|\omega_j' - \omega'\|^2 dt \to 0$, although d involves only the speeds. (ii) Conversely, if $\omega_j(s) \to \omega(s)$ for every $s \in [0, 1]$ and $\int_0^1 \|\omega_j' - \omega'\|^2 dt \to 0$, then $\omega_j \to \omega$ in Ω. (iii) If Φ is of class C^1 on an open set containing the curve $F([a, b])$, $F_j \to F$ uniformly on $[a, b]$, and the velocities F_j' are bounded in L^2 and converge to F' in L^2 (all curves differentiable off finite sets), then $(\Phi \circ F_j)' \to (\Phi \circ F)'$ in L^2. *Book:* not stated; it is what the continuity for d of the maps in the proof of Theorem 17.1 needs (p. 94). *Deviation (not in the book):* these are the analytic facts that make maps given pointwise by smooth formulas, such as the deformation of V10.18, continuous for d. *Proof:* (i) the energies converge (V10.2), and the velocities converge weakly in L^2 by integration by parts on the smooth pieces of ω and uniform convergence.

V10.17 (good balls, the exposition's Lemma 17.B; `Morse.IsMinJoin`, `Morse.minJoin`, `Morse.minGeo`, `Morse.IsGoodBall`, `Morse.IsGoodBall.eq_minJoin`, `Morse.IsGoodBall.minGeo_minGeo`,

`Morse.IsGoodBall.exists_smooth_minGeo`, `Morse.exists_isGoodBall_ball`, `Morse.exists_lebesgue_goodBa`
`Morse.exists_smooth_glue`, `Morse.exists_goodSet`, `Morse.JoinSystem`, `Morse.goodJoin`). For $y, z \in V$ the *minimal join vector* $v(y,z)$ is the unique $v \in T_yN$ with $\exp_y(v) = z$ and $\rho(y,z) = \|v\|$ for which the conclusion of Corollary 10.8 holds at some scale $\delta > \|v\|$ (0 if there is none), and $\sigma_{yz}(s) = \exp_y(s\,v(y,z))$. An open $O \subseteq V$ is a *good ball* if, for one $\delta > 0$, any two points of $N \cap O$ are at distance less than δ and joined as in Corollary 10.8 at the scale δ, and $\sigma_{yz}(s) \in N \cap O$ for $y, z \in N \cap O$ and $0 \le s \le 1$ (convexity). (i) For $x \in N$, the balls $\{\|y - x\| < r\}$ of V are good for all small $r > 0$. (ii) For compact $K \subseteq N$ there is $\varepsilon > 0$ such that the ball of radius ε about each point of K lies in a good ball. (iii) In a good ball, σ_{yz} is the unique minimal geodesic from y to z, its sub-arcs are minimal geodesics, and near each pair the map $(y, z, s) \mapsto \sigma_{yz}(s)$ is the restriction of a map smooth on an open set. (iv) Local smooth extensions are glued, by a partition of unity, into one map of (y, z, s), smooth for $-\varepsilon < s < 1 + \varepsilon$, that agrees with σ on a given compact set of pairs. So the balls $\{\|y - x\| < r\}$, $x \in N$, for which the concentric balls of radius up to $2r$ are good and $N \cap \{\|y - x\| \le r\}$ is compact, with the joins σ, form a *join system*: open good sets, convex for the joins, with $\sigma_{yz}(0) = y$, $\sigma_{yz}(1) = z$, $\sigma_{yy} = y$, one smooth map of (y, z, s) on each good set, and a Lebesgue number for every compact subset of N. *Book:* p. 93: for a continuous $f : M \to [0, \infty)$ with compact sets $f^{-1}([0, r])$, $\varepsilon_1(r)$ is the largest number such that any two points of $f^{-1}([0, r])$ at distance less than $\varepsilon_1(r)$ are joined by a unique minimal geodesic which varies differentiably with the end points. *Hidden hypotheses:* the modified proof of Theorem 17.1 needs regions that are convex for these geodesics (J. H. C. Whitehead's convex balls), one smooth map of (y, z, s) on each whole region, and a uniform scale; Milnor's construction provides none of these. *Deviation (not in the book):* the exposition's Lemma 17.B. *Deviation (restated):* good balls are balls of V intersected with N, not metric balls of ρ, and their convexity is proved with the ambient distance: along a geodesic γ, $(\|\gamma - x\|^2)'' = 2\|\gamma'\|^2 + 2\langle \gamma - x, \mathrm{II}(\gamma', \gamma')\rangle > 0$ while γ stays near x, so no normal coordinates or Gauss lemma are needed; smoothness in (y, z, s) uses the smoothness of exp on its whole domain (V11.1).

V10.18 (the exposition's Lemma 17.C; `Morse.dy`, `Morse.JoinSystem.IsGood`, `Morse.JoinSystem.YSet`,
`Morse.JoinSystem.isOpen_YSet`, `Morse.JoinSystem.YSet_mono`, `Morse.JoinSystem.exists_mem_YSet`,
`Morse.JoinSystem.XSet`, `Morse.JoinSystem.XSet_eq_preimage`, `Morse.JoinSystem.broken`,
`Morse.JoinSystem.interp`, `Morse.JoinSystem.continuous_interp_star`, `Morse.JoinSystem.continuous_bro`
`Morse.JoinSystem.continuous_interpΩ`, `Morse.JoinSystem.isHomotopyEquiv_toY`). Fix a join system (V10.17). For $k \ge 0$ let $Y_k \subseteq \Omega^*$ be the set of paths each of whose dyadic pieces $\omega([j2^{-k}, (j+1)2^{-k}])$ lies in a good set, and $X_k = i^{-1}(Y_k) \subseteq \Omega$. (i) The Y_k are open, increasing, and cover Ω^*. (ii) For $\omega \in Y_k$ let $b(\omega)$ be the broken geodesic with vertices $\omega(j2^{-k})$, joined by the joins, and let $H(\omega, u)(s) = \sigma_{\omega(s), b(\omega)(s)}(u)$ be the point at time u of the join from $\omega(s)$ to $b(\omega)(s)$; $H(\omega, 0) = \omega$ and $H(\omega, 1) = b(\omega)$. Then H is continuous from $Y_k \times [0,1]$ to Y_k for d^* and from $X_k \times [0,1]$ to X_k for d, and b is continuous from (Y_k, d^*) to (X_k, d). (iii) Hence $i : X_k \to Y_k$ is a homotopy equivalence with homotopy inverse b. *Book:* p. 94: $h(\omega)$, the broken geodesic agreeing with ω at $t = 0, A(\omega), \dots, kA(\omega)$ and 1; an argument similar to that in §16 shows $h \circ i \simeq \mathrm{id}$ and $i \circ h \simeq \mathrm{id}$. *Hidden hypotheses:* the deformation must stay where the joins are defined (convexity, V10.17); continuity for d of a map given by a smooth formula needs the velocities, not only the speeds, to converge in L^2 (V10.16); d being only a pseudometric, $b \circ i \simeq \mathrm{id}$ holds up to paths at distance 0, which is harmless. *Deviation (not in the book):* the dyadic sets Y_k and X_k, on each of which the subdivision is fixed. *Deviation (restated):* the deformation moves every point $\omega(s)$ along the join to $b(\omega)(s)$, instead of moving a break point along the path as Milnor's r_u (and the exposition's proof) does.

V10.19 (Theorem 17.1; `Morse.isHomotopyDirectLimit_preimage`, `Morse.theoremA_general`, `Morse.JoinSystem.isHomotopyEquiv_toPathC`, `Morse.theorem_17_1`). (i) If Y is paracompact and Hausdorff, $B_0 \subseteq B_1 \subseteq \cdots$ are subsets of Y such that every point lies in the interior of some B_n, and $g : X \to Y$ is continuous, then X is the homotopy direct limit (VA.2) of the sets $g^{-1}(B_n)$. (ii) For every join system, $i : \Omega(N;p,q) \to \Omega^*(N;p,q)$ is a homotopy equivalence (VA.1). (iii) **Theorem 17.1:** for every slice submanifold N of V, complete or not, and all p, q, the inclusion $i : \Omega \to \Omega^*$, with Milnor's metric d on Ω and the compact-open topology on Ω^*, is a homotopy equivalence. *Book:* p. 93, Theorem 17.1: the natural map i is a homotopy equivalence between Ω and Ω^*. Proof, pp. 93–94: with f and ε_1 as in V10.17, a continuous ε_2 with $0 < \varepsilon_2 < \varepsilon_1$, $\varepsilon(\omega) = \varepsilon_2(\max_t f(\omega(t)))$ and $F(\omega,\alpha) = (\alpha-1)\varepsilon(\omega) + \max_{|t-t'|\le\alpha} \rho(\omega(t),\omega(t'))$, the zero $\alpha = 2A(\omega)$ of $F(\omega,\cdot)$ defines a continuous $A : \Omega^* \to (0,1]$ such that points of ω at times less than $2A(\omega)$ apart are joined by a unique minimal geodesic; $h(\omega)$ is the path that agrees with ω at $t = 0, A(\omega), 2A(\omega), \dots, kA(\omega)$ (k the largest integer in $1/A(\omega)$) and at $t = 1$, and is a geodesic in between; "Clearly h is continuous"; and an argument similar to that in §16 gives $h \circ i \simeq \mathrm{id}$ and $i \circ h \simeq \mathrm{id}$. *A step that does not hold as we read it:* "Clearly h is continuous" (p. 94) fails for the metric d of Ω, the topology in which Theorem 17.1 is stated (h is continuous only for d^*). When $1/A(\omega)$ crosses an integer from below, $h(\omega)$ acquires a new last segment of duration $\delta \to 0$ and length $\ell \to 0$, whose energy ℓ^2/δ need not tend to 0; for a continuous path that is not Lipschitz near $t = 1$ it can tend to infinity, and since $\sqrt{E}$ is 1-Lipschitz for d (V10.2), h is then not continuous. The exposition's example: $M = \mathbb{R}$, $p = q = 0$, $f(x) = |x|$, $\varepsilon_2 \equiv 1$ and $\omega_\lambda(t) = \lambda((1-t)^{1/3} - (1-t))$; as λ decreases to $\lambda_* = \sqrt{3}/2$, $\omega_\lambda \to \omega_{\lambda_*}$ uniformly while $E(h(\omega_\lambda)) \to \infty$. This counterexample is machine-checked (`Morse.milnor_h_not_continuous`, `Morse.milnor_h_not_continuous'`): with Milnor's A and h defined exactly as on p. 94 for this choice, h is not continuous at ω_{λ_*} (it was first found by hand and numerically: `Exposition/Chapter10.md`, §5; `Exposition/Findings.md`). The theorem is true, and what is verified is the theorem itself, by the exposition's modified proof. (Also, A takes its values in $(0, \frac{1}{2}]$.) *Hidden hypotheses (of the modified proof):* good balls convex for minimal geodesics, with one smooth join on each (V10.17); the L^2 control of velocities by d (V10.16); Ω is not Hausdorff for d, so Example 1 of the Appendix (paracompact Hausdorff spaces) is applied to Ω^* and pulled back by i; Theorem A for open, not closed, pieces (VA.15). *Proof:* the exposition's, not Milnor's: the open sets Y_k exhaust the metric, hence paracompact, space Ω^*, which is therefore the homotopy direct limit of the Y_k (Example 1, VA.7), and Ω is that of the $X_k = i^{-1}(Y_k)$ by (i); i carries each X_k into Y_k by a homotopy equivalence (V10.18); Theorem A with no closedness hypothesis (VA.15) concludes. No completeness is used.

V10.20 (relative cell complexes and the tower; `Morse.IsRelCellComplexType`, `Morse.IsRelCellAttachment.tra` `Morse.IsRelCellComplexType.of_square`, `Morse.isRelCellComplexType_union_cells`, `Morse.isRelCellComplexType_sublevel_of_isCompact`, `Morse.isCountableCellComplexType_of_rel`, `Morse.isCountableCellComplexType_of_relSublevel`). (i) A map $\iota : A \to X$ is, *up to homotopy, the inclusion of A with cells $(\lambda_1, \dots, \lambda_k)$ attached* if there are a space K obtained from A by attaching a λ_1-cell, then a λ_2-cell, and so on (VA.18), with $j : A \to K$ the composite of the inclusions, and a homotopy equivalence $e : X \to K$ with $e \circ \iota \simeq j$. (ii) (Lemma 3.7, relative and iterated) Such an attachment $L \to K$ and a homotopy equivalence $L \to L'$ give an attachment $L' \to K'$ of the same cells and a homotopy equivalence $K \to K'$ extending $L \to L'$; and the notion in (i) is invariant under homotopy-commutative squares of homotopy equivalences. (iii) Disjoint embedded cells meeting a closed set exactly in their boundary spheres are attached to it one at a time (the exposition's Lemma D, relative). (iv) (the relative Theorem 3.5) For $f = F|N$ on a slice submanifold, F smooth, and levels $b \le a_0$, neither a critical value, with N^{a_0} compact and the critical points in N^{a_0} non-degenerate, the inclusion $N^b \to N^{a_0}$ is, up to homotopy, the inclusion

of N^b with one cell of dimension the index for each critical point in $f^{-1}(b, a_0)$, attached in order of non-decreasing value. (v) (the tower) If $E : X \to \mathbb{R}$ is continuous, $a_0 \le a_1 \le \cdots$ tend to ∞, $X^{a_0} = \{E \le a_0\}$ has the homotopy type of a finite cell complex with cells cells_0 (V2.11), and each inclusion $X^{a_n} \to X^{a_{n+1}}$ is, up to homotopy, the inclusion with cells cells_{n+1} attached, then X has the homotopy type of the countable cell complex with cells $\text{cells}_0, \text{cells}_1, \ldots$ (VA.18). *Book:* p. 95: by 16.2 together with 3.3 and 3.7, each Ω^{a_i} has the homotopy type of $\Omega^{a_{i-1}}$ with finitely many cells attached; "just as in the proof of 3.5", one constructs CW-complexes $K_0 \subset K_1 \subset \cdots$ with cells of the required description and homotopy equivalences $\Omega^{a_i} \to K_i$ forming a commutative diagram. *Hidden hypotheses:* "each extending the previous one": Lemma 3.7 extends a homotopy equivalence over the new cells, but $\Omega^{a_{i+1}}$ is only homotopy equivalent to Ω^{a_i} with cells attached, by retractions that move Ω^{a_i}, so the extension holds only up to homotopy (strict compatibility needs nested subdivisions, the exposition's Lemma 17.E); and the step needs a relative form of Theorem 3.5, for pairs of sublevel sets. *Deviation (restated):* the Appendix's comparison of telescopes needs only homotopy-commutative squares (VA.16 (ii), VA.19), so the tower is built up to homotopy and no nested subdivisions are needed.

V10.21 (the step from Ω^a to Ω^b; `Morse.PathSp.energyC`, `Morse.BrokenSetup.isRelCellComplexType_nodes`, `Morse.BrokenSetup.omegaEquivNodes`, `Morse.BrokenSetup.isRelCellComplexType_Omega`, `Morse.BrokenSetup.finite_geodesics`, `Morse.finite_geodesicEnergies`, `Morse.geodesicIndex`, `Morse.geodesicIndex_eq`). (i) For Milnor's data and levels $a \le b < c$, if neither a nor b is $E'(y)$ for a $y \in U$ with ω_y a geodesic, and every such ω_y with $E'(y) \le b$ has nullity 0, then the inclusion $\Omega^a \to \Omega^b$ is, up to homotopy (V10.20 (i)), the inclusion of Ω^a with one cell of dimension the index of E_{**} at ω_y for each such y with $a < E'(y) < b$, attached in order of non-decreasing energy; and likewise for the sublevel sets of E' in U. The homotopy equivalences $\Omega^t \simeq \{y \in U : E'(y) \le t\}$, $t < c$, are given by one formula, $\omega \mapsto$ the break points of $r_1(\omega)$, so they commute with the inclusions. (ii) If every ω_y, $y \in U$, that is a geodesic with $E'(y) \le b < c$ has nullity 0, there are finitely many such y; if N is closed in V, $p, q \in N$, and every geodesic in Ω has nullity 0, the energies at most b of the geodesics in Ω are finitely many, for every b. (iii) The index of a geodesic, as a natural number, equals the index of E_{**} at it (finite by Theorem 15.1, V9.24). *Book:* p. 95: by 16.2 together with 3.3 and 3.7, each Ω^{a_i} has the homotopy type of $\Omega^{a_{i-1}}$ with one λ-cell attached for each geodesic of index λ with energy in (a_{i-1}, a_i). *Hidden hypotheses:* the choice of levels needs the critical values of E to be discrete, which follows from Theorem 16.3 (finitely many geodesics below each level); the homotopy equivalences with the models must commute with the inclusions. *Deviation (restated):* each step uses its own manifold of broken geodesics, with an energy bound $c > b$, and any finite number of critical values between a and b.

V10.22 (Theorem 17.3, the fundamental theorem of Morse theory; `Morse.theorem_17_3_of_levels`, `Morse.theorem_17_3`, `Morse.theorem_17_3_nonconj`). Let N be closed in V and $p, q \in N$. (i) Suppose every geodesic in Ω has nullity 0, and let $a_0 \le a_1 \le \cdots$ be levels tending to ∞, with $a_0 < 0$, none of which is the energy of a geodesic. Then there are finite lists $\text{geo}_0 = \emptyset, \text{geo}_1, \text{geo}_2, \ldots$ of geodesics in Ω, geo_{n+1} consisting of geodesics with energy in (a_n, a_{n+1}), containing one geodesic that agrees on $[0, 1]$ with each geodesic of that energy and no two that agree on $[0, 1]$, ordered by energy, such that Ω, with Milnor's metric, has the homotopy type of the countable cell complex (VA.18) with one cell of dimension the index of E_{**} at γ for each entry γ. (ii) **Theorem 17.3:** if every geodesic in Ω has nullity 0, there are finite lists $\text{geo}_0, \text{geo}_1, \ldots$ of geodesics from p to q such that every geodesic from p to q agrees on $[0, 1]$ with exactly one entry of exactly one list, the energies increase strictly from one list to the next and weakly within a list, and $\Omega(N; p, q)$, with Milnor's metric, has the homotopy type of the countable cell complex with one cell of dimension

λ for each entry of index λ: one λ-cell for each geodesic from p to q of index λ, attached in order of increasing energy. (iii) The same holds if q is not conjugate to p along any geodesic (V11.4), which makes every nullity 0 (Theorem 14.1, V9.6). *Book:* p. 95, Theorem 17.3: if M is complete and p, q are not conjugate along any geodesic, then $\Omega(M; p, q)$ (or $\Omega^*(M; p, q)$) has the homotopy type of a countable CW-complex containing one cell of dimension λ for each geodesic from p to q of index λ. Proof, p. 95: levels $a_0 < a_1 < \cdots$, not critical values, with "precisely one" critical value in each interval and Ω^{a_0} empty; 16.2 with 3.3 and 3.7 at each step; the tower of CW-complexes (V10.20); the direct limit map $\Omega \to K$ induces isomorphisms of homotopy groups, hence is a homotopy equivalence by Whitehead's theorem, since Ω has the homotopy type of a CW-complex (17.2); and a reference to a different proof, not using 17.1, on p. 149. *Hidden hypotheses:* the a_i must tend to infinity, or the Ω^{a_i} do not exhaust Ω; "precisely one" critical value in each interval is impossible when there are finitely many geodesics, and "at most one" suffices; the compatibility of the tower (V10.20). *Deviation (narrower):* completeness is replaced by the closedness of N in V, which implies it (V7.30) and is used only through the compactness of the balls S (V10.3); the statement is for Ω, not for Ω^*; the "countable CW-complex" is the mapping telescope of a tower of cell attachments (VA.18). *Deviation (restated):* geodesics are counted by their values on $[0, 1]$; an interval between consecutive levels may contain several critical values, whose cells are attached in order of energy. *Proof:* Milnor's second route, Theorem A of the Appendix (VA.19), through V10.20 (v) and V10.21; neither Whitehead's theorem nor Corollary 17.2 is used.

V10.23 (the geodesics from p to q on S^n, and their indices; `Morse.greatCircle`, `Morse.isGeodesicOn_greatCirc` `Morse.energy_greatCircle`, `Morse.exists_eqOn_greatCircle`, `Morse.isJacobiField_sin_smul`, `Morse.isJacobiField_smul_deriv_greatCircle`, `Morse.conjMultiplicity_greatCircle`, `Morse.sphLen`, `Morse.sphGeod`, `Morse.mul_pi_lt_sphLen`, `Morse.sphLen_lt`, `Morse.energy_sphGeod`, `Morse.exists_eqOn_sphGeod`, `Morse.notMem_conjLocus_sph`, `Morse.formIndex_sphGeod`, `Morse.formIndex_eq_of_eqOn_sphGeod`, `Morse.formIndex_energyHessian_congr`). $S^n \subseteq \mathbb{R}^{n+1}$ is the unit sphere with the induced metric (V6.25), $n \geq 1$. (i) For unit vectors $p \perp u$ and $L \in \mathbb{R}$, $c(t) = \cos(Lt)\, p + \sin(Lt)\, u$ is a geodesic of S^n on $\mathbb{R}$, with energy L^2 on $[0, 1]$; every geodesic $\gamma \in \Omega(S^n; p, q)$ agrees on $[0, 1]$ with such a c, with $L \geq 0$. (ii) For every $e \perp p, u$, $t \mapsto \sin(Lt)\, e$ is a Jacobi field along c, and so is $t \mapsto t\, c'(t)$; for $L > 0$ and $0 < t \leq 1$, the multiplicity of 0 and t as conjugate parameters along c (V9.4) is $n - 1$ if Lt is a multiple of π, and 0 otherwise. (iii) Let $q \neq \pm p$, $\theta \in (0, \pi)$ the angle between p and q, and $u \perp p$ the unit vector with $q = \cos\theta\, p + \sin\theta\, u$; let γ_{2j} be the great circle along u with $L_{2j} = \theta + 2\pi j$, and γ_{2j+1} the one along $-u$ with $L_{2j+1} = 2\pi(j + 1) - \theta$. These are geodesics in $\Omega(S^n; p, q)$ with $E(\gamma_k) = L_k^2$ and $k\pi < L_k < (k + 1)\pi$; every geodesic in $\Omega(S^n; p, q)$ agrees on $[0, 1]$ with some γ_k; q is not conjugate to p along any geodesic (V11.4); and the index of E_{**} at γ_k, and at every geodesic that agrees with γ_k on $[0, 1]$, is $k(n - 1)$. *Book:* pp. 95–96, the example of the path space of S^n: for $q \neq p, p'$, p' the antipode of p, the geodesics from p to q are γ_0, the short great circle arc, γ_1, the long arc $pq'p'q$, $\gamma_2 = pqp'q'pq$, and so on; k is the number of times p or p' occurs in the interior of γ_k, and $\lambda(\gamma_k) = k(n - 1)$ since each of these points is conjugate to p with multiplicity $n - 1$; p. 80 (Example 2 of §14): antipodal points are conjugate along a great semicircle with multiplicity $n - 1$. *Hidden hypotheses:* that $q \neq p, p'$ makes p, q non-conjugate along every geodesic needs the conjugate points along every great circle through p and the list of all geodesics from p to q; the multiplicity is exactly $n - 1$ (Remark 14.2, V9.7) and no other interior point is conjugate to p; that every geodesic is a great circle uses the uniqueness of geodesics after extending the segment past $t = 0$ (V9.9 (ii)); "k occurrences" is the inequality $k\pi < L_k < (k + 1)\pi$; the index comes from Theorem 15.1 (V9.24). *Deviation (restated):* of Milnor's rephrasing of non-conjugacy as $q \neq p, p'$, the direction Theorem 17.3 needs is proved; the index does not depend on the values off $[0, 1]$.

Proof: the Jacobi fields are checked directly with $R(x,y)z = \langle x,z\rangle y - \langle y,z\rangle x$, not by rotations and Lemma 14.3; the fields $\sin(Lt)\,e$ and $t\,c'(t)$ span the n-dimensional space of Jacobi fields vanishing at 0 (V9.7).

V10.24 (Corollary 17.4 for $\Omega(S^n;p,q)$; `Morse.oneCellEach`, `Morse.omega_cells_of_unique`, `Morse.strictMono_energy_sphGeod`, `Morse.tendsto_energy_sphGeod`, `Morse.corollary_17_4`). (i) Let N be closed in V, $p,q \in N$, every geodesic in Ω of nullity 0, and suppose that the geodesics in Ω are, up to their values off $[0,1]$, a sequence $g_0, g_1, \ldots$ with strictly increasing energies tending to ∞, every geodesic agreeing with g_k on $[0,1]$ having index d_k. Then Ω has the homotopy type of the countable cell complex (VA.18) with no cell at stage 0 and one cell of dimension d_k at stage $k+1$. (ii) **Corollary 17.4:** for $n \geq 2$ and $p,q \in S^n$ with $q \neq \pm p$, $\Omega(S^n;p,q)$, with Milnor's metric, has the homotopy type of the countable cell complex with one cell in each of the dimensions $0, n-1, 2(n-1), \ldots$, attached in this order. *Book:* p. 96, Corollary 17.4: the loop space $\Omega(S^n)$ has the homotopy type of a CW-complex with one cell each in the dimensions $0, n-1, 2(n-1), 3(n-1), \ldots$, derived from Theorem 17.3 and the example (V10.23), whose end points are non-conjugate. *Hidden hypotheses:* in the loop space, p is conjugate to itself along every great circle; passing to loops needs that $\Omega^*(S^n;p,q)$ does not depend on q up to homotopy (the exposition's Lemma 17.D) and Theorem 17.1. *Deviation (narrower):* the path space $\Omega(S^n;p,q)$, $q \neq \pm p$, which is what Milnor's argument gives, not the loop space; $n \geq 2$ (the case $n = 1$, with every cell a point, is not stated); a countable cell complex in the sense of VA.18. *Proof:* V10.22 (i) with levels halfway between consecutive energies, one geodesic in each interval, and V10.23.

V10.25 (the homology of a path space read off from its geodesics; `Morse.Homology.subsingleton_subsetHomolo` `Morse.Homology.nonempty_linearEquiv_spaceHomology_of_exhaustion`, `Morse.Homology.bettiNumber_eq_` `Morse.Homology.exists_subset_sublevel_of_isCompact`, `Morse.Homology.sublevel_step_of_isCompact`, `Morse.BrokenSetup.finrank_subsetHomology_Omega`, `Morse.windowLevels`, `Morse.finrank_window`, `Morse.bettiNumber_omega_of_unique`). Homology is singular homology with coefficients in a field $\Bbbk$, and $b_i = \dim_{\Bbbk} H_i$ (V3.33). (i) Let $A_0 \subseteq A_1 \subseteq \cdots$ be subsets of a space X such that every compact subset of X lies in some A_k (for instance the sublevel sets of a continuous function at levels unbounded above). If $H_j(A_k, A_{k_0}) = 0$ for all $k \geq k_0$, then $H_j(X, A_{k_0}) = 0$; and if the steps $H_*(A_{k+1}, A_k)$ vanish in degrees i and $i+1$ for all $k \geq k_0$, then $b_i(X) = \dim H_i(A_{k_0})$. (ii) The homology of a slab (V3.33–V3.34) is computed in the same way for a slice submanifold that need not be compact, below a level whose sublevel set is compact. (iii) For Milnor's data and levels $a \leq b < c$, neither the energy of a geodesic, with at most one energy of a geodesic in (a,b) and every geodesic of energy at most b of nullity 0, $H_j(\Omega^b, \Omega^a)$ is finite-dimensional, of dimension the number of geodesics with energy in (a,b) and index j. (iv) In the situation of V10.24 (i), with levels halfway between consecutive energies, $H_j(\Omega^{a_{k+1}}, \Omega^{a_k})$ is $\Bbbk$ for $j = d_k$ and 0 otherwise; and if $d_k = kr$ with $r \geq 2$, then $b_i(\Omega) = 1$ when r divides i and 0 otherwise. *Book:* p. 96: for $n > 2$ the homology of $\Omega(S^n)$ "can be computed immediately" from the cells of Corollary 17.4. *Hidden hypotheses:* reading the homology off the cells is cellular homology, which needs cellular attaching maps; that the homology of Ω is the limit of that of the Ω^a needs every compact subset of Ω to have bounded energy, true for d since E is continuous (V10.2), false for d^*; the vanishing is needed in degrees i and $i+1$. *Deviation (restated):* the homology is computed from the filtration of Ω by the sets Ω^a, with the lacunary principle (Corollary 5.4, V3.15) on the finite stages, rather than from a cellular chain complex; the coefficients form a field.

V10.26 (the homology of $\Omega(S^n;p,q)$; `Morse.bettiNumber_sphere_paths`). For $n \geq 3$, $p,q \in S^n$ with $q \neq \pm p$, and every field $\Bbbk$, $\dim_{\Bbbk} H_i(\Omega(S^n;p,q);\Bbbk)$ is 1 if $n-1$ divides i and 0 otherwise. *Book:* p. 96: for $n > 2$ the homology of $\Omega(S^n)$ is computed from Corollary 17.4, and $\Omega(S^n)$ has

non-trivial homology in infinitely many dimensions. *Hidden hypotheses:* $n > 2$, so that no two cells have consecutive dimensions; for loops, Lemma 17.D and Theorem 17.1 as in V10.24. *Deviation (narrower):* the path space with $q \neq \pm p$; coefficients in a field, not the integers. *Proof:* V10.25 (iv) with $r = n - 1$, and V10.23.

V10.27 (Corollary 17.5 needs completeness; `Morse.isSliceSubmanifold_of_isOpen`, `Morse.tangentSubspace_c` `Morse.curvR_eq_zero_of_top`, `Morse.IsUnbrokenGeodesic.eqOn_segment_of_top`, `Morse.IsJacobiField.eqO` `Morse.not_isConjugate_of_top`, `Morse.formIndex_energyHessian_eq_zero_of_top`, `Morse.hessNullity_eq_` `Morse.unique_geodesic_of_top`, `Morse.puncturedSub`, `Morse.not_isGeodesicallyComplete_puncturedSub`, `Morse.not_isClosed_punctured`, `Morse.isPathConnected_punctured`, `Morse.puncturedHomotopyEquiv`, `Morse.remark_17_5_completeness`). (i) An open subset of V is a slice submanifold of V of codimension 0, with every tangent space equal to V. (ii) If every tangent space of N is V, the induced metric is flat ($P = 1$, II $= 0$, $R = 0$); every geodesic in $\Omega(N; p, q)$ agrees on $[0, 1]$ with the segment $t \mapsto p + t(q - p)$; Jacobi fields are affine, so no two parameters are conjugate along a curve in N, and every geodesic has index 0 and nullity 0; and if the segment from p to q lies in N, it is a geodesic in $\Omega(N; p, q)$ and every geodesic in $\Omega(N; p, q)$ agrees with it on $[0, 1]$. (iii) For $n \geq 1$, $N = \mathbb{R}^{n+1} \setminus \{0\}$ with the induced (flat) metric is homotopy equivalent to S^n and path-connected; it is neither closed in $\mathbb{R}^{n+1}$ nor geodesically complete ($\exp_q(-q)$ is undefined); no point of N is conjugate to another along any geodesic; and there are $p \neq q$ in N joined by exactly one geodesic. *Book:* p. 96, Corollary 17.5: if M has the homotopy type of S^n, $n > 2$, any two non-conjugate points of M are joined by infinitely many geodesics; completeness is not among its hypotheses, although the proof rests on Theorem 17.3, which assumes it (the Remark on p. 97 assumes it). *Hidden hypotheses:* the completeness of M. *Deviation (not in the book):* the example shows that this hypothesis cannot be dropped: for $n \geq 3$, N satisfies the hypotheses of Corollary 17.5 as printed and not its conclusion. That $\Omega(N; p, q)$ is not homotopy equivalent to a point, which would contradict Theorem 17.3 without completeness for every $n \geq 1$, is not formalized; what is proved is that N satisfies every hypothesis of the formal Theorem 17.3 (V10.22 (iii)) except closedness in the ambient space.

V10.28 (moving the end point, the exposition's Lemma 17.D with Theorem 17.1; `Morse.joined_of_riemDist_lt` `Morse.exists_joined_notMem_conjLocus`, `Morse.pathSpHomotopyEquiv`, `Morse.Homology.spaceHomologyEqui` `Morse.Homology.bettiNumber_eq_of_homotopyEquiv`). Let N be a slice submanifold of V and $p, q \in N$. (i) Two points of N at finite Riemannian distance are joined by a path in N, and q is joined by a path in N to a point $q' \in N$ which is not conjugate to p along any geodesic. (ii) A path in N from q to q' gives a homotopy equivalence $\Omega(N; p, q) \simeq \Omega(N; p, q')$ of path spaces with Milnor's metric. (iii) Homotopy equivalent spaces have isomorphic homology over a field, hence the same Betti numbers. *Book:* p. 96: the loop space of S^n is treated through the paths from p to a point $q \neq p, p'$, without comment; p. 106: Theorem 19.6 is stated for all p, q and proved for non-conjugate ones. *Hidden hypotheses:* that the homotopy type of the path space does not depend on the end point within a path component. *Deviation (not in the book):* (i) is the density of the non-conjugate points (V11.6) in a small ball about q; (ii) composes Theorem 17.1 (V10.19), the homeomorphism of Ω^* with Mathlib's space of paths, and composition of paths with the given one.

V10.29 (Corollary 17.2; `Morse.corollary_17_2`, `Morse.corollary_17_2_pathC`). Let N be closed in V and $p, q \in N$. Then $\Omega(N; p, q)$, with Milnor's metric, has the homotopy type of a countable cell complex (VA.18), and so does the space $\Omega^*(N; p, q)$ of continuous paths with the compact-open topology. *Book:* p. 95: "It is known that the space Ω^* has the homotopy type of a CW-complex" (Milnor, 1959); therefore (Corollary 17.2) Ω has the homotopy type of a CW-complex. *Hidden hy-*

potheses: none. *Deviation (restated):* Milnor's theorem of 1959 is not used: q is moved to a point q' not conjugate to p (V10.28), and Theorem 17.3 (V10.22) gives a countable cell complex for $\Omega(N; p, q')$. *Deviation (narrower):* N closed in V, hence complete, which Milnor's corollary does not assume; a countable cell complex in the sense of VA.18, with cells attached along arbitrary maps.

V10.30 (Corollary 17.4 for the loop space, and its homology; `Morse.exists_loop_homotopyEquiv_sph`, `Morse.corollary_17_4_loop`, `Morse.bettiNumber_sphere_loops`). Let $n \geq 2$ and $p \in S^n$. The loop space $\Omega(S^n; p, p)$ is homotopy equivalent to $\Omega(S^n; p, q)$ for some $q \neq \pm p$, and it has the homotopy type of the countable cell complex with one cell in each of the dimensions $0, n-1, 2(n-1), 3(n-1), \ldots$, attached in this order. For $n \geq 3$ and every field $\Bbbk$, $\dim H_i(\Omega(S^n; p, p); \Bbbk)$ is 1 if $n-1$ divides i and 0 otherwise. *Book:* p. 96, Corollary 17.4: the loop space $\Omega(S^n)$ has the homotopy type of a CW-complex with one cell each in the dimensions $0, n-1, 2(n-1), 3(n-1), \ldots$. *Hidden hypotheses:* p is conjugate to itself along every great circle, so Theorem 17.3 does not apply to loops; the end point must be moved (V10.28). *Deviation (narrower):* $n \geq 2$ (and $n \geq 3$ for the homology, over fields); a cell complex in the sense of VA.18. The path-space form is V10.24 and V10.26.

What is not verified in this chapter

- **Milnor's route to Corollary 17.2** (p. 95): his theorem that Ω^* has the homotopy type of a CW-complex (*On spaces having the homotopy type of a CW-complex*, 1959) is not in Mathlib. The corollary is proved for N closed in V by moving the end point (V10.29); for a manifold that is not complete it is not verified. Theorem 17.3 does not use it (V10.22).
- **Theorem 17.3 for** Ω^* (p. 95): not stated; it would follow from V10.22 and Theorem 17.1 (V10.19) by composing homotopy equivalences.
- **Corollary 17.4 for** $n = 1$ (p. 96): not stated (V10.24). The loop-space form for $n \geq 2$ is V10.30.
- **The homology of the path spaces of** S^2**, and integral homology**: for $n = 2$ there is a cell in every dimension and the lacunary principle does not apply; $H_j(\Omega S^2) \cong \mathbb{Z}$ for all j (through the Hopf fibration, quoted in the exposition) is not formalized. For $n \geq 3$ the homology is computed over fields only, not over $\mathbb{Z}$ (V10.26).
- **Corollary 17.5 itself** (p. 96): that two non-conjugate points of a complete M of the homotopy type of S^n, $n > 2$, are joined by infinitely many geodesics needs the homotopy invariance of Ω^* (the loop spaces of homotopy equivalent spaces are weakly equivalent, and have the same homology) and the homology of the countable cell complex of Theorem 17.3; only the counterexample without completeness is verified (V10.27).
- **Serre's remark** (p. 97): Serre's theorem on the homology of loop spaces (the spectral sequence of the path fibration) and the passage to the universal cover are not formalized. The Freudenthal suspension theorem, to which the remark is applied, is verified in Chapter 13 by the route of §22 (V13.9).
- **Milnor's construction in the proof of Theorem 17.1** (pp. 93–94) in general: ε_1, ε_2, $A(\omega)$ and h (the exposition's Lemma 17.A) are formalized only for $M = \mathbb{R}$ with $\varepsilon_2 \equiv 1$, where they serve the machine-checked counterexample to the continuity of h (V10.19).
- **Theorem 16.3 at a critical level** (p. 90): when a is the energy of a geodesic; the formal statements assume it is not (V10.12–V10.13).
- **The Remark on p. 92 for the length**: that the geodesic minimizing the energy is shortest is not stated (V10.14).
- **CW-complexes**: cellular approximation is not in Mathlib, so cells are attached along arbitrary maps (V2.11, VA.18); the exposition's Lemma 17.E (a strictly compatible tower from nested

subdivisions) is neither formalized nor needed (V10.20).

- **The exposition's interactive figure (§4) and its §7** (seismic surface waves on the round Earth): not formalized.
- **§§16–17 on an abstract Riemannian manifold**: only for submanifolds of a Euclidean space with the induced metric; Nash's embedding theorem is quoted, not proved.

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12. Chapter 11 (Milnor §§18–19): partly verified

Milnor's §18 (pp. 98–99) shows that $\exp_p(v)$ is conjugate to p along $t \mapsto \exp_p(tv)$ exactly when $\exp_p$ is critical at v (Theorem 18.1), so that by Sard's theorem almost every q is conjugate to p along no geodesic (Corollary 18.2); this justifies the hypothesis of Theorem 17.3. §19 (pp. 100–107) relates curvature and topology. If $\langle R(A,B)A, B\rangle \leq 0$, no two points are conjugate along any geodesic (Lemma 19.1); with §17 this makes a simply connected complete such manifold diffeomorphic to $\mathbb{R}^n$ by $\exp_p$ (Theorem 19.2, Cartan) and gives Corollary 19.3 on homotopy groups. If the Ricci curvature is at least $(n-1)/r^2$, every geodesic longer than πr contains conjugate points and is not minimal (Theorem 19.4, Myers), a complete such manifold is compact of diameter at most πr (Corollary 19.5), and on a compact manifold with positive definite Ricci tensor $\Omega(M; p, q)$ has the homotopy type of a CW-complex with finitely many cells in each dimension (Theorem 19.6). Verified, for a submanifold N of a Euclidean space with the induced metric, in the embedded setting of Chapters 6–9: the smoothness of exp on its whole domain, which Milnor uses and §10 does not provide (V11.1); Theorem 18.1 with the multiplicity, and Corollary 18.2, without completeness; the density of non-conjugate points; Lemma 19.1, with the steps of the proof of 19.2 that need only the index theorem and 18.1 (index 0, E_{**} positive definite, $\exp_p$ non-critical); the Ricci tensor, its symmetry and Milnor's Assertion; Theorem 19.4 and Corollary 19.5, for $n \geq 2$, a hypothesis the book leaves silent; the example of the unit sphere; and the index estimate in the proof of Theorem 19.6 (a geodesic longer than $k\pi r$ has index at least k). With §17 (Chapter 10) and the covering $\exp_p$: Theorem 19.2 (Cartan–Hadamard) and Milnor's description of the path space in curvature ≤ 0, the statement that each homotopy class of paths contains one geodesic, and the first half of Corollary 19.3 (V11.17–V11.21); and Theorem 19.6, for all end points, with the finiteness of the homology of the path space (V11.22–V11.24). Every Riemannian manifold is such an N by Nash's embedding theorem, which is quoted, not proved. **Not verified:** the second half of Corollary 19.3 (no torsion in π_1); the finiteness of π_1 under a positive Ricci bound; the optics paragraph, the examples and the remarks of §19; and the sectional curvature as a Gaussian curvature (the list at the end). Lean: `Morse/Chapter11/` — `ExpSmooth.lean` (695 lines), `NonConjugate.lean` (242), `Curvature.lean` (304), `Myers.lean` (619), `LongGeodesics.lean` (470), `ExpCovering.lean` (707), `CartanHadamard.lean` (687), `PositiveRicci.lean` (266), `RicciHomology.lean` (252), `PositiveRicciAll.lean` (73).

The embedded setting. As in Chapters 8 and 9: V is a finite-dimensional real inner product space and $N \subseteq V$ a slice submanifold modelled on Z with complement W (V1.6), with the induced metric; every tangent space T_qN, $q \in N$, has dimension $n = \dim Z$; R is Milnor's curvature tensor, with his sign (V6.16–V6.17), and the sectional curvature is that of V6.22. The exponential map, its domain and the tangent manifold TN are those of V7.7–V7.8; $\Omega(N; p, q)$, $T\Omega_\gamma$, the geodesics of Ω and E_{**} those of V8.1, V8.10 and V8.14; smooth curves in N on $[0,1]$, Jacobi fields, the space $\mathcal{J}(\gamma)$, conjugate parameters with their multiplicity, and the index and nullity of E_{**} those of V9.2–V9.4, V9.6 and V9.13. A *geodesic from* p is a smooth curve γ in N on $[0,1]$ with $\gamma(0) = p$ which is a

geodesic on $(0,1)$; "$\gamma(t)$ is conjugate to $\gamma(0)$" reads "t is conjugate to 0" (V9.4). $L(\gamma)$ is the length of γ on $[0,1]$ (V7.11), and ρ the distance of V7.18. N need not be closed, compact, connected or complete unless this is stated.

V11.1 (exp **is smooth on its whole domain;** `Morse.RaysGoodE`, `Morse.raysGoodE_of_isGeodesicOn`, `Morse.exists_retraction`, `Morse.expDom_subset_tangentBundleSet`, `Morse.mem_expDom_of_isGeodesicOn`, `Morse.exists_contDiffOn_exp`, `Morse.exists_isOpen_expDom`, `Morse.contMDiffOn_exp_expDom`**).** (i) Let E be a real normed space, $U_0 \subseteq E$ open, $u_0 \in U_0$, and $c, w : E \to V$ of class C^∞ on U_0 with $c(u) \in N$ and $w(u) \in T_{c(u)}N$ for $u \in U_0$. If $t \mapsto \exp_{c(u_0)}(t\,w(u_0))$ is a geodesic on $(-\delta, B+\delta)$, $B \geq 0$, $\delta > 0$, there are $\varepsilon, \eta > 0$ such that the ball of radius ε about u_0 lies in U_0, every $t \mapsto \exp_{c(u)}(t\,w(u))$ with u in this ball is a geodesic on $(-\eta, B+\eta)$, and $(u,t) \mapsto \exp_{c(u)}(t\,w(u))$ is C^∞ on the product of the ball with $(-\eta, B+\eta)$. (ii) Every point of N has an open neighbourhood U in V with a map r, C^∞ on U, with values in N and fixing the points of $N \cap U$. (iii) The domain of exp lies in TN; a point (y,w) of TN is in it as soon as $t \mapsto \exp_y(tw)$ is a geodesic on an open interval containing $[0,1]$; and near every point (p,v) of the domain there are an open set $O \subseteq V \times V$ and a map F, C^∞ on O, such that every point of $TN \cap O$ is in the domain and $\exp = F$ there. Hence the domain is $TN \cap O'$ for an open $O' \subseteq V \times V$, and $(q,v) \mapsto \exp_q(v)$ is C^∞ on it as a map on the manifold TN. *Book:* p. 98: a critical point is defined for a smooth map between manifolds of the same dimension and applied to $\exp = \exp_p : TM_p \to M$, so $\exp_p$ is treated as smooth on all of TM_p; M is assumed complete so that exp is everywhere defined, "although this assumption could easily be eliminated". *Hidden hypotheses:* the smoothness of exp away from the zero section, i.e. the smooth dependence of geodesics on their initial data over a compact time interval: §10 (Lemma 10.2) gives it only near the zero section (V7.8), and V7.23 (iv) only the continuity of exp on its domain. *Deviation (stronger):* the whole domain, without completeness. *Proof:* (i) continues the geodesics step by step with the local smooth flow (V9.9 (i)), as in V9.9 (iii), now with parameters in any normed space; (ii) from a slice chart σ, $r(y) = \sigma^{-1}(\sigma(y)_1, 0)$; (iii) is (i) with $E = V \times V$ and the data $(y,w) \mapsto (r(y), P_{r(y)}w)$, which agree with (y,w) on TN.

V11.2 ($\exp_q$ **and its derivative;** `Morse.expDomAt`, `Morse.zero_mem_expDomAt`, `Morse.smul_mem_expDomAt`, `Morse.exists_contDiffAt_expAt`, `Morse.isOpen_expDomAt`, `Morse.contDiffOn_expAt`, `Morse.dExpAt`, `Morse.isGeodesicOn_ray`, `Morse.isSmoothCurveIn_ray`, `Morse.expDomAt_eq_univ`**).** For $q \in N$ let $\mathcal{D}_q \subseteq T_qN$ be the set of v with (q,v) in the domain of exp. It contains 0, contains tv for every $v \in \mathcal{D}_q$ and $0 \leq t \leq 1$, is open in T_qN, and $\exp_q$ is C^∞ on it, as a map from the vector space T_qN to V. For $v \in T_qN$, $d(\exp_q)_v : T_qN \to V$ is the derivative of this map at v, and $\exp_q$ is *critical* at v if $d(\exp_q)_v$ is not one-to-one. For $v \in \mathcal{D}_q$, $\gamma_v(t) = \exp_q(tv)$ is a geodesic on an open interval containing $[0,1]$ and a smooth curve in N on $[0,1]$. If N is geodesically complete (V7.22), $\mathcal{D}_q = T_qN$. *Book:* p. 98: a smooth map f between manifolds of the same dimension is critical at x if $f_* : TN_x \to TM_{f(x)}$ is not one-to-one; this is applied to $\exp_p : TM_p \to M$, with $X \in T(TM_p)_v$ the tangent space at v of TM_p "considered as a manifold"; γ_v is the geodesic $t \mapsto \exp tv$ from p to $\exp v$. *Hidden hypotheses:* without completeness, that the domain of $\exp_p$ is open and star-shaped about 0 (V11.1). *Deviation (restated):* $T(TM_p)_v$ is identified with T_qN, as Milnor does on p. 99 (paths $v_i(u)$ in TM_p with $dv_i/du(0) = X_i$), and $d(\exp_q)_v$ takes values in V; whether it is one-to-one does not depend on the target. *Deviation (stronger):* no completeness.

V11.3 (Theorem 18.1; `Morse.expJacobi`, `Morse.expJacobi_zero`, `Morse.expJacobi_one`, `Morse.isJacobiField_expJacobi`, `Morse.covAlong_expJacobi_zero`, `Morse.expJacobi_mem_jacobiSpace`,

`Morse.expJacobi_injective`, `Morse.eq_expJacobi`, `Morse.conjSpace_eq_map_ker`, `Morse.theorem_18_1_mult` `Morse.theorem_18_1`**).** Let $q \in N$ and $v \in \mathcal{D}_q$. For $X \in T_qN$ let $W_X(t) = t\, d(\exp_q)_{tv}(X)$ for $t \in [0,1]$, constant off $[0,1]$. (i) $X \mapsto W_X$ is linear; each W_X is a Jacobi field along γ_v, in $\mathcal{J}(\gamma_v)$ (V9.3), with $W_X(0) = 0$, $\frac{DW_X}{dt}(0) = X$ and $W_X(1) = d(\exp_q)_v(X)$. (ii) $X \mapsto W_X$ is injective, and every field of $\mathcal{J}(\gamma_v)$ vanishing at 0 is a W_X. (iii) The fields of $\mathcal{J}(\gamma_v)$ vanishing at 0 and 1 are exactly the W_X with $X \in \ker d(\exp_q)_v$; so the multiplicity of the conjugate parameters 0 and 1 along γ_v (V9.4) is $\dim \ker d(\exp_q)_v$. (iv) **Theorem 18.1:** 0 and 1 are conjugate along γ_v if and only if $\exp_q$ is critical at v. *Book:* p. 98, Theorem 18.1: the point $\exp v$ is conjugate to p along the geodesic γ_v from p to $\exp v$ iff the mapping exp is critical at v. Proof (pp. 98–99): if $\exp_* X = 0$ with $X \neq 0$, a path $v(u)$ in TM_p with $v(0) = v$ and $\frac{dv}{du}(0) = X$ gives the variation $\alpha(u,t) = \exp tv(u)$ through geodesics, whose variation field W is a Jacobi field with $W(0) = 0$, $W(1) = \exp_* X = 0$ and $\frac{DW}{dt}(0) = \frac{D}{du}v(u)|_{u=0} \neq 0$; conversely, if $\exp_*$ is non-singular at v, independent $X_1, \dots, X_n$ give Jacobi fields $W_1, \dots, W_n$ vanishing at p whose values $\exp_*(X_i)$ at $\exp v$ are independent, and since n is the dimension of the space of Jacobi fields along γ_v vanishing at p, "clearly" no non-trivial Jacobi field vanishes at both p and $\exp v$. *Hidden hypotheses:* the smoothness of $\exp_p$ on its domain (V11.1); $\frac{DW}{dt}(0) = \frac{D}{du}v(u)$ needs the symmetry lemma 8.7 and the covariant derivative along the constant curve $u \mapsto \alpha(u,0) = p$; the dimension n is that of Remark 14.2 (V9.7). *Deviation (stronger):* the multiplicity equals $\dim \ker d(\exp_q)_v$, which Milnor does not state (the exposition's form of 18.1); no completeness. *Deviation (restated):* "$\exp v$ conjugate to p along γ_v" is the conjugacy of the parameters 0 and 1. *Proof:* Milnor's variation with the straight line $v(u) = v + uX$, whose variation field is W_X (V9.8); $\frac{DW_X}{dt}(0) = d(\exp_q)_0(X) = X$ by $d(\exp_q)_0 = \mathrm{id}$ (V7.8 (iii)), without the symmetry lemma; (ii) and (iii) use that a Jacobi field is determined by $J(0)$ and $\frac{DJ}{dt}(0)$ (V9.3) in place of the dimension count.

V11.4 (the conjugate locus is the set of critical values of $\exp_p$**;** `Morse.conjLocus`, `Morse.IsConjugate.congr_cur` `Morse.IsConjugate.congr_curve_iff`, `Morse.theorem_18_1_of_geodesic`, `Morse.conjLocus_eq`, `Morse.conjLocus_eq_of_complete`**).** For $p \in V$, the *conjugate locus* $C(p)$ is the set of q for which there is a geodesic γ from p with $\gamma(1) = q$ along which 0 and 1 are conjugate; conjugacy depends only on γ on $[0,1]$. Let $p \in N$. For every geodesic γ from p, its one-sided initial velocity v lies in $\mathcal{D}_p$, $\exp_p(v) = \gamma(1)$, and 0 and 1 are conjugate along γ iff $\exp_p$ is critical at v. Hence $C(p) = \exp_p(\{v \in \mathcal{D}_p : \exp_p \text{ is critical at } v\})$, the set of critical values of $\exp_p$; if N is geodesically complete, $\mathcal{D}_p = T_pN$. *Book:* p. 99: Corollary 18.2 "follows immediately from 18.1 together with Sard's theorem (6.1)". *Hidden hypotheses:* every geodesic from p is a radial geodesic $t \mapsto \exp_p(tv)$ (uniqueness of geodesics, and the extension of a geodesic segment past its end points); without completeness, that the domain of $\exp_p$ is open (V11.1). *Deviation (not in the book):* the conjugate locus is named and identified as a set. *Proof:* γ agrees on $[0,1]$ with $t \mapsto \exp_p(tv)$ (V9.9 (ii)), and Theorem 18.1 (V11.3) applies.

V11.5 (Corollary 18.2; `Morse.corollary_18_2`**).** Let $p \in N$, let σ be a slice chart of N (V1.6), and μ a Haar measure on Z. Then the image of $C(p) \cap \mathrm{dom}\,\sigma$ under the chart $y \mapsto \sigma(y)_1$ has μ-measure 0: $C(p)$ has measure zero in N. *Book:* p. 99, Corollary 18.2: for $p \in M$ and almost all $q \in M$, p is not conjugate to q along any geodesic. *Hidden hypotheses:* "measure zero" in a manifold is not defined in the book (Theorem 6.1, p. 33, uses the phrase for a manifold target); here it is read in charts. *Deviation (restated):* "almost all q" is "outside a set of measure zero in every slice chart, for every Haar measure on the model space". *Proof:* with $e : Z \to T_pN$ a linear isomorphism,

the map $z \mapsto \sigma(\exp_p(e(z)))_1$ is C^∞ on an open subset of Z, and the chart image of every point of $C(p) \cap \operatorname{dom}\sigma$ is a critical value of this map: a derivative that is not one-to-one stays so after composing, and a linear map of Z to itself is onto iff it is one-to-one. Sard's theorem for smooth maps (the companion project's `Milnor.sard`) applies.

V11.6 (non-conjugate points are dense, and non-degenerate; `Morse.dense_compl_conjLocus`, `Morse.hessNullity_eq_zero_of_notMem_conjLocus`). For $p \in N$, the points of N outside $C(p)$ are dense in N. If $q \notin C(p)$, then at every geodesic $\gamma \in \Omega(N; p, q)$ the nullity of E_{**} is 0. *Book:* p. 98: Theorem 17.3 describes $\Omega(M; p, q)$ when p and q are not conjugate along any geodesic, and "this section will justify this result by showing that such non-conjugate points always exist". *Hidden hypotheses:* from measure zero to existence: a null set has empty interior, since a non-empty open subset of Z has positive Haar measure. *Deviation (not in the book):* density, and the non-degeneracy that Theorems 16.3 and 17.3 use. *Proof:* V11.5 in the chart at a point of a given open set; Theorem 14.1 (V9.6).

V11.7 (Lemma 19.1; `Morse.NonposCurvature`, `Morse.nonposCurvature_iff_sectionalCurvature`, `Morse.hasDerivAt_inner_covAlong`, `Morse.continuousOn_covAlong`, `Morse.IsJacobiField.eqOn_zero_of_no`, `Morse.lemma_19_1`). N has *non-positive curvature* if $\langle R(A,B)A, B\rangle \le 0$ for all $A, B \in T_qN$ and all $q \in N$; equivalently, every sectional curvature of N is ≤ 0. (i) For a smooth curve γ in N on $[0,1]$ with velocity V and a Jacobi field J along it,

$$\frac{d}{dt}\Big\langle \frac{DJ}{dt}, J\Big\rangle = \Big\|\frac{DJ}{dt}\Big\|^2 - \langle R(V,J)V, J\rangle \quad \text{on } (0,1).$$

(ii) If N has non-positive curvature and γ is a smooth curve in N on $[0,1]$ which is a geodesic on $(0,1)$, a Jacobi field along γ vanishing at two parameters $a < b$ of $[0,1]$ vanishes on $[0,1]$; so no two parameters are conjugate along γ. *Book:* p. 100, Lemma 19.1: if $\langle R(A,B)A, B\rangle \le 0$ for every pair of vectors A, B in TM_p and every $p \in M$, no two points of M are conjugate along any geodesic; proof: $\langle \frac{D^2J}{dt^2}, J\rangle = -\langle R(V,J)V, J\rangle \ge 0$, so $\langle \frac{DJ}{dt}, J\rangle$ is monotonically increasing, strictly so where $\frac{DJ}{dt} \ne 0$; if $J(0) = J(t_0) = 0$ it vanishes on $[0, t_0]$, so $J(0) = \frac{DJ}{dt}(0) = 0$ and $J \equiv 0$. *Hidden hypotheses:* $\frac{DJ}{dt}(0) = 0$ comes from $\frac{DJ}{dt} = 0$ on $(0, t_0)$ by continuity. *Deviation (restated):* any two parameters $a < b$ in $[0,1]$ (Milnor: 0 and t_0); conjugate parameters for points (V9.4). *Deviation (stronger):* (i) along any smooth curve.

V11.8 (non-positive curvature: index 0, E_{**} positive definite, $\exp_p$ non-critical; `Morse.formIndex_eq_zero_of_n`, `Morse.energyHessian_pos_of_nonpos`, `Morse.injective_dExpAt_of_nonpos`). Let N have non-positive curvature. At every geodesic $\gamma \in \Omega(N; p, q)$ the index and the nullity of E_{**} are 0, and $E_{**}(W, W) > 0$ for every $W \in T\Omega_\gamma$ that does not vanish identically on $[0,1]$. For $p \in N$, $\exp_p$ is critical at no point of $\mathcal{D}_p$. *Book:* p. 102, proof of Theorem 19.2: "since there are no conjugate points, it follows from the index theorem that every geodesic from p to q has index $\lambda = 0$"; p. 103: "it follows from 18.1 that $\exp_p$ is non-critical everywhere". *Hidden hypotheses:* none. *Deviation (stronger):* positive definiteness of E_{**}, not in the book; neither completeness nor simple connectivity. *Proof:* the index theorem (V9.24) with Lemma 19.1; $E_{**} \ge 0$ from its integral form (V8.15), and a W with $E_{**}(W, W) = 0$ is a null vector of the semi-definite form, hence a Jacobi field vanishing at 0 and 1 (V9.5), hence 0; the last statement is V11.3 with Lemma 19.1.

V11.9 (the Ricci tensor and Milnor's Assertion; `Morse.ricci`, `Morse.ricci_eq_sum`, `Morse.ricci_comm`, `Morse.curvOp_self`, `Morse.ricci_eq_sum_erase`). For $q \in N$ and $U_1, U_2 \in T_qN$, $K(U_1, U_2)$ is the trace of the linear map $W \mapsto R(U_1, W)U_2$ of T_qN. For an orthonormal basis $E_1, \dots, E_n$

of T_qN, $K(U_1,U_2) = \sum_i \langle R(U_1,E_i)U_2,E_i\rangle$, and $K(U_1,U_2) = K(U_2,U_1)$. **Assertion:** for an orthonormal basis $U_1,\dots,U_n$ of T_qN, $K(U_n,U_n) = \sum_{i\neq n}\langle R(U_n,U_i)U_n,U_i\rangle$, the sum of the sectional curvatures of the $n-1$ planes spanned by U_n and U_i. *Book:* p. 104, Definition: the Ricci tensor at p is the bilinear pairing $K : TM_p \times TM_p \to \mathbb{R}$, $K(U_1,U_2)$ the trace of $W \mapsto R(U_1,W)U_2$; "it follows easily from 9.3 that K is symmetric"; Assertion: $K(U_n,U_n)$ is the sum of the sectional curvatures $\langle R(U_n,U_i)U_n,U_i\rangle$, $i = 1,\dots,n-1$, because the n-th diagonal term of the matrix $(\langle R(U_n,U_i)U_n,U_j\rangle)$ is zero. *Hidden hypotheses:* none. *Proof:* symmetry from the pair symmetry of Lemma 9.3 (4) (V6.21) in an orthonormal basis; the Assertion as Milnor's, with $R(U_n,U_n) = 0$.

V11.10 (a geodesic as a smooth curve with a parallel frame; `Morse.exists_smooth_clamp`, `Morse.exists_smooth_extension`, `Morse.exists_parallel_frame`). Let $\gamma \in \Omega(N;p,q)$ be a geodesic and $L = L(\gamma)$. (i) There are a C^∞ curve $c : \mathbb{R} \to N$ equal to γ on $[0,1]$ and a C^∞ field V along c, tangent to N, equal to γ' on $(0,1)$, with $\|V(t)\| = L$ for all t and $L^{-1}V$ parallel along c (V6.8). (ii) If $L > 0$, there are a C^∞ curve $c : \mathbb{R} \to N$ equal to γ on $[0,1]$, a C^∞ field V along c, tangent to N and equal to γ' on $(0,1)$, and $n-1$ parallel fields $P_1,\dots,P_{n-1}$ along c with $\|P_i(t)\| = 1$, such that, for every r for which $K(U,U) \geq (n-1)/r^2$ for all unit vectors U (V11.12), $\sum_{i=1}^{n-1}\langle R(V,P_i)V,P_i\rangle \geq L^2(n-1)/r^2$ at every t. *Book:* p. 104, proof of Theorem 19.4: for a geodesic $\gamma : [0,1] \to M$ of length L, choose parallel vector fields $P_1,\dots,P_n$ along γ which are orthonormal at one point, and hence everywhere, with P_n pointing along γ, so that $V = LP_n$ and $\frac{DP_i}{dt} = 0$. *Hidden hypotheses:* parallel fields along $\gamma : [0,1] \to M$ (Lemma 8.2 is formalized for curves on $\mathbb{R}$, V6.8, so γ is first extended past its ends); $V = LP_n$ needs the constant speed L of a geodesic on $[0,1]$ and that V itself is parallel; the frame must be orthonormal at every t, parallel translation being an isometry (V6.9), for the Assertion (V11.9) to apply. *Proof:* c is the radial geodesic of $\gamma'(0)$, defined near $[0,1]$ (V9.9 (ii)), composed with a smooth map $\mathbb{R} \to (-\eta,1+\eta)$ that is the identity on $[0,1]$; the P_i are the parallel translates of an orthonormal basis of $T_{\gamma(0)}N$ containing $V(0)/L$, and the sum is $L^2K(V/L,V/L)$ by the Assertion.

V11.11 (the test fields $\sin(\pi t)P(t)$; `Morse.integral_sin_sq_pi_mul`, `Morse.integral_cos_sq_pi_mul`, `Morse.energyHessian_sin_parallel`, `Morse.continuous_inner_curvR`, `Morse.exists_energyHessian_neg`). (i) Let $\gamma \in \Omega(N;p,q)$ agree on $[0,1]$ with a C^∞ curve $c : \mathbb{R} \to N$, P a parallel field along c with $\|P\| = 1$, and V a field equal to γ' on $(0,1)$. Then $W(t) = \sin(\pi t)P(t)$ lies in $T\Omega_\gamma$ and

$$\tfrac{1}{2}E_{**}(W,W) = \int_0^1 \left(\pi^2\cos^2(\pi t) - \sin^2(\pi t)\langle R(V,P)V,P\rangle\right) dt.$$

(ii) If $n \geq 2$, $r > 0$, $K(U,U) \geq (n-1)/r^2$ for every unit vector U, and $\gamma \in \Omega(N;p,q)$ is a geodesic with $L(\gamma) > \pi r$, then $E_{**}(X,X) < 0$ for some $X \in T\Omega_\gamma$. *Book:* p. 105: with $W_i(t) = \sin(\pi t)P_i(t)$, $\frac{1}{2}E_{**}(W_i,W_i) = -\int_0^1 \langle W_i, \frac{D^2W_i}{dt^2} + R(V,W_i)V\rangle dt = \int_0^1 (\sin\pi t)^2(\pi^2 - L^2\langle R(P_n,P_i)P_n,P_i\rangle)dt$; summing over $i = 1,\dots,n-1$ gives $\int_0^1 (\sin\pi t)^2((n-1)\pi^2 - L^2K(P_n,P_n))dt$, which is negative if $K(P_n,P_n) \geq (n-1)/r^2$ and $L > \pi r$, so $E_{**}(W_i,W_i) < 0$ for some i (the $\frac{1}{2}$, $\sum$ and $\int$ missing from the scan's display of the sum are dropped glyphs, `Exposition/Findings.md`). *Hidden hypotheses:* $n \geq 2$, so that there is a field P_i perpendicular to γ. *Deviation (restated):* the integral form $\int(\|\frac{DW}{dt}\|^2 - \langle R(V,W)V,W\rangle)$ (V8.15) gives $\pi^2\cos^2(\pi t)$ in place of $\pi^2\sin^2(\pi t)$; both integrate to $\pi^2/2$. *Proof:* $\frac{DW}{dt} = \pi\cos(\pi t)P$; with the frame of V11.10, $\sum_i \frac{1}{2}E_{**}(W_i,W_i) \leq (n-1)(\pi^2 - L^2/r^2)/2 < 0$.

V11.12 (Theorem 19.4, Myers; `Morse.RicciLowerBound`, `Morse.theorem_19_4`). Let $n \geq 2$, $r > 0$, and suppose $K(U,U) \geq (n-1)/r^2$ for every unit vector $U \in T_qN$ and every $q \in N$. Then every geodesic $\gamma \in \Omega(N;p,q)$ with $L(\gamma) > \pi r$ has a parameter $t \in (0,1)$ conjugate to 0, and is not minimal: some path of $\Omega(N;p,q)$ is shorter (V8.6). *Book:* p. 104, Theorem 19.4 (Myers): if $K(U,U) \geq (n-1)/r^2$ for every unit vector U at every point of M, r a positive constant, every geodesic on M of length $> \pi r$ contains conjugate points, and hence is not minimal; p. 105: $E_{**}(W_i,W_i) < 0$ makes the index of γ positive, and the index theorem gives conjugate points; a variation with field W_i decreases the energy. *Hidden hypotheses:* $n \geq 2$: for $n = 1$ the Ricci tensor vanishes, the hypothesis holds for every r, and a segment of a line longer than πr has no conjugate points and is minimal. *Deviation (restated):* "contains conjugate points" is "has a parameter in $(0,1)$ conjugate to 0"; minimality is among the paths of $\Omega(N;p,q)$ (V8.6). *Proof:* E_{**} is negative definite on the line through the X of V11.11, so the index (V9.13) is positive, and the index theorem (V9.24) gives the conjugate parameter; Lemma 13.6 (V8.18) excludes minimality.

V11.13 (Corollary 19.5; `Morse.IsMinimalGeodesic.isMinimalPath`, `Morse.corollary_19_5`). A minimal geodesic on $[0,1]$ in the sense of V7.24 is a geodesic of $\Omega(N;\gamma(0),\gamma(1))$ that is minimal in the sense of V8.6. Let N be connected, (N,ρ) complete (V7.22), $r > 0$ with $(n-1)/r^2 > 0$, and $K(U,U) \geq (n-1)/r^2$ for every unit vector U. Then $\rho(p,q) \leq \pi r$ for all $p,q \in N$, and N is compact. *Book:* p. 105, Corollary 19.5: if M is complete and $K(U,U) \geq (n-1)/r^2 > 0$ for all unit vectors U, then M is compact with diameter $\leq \pi r$; proof: a minimal geodesic from p to q has length $\leq \pi r$, and closed bounded sets in a complete manifold are compact. *Hidden hypotheses:* connectedness (Milnor's standing assumption), for minimal geodesics to exist; $n \geq 2$ is forced by $(n-1)/r^2 > 0$. *Proof:* Hopf–Rinow (V7.29) joins p and q by a minimal geodesic of length $\rho(p,q)$, which is $\leq \pi r$ by V11.12; N is then bounded and closed in itself, hence compact (V7.29).

V11.14 (the unit sphere; `Morse.ricci_sph`, `Morse.ricciLowerBound_sph`, `Morse.sphere_myers`). On the unit sphere $S^n \subseteq \mathbb{R}^{n+1}$ (V6.25), $K(U,U) = (n-1)\|U\|^2$, so the hypothesis of Theorem 19.4 holds with $r = 1$. For $n \geq 2$, every geodesic of $\Omega(S^n;p,q)$ of length $> \pi$ has a parameter in $(0,1)$ conjugate to 0 and is not minimal, and $\rho(p,q) \leq \pi$ for all $p,q \in S^n$. *Book:* p. 105, Example: on a sphere of radius r every sectional curvature is $1/r^2$, so $K(U,U) = (n-1)/r^2$, and by 19.4 every geodesic of length $> \pi r$ contains conjugate points: "a best possible result". *Hidden hypotheses:* none. *Deviation (narrower):* only the unit sphere, and not "best possible" (that a geodesic of length π has no interior conjugate point). *Proof:* on S^n, $R(U,X)U = \|U\|^2 X - \langle X,U\rangle U$ (V6.25), whose trace is $(n-1)\|U\|^2$; then V11.12, and V11.13 with the completeness of the closed set S^n (V7.30).

V11.15 (fields on disjoint pieces; `Morse.integral_eq_of_eqOn_pieces`, `Morse.integral_sin_sq_shift`, `Morse.integral_cos_sq_shift`, `Morse.sinBump`, `Morse.sinBump_mem_pathTangent`, `Morse.energyHessian_sin` `Morse.energyHessian_sinBump_disjoint`, `Morse.exists_sinBump_neg`). Let γ, c, V be as in V11.11 (i), with V smooth and tangent along c, P parallel along c, and $0 \leq a < b \leq 1$. The field X equal to $\sin(\kappa(t-a))P(t)$, $\kappa = \pi/(b-a)$, on $[a,b]$ and to 0 elsewhere lies in $T\Omega_\gamma$; if $\|P\| = 1$,

$$\tfrac{1}{2}E_{**}(X,X) = \int_a^b \left(\kappa^2\cos^2(\kappa(t-a)) - \sin^2(\kappa(t-a))\langle R(V,P)V,P\rangle\right) dt.$$

Two such fields on $[a,b]$ and $[a',b']$ with $b \leq a'$ satisfy $E_{**}(X,X') = 0$. If $n \geq 2$, $r > 0$, the P_i are unit parallel fields as in V11.10 (ii), and $L(\gamma)(b-a) > \pi r$, the field built on $[a,b]$ from some P_i has $E_{**}(X,X) < 0$. *Book:* p. 106, proof of Theorem 19.6: for a geodesic of length $> k\pi r$, "a similar argument" gives, for each $i = 1,\dots,k$, a vector field X_i along γ which vanishes outside the interval $(\frac{i-1}{k},\frac{i}{k})$ with $E_{**}(X_i,X_i) < 0$, and "clearly" $E_{**}(X_i,X_j) = 0$ for $i \neq j$. *Hidden*

hypotheses: the construction of the X_i (the computation of V11.11 on a piece of length $1/k$) and $n \geq 2$. *Deviation (not in the book):* stated for any piece $[a,b]$. *Proof:* the integrand of the integral form (V8.15) vanishes off $[a,b]$, and for two pieces off their common end point; on $[a,b]$ the sum over the P_i is at most $(n-1)(b-a)(\kappa^2 - L^2/r^2)/2 < 0$.

V11.16 (Theorem 19.6, the index estimate; `Morse.theorem_19_6_index`, `Morse.pathLength_le_of_formIndex_l` `Morse.le_finsum_conjMultiplicity_of_ricci`). Let $n \geq 2$, $r > 0$, $K(U,U) \geq (n-1)/r^2$ for every unit vector U, $\gamma \in \Omega(N;p,q)$ a geodesic and $k \geq 0$ an integer. If $L(\gamma) > k\pi r$, the index of E_{**} at γ is at least k, and the parameters $t \in (0,1)$ conjugate to 0, counted with multiplicity, number at least k. Equivalently, a geodesic of index $< k$ has length $\leq k\pi r$. *Book:* p. 106, proof of Theorem 19.6: the continuous function $K(U,U) > 0$ takes on a minimum $(n-1)/r^2 > 0$ on the compact space of unit vectors; then every geodesic $\gamma \in \Omega(M;p,q)$ of length $> \pi r$ has index $\lambda \geq 1$, and one of length $> k\pi r$ has index $\lambda \geq k$, since $X_1, \dots, X_k$ span a k-dimensional subspace of $T\Omega_\gamma$ on which E_{**} is negative definite. *Hidden hypotheses:* $n \geq 2$; that the X_i are linearly independent (they are non-zero with disjoint supports). *Deviation (narrower):* the bound $(n-1)/r^2$ is a hypothesis, not derived from the compactness of M and the positivity of the Ricci tensor (that step is V11.22 (i)); the rest of Theorem 19.6 is V11.22–V11.24. *Proof:* on the pieces $[\frac{i}{k}, \frac{i+1}{k}]$, V11.15 gives fields $X_0, \dots, X_{k-1}$ with $E_{**}(X_i, X_i) < 0$ and $E_{**}(X_i, X_j) = 0$ for $i \neq j$, so $E_{**}(\sum c_i X_i, \sum c_i X_i) = \sum c_i^2 E_{**}(X_i, X_i) < 0$ unless all $c_i = 0$; the count of conjugate parameters is the index theorem (V9.24).

V11.17 (Theorem 19.2, Cartan–Hadamard; `Morse.isGeodesicallyComplete_of_isClosed`, `Morse.injective_dExpAt_of_nonpos'`, `Morse.exists_eqOn_ray`, `Morse.injective_expToN_of_simplyConnect` `Morse.surjective_expToN`, `Morse.expDiffeomorph`, `Morse.theorem_19_2`). Let N be geodesically complete (V7.22; for instance closed in V), simply connected (as a topological space) and of non-positive curvature (V11.7). Then any two points $p, q \in N$ are joined by a geodesic $\gamma \in \Omega(N;p,q)$, and every geodesic of $\Omega(N;p,q)$ agrees with γ on $[0,1]$; and $\exp_p : T_pN \to N$ is a diffeomorphism, so N is diffeomorphic to $\mathbb{R}^n$, $n = \dim N$, for its manifold structure as a slice submanifold. *Book:* p. 102, Theorem 19.2 (Cartan): a simply connected complete M with $\langle R(A,B)A, B\rangle \leq 0$ has a unique geodesic between any two points and is diffeomorphic to $\mathbb{R}^n$; proof pp. 102–103: by 17.3 $\Omega(M;p,q)$ has the homotopy type of a connected 0-dimensional complex, a point, so the geodesic is unique, $\exp_p$ is one-one and onto, non-critical by 18.1, hence a diffeomorphism. *Hidden hypotheses:* that a simply connected space is connected (needed for Hopf–Rinow); in Milnor's route, that the components of Ω are the homotopy classes of paths (Theorem 17.1). *Deviation (restated):* the order of the argument is reversed: $\exp_p$ is a covering (V11.19), injective because N is simply connected (the image of a segment of T_pN with both ends over one point is a null-homotopic loop, whose lift is closed), onto by Hopf–Rinow; uniqueness of geodesics follows. Both routes rest on Lemma 19.1.

V11.18 (the path space in non-positive curvature; `Morse.omega_cells_of_nonpos`, `Morse.geodesic_iff_mem_fi` `Morse.nonempty_homotopyEquiv_pathC_fiber`, `Morse.omega_homotopyEquiv_fiber`, `Morse.contractibleSpac` `Morse.contractibleSpace_pathSp`, `Morse.omega_single_cell`). Let N have non-positive curvature. (i) If N is closed in V and $p, q \in N$, $\Omega(N;p,q)$ has the homotopy type of a countable cell complex (VA.18) all of whose cells are 0-cells, one for each geodesic from p to q. (ii) If N is geodesically complete, the fibre of $\exp_p$ over q is discrete and countable, its points v correspond bijectively (on $[0,1]$) to the geodesics $t \mapsto \exp_p(tv)$ from p to q, and $\Omega(N;p,q)$ is homotopy equivalent to it. (iii) If moreover N is simply connected, $\Omega(N;p,q)$ is contractible, and (for N closed) it has the homotopy type of a cell complex with exactly one cell, a 0-cell. *Book:* p. 102,

proof of 19.2: by 17.3 $\Omega(M;p,q)$ has the homotopy type of a 0-dimensional CW-complex with one vertex for each geodesic, connected because M is simply connected, hence a point. *Hidden hypotheses:* Theorem 17.1 for the connectedness. *Deviation (stronger):* (ii) for every complete N of curvature ≤ 0, simply connected or not, through the covering: lifting a path from $0 \in T_pN$ and taking the end point is a homotopy equivalence of Ω^* with the fibre, and $\Omega \simeq \Omega^*$ (V10.19). That a countable cell complex with only 0-cells (a telescope, VA.18) is homotopy equivalent to a discrete space is not formalized; (ii) reaches the discrete space directly.

V11.19 ($\exp_p$ is a covering map; `Morse.gauss_lemma_expDomAt`, `Morse.isLocalDiffeomorphAt_expToN`, `Morse.isLocalDiffeomorph_expToN`, `Morse.norm_expLift_le`, `Morse.exists_expLift`, `Morse.expLift_unique`, `Morse.isCoveringMap_expToN`, `Morse.isCoveringMap_expToN_of_nonpos`). (i) The Gauss lemma holds on the whole domain of $\exp_p$: $\langle d(\exp_p)_w(w), d(\exp_p)_w(x)\rangle = \langle w, x\rangle$ for $w \in \mathcal{D}_p$, $x \in T_pN$. (ii) $\exp_p : T_pN \to N$ is a local diffeomorphism at every point of its domain where $d(\exp_p)$ is one-to-one. Now let N be geodesically complete and $\exp_p$ nowhere critical. (iii) A lift Γ through $\exp_p$ of a smooth curve of speed $\leq L$ on $[0,T]$ satisfies $\|\Gamma(t)\| \leq \|\Gamma(0)\| + 1 + Lt$. (iv) Smooth curves in N lift through $\exp_p$, uniquely, from any point over the initial point. (v) $\exp_p : T_pN \to N$ is a covering map. In non-positive curvature (V11.8) all of this applies. *Book:* p. 103: by 18.1 $\exp_p$ is non-critical everywhere, "so that $\exp_p$ is locally a diffeomorphism"; for M not simply connected, Theorem 19.2 is applied to the universal covering $\tilde{M}$, which, "it is clear", inherits a complete metric of curvature ≤ 0. *Hidden hypotheses:* the completeness of the pulled-back metric (Milnor's "it is clear"), here (iii). *Deviation (not in the book):* the universal covering is $\exp_p : T_pN \to N$ itself, so no abstract covering space or lifted metric is needed; this is the standard proof of the Cartan–Hadamard theorem. The Gauss lemma is needed beyond a normal neighbourhood (Milnor states it only there, Lemma 10.5).

V11.20 (each homotopy class of paths contains exactly one geodesic; `Morse.pathOfMem`, `Morse.homotopy_class_unique_geodesic`). Let N be geodesically complete and of non-positive curvature, and $p, q \in N$. Every continuous path c in N from p to q is homotopic, relative to its end points, to a geodesic $\gamma \in \Omega(N;p,q)$; and two geodesics of $\Omega(N;p,q)$ both homotopic to c agree on $[0,1]$. *Book:* p. 103: for complete M of curvature ≤ 0, not simply connected, each homotopy class of paths from p to q contains precisely one geodesic (19.2 applied to $\tilde{M}$). *Hidden hypotheses:* as in V11.19. *Proof:* the geodesics from p are the images of the segments from 0 in T_pN, their lifts through the covering $\exp_p$; homotopic paths have lifts from 0 with the same end point.

V11.21 (Corollary 19.3, first half; `Morse.contractibleSpace_cube`, `Morse.isPreconnected_cube_face`, `Morse.genLoop_homotopic_const`, `Morse.corollary_19_3`). If N is geodesically complete and of non-positive curvature, then $\pi_i(N,x) = 0$ for every $i > 1$ and every $x \in N$ (Mathlib's homotopy groups); on the way, every map of the cube I^i, $i \geq 2$, sending the boundary to x is null-homotopic relative to the boundary. *Book:* p. 103, Corollary 19.3: for complete M with $\langle R(A,B)A, B\rangle \leq 0$, $\pi_i(M) = 0$ for $i > 1$, and $\pi_1(M)$ has no element of finite order other than the identity. *Hidden hypotheses:* "clearly $\pi_i(M) = \pi_i(\tilde{M})$". *Deviation (narrower):* only the first half; the torsion-freeness of π_1 (cohomology of groups and of finite cyclic groups) is not formalized. *Proof:* a map of the cube lifts through the covering $\exp_x$ (the cube is simply connected and locally path connected); the lift is 0 on the boundary because the fibre is discrete and the boundary is connected for $i \geq 2$; the straight-line contraction of the lift projects to the null-homotopy.

V11.22 (Theorem 19.6 for non-conjugate end points; `Morse.ricciMap`, `Morse.continuous_trace_ricciMap`,

`Morse.exists_ricciLowerBound`, `Morse.energy_le_of_geodesicIndex_le`, `Morse.theorem_19_6_of_ricciLow` `Morse.theorem_19_6`). (i) If N is compact, $n = \dim N \geq 2$ and the Ricci tensor is positive definite, there is $r > 0$ with $K(U,U) \geq (n-1)/r^2$ for every unit tangent vector U. (ii) Under such a bound, a geodesic of index $\leq d$ has energy $\leq ((d+1)\pi r)^2$. (iii) If N is closed in V with such a bound, or compact with positive definite Ricci tensor, and q is not conjugate to p along any geodesic ($q \notin C(p)$, V11.4), then $\Omega(N; p, q)$ has the homotopy type of a countable cell complex (VA.18) with only finitely many cells of each dimension. *Book:* p. 106, Theorem 19.6: for M compact with positive definite Ricci tensor, $\Omega(M; p, q)$ has the homotopy type of a CW-complex with only finitely many cells in each dimension; the proof treats p, q not conjugate along any geodesic, by 16.3 (finitely many geodesics of bounded length) and 17.3. *Hidden hypotheses:* $n \geq 2$; the continuity of $K(U,U)$ on the unit tangent bundle. *Deviation (restated):* "CW-complex" is a countable cell complex with arbitrary attaching maps (a telescope, VA.18); the general p, q are V11.24. *Proof:* the minimum of the continuous $K(U,U)$ on the compact unit tangent bundle; V11.16 bounds the length of a geodesic of index $< k$; Theorem 16.3 (V10.13) gives finitely many geodesics below each energy, and Theorem 17.3 (V10.22) the complex.

V11.23 (the homology of the path space in positive Ricci curvature; `Morse.slabEnergies`, `Morse.slab_homology`, `Morse.finiteDimensional_spaceHomology_omega`, `Morse.finiteDimensional_spaceHo` Let N be closed in V and every geodesic of $\Omega(N; p, q)$ of nullity 0. For levels $a \leq b$ that are not energies of geodesics, $H_j(\Omega^b, \Omega^a; \Bbbk)$ is finite-dimensional, and zero if no geodesic with energy in (a, b) has index j. Hence, under a Ricci bound $K(U,U) \geq (n-1)/r^2$, $n \geq 2$, and for $q \notin C(p)$ — in particular under the hypotheses of V11.22 (iii) — $H_i(\Omega(N; p, q); \Bbbk)$ is finite-dimensional for every i and every field $\Bbbk$. *Book:* not stated; p. 106 with the method of p. 96 (homology read off the cells). *Deviation (not in the book):* the homology is computed from the energy filtration, since the cell complexes of VA.18 have no cellular homology: above the energy $((i+2)\pi r)^2$ every geodesic has index $\geq i + 2$.

V11.24 (Theorem 19.6 for all end points; `Morse.exists_joined_notMem_conjLocus`, `Morse.pathSpHomotopyEqui` `Morse.IsCountableCellComplexType.of_homotopyEquiv`, `Morse.theorem_19_6_all`, `Morse.finiteDimensiona` Let N be compact, $n = \dim N \geq 2$, with positive definite Ricci tensor. Then for all $p, q \in N$, $\Omega(N; p, q)$ has the homotopy type of a countable cell complex (VA.18) with only finitely many cells of each dimension, and $H_i(\Omega(N; p, q); \Bbbk)$ is finite-dimensional for every i and every field $\Bbbk$. On the way: q is joined by a path in N to a point q' not conjugate to p along any geodesic, and then $\Omega(N; p, q) \simeq \Omega(N; p, q')$. *Book:* p. 106, Theorem 19.6, stated for all p, q. *Hidden hypotheses:* Milnor's proof treats only non-conjugate end points; the general case needs a nearby non-conjugate q' (Corollary 18.2, V11.5–V11.6) and the change of end point (the exposition's Lemma 17.D, through Theorem 17.1, V10.19). *Deviation (restated):* as in V11.22; no connectedness of N is needed, q' being chosen in a small Riemannian ball about q.

What is not verified in this chapter

- **Corollary 19.3, second half** (p. 103): $\pi_1(M)$ has no non-trivial element of finite order. It needs the cohomology of groups ($H^k(M) \cong H^k(\pi_1 M)$ for M with contractible universal covering) and the cohomology of finite cyclic groups. The first half is V11.21.
- **That a complex with only 0-cells is a discrete space up to homotopy** (p. 102): V11.18 (ii) reaches the discrete fibre of $\exp_p$ directly instead.
- **The finiteness of $\pi_1(M)$** (p. 105, after Corollary 19.5): the universal covering of a complete M with $K(U,U) \geq (n-1)/r^2 > 0$, with the lifted metric, is not formalized (in curvature

≤ 0 the covering $\exp_p$ of V11.19 plays its role).

- **The sectional curvature as a Gaussian curvature** (p. 100, the Remark after Lemma 19.1, citing Laugwitz): the Gaussian curvature of the surface $(u_1, u_2) \mapsto \exp_p(u_1 A + u_2 B)$ is not defined.
- **The optics paragraph and the examples** (pp. 101–102: apparent sizes and volumes, complete surfaces of curvature ≤ 0 in $\mathbb{R}^3$, the pseudo-sphere and Hilbert's theorem, compact surfaces of constant negative curvature), the sphere of radius $r \neq 1$ and "best possible" (p. 105), and the remarks of pp. 106–107 (complete non-compact M, $S^m \times S^k$, Synge's theorem): not formalized.
- **The exposition's application** (§6, shortest routes on the Earth) and its figure: not formalized.
- **§§18–19 on an abstract Riemannian manifold**: only for submanifolds of a Euclidean space with the induced metric; Nash's embedding theorem is quoted, not proved.

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13. Chapter 12 (Milnor §§20–21): partly verified

Milnor's §20 (pp. 109–111) defines symmetric spaces, shows that they are complete and that their curvature tensor is parallel (Lemma 20.1, Corollaries 20.2–20.4), and computes the conjugate points along a geodesic of a locally symmetric space from the eigenvalues of $K_V(W) = R(V,W)V$ (Theorem 20.5). §21 (pp. 112–117) shows that a Lie group with a left- and right-invariant metric is a symmetric space (Lemmas 21.1, 21.2), computes its curvature on left-invariant fields (Theorem 21.3, Corollary 21.4), draws topological consequences (Corollaries 21.5, 21.6), and proves Bott's theorem that the loop space of a compact simply connected Lie group has a CW structure with only even-dimensional cells (Theorem 21.7). Everything in §21 that is a computation in the Lie algebra is verified, for an abstract real Lie algebra L (Milnor's $\mathfrak{g}$ of left-invariant fields) with a bilinear form B (Milnor's inner product on $\mathfrak{g}$), every hypothesis on B — symmetric, positive definite, ad-invariant — being an explicit hypothesis of the theorem that uses it: Milnor's formula (d) and Theorem 21.3 (a)–(c) from the two geometric inputs of his proof, taken as hypotheses; Corollary 21.4; the algebraic steps of Corollaries 21.5 and 21.6; and the pairing of the eigenvalues of K_V that drives the proof of Theorem 21.7. For a compact group of matrices — a slice submanifold of the space of matrices, closed under products and adjoints, with a Lie algebra of skew-Hermitian matrices (V12.14) — the geometry of §21 is verified as well, in the embedding instead of through left-invariant fields: the one-parameter subgroups and their translates are the geodesics and the group is complete (Lemma 21.2, V12.15), the reflection $\sigma \mapsto \tau\sigma^{-1}\tau$ of Lemma 21.1 (V12.16), Theorem 21.3 (a)–(d) and Corollary 21.4 by the Gauss equation, which identifies the curvature with the algebraic one of V12.5 (V12.17), and the Jacobi equation of Theorem 20.5 along a one-parameter subgroup (V12.18). Not verified: §20 for general (locally) symmetric spaces, Lie groups not given as matrices, Myers's theorem in 21.5, the splitting of the group in 21.6, and Theorem 21.7 itself. Lean: `Morse/Chapter12/BiInvariant.lean` (557 lines), `MatrixInner.lean` (359), `MatrixGroups.lean` (793), `GroupGeodesics.lean` (593), `GroupJacobi.lean` (276); `SUMinimal.lean`, `SUJacobi.lean` and `SUIndex.lean` serve §23 and are reported in Chapter 14; one counting step of Theorem 21.7 (V12.13) sits in `Morse/Chapter14/IndexCount.lean`.

V12.1 (the algebraic model of a bi-invariant metric; `Morse.IsAdInvariant`, `Morse.isAdInvariant_iff_lieInva` `Morse.IsLeviCivita`). L is a real Lie algebra and $B : L \times L \to \mathbb{R}$ a bilinear form. B is *ad-invariant* if $B([X,Y],Z) = B(X,[Y,Z])$ for all X, Y, Z; this is Mathlib's `lieInvariant`. A connection on left-invariant fields is a bilinear map $D : L \times L \to L$, $D(X,Y)$ standing for $X \vdash Y$; it

is *Levi-Civita for* B (`Morse.IsLeviCivita`) if $D(X,Y) - D(Y,X) = [X,Y]$ (Milnor's 8.5) and $B(D(X,Y),Z)+B(Y,D(X,Z)) = 0$ (Corollary 8.4 for fields whose inner products are constant), for all X, Y, Z. *Book:* p. 113: the Lie algebra $\mathfrak{g}$ of left-invariant vector fields with the bracket, on a Lie group with a left and right invariant metric; 21.3 (a) is the identity $\langle [X,Y], Z\rangle = \langle X, [Y,Z]\rangle$. *Hidden hypotheses:* that $X \vdash Y$ is left-invariant when X and Y are, so that the connection restricts to a map $\mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$, which Milnor's computation of R with (d) needs. *Deviation (restated):* the Lie group, its left-invariant fields and its metric are replaced by an abstract Lie algebra with a bilinear form; the geometric facts are hypotheses (V12.2, V12.3).

V12.2 (Milnor's formula (d); `Morse.IsLeviCivita.conn_eq`). Let D be Levi-Civita for a bilinear form B (no condition on B), and suppose $D(X,X) = 0$ for every X. Then $D(X,Y) = \frac{1}{2}[X,Y]$ for all X, Y. *Book:* pp. 113–114: $X \vdash X = 0$ because the integral curves of a left-invariant X are left translates of one-parameter subgroups, hence geodesics; expanding $(X+Y) \vdash (X+Y) = 0$ and adding $X \vdash Y - Y \vdash X = [X,Y]$ gives (d), $2X \vdash Y = [X,Y]$. *Hidden hypotheses:* $X \vdash X = 0$ rests on Lemma 21.2 and on left translations being isometries. *Deviation (restated):* $X \vdash X = 0$ is a hypothesis. *Proof:* as in the book; only the symmetry of D is used.

V12.3 (Theorem 21.3 (a); `Morse.IsLeviCivita.isAdInvariant`). If D is Levi-Civita for B and $D(X,X) = 0$ for every X, then B is ad-invariant. *Book:* p. 113, Theorem 21.3 (a); proof on p. 114: $Y\langle X,Z\rangle = \langle Y \vdash X, Z\rangle + \langle X, Y \vdash Z\rangle$ (8.4), the left side vanishes because $\langle X,Z\rangle$ is constant, and (d) is substituted. *Hidden hypotheses:* the constancy of $\langle X,Z\rangle$ for left-invariant X, Z (left invariance of the metric), here built into the compatibility condition of V12.1. *Deviation (restated):* as in V12.2. *Proof:* as in the book.

V12.4 (the converse route: the Koszul formula; `Morse.IsLeviCivita.koszul`, `Morse.IsLeviCivita.conn_eq_of` `Morse.halfLie`, `Morse.halfLie_apply`, `Morse.isLeviCivita_halfLie`). If B is symmetric and D is Levi-Civita for B, then $2B(D(X,Y),Z) = B([X,Y],Z) - B([Y,Z],X) + B([Z,X],Y)$. If moreover B is ad-invariant and non-degenerate, then $D(X,Y) = \frac{1}{2}[X,Y]$. Conversely, for every ad-invariant B the map $D(X,Y) = \frac{1}{2}[X,Y]$ is Levi-Civita for B. *Book:* not in Milnor's text. *Deviation (not in the book):* this derives (d) from the ad-invariance of the metric instead of from $X \vdash X = 0$: on left-invariant fields the Levi-Civita connection of a bi-invariant metric is forced to be $\frac{1}{2}[\,\cdot\,,\cdot\,]$, which is the uniqueness of Lemma 8.6 in this algebraic model.

V12.5 (Theorem 21.3 (b); `Morse.curv`, `Morse.curv_halfLie`). For a bilinear $D : L \times L \to L$ put $R(X,Y)Z = -D(X,D(Y,Z)) + D(Y,D(X,Z)) + D([X,Y],Z)$, Milnor's sign. In any real Lie algebra, with $D = \frac{1}{2}[\,\cdot\,,\cdot\,]$, $R(X,Y)Z = \frac{1}{4}[[X,Y],Z]$. *Book:* p. 113, Theorem 21.3 (b); proof on p. 114, by substituting (d) into the definition of R and using the Jacobi identity. *Hidden hypotheses:* the left invariance of $Y \vdash Z$ (V12.1). *Proof:* as in the book.

V12.6 (Theorem 21.3 (c); `Morse.curv_inner`). If B is ad-invariant, then $B(R(X,Y)Z,W) = \frac{1}{4}B([X,Y],[Z,W])$ for $D = \frac{1}{2}[\,\cdot\,,\cdot\,]$. *Book:* p. 113, Theorem 21.3 (c), which "follows from (a) and (b)" (p. 114). *Hidden hypotheses:* none. *Proof:* as in the book.

V12.7 (Cartan's 3-form; `Morse.cartan_form_swap` , `Morse.cartan_form_swap`). For every bilinear B, $B([Y,X],Z) = -B([X,Y],Z)$; if B is ad-invariant, $B([X,Z],Y) = -B([X,Y],Z)$. So $(X,Y,Z) \mapsto B([X,Y],Z)$ is alternating. *Book:* footnote on p. 114: this trilinear function is skew-symmetric in all three variables, giving a left-invariant 3-form, from which Cartan proved $H^3(G) \neq 0$ for a non-abelian compact connected Lie group. *Hidden hypotheses:* none. *Deviation (narrower):* only the skew-symmetry; the differential form, its de Rham class and Cartan's theorem are not formalized.

V12.8 (Corollary 21.4; `Morse.sectional`, `Morse.sectional_nonneg`, `Morse.sectional_eq_zero_iff`, `Morse.sectional_std`). If B is ad-invariant, then for $D = \frac{1}{2}[\,\cdot\,,\cdot\,]$ and all X, Y, $B(R(X,Y)X,Y) = \frac{1}{4}B([X,Y],[X,Y])$; if B is also positive definite, this is ≥ 0, with equality iff $[X,Y] = 0$. With the standard sign, $B(-R(X,Y)Y,X) = \frac{1}{4}B([X,Y],[X,Y])$. *Book:* p. 115, Corollary 21.4: the sectional curvature $\langle R(X,Y)X,Y\rangle = \frac{1}{4}\langle [X,Y],[X,Y]\rangle$ is always ≥ 0, with equality iff $[X,Y] = 0$. *Hidden hypotheses:* none (positive definiteness of B, needed for the inequality, is part of "Riemannian metric"). *Deviation (stronger):* for all X, Y, not only orthonormal pairs.

V12.9 (Corollary 21.5, the algebraic step: Ricci curvature; `Morse.jacobiOp`, `Morse.curv_halfLie_eq_jacobiOp`, `Morse.trace_jacobiOp`, `Morse.trace_eq_sum_of_isOrtho`, `Morse.trace_jacobiOp_pos`, `Morse.killingForm_neg`). Let $K_V = -\frac{1}{4}\operatorname{ad}_V^2$; then $K_V(W) = R(V,W)V$ for $D = \frac{1}{2}[\,\cdot\,,\cdot\,]$, and in every real Lie algebra $\operatorname{tr} K_V = -\frac{1}{4}\kappa(V,V)$, κ Mathlib's Killing form. If L is finite-dimensional, B symmetric, positive definite and ad-invariant, and the centre of L is zero, then $\operatorname{tr} K_V > 0$ for every $V \neq 0$; equivalently $\kappa(V,V) < 0$. *Book:* p. 115, Corollary 21.5 (trivial centre implies G compact with finite fundamental group); in the proof, the Ricci curvature $K(X_1,X_1) = \sum_i \langle R(X_1,X_i)X_1,X_i\rangle$, over an orthonormal basis, is strictly positive since $[X_1,X_i] \neq 0$ for some i, and bounded away from zero; Myers's theorem (Corollary 19.5) concludes. *Hidden hypotheses:* the termwise inequality $\langle R(X_1,X_i)X_1,X_i\rangle \geq 0$ (V12.8), used without comment. *Deviation (narrower):* only the positivity of the Ricci curvature; the uniform bound and Myers's theorem are not formalized. *Deviation (not in the book):* the identification with the Killing form. *Proof:* the trace is computed in a B-orthogonal basis, where its diagonal terms are $B(K_V b_i, b_i)/B(b_i,b_i) \geq 0$, one of them positive.

V12.10 (Corollary 21.6, the algebraic step; `Morse.centerCompl`, `Morse.mem_centerCompl`, `Morse.isCompl_center_centerCompl`, `Morse.eq_zero_of_mem_centerCompl`). For ad-invariant B, the set $\mathfrak{g}' = \{X : B(C,X) = 0 \text{ for every } C \text{ in the centre } \mathfrak{c}\}$ is a Lie ideal of L. If L is finite-dimensional and B symmetric, positive definite and ad-invariant, then $L = \mathfrak{c} \oplus \mathfrak{g}'$ as vector spaces, and $\mathfrak{g}'$ has trivial centre: an element of $\mathfrak{g}'$ commuting with all of $\mathfrak{g}'$ is zero. *Book:* pp. 115–116, Corollary 21.6: a simply connected Lie group with a bi-invariant metric splits as $G' \times \mathbb{R}^k$ with G' compact and the Lie algebra of G' with trivial centre; in the proof, the orthogonal complement $\mathfrak{g}'$ of $\mathfrak{c}$ is a subalgebra, since $\langle [X,Y],C\rangle = \langle X,[Y,C]\rangle = 0$, and $\mathfrak{g} = \mathfrak{g}' \oplus \mathfrak{c}$. *Hidden hypotheses:* none at this level. *Deviation (stronger):* $\mathfrak{g}'$ is an ideal (for a complement of the centre this is the same as a subalgebra, and it is what the splitting as Lie algebras uses). *Deviation (narrower):* the splitting of the group and the compactness of G' are not formalized.

V12.11 (the operator K_V in the proof of 21.7; `Morse.jacobiOp_apply`, `Morse.ad_skew`, `Morse.jacobiOp_symm`, `Morse.jacobiOp_self`, `Morse.jacobiOp_self_nonneg`, `Morse.jacobiOp_eq_zero_iff`, `Morse.eigenvalue_nonneg`, `Morse.ad_eigenvalue_eq_zero`). Let B be ad-invariant. Then ad_V is skew, $B([V,X],Y) = -B(X,[V,Y])$; $K_V(W) = -\frac{1}{4}[V,[V,W]]$ is self-adjoint for B, and $B(K_V W,W) = \frac{1}{4}B([V,W],[V,W])$. If B is also positive definite: K_V is positive semidefinite, $K_V W = 0$ iff $[V,W] = 0$, every eigenvalue of K_V is ≥ 0, and the only real eigenvalue of ad_V is 0. *Book:* pp. 116–117: $K_V(W) = R(V,W)V = \frac{1}{4}[[V,W],V]$, $K_V = -\frac{1}{4}(\operatorname{Ad} V)\circ(\operatorname{Ad} V)$, and $\operatorname{Ad} V$ is skew-symmetric by 21.3 (a). *Hidden hypotheses:* none (positive definiteness is part of "Riemannian metric"). *Deviation (stronger):* the kernel of K_V and the real spectrum of ad_V are stated.

V12.12 (the non-zero eigenvalues of K_V occur in pairs; `Morse.jacobiOp_of_lie_lie`, `Morse.lie_lie_of_jacobiOp`, `Morse.lie_mem_eigenspace`, `Morse.eigenJ`, `Morse.eigenJ_comp_self`, `Morse.eigenJ_isometry`, `Morse.even_finrank_of_comp_self_eq_neg_id`, `Morse.even_finrank_eigenspace`).

$K_V W = \mu W$ iff $[V,[V,W]] = -4\mu W$, and ad_V maps each eigenspace E_μ of K_V into itself. For $\mu > 0$ the map $J = (2\sqrt{\mu})^{-1}\,\mathrm{ad}_V$ on E_μ satisfies $J^2 = -1$, and it preserves B when B is ad-invariant. A finite-dimensional real vector space with such a J has even dimension. Hence, for L finite-dimensional and B ad-invariant and positive definite, every eigenspace E_μ of K_V with $\mu \neq 0$ has even dimension. *Book:* p. 117: an orthonormal basis puts $\mathrm{Ad}\,V$ in block form with blocks $\begin{pmatrix} 0 & a_i \\ -a_i & 0 \end{pmatrix}$, so $(\mathrm{Ad}\,V)^2$ is diagonal with entries $-a_1^2, -a_1^2, -a_2^2, \ldots$, and the non-zero eigenvalues of K_V are positive and occur in pairs. *Hidden hypotheses:* the normal form of a skew-symmetric map is quoted without proof. *Deviation (restated):* the normal form is replaced by the complex structure J on each eigenspace, which is what "occur in pairs" needs.

V12.13 (even index from paired eigenvalues; `Morse.even_spectralIndex`, `Morse.spectralIndex`, `Morse.conjParams`). Let $e_1, \ldots, e_N$ be real numbers (a spectrum of K_V with multiplicity) and define the index of the spectrum as the sum over $t \in (0,1)$ of the number of i with $e_i > 0$ and $t \in \pi\mathbb{Z}/\sqrt{e_i}$ — Theorems 20.5 and 15.1 written as a definition, stated in full in Chapter 14 (V14.7, V14.8). If every positive value occurs among the e_i an even number of times, this index is even. *Book:* p. 117: by 20.5 the conjugate points occur in pairs, every conjugate point has even multiplicity, and with the Index Theorem the index of every geodesic from p to q is even. *Hidden hypotheses:* none. *Deviation (restated):* the index is the number defined from the spectrum; Theorems 20.5 and 15.1 are not proved for a general bi-invariant metric (for $SU(n)$ the index of a geodesic from I to $-I$ is computed from them, V14.18–V14.19). The bridge from V12.12 (even dimension of each eigenspace) to the hypothesis here (even multiplicity in a list of eigenvalues) is the spectral theorem for the self-adjoint K_V, which is not formalized in this form, so the two items are not joined into one theorem.

Matrix groups. From here on $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, and $M_n(\mathbb{K})$, the $n \times n$ matrices indexed by a finite set, carries the real inner product $\langle A, B\rangle = \mathrm{Re}\,\mathrm{tr}(A^*B)$ (Milnor's $\mathrm{Re}\,\mathrm{tr}(AB^*)$ up to the order of the factors) and the Frobenius norm; exp is the matrix exponential, and $[X,Y] = XY - YX$. The groups are slice submanifolds of $M_n(\mathbb{K})$ (V1.6) with the induced metric, so that geodesics (V7.1), the connection (V6.4), the curvature (V6.17) and Jacobi fields (V9.2) are those of Chapters 6–9.

V12.14 (matrix groups and their bi-invariant metric; `Morse.MatrixInner.innerProductSpace`, `Morse.inner_def`, `Morse.inner_mul_left_of_unitary`, `Morse.inner_mul_right_of_unitary`, `Morse.inner_eq_zero_of_hermitian_of_skew`, `Morse.contDiff_exp_matrix`, `Morse.exp_conjTranspose_mul_`, `Morse.skewHermitian`, `Morse.IsMatrixGroup`, `Morse.IsMatrixGroup.tangentSubspace_eq`, `Morse.IsMatrixGroup.mem_tangentSubspace_iff`, `Morse.IsMatrixGroup.mulLeftIso_image`, `Morse.IsMatrixGroup.mulRightIso_image`, `Morse.frobForm`, `Morse.isAdInvariant_frobForm`, `Morse.frobForm_pos`). Multiplication on either side by a unitary matrix is a linear isometry of $M_n(\mathbb{K})$; Hermitian and skew-Hermitian matrices are orthogonal; exp is smooth and maps skew-Hermitian matrices to unitary ones. A *matrix group* is a pair $(G, \mathfrak{g})$ of a slice submanifold $G \subseteq M_n(\mathbb{K})$ and a real Lie subalgebra $\mathfrak{g}$ of $M_n(\mathbb{K})$ such that every $g \in G$ is unitary, G is closed under products and $g \mapsto g^*$, every $X \in \mathfrak{g}$ is skew-Hermitian, $g\mathfrak{g}g^* \subseteq \mathfrak{g}$ and $\exp\mathfrak{g} \subseteq G$, and $\dim G = \dim\mathfrak{g}$. Then $T_gG = g\mathfrak{g}$ for every $g \in G$; left and right translations by elements of G are linear isometries of $M_n(\mathbb{K})$ carrying G onto G, so the induced metric is bi-invariant; and the inner product restricted to $\mathfrak{g}$ is ad-invariant and positive definite, so V12.1–V12.13 apply to it. *Book:* p. 112: a Lie group with a metric invariant under left and right translations; for compact G such a metric exists by averaging with the Haar measure; p. 125 (in §23): the tangent space of $U(n)$ at I is the space of skew-Hermitian matrices. *Hidden hypotheses:* none at this level. *Deviation (narrower):* only groups given as matrices, with the induced metric; Milnor's abstract Lie groups,

left-invariant fields and Haar averaging are not formalized. (Every compact Lie group is a matrix group by Peter–Weyl, which is not needed for §§23–24.) *Proof:* $g\mathfrak{g} \subseteq T_gG$ through the curves $t \mapsto g\exp(tX)$, and the dimensions agree.

V12.15 (Lemma 21.2 for matrix groups, and completeness; `Morse.IsMatrixGroup.isGeodesicOn_mul_exp_smul`, `Morse.IsMatrixGroup.isGeodesicOn_exp_smul`, `Morse.IsMatrixGroup.eqOn_mul_exp_smul`, `Morse.IsMatrixGroup.lemma_21_2`, `Morse.IsMatrixGroup.eq_exp_of_oneParam`, `Morse.IsMatrixGroup.isGeo` `Morse.IsMatrixGroup.exp_eq`, `Morse.IsMatrixGroup.isGeodesicallyComplete`). Let $(G, \mathfrak{g})$ be a matrix group. For $g \in G$ and $X \in \mathfrak{g}$, $t \mapsto g\exp(tX)$ is a geodesic of G on $\mathbb{R}$. Conversely a geodesic γ of G on a preconnected open set $s \ni t_0$ is $\gamma(t) = \gamma(t_0)\exp((t-t_0)X)$ on s with $X = \gamma(t_0)^*\gamma'(t_0) \in \mathfrak{g}$. (i) A geodesic $\gamma : \mathbb{R} \to G$ with $\gamma(0) = I$ satisfies $\gamma'(0) \in \mathfrak{g}$, $\gamma(t) = \exp(t\gamma'(0))$ and $\gamma(s+t) = \gamma(s)\gamma(t)$. (ii) A differentiable $\theta : \mathbb{R} \to G$ with $\theta(0) = I$ and $\theta(s+t) = \theta(s)\theta(t)$ is a geodesic. The Riemannian exponential at g is $\exp_g(gX) = g\exp X$, and G is geodesically complete (V7.22). *Book:* p. 113, Lemma 21.2: the geodesics γ with $\gamma(0) = e$ are precisely the one-parameter subgroups; p. 110, Corollary 20.2 (a symmetric space is complete). *Hidden hypotheses:* in Milnor's proof of (i), the passage from commensurable parameters to all of $\mathbb{R}$ "by continuity" needs the continuity of multiplication. *Deviation (restated):* for matrix groups; the proof does not go through Lemma 20.1. *Proof:* the acceleration $g\exp(tX)X^2$ is normal, X^2 being Hermitian; uniqueness of geodesics (V7.5) gives the converse; (ii) $\exp(-tX)\theta(t)$ has derivative 0.

V12.16 (Lemma 21.1 for matrix groups; `Morse.IsMatrixGroup.symmIso`, `Morse.IsMatrixGroup.symmIso_mem`, `Morse.IsMatrixGroup.symmIso_self`, `Morse.IsMatrixGroup.symmIso_mul_exp_smul`). For a unitary τ, $I_\tau(A) = \tau A^*\tau$ is an involutive linear isometry of $M_n(\mathbb{K})$; on G it is $\sigma \mapsto \tau\sigma^{-1}\tau$. For $\tau \in G$ it maps G into G, fixes τ, and reverses the geodesics through τ: $I_\tau(\tau\exp(tX)) = \tau\exp(-tX)$ for $X \in \mathfrak{g}$. *Book:* p. 112, Lemma 21.1: with a bi-invariant metric, $I_\tau(\sigma) = \tau\sigma^{-1}\tau$ is an isometry fixing τ that reverses geodesics through τ, so G is a symmetric space. *Hidden hypotheses:* "symmetric space" includes connectedness, so the last clause needs G connected. *Deviation (narrower):* the definition of a symmetric space is not formalized, so "G is a symmetric space" is not a statement; by V12.15 every geodesic through τ is a $\tau\exp(tX)$, so all of them are reversed.

V12.17 (Theorem 21.3 and Corollary 21.4 for matrix groups; `Morse.trace_gauss_identity`, `Morse.inner_gauss_identity`, `Morse.inner_lie_left`, `Morse.algProj`, `Morse.algProj_mul`, `Morse.IsMatrixGroup.tanProj_eq`, `Morse.IsMatrixGroup.secondFundForm_mul`, `Morse.IsMatrixGroup.covAt` `Morse.IsMatrixGroup.curvOp_mul`, `Morse.IsMatrixGroup.inner_curvOp_mul`, `Morse.IsMatrixGroup.section` `Morse.IsMatrixGroup.sectional_mul_nonneg`, `Morse.IsMatrixGroup.curvOp_eq_curv_halfLie`, `Morse.IsMatrixGroup.coe_curvTensor_eq_curv_halfLie`, `Morse.IsMatrixGroup.curvOp_eq_jacobiOp`). Let $(G, \mathfrak{g})$ be a matrix group, $g \in G$, $X, Y, Z, W \in \mathfrak{g}$. (a) $\langle [X,Y], Z\rangle = \langle X, [Y,Z]\rangle$. (d) The covariant derivative of the left-invariant field $q \mapsto qY$ in the direction gX is $g \cdot \frac{1}{2}[X,Y]$. The orthogonal projection onto T_gG is $M \mapsto gP_{\mathfrak{g}}(g^*M)$, and the second fundamental form is $\mathrm{II}_g(gX, gY) = \frac{1}{2}g(XY + YX)$. (b) $R(gX,gY)gZ = \frac{1}{4}g[[X,Y],Z]$, Milnor's sign; this is g times the algebraic curvature of V12.5 for $D = \frac{1}{2}[\,\cdot\,,\cdot\,]$, also for the trilinear tensor of Lemma 9.1, and $R(gA,gW)gA = g\,K_A(W)$ with K_A of V12.9. (c) $\langle R(gX,gY)gZ, gW\rangle = \frac{1}{4}\langle [X,Y],[Z,W]\rangle$. Corollary 21.4: $\langle R(gX,gY)gX, gY\rangle = \frac{1}{4}\|[X,Y]\|^2 \geq 0$, with equality iff $XY = YX$. *Book:* p. 113, Theorem 21.3 (a)–(c), with (d) inside the proof on p. 114; p. 115, Corollary 21.4. *Hidden hypotheses:* the left invariance of $X \vdash Y$ and $X \vdash X = 0$ (V12.1–V12.3), here theorems. *Deviation (restated):* for matrix groups, by the Gauss equation (V6.18) instead of substituting (d) into the definition of R: the second fundamental form gives $\frac{1}{4}(\langle XZ + ZX, YW + WY\rangle - \langle XW + WX, YZ + ZY\rangle)$,

which a trace identity turns into $\frac{1}{4}\langle[[X,Y],Z],W\rangle$. This removes, for matrix groups, the hypotheses of V12.1–V12.3.

V12.18 (Theorem 20.5 for matrix groups: the Jacobi equation along a one-parameter subgroup; `Morse.halfExp`, `Morse.jacK`, `Morse.IsMatrixGroup.covD_conj`, `Morse.IsMatrixGroup.curvR_conj`, `Morse.IsMatrixGroup.isJacobiField_conj`, `Morse.IsMatrixGroup.covAlong_conj_zero`). Let $(G,\mathfrak{g})$ be a matrix group, $A\in\mathfrak{g}$, $\gamma(t)=\exp(tA)$ and $h(t)=\exp(tA/2)$. For a differentiable $\mathfrak{g}$-valued U, the covariant derivative of $t\mapsto h(t)U(t)h(t)$ along γ is $hU'h$, and $R(\gamma',hMh)\gamma'=h\,K_A(M)\,h$ for $M\in\mathfrak{g}$, $K_A(M)=\frac{1}{4}[[A,M],A]$. Hence, if U is a smooth $\mathfrak{g}$-valued curve with $U''+K_A(U)=0$, then $J=hUh$ is a Jacobi field along γ, with $\frac{DJ}{dt}(0)=U'(0)$. *Book:* p. 110, Theorem 20.5: in a locally symmetric space the conjugate points along γ are the $\gamma(\pi k/\sqrt{e_i})$, $e_i>0$ an eigenvalue of $K_V(W)=R(V,W)V$, with multiplicity the number of e_i for which t is a multiple of $\pi/\sqrt{e_i}$; proof p. 111: in a parallel eigenframe the Jacobi equation becomes $w_i''+e_iw_i=0$. *Hidden hypotheses:* that K_V is self-adjoint, so that an orthonormal eigenbasis exists (Lemma 9.3). *Deviation (narrower):* only along one-parameter subgroups of matrix groups, and only the implication from the constant-coefficient equation to Jacobi fields; the parallel frame is $U\mapsto hUh$, so no eigenbasis is chosen. The count of conjugate points with their multiplicities is proved for $SU(n)$ only (V14.18).

What is not verified in this chapter

- **§20 for general symmetric spaces** (pp. 109–111): the definition of a symmetric space, Lemma 20.1, Corollaries 20.2–20.4, locally symmetric spaces, and Theorem 20.5 beyond one-parameter subgroups of matrix groups (V12.18) and $SU(n)$ (V14.18). The project's geodesics, curvature and Jacobi fields are those of submanifolds of a Euclidean space (Chapters 6–9); isometries of an abstract manifold and the parallel eigenframe of Milnor's proof are not formalized.
- **The passage from an abstract Lie group to its Lie algebra:** left-invariant fields, the existence of a bi-invariant metric on a compact group by averaging with Haar measure (p. 112), and Lemmas 21.1–21.2 and Theorem 21.3 for groups not given as matrices. For matrix groups they are V12.14–V12.17; for an abstract group the hypotheses of V12.1–V12.3 remain hypotheses.
- **The topology of Corollaries 21.5 and 21.6:** the application of Myers's theorem (Corollary 19.5, V11.13) to G, the compactness and finite fundamental group of G; the splitting of a simply connected G as $G'\times\mathbb{R}^k$. (For 21.5 the exposition also records that G must be connected.)
- **Theorem 21.7 and its Remarks** (pp. 116–117): the CW structure of $\Omega(G)$ with only even cells, finitely many in each dimension (it would combine Theorem 17.3, V10.22, with Theorem 20.5 and the index theorem for a general compact simply connected G), Remark 1 (the homology of ΩS^3) and Remark 2 (Iwasawa's theorem). Only the algebra of V12.11–V12.13 is checked.
- **The normal form of a skew-symmetric map** (p. 117), replaced by V12.12.

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14. Chapter 13 (Milnor §22): partly verified

Milnor's §22 (pp. 118–123) shows that when every non-minimal geodesic from p to q has index at least λ_0, the space Ω^d of minimal geodesics carries the homotopy groups of the path space below λ_0-1 (Theorem 22.1), so that $\pi_i\Omega^d\cong\pi_{i+1}M$ for $i\le\lambda_0-2$ (Corollary 22.2); for antipodal points of S^{n+1} this is the Freudenthal suspension theorem (Corollary 22.3). The proof perturbs a function whose minimum set is a manifold without lowering the indices of its other critical

points (Lemma 22.4), pushes maps past the critical levels (Lemma 22.5), and runs this on the finite-dimensional model of Chapter 10. Verified, for a slice submanifold N of a finite-dimensional inner product space with the induced metric and Milnor's path space $\Omega = \Omega(N; p, q)$ with his metric (V10.1): Theorem 22.1 in a strong form (every map of a compact space of small dimension into Ω is homotopic, relative to the part already in Ω^d, to a map into Ω^d), with **the hypothesis "Ω^d is a topological manifold" replaced by "Ω^d is a retract of an open neighbourhood in Ω"**, which is what Milnor's proof uses of it through Hanner's theorem, not formalized; Milnor's corollary on the inclusion $\Omega^d \to \Omega$, including π_0 and the surjectivity one degree higher; Corollary 22.2 as a bijection and, for $i \geq 1$, as a group isomorphism, with the change of basepoint it hides; Lemmas 22.4 and 22.5 and the exposition's Lemma D; the exposition's Lemmas A and B for the sphere, the neighbourhood retraction written down there, and Corollary 22.3 with its second half ($i = 2n - 1$, a surjection), as sets and as groups; and the variant of Theorem 22.1 for an open and closed set of paths that §24 uses (p. 147). Not verified: Hanner's theorem, the homological form of the corollary (relative Hurewicz), and the setting of an abstract Riemannian manifold. Lean: `Morse/Chapter13/` — `Perturbation.lean` (364 lines), `Compression.lean` (358), `ManifoldOfMinima.lean` (338), `MinimalDeform.lean` (210), `Theorem22.lean` (306), `Corollaries.lean` (141), `CorollaryGroups.lean` (142), `SphereMinimal.lean` (389), `Freudenthal.lean` (257), `Componentwise.lean` (385), `SUInstance.lean` (158, reported in Chapter 14); change of basepoint as a homomorphism in `Morse/Homotopy/ChangeOfBaseGroup.lean` (137).

The setting. V is a finite-dimensional real inner product space and $N \subseteq V$ a slice submanifold (V1.6) with the induced metric. $\Omega = \Omega(N; p, q)$ is the space of piecewise smooth paths with Milnor's metric (V10.1); paths are functions on $\mathbb{R}$ whose values off $[0, 1]$ play no role, so this metric is only a pseudometric, and two paths agreeing on $[0, 1]$ are at distance 0 (V10.5). E is the energy (V8.5), a geodesic of Ω is an unbroken geodesic (V8.10), and its index is the index of E_{**} (V8.14, V9.13). ρ is the Riemannian distance (V7.18); "the closed balls about p are compact" stands for Milnor's completeness (Hopf–Rinow, V7.29). $\pi_i(X, x)$ is Mathlib's homotopy group, the classes of maps $I^i \to X$ sending ∂I^i to x (π_0 a pointed set); a continuous map induces maps of homotopy groups (`Morse.Homotopy.homotopyGroupMap`), which are homomorphisms for $i \geq 1$ (`Morse.Homotopy.homotopyGroupHom`). *Pushing maps* (V13.12) replaces relative homotopy groups: "$\pi_r(X, A) = 0$" is read as "every map $(I^r, \partial I^r) \to (X, A)$ is homotopic relative to ∂I^r to a map into A", the exposition's (H1), and the theorems state this for every compact space P injected continuously into a real vector space of dimension less than the bound, for instance $P = I^r$ or $P = I \times I^r$.

V13.1 (the space Ω^d; `Morse.minPaths`, `Morse.isUnbrokenGeodesic_of_minimal`). Ω^d is the set of paths $\gamma \in \Omega$ whose energy is at most that of every path of Ω. A path of least energy is an unbroken geodesic. *Book:* p. 118: Ω^d denotes the space of minimal geodesics from p to q, $d = \rho(p, q)^2$. *Hidden hypotheses:* that the minimal geodesics, parametrized proportionally to arc length, are exactly the paths of least energy d (Lemma 12.1, V8.6); that one exists (Hopf–Rinow). *Deviation (restated):* Ω^d is defined by least energy; d is not named. *Proof:* a minimum of E is a critical point (every variation gives a function of u with a minimum at 0), and Corollary 12.3 (V8.11) applies.

V13.2 (Theorem 22.1; `Morse.theorem_22_1`, `Morse.compresses_minPaths`, `Morse.minPathsIncl`). Let $p, q \in N$, let the closed Riemannian balls about p be compact, and let $\lambda_0 \geq 0$ be an integer such that every geodesic $\gamma \in \Omega$ which is not of least energy has index $\geq \lambda_0$. Suppose Ω^d is a retract of an open neighbourhood: there are an open $O \supseteq \Omega^d$ and a map $\rho : O \to \Omega^d$, continuous on O, equal to the identity on Ω^d. Let P be a compact space with a continuous injection into a real vector space of dimension $< \lambda_0$. Then every map $h : P \to \Omega$ is homotopic, relative to

$h^{-1}(\Omega^d)$, to a map into Ω^d; equivalently, the inclusion $\Omega^d \to \Omega$ compresses maps from P. *Book:* p. 118, Theorem 22.1: if Ω^d is a topological manifold and every non-minimal geodesic from p to q has index $\geq \lambda_0$, then $\pi_i(\Omega, \Omega^d) = 0$ for $0 \leq i < \lambda_0$; proof pp. 122–123. *Hidden hypotheses:* M complete (the standing assumption of §22); the meaning of $\pi_i(\Omega, \Omega^d) = 0$ for $i = 0, 1$, where these are sets. *Deviation (narrower):* the manifold hypothesis is replaced by the neighbourhood retraction, which is all the proof uses of it; for a manifold it follows from Hanner's theorem (in the metrizable manifold B of broken geodesics, composed with Milnor's r_1), which is not formalized. For the sphere and for $SU(2m)$ the retraction is constructed (V13.8, V14.21). *Deviation (stronger):* relative to the whole part of P already mapped into Ω^d, at all base points, for every such P. *Proof:* Milnor's, through the finite-dimensional model (V10.4–V10.11): the energy is bounded on the compact image; Milnor's deformation r_u (V10.7) moves h into the broken geodesics, fixing Ω^d, which consists of unbroken geodesics (V13.1); the manifold of break points carries the energy as a smooth function with compact sublevel sets, whose critical points off the minimum have index $\geq \lambda_0$ by Theorem 16.2 (V10.11); Lemma 22.5 below a level (V13.14) with the neighbourhood of V13.15 pushes the map into the minimum; the result is moved back to Ω, corrected on paths at distance 0. No reparametrization $F \circ E$ is needed, since V13.14 needs only one compact sublevel set.

V13.3 (the inclusion $\Omega^d \to \Omega$ on homotopy groups; `Morse.theorem_22_1_bijective`, `Morse.theorem_22_1_surjective`). Under the hypotheses of V13.2, at every $\gamma \in \Omega^d$ the inclusion induces a bijection $\pi_i(\Omega^d, \gamma) \to \pi_i(\Omega, \gamma)$ for $i \leq \lambda_0 - 2$ and a surjection for $i \leq \lambda_0 - 1$, π_0 included. For $i \geq 1$ the induced map is a homomorphism, hence an isomorphism of groups. *Book:* p. 118: it follows that the inclusion homomorphism $\pi_i\Omega^d \to \pi_i\Omega$ is an isomorphism for $i \leq \lambda_0 - 2$ (from the exact sequence of the pair), and onto for $i = \lambda_0 - 1$. *Hidden hypotheses:* the exact sequence ends with pointed sets; $i = 0$ needs a direct argument (exposition, §1). *Deviation (restated):* no exact sequence of the pair: compressing cubes gives surjectivity and compressing cylinders $I \times I^i$ injectivity, uniformly in i.

V13.4 (Corollary 22.2 as a bijection; `Morse.corollary_22_2`, `Morse.pathCHomeomorph`, `Morse.contractLoop`, `Morse.contractPath`, `Morse.corollary_22_2_surjective`). Under the hypotheses of V13.2, for every $\gamma \in \Omega^d$ and $i + 2 \leq \lambda_0$ there is a bijection $\pi_i(\Omega^d, \gamma) \cong \pi_{i+1}(N, p)$; for $i + 1 \leq \lambda_0$ there is a surjection $\pi_i(\Omega^d, \gamma) \to \pi_{i+1}(N, p)$. *Book:* p. 118, Corollary 22.2: $\pi_i\Omega^d \cong \pi_{i+1}M$ for $0 \leq i \leq \lambda_0 - 2$, since $\pi_i\Omega \cong \pi_{i+1}M$. *Hidden hypotheses:* the base point: $\pi_i(\Omega, \gamma)$ is taken at a minimal geodesic, while $\pi_i(\Omega M) \cong \pi_{i+1}M$ concerns the component of the constant loop; and Theorem 17.1 (V10.19) for $\Omega \simeq \Omega^*$. *Deviation (stronger):* no connectedness of N is needed, since γ itself is used to pass from paths $p \to q$ to loops at p. *Proof:* six bijections: V13.3; $\Omega \to \Omega^*$ (Theorem 17.1); Ω^* is homeomorphic to Mathlib's paths from p to q in N; $\delta \mapsto \delta \cdot \gamma^{-1}$, a homotopy equivalence onto the loops at p, sending γ to $\gamma \cdot \gamma^{-1}$; change of basepoint along the contraction $s \mapsto \gamma|_{[0,s]} \cdot \gamma|_{[0,s]}^{-1}$ to the constant loop; and $\pi_i(\Omega N) \cong \pi_{i+1}N$.

V13.5 (change of basepoint is a homomorphism, and Corollary 22.2 as groups; `Morse.Homotopy.changeBaseMap`, `Morse.Homotopy.changeBaseMulEquiv`, `Morse.Homotopy.evalZeroHomotopyEquiv`, `Morse.nonempty_mulEquiv`, `Morse.corollary_22_2_mulEquiv`, `Morse.corollary_22_2_surjective_hom`). For a path ω from y_0 to y_1 in any space Y and $i \geq 1$, the change of basepoint $\pi_i(Y, y_0) \to \pi_i(Y, y_1)$ along ω is a group isomorphism. Hence, for every $\gamma \in \Omega$ and $j \geq 0$, $\pi_{j+1}(\Omega, \gamma) \cong \pi_{j+2}(N, p)$ as groups; under the hypotheses of V13.2, $\pi_i(\Omega^d, \gamma) \cong \pi_{i+1}(N, p)$ as groups for $1 \leq i \leq \lambda_0 - 2$, and there is a surjective homomorphism for $1 \leq i \leq \lambda_0 - 1$. *Book:* p. 118, Corollary 22.2 (the isomorphisms are of groups for $i \geq 1$). *Hidden hypotheses:* that the change of basepoint hidden in "$\pi_i\Omega \cong \pi_{i+1}M$" is a homomorphism. *Deviation (not in the book):* the homomorphism property is proved, not quoted.

Proof: on the free path space $C(I, Y)$, evaluation at 1 is homotopic to evaluation at 0 along the tracks $s \mapsto \omega(s)$, so $(\mathrm{ev}_1)_* = c_\omega \circ (\mathrm{ev}_0)_*$; $(\mathrm{ev}_0)_*$ is onto (it is a homotopy equivalence) and both are homomorphisms, so c_ω is one; the other five steps of V13.4 are induced by maps or are Mathlib's loop-space isomorphism.

V13.6 (the exposition's Lemma A: the geodesics from p to $-p$ on S^{n+1}, and the minimal ones; `Morse.exists_eqOn_greatCircle_antipode`, `Morse.greatCircle_mem_pathSpace_antipode`, `Morse.pi_sq_le_energy_sph`, `Morse.equator`, `Morse.semicircle`, `Morse.semicircle_mem_minPaths`, `Morse.exists_eqOn_semicircle`, `Morse.continuous_semicircle`, `Morse.sphMinPathsHomotopyEquiv`, `Morse.sphMinPathsHomeomorph`). On the unit sphere $S^{n+1} \subseteq \mathbb{R}^{n+2}$ (V6.25), for $p \in S^{n+1}$: every geodesic from p to $-p$ agrees on $[0, 1]$ with $t \mapsto \cos(Lt)p + \sin(Lt)u$ for a unit $u \perp p$ and $L = (2m + 1)\pi$, $m \geq 0$, and each of these is one; every path from p to $-p$ has energy $\geq \pi^2$; the paths of least energy are, on $[0, 1]$, the semicircles $\gamma_u(t) = \cos(\pi t)p + \sin(\pi t)u$, u in the equator $\{u : \|u\| = 1, u \perp p\}$; $u \mapsto \gamma_u$ is continuous for Milnor's metric and is a homotopy equivalence from the equator to Ω^d, with homotopy inverse $\gamma \mapsto \gamma(\frac{1}{2})$, and a homeomorphism of the equator onto the separation quotient of Ω^d (paths at distance 0 identified). *Book:* p. 119, proof of Corollary 22.3: the space Ω^{π^2} of minimal geodesics from p to $-p$ "can be identified with" the equator S^n. *Hidden hypotheses:* the list of geodesics from p to $-p$, and that $d = \pi^2$ (a minimal geodesic exists and is one of them). *Deviation (restated):* "homeomorphic" holds only up to the separation quotient, because the pseudometric does not see values off $[0, 1]$; the homotopy equivalence is what the homotopy groups need. *Proof:* the speeds of the semicircles are all π, so the L^2 part of Milnor's metric vanishes on them, and the uniform part is controlled by pointwise convergence at bounded energy.

V13.7 (the exposition's Lemma B: the indices; `Morse.formIndex_greatCircle_antipode`, `Morse.le_formIndex_of_not_minimal_sph`). The geodesic $t \mapsto \cos((2m + 1)\pi t)p + \sin((2m + 1)\pi t)u$ from p to $-p$ on S^{n+1} has index $2mn$. Every geodesic from p to $-p$ that is not of least energy has index $\geq 2n$. *Book:* p. 119: every non-minimal geodesic winds one and a half times around S^{n+1} and therefore has index $\geq 2n$ (two interior conjugate points of multiplicity n). *Hidden hypotheses:* the conjugate points along a great circle and their multiplicity n (V10.23); the end point $t = 1$ is conjugate but does not count. *Proof:* the index theorem (V9.24): the parameters $t = j/(2m + 1)$, $1 \leq j \leq 2m$, each of multiplicity n.

V13.8 (the neighbourhood retraction for S^{n+1}; `Morse.eqComp`, `Morse.sphNbhd`, `Morse.isOpen_sphNbhd`, `Morse.sphRetrPath`, `Morse.continuousOn_sphRetrPath`, `Morse.exists_retraction_minPaths_sph`). The set O of paths $\omega \in \Omega(S^{n+1}; p, -p)$ with $\omega(\frac{1}{2}) \neq \pm p$ is open and contains Ω^d, and $\omega \mapsto$ the semicircle through the normalized component of $\omega(\frac{1}{2})$ orthogonal to p, changed to the identity on Ω^d, is a retraction of O onto Ω^d, continuous on O. *Book:* p. 119: "evidently the hypotheses of Theorem 22.1 are satisfied" ($\Omega^d \cong S^n$ is a manifold). *Deviation (not in the book):* the retraction that Hanner's theorem would provide is written down, so the hypothesis of V13.2 is proved, not assumed. *Proof:* on Ω^d the semicircle agrees with the path on $[0, 1]$, at distance 0, which keeps the modified map continuous.

V13.9 (Corollary 22.3, Freudenthal; `Morse.equatorHomeomorph`, `Morse.hball_sph`, `Morse.hidx_sph`, `Morse.corollary_22_3`, `Morse.corollary_22_3_surjective`, `Morse.corollary_22_3_mulEquiv`, `Morse.corollary_22_3_surjective_hom`). For all $x \in S^n$ and $y \in S^{n+1}$: $\pi_i(S^n, x) \cong \pi_{i+1}(S^{n+1}, y)$ for $i \leq 2n - 2$, as sets, and as groups for $1 \leq i \leq 2n - 2$; and for $i \leq 2n - 1$ there is a surjection $\pi_i(S^n, x) \to \pi_{i+1}(S^{n+1}, y)$, a homomorphism for $i \geq 1$. *Book:* p. 119, Corollary 22.3: $\pi_i S^n \cong \pi_{i+1} S^{n+1}$ for $i \leq 2n - 2$. *Hidden hypotheses:* the base points (the pole $p = y$ is

arbitrary, and every point of S^n is the midpoint of a semicircle). *Deviation (stronger):* the second half of Freudenthal's theorem, $i = 2n - 1$, which Milnor does not state in 22.3 (it follows from his remark on $\pi_{\lambda_0 - 1}$). Not formalized: that the isomorphism is the suspension homomorphism (the exposition's remark). *Proof:* V13.4–V13.5 with $\lambda_0 = 2n$ (V13.7), the compact balls of the closed sphere, the retraction V13.8, V13.6, and the equator $\cong S^n$ by an isometry of $p^\perp$ with $\mathbb{R}^{n+1}$.

V13.10 (Lemma 22.4; `Morse.exists_negDef_margin`, `Morse.le_index_of_norm_sub_lt`, `Morse.lemma_22_4`, `Morse.fderiv_hessian_add`, `Morse.lemma_22_4_add`). Let E be a finite-dimensional real normed space, $U \subseteq E$ open, $K \subseteq U$ compact, $f : E \to \mathbb{R}$ of class C^2 on U, and suppose every critical point of f in K has index $\geq \lambda_0$ (critical points may be degenerate). Then there is $\varepsilon > 0$ such that for every $g : E \to \mathbb{R}$ with $\|dg_x - df_x\| < \varepsilon$ and $\|d^2 g_x - d^2 f_x\| < \varepsilon$ (operator norms) at every $x \in K$, every critical point of g in K has index $\geq \lambda_0$. The same with $g = f + h$, h of class C^2 on U with its first two derivatives smaller than ε on K. On the way: a form negative definite on a subspace W is $\leq -\mu\|v\|^2$ there for some $\mu > 0$, and every form within μ of it has index $\geq \dim W$. *Book:* p. 119, Lemma 22.4, for $K \subseteq U \subseteq \mathbb{R}^n$, f, g smooth, closeness of the first and second partial derivatives uniformly on K; proof pp. 120–121, through the continuity of the λ-th eigenvalue and "a uniform continuity argument which will be left to the reader". *Hidden hypotheses:* that f may have degenerate critical points (needed in V13.2: non-minimal geodesics may be degenerate). *Deviation (restated):* operator norms instead of partial derivatives (equivalent up to constants); any finite-dimensional E; f only C^2. *Proof:* no eigenvalues: the index is the largest dimension of a negative definite subspace (V1.3, `Morse.index_isGreatest`), negative definiteness on a fixed subspace survives small perturbations, and a non-critical point stays non-critical for nearby g; compactness of K gives one ε (Milnor's left-to-the-reader step).

V13.11 (Lemma 22.4 on a submanifold; `Morse.exists_perturb_slice`, `Morse.lemma_22_4_submanifold`). Let $N \subseteq V$ be a slice submanifold, $F : V \to \mathbb{R}$ smooth and $K \subseteq N$ compact, and suppose every critical point of $F|N$ in K has index $\geq \lambda_0$. Then there is $\delta > 0$ such that for every linear form a on V with $\|a\| < \delta$, every critical point of $(F + a)|N$ in K has index $\geq \lambda_0$. *Book:* p. 122, in the proof of Lemma 22.5: g is chosen with derivatives so close to those of f that 22.4 applies "in a coordinate neighbourhood" of each point of the compact set. *Deviation (narrower):* only linear perturbations, the ones by which Corollary 6.8 (V4.22, `Morse.ae_isMorseFunction_add_inner_comp`) produces Morse functions. *Proof:* V13.10 in each slice chart, the first two derivatives of the parametrization being bounded on a closed ball; finitely many charts cover K.

V13.12 (pushing maps, and pushing off cells; `Morse.PushesInto`, `Morse.pushesInto_self`, `Morse.PushesInto.mono`, `Morse.PushesInto.trans`, `Morse.pushesInto_of_isStrongDeformationRetract`, `Morse.pushesInto_union_cell`, `Morse.pushesInto_union_cells`). For subsets A, X of a space T and a space P: *A pushes maps from P out of X* if every map $h : P \to T$ with values in X is homotopic in T, relative to $h^{-1}(A)$, to a map into A. This is transitive ($A \subseteq B$, A pushes out of B, B out of X give A out of X) and monotone in X; a strong deformation retract A of X pushes maps from every P out of X. If T is Hausdorff, A closed, P compact with a continuous injection into a real vector space of dimension d, and $\chi_1, \dots, \chi_k$ are continuous maps of closed unit balls of dimensions $n_i > d$, injective, with pairwise disjoint images, each meeting A exactly in the image of the boundary sphere, then A pushes maps from P out of $A \cup \chi_1(D^{n_1}) \cup \cdots \cup \chi_k(D^{n_k})$. *Book:* p. 122: "since $r < \lambda_0$ it follows that h is homotopic … to a map into $g^{-1}(-\infty, 2\delta]$". *Hidden hypotheses:* the push-off lemma of cellular approximation (Hatcher, Lemma 4.10), the exposition's (T5); deformations fixing the lower level. *Deviation (not in the book):* the notion of pushing replaces $\pi_r(X, A) = 0$. *Proof:* one cell is the compression lemma for attaching a cell (Part XII of the proof-by-proof edition); several cells by induction, the partial union being closed.

V13.13 (the exposition's Lemma D: passing critical levels of high index; `Morse.exists_gap_of_finite`, `Morse.lemma_D`). Let $N \subseteq V$ be a slice submanifold, $F : V \to \mathbb{R}$ smooth, $f = F|N$, with $f^{-1}(-\infty, a_0]$ compact; let a be a regular value of f, and suppose every critical point q with $a < f(q) \le a_0$ is non-degenerate of index $\ge \lambda_0$. Let P be compact with a continuous injection into a real vector space of dimension $< \lambda_0$. Then for $a \le b < a_0$, $N^a = f^{-1}(-\infty, a]$ pushes maps from P out of N^b. *Book:* p. 122: $g^{-1}(-\infty, c+\delta]$ "has the homotopy type of" $g^{-1}(-\infty, 2\delta]$ with cells of dimension $\ge \lambda_0$ attached. *Hidden hypotheses:* the critical points above a are finitely many (isolated in a compact set, and not accumulating at the regular level a); the deformations of Theorem 3.1 and Remark 3.3 fix the lower level, which an abstract homotopy type would not provide. *Deviation (restated):* the exposition's lemma, with b not required to be regular. *Proof:* induction on the number of critical points below b: Theorem 3.1 on a submanifold (V2.3, `Morse.theorem_3_1_submanifold`) between critical levels, Remark 3.3 (V2.9, `Morse.remark_3_3_submanifold`) at a critical level, V13.12 for the cells.

V13.14 (Lemma 22.5; `Morse.pushesInto_of_deformation`, `Morse.criticalPoint_hessianM_congr`, `Morse.lemma_22_5_sublevel`, `Morse.lemma_22_5`). Let $N \subseteq V$ be a slice submanifold, $F : V \to \mathbb{R}$ smooth, $f = F|N \ge d$, $M^0 = f^{-1}(d)$, and P compact with a continuous injection into a real vector space of dimension $< \lambda_0$. Suppose M^0 has an open neighbourhood U such that M^0 pushes maps from P out of U (for instance, U deforms into M^0 within N, fixing M^0). (i) If $f^{-1}(-\infty, a_0]$ is compact and every critical point q with $d < f(q) < a_0$ has index $\ge \lambda_0$, then M^0 pushes maps from P out of $f^{-1}(-\infty, b]$ for every $b < a_0$. (ii) If all sublevel sets are compact and every critical point outside M^0 has index $\ge \lambda_0$, then M^0 pushes maps from P out of N. *Book:* p. 121, Lemma 22.5: for $f : M \to \mathbb{R}$ smooth with minimum 0 and compact M^c, if M^0 is a manifold and every critical point in $M - M^0$ has index $\ge \lambda_0$, then $\pi_r(M, M^0) = 0$ for $0 \le r < \lambda_0$; proof pp. 121–122. *Hidden hypotheses:* M^0 compact; the approximation g must live on an open set, where f is bounded; regular levels. *Deviation (narrower):* the conclusion of Milnor's first step (Hanner's theorem and short geodesics: a neighbourhood deforming into M^0) is the hypothesis on U. *Deviation (stronger):* (i) needs only one compact sublevel set, the form used in V13.2. *Proof:* $g = F + \rho\ell$, ℓ a small linear form making $(F+\ell)|N$ a Morse function (Corollary 6.8) and ρ a bump function equal to 1 near the compact $f^{-1}(-\infty, a_0]$, so that $|g - f|$ is small on all of N; Lemma 22.4 (V13.11) keeps the indices; a regular level of g just above d; Lemma D (V13.13) for g; then U.

V13.15 (deforming a neighbourhood of the minima in the model; `Morse.BrokenSetup.mem_ball_of_mem_nodes`, `Morse.BrokenSetup.joinVec_self`, `Morse.BrokenSetup.exists_pushesInto_nodes`). In the finite-dimensional model of Chapter 10 (a broken-geodesic setup with energy bound c, V10.3–V10.4), let $N^m \cap O$ (O open) consist of admissible break points and $d < c$. If $\{\gamma \in \Omega : E(\gamma) \le d\}$ is a retract of an open set of Ω containing it, then $\{y \in N^m \cap O : E'(y) \le d\}$ has an open neighbourhood in $N^m \cap O$ from which it pushes maps from every P. *Book:* p. 121: "taking U small, we may assume that each point of U is joined to its image under the retraction by a unique minimal geodesic; hence U can be deformed into M^0 within M". *Hidden hypotheses:* the deformation must stay in the open manifold $N^m \cap O$ (true near M^0 by the tube lemma); the retraction in Ω is transported to break points. *Proof:* each break point is moved to the corresponding break point of the retracted path along the short geodesic of Corollary 10.8 (V7.20); this fixes M^0.

V13.16 (Theorem 22.1 for an open and closed set of paths, the exposition's Lemma 6.2 of Chapter 15; `Morse.PathSp.mem_of_edist_eq_zero`, `Morse.theorem_22_1_clopen`, `Morse.clopenIncl`, `Morse.compresses_clopenIncl`, `Morse.theorem_22_1_clopen_bijective`, `Morse.theorem_22_1_clopen_bijective_minPathsIncl`, `Morse.Homotopy.bijective_homotopyGroupMap_clop`

In the setting of V13.2 (compact balls, neighbourhood retraction), let $\Omega' \subseteq \Omega$ be open and closed, and suppose every geodesic *in* Ω' that is not of least energy has index $\geq \lambda_0$. Then every map $h : P \to \Omega'$ (P as in V13.2) is homotopic, relative to $h^{-1}(\Omega^d)$, to a map into $\Omega^d \cap \Omega'$. Hence the inclusion $\Omega^d \cap \Omega' \to \Omega'$ is bijective on π_i for $i \leq \lambda_0 - 2$, π_0 included, and the inclusion $\Omega^d \to \Omega$ is bijective on π_i, $1 \leq i \leq \lambda_0 - 2$, at every point of $\Omega^d \cap \Omega'$. On the way: the inclusion of an open and closed set containing the base point is bijective on π_i, $i \geq 1$, in any space. *Book:* p. 147 (§24, case $k \equiv 2 \pmod 4$): "a variant of Theorem 22.1 for the component $\Omega^{(c)}$ of the path space"; the exposition states it as Lemma 6.2. *Hidden hypotheses:* where the proof of 22.1 must see Ω' (the deformation r_u keeps each path in its component, the manifold of break points is cut down to the paths in Ω', and the last homotopy stays there); d is the least energy on all of Ω, and Ω' need not contain a minimal geodesic. *Deviation (stronger):* any open and closed set, not only a path component.

What is not verified in this chapter

- **Hanner's theorem** (p. 121): a topological manifold is a neighbourhood retract. Theorem 22.1 (V13.2) and Lemma 22.5 (V13.14) assume its conclusion; for the sphere and for $SU(2m)$ the retraction is constructed (V13.8, V14.21), so Corollary 22.3 and Theorem 23.3 are unconditional.
- **The homological form** (p. 118): $H_i(\Omega, \Omega^d) = 0$ for $i < \lambda_0$ and $H_i\Omega^d \cong H_i\Omega$ for $i \leq \lambda_0 - 2$, by the relative Hurewicz theorem; and $H_i(\Omega S^{n+1}) \cong H_i S^n$ for $i \leq 2n - 2$.
- **Relative homotopy groups and the exact sequence of the pair** (Ω, Ω^d): replaced by pushing maps (V13.12) and compression (V13.3).
- **Milnor's eigenvalue argument** for Lemma 22.4 (continuity of the λ-th eigenvalue, his footnote's minimax): not needed (V13.10).
- **That the isomorphism of Corollary 22.3 is the suspension homomorphism** (exposition), and the exposition's application (§5: the stable stems, Toda's table). The first statement of that application, $\pi_j S^N = 0$ for $j < N$, is verified, though not by the induction through Corollary 22.3 given there: `Morse.Homotopy.subsingleton_pi_sphere`, by pushing maps off a point (see V14.25).
- **§22 on an abstract Riemannian manifold**: only for submanifolds of a Euclidean space with the induced metric (Nash's theorem quoted).

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15. Chapter 14 (Milnor §23): partly verified

Milnor's §23 (pp. 124–132) proves Bott periodicity for the unitary group. He reviews $U(n)$, its exponential and its bi-invariant metric $\langle A, B\rangle = \operatorname{Re}\operatorname{tr}(AB^*)$ (pp. 124–126), finds the geodesics from I to $-I$ and the minimal ones (p. 127), identifies the minimal geodesics in $SU(2m)$ with the Grassmannian $G_m(\mathbb{C}^{2m})$ (Lemma 23.1), shows that every other geodesic has index at least $2m+2$ (Lemma 23.2, proved on pp. 130–132), deduces from §22 that $G_m(\mathbb{C}^{2m}) \to \Omega(SU(2m); I, -I)$ is an isomorphism on π_i for $i \leq 2m$ (Theorem 23.3), and with the fibrations of Lemma 23.4 obtains $\pi_{i-1}U \cong \pi_{i+1}U$ and the stable homotopy groups of U (Theorem 23.5). What is verified first is the computation on which Lemma 23.2 rests, for the diagonal velocity $A = \operatorname{diag}(i\pi k_1, \dots, i\pi k_n)$: the matrix facts ($\exp A = -I$ exactly for odd k_j, the traces of AA^* and of A, Lemma 23.1 as a statement about projections, the entries of $K_A(W) = \frac{1}{4}[[A, W], A]$, Milnor's eigenvectors and the fact that they span), the count of conjugate points produced by an eigenvalue, the index formula

$\lambda = \sum_{k_i > k_j}(k_i - k_j - 2)$ from Milnor's spectrum, and Lemma 23.2 for that formula; there Theorems 20.5 and 15.1 enter as definitions (V14.7, V14.8). Then the geometry and the topology (V14.13–V14.30): $U(n)$ and $SU(n)$ as compact submanifolds of the space of matrices with the induced bi-invariant metric; every geodesic from I to $-I$ in $SU(n)$ is $t \mapsto \exp(tA)$ with A unitarily conjugate to Milnor's diagonal form; the distance from I to $-I$ in $SU(2m)$ and the minimal geodesics; Lemma 23.1 as a homeomorphism of the set of initial velocities with the Grassmannian, and as a homotopy equivalence (a homeomorphism after identifying paths that agree on $[0,1]$) with the space Ω^d of minimal paths; Theorem 20.5 for $SU(n)$, so that the index of every geodesic from I to $-I$ is the formula above, by the project's Index Theorem, and Lemma 23.2 for geodesics; the neighbourhood retraction that Theorem 22.1 (Chapter 13) assumes; Theorem 23.3; the bundles of Lemma 23.4 on concrete Stiefel manifolds and Grassmannians, with their exact sequences; Lemma 23.4; the periodicity theorem; and Theorem 23.5 — each as a bijection of pointed sets and, where both sides are groups, as a group isomorphism. The stable groups $\pi_i U$ are not defined as a limit: the statements are about $\pi_j U(m)$ in the stable range, and the periodicity theorem is stated for $1 \le i \le 2m-2$. Not verified: the linear independence of Milnor's eigenvectors, the space form of Bott's theorem (the exposition's Corollary 23.6), and that $\sum k_j$ is constant on the components of the path space of $U(n)$. Lean: `Morse/Chapter14/` — `SUMatrices.lean` (318 lines), `IndexCount.lean` (710; it also contains the counts of §24, reported in Chapter 15, and the evenness step of Theorem 21.7, reported in Chapter 12), `MinimalPaths.lean` (311), `MinimalRetract.lean` (337), `Periodicity.lean` (198), `PeriodicityGroups.lean` (139), `CircleGroup.lean` (93); `Morse/Chapter12/` — `MatrixInner.lean`, `MatrixGroups.lean` (the groups), `SUMinimal.lean` (444), `SUJacobi.lean` (370), `SUIndex.lean` (530); `Morse/Chapter13/SUInstance.lean` (158); `Morse/Homotopy/` — `Homogeneous.lean` (137), `Stiefel.lean` (530), `Grassmannian.lean` (595), `BundleHomotopy.lean` (408), `BundlesUnitary.lean` (595), on top of the homotopy groups, Serre fibrations and exact sequences of Part XII of the proof-by-proof edition.

Conventions. Matrices are complex $n \times n$ matrices indexed by a finite set, A^* is the conjugate transpose, exp is Mathlib's matrix exponential, and $k = (k_1, \dots, k_n)$ is a family of integers indexed by a finite set.

V14.1 (exp $A = -I$ for diagonal A; `Morse.exp_diagonal_eq_neg_one_iff`, `Morse.exp_ofReal_mul_I_eq_neg_o`
For a real number a, $e^{ia} = -1$ iff $a = \pi k$ with k an odd integer. Hence for real $a_1, \dots, a_n$, $\exp \operatorname{diag}(ia_1, \dots, ia_n) = -I$ iff every a_j is π times an odd integer. *Book:* p. 127: for diagonal $A = \operatorname{diag}(ia_1, \dots, ia_n)$, $\exp A = \operatorname{diag}(e^{ia_1}, \dots, e^{ia_n})$, so $\exp A = -I$ iff $A = \operatorname{diag}(k_1 i\pi, \dots, k_n i\pi)$ with odd integers k_j. *Hidden hypotheses:* none. *Deviation (narrower):* only diagonal A; the reduction of a general skew-Hermitian A by $\exp(TAT^{-1}) = T(\exp A)T^{-1}$ (pp. 126–127) is V14.14.

V14.2 (length and trace of the velocity; `Morse.suDiag`, `Morse.trace_suDiag_mul_conjTranspose`, `Morse.trace_suDiag`). For integers k_j and $A = \operatorname{diag}(i\pi k_1, \dots, i\pi k_n)$, $\operatorname{tr}(AA^*) = \pi^2 \sum_j k_j^2$ and $\operatorname{tr} A = i\pi \sum_j k_j$. *Book:* p. 127: the geodesic $t \mapsto \exp tA$, $0 \le t \le 1$, has length $|A| = \sqrt{\operatorname{tr} AA^*} = \pi\sqrt{k_1^2 + \cdots + k_n^2}$; p. 130: A in the Lie algebra of $SU(n)$ gives a geodesic from I to $-I$ iff its eigenvalues are $i\pi k_1, \dots, i\pi k_n$ with odd k_j of sum zero. *Hidden hypotheses:* none. *Deviation (restated):* only the trace identities; that $|A|$ is the length of the geodesic is V14.15.

V14.3 (minimal geodesics, arithmetically; `Morse.card_le_sum_sq`, `Morse.sum_sq_eq_card_iff`, `Morse.card_eq_of_minimal`). For odd integers $k_1, \dots, k_n$, $\sum_j k_j^2 \ge n$, with equality iff every $k_j = \pm 1$. If $n = 2m$, every $k_j = \pm 1$ and $\sum_j k_j = 0$, then exactly m of the k_j equal 1 and m equal -1. *Book:* p. 127: A determines a minimal geodesic iff each $k_j = \pm 1$, the length then being $\pi\sqrt{n}$;

in $SU(2m)$ the condition $\operatorname{tr} A = 0$ makes Eigen$(i\pi)$ m-dimensional. *Hidden hypotheses:* none. *Deviation (restated):* "minimal" is read through V14.2 as "of least $\sum k_j^2$"; that this is minimality of the geodesic among all geodesics from I to $-I$ (all of which are conjugate to diagonal ones) is V14.15.

V14.4 (Lemma 23.1 as a statement about projections; `Morse.sq_eq_neg_pi_sq_iff`, `Morse.trace_pi_smul_proj`). For a skew-Hermitian matrix A: $A^2 = -\pi^2 I$ iff $A = i\pi(2P - I)$ for some Hermitian projection P ($P^* = P = P^2$). For every P, $\operatorname{tr}(i\pi(2P - I)) = i\pi(2\operatorname{tr} P - n)$, so in $SU(n)$ the condition $\operatorname{tr} A = 0$ reads $\operatorname{tr} P = n/2$. *Book:* p. 127: a minimal A is completely determined by Eigen$(i\pi)$, an arbitrary subspace of $\mathbb{C}^n$ (of dimension m in $SU(2m)$); p. 128, Lemma 23.1: the space of minimal geodesics from I to $-I$ in $SU(2m)$ is homeomorphic to $G_m(\mathbb{C}^{2m})$. *Hidden hypotheses:* "completely determined by Eigen$(i\pi)$" uses that a skew-Hermitian matrix is normal, so that its eigenspaces are orthogonal; the formal proof produces the projection without the spectral theorem. *Deviation (narrower):* a correspondence of sets of matrices only. Not formalized: that P is the projection onto Eigen$(i\pi)$ (immediate from $A = i\pi(2P - I)$) and that $\operatorname{tr} P$ is the dimension of the range of P (the Grassmannian is defined by the trace, V14.16). That $A^2 = -\pi^2 I$ is equivalent to minimality is V14.15, and the topology is V14.16–V14.17.

V14.5 (the operator K_A for diagonal A; `Morse.jacobiMat`, `Morse.commutator_diagonal_apply`, `Morse.jacobiMat_suDiag_apply`). Put $K_A(W) = \frac{1}{4}[[A, W], A]$ for matrices A, W, with $[A, W] = AW - WA$. For a diagonal matrix $\operatorname{diag}(d)$, the (j, l) entry of $[\operatorname{diag}(d), W]$ is $(d_j - d_l)w_{jl}$; for $A = \operatorname{diag}(i\pi k_1, \dots, i\pi k_n)$ and every W, the (j, l) entry of $K_A(W)$ is $\frac{\pi^2}{4}(k_j - k_l)^2 w_{jl}$. *Book:* p. 130: by Theorem 20.5 the conjugate points are determined by $K_A(W) = R(A, W)A = \frac{1}{4}[[A, W], A]$; "a short computation" gives $[A, W] = (i\pi(k_j - k_l)w_{jl})$, $[A, [A, W]] = (-\pi^2(k_j - k_l)^2 w_{jl})$, and (p. 131) $K_A(W) = (\frac{\pi^2}{4}(k_j - k_l)^2 w_{jl})$. *Hidden hypotheses:* none. *Deviation (restated):* K_A is defined by the matrix formula. That it is $W \mapsto R(A, W)A$ for the bi-invariant metric of $SU(n)$ is Theorem 21.3 (b) applied to the Lie algebra of skew-Hermitian matrices with the commutator as bracket (p. 124); that application is V12.17 (for every matrix group).

V14.6 (Milnor's eigenvectors, and that they span; `Morse.eReal`, `Morse.eImag`, `Morse.jacobiMat_eReal`, `Morse.jacobiMat_eImag`, `Morse.jacobiMat_diagonal`, `Morse.eReal_conjTranspose`, `Morse.eImag_conjTranspose`, `Morse.trace_eReal`, `Morse.trace_eImag`, `Morse.eq_diagonal_add_sum_eReal_eImag`). Let E_{jl} have $+1$ at (j, l) and -1 at (l, j), and E'_{jl} have i at both places. Both are skew-Hermitian, of trace 0 when $j \neq l$, and eigenvectors of K_A (V14.5) with eigenvalue $\frac{\pi^2}{4}(k_j - k_l)^2$; every diagonal matrix lies in the kernel of K_A. Every skew-Hermitian W equals its diagonal part plus $\sum_{j<l}(\operatorname{Re} w_{jl}\, E_{jl} + \operatorname{Im} w_{jl}\, E'_{jl})$, for any linear order of the index set. *Book:* p. 131: a basis of the Lie algebra of $SU(n)$ consisting of eigenvectors of K_A: E_{jl} and E'_{jl} for $j < l$, with eigenvalue $\frac{\pi^2}{4}(k_j - k_l)^2$, and the diagonal matrices of the Lie algebra, with eigenvalue 0; so the non-zero eigenvalues are the $\frac{\pi^2}{4}(k_j - k_l)^2$ with $k_j > k_l$, each counted twice. *Hidden hypotheses:* Milnor does not say why these vectors span. *Deviation (narrower):* spanning is proved, linear independence (the count $n(n-1) + (n-1) = \dim \mathfrak{su}(n)$) is not; so "the spectrum of K_A with multiplicity is Milnor's list" is not a formal theorem, and Milnor's list is the input of V14.9. (The index computation of V14.18–V14.19 needs no eigenbasis.)

V14.7 (conjugate parameters of an eigenvalue: Theorem 20.5 as a definition; `Morse.conjParams`, `Morse.conjParams_finite`, `Morse.conjParams_sq`, `Morse.ncard_conjParams_sq`). For a real number e, the set of conjugate parameters of e is $\{t : 0 < t < 1,\ e > 0,\ t = \pi s/\sqrt{e}$ for some integer $s\}$, empty when $e \leq 0$. It is finite, and for $e = (\pi d)^2$ with $d \geq 1$

an integer it is $\{1/d, 2/d, \dots, (d-1)/d\}$, with $d-1$ elements. *Book:* p. 110, Theorem 20.5: the conjugate points to p along γ are the $\gamma(\pi k/\sqrt{e_i})$, k a non-zero integer and e_i a positive eigenvalue of K_V; p. 131: each eigenvalue $e > 0$ gives conjugate points at $t = \pi/\sqrt{e}, 2\pi/\sqrt{e}, \dots$, here $t = 2/(k_j - k_l), 4/(k_j - k_l), \dots$, and the number in $(0,1)$ is "evidently" $(k_j - k_l)/2 - 1$. *Hidden hypotheses:* the count needs $k_j - k_l$ even (both are odd), so that $\frac{\pi^2}{4}(k_j - k_l)^2 = (\pi d)^2$ with $d = (k_j - k_l)/2$ an integer. *Deviation (restated):* Theorem 20.5 is not proved here; this set is its conclusion, for a geodesic parametrized on $[0,1]$, taken as a definition (for $SU(n)$ Theorem 20.5 is V14.18, which shows that this definition gives the multiplicities). The reader should confirm that it says what 20.5 says.

V14.8 (the index of a spectrum: Theorems 20.5 and 15.1 as a definition; `Morse.conjMult`, `Morse.spectralIndex`, `Morse.spectralIndex_eq`, `Morse.spectralIndex_double`). For a finite family $e = (e_i)$ of real numbers (the eigenvalues of K_V with multiplicity), the multiplicity of t is the number of i such that t is a conjugate parameter of e_i (Theorem 20.5), and the index of e is the sum of these multiplicities over all t (only $t \in (0,1)$ contribute; the Index Theorem 15.1). It equals $\sum_i \#\{\text{conjugate parameters of } e_i\}$. A family in which every value is repeated twice has twice the index. *Book:* p. 110, Theorem 20.5: the multiplicity of $\gamma(t)$ as a conjugate point is the number of e_i such that t is a multiple of $\pi/\sqrt{e_i}$; p. 131: applying the Index Theorem and "adding over all j, l". *Hidden hypotheses:* "adding over all j, l" exchanges the sum over conjugate points t with the sum over eigenvalues. *Deviation (restated):* a definition; that it is the Morse index of the energy at the geodesic, through Theorems 15.1 and 20.5, is V14.19.

V14.9 (the index formula; `Morse.bottIndex`, `Morse.pairSpectrum`, `Morse.suSpectrum`, `Morse.two_mul_ncard_conjParams`, `Morse.two_mul_spectralIndex_pairSpectrum`, `Morse.spectralIndex_suS`
Put $\mathrm{bott}(k) = \sum(k_i - k_j - 2)$, the sum over the ordered pairs (i,j) with $k_i > k_j$. For odd integers $a > b$, the eigenvalue $\frac{\pi^2}{4}(a-b)^2$ has $(a-b)/2 - 1$ conjugate parameters (V14.7). Hence, if all k_j are odd, the index (V14.8) of Milnor's spectrum — the number $\frac{\pi^2}{4}(k_i - k_j)^2$ for each pair with $k_i > k_j$, each counted twice — is $\mathrm{bott}(k)$; with each counted once it is $\frac{1}{2}\,\mathrm{bott}(k)$. *Book:* p. 131: each pair $k_j > k_l$ gives two copies of $\frac{\pi^2}{4}(k_j - k_l)^2$ and contributes $2((k_j - k_l)/2 - 1)$, whence $\lambda = \sum_{k_j > k_l}(k_j - k_l - 2)$. *Hidden hypotheses:* as in V14.7 and V14.8. *Deviation (restated):* the conclusion concerns the index of Milnor's spectrum; that it is the Morse index of the geodesic $t \mapsto \exp tA$ is V14.19.

V14.10 (elementary properties of the formula; `Morse.bottIndex_nonneg`, `Morse.bottIndex_neg`, `Morse.bottIndex_comp_equiv`, `Morse.bottIndex_eq_zero_of_minimal`). If all k_j are odd, $\mathrm{bott}(k) \geq 0$ (every term is). For all k, $\mathrm{bott}(-k) = \mathrm{bott}(k)$ and bott does not change when the k_j are reordered. If every $k_j = \pm 1$, $\mathrm{bott}(k) = 0$. *Book:* pp. 131–132: for a minimal geodesic all $k_j = \pm 1$, hence $\lambda = 0$; the reordering is Milnor's normalization $k_1 \geq \dots \geq k_n$ (p. 130), and the symmetry is his "(say) negative" (p. 132). *Hidden hypotheses:* none.

V14.11 (Lemma 23.2; `Morse.bottIndex_ge`, `Morse.exists_pos_neg_data`, `Morse.sum_pos_neg_le_bottIndex`, `Morse.sum_pos_neg_eq`). Let $k_1, \dots, k_{2m}$ be odd integers with $\sum_j k_j = 0$, not all equal to ± 1. Then $\mathrm{bott}(k) \geq 2m + 2$. *Book:* p. 128, Lemma 23.2: every non-minimal geodesic from I to $-I$ in $SU(2m)$ has index $\geq 2m + 2$; proof on p. 132: Case 1, at least $m + 1$ of the k_j (say) negative, so some positive $k_j \geq 3$ and $\lambda \geq 2(m+1)$; Case 2, m positive and m negative, not all ± 1, so one is ≥ 3 and one ≤ -3, and $\lambda \geq 4m \geq 2(m+1)$. *Hidden hypotheses:* "(say) negative" needs the symmetry $k \mapsto -k$ (V14.10); the claims that some positive k_j is ≥ 3 (Case 1) and that one k_j is ≥ 3 and one ≤ -3 (Case 2) use $\sum k_j = 0$, and Case 2 also the equal numbers of each sign. *Deviation*

(restated): a statement about the formula; for the index of a geodesic it is V14.20. *Proof:* both cases through the sum over the pairs (positive k_i, negative k_j), which equals $NS_+ + PS_- - 2PN$ (P, N the numbers of positive and negative k_j, S_+ and S_- the sums of the positive k_j and of the $|k_j|$ with $k_j < 0$); this reproduces Milnor's bounds $2N$ and $4m$.

V14.12 (why not $U(n)$; `Morse.remarkK`, `Morse.bottIndex_remarkK`, `Morse.not_minimal_remarkK`, `Morse.sum_remarkK`, `Morse.abs_sum_le_card_of_minimal`). For $k = (3, 1, \dots, 1) \in \mathbb{Z}^{n+1}$: $\mathrm{bott}(k) = 0$, not every k_j is ± 1, and $\sum_j k_j = (n+1) + 2$; whereas every family of entries ± 1 has $|\sum_j k_j|$ at most the number of entries. *Book:* p. 127: the space of minimal geodesics in $U(n)$ is "rather inconvenient", having components of varying dimensions, which is why Milnor passes to $SU(2m)$. *Deviation (not in the book):* the exposition's stronger reason: in $U(n+1)$ the non-minimal geodesic with $k = (3, 1, \dots, 1)$ has index 0 (by V14.9's formula) and lies in a component of the path space containing no minimal geodesic, so Theorem 22.1 cannot be applied to $U(n)$. That $\sum k_j$, the degree of det along the geodesic, is constant on the components of the path space is topology and is not formalized.

The geometry and the topology. From here on $U(n)$, $SU(n)$ are the matrix groups of Chapter 12 (V12.14) inside $M_n(\mathbb{C})$ with the Frobenius inner product, $\Omega = \Omega(SU(n); I, -I)$ is Milnor's path space with his metric (V10.1, Chapter 13), Ω^d its subset of paths of least energy (V13.1), and homotopy groups and pushing maps are as in Chapter 13. A *Hermitian projection* is a matrix P with $P^* = P = P^2$; $G_m(\mathbb{C}^n)$ is the set of Hermitian projections of trace m with the subspace topology (the projections onto the m-dimensional subspaces). A *Stiefel manifold* $V_k(\mathbb{K}^n)$ is the set of $n \times k$ matrices F with $F^*F = I$ (orthonormal k-frames), with the subspace topology. Locally trivial maps, Serre fibrations and their exact homotopy sequences are those of Part XII of the proof-by-proof edition (`Morse.Homotopy.IsLocallyTrivial`, `Morse.Homotopy.IsSerreFibration`).

V14.13 ($U(n)$ and $SU(n)$; `Morse.unitaryMap`, `Morse.isRegularValue_unitaryMap`, `Morse.isSliceSubmanifold` `Morse.isMatrixGroup_unitaryGroup`, `Morse.isCompact_unitaryGroup`, `Morse.specialSkewHermitian`, `Morse.suMap`, `Morse.isRegularValue_suMap`, `Morse.isSliceSubmanifold_specialUnitaryGroup`, `Morse.isMatrixGroup_specialUnitaryGroup`, `Morse.isCompact_specialUnitaryGroup`, `Morse.fderiv_det_mul`, `Morse.det_exp_smul_of_trace_eq_zero`). $U(n) = \{A : A^*A = I\}$ is a compact slice submanifold of $M_n(\mathbb{C})$ modelled on $\mathfrak{u}(n)$, the skew-Hermitian matrices, and $(U(n), \mathfrak{u}(n))$ is a matrix group; for $n \geq 1$, $SU(n) = \{A \in U(n) : \det A = 1\}$ is a compact slice submanifold modelled on $\mathfrak{su}(n)$, the traceless skew-Hermitian matrices, and $(SU(n), \mathfrak{su}(n))$ is a matrix group. Jacobi's formula $d(\det)_A(AX) = \det A \cdot \mathrm{tr}\, X$ holds, and $\det \exp(tX) = 1$ when $\mathrm{tr}\, X = 0$. (The same file gives $O(n)$ and $SO(n)$, used in Chapter 15.) *Book:* pp. 124–126: $U(n)$ is a compact smooth submanifold of $M_n(\mathbb{C})$ and a Lie group, with tangent space $\mathfrak{u}(n)$ at I ((5), (6)), Lie bracket the commutator, the bi-invariant metric $\mathrm{Re}\,\mathrm{tr}(AB^*)$; $\det \exp A = e^{\mathrm{tr}\, A}$, and $SU(n)$ with Lie algebra $\mathfrak{su}(n)$. *Hidden hypotheses:* $n \geq 1$ for $SU(n)$. *Deviation (restated):* $U(n)$ is a regular level set of $A \mapsto A^*A - I$ (Hermitian values) rather than a submanifold by the chart exp; $SU(n)$ is not a regular level set of $(A^*A - I, \det A - 1)$, and is obtained as the open piece $\mathrm{Re}\det > 0$ of the regular level set $\{A^*A = I, \mathrm{Im}\det A = 0\}$; only the traceless case of $\det \exp A = e^{\mathrm{tr}\, A}$ is proved. The bracket of left-invariant fields is not used (V12.17). Not formalized: that exp is onto and that $SU(n)$ is connected (not needed).

V14.14 (every geodesic from I to $-I$ in $SU(n)$ is of Milnor's form; `Morse.eqOn_exp_smul_of_isUnbrokenGeodes` `Morse.exists_unitary_suDiag`, `Morse.exp_conj_suDiag`, `Morse.conj_suDiag_mem`, `Morse.mem_pathSpace_exp` `Morse.isUnbrokenGeodesic_exp`). Every geodesic $\gamma \in \Omega(SU(n); I, q)$ agrees on $[0, 1]$ with $t \mapsto \exp(tA)$ for some $A \in \mathfrak{su}(n)$. If $A \in \mathfrak{su}(n)$ and $\exp A = -I$, there are a unitary T and

odd integers k_j with $\sum k_j = 0$ and $A = T\,\mathrm{diag}(i\pi k_1, \dots, i\pi k_n)T^*$; conversely every such A lies in $\mathfrak{su}(n)$ and has $\exp A = -I$, and then $t \mapsto \exp(tA)$ is a geodesic in $\Omega(SU(n); I, -I)$. *Book:* pp. 126–127: after conjugation by a unitary matrix, a skew-Hermitian A is diagonal, $\exp A = -I$ iff A has eigenvalues $i\pi k_j$ with k_j odd; p. 130: in $SU(n)$, moreover $\sum k_j = 0$. *Hidden hypotheses:* that the geodesics from I are the $\exp(tA)$ (Lemma 21.2, V12.15) and the spectral theorem. *Proof:* Mathlib's spectral theorem for the Hermitian matrix iA, and V14.1.

V14.15 (the distance from I to $-I$ and the minimal geodesics; `Morse.norm_conj_suDiag_sq`, `Morse.pathLength_exp_smul`, `Morse.pi_sqrt_card_le_norm`, `Morse.kHalf`, `Morse.riemDist_one_neg_one`, `Morse.sq_conj_suDiag_iff`, `Morse.isMinimalGeodesic_exp_iff`). For skew-Hermitian A, the length of $t \mapsto \exp(tA)$ on $[0,1]$ is $\|A\|$, and for $A = T\,\mathrm{diag}(i\pi k_j)T^*$ this is $\pi(\sum k_j^2)^{1/2} \geq \pi\sqrt{n}$ (odd k_j). In $SU(2m)$, $m \geq 1$, the distance $\rho(I, -I)$ is $\pi\sqrt{2m}$; for $A \in \mathfrak{su}(2m)$ with $\exp A = -I$, the geodesic $t \mapsto \exp(tA)$ is minimal (V7.24) iff $A^2 = -\pi^2 I$, iff all $k_j = \pm 1$. *Book:* p. 127: the geodesic has length $|A| = \pi(k_1^2 + \cdots + k_n^2)^{1/2}$, and it is minimal iff every $k_j = \pm 1$. *Hidden hypotheses:* that a minimal geodesic exists (Hopf–Rinow, V7.26, in the compact $SU(2m)$) and that every geodesic from I is an $\exp(tA)$ (V12.15), without which "length $\pi\sqrt{n}$" would not imply "minimal". *Deviation (narrower):* for $SU(2m)$, the case used; $U(n)$ is not treated.

V14.16 (Lemma 23.1 for the initial velocities; `Morse.minVel`, `Morse.grassmannian`, `Morse.exp_eq_neg_one_of_s` `Morse.mem_minVel_iff`, `Morse.lemma_23_1`). Let $|\iota| = 2m$ and $\mathrm{minVel} = \{A \in \mathfrak{su}(2m) : A^2 = -\pi^2 I\}$. Then $A \in \mathrm{minVel}$ iff $A \in \mathfrak{su}(2m)$, $\exp A = -I$ and $t \mapsto \exp(tA)$ is a minimal geodesic; and $A \mapsto \frac{1}{2}(I - \frac{i}{\pi}A)$ is a homeomorphism of minVel onto $G_m(\mathbb{C}^{2m})$, with inverse $P \mapsto i\pi(2P - I)$. *Book:* p. 128, Lemma 23.1: the space of minimal geodesics from I to $-I$ in $SU(2m)$ is homeomorphic to $G_m(\mathbb{C}^{2m})$; p. 127: a minimal A is completely determined by $\mathrm{Eigen}(i\pi)$. *Hidden hypotheses:* the topology (Milnor matches the spaces only as sets, `Exposition/Findings.md`). *Deviation (restated):* the space of minimal geodesics is represented here by their initial velocities; the path space is V14.17. G_m is a set of projections.

V14.17 (Lemma 23.1 in the path space; `Morse.exp_half_smul_of_mem_minVel`, `Morse.pi_sq_le_energy`, `Morse.expPath`, `Morse.expPath_mem_minPaths`, `Morse.exists_minVel_of_mem_minPaths`, `Morse.continuous_expPath`, `Morse.minVelToPaths`, `Morse.pathsToMinVel`, `Morse.minPathsHomotopyEquiv`, `Morse.minVelHomeomorph`). Let $|\iota| = 2m$. Every path from I to $-I$ in $SU(2m)$ has energy $\geq 2m\pi^2$; for $A \in \mathrm{minVel}$, $t \mapsto \exp(tA)$ lies in Ω^d and $\exp(A/2) = A/\pi$; every $\gamma \in \Omega^d$ agrees on $[0,1]$ with such a path, with $A = \pi\gamma(\frac{1}{2})$. The map $A \mapsto (t \mapsto \exp tA)$ is continuous for Milnor's metric, and it is a homotopy equivalence $\mathrm{minVel} \simeq \Omega^d$, with homotopy inverse $\gamma \mapsto \pi\gamma(\frac{1}{2})$, and a homeomorphism of minVel onto the separation quotient of Ω^d. *Book:* p. 128, Lemma 23.1 (as above). *Hidden hypotheses:* the continuity for Milnor's metric (the exposition's Lemma 2.3). *Deviation (restated):* "homeomorphic" holds only after identifying paths that agree on $[0,1]$, as for the sphere (V13.6); the homotopy equivalence is what the homotopy groups use. With V14.16 this is Milnor's $\Omega^d \cong G_m(\mathbb{C}^{2m})$.

V14.18 (Theorem 20.5 for $SU(n)$; `Morse.jacSin`, `Morse.hadSin`, `Morse.diagI`, `Morse.jacK_diagI_apply`, `Morse.suSol`, `Morse.suSol_ode`, `Morse.suJac`, `Morse.isJacobiField_suJac`, `Morse.covAlong_suJac_zero`, `Morse.suJac_eq_zero_iff`, `Morse.hadKer`, `Morse.conjMultiplicity_su`, `Morse.finrank_hadKer`). Let $a \in \mathbb{R}^n$, T unitary, $A = T\,\mathrm{diag}(ia_j)T^* \in \mathfrak{su}(n)$, $s_b(t) = \sin(bt)/b$ ($= t$ for $b = 0$), and $S(t) \odot W$ the matrix with entries $s_{(a_j - a_l)/2}(t)w_{jl}$. For $C \in \mathfrak{su}(n)$, $J_C(t) = e^{tA/2}\,T(S(t) \odot T^*CT)T^*\,e^{tA/2}$ is a Jacobi field along $t \mapsto \exp(tA)$ with $J_C(0) = 0$ and $\frac{DJ_C}{dt}(0) = C$. The multiplicity (V9.4) of $t_0 \in [0,1]$ as a parameter conjugate to 0 is

$\dim\{W \in \mathfrak{su}(n) : S(t_0) \odot W = 0\}$; for $a = \pi k$, k integers, and $0 < t_0 < 1$ this dimension is the multiplicity of t_0 computed from Milnor's spectrum (V14.8, V14.9): twice the number of pairs $k_j > k_l$ such that t_0 is a conjugate parameter of $\frac{\pi^2}{4}(k_j - k_l)^2$. *Book:* p. 110, Theorem 20.5, applied on p. 130 to $K_A(W) = \frac{1}{4}[[A, W], A]$; p. 131, "adding over all j, l". *Hidden hypotheses:* the multiplicity of a conjugate point is the dimension of a space of Jacobi fields, so all Jacobi fields vanishing at 0 must be described; Milnor does so in a parallel eigenframe. *Deviation (restated):* for $SU(n)$ only, without an eigenbasis (V14.6): the Jacobi equation is solved entrywise in the frame of V12.18, and the reduction to diagonal A is replaced by carrying T through the computation.

V14.19 (the index formula in $SU(n)$; `Morse.formIndex_su_geodesic`, `Morse.index_su_eq_bottIndex`). For integers k_j with $\sum k_j = 0$, T unitary and $A = T\,\mathrm{diag}(i\pi k_j)T^*$ such that $t \mapsto \exp(tA)$ is a path from I to $-I$, the index of E_{**} at this geodesic is the index of Milnor's spectrum (V14.8). Hence every geodesic $t \mapsto \exp(tA)$ from I to $-I$ in $SU(n)$ has index $\mathrm{bott}(k) = \sum_{k_i > k_j}(k_i - k_j - 2)$, for the odd k_j of V14.14. *Book:* p. 131: by the Index Theorem the index of $t \mapsto \exp tA$, $0 \le t \le 1$, is $\sum_{k_j > k_l}(k_j - k_l - 2)$. *Hidden hypotheses:* as in V14.7–V14.9 and V14.18. *Proof:* the Index Theorem (V9.24) and V14.18, with V14.9.

V14.20 (Lemma 23.2 for the geodesics of $SU(2m)$; `Morse.lemma_23_2_index`, `Morse.energy_exp_smul`, `Morse.le_formIndex_of_not_minimal_su`). Let $|\iota| = 2m$. A geodesic $t \mapsto \exp(tA)$ from I to $-I$ in $SU(2m)$ has index 0 if all $k_j = \pm 1$ (a minimal geodesic) and index $\ge 2m + 2$ otherwise. Every geodesic in $\Omega(SU(2m); I, -I)$ that is not of least energy has index $\ge 2m + 2$: the index hypothesis of Theorem 22.1 (V13.2) with $\lambda_0 = 2m + 2$. *Book:* p. 128, Lemma 23.2: every non-minimal geodesic from I to $-I$ in $SU(2m)$ has index $\ge 2m + 2$; proof pp. 130–132. *Hidden hypotheses:* "non-minimal" in Theorem 22.1 is "not of least energy"; a geodesic with all $k_j = \pm 1$ has least energy ($E = \|A\|^2 = L^2 \le L(\omega)^2 \le E(\omega)$, V8.5). *Proof:* V14.11 applied to V14.19, with V14.14 and V14.15.

V14.21 (the neighbourhood retraction for $SU(2m)$, and Corollary 22.2 for $SU(2m)$; `Morse.specProj`, `Morse.continuousOn_specProj`, `Morse.hermOf`, `Morse.minProj`, `Morse.minNbhd`, `Morse.retrPath`, `Morse.exists_retraction_minPaths_su`, `Morse.corollary_22_2_su`, `Morse.corollary_22_2_su'`, `Morse.corollary_22_2_su_mulEquiv`). Let $|\iota| = 2m$. For Hermitian H let $Q(H)$ be the spectral projection onto the eigenvalues $> \frac{1}{2}$; it depends continuously on H where $\frac{1}{2}$ is not an eigenvalue. The set O of paths $\omega \in \Omega$ such that $\frac{1}{2}$ is not an eigenvalue of $H(\omega)$, the Hermitian part of $\frac{1}{2}(I - i\omega(\frac{1}{2}))$, and $|\operatorname{tr} Q(H(\omega)) - m| < \frac{1}{2}$, is open and contains Ω^d, and $\omega \mapsto (t \mapsto \exp(t\, i\pi(2Q(H(\omega)) - I)))$, changed to the identity on Ω^d, is a retraction of O onto Ω^d. Hence (V13.4, V13.5, V14.20) for every $\gamma \in \Omega^d$, $\pi_i(\Omega^d, \gamma) \cong \pi_{i+1}(SU(2m), I)$ for $i \le 2m$, as groups for $1 \le i \le 2m$. *Book:* p. 128: by Lemma 23.1 Ω^d is a manifold, so Theorem 22.1 applies. *Deviation (not in the book):* the retraction that Hanner's theorem would give is constructed, so the result does not depend on the hypothesis of V13.2; the trace is locally constant on projections, which keeps the rank m.

V14.22 (Theorem 23.3; `Morse.theorem_23_3`, `Morse.theorem_23_3_mulEquiv`). Let $|\iota| = 2m$. For every $P \in G_m(\mathbb{C}^{2m})$ and $i \le 2m$ there is a bijection $\pi_i(G_m(\mathbb{C}^{2m}), P) \cong \pi_{i+1}(SU(2m), I)$, a group isomorphism for $1 \le i \le 2m$. *Book:* p. 128, Theorem 23.3 (Bott): the inclusion $G_m(\mathbb{C}^{2m}) \to \Omega(SU(2m); I, -I)$ induces isomorphisms of homotopy groups in dimensions $\le 2m$; hence $\pi_i G_m(\mathbb{C}^{2m}) \cong \pi_{i+1} SU(2m)$ for $0 \le i \le 2m$. *Hidden hypotheses:* the base points (as in V13.4); $i = 0$ (pointed sets). *Deviation (restated):* the statement about the inclusion is V13.3 applied with V14.20–V14.21 (for Ω^d, identified with G_m by V14.16–V14.17); the isomorphism is

the composite of V14.16, V14.17 and V14.21, a bijection, and a group isomorphism for $i \geq 1$.

V14.23 (local triviality from orbit sections; the Stiefel bundles; `Morse.Homotopy.orbitTriv`, `Morse.Homotopy.exists_triv_of_orbitSection`, `Morse.Homotopy.Stiefel`, `Morse.Homotopy.secMat`, `Morse.Homotopy.secMat_mem_unitaryGroup`, `Morse.Homotopy.secMat_mul_baseFrame`, `Morse.Homotopy.exists_unitary_smul_baseFrame`, `Morse.Homotopy.stiefelRestrict_locallyTrivial`, `Morse.Homotopy.unitaryOrbit_locallyTrivial`, `Morse.Homotopy.stiefelSphere`, `Morse.Homotopy.isLocall`
(i) If a topological group G acts continuously on E and B, $p : E \to B$ is continuous and equivariant, G acts transitively on B, and the orbit map $g \mapsto gb_0$ has a continuous section over a neighbourhood of b_0, then p is locally trivial with fibre $p^{-1}(b_0)$. (ii) $U(n)$ acts transitively on $V_k(\mathbb{K}^n)$, and near the base frame F_0 (the last k standard vectors) an explicit unitary matrix $s(F)$ with $s(F)F_0 = F$ depends continuously on F. (iii) Restricting a frame to its last vectors, $V_{k+l}(\mathbb{K}^n) \to V_l(\mathbb{K}^n)$, is locally trivial with fibre $V_k(\mathbb{K}^{n-l})$, and $g \mapsto gF_0$, $U(n) \to V_k(\mathbb{K}^n)$, is locally trivial with fibre $U(n-k)$, embedded as $S \mapsto \operatorname{diag}(S, I)$. $V_1(\mathbb{K}^n)$ is the unit sphere, so $U(m+1) \to S^{2m+1}$ is locally trivial with fibre $U(m)$. *Book:* pp. 128–129: $U(m+1)/U(m) \cong S^{2m+1}$, the complex Stiefel manifold $U(2m)/U(m)$, and the fibrations $U(m) \to U(m+1) \to S^{2m+1}$, $U(m) \to U(2m) \to U(2m)/U(m)$, quoted from Steenrod. *Hidden hypotheses:* the bundle structure of a coset map (local sections). *Deviation (restated):* concrete spaces of matrices instead of coset spaces; the local section is written down.

V14.24 (the Grassmannian bundle; `Morse.Homotopy.Grassmannian`, `Morse.Homotopy.eq_of_mul_eq_of_trace_`, `Morse.Homotopy.frameSpan`, `Morse.Homotopy.exists_frame_of_mem_grassmannian`, `Morse.Homotopy.gsFrame`, `Morse.Homotopy.continuousAt_gsFrame`, `Morse.Homotopy.exists_unitary_smul_grassBase`, `Morse.Homotopy.frameSpan_locallyTrivial`, `Morse.Homotopy.grassmannian_eq_Grassmannian`).
Every $P \in G_m(\mathbb{K}^n)$ is FF^* for an orthonormal m-frame F; $F \mapsto FF^*$, $V_m(\mathbb{K}^n) \to G_m(\mathbb{K}^n)$, is continuous, equivariant under $U(n)$ acting on G_m by $P \mapsto gPg^*$, transitively, and locally trivial with fibre $U(m)$. Near the base point the section is the Gram–Schmidt process applied to the columns of PF_0, which is continuous at independent families. The Grassmannian of Lemma 23.1 (V14.16) is this one. *Book:* p. 129: $G_m(\mathbb{C}^{2m})$ "can be identified with" $U(2m)/U(m) \times U(m)$, and there is a fibration $U(m) \to U(2m)/U(m) \to G_m(\mathbb{C}^{2m})$. *Hidden hypotheses:* as in V14.23. *Deviation (restated):* G_m as projections, the bundle as the frame-to-span map. *Proof:* two Hermitian projections with $PQ = Q$ and equal traces are equal.

V14.25 (Lemma 23.4: connectivity of Stiefel manifolds and stability; `Morse.Homotopy.SphereConnected`, `Morse.Homotopy.sphereConnected_complex`, `Morse.Homotopy.subsingleton_pi_stiefel`, `Morse.Homotopy.subsingleton_pi_stiefel_complex`, `Morse.Homotopy.unitaryInclC`, `Morse.Homotopy.bijective_homotopyGroupMap_unitaryIncl`, `Morse.Homotopy.bijective_homotopyGroupMap`, `Morse.Homotopy.surjective_homotopyGroupMap_unitaryIncl_complex`). $\pi_i S^{2n-1} = 0$ for $i < 2n-1$, at every base point. Hence $\pi_i V_k(\mathbb{C}^n) = 0$ for $k \leq n$ and $i \leq 2(n-k)$; and the inclusion $U(m) \to U(m+k)$, $S \mapsto \operatorname{diag}(S, I_k)$, induces bijections on π_i for $i < 2m$ and surjections for $i \leq 2m$, at every base point. (The same holds over $\mathbb{R}$ with the bounds of S^{n-1}, Chapter 15.) *Book:* pp. 128–129, Lemma 23.4 and its proof: $\pi_{i-1}U(m) \cong \pi_{i-1}U(m+1) \cong \cdots$ for $i \leq 2m$, by the inclusions; $\pi_{2m}U(m) \to \pi_{2m}U(m+1)$ is onto; $\pi_i(U(2m)/U(m)) = 0$ for $i \leq 2m$. *Hidden hypotheses:* $\pi_i S^N = 0$ for $i < N$ (Part XII, quoted by Milnor). *Deviation (stronger):* every $V_k(\mathbb{C}^n)$, by induction on k through the restriction bundles, instead of Milnor's exact-sequence argument for $V_m(\mathbb{C}^{2m})$ alone.

V14.26 (Lemma 23.4: $\pi_i G_m(\mathbb{C}^{2m}) \cong \pi_{i-1}U(m)$; `Morse.Homotopy.grassmannianEquiv`, `Morse.Homotopy.grassmannianEquiv_complex`, `Morse.Homotopy.nonempty_mulEquiv_grassmannian_complex`, `Morse.Homotopy.grassmannianPiEquiv`, `Morse.Homotopy.nonempty_mulEquiv_grassmannianPi`,

`Morse.Homotopy.minVelPiEquiv`, `Morse.Homotopy.nonempty_mulEquiv_minVelPi`). For $1 \le i \le 2m$ and every $P \in G_m(\mathbb{C}^{2m})$ there is a bijection $\pi_i(G_m(\mathbb{C}^{2m}), P) \cong \pi_{i-1}(U(m), I)$, a group isomorphism for $2 \le i \le 2m$; the same for the space minVel of V14.16. *Book:* p. 129, Lemma 23.4: $\pi_i G_m(\mathbb{C}^{2m}) \cong \pi_{i-1}U(m)$ for $1 \le i \le 2m$. *Hidden hypotheses:* base points; π_0 is not a group. *Deviation (restated):* for $i = 1$ a bijection of pointed sets ($\pi_1 G_m$ and $\pi_0 U(m)$ are both trivial, which is not separately stated). *Proof:* the boundary map of the exact sequence of $U(m) \to V_m(\mathbb{C}^{2m}) \to G_m$, whose total space is $2m$-connected (V14.25), moved to any P by a unitary motion.

V14.27 (Lemma 23.4: $SU(n)$ and $U(n)$; `Morse.Homotopy.detCircle`, `Morse.Homotopy.diagCircle`, `Morse.Homotopy.unitaryProd`, `Morse.Homotopy.isLocallyTrivial_detCircle`, `Morse.Homotopy.subsingletc` `Morse.Homotopy.suInclC`, `Morse.Homotopy.bijective_homotopyGroupMap_suInclC`). $g \mapsto (\det g, gD(\det g)^{-1})$, with $D(z) = \mathrm{diag}(z, 1, \dots, 1)$, is a homeomorphism $U(n) \cong S^1 \times SU(n)$; $\pi_j S^1 = 0$ for $j \ge 2$; and the inclusion $SU(n) \to U(n)$ ($n \ge 1$) induces bijections on π_j, $j \ge 2$, at every point of $SU(n)$. *Book:* p. 129, Lemma 23.4, last clause: $\pi_j U(m) \cong \pi_j SU(m)$ for $j \ne 1$ (from $SU(m) \to U(m) \to S^1$). *Deviation (narrower):* $j = 0$ (both groups connected) is not formalized; it is not used.

V14.28 (Milnor's chain and the periodicity theorem; `Morse.nonempty_equiv_pi_unitary_two_mul`, `Morse.periodicity_theorem`, `Morse.nonempty_mulEquiv_pi_unitary_two_mul`, `Morse.periodicity_theorem_` For $1 \le i \le 2m$, $\pi_{i-1}(U(m), I) \cong \pi_{i+1}(U(2m), I)$, as groups for $2 \le i \le 2m$. For $1 \le i \le 2m-2$, $\pi_{i-1}(U(m), I) \cong \pi_{i+1}(U(m), I)$, as groups for $2 \le i \le 2m-2$. *Book:* p. 129: $\pi_{i-1}U = \pi_{i-1}U(m) \cong \pi_i G_m(\mathbb{C}^{2m}) \cong \pi_{i+1}SU(2m) \cong \pi_{i+1}U(2m) = \pi_{i+1}U$ (Lemma 23.4, Theorem 23.3, Lemma 23.4), the periodicity theorem $\pi_{i-1}U \cong \pi_{i+1}U$. *Hidden hypotheses:* the stable groups $\pi_j U$ are the $\pi_j U(m)$ for $2m - 1 \ge j$ (the exposition's definition); the chain's base points. *Deviation (restated):* no stable group $\pi_j U$ is defined; periodicity is stated inside one $U(m)$, which needs $i + 1 < 2m$ for stability back from $U(2m)$, hence the range $i \le 2m - 2$; for $i = 1$ a bijection of pointed sets. Every stable group is reached for m large.

V14.29 (Theorem 23.5; `Morse.Homotopy.zerothHomotopyEquivOfDiscrete`, `Morse.Homotopy.nonempty_pi_one_` `Morse.Homotopy.unitaryOneHomeomorph`, `Morse.Homotopy.nonempty_pi_unitary_equiv_circle`, `Morse.theorem_23_5`). For $j \le 2m - 1$: $\pi_j(U(m), I)$ is trivial if j is even, and in bijection with $\mathbb{Z}$ if j is odd. On the way: $\pi_1(S^1) \cong \mathbb{Z}$ as sets, by the boundary map of the covering $\mathbb{Z} \to \mathbb{R} \to S^1$; $\det : U(1) \to S^1$ is a homeomorphism; and $\pi_i U(m) \cong \pi_i S^1$ for $i \le 1$, $m \ge 1$. *Book:* p. 130, Theorem 23.5 (Bott): the stable groups $\pi_0 U \cong \pi_2 U \cong \cdots$ are zero and $\pi_1 U \cong \pi_3 U \cong \cdots$ are infinite cyclic. *Hidden hypotheses:* $\pi_1 S^1 \cong \mathbb{Z}$ (quoted); the stable range as in V14.28. *Deviation (restated):* in the stable range of $U(m)$, not for a limit group; for odd j a bijection here, a group isomorphism in V14.30.

V14.30 (Theorem 23.5 as groups; `Morse.Homotopy.piOneCircleMulEquiv`, `Morse.Homotopy.piMulEquivOfHomeo` `Morse.Homotopy.nonempty_mulEquiv_pi_one_unitary_circle`, `Morse.nonempty_mulEquiv_pi_unitary_odd`, `Morse.nonempty_mulEquiv_pi_unitary_even`, `Morse.theorem_23_5_mulEquiv`). $\pi_1(S^1, 1) \cong \mathbb{Z}$ as groups. For $2k + 2 \le 2m$, $\pi_{2k+1}(U(m), I) \cong \mathbb{Z}$ as groups; in the stable range $\pi_j U(m) \cong \pi_1 U(m)$ for odd j and $\pi_j U(m) \cong \pi_2 U(m)$ for even $j \ge 2$, as groups. *Book:* p. 130, Theorem 23.5. *Hidden hypotheses:* as in V14.29. *Deviation (not in the book):* $\pi_1 S^1 \cong \mathbb{Z}$ is proved, through Mathlib's covering-space theory (the monodromy of $t \mapsto e^{it}$, whose deck group $2\pi\mathbb{Z}$ is abelian) and Mathlib's identification of π_1 with the fundamental group.

What is not verified in this chapter

- **The reduction to diagonal form by an isometry** (pp. 126–127, 130): that conjugation by $T \in U(n)$ is an isometry preserving conjugate points and indices. Not needed: T is carried through the computation (V14.14, V14.18).
- **The linear independence of Milnor's eigenvectors** (V14.6); the formal index computation needs no eigenbasis.
- $U(n)$ **beyond what** $SU(2m)$ **needs** (p. 127): the distance $\pi\sqrt{n}$ in $U(n)$, its minimal geodesics as all subspaces of $\mathbb{C}^n$, the surjectivity of exp, the connectedness of $U(n)$ and $SU(n)$; and that $\sum k_j$ is constant on the components of $\Omega(U(n); I, -I)$ (V14.12).
- π_0 **in Lemma 23.4**: $\pi_0 U(m) \cong \pi_0 SU(m)$ (both connected) and the identification of $\pi_1 G_m$ and $\pi_0 U(m)$ as trivial.
- **The stable groups** $\pi_i U$ **as homotopy groups of a limit**, and the space form of Bott's theorem ($\Omega SU \simeq BU$, $\Omega U \simeq \mathbb{Z} \times BU$, the exposition's Corollary 23.6): colimit topologies and CW homotopy types are not formalized.
- **Lemma 23.1 as a homeomorphism of path spaces on the nose** (V14.17): paths are functions on $\mathbb{R}$, so only the separation quotient is homeomorphic to $G_m(\mathbb{C}^{2m})$.
- **The exposition's application** (§5, complex vector bundles over spheres).

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16. Chapter 15 (Milnor §24): partly verified

Milnor's §24 (pp. 133–148) proves Bott periodicity for the orthogonal group. Complex structures on $\mathbb{R}^n$ form $\Omega_1(n)$, the space of minimal geodesics from I to $-I$ in $O(n)$ (Lemma 24.1); other geodesics have index at least $2m-2$ in $O(2m)$ (Lemma 24.2), so $\pi_i\Omega_1(n) \cong \pi_{i+1}O(n)$ for $i \leq n-4$ (Theorem 24.3). Iterating with anticommuting complex structures $J_1, J_2, \ldots$ gives a chain $\Omega_k(n) \subset \cdots \subset \Omega_1(n) \subset O(n)$ of compact, totally geodesic submanifolds, each the space of minimal geodesics of the previous one (Lemma 24.4); in the limit $\Omega_{k+1} \to \Omega\Omega_k$ is a homotopy equivalence (Theorem 24.5), and the identification of $\Omega_8(16r)$ with $O(r)$ (Lemma 24.6) gives $\pi_i O \cong \pi_{i+8}O$ (Theorem 24.7). The proof of 24.5 rests on minimal Clifford modules and Lemma 24.8 (Bott–Shapiro), with the dimensions m_k, and Milnor remarks that these yield vector fields on spheres. What is verified first (V15.1–V15.20) is the algebra: complex structures; $\Omega_k(n)$ as sets, with their inclusions and compactness; the algebraic parts of Lemma 24.4 (the reflection, the "if" halves of Assertions 1 and 2, the decomposition $J' = J_k A$ and the semicircle through J'); Milnor's imbedding $\Omega_k(n) \to \Omega_k(n+n')$; explicit anticommuting complex structures on $\mathbb{R}^{16r}$; every product identity Milnor uses on pp. 139–143; the odd steps of Lemma 24.6 as bijections with involutions, and its dimensions; the dimensions m_k and Lemma 24.8 for $k \leq 8$ (uniqueness up to linear isomorphism); the orthonormal vector fields on spheres; and the index counts of Lemma 24.2 and of the proof of Theorem 24.5, from Milnor's spectra. Then the geometry and the topology (V15.21–V15.46), for each n: the strata of $\Omega_k(n)$ (the sets on which Milnor's tangent space has a fixed dimension) as compact totally geodesic submanifolds of the space of matrices, with both halves of Assertions 1 and 2; Lemma 24.1 and the second half of Lemma 24.4, that is, the distance $\pi\sqrt{n}$ from J_k to $-J_k$, the minimal geodesics as the semicircles through the points of $\Omega_{k+1}(n)$, and a homotopy equivalence of $\Omega_{k+1}(n)$ with the space Ω^d of minimal paths (a homeomorphism after identifying paths that agree on $[0,1]$); the neighbourhood retraction that Theorem 22.1 (Chapter 13) assumes; the index bounds for the geodesics themselves (Lemma 24.2 and Milnor's Assertions of pp. 146 and 148), obtained from Killing fields and the project's Index Theorem instead of the curvature of $\Omega_k(n)$; Theorem 24.3; Theorem 24.5 for each n and

$k \le 7$, as bijections of homotopy groups in a range that grows with n, class by class for $k = 2, 6$, where the winding number of $\det_{\mathbb{C}}$ is shown to be an integer; the compatibility with Milnor's stabilization; $\Omega_8(16r) \cong O(r)$ (Lemma 24.6 (8)); the eight-step chain; and Theorem 24.7 in the stable range, $\pi_i O(n) \cong \pi_{i+8} O(n')$ for $n \ge i+2$ and $n' \ge i+10$. Every isomorphism of V15.21–V15.46 is a bijection of pointed sets. Not verified: the limit spaces Ω_k and O and the homotopy equivalences of Theorems 24.5 and 24.7; the group structure along the chain; the table of $\pi_i O$ and $\pi_i Sp$; the identifications of $\Omega_1, \dots, \Omega_7$ with coset spaces and Grassmannians; the symmetric-space structure; and the exact index of a geodesic. Lean: `Morse/Chapter15/` — for V15.1–V15.18, `Relations.lean` (189 lines), `ComplexStructure.lean` (455), `Clifford.lean` (372), `CliffordModules.lean` (595), `MinimalModules.lean` (643); Lemma 24.2 and the counts of the proof of 24.5 (V15.3, V15.19, V15.20) are in `Morse/Chapter14/IndexCount.lean`; for V15.21–V15.46, `TotallyGeodesic.lean` (344), `OmegaManifold.lean` (554), `OmegaMinimal.lean` (611), `OmegaRetract.lean` (298), `OmegaKilling.lean` (272), `OmegaIntertwiners.lean` (298), `InvariantSplitting.lean` (181), `SplittingClifford.lean` (527), `SplittingData.lean` (275), `IndexBound.lean` (270), `FiniteStage.lean` (257), `Winding.lean` (396), `OmegaWinding.lean` (188), `OmegaComponents.lean` (488), `SplittingTwo.lean` (283), `SplittingTwoData.lean` (295), `KillingTwo.lean` (291), `IndexBoundTwo.lean` (360), `FiniteStageTwo.lean` (178), `Stabilization.lean` (193), `EightCommutant.lean` (211), `OmegaEight.lean` (324), `BottChain.lean` (146), `StablePeriodicity.lean` (182), on top of Theorem 22.1 (Chapter 13) and the homotopy groups of Chapter 14.

Conventions. Matrices are real $n \times n$ matrices indexed by a finite set, $O(n)$ is Mathlib's orthogonal group, and $J_1, J_2, \dots$ is a fixed sequence of matrices (the index 0 is unused). For the abstract statements (V15.10, V15.12–V15.17) V is a real vector space, finite-dimensional where dimensions are counted, and the J_α are linear maps of V; no inner product is used there. From V15.21 on, $M_n(\mathbb{R})$ carries the inner product $\langle A, B\rangle = \operatorname{tr}(AB^{\mathsf{T}})$, whose norm $\|A\|$ is the Frobenius norm, and a submanifold N of $M_n(\mathbb{R})$ carries the induced metric, as in Chapters 6–11 and 14; $\Omega(N; p, q)$ is the space of piecewise smooth paths with Milnor's metric (Chapter 10), Ω^d the set of its paths of least energy (V13.1), and the index of a geodesic is that of E_{**} (Chapter 9). "An anticommuting family $J_1, \dots, J_k$" is as in V15.4; in statements about $\Omega_k(n)$ it includes the base point $J_k \in \Omega_k(n)$, whose existence is Milnor's standing assumption (p. 136), and $J_0 = I$ is the base point of $\Omega_0(n) = O(n)$. Homotopy groups are Mathlib's π_i, $i \ge 0$, of a pointed space, and a bijection between them is one of sets.

V15.1 (complex structures; `Morse.IsComplexStructure`, `Morse.isComplexStructure_iff`, `Morse.IsComplexStructure.even_card`, `Morse.IsComplexStructure.trace_transpose_mul_self`). A complex structure on $\mathbb{R}^n$ is a matrix $J \in O(n)$ with $J^2 = -I$. Equivalently, $J^{\mathsf{T}} = -J$ and $J^2 = -I$. A complex structure exists only if n is even, and every complex structure has $\operatorname{tr}(J^{\mathsf{T}} J) = n$. *Book:* p. 133: the definition, after "Now suppose that n is even"; pp. 134–135, in the proof of Lemma 24.1: a skew A with $A^2 = -I$ is a complex structure, and $JJ^* = I$, $JJ = -I$ give $J^* = -J$. *Hidden hypotheses:* none. *Deviation (stronger):* the parity of n is derived rather than assumed.

V15.2 (the geodesic $t \mapsto \exp(\pi t J)$; `Morse.exp_smul_of_mul_self_eq_neg_one`, `Morse.exp_pi_smul`). If $J^2 = -I$, then $\exp(\theta J) = \cos\theta\, I + \sin\theta\, J$ for every real θ; in particular $\exp(\pi J) = -I$. *Book:* pp. 134–135, proof of Lemma 24.1: in a suitable basis $\exp(\pi A)$ is block diagonal with rotation blocks, and for a complex structure J, $\exp \pi J = -I$. *Hidden hypotheses:* none. *Deviation (narrower):* only this matrix identity; that $t \mapsto \exp(\pi t J)$ is a minimal geodesic of $O(n)$, and Lemma 24.1 itself, are V15.24–V15.26.

V15.3 (Lemma 24.2; `Morse.bottIndex0`, `Morse.oSpectrum`, `Morse.spectralIndex_oSpectrum`, `Morse.bottIndex0_eq_zero`, `Morse.bottIndex0_ge`). For integers $a_1, \dots, a_m$ put $\lambda(a) = \sum_{i<j}(a_i + a_j - 2) + \sum_{a_i > a_j}(a_i - a_j - 2)$, the second sum over ordered pairs. Let all a_j be odd and positive. (i) The index, in the sense of Chapter 14 (V14.8), of the spectrum consisting of $\frac{\pi^2}{4}(a_i + a_j)^2$ for $i < j$ and $\frac{\pi^2}{4}(a_i - a_j)^2$ for $a_i > a_j$, each counted twice, is $\lambda(a)$. (ii) If all $a_j = 1$, $\lambda(a) = 0$. (iii) If not all $a_j = 1$, $\lambda(a) \geq 2m - 2$. *Book:* p. 135, Lemma 24.2: any non-minimal geodesic from I to $-I$ in $O(2m)$ has index $\geq 2m - 2$; in the proof the geodesic is $t \mapsto \exp(\pi t A)$ with A block diagonal with blocks $a_j \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, $a_1 \geq \dots \geq a_m > 0$ odd; the non-zero eigenvalues of $K_A = -\frac{1}{4}(\operatorname{Ad} A)^2$ are $(a_i + a_j)^2/4$ for $i < j$ and $(a_i - a_j)^2/4$ for $i < j$ with $a_i \neq a_j$, each counted twice, whence the formula for λ, which vanishes for the minimal geodesic and is $\geq \sum_{j=2}^{m}(3+1-2) = 2m-2$ otherwise. *Hidden hypotheses:* the conjugate points of $t \mapsto \exp(\pi t A)$ are governed by the eigenvalues of $K_{\pi A} = \pi^2 K_A$, the velocity being πA; Milnor's formula is the right one for them. *Deviation (restated):* Milnor's spectrum is the input; its computation for block-diagonal A, the reduction of a geodesic to that form, and Theorem 20.5 for $O(n)$ are not formalized. The bound of Lemma 24.2 for the geodesics themselves is V15.30, proved without the spectrum.

V15.4 (the spaces $\Omega_k(n)$; `Morse.Omega`, `Morse.omega_succ_subset`, `Morse.isClosed_omega`, `Morse.isCompact_omega`, `Morse.IsAnticommFamily`, `Morse.IsAnticommFamily.mem_omega`). $\Omega_0(n) = O(n)$, and for $k \geq 1$, $\Omega_k(n)$ is the set of complex structures J with $J_\alpha J = -J J_\alpha$ for $1 \leq \alpha \leq k-1$. For any fixed matrices J_α, $\Omega_{k+1}(n) \subseteq \Omega_k(n)$, and every $\Omega_k(n)$ is closed and compact. $J_1, \dots, J_k$ form an anticommuting family if each is a complex structure and $J_\alpha J_\beta = -J_\beta J_\alpha$ for $\alpha \neq \beta$; then $J_k \in \Omega_k(n)$ for $k \geq 1$. *Book:* p. 136: with fixed anticommuting complex structures $J_1, \dots, J_{k-1}$, $\Omega_k(n)$ is the set of complex structures anticommuting with them; $\Omega_k(n) \subset \cdots \subset \Omega_1(n) \subset O(n)$; "Clearly each $\Omega_k(n)$ is a compact set"; $\Omega_0(n) = O(n)$. *Hidden hypotheses:* none. *Deviation (stronger):* the inclusions and the compactness hold for any fixed matrices J_α.

V15.5 (Lemma 24.4, the reflection; `Morse.reflection_mem_omega`). For $k \geq 1$ and $P, J' \in \Omega_k(n)$, $-PJ'P \in \Omega_k(n)$. Since $J'^{-1} = -J'$ for a complex structure, $-PJ'P = PJ'^{-1}P$ is the image of J' under $\sigma \mapsto P\sigma^{-1}P$, the reflection of $O(n)$ in P (Lemma 21.1). *Book:* p. 136: each component of $\Omega_k(n)$ is a symmetric space, because the isometric reflection of $O(n)$ in a point of $\Omega_k(n)$ carries $\Omega_k(n)$ to itself. *Hidden hypotheses:* none. *Deviation (narrower):* only the algebraic invariance; that the reflection is an isometry and that the components of $\Omega_k(n)$ are symmetric spaces are not formalized (the manifold structure is V15.23).

V15.6 (Assertions 1 and 2 in the proof of Lemma 24.4, "if" halves; `Morse.isComplexStructure_mul_exp`, `Morse.anticomm_mul_exp`). If J is a complex structure and A is skew with $AJ = -JA$, then $J \exp A$ is a complex structure. If $J_\alpha J = -J J_\alpha$ and A commutes with J_α, then J_α anticommutes with $J \exp A$. *Book:* p. 137, Assertion 1: $J \exp A$ is a complex structure iff A anticommutes with J; Assertion 2: $J \exp A$ anticommutes with $J_1, \dots, J_{k-1}$ iff A commutes with them; here A is small and skew. *Hidden hypotheses:* the "only if" halves need A small (Milnor uses "Since A is small"). *Deviation (stronger):* the "if" halves hold for every skew A (for Assertion 2, for every A). The "only if" halves, which need the injectivity of exp near 0, are V15.22, and their consequence that $\Omega_k(n)$ is a smooth totally geodesic submanifold is V15.23.

V15.7 (Lemma 24.4, the step from Ω_k to Ω_{k+1}, algebraically, and the Remark; `Morse.omega_succ_decomp`, `Morse.mul_mem_omega_succ`, `Morse.semicircle_mem_omega`, `Morse.mul_exp_eq_semicircle`,

`Morse.trace_transpose_mul_eq_zero`). Let $J_1, \dots, J_k$ ($k \geq 1$) be an anticommuting family. (i) If $J' \in \Omega_{k+1}(n)$, then $A = -J_k J'$ ($= J_k^{-1} J'$) is a complex structure that anticommutes with J_k and commutes with $J_1, \dots, J_{k-1}$, and $J' = J_k A$. (ii) Conversely, for every such A, $J_k A \in \Omega_{k+1}(n)$. (iii) For $J' \in \Omega_{k+1}(n)$ and every real t, $\cos t\, J_k + \sin t\, J' \in \Omega_k(n)$, and $J_k \exp(tA) = \cos t\, J_k + \sin t\, J'$ for $A = -J_k J'$: the curve $t \mapsto J_k \exp(tA)$ stays in $\Omega_k(n)$, passes through J' at $t = \pi/2$ and reaches $-J_k$ at $t = \pi$. (iv) A complex structure J' anticommuting with a matrix J satisfies $\mathrm{tr}(J'^{\mathsf{T}} J) = 0$. *Book:* pp. 137–138: choosing $J_k \in \Omega_k(n)$ and a complex structure $J = J_k A$ anticommuting with $J_1, \dots, J_k$, A is a complex structure anticommuting with J_k and commuting with $J_1, \dots, J_{k-1}$, so $t \mapsto J_k \exp(\pi t A)$ is a minimal geodesic from J_k to $-J_k$ in $\Omega_k(n)$, and conversely; Remark, p. 138: $J_k A$ is the midpoint $\gamma(\frac{1}{2})$. *Hidden hypotheses:* Milnor's point J_k is, in his later use, the next member of the fixed anticommuting family, as here, and the lemma needs $\Omega_{k+1}(n) \neq \emptyset$. *Deviation (narrower):* (iii) concerns the curve (Milnor's, reparametrized by πt), not geodesics: that it is a minimal geodesic of $\Omega_k(n)$, and that every minimal geodesic from J_k to $-J_k$ in $\Omega_k(n)$ arises so (the homeomorphism with $\Omega_{k+1}(n)$), are V15.25–V15.26. *Deviation (not in the book):* (iv), the orthogonality of J' and J for the trace inner product.

V15.8 (Milnor's imbedding $\Omega_k(n) \to \Omega_k(n+n')$; `Morse.IsAnticommFamily.fromBlocks`, `Morse.fromBlocks_mem_omega`). If $J_1, \dots, J_k$ and $J'_1, \dots, J'_k$ are anticommuting families on $\mathbb{R}^n$ and $\mathbb{R}^{n'}$, the block matrices $J_\alpha \oplus J'_\alpha$ form an anticommuting family on $\mathbb{R}^n \oplus \mathbb{R}^{n'}$. For any fixed matrices and every k: if $P \in \Omega_k(n)$ for the J_α and $P' \in \Omega_k(n')$ for the J'_α, then $P \oplus P' \in \Omega_k(n+n')$ for the $J_\alpha \oplus J'_\alpha$. With $P' = J'_k$ (V15.4) this is Milnor's map $J \mapsto J \oplus J'_k$. *Book:* p. 138: choose anticommuting complex structures $J'_1, \dots, J'_k$ on $\mathbb{R}^{n'}$; each $J \in \Omega_k(n)$ determines $J \oplus J'_k$, which anticommutes with $J_\alpha \oplus J'_\alpha$. *Hidden hypotheses:* $\mathbb{R}^{n'}$ must carry k anticommuting complex structures, which the choice presupposes (V15.9 supplies them for $n' = 16r'$, $k \leq 8$). *Deviation (narrower):* the direct limit Ω_k and its topology are not formalized; the compatibility of the imbedding with φ is V15.41.

V15.9 (explicit anticommuting complex structures; `Morse.bott4`, `Morse.bott8`, `Morse.bott16`, `Morse.isAnticommFamily_bott2`, `Morse.isAnticommFamily_bott4`, `Morse.isAnticommFamily_bott8`, `Morse.isAnticommFamily_bott16`, `Morse.IsAnticommFamily.blockDiagonal`, `Morse.IsAnticommFamily.reind`, `Morse.isAnticommFamily_bott16r`). With $\sigma = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, $\tau = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $\kappa = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, explicit Kronecker products give one complex structure on $\mathbb{R}^2$ (σ), three anticommuting ones on $\mathbb{R}^4$ ($1 \otimes \sigma$, $\sigma \otimes \tau$, $\sigma \otimes \kappa$), seven on $\mathbb{R}^8$, and eight on $\mathbb{R}^{16}$ (the seven tensored with τ, and $1 \otimes 1 \otimes 1 \otimes \sigma$). Renaming the coordinates and repeating block-diagonally ($J_\alpha \oplus \cdots \oplus J_\alpha$) preserve anticommuting families; so $\mathbb{R}^{16r}$ carries eight anticommuting complex structures for every r. *Book:* p. 136: assume that n is divisible by a high power of 2, and let $J_1, \dots, J_{k-1}$ be fixed anticommuting complex structures; p. 139: it is convenient to set $n = 16r$. Milnor never writes the structures down. *Hidden hypotheses:* their existence. *Deviation (not in the book):* explicit structures, which make the standing assumptions of §24 non-vacuous for $n = 16r$ and $k \leq 8$.

V15.10 (products of anticommuting complex structures; `Morse.IsClifford`, `Morse.IsAnticommFamily.isCliff`, `Morse._spec`, `Morse._anticommute`, `Morse._spec`, `Morse._spec`, `Morse._spec`, `Morse.J J _spec`, `Morse.J J _spec`, `Morse._spec`). In any ring, call $J_1, \dots, J_k$ Clifford if $J_\alpha^2 = -1$ and $J_\alpha J_\beta = -J_\beta J_\alpha$ for $\alpha \neq \beta$ ($1 \leq \alpha, \beta \leq k$); an anticommuting family of matrices is Clifford. Put $\theta_3 = J_1 J_2 J_3$, $\theta_5 = J_1 J_4 J_5$, $\theta_6 = J_2 J_4 J_6$, $\theta_7 = J_1 J_6 J_7$ and $\omega_7 = J_1 J_2 \cdots J_7$. Then, whenever k is large enough for the indices involved: $\theta_3^2 = 1$, θ_3 commutes with J_1, J_2, J_3 and anticommutes with J_α for $4 \leq \alpha \leq k$; $J_3 J_4$ anticommutes with θ_3 and commutes with J_1, J_2;

$\theta_5^2 = 1$, θ_5 commutes with J_1, J_4, J_5, θ_3 and anticommutes with J_2, J_3; $\theta_6^2 = 1$, θ_6 commutes with $J_2, J_4, J_6, \theta_3, \theta_5$ and anticommutes with J_1, J_3, J_5; $\theta_7^2 = 1$, θ_7 commutes with $J_1, J_6, J_7, \theta_3, \theta_5, \theta_6$ and anticommutes with $J_2, \dots, J_5$; $J_7 J_8$ commutes with $\theta_3, \theta_5, \theta_6$ and anticommutes with θ_7; $\omega_7 = -\theta_3\theta_5\theta_7$, and it commutes with $J_1, \dots, J_7$. *Book:* pp. 139–143, the facts on which Lemma 24.6 and the discussion of Lemma 24.8 run: $J_1J_2J_3$ has square $+I$ and commutes with J_1, J_2 (p. 139); J_3J_4 anticommutes with $J_1J_2J_3$ and commutes with J_1, J_2 (pp. 139–140); the corresponding statements for $J_1J_4J_5$ (p. 140), $J_2J_4J_6$ (pp. 140–141), $J_1J_6J_7$ and J_7J_8 (p. 141); and, for $k = 7$, the two minimal spaces distinguished by $J_7 = \pm J_1 \cdots J_6$ (p. 143). *Hidden hypotheses:* none; Milnor states these without computation. *Deviation (stronger):* the complete commutation table, in any ring.

V15.11 (Lemma 24.6, the odd steps; `Morse.theta_of_mem_omega`, `Morse.mem_omega_of_theta`, `Morse.mul_theta_mul`). Let $J_1, \dots, J_k$ be an anticommuting family and $1 \le a < b \le k$. For $J' \in \Omega_{k+1}(n)$, $\theta = J_aJ_bJ'$ is symmetric with $\theta^2 = I$, commutes with J_a and J_b, and anticommutes with every other J_c, $c \le k$. Conversely, for every such θ, $J' = J_bJ_a\theta$ lies in $\Omega_{k+1}(n)$ and $J_aJ_bJ' = \theta$; and $J_bJ_a(J_aJ_bJ') = J'$ for every J'. So $J' \mapsto J_aJ_bJ'$ is a bijection of $\Omega_{k+1}(n)$ onto the symmetric involutions with this commutation pattern. With $(a, b, k) = (1, 2, 2), (1, 4, 4), (2, 4, 5), (1, 6, 6)$ it is the passage $J_3 \mapsto \theta_3$, $J_5 \mapsto \theta_5$, $J_6 \mapsto \theta_6$, $J_7 \mapsto \theta_7$ of 24.6 (3), (5), (6), (7). *Book:* pp. 139–141, Lemma 24.6 (3), (5), (6), (7): $\Omega_3(16r)$ is the quaternionic Grassmannian of $\mathbb{H}^{4r}$, through the $+1$-eigenspace V_1 of $J_1J_2J_3$; $\Omega_5(16r)$ the subspaces $W \subset V_1$ closed under J_1 with $V_1 = W \oplus J_2W$; $\Omega_6(16r)$ the real subspaces $X \subset W$ with $W = X \oplus J_1X$; $\Omega_7(16r)$ the real Grassmannian of $X \cong \mathbb{R}^{2r}$. For (3) Milnor writes J_3 back in terms of V_1; for (5), (6), (7) he asserts without proof that the subspace determines J_5, J_6, J_7 uniquely. *Hidden hypotheses:* none; the uniqueness claims are the inverse maps $\theta \mapsto J_bJ_a\theta$. *Deviation (narrower):* a symmetric involution is the same as its $+1$-eigenspace, but the passage from involutions of $\mathbb{R}^{16r}$ to Milnor's subspaces of V_1, W, X, and the identification with Grassmann manifolds as spaces, are not formalized. Of the even steps, (4), $\Omega_4(16r) \cong Sp(2r)$ through J_3J_4, is covered only by V15.7 (i)–(ii), not as an identification with a group; (8), $\Omega_8(16r) \cong O(r)$ through J_7J_8, is V15.42.

V15.12 (Lemma 24.6, the dimensions; `Morse.eigV` , `Morse.eigW`, `Morse.eigX`, `Morse.jointEigen`, `Morse.finrank_eq_two_mul_of_anticomm`, `Morse.finrank_chain`, `Morse.finrank_eq_sixteen_mul`, `Morse.finrank_jointEigen_bott16r`). On a finite-dimensional real vector space V, if $\theta^2 = 1$ and an injective P anticommutes with θ, then $\dim V = 2\dim\{\theta = 1\}$ (P exchanges the ±1-eigenspaces of θ). Hence, for Clifford $J_1, \dots, J_8$ on V, with $V_1 = \{\theta_3 = 1\}$, $W = V_1 \cap \{\theta_5 = 1\}$, $X = W \cap \{\theta_6 = 1\}$ and $X_1 = X \cap \{\theta_7 = 1\}$: $\dim V = 2\dim V_1 = 4\dim W = 8\dim X = 16\dim X_1$, the halvings being effected by J_4, J_2, J_1 and J_7J_8. For $\mathbb{R}^{16r}$ with the structures of V15.9, $\dim V_1 = 8r$, $\dim W = 4r$, $\dim X = 2r$ and $\dim X_1 = r$. *Book:* pp. 139–141: the splittings $V_1 \oplus V_2$, $W \oplus J_2W$, $X \oplus J_1X$, $X_1 \oplus X_2$; Milnor restricts to the component of $\Omega_3(16r)$ with $\dim_{\mathbb{H}} V_1 = \dim_{\mathbb{H}} V_2 = 2r$ (p. 139) and to the component of $\Omega_7(16r)$ with $\dim X_1 = \dim X_2 = r$ (p. 141), and writes $X \cong \mathbb{R}^{2r}$. *Hidden hypotheses:* that each involution splits the space into halves of equal dimension needs an invertible map anticommuting with it (J_4, J_2, J_1, J_7J_8), which Milnor uses without comment. *Deviation (stronger):* the equal dimensions are forced as soon as the later structures exist (J_4 for V_1, J_8 for X_1), which Milnor's later steps presuppose; the restriction to components is then automatic.

V15.13 (minimal spaces and the numbers m_k; `Morse.IsMinimalCliffordModule`, `Morse.milnorM`). A real vector space V with linear maps $J_1, J_2, \dots$ is minimal for $J_1, \dots, J_k$ if $V \neq 0$ and every subspace closed under $J_1, \dots, J_k$ is 0 or V. The numbers m_k are defined by the table $m_0, \dots, m_8 = 1, 2, 4, 4, 8, 8, 8, 8, 16$ and $m_{k+8} = 16m_k$. *Book:* p. 143: V is a minimal $(J_1, \dots, J_k)$-

space if no proper non-trivial subspace is closed under $J_1, \dots, J_k$; two minimal spaces are isomorphic if there is an isometry between them commuting with the J_α; m_k is the dimension of a minimal space, with the table "from the above discussion", and $m_k = 16m_{k-8}$ for $k > 8$ "can be shown". *Hidden hypotheses:* $V \neq 0$. *Deviation (restated):* m_k is defined by the table; that it is the dimension of every minimal space is V15.16, for $k \leq 8$. For $k > 8$ the recursion is only a definition.

V15.14 (m_k divides the dimension, and is the least one; `Morse.two_dvd_finrank_of_sq`, `Morse.four_dvd_finrank_of_anticomm`, `Morse.eight_dvd_finrank`, `Morse.milnorM_dvd_finrank`, `Morse.milnorM_dvd_card`, `Morse.exists_isAnticommFamily_fin`, `Morse.isLeast_milnorM`). On a real vector space: one J with $J^2 = -1$ makes the dimension even (V is a $\mathbb{C}$-module); two anticommuting ones make it divisible by 4 (V is an $\mathbb{H}$-module, through Mathlib's quaternion algebra); four Clifford maps on a finite-dimensional V make it divisible by 8; and for $k \leq 8$, Clifford $J_1, \dots, J_k$ on a finite-dimensional V force $m_k \mid \dim V$. For matrices: if $\mathbb{R}^n$ carries $k \leq 8$ anticommuting complex structures then $m_k \mid n$, and $\mathbb{R}^{m_k}$ carries k of them; so m_k is the least positive n for which $\mathbb{R}^n$ carries k anticommuting complex structures. *Book:* p. 143: the values $m_0, \dots, m_8$, from the description of the minimal spaces as lines over $\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$, and as H, $H \oplus H'$, L, $L \oplus L'$. *Hidden hypotheses:* the halving of V15.12. *Deviation (stronger):* divisibility for every space, not only for minimal ones, and explicit existence on $\mathbb{R}^{m_k}$.

V15.15 (the Clifford algebra; `Morse.cliffQ`, `Morse.cliffordLift`, `Morse.cliffordLift_`, `Morse.finrank_cliffordAlgebra`, `Morse.finrank_le_two_pow`). Let C_k be Mathlib's Clifford algebra of the quadratic form $Q(x) = -\sum x_i^2$ on $\mathbb{R}^k$. Clifford elements $J_1, \dots, J_k$ of a real algebra A determine an algebra homomorphism $C_k \to A$ sending the i-th basis vector to J_i; $\dim C_k = 2^k$; and a minimal $(J_1, \dots, J_k)$-space has dimension at most 2^k, being the image of C_k under $c \mapsto c \cdot u$ for any $u \neq 0$. *Book:* p. 136, footnote: these structures make $\mathbb{R}^n$ a module over a suitable Clifford algebra; the Clifford algebras are suppressed in the presentation. *Hidden hypotheses:* none. *Deviation (not in the book, beyond the footnote):* the bound 2^k is the tool of V15.16; given the splitting of $\mathbb{R}^n$ into minimal summands (not formalized), it alone gives what the proof of 24.5 needs from m_{k+1} — that the number of summands, at least $n/2^{k+1}$, tends to infinity with n.

V15.16 (Lemma 24.8, the dimensions; `Morse.finrank_eq_milnorM_of_isMinimal`, `Morse.isMinimal_of_finran` `Morse.milnorM_le_finrank_le`, `Morse.finrank_eq_milnorM_of_le_two`, `Morse.central_involution`, `Morse.minimal_three`, `Morse.minimal_seven`). Let $k \leq 8$ and let $J_1, \dots, J_k$ be Clifford on a finite-dimensional real V. If V is minimal, $\dim V = m_k$; conversely, if $\dim V = m_k$, V is minimal; a minimal V has $m_k \leq \dim V \leq 2^k$. An involution commuting with all J_α acts on a minimal space as 1 or as -1; hence a minimal space for $k = 3$ (resp. 7) is minimal for J_1, J_2 (resp. $J_1, \dots, J_6$), the involution being $J_1J_2J_3$ (resp. $J_1 \cdots J_7$). *Book:* p. 143: the proof of 24.8 "follows that of 24.6": a minimal space is a line over $\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$ for $k \leq 2$; for $k = 3$ a quaternionic line, with $J_3 = \pm J_1J_2$; $H \oplus H'$ for $k = 4, 5, 6$; for $k = 7$, $J_7 = \pm J_1 \cdots J_6$; $L \oplus L'$ for $k = 8$; for $k > 8$ the situation repeats more or less periodically, and $k \leq 8$ suffices. *Hidden hypotheses:* that $J_1J_2J_3$ and $J_1 \cdots J_7$ act as ± 1 on a minimal space, because they commute with every J_α and square to 1 (Milnor states $J_3 = \pm J_1J_2$ directly). *Deviation (narrower):* $k \leq 8$. *Deviation (stronger):* no inner product, and only the Clifford relations. *Proof:* replaces "follows that of 24.6": a minimal space is the image of C_k under $c \mapsto c \cdot u$, $u \neq 0$ (V15.15); taking u a joint $+1$-eigenvector of the available involutions θ, it is the image of the subspace of C_k fixed by right multiplication by the θ's, which has dimension m_k for $k = 4, 5, 6, 8$ by the halving of V15.12 applied to C_k (all of C_k for $k \leq 2$); with $m_k \mid \dim V$ (V15.14) this forces $\dim V = m_k$.

V15.17 (Lemma 24.8, uniqueness; `Morse.exists_iso_of_isMinimal`, `Morse.cyclicMap`, `Morse.range_cyclicMap_eq_top`, `Morse.minimal_of_fixed`, `Morse.exists_iso_of_equiv`). Let $k \le 8$ with $k \not\equiv 3 \pmod 4$, and let V, V' be finite-dimensional real spaces with Clifford $J_1, \dots, J_k$ and $J'_1, \dots, J'_k$, both minimal. Then there is a linear isomorphism $T : V \to V'$ with $T(J_\alpha v) = J'_\alpha(Tv)$ for all $\alpha \le k$ and $v \in V$. *Book:* p. 143, Lemma 24.8 (Bott and Shapiro): for $k \not\equiv 3 \pmod 4$ any two minimal $(J_1, \dots, J_k)$-spaces are isomorphic, by an isometry commuting with the J_α. *Hidden hypotheses:* a cyclic-vector argument (a minimal space is generated by one joint eigenvector), which "follows that of 24.6" does not supply. *Deviation (narrower):* a linear isomorphism rather than an isometry (for orthogonal J_α an isometry would follow by polar decomposition, which is not formalized), and $k \le 8$. *Deviation (stronger):* no inner product is needed. *Proof:* both spaces are isomorphic to the same fixed subspace of C_k by orbit maps (V15.16), hence to each other.

V15.18 (vector fields on spheres; `Morse.IsAnticommFamily.dotProduct_mulVec`, `Morse.IsComplexStructure.d` `Morse.bott16r_vectorFields`). If $J_1, \dots, J_k$ are anticommuting complex structures on $\mathbb{R}^n$, then for every u, $\langle J_\alpha u, J_\beta u\rangle = \delta_{\alpha\beta}|u|^2$ and $\langle J_\alpha u, u\rangle = 0$; so for $|u| = 1$ the $J_\alpha u$ are k orthonormal vectors tangent to the unit sphere at u. This is written out for the eight structures of V15.9 on $\mathbb{R}^{16r}$ (eight fields on S^{16r-1}), and the general statement applies to the block-diagonal families on $\mathbb{R}^{4r}$ and $\mathbb{R}^{8r}$ built from V15.9 (three and seven fields). *Book:* pp. 143–144, Remark: for anticommuting $J_1, \dots, J_k$ on a space of dimension rm_k, the vectors $J_1 u, \dots, J_k u$ are perpendicular to each other and to u, giving k linearly independent vector fields on the $(rm_k - 1)$-sphere: 3 on S^{4r-1}, 7 on S^{8r-1}, 8 on S^{16r-1} (Hurwitz–Radon; Adams showed these are best possible). *Hidden hypotheses:* none. *Deviation (narrower):* the identities only (the fields are linear in u, hence continuous); the general Hurwitz–Radon count and Adams's theorem are not formalized.

V15.19 (the proof of Theorem 24.5 for $k \not\equiv 2 \pmod 4$: the index bound; `Morse.card_lt_le_spectralIndex`, `Morse.card_sub_one_le_spectralIndex`). The index of a spectrum (in the sense of Chapter 14, V14.8) is at least the number of its eigenvalues greater than π^2. Hence, for integers $a_1, \dots, a_s \ge 1$ with some $a_{h_0} \ge 3$, the $s-1$ eigenvalues $\frac{\pi^2}{4}(a_{h_0} + a_j)^2$, $j \ne h_0$, already have index at least $s-1$. *Book:* pp. 144–146: K_A has at least $s-1$ eigenvalues $\ge (3+1)^2/4 = 4$, namely $(a_h + a_j)^2/4$ (Assertion, p. 145); each eigenvalue e^2 of K_A gives conjugate points at $t = e^{-1}, 2e^{-1}, \dots$, so $e^2 \ge 4$ gives an interior conjugate point; hence the index of a non-minimal geodesic from J to $-J$ in $\Omega_k(n)$ is at least $n/m_{k+1} - 1$ (Assertion, p. 146). *Hidden hypotheses:* as in V15.3, the conjugate points of $t \mapsto J\exp(\pi t A)$ come from $K_{\pi A} = \pi^2 K_A$, whose eigenvalues $\pi^2 e^2$ are the ones used here (Milnor's $t = e^{-1}$ is the count for them). *Deviation (restated):* the eigenvalues are input; the formula for K_A on the tangent space of the totally geodesic $\Omega_k(n)$ is not formalized. The splitting of $\mathbb{R}^n$ into minimal summands M_h and the isomorphisms $M_h \to M_j$ given by Lemma 24.8 are V15.29, and the resulting bound for the index of a geodesic is V15.30, proved through Killing fields instead of the eigenvectors B. *Deviation (stronger):* oddness of the a_j is not needed.

V15.20 (the proof of Theorem 24.5 for $k \equiv 2 \pmod 4$: the index bound on a component; `Morse.card_sub_abs_sum_le_two_mul_bottIndex`, `Morse.card_sub_abs_sum_le_four_mul_spectralIndex`). Let $a_1, \dots, a_r$ be odd integers, some $|a_h| \ge 3$, and $c = \sum_h a_h$. Then $\sum_{a_h > a_j}(a_h - a_j - 2) \ge (r - |c|)/2$; equivalently, the index (V14.8) of the spectrum with one eigenvalue $\frac{\pi^2}{4}(a_h - a_j)^2$ for each pair $a_h > a_j$ is at least $(r - |c|)/4$. *Book:* pp. 147–148: on a component of $\Omega\Omega_k(n)$, where $\operatorname{tr} A = icm_k/2$ is fixed, the a_h are odd with $\sum a_h = c$ and $\max|a_h| \ge 3$; the pairs $a_h > a_j$ give eigenvalues $(a_h - a_j)^2/4$ and $2\lambda \ge \sum_{a_h > a_j}(a_h - a_j - 2)$; Milnor then treats, "for example",

the case of some a_h equal to -3: with p and $-q$ the sums of the positive and of the negative a_h, $p-q=c$, $p+q\geq r$, and the displayed chain $2\lambda \geq \sum(a_h-a_j-2) > \sum_{a_h>0}(a_h-(-3)-3) = p$ gives $4\lambda \geq 2p \geq r+c$, with $r = n/m_k \to \infty$. *Hidden hypotheses:* Milnor treats only the example of an a_h equal to -3; the strict inequality of his chain fails when no a_h is positive (for $a=(-1,-3)$ both sides are 0), although his final bound still holds then. *Deviation (value differs):* the formal bound is $4\lambda \geq r-|c|$, whichever sign the large $|a_h|$ has; Milnor's $4\lambda \geq r+c$ is for his example, and some $a_h \geq 3$ gives the bound with $r-c$ instead. Since c is fixed on a component and $r\to\infty$, either suffices. *Deviation (restated):* as in V15.19, the eigenvalues are input; the bound for the geodesics themselves is V15.38.

V15.21 (totally geodesic sets of orthogonal matrices; `Morse.isSliceSubmanifold_of_param`, `Morse.IsTotGeo`, `Morse.IsTotGeo.tangentSubspace_eq`, `Morse.IsTotGeo.isGeodesicOn_mul_exp_smul`, `Morse.IsTotGeo.eqOn_of_isUnbrokenGeodesic`, `Morse.IsTotGeo.exp_eq`, `Morse.IsTotGeo.isGeodesicallyCo` Let $N\subseteq M_n(\mathbb{R})$ be a slice submanifold consisting of orthogonal matrices, and for each $x\in N$ let T_x be a space of skew matrices with $\dim T_x = \dim N$, such that $x\exp A\in N$ and $A\in T_{x\exp A}$ whenever $A\in T_x$. Then: (i) the tangent space of N at x is xT_x; (ii) for $A\in T_x$ the curve $t\mapsto x\exp(tA)$ is a geodesic of N on all of $\mathbb{R}$; (iii) every unbroken geodesic $\gamma\in\Omega(N;p,q)$ agrees on $[0,1]$ with $s\mapsto p\exp(sV)$ for some $V\in T_p$; (iv) the Riemannian exponential of N is $\exp_x(xA)=x\exp A$ for $A\in T_x$, and N is geodesically complete. Also, as a tool for V15.23: let S be a subset of a real normed space V such that near each $x\in S$ there is a smooth $\Phi: Z\times W\to V$ (Z, W finite-dimensional) with $\Phi(0)=x$ and invertible derivative at 0, for which, near 0, the points $\Phi(z,0)$ lie in S and every point $\Phi(q)\in S$ equals $\Phi(g(q),0)$ with $g(q)\to 0$ as $q\to 0$; then S is a slice submanifold modelled on Z. *Book:* p. 136, footnote: a submanifold is totally geodesic if each of its geodesics is a geodesic of the larger manifold; p. 137: the description of $\Omega_k(n)$ near J "clearly implies" that it is a totally geodesic submanifold of $O(n)$. *Hidden hypotheses:* "clearly" needs an argument (the exposition's Lemma 2.2). *Deviation (restated):* "totally geodesic" is not used as a definition, and the geodesics of N are not compared with those of $O(n)$; what Milnor draws from it, the list (ii)–(iii) of the geodesics of N, is proved directly. *Proof:* the acceleration of $t\mapsto x\exp(tA)$ is yA^2, $y=x\exp(tA)$; A^2 is symmetric, hence yA^2 is orthogonal to the tangent space yT_y, which consists of y times skew matrices; so the curve is a geodesic of the submanifold N, and (iii) follows by the uniqueness of geodesics.

V15.22 (the space $\mathcal{T}_J$, and Assertions 1 and 2 in both directions; `Morse.tanAlg`, `Morse.mem_tanAlg`, `Morse.tanAlg_zero`, `Morse.mul_exp_mem_omega`, `Morse.mem_tanAlg_mul_exp`, `Morse.exists_exp_injective`, `Morse.mem_tanAlg_of_mul_exp_mem`, `Morse.exists_local_log`). For a matrix x let $\mathcal{T}_x$ be the space of skew matrices A with $Ax=-xA$ and $AJ_\alpha = J_\alpha A$ for $1\leq\alpha\leq k-1$; for $k=0$ the first condition is dropped, so that $\mathcal{T}_x$ is the space $\mathfrak{o}(n)$ of all skew matrices, the same for every x. (i) For any fixed matrices J_α: if $x\in\Omega_k(n)$ and $A\in\mathcal{T}_x$, then $x\exp A\in\Omega_k(n)$ and $A\in\mathcal{T}_{x\exp A}$. (ii) There is $\varepsilon>0$ such that exp is injective on the matrices of norm $<\varepsilon$. For every such ε, every anticommuting family $J_1,\dots,J_k$, every $x\in\Omega_k(n)$ and every matrix C with $\|C\|<\varepsilon$: if $x\exp C\in\Omega_k(n)$, then $C\in\mathcal{T}_x$. (iii) Hence, for an anticommuting family and $x\in\Omega_k(n)$, there is a map L with $L(y)\to 0$ as $y\to x$, such that every $y\in\Omega_k(n)$ near x satisfies $L(y)\in\mathcal{T}_x$ and $y=x\exp(L(y))$. *Book:* p. 137: Assertion 1 ($J\exp A$ is a complex structure if and only if A anticommutes with J) and Assertion 2 ($J\exp A$ anticommutes with $J_1,\dots,J_{k-1}$ if and only if A commutes with them), for small skew A; hence "a neighborhood of J in $\Omega_k(n)$ consists of all points $J\exp A$ where A ranges over all small matrices in a linear subspace of the Lie algebra". *Hidden hypotheses:* "small" must be measured by a norm invariant under transposition and under conjugation by orthogonal matrices, since the "only if" halves apply the injectivity of exp to C^{T},

$x^{\mathsf T}Cx$ and $J_\alpha C J_\alpha^{\mathsf T}$ as well as to $\pm C$; the Frobenius norm is. *Deviation (stronger):* in (ii) C is any small matrix; that it is skew is part of the conclusion. The "if" halves are V15.6, here for all $k \geq 0$.

V15.23 (Lemma 24.4, first half: the strata of $\Omega_k(n)$ are compact totally geodesic submanifolds; `Morse.omegaStratum`, `Morse.finrank_tanAlg_mul_exp`, `Morse.eventually_mem_omegaStratum`, `Morse.isClosed_omegaStratum`, `Morse.isCompact_omegaStratum`, `Morse.isSliceSubmanifold_omegaStratum`, `Morse.isTotGeo_omegaStratum`, `Morse.omegaStratum_zero`). The *stratum* of $\Omega_k(n)$ through a matrix p is the set of $x \in \Omega_k(n)$ with $\dim \mathcal{T}_x = \dim \mathcal{T}_p$. (i) $\dim \mathcal{T}_{x\exp A} = \dim \mathcal{T}_x$ for $A \in \mathcal{T}_x$. Let $J_1, \dots, J_k$ be an anticommuting family. (ii) The stratum is open in $\Omega_k(n)$ (each of its points has a neighbourhood in $M_n(\mathbb{R})$ whose points in $\Omega_k(n)$ belong to it), closed in $M_n(\mathbb{R})$, and compact. (iii) It is a slice submanifold of $M_n(\mathbb{R})$ modelled on $\mathcal{T}_p$, and it satisfies the hypotheses of V15.21 with $T_x = \mathcal{T}_x$. Consequently its tangent space at x is $x\mathcal{T}_x$, its geodesics are the curves $t \mapsto x\exp(tA)$ with $A \in \mathcal{T}_x$, and it is geodesically complete. For $k = 0$ the stratum through any p is $O(n)$. *Book:* p. 136, Lemma 24.4: each $\Omega_k(n)$ is a smooth, totally geodesic submanifold of $O(n)$; p. 139: $\Omega_3(16r)$ "contains components of varying dimension" (and p. 141 for $\Omega_7(16r)$). *Hidden hypotheses:* as the components of $\Omega_k(n)$ may have different dimensions, the lemma must be read on each component separately. *Deviation (restated):* on the stratum through p, which is open and closed in $\Omega_k(n)$, hence a union of components, among them that of p when $p \in \Omega_k(n)$; as a submanifold of $M_n(\mathbb{R})$ rather than of $O(n)$; and with "totally geodesic" in the sense of V15.21. *Proof:* near a point x of the stratum, $(a, w) \mapsto x(\exp a + w)$, $a \in \mathcal{T}_x$, $w \in \mathcal{T}_x^\perp$, is a parametrization as in V15.21, by V15.22 (iii) and (i).

V15.24 (Lemma 24.1, the length of a geodesic from I to $-I$; `Morse.pi_sq_le_of_sq_mul`, `Morse.pi_sq_mul_card_le_norm_sq`, `Morse.mul_self_eq_of_norm_sq_le`). Let V be a skew matrix with $\exp V = -I$. If $E \neq 0$ is a matrix and $V^2E = -\lambda E$ for a real λ, then $\lambda \geq \pi^2$. Hence $\|V\|^2 \geq \pi^2 n$, and if $\|V\|^2 \leq \pi^2 n$ then $V^2 = -\pi^2 I$. *Book:* p. 134, proof of Lemma 24.1: for A in the block form with blocks $a_j\left(\begin{smallmatrix}0 & 1\\ -1 & 0\end{smallmatrix}\right)$, $\exp(\pi A) = -I$ if and only if the a_j are odd integers; $\langle A, A\rangle = 2(a_1^2 + \cdots + a_m^2)$; so the geodesic $t \mapsto \exp(\pi t A)$ is minimal if and only if $a_1 = \cdots = a_m = 1$, and then $A^2 = -I$. *Hidden hypotheses:* none. *Deviation (restated):* $V = \pi A$, and no normal form: the symmetric matrix $V^{\mathsf T}V = -V^2$ is diagonalized by Mathlib's spectral theorem, and for an eigenvalue $\lambda > 0$ the solution $\cos(\sqrt{\lambda}\,t)E + \lambda^{-1/2}\sin(\sqrt{\lambda}\,t)VE$ of $c' = Vc$ must be $-E$ at $t = 1$, which forces $\sqrt{\lambda} \geq \pi$. That this bound describes the minimal geodesics is V15.25.

V15.25 (the distance from p to $-p$, and the minimal geodesics; `Morse.minVel0`, `Morse.omegaGeo`, `Morse.neg_mem_omegaStratum`, `Morse.riemDist_omegaStratum`, `Morse.pi_sq_card_le_energy`, `Morse.omegaPath_mem_minPaths`, `Morse.exists_minVel0_of_mem_minPaths`, `Morse.omegaGeo_half`). Let $J_1, \dots, J_k$ be an anticommuting family, $p \in \Omega_k(n)$, N the stratum through p, and $\mathcal{C}_p = \{A \in \mathcal{T}_p : A^2 = -I\}$; suppose $\mathcal{C}_p \neq \emptyset$. For $A \in \mathcal{C}_p$ put $\gamma_A(t) = p\exp(\pi t A)$. Then: (i) $-p \in N$; (ii) the Riemannian distance in N is $\rho(p, -p) = \pi\sqrt{n}$, and every path in $\Omega(N; p, -p)$ has energy $\geq \pi^2 n$; (iii) every γ_A, $A \in \mathcal{C}_p$, is a path of least energy in $\Omega(N; p, -p)$, and its midpoint is $\gamma_A(\frac{1}{2}) = pA$; (iv) every path of least energy agrees on $[0, 1]$ with γ_A for some $A \in \mathcal{C}_p$. *Book:* pp. 137–138, proof of Lemma 24.4: the geodesic $t \mapsto J_k\exp(\pi t A)$ "is minimal in $O(n)$", hence "certainly minimal in $\Omega_k(n)$"; conversely a minimal geodesic from J_k to $-J_k$ in $\Omega_k(n)$ is $J_k\exp(\pi t A)$ with A a complex structure (by 24.1) commuting with $J_1, \dots, J_{k-1}$ and anticommuting with J_k (by Assertions 1, 2); Remark, p. 138: J_kA is the midpoint $\gamma(\frac{1}{2})$. *Hidden hypotheses:* that a minimal geodesic exists; in the converse, that a minimal geodesic of $\Omega_k(n)$ is minimal in $O(n)$, which needs the distance from J_k to $-J_k$ in $O(n)$. *Deviation (restated):* distances

in $\Omega_k(n)$ and in $O(n)$ are not compared. N is complete (V15.23), so by the Hopf–Rinow theorem (V7.26) a shortest geodesic from p to $-p$ exists; it is $t \mapsto p\exp(tV)$ with $V \in \mathcal{T}_p$ and $\exp V = -I$ (V15.23), of length $\|V\| \geq \pi\sqrt{n}$ (V15.24). "Minimal" is read as "of least energy" (V13.1). *Deviation (stronger):* any $p \in \Omega_k(n)$ with $\mathcal{C}_p \neq \emptyset$, not only J_k.

V15.26 (Lemma 24.1, and Lemma 24.4, second half: the minimal geodesics and $\Omega_{k+1}(n)$; `Morse.basePt`, `Morse.mem_minVel0_iff`, `Morse.minVel0Homeomorph`, `Morse.omegaGeo_eq_semicircle`, `Morse.omegaMinPathsHomotopyEquiv`, `Morse.omegaMinPathsHomeomorph`, `Morse.omegaPhi`, `Morse.omegaPhi_apply`, `Morse.isHomotopyEquiv_omegaPhi_minPaths`). Let $J_1, \dots, J_k$ be an anticommuting family, with base point J_k ($J_0 = I$). (i) $A \in \mathcal{C}_{J_k}$ if and only if $J_k A \in \Omega_{k+1}(n)$; $A \mapsto J_k A$ is a homeomorphism of $\mathcal{C}_{J_k}$ onto $\Omega_{k+1}(n)$, with inverse $J' \mapsto J_k^{\mathsf{T}} J'$; and for $J' \in \Omega_{k+1}(n)$ and $A = J_k^{\mathsf{T}} J'$, $\gamma_A(t) = \cos(\pi t)\, J_k + \sin(\pi t)\, J'$ for all t. (ii) In the setting of V15.25, $A \mapsto \gamma_A$ is a homotopy equivalence of $\mathcal{C}_p$ with $\Omega^d(N; p, -p)$, and a homeomorphism of $\mathcal{C}_p$ onto the quotient of Ω^d in which paths at distance 0 are identified. (iii) If $J_1, \dots, J_{k+1}$ is an anticommuting family and N is the stratum of $\Omega_k(n)$ through J_k, the map $\varphi : \Omega_{k+1}(n) \to \Omega(N; J_k, -J_k)$, $\varphi(J')(t) = \cos(\pi t)\, J_k + \sin(\pi t)\, J'$, is continuous, and it is the inclusion $\Omega^d \to \Omega$ composed with a homotopy equivalence $\Omega_{k+1}(n) \to \Omega^d$. For $k = 0$, $N = O(n)$ and $\Omega_1(n)$ is the set of all complex structures: this is Lemma 24.1. *Book:* p. 133, Lemma 24.1: the space of minimal geodesics from I to $-I$ in $O(n)$ is homeomorphic to the space $\Omega_1(n)$ of complex structures; p. 136, Lemma 24.4: the space of minimal geodesics from J_ℓ to $-J_\ell$ in $\Omega_\ell(n)$ is homeomorphic to $\Omega_{\ell+1}(n)$, for $0 \leq \ell < k$. *Hidden hypotheses:* $\Omega_{\ell+1}(n) \neq \emptyset$, here the existence of J_{k+1}; the topology of the space of minimal geodesics. *Deviation (narrower):* paths are functions on $\mathbb{R}$, and two paths at distance 0 are two that agree on $[0, 1]$; so Ω^d itself is only homotopy equivalent to $\Omega_{k+1}(n)$, and homeomorphic to it after that identification (as for $SU(2m)$, V14.17).

V15.27 (the neighbourhood retraction for $\Omega_k(n)$; `Morse.midDir`, `Morse.polarDir`, `Morse.polarDir_mem`, `Morse.omegaNbhd`, `Morse.isOpen_omegaNbhd`, `Morse.exists_retraction_minPaths_omega`). In the setting of V15.25, $\Omega^d(N; p, -p)$ is a retract of an open neighbourhood: there are an open $O \supseteq \Omega^d$ in $\Omega(N; p, -p)$ and a map $\rho : O \to \Omega^d$, continuous on O and equal to the identity on Ω^d. Here, for a matrix Y, let Y_1 be the orthogonal projection of $p^{\mathsf{T}}Y$ onto $\mathcal{T}_p$; if $\det(Y_1^{\mathsf{T}} Y_1) \neq 0$, then $Y_1 (Y_1^{\mathsf{T}} Y_1)^{-1/2} \in \mathcal{C}_p$. O is the set of paths ω whose midpoint $Y = \omega(\frac{1}{2})$ has $\det(Y_1^{\mathsf{T}} Y_1) \neq 0$, an open set, and off Ω^d the retraction is $\rho(\omega) = \gamma_A$ with $A = Y_1 (Y_1^{\mathsf{T}} Y_1)^{-1/2}$. *Book:* p. 118, Theorem 22.1 assumes that Ω^d is a topological manifold; in §24 this is the statement that $\Omega_{k+1}(n)$ is a manifold (pp. 135–136: "together with the statement that $\Omega_1(n)$ is a manifold"). *Hidden hypotheses:* Hanner's theorem, which turns "manifold" into the retraction that the proof of 22.1 uses (V13.2). *Deviation (not in the book):* an explicit retraction, the third of the project (after V13.8 and V14.21): the nearest complex structure in $\mathcal{T}_p$ to the midpoint, by projection and polar decomposition. It supplies the hypothesis of V13.2 for every stratum.

V15.28 (Killing fields are Jacobi fields, and a lower bound for the index; `Morse.isoAlg`, `Morse.killing`, `Morse.isJacobiField_killing`, `Morse.finrank_le_conjMultiplicity`, `Morse.sum_finrank_le_formIndex`). Let $J_1, \dots, J_k$ be an anticommuting family, $p \in \Omega_k(n)$, N the stratum through p, and $V \in \mathcal{T}_p$. The *isotropy algebra* of p is the space of skew matrices commuting with p and with $J_1, \dots, J_{k-1}$. (i) For X in the isotropy algebra, $K(t) = X\gamma(t) - \gamma(t)X$ is a Jacobi field along $\gamma(t) = p\exp(tV)$ in N. (ii) Let $\gamma \in \Omega(N; p, q)$ be an unbroken geodesic equal to $t \mapsto p\exp(tV)$ on $[0, 1]$, let $\tau \in [0, 1]$, and let $\mathcal{X}$ be a subspace of the isotropy algebra whose elements commute with $\exp(\tau V)$ and none of whose non-zero elements commutes with V. Then $\gamma(\tau)$ is conjugate to $\gamma(0)$ along γ with multiplicity (V9.4) at least $\dim \mathcal{X}$. (iii) For a finite

set $T \subset (0,1)$ and such subspaces $\mathcal{X}_\tau$, $\tau \in T$, the index of γ is at least $\sum_{\tau\in T} \dim \mathcal{X}_\tau$. *Book:* pp. 144, 146: since $\Omega_k(n)$ is totally geodesic in $O(n)$, K_A is computed by $K_A B = -\frac{1}{4}[A,[A,B]]$ on the space T of Milnor's conditions 1)–3); each eigenvalue e^2 of K_A gives conjugate points at $t = e^{-1}, 2e^{-1}, \dots$ by 20.5; then the index theorem. *Hidden hypotheses:* that the curvature of a totally geodesic submanifold is that of the ambient manifold, and Theorem 20.5 for the components of $\Omega_k(n)$. *Deviation (restated):* neither is formalized, and neither is needed for a lower bound, which only asks for Jacobi fields vanishing at 0 and at τ. These come from the isometries $y \mapsto gyg^{-1}$, $g = \exp(uX)$, of N, which fix p: K is the variation field of the geodesics $t \mapsto g\,p\exp(tV)\,g^{-1}$ (Lemma 14.3, V9.8), and it vanishes at τ when X commutes with $\exp(\tau V)$. (iii) is the Index Theorem (V9.24).

V15.29 (the splitting into minimal pieces, and Lemma 24.8 between two pieces; `Morse.IsInvar`, `Morse.IsMinInvar`, `Morse.IsStarSigned`, `Morse.exists_minInvar_decomp`, `Morse.IsMinInvar.exists_scalar` `Morse.IsOmegaOps`, `Morse.omegaGens`, `Morse.IsOmegaOps.exists_sq_eq`, `Morse.one_le_of_minInvar`, `Morse.IsOmegaOps.isClifford_cliffFam`, `Morse.IsOmegaOps.isMinimal_cliffFam`, `Morse.IsOmegaOps.exist` `Morse.IsOmegaOps.exists_piece_gt`, `Morse.IsOmegaOps.exists_omegaPieceData`**).** Let E be a finite-dimensional real inner product space and $\mathcal{G}$ a set of operators on E, each self-adjoint or skew-adjoint. A subspace is *minimal invariant* if it is non-zero, invariant under $\mathcal{G}$, and contains no invariant subspace other than 0 and itself. (i) Every invariant subspace U is spanned by a list of pairwise orthogonal minimal invariant subspaces of U whose dimensions add up to $\dim U$; and a self-adjoint operator that commutes with $\mathcal{G}$ and preserves a minimal invariant M is a scalar on M. (ii) Let A be skew-adjoint, let $J, J_1, \dots, J_{k-1}$ be skew-adjoint operators with square -1, anticommuting with one another, and let A anticommute with J and commute with the J_α (for $k = 0$: $J = 1$ and there are no J_α); put $\mathcal{G} = \{A, J, J_1, \dots, J_{k-1}\}$. On each minimal invariant M, $A^2 = -c$ for a real c; and for real s with $s^2 c = 1$ the restrictions of $J_1, \dots, J_{k-1}, J, sJA$ to M satisfy the Clifford relations (V15.10) and make M a minimal space for $k+1$ structures (V15.13). For matrices: if A is skew, $\exp(\pi A) = -I$, and $A^2 = -c$ on a non-zero subspace of $\mathbb{R}^n$, then $c \geq 1$. (iii) If $k \leq 7$, $k \not\equiv 2 \pmod 4$, and M_0, M_1 are minimal invariant with $A^2 = -a_0^2$ on M_0 and $A^2 = -b^2$ on M_1, $a_0, b > 0$, there is a linear isomorphism $T : M_0 \to M_1$ that commutes with J and the J_α and carries $a_0^{-1}JA$ to $-b^{-1}JA$. (iv) If moreover $c \geq 1$ on every minimal invariant subspace and $A^2 \neq -1$, then in a splitting (i) of E some piece M_0 has $c > 1$; and there are an integer s with $(s+1)m_{k+1} = \dim E$, reals $a_0 > 1$ and $b_1, \dots, b_s \geq 1$, and operators $D_1, \dots, D_s$ on E (each the projection onto M_0 followed by an isomorphism (iii) onto another piece) which commute with J and the J_α, satisfy $b_j D_j A = -a_0 A D_j$, $A^2 D_j = -b_j^2 D_j$ and $D_j A^2 = -a_0^2 D_j$, and for which the AD_j are separated by projections (for $k = 0$ there are two for each j, D_j and $D_j A$). *Book:* pp. 144–145: $\mathbb{R}^n = M_1 \oplus \cdots \oplus M_s$, mutually orthogonal subspaces closed and minimal under $J_1, \dots, J_{k-1}$, J and A; the eigenvalues of A on M_h are "all equal, except for sign", $\pm i a_h$ with a_h odd and positive; $J' = a_h^{-1} JA|M_h$ makes M_h a minimal $(J_1, \dots, J_{k-1}, J, J')$-space, so $\dim M_h = m_{k+1}$ and $s = n/m_{k+1}$; for $h \neq j$ an isometry $M_h \to M_j$ "exists by 24.8". *Hidden hypotheses:* that the orthogonal complement of an invariant subspace is invariant (the operators are skew); $k + 1 \leq 8$, the range in which Lemma 24.8 is proved (V15.16–V15.17). *Deviation (restated):* $a_h = \sqrt{c}$; that the a_h are odd integers is not used, only $c \geq 1$. Milnor changes the sign of J on M_j; changing the sign of the last structure instead is equivalent, and keeps T commuting with J. *Deviation (narrower):* T is a linear isomorphism, not an isometry (V15.17); this suffices, because only the skew matrix $D - D^{\mathsf{T}}$ is used.

V15.30 (the index of a non-minimal geodesic for $k \not\equiv 2 \pmod 4$, and Lemma 24.2 for geodesics; `Morse.IsInterRel`, `Morse.IsInterRel.commute_exp`, `Morse.exists_killing_spaces`, `Morse.exists_killing_`

`Morse.le_formIndex_of_not_minimal_omega`, `Morse.omega_hidx`). Let $J_1, \dots, J_k$ be an anticommuting family on $\mathbb{R}^n$ with $\mathcal{C}_{J_k} \neq \emptyset$ (equivalently $\Omega_{k+1}(n) \neq \emptyset$, V15.26), let $k \leq 7$ and $k \not\equiv 2 \pmod 4$, and let N be the stratum through J_k. If $\gamma \in \Omega(N; J_k, -J_k)$ is an unbroken geodesic which is not of least energy, then m_{k+1} divides n and the index of γ is at least $n/m_{k+1} - 1$; for $k = 0$ it is at least $2(n/2 - 1) = n - 2$. *Book:* p. 135, Lemma 24.2: any non-minimal geodesic from I to $-I$ in $O(2m)$ has index $\geq 2m - 2$; p. 146, Assertion: the index of a non-minimal geodesic from J to $-J$ in $\Omega_k(n)$ is $\geq n/m_{k+1} - 1$ (for $k \not\equiv 2 \pmod 4$), from the $s - 1$ eigenvalues $(a_h + a_j)^2/4 \geq 4$ of K_A (p. 145). *Hidden hypotheses:* the linear independence of the eigenvectors belonging to different pairs (h, j); $k + 1 \leq 8$. *Deviation (restated):* by V15.23 and V15.25, γ is $t \mapsto J_k \exp(\pi t A)$ on $[0, 1]$ with $A \in \mathcal{T}_{J_k}$, $\exp(\pi A) = -I$ and $A^2 \neq -I$. With the pieces and the operators D_j of V15.29, $X_j = D_j - D_j^{\mathsf{T}}$ lies in the isotropy algebra of J_k and commutes with $\exp(\tau_j \pi A)$ for $\tau_j = 2/(a_0 + b_j) \in (0, 1)$, and the $[X_j, A]$ are linearly independent; V15.28 (iii) gives the bound. Milnor's eigenvector B is $[X_j, A]$ up to a factor. For $k = 0$ each pair of pieces gives two fields, from D_j and $D_j A$: Milnor's eigenvalues "counted twice". "Non-minimal" is read as "not of least energy". *Deviation (narrower):* only the lower bound; the formula of Lemma 24.2 for the index of a geodesic of $O(2m)$ is not proved for geodesics (V15.3 derives it from Milnor's spectrum).

V15.31 (Theorem 24.5 for each n, $k \not\equiv 2 \pmod 4$; `Morse.omega_theorem_22_1_bijective`, `Morse.omegaPhi_bijective`, `Morse.stratumIn`, `Morse.isClopen_stratumIn`, `Morse.nonempty_pi_stratum_equ` `Morse.omega_pi_equiv`). Let $J_1, \dots, J_{k+1}$ be an anticommuting family on $\mathbb{R}^n$, $k \leq 7$, $k \not\equiv 2 \pmod 4$, let N be the stratum of $\Omega_k(n)$ through J_k, and let $i + 2 \leq n/m_{k+1} - 1$ (for $k = 0$: $i + 2 \leq n - 2$). Then: (i) the inclusion $\Omega^d \to \Omega(N; J_k, -J_k)$ induces a bijection on π_i at every minimal path; (ii) for every $J' \in \Omega_{k+1}(n)$, the map φ of V15.26 induces a bijection $\pi_i(\Omega_{k+1}(n), J') \to \pi_i(\Omega(N; J_k, -J_k), \varphi(J'))$; (iii) for every $J' \in \Omega_{k+1}(n)$ there is a bijection $\pi_i(\Omega_{k+1}(n), J') \cong \pi_{i+1}(\Omega_k(n), J_k)$. On the way: N is open and closed in $\Omega_k(n)$, and $\pi_{j+1}(N, x) \cong \pi_{j+1}(\Omega_k(n), x)$ for $x \in N$ and $j \geq 0$. *Book:* p. 146: the inclusion $\Omega_{k+1}(n) \to \Omega\Omega_k(n)$ induces isomorphisms of homotopy groups in dimensions $\leq n/m_{k+1} - 3$; p. 138, Theorem 24.5: $\pi_h O \cong \pi_{h-1}\Omega_1 \cong \pi_{h-2}\Omega_2 \cong \cdots \cong \pi_1 \Omega_{h-1}$. *Hidden hypotheses:* base points; $k + 1 \leq 8$; the case $i = 0$. *Deviation (restated):* V13.2–V13.4 for the stratum, with V15.27 and V15.30. Milnor's $\Omega\Omega_k(n)$ is here the space of piecewise smooth paths from J_k to $-J_k$ in N with Milnor's metric; the passage to continuous paths and loops is inside V13.4. Bijections of pointed sets, π_0 included, at every base point J'. The quotient n/m_{k+1} is exact, since J_{k+1} exists (V15.14).

V15.32 (Theorem 24.3; `Morse.theorem_24_3`). Suppose $\mathbb{R}^n$ carries a complex structure (so n is even), and let $i + 4 \leq n$. For every complex structure J' on $\mathbb{R}^n$, the map $\varphi : \Omega_1(n) \to \Omega(O(n); I, -I)$, $\varphi(J)(t) = \cos(\pi t)\, I + \sin(\pi t)\, J$, induces a bijection on π_i at J', and there is a bijection $\pi_i(\Omega_1(n), J') \cong \pi_{i+1}(O(n), I)$. *Book:* p. 136, Theorem 24.3 (Bott): the inclusion map $\Omega_1(n) \to \Omega O(n)$ induces isomorphisms of homotopy groups in dimensions $\leq n - 4$; hence $\pi_i \Omega_1(n) \cong \pi_{i+1} O(n)$ for $i \leq n - 4$. *Hidden hypotheses:* base points. *Deviation (restated):* the case $k = 0$ of V15.31, where the stratum is $O(n)$; bijections of pointed sets, at every J'.

V15.33 (angles, and the sets of paths with a given end angle; `Morse.IsAngleLift`, `Morse.IsAngleLift.unique`, `Morse.exists_isAngleLift`, `Morse.IsAngleLift.perturb`, `Morse.windSet`, `Morse.isClopen_windSet`, `Morse.disjoint_windSet`). Let F be a complex-valued function on a finite-dimensional real inner product space V and $\omega : \mathbb{R} \to V$. An *angle* of $F \circ \omega$ is a continuous $\theta : [0, 1] \to \mathbb{R}$ with $\theta(0) = 0$ and $F(\omega(t)) = e^{i\theta(t)}$ for $t \in [0, 1]$. An angle is unique, and it exists whenever $t \mapsto F(\omega(t))$ is continuous on $[0, 1]$, of modulus 1, with $F(\omega(0)) = 1$. For $N \subseteq V$, points p, q and a real

w, let $\Omega_w \subseteq \Omega(N;p,q)$ be the set of paths ω for which $F\circ\omega$ has an angle with $\theta(1) = w$. If F is continuous and $|F| = 1$ on a compact set containing N, every Ω_w is open and closed in $\Omega(N;p,q)$; and Ω_w, $\Omega_{w'}$ are disjoint for $w \neq w'$. *Book:* p. 147: f maps the geodesic into a closed loop on S^1 "which is completely determined by the trace of A", and "this trace is invariant under homotopy of the geodesic within the path space"; p. 148: "restrict attention to some fixed component of $\Omega\Omega_k(n)$". *Hidden hypotheses:* Theorem 22.1 is applied to one class of paths, which must then be open and closed in the path space (V13.16). *Deviation (restated):* the classes Ω_w in place of components. *Proof:* paths that are close in Milnor's metric are uniformly close on $[0,1]$; then $|F\circ\omega' - F\circ\omega| < 1$, the quotient $F\circ\omega'/F\circ\omega$ has positive real part, and adding its argument to the angle of ω gives the angle of ω', with the same end value.

V15.34 ($\det_{\mathbb{C}}$ of a real matrix; `Morse.cplx`, `Morse.projP`, `Morse.projM`, `Morse.cdet`, `Morse.cdet_mul`, `Morse.cdet_one`, `Morse.norm_cdet`, `Morse.cdet_exp`). For real $n\times n$ matrices $\mathbf{i}$ and M put $\det_{\mathbb{C}} M = \det(M_{\mathbb{C}}P_+ + P_-)$, a complex number, where $M_{\mathbb{C}}$ is M read as a complex matrix and $P_\pm = \frac{1}{2}(1 \mp \sqrt{-1}\,\mathbf{i}_{\mathbb{C}})$. Then $\det_{\mathbb{C}} I = 1$, and, if $\mathbf{i}^2 = -I$: $\det_{\mathbb{C}}(MN) = \det_{\mathbb{C}} M \det_{\mathbb{C}} N$ whenever N commutes with $\mathbf{i}$; $|\det_{\mathbb{C}} M| = 1$ when $\mathbf{i}$ is skew and M is orthogonal and commutes with $\mathbf{i}$; and $\det_{\mathbb{C}}\exp(tV) = e^{-\sqrt{-1}\,t\,\mathrm{tr}(V\mathbf{i})/2}$ for every real t, when V commutes with $\mathbf{i}$ and $\mathrm{tr}\,V = 0$. *Book:* p. 146: $\mathbb{R}^n$ is made a complex vector space by $iv = J_1J_2\cdots J_{k-1}v$, and a unitary complex linear transformation "has a well defined complex determinant"; p. 147: $f(J\exp(\pi tA)) = \det(\exp(\pi tA)) = e^{\pi t\,\mathrm{trace}\,A}$, the trace being the complex one. *Hidden hypotheses:* none. *Deviation (restated):* no complex basis is chosen. $M_{\mathbb{C}}P_+ + P_-$ is M on the $\sqrt{-1}$-eigenspace of $\mathbf{i}_{\mathbb{C}}$, a copy of the complex vector space $(\mathbb{R}^n,\mathbf{i})$, and the identity on the other eigenspace; and the complex trace of a real V commuting with $\mathbf{i}$ is $\frac{1}{2}(\mathrm{tr}\,V - \sqrt{-1}\,\mathrm{tr}(V\mathbf{i}))$.

V15.35 (Milnor's i and f, and the sets $\Omega^{(c)}$; `Morse.IsCentralI`, `Morse.iWordM`, `Morse.isCentralI_two`, `Morse.isCentralI_six`, `Morse.omegaF`, `Morse.norm_omegaF`, `Morse.omegaF_self`, `Morse.omegaF_geo`, `Morse.omegaComp`, `Morse.isClopen_omegaComp`, `Morse.disjoint_omegaComp`, `Morse.mem_omegaComp_geo_iff`). Call a matrix $\mathbf{i}$ *central for k* if $\mathbf{i}^2 = -I$, $\mathbf{i}^{\mathsf{T}} = -\mathbf{i}$, $\mathbf{i}$ commutes with $J_1,\dots,J_{k-1}$ and with every matrix that commutes with all of them, and $\mathbf{i}$ anticommutes with every matrix that anticommutes with all of them. The product $\mathbf{i} = J_1J_2\cdots J_{k-1}$ is central for $k = 2$ when J_1 is a complex structure, and for $k = 6$ when $J_1,\dots,J_5$ is an anticommuting family. Let $\mathbf{i}$ be central for $k \geq 1$, $p \in \Omega_k(n)$, and $f(X) = \det_{\mathbb{C}}(p^{\mathsf{T}}X)$ (V15.34 with this $\mathbf{i}$). Then $|f| = 1$ on $\Omega_k(n)$, $f(p) = 1$, and $f(p\exp(tV)) = e^{-\sqrt{-1}\,t\,\mathrm{tr}(V\mathbf{i})/2}$ for $V \in \mathcal{T}_p$. Now let $J_1,\dots,J_k$ be an anticommuting family, $k \geq 1$, $p = J_k$, N the stratum through J_k, and let $\Omega_w \subseteq \Omega(N;J_k,-J_k)$ be the set of paths whose f-angle ends at w (V15.33). Each Ω_w is open and closed, different w give disjoint sets, and a path that agrees on $[0,1]$ with $t \mapsto J_k\exp(tV)$, $V \in \mathcal{T}_{J_k}$, lies in Ω_w if and only if $w = -\mathrm{tr}(V\mathbf{i})/2$. For an integer c, $\Omega^{(c)}$ denotes Ω_w with $w = \pi m_{k+1}c/2$. *Book:* pp. 146–148: $iv = J_1J_2\cdots J_{k-1}v$, where "the condition $k \equiv 2 \pmod 4$ guarantees that $i^2 = -1$, and that $J_1, J_2, \dots, J_{k-1}$ commute with i"; $f(J') = \det_{\mathbb{C}}(J^{-1}J')$; a component of the path space is fixed by trace $A = icm_k/2$. *Hidden hypotheses:* that every $J' \in \Omega_k(n)$ anticommutes with i and every $A \in \mathcal{T}$ commutes with it, so that $J^{-1}J'$ and A are complex linear. *Deviation (restated):* the properties of i that are used are taken as the definition of "central" and proved for $k = 2$ and $k = 6$, the cases that occur for $k \leq 7$. For these $m_{k+1} = m_k$, and for the geodesic $J_k\exp(\pi tA)$ Milnor's trace A is $-\frac{\sqrt{-1}}{2}\,\mathrm{tr}(A\mathbf{i})$; so $\Omega^{(c)}$ contains exactly the geodesics with trace $A = icm_k/2$, as in the book.

V15.36 (every path lies in exactly one $\Omega^{(c)}$; `Morse.existsUnique_mem_omegaComp`, `Morse.exists_mem_omegaComp`, `Morse.exists_mem_omegaComp_six`, `Morse.pdet`, `Morse.pdet_split`, `Morse.omegaF_eq_pow_four`). Let $k = 2$ or $k = 6$, let $J_1,\dots,J_k$ be an anticommuting family, $\mathbf{i} = J_1\cdots J_{k-1}$ and N the stratum

through J_k. Every path in $\Omega(N; J_k, -J_k)$ lies in $\Omega^{(c)}$ for exactly one integer c. For $k = 6$ the proof uses that $f = G^4$ on $\Omega_6(n)$, where $G(X) = \det((J_6^{\mathsf{T}} X)_{\mathbb{C}} Q + 1 - Q)$ and Q is the product of the projections P_+ of V15.34 for $\mathbf{i}$, $J_1 J_2$ and $J_3 J_4$. *Book:* p. 148: "trace $A = icm_k/2$ where c is some constant integer". *Hidden hypotheses:* that the class of an arbitrary path, not only of a geodesic, is given by an integer c. The end angle is a multiple of 2π because $f(-J_k) = \det_{\mathbb{C}}(-I) = 1$, which holds since 4 divides n (V15.14); this is Milnor's "closed loop" (p. 147). For $k = 2$ that is the claim, as $\pi m_3/2 = 2\pi$. For $k = 6$ the normalization asks for multiples of $\pi m_7/2 = 4\pi$, which needs f to be the fourth power of a continuous function G with $G(-J_6) = \pm 1$. *Deviation (not in the book):* the statement, and the fourth-power argument: the commutant of $J_1, \dots, J_5$ acts on the $+\sqrt{-1}$-eigenspace of $\mathbf{i}$ as four copies of one representation, split off by $J_1 J_2$ and $J_3 J_4$.

V15.37 (the pieces for $k \equiv 2 \pmod 4$; `Morse.iWord`, `Morse.IsIOps`, `Morse.IsOmegaOps.exists_eq_smul_iWord`, `Morse.exists_odd_of_mulVec`, `Morse.exists_decomp_two`, `Morse.exists_inter_two`). In the setting of V15.29 (ii) with $k \geq 1$, let $\mathbf{i} = J_1 \cdots J_{k-1}$ and suppose that $\mathbf{i}$ is skew-adjoint with $\mathbf{i}^2 = -1$, commutes with A and the J_α, and anticommutes with J. (i) On each minimal invariant M, $A = c\,\mathbf{i}$ for a real c. (ii) For matrices: if $\mathbf{i}$ is skew with $\mathbf{i}^2 = -I$ and commutes with A, $\exp(\pi A) = -I$, and $Aw = c\,\mathbf{i}w$ for a vector $w \neq 0$, then c is an odd integer. (iii) If $k \leq 7$ and the scalars c of (i) are odd integers, then there are pairwise orthogonal minimal invariant subspaces $M_1, \dots, M_\nu$ of E with $A = a_h \mathbf{i}$ on M_h, a_h an odd integer, $\nu m_{k+1} = \dim E$, $\operatorname{tr}(A\mathbf{i}) = -m_{k+1} \sum_h a_h$, and $A^2 = -1$ if every $a_h = \pm 1$. (iv) If $k \leq 8$, $k \not\equiv 3 \pmod 4$, and M_0, M_1 are minimal invariant with $A = c_0 \mathbf{i}$ on M_0 and $A = c_1 \mathbf{i}$ on M_1, $c_0, c_1 \neq 0$, there is a linear isomorphism $M_0 \to M_1$ that commutes with the J_α and anticommutes with J. *Book:* p. 147: on M_h the complex linear transformation $A|M_h$ "can have only one eigenvalue, say ia_h", so $A|M_h = a_h J_1 J_2 \cdots J_{k-1}|M_h$; M_h is minimal under $J_1, \dots, J_{k-1}, J$, of complex dimension $m_k/2$, so trace $A = i(a_1 + \cdots + a_r)m_k/2$; by 24.8 there is an isometry $M_h \to M_j$ which commutes with $J_1, \dots, J_{k-1}$ and anticommutes with J. *Hidden hypotheses:* why $A|M_h$ has only one eigenvalue. *Deviation (restated):* $A\mathbf{i}$ is self-adjoint and commutes with $\mathcal{G}$, hence is a scalar on M (V15.29 (i)); complex eigenspaces are not used. The count is in terms of m_{k+1}, which is m_k for $k = 2, 6$. *Deviation (narrower):* a linear isomorphism, not an isometry, as in V15.29.

V15.38 (the index bound on $\Omega^{(c)}$; `Morse.IsInterRelI`, `Morse.IsInterRelI.commute_exp`, `Morse.exists_killing_spaces_two`, `Morse.le_formIndex_of_not_minimal_omega_two`, `Morse.omega_hidx_two`). Let $J_1, \dots, J_k$ be an anticommuting family on $\mathbb{R}^n$ with $1 \leq k \leq 7$, $k \not\equiv 3 \pmod 4$, such that $\mathbf{i} = J_1 \cdots J_{k-1}$ is central for k and $\mathcal{C}_{J_k} \neq \emptyset$, and let c be an integer. If $\gamma \in \Omega^{(c)}$ is an unbroken geodesic which is not of least energy, then $n = \nu m_{k+1}$ for an integer ν, and the index of γ is at least $\lfloor (\nu - |c|)/4 \rfloor$ (at least 0 if $|c| > \nu$). *Book:* pp. 147–148: $2\lambda \geq \sum_{a_h > a_j} (a_h - a_j - 2)$; then, supposing "for example" that some a_h is -3, $4\lambda \geq 2p \geq r + c$, where $r = n/m_k$. *Hidden hypotheses:* as in V15.20, and the linear independence of the eigenvectors of different pairs. *Deviation (value differs):* $4\lambda \geq \nu - |c|$ up to rounding, whichever sign the large $|a_h|$ has (V15.20). *Deviation (restated):* as in V15.30. The geodesic is $J_k \exp(\pi t A)$ with $A = a_h \mathbf{i}$ on the pieces M_h and $\sum_h a_h = c$ (V15.35, V15.37), and some $|a_{h_0}| \geq 3$. If $a_{h_0} \geq 3$, the isomorphisms $T_j : M_{h_0} \to M_j$ of V15.37 (iv), for the j with $a_j < 0$, give operators D_j and elements $X_j = D_j \mathbf{i} - (D_j \mathbf{i})^{\mathsf{T}}$ of the isotropy algebra which commute with $\exp(\tau \pi A)$ at the $(a_{h_0} - a_j)/2 - 1$ parameters $\tau = 2s/(a_{h_0} - a_j)$ in $(0, 1)$; V15.28 (iii) and the count of V15.20 give the bound. If $a_{h_0} \leq -3$, the same is done for $-A$. Only the pairs (h_0, j) with a_j of the sign opposite to a_{h_0} are used.

V15.39 (Theorem 24.5 for each n, $k = 2, 6$; `Morse.isCentralI_iWordM`, `Morse.omegaPhi_mem_omegaComp`,

`Morse.omega_minPaths_bijective_two`, `Morse.omegaPhi_bijective_two`, `Morse.theorem_24_5_two_six`, `Morse.omega_pi_equiv_two`). Let $k = 2$ or $k = 6$, let $J_1, \dots, J_{k+1}$ be an anticommuting family on $\mathbb{R}^n$, $\nu = n/m_{k+1}$, and c an integer. (i) For $J' \in \Omega_{k+1}(n)$, $\varphi(J')$ lies in Ω_w with $w = -\pi \operatorname{tr}(J_k^{\mathsf{T}} J' \mathbf{i})/2$; so $\varphi(J') \in \Omega^{(c)}$ exactly when $-\operatorname{tr}(J_k^{\mathsf{T}} J' \mathbf{i}) = m_{k+1} c$. (ii) For $j+2 \le \lfloor (\nu - |c|)/4 \rfloor$, the inclusion $\Omega^d \cap \Omega^{(c)} \to \Omega^{(c)}$ induces a bijection on π_j at every point. (iii) For $j+3 \le \lfloor (\nu - |c|)/4 \rfloor$ and $J' \in \Omega_{k+1}(n)$ with $\varphi(J') \in \Omega^{(c)}$, φ induces a bijection on π_{j+1} at J', and there is a bijection $\pi_{j+1}(\Omega_{k+1}(n), J') \cong \pi_{j+2}(\Omega_k(n), J_k)$. *Book:* p. 148: "the component of $\Omega\Omega_k(n)$ is approximated up to higher and higher dimensions by the corresponding component of $\Omega_{k+1}(n)$, as $n \to \infty$". *Hidden hypotheses:* Theorem 22.1 for one class of paths (V13.16); which part of $\Omega_{k+1}(n)$ corresponds to $\Omega^{(c)}$. *Deviation (restated):* V13.16 with V15.27, V15.35 and V15.38, with explicit ranges; bijections of pointed sets. (ii) includes π_0; (iii) is stated in degrees ≥ 1, where the open and closed set $\Omega^{(c)}$ has the homotopy groups of the whole path space. The path components are V15.40.

V15.40 (Theorem 24.5 for each n, $k = 2, 6$: path components; `Morse.theorem_24_5_pi0_two_six`). In the setting of V15.39, let $\lfloor (\nu - |c|)/4 \rfloor \ge 2$. (i) Every path in $\Omega^{(c)}$ can be joined, by a path in $\Omega(N; J_k, -J_k)$, to $\varphi(J')$ for some $J' \in \Omega_{k+1}(n)$. (ii) If $\varphi(J') \in \Omega^{(c)}$, then for every $J'' \in \Omega_{k+1}(n)$: J' and J'' can be joined in $\Omega_{k+1}(n)$ if and only if $\varphi(J')$ and $\varphi(J'')$ can be joined in $\Omega(N; J_k, -J_k)$. *Book:* p. 146: $\Omega\Omega_k(n)$ has infinitely many components, while the approximating subspace $\Omega_{k+1}(n)$ has only finitely many; p. 148: passing to the direct limit, "a homotopy equivalence on each component". *Hidden hypotheses:* that in the limit φ is a bijection on path components. *Deviation (narrower):* for a fixed n only the classes with $|c| \le \nu - 8$ are controlled. With V15.36 (every path has a class) and V15.41 (the class is stable) this is what the limit needs; the limit itself is not formed.

V15.41 (the structures on $\mathbb{R}^{16r}$, and stabilization; `Morse.bottJ`, `Morse.isAnticommFamily_bottJ`, `Morse.stabEquiv`, `Morse.stab`, `Morse.stab_mem_omega`, `Morse.stab_basePt`, `Morse.omegaPhi_stab`, `Morse.omegaLabel_stab`, `Morse.omegaPhi_bottJ_mem_omegaComp`). On $\mathbb{R}^{16r} = \mathbb{R}^{16} \otimes \mathbb{R}^r$ let $J_\alpha = J'_\alpha \oplus \cdots \oplus J'_\alpha$ (r summands), where $J'_1, \dots, J'_8$ are the structures of V15.9 on $\mathbb{R}^{16}$; $J_1, \dots, J_8$ is an anticommuting family. Identify $\mathbb{R}^{16r} \oplus \mathbb{R}^{16}$ with $\mathbb{R}^{16(r+1)}$, the new block last, and put $s(X) = X \oplus J'_k$ (for $k = 0$: $X \oplus I_{16}$). Then: (i) for $k \le 8$, s maps $\Omega_k(16r)$ into $\Omega_k(16r+16)$ and carries J_k to J_k; (ii) for $k \le 7$ and $J' \in \Omega_{k+1}(16r)$, $\varphi(J' \oplus J'_{k+1})(t) = \varphi(J')(t) \oplus (\cos(\pi t)\, J'_k + \sin(\pi t)\, J'_{k+1})$ for all t; (iii) for $k = 2, 6$ and every matrix X on $\mathbb{R}^{16r}$, $\operatorname{tr}(J_k^{\mathsf{T}}(X \oplus J'_{k+1})\,\mathbf{i}) = \operatorname{tr}(J_k^{\mathsf{T}} X \mathbf{i})$, the left side computed on $\mathbb{R}^{16(r+1)}$, so that by V15.39 (i) the class c of $\varphi(J')$ does not change under stabilization; (iv) for $k = 2, 6$, $\varphi(J_{k+1}) \in \Omega^{(0)}$. *Book:* p. 138: $\Omega_k(n)$ is imbedded in $\Omega_k(n+n')$ by $J \mapsto J \oplus J'_k$; "It is not difficult to see that the inclusions $\Omega_{k+1}(n) \to \Omega\Omega_k(n)$ give rise, in the limit, to inclusions $\Omega_{k+1} \to \Omega\Omega_k$"; p. 139: $n = 16r$. *Hidden hypotheses:* the compatibility (ii) of φ with the imbeddings; for $k = 2, 6$, that the class is stable (iii). *Deviation (narrower):* the direct limits Ω_k, the limit map, and the naturality under stabilization of the bijections of V15.31 and V15.39 are not formalized; V15.45 does not need them.

V15.42 (Lemma 24.6 (8): $\Omega_8(16r)$ is homeomorphic to $O(r)$; `Morse.W16`, `Morse.eq_of_commute_bott16`, `Morse.qOf`, `Morse.eq_qOf_of_commute`, `Morse.omegaEightHomeomorph`, `Morse.omegaEightHomeomorph_basePt`). (i) A 16×16 matrix that commutes with $J'_1, \dots, J'_7$ is $b\, I + c\, W$ for real b, c, where $W = 1 \otimes 1 \otimes 1 \otimes \tau$ in the notation of V15.9. (ii) With $P_\pm = \frac{1}{2}(1 \pm W)$ and $q(U, V) = P_+ \otimes U + P_- \otimes V$ for $r \times r$ matrices U, V: every matrix on $\mathbb{R}^{16r}$ that commutes with $J_1, \dots, J_7$ (V15.41) is $q(U, V)$ for some U, V. (iii) $\Omega_8(16r)$, for the structures of V15.41, is homeomorphic to $O(r)$: J corresponds to

the U with $-J_8 J = q(U, U^{\mathsf{T}})$, and U to $J_8\, q(U, U^{\mathsf{T}})$; the base point J_8 corresponds to I. *Book:* pp. 141–142, Lemma 24.6 (8): $\Omega_8(16r)$ can be identified with the set of all real isometries from X_1 to X_2, through $J_7 J_8$; "Thus we see that $\Omega_8(16r)$ is diffeomorphic to the orthogonal group $O(r)$". *Hidden hypotheses:* $\dim X_1 = \dim X_2 = r$ (V15.12), and that the isometry "determines J_8 uniquely". *Deviation (restated):* Milnor's X_1, X_2 are replaced by the commutant of $J_1, \dots, J_7$: for $J \in \Omega_8(16r)$, $-J_8 J$ commutes with $J_1, \dots, J_7$, and the conditions on $q(U, V)$ are that U be orthogonal and $V = U^{\mathsf{T}}$. *Deviation (narrower):* a homeomorphism, not a diffeomorphism, and for the explicit structures. The other parts of Lemma 24.6 remain as in V15.11–V15.12. *Proof of (i):* the 256 Kronecker words in $1, \sigma, \tau, \kappa$ form a basis of the 16×16 matrices, each word commutes or anticommutes with each J'_α, and a finite check shows that only 1 and W commute with all seven.

V15.43 (Bott's chain at $n = 16r$; `Morse.bott_chain`, `Morse.bott_chain_of_le`, `Morse.bott_periodicity_finit` `Morse.bott_periodicity_finite_of_le`). For the structures of V15.41, every $i \geq 0$ and every $r \geq 2i + 7$, there are bijections $\pi_i(\Omega_8(16r), J_8) \cong \pi_{i+8}(O(16r), I)$ and $\pi_i(O(r), I) \cong \pi_{i+8}(O(16r), I)$. *Book:* p. 138, Theorem 24.5: $\pi_h O \cong \pi_{h-1}\Omega_1 \cong \pi_{h-2}\Omega_2 \cong \cdots \cong \pi_1 \Omega_{h-1}$; p. 142: "Combining this fact with Theorem 24.5", for $\Omega_8 \cong O$. *Hidden hypotheses:* the base points along the chain; for $k = 2, 6$ the base point J_{k+1} must lie in the class $c = 0$ (V15.41 (iv)). *Deviation (restated):* eight steps at the finite stage $n = 16r$, each at the base points: V15.31 for $k = 7, 5, 4, 3, 1$; V15.39 with $c = 0$ for $k = 6$ and $k = 2$, used in degrees ≥ 1 only; V15.32 for $k = 0$; then V15.42. The ranges they need, $i + 2 \leq r - 1$, $i + 3 \leq \lfloor r/2 \rfloor$, $i + 6 \leq 2r - 1$, $i + 7 \leq r$, $i + 8 \leq 4r - 1$ and $i + 11 \leq 16r$, all follow from $2i + 7 \leq r$. Bijections of sets.

V15.44 (stability of the homotopy groups of $O(n)$ and $U(n)$; `Morse.Homotopy.sphereConnected_real`, `Morse.Homotopy.nonempty_pi_unitary_equiv_of_le`, `Morse.Homotopy.nonempty_pi_unitary_equiv_stable`). Let $\mathbb{K} = \mathbb{R}$ with $d = 1$, or $\mathbb{K} = \mathbb{C}$ with $d = 2$, and let $U_{\mathbb{K}}(m)$ be $O(m)$, resp. $U(m)$. If $i + 2 \leq dm + d - 1$ and $i + 2 \leq dm' + d - 1$, there is a bijection $\pi_i(U_{\mathbb{K}}(m), I) \cong \pi_i(U_{\mathbb{K}}(m'), I)$. So $\pi_i O(n) \cong \pi_i O(n')$ whenever $i + 2 \leq n$ and $i + 2 \leq n'$, and $\pi_i U(n) \cong \pi_i U(n')$ whenever $i < 2n$ and $i < 2n'$. *Book:* for U, pp. 128–129, proof of Lemma 23.4: $\pi_{i-1}U(m) \cong \pi_{i-1}U(m+1)$ for $i \leq 2m$, from the fibration $U(m) \to U(m+1) \to S^{2m+1}$. For O the book has no such statement: $\pi_i O$ is the homotopy group of the direct limit (p. 138). *Hidden hypotheses:* for O, the fibration $O(n) \to O(n+1) \to S^n$ and $\pi_i S^n = 0$ for $i < n$. *Deviation (not in the book, for O):* it is what allows Theorem 24.7 to be stated without the limit (V15.45). The proof is that of V14.25, over $\mathbb{R}$ and $\mathbb{C}$ at once.

V15.45 (Theorem 24.7 in the stable range; `Morse.theorem_24_7`). For every $i \geq 0$, $n \geq i + 2$ and $n' \geq i + 10$ there is a bijection $\pi_i(O(n), I) \cong \pi_{i+8}(O(n'), I)$. *Book:* p. 142, Theorem 24.7 (Bott): the infinite orthogonal group O has the same homotopy type as its own 8-th loop space; hence $\pi_i O$ is isomorphic to $\pi_{i+8} O$ for $i \geq 0$. *Hidden hypotheses:* for the homotopy equivalence, that Ω_{k+1} and $\Omega\Omega_k$ have the homotopy type of a CW-complex (p. 146, "it can be shown"). *Deviation (narrower):* a bijection between homotopy groups of finite orthogonal groups in the stable range, where $\pi_i O(n)$ does not depend on $n \geq i + 2$ (V15.44) and is Milnor's $\pi_i O$. The limit group O, the homotopy equivalence $O \simeq \Omega^8 O$, the statement that the bijection is a group isomorphism, and the table of $\pi_i O$ (p. 142) are not formalized. *Proof:* with $r = 2i + 7$, $\pi_i O(n) \cong \pi_i O(r) \cong \pi_{i+8} O(16r) \cong \pi_{i+8} O(n')$, by V15.44, V15.43 and V15.44.

V15.46 (the periodicity theorem for the unitary group in the stable range; `Morse.periodicity_theorem_stable`). For $i < 2n$ and $i + 2 < 2n'$ there is a bijection $\pi_i(U(n), I) \cong \pi_{i+2}(U(n'), I)$. *Book:* p. 129, the Periodicity Theorem of §23: $\pi_{i-1}U \cong \pi_{i+1}U$ for $i \geq 1$, the stable groups being $\pi_{i-1}U = \pi_{i-1}U(m)$ for $i \leq 2m$. *Hidden hypotheses:* none. *Deviation (restated):* V14.28 gives $\pi_{i-1}U(m) \cong \pi_{i+1}U(2m)$ for $1 \leq i \leq 2m$; with the stability of V15.44 on both sides the statement no longer refers to a

single m, and holds between any two groups in the stable range. A bijection of sets; the group isomorphisms of Chapter 14 are V14.28 and V14.30.

What is not verified in this chapter

- **The limit spaces and the homotopy equivalences** (pp. 138, 142, 146): the direct limits Ω_k and O with their topology; the limit maps $\Omega_{k+1} \to \Omega\Omega_k$; that these spaces have the homotopy type of CW-complexes; Whitehead's theorem (in Mathlib only in an abstract model-category form); hence Theorem 24.5 as a homotopy equivalence and Theorem 24.7 as $O \simeq \Omega^8 O$. The naturality under stabilization of the bijections of V15.31 and V15.39 is not formalized either; Theorem 24.7 is obtained in the stable range instead (V15.45). - **Group structures:** the bijections of V15.31, V15.32, V15.39, V15.43 and V15.45 are not shown to be homomorphisms. - **The table** of $\pi_i O$ and $\pi_i Sp$ (p. 142), which Milnor leaves to the reader, and the statements about Sp on that page. - **Lemma 24.4 beyond V15.23, V15.25 and V15.26:** $\Omega_k(n)$ as a submanifold of $O(n)$ (each stratum is a submanifold of $M_n(\mathbb{R})$); that the reflection is an isometry and the components are symmetric spaces (V15.5 is the algebraic part). Of Lemma 24.1: the block normal form of a skew matrix (p. 134), which is not needed, and the Remark $\Omega_1(n) \cong O(n)/U(n/2)$ (p. 133). - **The rest of Lemma 24.6** (pp. 139–142): the identifications of $\Omega_2(n)$ with $U(n/2)/Sp(n/4)$ and of $\Omega_3, \dots, \Omega_7(16r)$ with Grassmannians, $Sp(2r)$, $Sp(2r)/U(2r)$ and $U(2r)/O(2r)$, the Assertion on the largest component of $\Omega_7(16r)$, and the footnote $\Omega_k(16r) \cong \Omega_{k-8}(r)$ for $k > 8$ (p. 142). V15.11–V15.12 are their algebraic parts; (8) is V15.42. - **The exact index of a geodesic:** the eigenvalues of K_A on $\Omega_k(n)$ (which need the curvature of a totally geodesic submanifold and Theorem 20.5 there) and the formula of Lemma 24.2 for the geodesics of $O(2m)$. For geodesics only the lower bounds V15.30 and V15.38 are proved; V15.3, V15.19 and V15.20 remain counts from Milnor's spectra. - **Theorem 24.5 for $k = 2, 6$ at a fixed n beyond the classes** $|c| \leq n/m_{k+1} - 8$ (V15.40); and Milnor's remark that $\Omega_k(n)$ has an infinite cyclic fundamental group (p. 146), which is not needed. - **Lemma 24.8 beyond V15.16–V15.17:** the two non-isomorphic classes for $k = 3$ and $k = 7$, the cases $k > 8$ and the relation $m_{k+8} = 16m_k$ as a theorem, and the isometric form of the isomorphism; the Hurwitz–Radon count and Adams's theorem (p. 144).

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17. Appendix (Milnor pp. 149–153): partly verified

Milnor's Appendix gives an alternative final step for the proof of Theorem 17.3. For subspaces $X_0 \subseteq X_1 \subseteq \cdots$ of a space X it defines the telescope $X_\Sigma \subseteq X \times \mathbb{R}$ and calls X the homotopy direct limit of $\{X_i\}$ if the projection $X_\Sigma \to X$ is a homotopy equivalence (p. 149). Examples 1–3 (pp. 149–150) give two sufficient conditions (the interiors of the X_i cover a paracompact X; the X_i are subcomplexes of a CW-complex) and a counterexample. Theorem A (p. 150) says that a map of homotopy direct limits that carries each X_i into Y_i by a homotopy equivalence is a homotopy equivalence; it is proved by reduction to the map of telescopes, in two cases (pp. 150–153), and gives the final step of 17.3. The Corollary (p. 153) says that a homotopy direct limit of spaces of the homotopy type of CW-complexes has that homotopy type. Verified, for arbitrary topological spaces: telescopes, homotopy direct limits, and the facts Milnor uses about them without comment ((T1)–(T3) and Lemmas 2.1–2.2 of the exposition); Examples 1 and 3; Milnor's proof of Theorem A in both cases, with the misprint $(4 - 3t)u$ corrected to $(4t - 3)u$; **Theorem A exactly as Milnor states it, with no closedness hypothesis** (VA.15); the Corollary, with a mapping telescope in place of a CW-complex; and the topological part of Theorem 17.3, for a space exhausted by sublevel

sets of a continuous function. The proof departs from Milnor's in one respect. **Milnor's maps are defined strip by strip**, which makes them continuous only when the sets are closed (T3): Case 1 as written needs the X_i closed; Case 2 applies Case 1 only to truncations of telescopes, which are always closed, and pastes only one map, G, on the telescope of the Y_i, so his proof of Theorem A needs only the Y_i closed (VA.12); and the same proof run on mapping telescopes, where pasting is automatic, needs no closedness at all (VA.15). **Not verified:** Example 2 (Whitehead's theorem), a counterexample for the Remark, and the CW structure of mapping telescopes (cellular approximation), so that "CW-complex" becomes a mapping telescope of iterated adjunction spaces (the list at the end). Lean: `Morse/Appendix/` — `Telescope.lean` (630 lines), `Examples.lean` (239), `TheoremA.lean` (765), `MappingTelescope.lean` (517), `MonotoneUnion.lean` (569).

Conventions. Spaces are arbitrary topological spaces, with no separation axiom unless one is stated; maps are continuous; subsets carry the subspace topology; $I = [0,1]$. Sequences are indexed from 0. A sequence of subsets $X_0, X_1, \ldots$ of X is *monotone* if $X_0 \subseteq X_1 \subseteq \cdots$; Milnor's sequences always are, and most statements below assume it. Homotopies of maps into a subset take values in that subset.

VA.1 (homotopy equivalences; `Morse.IsHomotopyEquiv`, `Morse.IsHomotopyEquiv.comp`, `Morse.IsHomotopyEquiv.of_homotopic`, `Morse.IsHomotopyEquiv.of_inverse`, `Morse.IsHomotopyEquiv.of_le` `Morse.restrictTo`). A map $f : X \to Y$ is a *homotopy equivalence* if there is a map $g : Y \to X$ with $gf \simeq \mathrm{id}_X$ and $fg \simeq \mathrm{id}_Y$. A composite of homotopy equivalences is one; a map homotopic to a homotopy equivalence is one; a homotopy inverse is one; and a map with a left homotopy inverse and a right homotopy inverse is a homotopy equivalence. If f carries $A \subseteq X$ into $B \subseteq Y$, $f|A : A \to B$ is its restriction. *Book:* p. 22, Assertion: if a map F has a left homotopy inverse L and a right homotopy inverse R, then F is a homotopy equivalence; p. 153 refers to it ("compare page 22"). *Hidden hypotheses:* none. *Deviation (not in the book):* the closure properties, used throughout without comment.

VA.2 (the telescope and homotopy direct limits; `Morse.telescope`, `Morse.mem_telescope_iff`, `Morse.mem_telescope_of_le`, `Morse.stripMap`, `Morse.telescopeProj`, `Morse.IsHomotopyDirectLimit`). For subsets $X_0, X_1, \ldots$ of X, the *telescope* is $X_\Sigma = \bigcup_n X_n \times [n, n+1] \subseteq X \times \mathbb{R}$, with the subspace topology; the n-th *strip* is the map $X_n \times [0,1] \to X_\Sigma$, $(x,t) \mapsto (x, n+t)$; $p : X_\Sigma \to X$ is $p(x,\tau) = x$; and X is the *homotopy direct limit* of $\{X_i\}$ if p is a homotopy equivalence. **(T1)** For a monotone sequence, $(x,\tau) \in X_\Sigma$ iff $x \in X_n$ for some integer $n \geq 0$ with $n \leq \tau$; so $(x,\tau) \in X_\Sigma$ and $\tau' \geq \tau$ give $(x,\tau') \in X_\Sigma$. *Book:* p. 149: for subspaces $X_0 \subset X_1 \subset \cdots$, the infinite union $X_\Sigma = X_0 \times [0,1] \cup X_1 \times [1,2] \cup \cdots$, "to be topologized as a subset of $X \times \mathbb{R}$"; Definition: X is the homotopy direct limit of $\{X_i\}$ if $p : X_\Sigma \to X$, $p(x,\tau) = x$, is a homotopy equivalence. *Hidden hypotheses:* none. *Deviation (not in the book):* (T1), used without comment.

VA.3 ((T3) pasting along the strips; `Morse.continuous_of_stripMap`, `Morse.telescopeLift`, `Morse.telescopeDesc`, `Morse.telescopeHomotopy`). If every X_n is closed in X, then for every space P a function F on $P \times X_\Sigma$ is continuous as soon as each $(p,x,t) \mapsto F(p,(x,n+t))$ is continuous on $P \times X_n \times [0,1]$. Hence, for a monotone sequence of closed sets, maps $\Phi_n : P \times X_n \times [0,1] \to Z$ with $\Phi_{n+1}(p,x,0) = \Phi_n(p,x,1)$ define a continuous map $P \times X_\Sigma \to Z$, $(p,(x,n+t)) \mapsto \Phi_n(p,x,t)$; in particular maps $X_\Sigma \to Z$, and homotopies between them, may be given strip by strip. *Book:* every map and homotopy on a telescope in the proof of Theorem A (pp. 151–153) is given strip by strip, by formulas in $(x, n+t)$, $0 \leq t \leq 1$. *Hidden hypotheses:* closedness of the X_n, which Milnor never states; then the strips are closed in X_Σ and form a locally finite family. *Deviation (not in the book):* stated and proved as a lemma.

VA.4 (the map of telescopes, and "clearly sufficient"; `Morse.telescopeMap`, `Morse.telescopeProj_comp_telesc` `Morse.IsHomotopyEquiv.of_telescopeMap`). If $f : X \to Y$ carries each X_n into Y_n, then $f_\Sigma(x,\tau) = (f(x),\tau)$ is a map $X_\Sigma \to Y_\Sigma$ with $p_Y f_\Sigma = f p_X$. If X and Y are the homotopy direct limits of $\{X_i\}$ and $\{Y_i\}$ and f_Σ is a homotopy equivalence, then so is f. *Book:* p. 150: define $f_\Sigma(x,t) = (f(x),t)$; "it is clearly sufficient to prove that f_Σ is a homotopy equivalence". *Hidden hypotheses:* none; neither closedness nor monotonicity is needed. *Proof:* $f \simeq f p_X s = p_Y f_\Sigma s$ for a homotopy inverse s of p_X.

VA.5 ((T2) truncations; `Morse.truncation`, `Morse.isClosed_truncation`, `Morse.monotone_truncation`, `Morse.truncIncl`, `Morse.truncProj`, `Morse.isHomotopyEquiv_truncIncl`, `Morse.isHomotopyEquiv_truncProj` `Morse.level_le_of_stripMap`). The truncation $X^n_\Sigma = \{(x,\tau) \in X_\Sigma : \tau \le n\}$ is closed in X_Σ, and $X^0_\Sigma \subseteq X^1_\Sigma \subseteq \cdots$, whatever the X_i. For a monotone sequence, $x \mapsto (x,n)$ is a homotopy equivalence $X_n \to X^n_\Sigma$ with homotopy inverse $(x,\tau) \mapsto x$, the homotopy raising τ to n; and a function F on X_Σ that sends the n-th strip to levels $\le n+1$ and $X_n \times \{n\}$ to levels $\le n$ sends each X^m_Σ to levels $\le m$. *Book:* p. 152: $X^n_\Sigma = X_0 \times [0,1] \cup \cdots \cup X_{n-1} \times [n-1,n] \cup X_n \times [n]$, which "contains $X_n \times [n]$ as deformation retract"; $G f_\Sigma$ carries X^n_Σ into itself. *Hidden hypotheses:* none; no closedness. *Deviation (not in the book):* proofs of the two facts Milnor states.

VA.6 (Lemmas 2.1 and 2.2 of the exposition; `Morse.isHomotopyDirectLimit_of_level`, `Morse.isHomotopyDirectLimit_truncation`). (i) For a monotone sequence, if $f : X \to \mathbb{R}$ is continuous, $f \ge 0$, and $f(x) \ge n+1$ whenever $x \notin X_n$, then X is the homotopy direct limit of $\{X_i\}$. (ii) For any sequence, X_Σ is the homotopy direct limit of its truncations $\{X^n_\Sigma\}$. *Book:* p. 149, in Example 1: $x \mapsto (x, f(x))$ maps X homeomorphically onto a subset of X_Σ "which is clearly a deformation retract"; (ii) is used without mention at the end of Case 2 (p. 153). *Hidden hypotheses:* (ii), for Case 2 (VA.11). *Deviation (not in the book):* (i) needs no paracompactness once f is given. *Proof:* (i) the homotopy $(x, (1-s)f(x) + s\tau)$ stays in X_Σ by (T1); (ii) is (i) with $f(x,\tau) = \tau + 1$.

VA.7 (Example 1, and sublevel sets; `Morse.example1`, `Morse.isHomotopyDirectLimit_sublevel`). (i) Let X be paracompact and Hausdorff, and $X_0 \subseteq X_1 \subseteq \cdots$ with every point of X in the interior of some X_i. Then X is the homotopy direct limit of $\{X_i\}$. (ii) Let $E : X \to \mathbb{R}$ be continuous, on any space X, and $a_0 \le a_1 \le \cdots$ with $a_i \to \infty$. Then X is the homotopy direct limit of the sublevel sets $\{x : E(x) \le a_i\}$. *Book:* p. 149, Example 1: if each point of X lies in the interior of some X_i and X is paracompact, a partition of unity gives $f : X \to \mathbb{R}$ with $f(x) \ge i+1$ for $x \notin X_i$ and $f \ge 0$, and X is a homotopy direct limit. *Hidden hypotheses:* Hausdorff: Mathlib's partitions of unity need a normal space, and Milnor's "paracompact" is read with the Hausdorff axiom. *Deviation (stronger):* (ii) needs no paracompactness; it covers the sublevel sets of the energy in Theorem 17.3, where Milnor uses Example 1 and the paracompactness of a metric space. *Proof:* (i) $f = \sum_i i\,\varphi_i$ for a partition of unity $\{\varphi_i\}$ subordinate to $\{\operatorname{int} X_i\}$, then VA.6 (i); (ii) $f = \sum_j \min(1, \max(0, E - a_j + 1))$, locally a finite sum.

VA.8 (Example 3; `Morse.example3Set`, `Morse.monotone_example3Set`, `Morse.isClosed_example3Set`, `Morse.iUnion_example3Set`, `Morse.example3`). Let $X_n = \{0\} \cup [\frac{1}{n+1}, 1] \subseteq [0,1]$. The X_n are closed and increasing, with union $[0,1]$, and $[0,1]$ is not the homotopy direct limit of $\{X_n\}$. *Book:* p. 150, Example 3: the unit interval $[0,1]$ is not the homotopy direct limit of the sequence of closed subsets $[0] \cup [1/i, 1]$; no proof. *Hidden hypotheses:* none. *Deviation (restated):* indexed from 0. *Proof:* $\{0\} \times [0,\infty)$ is open and closed in X_Σ, so X_Σ is not connected, while a homotopy inverse s of p would join $(0,0)$ to $s(0)$, $s(0)$ to $s(1)$ and $s(1)$ to $(1,0)$ by paths.

VA.9 (Milnor's maps h_1 and their inverse g; `Morse.reparam`, `Morse.reparam_homotopic`, `Morse.scaleMap`, `Morse.IsMilnorMap`, `Morse.IsMilnorMap.fix`, `Morse.milnorInv`, `Morse.loopFun`,

`Morse.milnorH`, `Morse.milnorK`, `Morse.isHomotopyEquiv_of_milnorMap`). Let $X_0 \subseteq X_1 \subseteq \cdots$ be closed in X. (i) For a map $\rho : I \to I$ with $\rho(0) = 0$ and $\rho(1) = 1$, $(x, n+t) \mapsto (x, n + \rho(t))$ is homotopic to the identity of X_Σ; in particular so is $R_c(x, n+t) = (x, n + \min(ct, 1))$, $c \geq 1$. (ii) Let $\psi : X_\Sigma \to X_\Sigma$ be a map with $\psi(x, n+t) = (x, n+2t)$ for $t \leq \frac{1}{2}$ and $\psi(x, n+t)$ at level $n+1$ for $t \geq \frac{1}{2}$. Then ψ fixes the points at integer levels, and it is a homotopy equivalence, with homotopy inverse Milnor's g: $g(x, n+t) = (x, n+2t)$ for $t \leq \frac{1}{2}$ and $\psi(x, n + \frac{3}{2} - t)$ for $t \geq \frac{1}{2}$. *Book:* p. 151: any map h_1 with $h_1(x, n+t) = (x, n+2t)$ for $t \leq \frac{1}{2}$ and $h_1(x, n+t) \in X_{n+1} \times [n+1]$ for $t \geq \frac{1}{2}$ is a homotopy equivalence, with homotopy inverse g, well defined since $h_1(x, n+\frac{1}{2}) = h_1(x, n+1) = (x, n+1)$; $h_1 g \simeq \mathrm{id}$ by a homotopy H_u, $0 \leq u \leq \frac{1}{2}$, ending at $H_{1/2}(x, n+t) = (x, n + \min(4t, 1))$, "clearly" homotopic to the identity; p. 152: $gh_1 \simeq \mathrm{id}$ "by a completely analogous argument". *Hidden hypotheses:* the X_n closed (VA.3); the formula for $h_1 g$ uses that h_1 fixes the integer levels. *Proof:* with u rescaled to $[0, 1]$, Milnor's H_u is the single formula $(x, n+4t)$ for $t \leq \frac{1}{4}$ and $\psi(x, n + \frac{1}{2} + \frac{\mu}{2})$ for $t \geq \frac{1}{4}$, $\mu = \min(4t-1, 1-u, 2-2t)$; the analogous homotopy for $g\psi$ takes $\mu = \max(2-4t, 2t-1, u)$; both end at R_4, homotopic to the identity by (i).

VA.10 (Case 1 of Theorem A; `Morse.case1Fun`, `Morse.case1Family`, `Morse.case1Map_zero`, `Morse.case1Map_one`, `Morse.theoremA_case1`**).** Let $X_0 \subseteq X_1 \subseteq \cdots$ be closed in X, and $\varphi : X \to X$ a map carrying each X_n into itself by a map φ_n homotopic, in X_n, to the identity of X_n. Then φ_Σ is a homotopy equivalence. The steps: with homotopies h^n_u from φ_n to the identity, $h_u(x, n+t)$ is $(h^n_u(x), n+2t)$ for $t \leq \frac{1}{2}$, $(h^n_{(3-4t)u}(x), n+1)$ for $\frac{1}{2} \leq t \leq \frac{3}{4}$ and $(h^{n+1}_{(4t-3)u}(x), n+1)$ for $\frac{3}{4} \leq t \leq 1$; $(u, z) \mapsto h_u(z)$ is continuous; $h_0 = \varphi_\Sigma \circ R_2$; and h_1 is a map as in VA.9 (ii). *Book:* pp. 150–152, Case 1: $X_i = Y_i$ and each $f_i : X_i \to Y_i$ homotopic to the identity; the third line of h_u is printed $h^{n+1}_{(4-3t)u}$; "taking $u = 0$ this defines a map h_0 which is clearly homotopic to f_Σ". *Hidden hypotheses:* the X_n closed, for the continuity of h_u and of g (VA.3); without it Milnor's g can be discontinuous (the exposition's example: $X = [0, 1]$, $X_0 = (0, 1]$). This part does not use that X is a homotopy direct limit. *Deviation (value differs):* the third line is $h^{n+1}_{(4t-3)u}$; the printed $(4-3t)u$ is a misprint, giving $h^{n+1}_{7u/4}$ at $t = \frac{3}{4}$ where continuity needs h^{n+1}_0. *Proof:* Milnor's, with $h_0 \simeq \varphi_\Sigma$ by VA.9 (i).

VA.11 (Case 2: f_Σ is a homotopy equivalence; `Morse.isHomotopyEquiv_of_truncation`, `Morse.homotopic_square`, `Morse.case2Inv`, `Morse.case2Inv_level`, `Morse.theoremA_telescope`**).** (i) Let $X_0 \subseteq X_1 \subseteq \cdots$ be a monotone sequence, closed or not, and $F : X_\Sigma \to X_\Sigma$ a map carrying each truncation X^m_Σ into itself, with $F(x, m) = (e_m(x), m)$ for maps $e_m : X_m \to X_m$ homotopic to the identity. Then F is a homotopy equivalence. (ii) Let $X_0 \subseteq X_1 \subseteq \cdots$ in X and $Y_0 \subseteq Y_1 \subseteq \cdots$ in Y, the Y_n closed, and let $f : X \to Y$ carry each X_n into Y_n by a homotopy equivalence f_n. Then $f_\Sigma : X_\Sigma \to Y_\Sigma$ is a homotopy equivalence. *Book:* pp. 152–153, Case 2: "now let X and Y be arbitrary"; with homotopy inverses g_n of the f_n, $i_n g_n \simeq g_{n+1} f_{n+1} i_n g_n = g_{n+1} j_n f_n g_n \simeq g_{n+1} j_n$; with a homotopy h^n_u from $i_n g_n$ to $g_{n+1} j_n$, $G(y, n+t) = (g_n(y), n+2t)$ for $t \leq \frac{1}{2}$ and $(h^n_{2t-1}(y), n+1)$ for $t \geq \frac{1}{2}$; Gf_Σ carries X^n_Σ into itself by a map homotopic to the identity, so "we can apply Case 1 to the sequence $\{X^n_\Sigma\}$, and conclude that Gf_Σ is a homotopy equivalence"; similarly $f_\Sigma G$; so f_Σ has a left and a right homotopy inverse (VA.1). *Hidden hypotheses:* the Y_n closed, for the continuity of G (VA.3); the return from Case 1, which is about the induced map of the telescope of $\{X^n_\Sigma\}$, to Gf_Σ itself needs X_Σ to be the homotopy direct limit of its truncations (VA.6 (ii)); that Gf_Σ on X^m_Σ is homotopic to the identity is the chain $\varphi \simeq \varphi \iota r = \iota(g_m f_m) r \simeq \iota r \simeq \mathrm{id}$, with ι, r the maps of (T2) (VA.5). *Deviation (narrower):* the Y_n closed, where Milnor's Case 2 has no hypothesis. *Deviation (stronger):* no homotopy direct limit hypothesis, and no closedness of the

X_n: Case 1 is applied only to the truncations X_Σ^m and Y_Σ^m, which are always closed (VA.5).

VA.12 (Theorem A, by Milnor's proof; `Morse.theoremA`). Let X be the homotopy direct limit of $X_0 \subseteq X_1 \subseteq \cdots$ and Y that of $Y_0 \subseteq Y_1 \subseteq \cdots$, the Y_i closed in Y, and let $f : X \to Y$ carry each X_i into Y_i by a homotopy equivalence $f_i = f|X_i : X_i \to Y_i$. Then f is a homotopy equivalence. *Book:* p. 150, Theorem A, with no closedness hypothesis. *Hidden hypotheses:* Milnor's proof needs the Y_i closed (VA.11), and only these. *Deviation (narrower):* the Y_i closed; this is removed in VA.15. *Proof:* VA.11 (ii) and VA.4.

VA.13 (the mapping telescope; `Morse.MapTelRel`, `Morse.MappingTelescope`, `Morse.MappingTelescope.mk`, `Morse.MappingTelescope.mk_one`, `Morse.MappingTelescope.desc`, `Morse.MappingTelescope.lift`, `Morse.MappingTelescope.homotopy`, `Morse.MappingTelescope.level`, `Morse.MappingTelescope.trunc`, `Morse.MappingTelescope.isClosed_trunc`, `Morse.MappingTelescope.isHomotopyDirectLimit_trunc`, `Morse.MappingTelescope.push`, `Morse.MappingTelescope.mk_zero_injective`, `Morse.MappingTelescope.isI` `Morse.MappingTelescope.levelIncl`, `Morse.MappingTelescope.levelRetract`, `Morse.MappingTelescope.isH`). For a sequence of spaces and maps $K_0 \xrightarrow{c_0} K_1 \xrightarrow{c_1} \cdots$, the *mapping telescope* T is the quotient of the disjoint union of the blocks $K_n \times [0,1]$ by $(k,1)_n \sim (c_n(k),0)_{n+1}$, with the quotient topology; $(k,t)_n$ is at level $n+t$, a continuous function on T. Maps out of T, and out of $P \times T$ for a locally compact P (for homotopies, $P = I$), may be given block by block, compatibly with the gluing, with no closedness condition. The truncations $T_n = \{\text{level} \le n\}$ are closed and increasing, and T is the homotopy direct limit of $\{T_n\}$; $k \mapsto (k,0)_n$ is an embedding $K_n \to T$ and a homotopy equivalence $K_n \to T_n$. *Book:* not in the book; the exposition's proof of the Corollary uses it. *Hidden hypotheses:* none. *Deviation (not in the book):* the device that removes the closedness hypotheses (VA.15) and replaces a union of complexes that are not subspaces of one space (VA.16, VA.17). *Proof:* the product of a quotient map with a locally compact space is a quotient map; the homotopy inverse of $K_n \to T_n$ reads off the point reached by pushing each point of T_n up to level n along the maps c_m, and is continuous because $K_n \to T$ is an embedding; T is a homotopy direct limit by VA.6 (i) with $f = \text{level} + 1$.

VA.14 (the mapping telescope of the inclusions; `Morse.subspaceFunctor`, `Morse.telescopeComparison`, `Morse.telescopeComparisonHomeomorph`, `Morse.isHomotopyEquiv_telescopeComparison`, `Morse.telescopeProj'`, `Morse.isHomotopyEquiv_telescopeProj'`). Let $X_0 \subseteq X_1 \subseteq \cdots$ be subsets of X, not necessarily closed, and T the mapping telescope of the inclusions. The map $T \to X_\Sigma$, $(x,t)_n \mapsto (x, n+t)$, is a homotopy equivalence, and a homeomorphism when the X_n are closed. If X is the homotopy direct limit of $\{X_i\}$, the map $T \to X$, $(x,t)_n \mapsto x$, is a homotopy equivalence. *Book:* not in the book. *Hidden hypotheses:* none. *Deviation (not in the book). Proof:* the map carries T_m into X_Σ^m by a map homotopic to the composite of the homotopy equivalences $T_m \to X_m \to X_\Sigma^m$ (VA.13, VA.5), so VA.12 applies to the closed truncations, with VA.6 (ii) and VA.13; the inverse of the homeomorphism is given strip by strip (VA.3).

VA.15 (Theorem A with no closedness hypothesis; `Morse.mappingTelescopeMap`, `Morse.mapsTo_mappingTelesc` `Morse.theoremA_general`). Let X be the homotopy direct limit of $X_0 \subseteq X_1 \subseteq \cdots$ and Y that of $Y_0 \subseteq Y_1 \subseteq \cdots$, arbitrary subsets, and let $f : X \to Y$ carry each X_i into Y_i by a homotopy equivalence $f_i = f|X_i : X_i \to Y_i$. Then f is a homotopy equivalence. *Book:* p. 150, Theorem A: suppose that X is the homotopy direct limit of $\{X_i\}$ and Y the homotopy direct limit of $\{Y_i\}$, and let $f : X \to Y$ be a map which carries each X_i into Y_i by a homotopy equivalence; then f itself is a homotopy equivalence. *Hidden hypotheses:* none. *Proof:* departs from Milnor's in the space on which his proof is run. The map $(x,t)_n \mapsto (f(x),t)_n$ of the mapping telescopes of the inclusions carries the m-th truncation into the m-th by a map homotopic to the composite of

homotopy equivalences $T_m \to X_m \xrightarrow{f_m} Y_m \to T_m$ (VA.13), and the truncations are closed, so VA.12 (Milnor's proof, Cases 1 and 2) makes it a homotopy equivalence; the mapping telescopes are homotopy equivalent to X and Y by maps commuting with f (VA.14).

VA.16 (monotone unions, and the final step of Theorem 17.3; `Morse.nonempty_homotopyEquiv_telescope`, `Morse.homotopic_inverse_square`, `Morse.compareMap`, `Morse.compareMap_mapsTo`, `Morse.isHomotopyEquiv_c` `Morse.nonempty_homotopyEquiv_mappingTelescope`). (i) If X is the homotopy direct limit of $X_0 \subseteq X_1 \subseteq \cdots$, the $Y_i \subseteq Y$ are closed and increasing, and $f : X \to Y$ carries each X_i into Y_i by a homotopy equivalence, then X is homotopy equivalent to the telescope Y_Σ. (ii) If X is the homotopy direct limit of $X_0 \subseteq X_1 \subseteq \cdots$ (closed or not), $K_0 \xrightarrow{c_0} K_1 \xrightarrow{c_1} \cdots$ is a sequence of spaces and maps, and $g_n : X_n \to K_n$ are homotopy equivalences with $c_n g_n \simeq g_{n+1}$ on X_n, then X is homotopy equivalent to the mapping telescope of the K_n. *Book:* p. 150: in the proof of Theorem 17.3 homotopy equivalences $\Omega^{a_i} \to K_i$ form a commutative diagram with $K_0 \subset K_1 \subset \cdots$; since $\Omega = \bigcup \Omega^{a_i}$ and $K = \bigcup K_i$ are homotopy direct limits (Examples 1 and 2), the limit mapping $\Omega \to K$ is a homotopy equivalence. *Hidden hypotheses:* the limit mapping exists only because each map extends the previous one; K is the homotopy direct limit of $\{K_i\}$ by Example 2, which is not verified. *Deviation (restated):* the conclusion is a homotopy equivalence with Y_Σ, or with the mapping telescope of the K_n, which stands for the union K with the weak topology; the two are homotopy equivalent by Example 2 (Whitehead's theorem). *Deviation (stronger):* in (ii) the squares need only commute up to homotopy, and the K_n need not be subspaces of one space. *Proof:* (ii) Milnor's map G of Case 2, built from homotopy inverses of the g_n and defined from the mapping telescope of the K_n to X_Σ, where it is continuous block by block, is a homotopy equivalence by VA.12 for the closed truncations.

VA.17 (the Corollary; `Morse.corollary_appendix`, `Morse.corollary_cellComplex`). (i) If X is the homotopy direct limit of $X_0 \subseteq X_1 \subseteq \cdots$ and the X_n are homotopy equivalent to spaces K_n, then there are maps $K_0 \to K_1 \to \cdots$ (in the proof $e_{n+1}\iota_n e'_n$, with $e_n : X_n \to K_n$ a homotopy equivalence, e'_n a homotopy inverse and ι_n the inclusion) such that X is homotopy equivalent to their mapping telescope. (ii) If each X_n has the homotopy type of a finite cell complex with cells cells_n (V2.11), X is homotopy equivalent to the mapping telescope of a sequence of iterated adjunction spaces K_n with these cells. No closedness of the X_n is assumed. *Book:* p. 153, Corollary: if X is the homotopy direct limit of $\{X_i\}$ and each X_i has the homotopy type of a CW-complex, then X itself has the homotopy type of a CW-complex; "the proof is not difficult". *Hidden hypotheses:* that a telescope of CW-complexes is a CW-complex needs cellular maps (cellular approximation) and the CW structure of mapping telescopes. *Deviation (narrower):* the conclusion is a mapping telescope of the K_n, not a CW-complex; for finite complexes, iterated adjunction spaces (V2.11) replace CW-complexes. *Proof:* VA.16 (ii).

VA.18 (countable cell complexes; `Morse.IsRelCellAttachment`, `Morse.IsCellAttachment.of_rel`, `Morse.CellTower`, `Morse.CellTower.isCellAttachment`, `Morse.CellTower.ofSequence`, `Morse.IsCountableCellComplexType`, `Morse.IsCountableCellComplexType.of_homotopyEquiv`). A space K is obtained from L by attaching cells of dimensions $(\lambda_1, \dots, \lambda_k)$ if it is obtained by attaching a λ_1-cell to L, then a λ_2-cell, and so on, each along an arbitrary map of its boundary sphere into the space built so far (V1.2); the map $L \to K$ is the composite of the inclusions. A *cell tower* with cells $\text{cells}_0, \text{cells}_1, \dots$ (finite lists of dimensions) is a sequence $K_0 \to K_1 \to \cdots$ in which K_0 is an iterated adjunction space with cells cells_0 (V2.11) and K_{n+1} is obtained from K_n by attaching the cells cells_{n+1}, the map $K_n \to K_{n+1}$ being the inclusion; each K_n is then an iterated adjunction space with the cells $\text{cells}_0, \dots, \text{cells}_n$. A space X *has the homotopy type of a countable cell*

complex with cells $\text{cells}_0, \text{cells}_1, \ldots$ if it is homotopy equivalent to the mapping telescope (VA.13) of such a tower; this is invariant under homotopy equivalence. *Book:* p. 149: each Ω^{a_i} has the homotopy type of "a certain CW-complex", and so should the union; p. 150: $K = \bigcup K_i$. *Hidden hypotheses:* none. *Deviation (narrower):* weaker than a CW-complex: the attaching maps need not be cellular (V2.11), and the mapping telescope of the tower stands for the union of the K_n with the weak topology, to which it is homotopy equivalent by Example 2, not verified.

VA.19 (the topological part of Theorem 17.3; `Morse.isCountableCellComplexType_of_tower`, `Morse.isCountableCellComplexType_of_sublevel`, `Morse.isCountableCellComplexType_of_sublevel'`). (i) If X is the homotopy direct limit of $X_0 \subseteq X_1 \subseteq \cdots$ and $g_n : X_n \to K_n$ are homotopy equivalences to the stages of a cell tower with $c_n g_n \simeq g_{n+1}$ on X_n, then X has the homotopy type of the countable cell complex. (ii) Let $E : X \to \mathbb{R}$ be continuous, $a_0 \le a_1 \le \cdots$ with $a_n \to \infty$, and $g_n : \{E \le a_n\} \to K_n$ homotopy equivalences to the stages of a cell tower (given as a tower, or by its stages K_n and the inclusions $c_n : K_n \to K_{n+1}$ attaching the cells cells_{n+1}) with $c_n g_n = g_{n+1}$ on $\{E \le a_n\}$. Then X has the homotopy type of the countable cell complex with cells $\text{cells}_0, \text{cells}_1, \ldots$. *Book:* p. 150, the alternative final step of Theorem 17.3 (VA.16). *Hidden hypotheses:* $a_i \to \infty$, so that the sublevel sets exhaust X; each g_{n+1} extends g_n, for the limit map. *Deviation (stronger):* no paracompactness (VA.7 (ii)). *Deviation (restated):* "CW-complex" in the sense of VA.18. *Proof:* VA.16 (ii), with VA.7 (ii) in (ii).

What is not verified in this chapter

- **Example 2** (p. 149): a CW-complex is the homotopy direct limit of subcomplexes with union X. It needs Whitehead's theorem, the fact that compact subsets of a CW-complex lie in finite subcomplexes, and the CW topology of products (Lemma 2.3 and Facts (a)–(c) of the exposition), none of which is in Mathlib. Hence the mapping telescope of a cell tower is not compared with the union of its stages in the weak topology (VA.16, VA.18).
- **The Remark** (p. 150): that under the hypotheses of Case 1 f_Σ need not be homotopic to the identity ("counterexamples can be given"). The exposition's phantom-map counterexample needs cellular cohomology and $\mathbb{CP}^\infty$; it is not formalized.
- **The CW structure of mapping telescopes**: the Corollary's conclusion that X has the homotopy type of a CW-complex needs cellular approximation, to make the maps $K_n \to K_{n+1}$ cellular, and the CW structure of the mapping telescope of cellular maps; neither is in Mathlib, so the conclusion is a mapping telescope (VA.17).
- **The application to the path space** (p. 150) is not part of this chapter: the sublevel sets Ω^{a_i}, Milnor's metric and the tower of complexes are verified in Chapter 10, where VA.19 gives Theorem 17.3 (V10.20–V10.22) and the homology of $\Omega(S^n; p, q)$ is computed level by level (V10.25–V10.26).

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18. Sign-off

The statements below are the ones the reader is asked to confirm say what Milnor means. Each is a summary of the V-item it names; the standing hypotheses of that item are not repeated.

1. (V1.2) $X \cup_\varphi e^n$ is the quotient of $X \sqcup D^n$ identifying each $u \in S^{n-1}$ with $\varphi(u)$, that is, the pushout of $X \leftarrow S^{n-1} \to D^n$.
2. (V1.3) In coordinates, a critical point is a zero of the derivative, the Hessian is the second

derivative, the index is the largest dimension of a subspace on which the Hessian is negative definite, the nullity is the dimension of its null space, and non-degenerate means nullity 0, equivalently a non-singular matrix of second partial derivatives.

3. (V1.4) On a manifold, with no Hausdorff hypothesis, the same notions are read in the preferred chart at p, critical points agree with those of the companion volume, and a Morse function is a smooth function all of whose critical points are non-degenerate.
4. (V1.5) At a critical point of a function C^2 there, criticality, the Hessian (transformed as a bilinear form by the derivative of the change of coordinates), non-degeneracy, index and nullity do not depend on the coordinate system.
5. (V1.6) For $f = F|N$ on a slice submanifold N of V, a point q is critical exactly when dF_q vanishes on T_qN (for an inner product, when $\nabla F(q) \perp T_qN$), the Hessian, index and nullity can be computed in any slice parametrization, and for L_p Milnor's criterion and Hessian formula of pp. 35–36 hold.
6. (V1.7) A function smooth on an open convex neighbourhood V of 0 in $\mathbb{R}^n$ with $f(0) = 0$ is $\sum_i x_i g_i(x)$ with g_i smooth on V and $g_i(0) = \partial f/\partial x_i(0)$; on an open star-shaped V in any normed space it is $L(x)x$ with L smooth and $L(0) = df_0$, and if also $df_0 = 0$ it is $B(x)(x,x)$ with B smooth and symmetric and $B(0) = \frac{1}{2}d^2f_0$.
7. (V1.8) Near a non-degenerate critical point p of index λ of a function smooth near p, on $\mathbb{R}^n$ or on any smooth manifold, there are coordinates with $f = f(p) - \sum_{i\le\lambda}(y^i)^2 + \sum_{i>\lambda}(y^i)^2$.
8. (V1.9) If f has such a normal form with k minus signs in some coordinate system around p, then p is a non-degenerate critical point of index k.
9. (V1.10) A non-degenerate critical point of a function C^2 at that point has a neighbourhood containing no other critical point.
10. (V1.11) A C^2 function whose critical points are all non-degenerate has only finitely many critical points in each compact set.
11. (V1.12) The origin is the only critical point of x^3 and of $x^3 - 3xy^2$, and is degenerate; the critical points of x^2 (the y axis) and of x^2y^2 (the two axes) are all degenerate.
12. (V1.13) A one-parameter group of diffeomorphisms of E is a jointly smooth φ with each φ_t a diffeomorphism and $\varphi_{t+s} = \varphi_t \circ \varphi_s$; X generates it when $X(x)$ is the velocity of $t \mapsto \varphi_t(x)$ at $t = 0$, and then its orbits are integral curves of X.
13. (V1.14) A C^n ($1 \le n \le \infty$) compactly supported vector field on a Banach space has a global flow, C^n in (t,x), each φ_t being a C^n diffeomorphism.
14. (V1.15) A smooth vector field on E vanishing outside a compact set generates exactly one one-parameter group of diffeomorphisms of E.
15. (V2.1) A is a strong deformation retract of X if some homotopy from the identity of X to a map into A fixes A at all times; the inclusion of such an A is a homotopy equivalence.
16. (V2.2) On $\mathbb{R}^n$ with an inner product, if $f^{-1}[a,b]$ is compact and free of critical points, a diffeomorphism of $\mathbb{R}^n$ carries M^a onto M^b, and M^a is a strong deformation retract of M^b.
17. (V2.3) The same holds for $f = F|N$ on a slice submanifold $N \subseteq \mathbb{R}^K$, closed or not, with a diffeomorphism of N.
18. (V2.4) The same holds on a compact Hausdorff manifold, where $f^{-1}[a,b]$ need only be free of critical points.
19. (V2.5) On $\mathbb{C}\setminus\{i\}$ with $f = |z|^2$, the set $f^{-1}[\frac{1}{4},4]$ contains no critical points but is not compact, and $M^{1/4}$ and M^4 are not homotopy equivalent.
20. (V2.6) On $\mathbb{R}^n$, under the hypotheses of Theorem 3.2 and for all small ϵ, $M^{c+\epsilon}$ strongly deformation retracts onto $M^{c-\epsilon}$ together with an embedded λ-cell meeting $M^{c-\epsilon}$ in its boundary sphere, and so is homotopy equivalent to $M^{c-\epsilon}$ with a λ-cell attached.

21. (V2.7) In a Hausdorff space, a closed set A together with an embedded closed disk meeting A exactly in its boundary sphere is A with a cell attached along the restriction of the embedding to the sphere.
22. (V2.8) Theorem 3.2 holds for $f = F|N$ on a slice submanifold $N \subseteq \mathbb{R}^K$, and its topological half (given Assertion 3) holds in any Morse coordinate patch of a Hausdorff space.
23. (V2.9) With finitely many non-degenerate critical points on the level c and none other near it, $N^{c+\epsilon}$ strongly deformation retracts onto $N^{c-\epsilon}$ together with pairwise disjoint cells, one of dimension λ_p for each critical point p.
24. (V2.10) On $\mathbb{R}^n$, M^c is a strong deformation retract of $M^{c+\epsilon}$, and $M^{c-\epsilon} \cup e^\lambda$ is one of M^c.
25. (V2.11) An iterated adjunction space is built from the empty space by attaching cells one at a time along arbitrary continuous maps; it replaces Milnor's CW-complex, without the cellular condition.
26. (V2.12) A compact slice submanifold on which $F|N$ has only non-degenerate critical points has the homotopy type of an iterated adjunction space with exactly C_λ cells of dimension λ, attached in order of critical value.
27. (V2.13) Attaching an n-cell along homotopic maps gives homotopy equivalent spaces, by a homotopy equivalence which is the identity on X.
28. (V2.14) A homotopy equivalence $f : X \to Y$, extended by the identity on the cell, is a homotopy equivalence $X \cup_\varphi e^n \to Y \cup_{f\varphi} e^n$.
29. (V2.15) For a not a critical value, N^a has the homotopy type of an iterated adjunction space with one λ-cell for each critical point of index λ below a.
30. (V3.1) A local extremum is a critical point, and at a non-degenerate one the index is 0 for a minimum and dim M for a maximum.
31. (V3.2) Near a non-degenerate minimum $f = f(p) + \|y\|^2$ in smooth coordinates, and when the minimum is strict and the space compact, each small sublevel set $\{f \leq f(p) + \varepsilon\}$ is the image of the closed unit ball, with the unit sphere going onto the level set.
32. (V3.3) On a compact manifold of positive dimension, a smooth function with exactly two critical points, both non-degenerate, has them at its strict minimum and its strict maximum, of indices 0 and dim M.
33. (V3.4) If $f^{-1}[a, b]$ is compact and free of critical points, a diffeomorphism carries N^a onto N^b and the level $f^{-1}(a)$ onto $f^{-1}(b)$.
34. (V3.5) A Hausdorff space that is the union of two embedded closed n-balls meeting exactly in their common boundary sphere is homeomorphic to S^n.
35. (V3.6) A compact submanifold of a finite-dimensional space carrying a smooth function with exactly two critical points, both non-degenerate, is homeomorphic to S^n, n its dimension.
36. (V3.7) A function S on pairs $a \leq b$ of a preordered set is subadditive if $S(a, c) \leq S(a, b) + S(b, c)$ for $a \leq b \leq c$, additive if equality holds.
37. (V3.8) For subadditive S and a chain $X_0 \leq \ldots \leq X_n$ with $n \geq 1$, $S(X_0, X_n) \leq \sum_k S(X_k, X_{k+1})$, with equality for additive S.
38. (V3.9) A homology of a filtration is a family of finite-dimensional vector spaces $H_i(a, b)$, $a \leq b$, with a long exact sequence for every triple ending in 0; nothing else is assumed.
39. (V3.10) For such a structure dim H_λ is subadditive, and the Euler characteristic truncated at N is additive when H_{N+1} vanishes.
40. (V3.11) For such a structure and a chain, $R_\lambda \leq C_\lambda$, and the alternating sums of the R_i and C_i up to N agree when the slabs have no homology in degree $N + 1$; under the cell hypothesis C_λ is the number of critical points of index λ.
41. (V3.12) $S_\lambda = R_\lambda - R_{\lambda-1} + \cdots \pm R_0$ is subadditive, by Milnor's rank computation.

42. (V3.13) $R_\lambda - R_{\lambda-1} + \cdots \pm R_0 \le C_\lambda - C_{\lambda-1} + \cdots \pm C_0$ for every λ.
43. (V3.14) The strong inequalities imply the weak ones, become equalities when $C_{\lambda+1} = 0$, and give $\sum(-1)^i R_i = \sum_p (-1)^{\mathrm{ind}(p)}$.
44. (V3.15) If $C_{\lambda+1} = C_{\lambda-1} = 0$ then $R_\lambda = C_\lambda$ and $R_{\lambda\pm1} = 0$; if no two consecutive C_i are nonzero, $R_i = C_i$ for all i.
45. (V3.16) $C_\lambda(f)$ is the number of critical points of f of index λ, and the sum of $(-1)^{\mathrm{index}}$ over the critical points is $\sum_\lambda (-1)^\lambda C_\lambda(f)$.
46. (V3.17) An Euler number of a compact nonempty $N \subseteq \mathbb{R}^{K+1}$ is an integer equal to the index sum of every smooth tangent field with finitely many zeros; one exists for $K \ge 1$.
47. (V3.18) For a non-degenerate symmetric form $H(v,w) = \langle Av, w\rangle$, $\operatorname{sgn}\det A = (-1)^{\mathrm{index}\, H}$.
48. (V3.19) At a non-degenerate critical point of index λ of $F|_N$, the gradient field of $F|_N$ for the induced metric has index $(-1)^\lambda$.
49. (V3.20) On a compact $N \subseteq \mathbb{R}^{K+1}$, $K \ge 1$, $\dim N \ge 1$, every Morse function has $\sum_\lambda (-1)^\lambda C_\lambda = \chi(N)$, the Poincaré–Hopf integer; so this sum does not depend on the Morse function.
50. (V3.21) Every Morse function on S^n, $n \ge 1$, has $\sum_\lambda (-1)^\lambda C_\lambda = 1 + (-1)^n$; the height function has exactly two critical points, of indices 0 and n.
51. (V3.22) For $0 < \rho < R$ the torus of revolution is a compact smooth surface in $\mathbb{R}^3$, the regular zero set of a quartic polynomial.
52. (V3.23) The height of the upright torus above its tangent plane has exactly four critical points, all non-degenerate, of indices $0, 1, 1, 2$, at heights $0, 2\rho, 2R, 2R + 2\rho$.
53. (V3.24) T^a is empty for $a < 0$ and all of T for $a \ge 2R + 2\rho$, and between consecutive critical values Theorem 3.1 holds on T.
54. (V3.25) The Euler number of the torus is 0, by Morse theory and by a nowhere-zero field, so every Morse function on it has $C_0 - C_1 + C_2 = 0$.
55. (V3.26) Relative singular homology $H_n(X, A)$ is functorial in pairs, has the exact sequence of a pair, is homotopy invariant, and $H_n(B, A) \cong H_n(X, A)$ when $B \supseteq A$ is a strong deformation retract of X.
56. (V3.27) A triple $Z \subseteq Y \subseteq X$ has an exact sequence $\cdots \to H_n(Y, Z) \to H_n(X, Z) \to H_n(X, Y) \to H_{n-1}(Y, Z) \to \cdots \to H_0(X, Y) \to 0$.
57. (V3.28) If the closure of Z lies in the interior of A, then $(X \setminus Z, A \setminus Z) \to (X, A)$ induces isomorphisms in homology.
58. (V3.29) If the interiors of A and B cover X, the Mayer–Vietoris sequence of $A \cap B$, A and B, and X is exact.
59. (V3.30) $H_i(D^n, S^{n-1})$ is the coefficient ring for $i = n$ and 0 otherwise, and $H_i(S^n)$, $n \ge 1$, is the coefficient ring for $i = 0$ and $i = n$ and 0 otherwise.
60. (V3.31) Attaching disjoint cells to a closed subset Y of a Hausdorff space gives $H_i(X, Y) \cong R^{c_i}$, c_i the number of cells of dimension i.
61. (V3.32) For a filtration $X_0 \subseteq \cdots \subseteq X_n$ whose steps have finite-dimensional homology, the Betti numbers of (X_n, X_0) satisfy the weak and strong Morse inequalities and the Euler characteristic formula against the sums C_i over the steps.
62. (V3.33) If A, $B \cup \bigcup_j e_j$ and X' are strong deformation retracts of B, X' and X respectively, then $\dim H_i(X, A)$ is the number of cells e_j of dimension i.
63. (V3.34) For a function with non-degenerate critical points on a compact submanifold, a slab between two non-critical levels containing one critical value has homology of dimension the number of critical points of each index in it; $H_*(N)$ is finite-dimensional.
64. (V3.35) For such a function on a compact N, over every field, $b_\lambda(N) \le C_\lambda$, $\sum(-1)^i b_i =$

$\sum(-1)^\lambda C_\lambda$, and $b_i = 0$ above $\dim N$.

65. (V3.36) For such a function, $b_\lambda - b_{\lambda-1} + \cdots \pm b_0 \le C_\lambda - C_{\lambda-1} + \cdots \pm C_0$.
66. (V3.37) If $C_{\lambda+1} = C_{\lambda-1} = 0$, then $b_\lambda = C_\lambda$ and $b_{\lambda\pm1} = 0$; if no two consecutive C_i are nonzero, then $b_i = C_i$ for all i.
67. (V3.38) The Poincaré–Hopf integer of a compact submanifold, and $\chi(M)$ of a compact manifold, equal $\sum_i (-1)^i \dim H_i(\,\cdot\,;\mathbb{k})$ for every field $\mathbb{k}$.
68. (V3.39) On every compact Hausdorff manifold and for every Morse function, the weak and strong inequalities, formula (3) and the lacunary principle hold for the Betti numbers over every field.
69. (V3.40) Over every field: a function with exactly two critical points, both non-degenerate, on M^n, $n \ge 2$, forces $b_0 = b_n = 1$ and $b_i = 0$ otherwise; S^n has Euler characteristic $1 + (-1)^n$; the torus has $b_0 - b_1 + b_2 = 0$.
70. (V3.41) The Hermitian projections of trace 1 in $M_{n+1}(\mathbb{C})$ form a compact submanifold of dimension $2n$, which is $\mathbb{C}P^n$, and Milnor's functions x_j, y_j on U_k are coordinates on it.
71. (V3.42) For distinct c_j, $f = \sum c_j |z_j|^2$ on $\mathbb{C}P^n$ has exactly the $n+1$ critical points p_k, all non-degenerate, p_k of index $2\,\#\{j : c_j < c_k\}$; so there is one critical point of each index $0, 2, \dots, 2n$.
72. (V3.43) $\mathbb{C}P^n$ has the homotopy type of a space with one cell in each dimension $0, 2, \dots, 2n$, and over every field its Betti numbers are 1 in those degrees and 0 elsewhere.
73. (V4.1) For a k-dimensional submanifold $M \subseteq \mathbb{R}^n$, the normal bundle N is an n-dimensional submanifold of $\mathbb{R}^n \times \mathbb{R}^n$, and $E(q, v) = q + v$.
74. (V4.2) e is a focal point of (M, q) of multiplicity μ iff $(q, e-q) \in N$ and dE has nullity $\mu > 0$ there; the focal points of M are the critical values of E.
75. (V4.3) For a smooth map from a second countable manifold into a finite-dimensional vector space, the image of the points where the derivative is not onto has measure zero.
76. (V4.4) Almost no point of $\mathbb{R}^n$ is a focal point of M.
77. (V4.5) In the slice-chart coordinates x at q, $g = \langle dx\,\cdot, dx\,\cdot\rangle$ and $\ell_v = \langle v, d^2x(\cdot,\cdot)\rangle$ are Milnor's first and second fundamental forms.
78. (V4.6) The principal curvatures in a normal direction v are the eigenvalues, with multiplicity, of the self-adjoint shape operator S_v on T_qM.
79. (V4.7) The focal points of (M, q) on the normal line $q + tv$ are the points $q + K^{-1}v$, $K \ne 0$ a principal curvature, with the multiplicity of K; at most k of them counted with multiplicity.
80. (V4.8) The nullity of dE at $(q, v) \in N$ equals the nullity of $g - \ell_v$.
81. (V4.9) $q + tv$ is a focal point of (M, q) of multiplicity μ iff $g - t\ell_v$ has nullity $\mu > 0$; q itself is never one.
82. (V4.10) q is a critical point of L_p iff $p - q$ is normal to M at q, and the Hessian of L_p in the coordinates x is $2(g - \ell_{p-q})$.
83. (V4.11) q is a degenerate critical point of L_p iff p is a focal point of (M, q), the nullity being the multiplicity.
84. (V4.12) For almost every $p \in \mathbb{R}^n$, L_p is a Morse function on M.
85. (V4.13) A closed submanifold of $\mathbb{R}^n$ carries a Morse function all of whose sublevel sets are compact.
86. (V4.14) The image of a smooth topological embedding with injective derivative is a submanifold, and the embedding is a diffeomorphism onto it.
87. (V4.15) Critical points, Hessians, non-degeneracy, index, nullity and the numbers C_λ are preserved by diffeomorphisms.
88. (V4.16) Every compact Hausdorff manifold is diffeomorphic to a compact submanifold of

some $\mathbb{R}^{K+1}$, $K \geq 1$.

89. (V4.17) Every compact Hausdorff manifold carries a Morse function.
90. (V4.18) $\chi(M)$ is defined as $\sum_\lambda (-1)^\lambda C_\lambda(f)$ for a chosen Morse function f.
91. (V4.19) On a compact Hausdorff manifold every Morse function gives the same $\sum_\lambda (-1)^\lambda C_\lambda = \chi(M)$, which is the index sum of every tangent field on every embedded copy.
92. (V4.20) For every smooth f on a submanifold of $\mathbb{R}^n$, $f + \langle a, \cdot \rangle$ is a Morse function for almost every a.
93. (V4.21) Every smooth function on a submanifold of $\mathbb{R}^n$, bounded or not, is a uniform limit of Morse functions, with derivatives up to any order converging on compact pieces of coordinate charts.
94. (V4.22) The same holds on every compact Hausdorff manifold.
95. (V4.23) The index of L_p at a non-degenerate critical point q is the number of focal points of (M, q) on the segment from q to p, counted with multiplicity.
96. (V4.24) A compact Hausdorff manifold carrying a smooth function with exactly two critical points, both non-degenerate, is homeomorphic to S^n.
97. (V5.1) The Hessian is the second derivative, defined at every point; the index of a bilinear form is the largest dimension of a subspace on which it is negative definite, and the nullity is the dimension of its null space.
98. (V5.2) Re B of a complex bilinear form is Milnor's Q', multiplication by i is an orthogonal map with square -1, and it changes the sign of Re B.
99. (V5.3) The eigenvalues of the matrix of Q' come in pairs $\pm e$ of equal multiplicity.
100. (V5.4) If G is J-invariant and L is J-anti-invariant, then $G - tL$ and $G + tL$ have the same nullity, and so do $L - eG$ and $L + eG$.
101. (V5.5) A vector whose real inner product with every vector of a complex subspace vanishes is Hermitian-orthogonal to that subspace.
102. (V5.6) In holomorphic coordinates the Hessian of $\|w(\cdot) - p\|^2$ is twice the first fundamental form plus twice Re$\langle w(q) - p, d^2 w_q \rangle$.
103. (V5.7) In a holomorphic chart, the Hessians of L_p for $p = w(q) \pm tn$ have the same nullity (Assertion 2, before the translation into focal points).
104. (V5.8) If $H = G - L$ on the diagonal with G J-invariant and L J-anti-invariant, then $H(Jv, Jv) = 2G(v, v) - H(v, v)$.
105. (V5.9) If $H(v, v) + H(Jv, Jv) \geq 0$ for an injective linear J, the index of H is at most half the real dimension.
106. (V5.10) A real form $G -$ Re B on $\mathbb{C}^k$, with $G \geq 0$ invariant under $v \mapsto iv$ and B complex bilinear, has index at most k.
107. (V5.11) For w holomorphic near q on a k-dimensional complex space, the Hessian of $\|w(\cdot) - p\|^2$ at q has index at most k, whether or not q is critical or non-degenerate.
108. (V5.12) If moreover G is positive definite and the space is non-zero, the form takes a positive value.
109. (V5.13) On a submanifold of $\mathbb{R}^{2m}$ whose tangent spaces are complex subspaces, of real dimension $2k$, every critical point of $L_p = \| \cdot - p\|^2$ has index at most k.
110. (V5.14) If no critical point between two regular levels $a < b$ has index j, and the slab between them is compact with non-degenerate critical points, then $H_j(N^b, N^a) = 0$ over a field.
111. (V5.15) Theorem 7.2: a closed submanifold of $\mathbb{R}^{2m}$ of dimension $2k$ with complex tangent spaces has the homotopy type of a countable cell complex with cells of dimension at most k.
112. (V5.16) Theorem 7.1 over a field: such a submanifold has no homology above degree k.

113. (V5.17) If X is compact, $X \setminus A$ is such a submanifold going to infinity at A, and a neighbourhood of A deforms into A keeping A fixed, then $H_r(X, A) = 0$ for $r < k$ over a field, and $H_r(A) \to H_r(X)$ is bijective for $r < k-1$ and surjective for $r = k-1$.
114. (V5.18) Corollary 7.3 over a field: this holds for a closed $V \subseteq \mathbb{C}P^n$ and the hyperplane $P = \{z_0 = 0\}$, when the affine part of V is a complex submanifold of complex dimension k and a neighbourhood of $V \cap P$ deforms into it.
115. (V5.19) Theorem 7.4: under the same hypotheses every map $I^r \to V$, $r < k$, deforms into $V \cap P$ relative to the part already there; so $\pi_i(V \cap P) \to \pi_i(V)$ is bijective for $i < k-1$ and surjective for $i = k-1$.
116. (V6.1) In the embedded setting, T_qN is the range of the derivative of the inclusion, P_q the orthogonal projection onto it, and a vector field is a map $N \to V$ with tangent values.
117. (V6.2) $q \mapsto P_q$ is smooth, so every tangent vector u at a point extends to the smooth vector field $q \mapsto P_q u$.
118. (V6.3) A Mathlib covariant derivative satisfies Milnor's rules (2)–(4) at points where the sections involved are differentiable; his (1) is a separate smoothness class.
119. (V6.4) $X \vdash Y = P(dY(X))$ is tangent and satisfies Milnor's rules (1)–(4).
120. (V6.5) $DX/dt = P(X')$ satisfies Milnor's axioms (a), (b), (c).
121. (V6.6) Every operation satisfying (a), (b), (c) along a smooth curve equals $P(X')$ on smooth fields along the curve.
122. (V6.7) Linear equations $x' = A(t)x$ have unique solutions, and global ones when $A(t)$ is skew-adjoint.
123. (V6.8) Along a smooth curve, through every tangent vector at any time there is exactly one parallel field, defined for all t and smooth.
124. (V6.9) Parallel fields have constant inner products, and parallel translation is a linear isometry between tangent spaces, the identity at equal times and transitive.
125. (V6.10) $\frac{d}{dt}\langle X, Y\rangle = \langle DX/dt, Y\rangle + \langle X, DY/dt\rangle$ for fields along any curve.
126. (V6.11) $v\langle Y, Y'\rangle = \langle v \vdash Y, Y'\rangle + \langle Y, v \vdash Y'\rangle$; on an abstract manifold this identity is the definition of compatibility, equal to Mathlib's.
127. (V6.12) $\mathrm{II}_q(v, w) = D_v w$ is the normal part of $dW_q(v)$ for any field W with $W(q) = w$; it is normal, symmetric, and D_v is self-adjoint.
128. (V6.13) The ambient bracket of tangent fields is tangent and is Milnor's Poisson bracket, and the induced connection is symmetric; on an abstract manifold symmetric means torsion-free.
129. (V6.14) $P(dY(X))$ is symmetric and compatible, and any symmetric compatible operation agrees with it on smooth fields; on an abstract manifold two symmetric compatible connections agree on differentiable fields.
130. (V6.15) $\frac{D}{\partial x}\frac{\partial s}{\partial y} = \frac{D}{\partial y}\frac{\partial s}{\partial x}$ for every smooth surface in N.
131. (V6.16) Milnor's $R(X, Y)Z = -X \vdash (Y \vdash Z) + Y \vdash (X \vdash Z) + [X, Y] \vdash Z$ is the negative of Nomizu's curvature.
132. (V6.17) For smooth vector fields on N, $R(X, Y)Z = \mathrm{II}(Y, \mathrm{II}(X, Z)) - \mathrm{II}(X, \mathrm{II}(Y, Z))$, so R is a tensor.
133. (V6.18) $\langle R(X, Y)Z, W\rangle = \langle \mathrm{II}(X, Z), \mathrm{II}(Y, W)\rangle - \langle \mathrm{II}(X, W), \mathrm{II}(Y, Z)\rangle$, with Milnor's sign.
134. (V6.19) $\frac{D}{\partial y}\frac{D}{\partial x}W - \frac{D}{\partial x}\frac{D}{\partial y}W = R(\partial s/\partial x, \partial s/\partial y)W$ for every smooth tangent field along a smooth surface.
135. (V6.20) The curvature of N satisfies Lemma 9.3 (1)–(3), pointwise and for smooth vector fields.
136. (V6.21) Skew-symmetry in each pair and the cyclic identity imply pair symmetry (Lemma 9.3 (4)).

137. (V6.22) The sectional curvature $K(X, Y)$ depends only on the plane, and is the same in Milnor's and in the standard sign convention.
138. (V6.23) A curvature tensor is determined by its sectional curvatures.
139. (V6.24) Constant sectional curvature κ means $R(X, Y)Z = \kappa(\langle X, Z\rangle Y - \langle Y, Z\rangle X)$ in Milnor's sign.
140. (V6.25) The unit sphere has $\mathrm{II}_q(v, w) = -\langle v, w\rangle q$ and constant sectional curvature 1.
141. (V7.1) A geodesic of N on an open interval is a smooth curve in N whose acceleration in V is normal to N, that is, $\frac{D}{dt}\frac{d\gamma}{dt} = 0$.
142. (V7.2) A geodesic on an interval has constant speed, its length on $[a, b]$ is its speed times $b-a$, and $t \mapsto \gamma(at + b)$ is again a geodesic.
143. (V7.3) A smooth curve in N is a geodesic iff $\gamma'' = \mathrm{II}(\gamma', \gamma')$, iff (γ, γ') is an integral curve of the spray $(x, v) \mapsto (v, \mathrm{II}_x(v, v))$; solutions of the chart equation $u'' = -\Gamma(u)(u', u')$ are geodesics.
144. (V7.4) $u'' = F(u, u')$, F smooth, has solutions for a uniform time, depending smoothly on (u_0, v_0, t), and two solutions with the same initial data agree on every common interval.
145. (V7.5) Two geodesics on an interval with the same position and velocity at one time coincide.
146. (V7.6) Near each point there are U and ε such that every tangent vector v at $p \in U$ with $\|v\| < \varepsilon$ is the initial velocity of a unique geodesic on $(-2, 2)$, depending smoothly on (p, v, t).
147. (V7.7) $\exp_q(v) = \gamma(1)$ for the geodesic γ, defined near $[0, 1]$, with $\gamma(0) = q$ and $\gamma'(0) = v$; it is well defined, $\exp_q(0) = q$, and $\gamma(t) = \exp_q(tv)$.
148. (V7.8) exp is defined and smooth on a neighbourhood of the zero section of TN, and the derivative of $\exp_q$ at 0 is the inclusion of T_qN.
149. (V7.9) ε is a normal radius at q when $\exp_q$ maps the open ε-ball of T_qN diffeomorphically onto an open subset of N.
150. (V7.10) Every point has a neighbourhood W and an $\varepsilon > 0$ such that for $q_1, q_2 \in W$ exactly one v with $\|v\| < \varepsilon$ has $\exp_{q_1}(v) = q_2$, v depends smoothly on (q_1, q_2), and ε is a normal radius at every point of W.
151. (V7.11) Lengths of piecewise smooth paths are additive, unchanged by reversal and translation, at least the distance in V of the endpoints, and zero only for constant paths.
152. (V7.12) $\langle d(\exp_q)_w(w), d(\exp_q)_w(x)\rangle = \langle w, x\rangle$ inside a normal ball; in particular the radial geodesics are orthogonal to the images of the spheres.
153. (V7.13) Along a path in a punctured normal neighbourhood, the length is at least the increase of the radius r, and with equality the direction v is constant.
154. (V7.14) A path from q that leaves the open normal ball of radius s has a segment, from its last visit of q to its first arrival on the shell of radius s, of length at least s.
155. (V7.15) The geodesic $t \mapsto \exp_{q_1}(tv)$ of Lemma 10.3 is not longer than any piecewise smooth path in N with the same endpoints, and equality forces the same image.
156. (V7.16) A path from q to $\exp_q(v)$ of length $\|v\|$ is $t \mapsto \exp_q(\rho(t)v)$ with ρ continuous and nondecreasing from 0 to 1.
157. (V7.17) A piecewise smooth path parametrized by arc length and not longer than any path with the same endpoints is a geodesic, smooth at its break points, and extends to a geodesic on an open interval.
158. (V7.18) The infimum ρ of the lengths of joining paths is a metric on N with values in $[0, \infty]$.
159. (V7.19) Within a normal radius, $\rho(p, \exp_p v) = \|v\|$ and the metric balls are the normal balls; the metric balls form neighbourhood bases of the topology of N.
160. (V7.20) For compact $K \subseteq N$ there is $\delta > 0$ such that points of K at distance $< \delta$ are joined by

a unique geodesic of length $< \delta$, which is minimal and depends smoothly on its endpoints.

161. (V7.21) ρ is at least the distance in V, continuous on $N \times N$, finite when N is connected, and makes N a metric space with its own topology.
162. (V7.22) (N, ρ) is complete if every ρ-Cauchy sequence converges in N; N is geodesically complete if $\exp_q$ is defined on all of T_qN for every q, which holds iff every geodesic on an open interval extends to a geodesic on $\mathbb{R}$.
163. (V7.23) Every geodesic is $t \mapsto \exp_q(tv)$; near each point geodesics of bounded speed exist for a uniform time and can be continued, and exp is continuous on its domain, which is open in TN.
164. (V7.24) A minimal geodesic on $[a, b]$ is a geodesic no longer than any piecewise smooth path in N with the same endpoints; $t \mapsto \exp_p(tv)$ on $[0, 1]$ is one when $\rho(p, \exp_p v) = \|v\|$.
165. (V7.25) For $0 < \delta < \varepsilon$ and $\delta < \rho(p, q) < \infty$ some point p_0 of the sphere of radius δ about p has $\rho(p, q) = \delta + \rho(p_0, q)$; a broken path made of two unit-speed geodesics and of minimal length is unbroken.
166. (V7.26) If $\exp_p$ is defined on all of T_pN, every point at finite distance from p is joined to p by a minimal geodesic of length $\rho(p, q)$; in a connected geodesically complete N any two points are.
167. (V7.27) If N is connected and $\exp_p$ is defined on all of T_pN for one p, every bounded set has compact closure in N, and then (N, ρ) is complete.
168. (V7.28) If (N, ρ) is complete, N is geodesically complete.
169. (V7.29) For connected nonempty N, metric completeness, geodesic completeness, completeness of $\exp_p$ at one point, and compactness of the closures of bounded sets are equivalent; then closed bounded sets are compact.
170. (V7.30) A connected submanifold that is closed in V, for instance a compact one, is complete in all these senses; any two points of S^n, $n \geq 1$, are joined by a minimal geodesic.
171. (V8.1) $\Omega(N; p, q)$ is the set of piecewise smooth paths in N on $[0, 1]$ from p to q, and $T\Omega_\omega$ the vector space of piecewise smooth tangent fields along ω vanishing at 0 and 1.
172. (V8.2) A variation is a map $\alpha(u, t)$, u in an open set around 0, with values in N for $t \in [0, 1]$, fixed end points, and smooth on the closed strips of a subdivision; its variation field lies in $T\Omega_\omega$.
173. (V8.3) Every finite family of fields in $T\Omega_\omega$ is the family of variation fields of the variation $\exp_{\omega(t)}(\sum_j u_j W_j(t))$, $\|u\|$ small, without completeness.
174. (V8.4) ω is a critical path of E if every variation of ω has $\frac{d}{du}E(\bar{\alpha}(u)) = 0$ at $u = 0$.
175. (V8.5) $E_a^b = \int_a^b \|c'\|^2$, and $(L_a^b)^2 \leq (b - a)E_a^b$, with equality iff the speed is constant off the break points.
176. (V8.6) A geodesic that is no longer than any path of Ω has energy at most that of every path of Ω.
177. (V8.7) The first variation of ω in the direction W is $-\sum_t \langle W_t, \Delta_t V\rangle - \int_0^1 \langle W_t, A_t\rangle\, dt$, $\Delta_t V$ the jump of the velocity and A the acceleration.
178. (V8.8) On closed strips the mixed partials commute up to the edges, one may differentiate under the integral sign, and integration by parts over the pieces produces the jump terms.
179. (V8.9) For every variation and every u, $\frac{1}{2}\frac{d}{du}E(\bar{\alpha}(u))$ is the first variation of $\bar{\alpha}(u)$ in the direction $\frac{\partial \alpha}{\partial u}(u, \cdot)$; at $u = 0$ this is Theorem 12.2.
180. (V8.10) A geodesic of Ω is a path smooth on $[0, 1]$ with zero acceleration; a path of Ω is critical iff its first variation vanishes on $T\Omega_\omega$.
181. (V8.11) A path of Ω is a critical path of the energy iff it is a geodesic.

182. (V8.12) $\frac{D}{\partial u}\frac{D}{\partial t}\frac{\partial F}{\partial t} = \frac{D}{\partial t}\frac{D}{\partial t}\frac{\partial F}{\partial u} + R(\frac{\partial F}{\partial t}, \frac{\partial F}{\partial u})\frac{\partial F}{\partial t}$ with Milnor's R, and the jumps of $\partial\alpha/\partial t$ can be differentiated in u.
183. (V8.13) For a 2-parameter variation of a geodesic, $\frac{1}{2}\frac{\partial^2 E}{\partial u_1 \partial u_2}(0,0)$ exists and equals $-\sum_t \langle W_2, \Delta_t \frac{DW_1}{dt}\rangle - \int_0^1 \langle W_2, \frac{D^2 W_1}{dt^2} + R(V, W_1)V\rangle\, dt$.
184. (V8.14) E_{**}, defined on $T\Omega_\gamma$ by this formula, is Milnor's Hessian: at a geodesic every 2-parameter variation computes it, and every pair of fields comes from one.
185. (V8.15) $\frac{1}{2}E_{**}(W_1, W_2) = \int_0^1 (\langle \frac{DW_1}{dt}, \frac{DW_2}{dt}\rangle - \langle R(V, W_1)V, W_2\rangle)\, dt$.
186. (V8.16) E_{**} is a symmetric bilinear form on $T\Omega_\gamma$.
187. (V8.17) For a 1-parameter variation of a geodesic with variation field W, $\frac{d^2 E(\bar\alpha(u))}{du^2}(0) = E_{**}(W, W)$.
188. (V8.18) At a minimal geodesic, $E_{**}(W, W) \geq 0$ for every $W \in T\Omega_\gamma$.
189. (V9.1) Linear equations $x' = A(t)x$ with continuous coefficients have solutions on all of $\mathbb{R}$ through every initial value, unique on closed intervals and smooth where A is.
190. (V9.2) A Jacobi field is a smooth tangent field with $\frac{D^2 J}{dt^2} + R(V, J)V = 0$; one exists for all tangent initial values $(J(s), \frac{DJ}{dt}(s))$, and it is determined by them.
191. (V9.3) The Jacobi fields along γ form a $2n$-dimensional space, isomorphic to $T_{\gamma(s)}N \times T_{\gamma(s)}N$ by $J \mapsto (J(s), \frac{DJ}{dt}(s))$ for every s.
192. (V9.4) Parameters $a \neq b$ are conjugate if a non-zero Jacobi field vanishes at both; the multiplicity is the dimension of the space of such fields.
193. (V9.5) At a geodesic, $W_1 \in T\Omega_\gamma$ is a null vector of E_{**} iff it is a Jacobi field.
194. (V9.6) At a geodesic, the nullity of E_{**} is the multiplicity of 0 and 1 as conjugate parameters, and E_{**} is degenerate iff they are conjugate.
195. (V9.7) At a geodesic, the nullity of E_{**} is less than $n = \dim N$ when $n \geq 1$.
196. (V9.8) The variation field of a smooth variation through geodesics is a Jacobi field.
197. (V9.9) Every Jacobi field along a geodesic $\gamma : [0,1] \to N$ is the variation field of a smooth variation of γ through geodesics.
198. (V9.10) At two non-conjugate parameters, the values of a Jacobi field determine it and can be prescribed.
199. (V9.11) Along the short geodesic joining two points of a neighbourhood of Lemma 10.3, Jacobi fields are determined by their end values, which can be prescribed.
200. (V9.12) Along any smooth curve, no parameter in some interval $(0, \delta)$ is conjugate to 0.
201. (V9.13) The index of a bilinear form is the supremum of the dimensions of the finite-dimensional subspaces on which it is negative definite.
202. (V9.14) A perpendicular summand on which the form is positive definite changes neither the index nor the null space; definiteness on a finite-dimensional subspace persists under continuous perturbation; and Milnor's compound matrix gives index at least $\lambda + \nu$.
203. (V9.15) If X is piecewise a Jacobi field, $E_{**}(X, Y)$ is -2 times the sum of the jumps of $\frac{DX}{dt}$ paired with the values of Y.
204. (V9.16) If a field has a zero within h_0 before every point, $\frac{1}{2}E_{**}(X, X) \geq \int_0^1 \|X\|^2$; hence parameters at most h_0 apart are never conjugate.
205. (V9.17) For a fine subdivision, broken Jacobi fields along $\gamma|[0,\tau]$ correspond bijectively to their values at the division points.
206. (V9.18) Lemma 15.3: the fields along $\gamma|[0,\tau]$ split into the broken Jacobi fields and T', perpendicular for E_{**}, with E_{**} positive definite on T'.
207. (V9.19) Lemma 15.4: the index and the null space of E_{**} along $\gamma|[0,\tau]$ are those on the broken

Jacobi fields; the index is finite and the nullity is $\nu(\tau)$.

208. (V9.20) $\lambda(\tau)$, the index of E_{**} on the fields vanishing outside $[0,\tau]$, is non-decreasing and 0 for small τ, and $\lambda(1)$ is the index of E_{**}.
209. (V9.21) $\lambda(\tau)$ and $\nu(\tau)$ are the index and the nullity of a form H_τ on a fixed finite-dimensional space, continuous in τ.
210. (V9.22) λ is continuous from the left.
211. (V9.23) λ jumps by $\nu(\tau)$ just after τ, and $\lambda(\tau') \geq \lambda(\tau) + \nu(\tau)$ for every $\tau' > \tau$.
212. (V9.24) At a geodesic, the index of E_{**} is the number of parameters in $(0,1)$ conjugate to 0, counted with multiplicity, and is finite.
213. (V9.25) A geodesic segment has only finitely many parameters conjugate to 0.
214. (V10.1) Milnor's distance d, the largest distance of corresponding points plus the L^2 distance of the speeds, is a pseudometric on paths, vanishing for paths that agree on $[0,1]$.
215. (V10.2) $\sqrt{E_a^b}$ is 1-Lipschitz for d, so the energy is continuous, and a convergent sequence of paths has uniformly small energy on short intervals.
216. (V10.3) Paths of energy at most c lie in the ball S of radius $\sqrt{c}$ about p, which is compact when N is closed (or connected and complete), and all subdivisions with $t_{i+1} - t_i < \delta^2/c$ are fine enough.
217. (V10.4) Lemma 16.1: the broken geodesics of energy less than c correspond bijectively to their break points, which form an open subset U of $N \times \cdots \times N$, and their energy is $\sum_i \rho(x_i, x_{i+1})^2/(t_{i+1} - t_i)$.
218. (V10.5) This correspondence and its inverse are continuous, and identify each B^a with its set of break points up to homotopy.
219. (V10.6) E' is smooth on U, and its sublevel sets B^a, $a < c$, are compact.
220. (V10.7) Milnor's deformation r_u does not increase the energy, runs from the identity to the retraction onto B, fixes B, and is continuous in both variables.
221. (V10.8) For $a < c$, B^a is a strong deformation retract of Ω^a.
222. (V10.9) The critical points of E' are exactly the break points of the unbroken geodesics of energy less than c, and every such geodesic is a broken geodesic.
223. (V10.10) For the subdivision of §16, broken Jacobi fields take arbitrary values at the break points and are determined by them, E_{**} is positive definite on the fields vanishing there, and Lemma 15.4 holds.
224. (V10.11) At a geodesic, the Hessian of E' is E_{**} on the broken Jacobi fields, so the two have the same index and the same nullity.
225. (V10.12) If $a < c$ is not the energy of a geodesic and the geodesics of energy at most a are non-degenerate, they are finitely many, and B^a has the homotopy type of a cell complex with one λ-cell for each of them of index λ.
226. (V10.13) Theorem 16.3: under the same hypotheses, Ω^a has that homotopy type and contains finitely many geodesics.
227. (V10.14) If some path has energy less than c, a minimum of E' is a geodesic minimizing the energy on all of Ω.
228. (V10.15) Ω^* is the space of continuous paths with the topology of uniform convergence, and the inclusion $\Omega \to \Omega^*$ is continuous since $d^* \leq d$.
229. (V10.16) Convergence for d implies L^2 convergence of the velocities, and, together with pointwise convergence, is implied by it.
230. (V10.17) Small ambient balls are good: two of their points are joined by a unique minimal geodesic, which stays in the ball and depends smoothly on its end points, and every compact set has a Lebesgue number for good balls.

231. (V10.18) On the open set Y_k of paths whose dyadic pieces of level k lie in good balls, the dyadic broken geodesic is a homotopy inverse of the inclusion of $X_k = Y_k \cap \Omega$.
232. (V10.19) Theorem 17.1: the inclusion of Ω, with Milnor's metric, into Ω^* is a homotopy equivalence; Milnor's map h is not continuous for d, and the proof verified is the exposition's modified proof.
233. (V10.20) Relative cell attachments carry over along homotopy equivalences, the relative Theorem 3.5 holds below a compact sublevel set, and a tower of relative cell attachments over sublevel sets that exhaust a space gives a countable cell complex.
234. (V10.21) Between two levels below c that are not energies of geodesics, Ω^b is, up to homotopy, Ω^a with one λ-cell attached for each geodesic of index λ with energy in between.
235. (V10.22) Theorem 17.3: for N closed in V and q not conjugate to p along any geodesic, $\Omega(N; p, q)$ has the homotopy type of a countable cell complex with one λ-cell for each geodesic from p to q of index λ.
236. (V10.23) For $q \neq \pm p$ on S^n, the geodesics from p to q are the great-circle arcs γ_k with $k\pi < L(\gamma_k) < (k+1)\pi$, q is not conjugate to p along any of them, and γ_k has index $k(n-1)$.
237. (V10.24) Corollary 17.4 for $\Omega(S^n; p, q)$, $q \neq \pm p$, $n \geq 2$: one cell in each of the dimensions $0, n-1, 2(n-1), \ldots$.
238. (V10.25) A path space with one geodesic of index kr, $r \geq 2$, in the k-th energy window has Betti numbers 1 in the degrees divisible by r and 0 in the others.
239. (V10.26) For $n \geq 3$ and $q \neq \pm p$, $H_i(\Omega(S^n; p, q))$ over a field has dimension 1 if $n-1$ divides i and 0 otherwise.
240. (V10.27) $\mathbb{R}^{n+1} \setminus \{0\}$ has the homotopy type of S^n, no conjugate points and two points joined by a single geodesic, so Corollary 17.5 needs completeness.
241. (V10.28) Every q can be joined in N to a point not conjugate to p, and a path from q to q' makes $\Omega(N; p, q)$ and $\Omega(N; p, q')$ homotopy equivalent.
242. (V10.29) Corollary 17.2: for N closed in a Euclidean space, $\Omega(N; p, q)$ and $\Omega^*(N; p, q)$ have the homotopy type of a countable cell complex.
243. (V10.30) Corollary 17.4 for loops: for $n \geq 2$ the loop space of S^n has the homotopy type of a cell complex with one cell in each dimension $k(n-1)$; for $n \geq 3$ its Betti numbers over a field are 1 in the degrees divisible by $n-1$ and 0 otherwise.
244. (V11.1) The exponential map is defined on an open subset of TN and is smooth there, without completeness.
245. (V11.2) The domain of $\exp_q$ in T_qN is open and star-shaped, $\exp_q$ is smooth on it, and $\exp_q$ is critical at v when $d(\exp_q)_v$ is not one-to-one.
246. (V11.3) Theorem 18.1: $\exp_q(v)$ is conjugate to q along $t \mapsto \exp_q(tv)$ iff $\exp_q$ is critical at v, the multiplicity being $\dim \ker d(\exp_q)_v$.
247. (V11.4) The points conjugate to p along some geodesic are exactly the critical values of $\exp_p$.
248. (V11.5) Corollary 18.2: these points form a set of measure zero in every chart of N.
249. (V11.6) The points not conjugate to p along any geodesic are dense, and every geodesic from p to such a point has nullity 0.
250. (V11.7) Lemma 19.1: with $\langle R(A,B)A, B\rangle \leq 0$, no two parameters are conjugate along any geodesic.
251. (V11.8) With $\langle R(A,B)A, B\rangle \leq 0$, every geodesic has index and nullity 0, E_{**} is positive definite, and $\exp_p$ is nowhere critical.
252. (V11.9) The Ricci tensor $K(U_1, U_2)$ is the trace of $W \mapsto R(U_1, W)U_2$; it is symmetric, and $K(U_n, U_n)$ is the sum of the sectional curvatures of the planes through U_n and U_i.

253. (V11.10) A geodesic segment extends to a smooth curve on $\mathbb{R}$ along which its velocity is parallel, with parallel unit fields $P_1, \dots, P_{n-1}$ such that $\sum \langle R(V,P_i)V, P_i\rangle \geq L^2(n-1)/r^2$ under the Ricci bound.
254. (V11.11) $\frac{1}{2}E_{**}(\sin(\pi t)P, \sin(\pi t)P) = \int_0^1 (\pi^2\cos^2\pi t - \sin^2\pi t\,\langle R(V,P)V,P\rangle)\,dt$, and under the Ricci bound a geodesic longer than πr has a field with $E_{**}(X,X) < 0$.
255. (V11.12) Theorem 19.4: for $n \geq 2$ and $K(U,U) \geq (n-1)/r^2$, a geodesic longer than πr contains a conjugate point and is not minimal.
256. (V11.13) Corollary 19.5: if moreover N is connected and complete, its diameter is at most πr and N is compact.
257. (V11.14) The unit sphere S^n has $K(U,U) = (n-1)\|U\|^2$, so for $n \geq 2$ its geodesics longer than π are not minimal and its diameter is at most π.
258. (V11.15) Test fields supported in disjoint intervals are E_{**}-orthogonal, and on a piece $[a,b]$ with $L(b-a) > \pi r$ one of them has $E_{**}(X,X) < 0$.
259. (V11.16) Under the Ricci bound, a geodesic of length $> k\pi r$ has index at least k.
260. (V11.17) Theorem 19.2: a simply connected complete N of curvature ≤ 0 has exactly one geodesic between any two points, and $\exp_p : T_pN \to N$ is a diffeomorphism.
261. (V11.18) In curvature ≤ 0, $\Omega(N;p,q)$ has the homotopy type of the discrete set of geodesics from p to q, and is contractible when N is simply connected.
262. (V11.19) For complete N with $\exp_p$ nowhere critical (e.g. curvature ≤ 0), $\exp_p : T_pN \to N$ is a covering map.
263. (V11.20) In a complete N of curvature ≤ 0, each homotopy class of paths from p to q contains exactly one geodesic.
264. (V11.21) Corollary 19.3, first half: in a complete N of curvature ≤ 0, $\pi_i(N) = 0$ for $i > 1$.
265. (V11.22) For N compact with positive definite Ricci tensor and q not conjugate to p, $\Omega(N;p,q)$ has the homotopy type of a countable cell complex with finitely many cells in each dimension.
266. (V11.23) Under the same hypotheses its homology over any field is finite-dimensional in each degree.
267. (V11.24) Theorem 19.6 for all p, q: the same conclusions without the non-conjugacy hypothesis.
268. (V12.1) A bi-invariant metric is modelled by a bilinear form on a Lie algebra, and its Levi-Civita connection on left-invariant fields by a bilinear D with $D(X,Y) - D(Y,X) = [X,Y]$ and $B(D(X,Y),Z) + B(Y,D(X,Z)) = 0$.
269. (V12.2) If moreover $D(X,X) = 0$ for all X, then $D(X,Y) = \frac{1}{2}[X,Y]$.
270. (V12.3) Under the same hypotheses, $B([X,Y],Z) = B(X,[Y,Z])$.
271. (V12.4) For a symmetric, non-degenerate, ad-invariant B, the Levi-Civita connection on the Lie algebra must be $\frac{1}{2}[\,\cdot\,,\cdot\,]$, and it is one.
272. (V12.5) With $D = \frac{1}{2}[\,\cdot\,,\cdot\,]$, Milnor's curvature is $R(X,Y)Z = \frac{1}{4}[[X,Y],Z]$.
273. (V12.6) For ad-invariant B, $B(R(X,Y)Z,W) = \frac{1}{4}B([X,Y],[Z,W])$.
274. (V12.7) For ad-invariant B, $B([X,Y],Z)$ is alternating in X, Y, Z.
275. (V12.8) $B(R(X,Y)X,Y) = \frac{1}{4}B([X,Y],[X,Y])$, which is ≥ 0 and vanishes iff $[X,Y] = 0$ when B is positive definite.
276. (V12.9) With trivial centre and a positive definite ad-invariant B, the Ricci curvature tr K_V is positive for $V \neq 0$; it equals $-\frac{1}{4}$ of the Killing form.
277. (V12.10) The orthogonal complement of the centre is an ideal, complementary to the centre, with trivial centre.
278. (V12.11) $K_V = -\frac{1}{4}\operatorname{ad}_V^2$ is self-adjoint and positive semidefinite, with kernel the centralizer

of V.

279. (V12.12) Every eigenspace of K_V for a non-zero eigenvalue carries a complex structure and so has even dimension.
280. (V12.13) A spectrum in which every positive eigenvalue has even multiplicity has even index, the index being defined by Theorems 20.5 and 15.1.
281. (V12.14) A matrix group G with Lie algebra $\mathfrak{g}$ has $T_gG = g\mathfrak{g}$, its induced metric is bi-invariant, and the inner product on $\mathfrak{g}$ is ad-invariant.
282. (V12.15) The geodesics of a matrix group are the curves $t \mapsto g\exp(tX)$; those through I are the one-parameter subgroups, and the group is complete.
283. (V12.16) $\sigma \mapsto \tau\sigma^{-1}\tau$ is an isometry of G fixing τ and reversing the geodesics through τ.
284. (V12.17) On a matrix group, $X \vdash Y = \frac{1}{2}[X,Y]$, $R(X,Y)Z = \frac{1}{4}[[X,Y],Z]$, and the sectional curvature is $\frac{1}{4}\|[X,Y]\|^2 \geq 0$.
285. (V12.18) Along $\exp(tA)$, $\exp(tA/2)U(t)\exp(tA/2)$ is a Jacobi field whenever $U'' + \frac{1}{4}[[A,U],A] = 0$ in $\mathfrak{g}$.
286. (V13.1) Ω^d is the set of paths of least energy, and these are geodesics.
287. (V13.2) Theorem 22.1: if the closed balls about p are compact, every geodesic not of least energy has index $\geq \lambda_0$, and Ω^d is a retract of an open neighbourhood, then every map of a compact P of dimension $< \lambda_0$ into Ω is homotopic, relative to the part already in Ω^d, to a map into Ω^d.
288. (V13.3) The inclusion $\Omega^d \to \Omega$ is bijective on π_i for $i \leq \lambda_0 - 2$ and onto for $i \leq \lambda_0 - 1$, at every point of Ω^d.
289. (V13.4) Corollary 22.2: $\pi_i(\Omega^d, \gamma) \cong \pi_{i+1}(N,p)$ for $i \leq \lambda_0 - 2$ (bijection), onto for $i \leq \lambda_0 - 1$.
290. (V13.5) Change of basepoint is a group isomorphism, and Corollary 22.2 holds as groups for $1 \leq i \leq \lambda_0 - 2$.
291. (V13.6) The geodesics from p to $-p$ on S^{n+1} are the great circles of length $(2m+1)\pi$, the minimal ones are the semicircles, and these form a space homotopy equivalent to the equator, and homeomorphic to it after identifying paths that agree on $[0,1]$.
292. (V13.7) The great circle of length $(2m+1)\pi$ from p to $-p$ has index $2mn$; the non-minimal ones have index $\geq 2n$.
293. (V13.8) $\Omega^d(S^{n+1}; p, -p)$ is a retract of the open set of paths whose midpoint is not $\pm p$.
294. (V13.9) Corollary 22.3: $\pi_i S^n \cong \pi_{i+1} S^{n+1}$ for $i \leq 2n-2$ (groups for $i \geq 1$), and onto for $i = 2n-1$.
295. (V13.10) Lemma 22.4: a C^2-small perturbation of f near a compact K has no critical points in K of index below λ_0 if f has none.
296. (V13.11) On a submanifold, small linear perturbations keep the indices of critical points in a compact set $\geq \lambda_0$.
297. (V13.12) Pushing maps is transitive, holds for strong deformation retracts, and holds for a closed set with finitely many disjoint cells of dimension $>$ dim P attached.
298. (V13.13) Lemma D: across critical levels whose critical points are non-degenerate of index $\geq \lambda_0$, N^a pushes maps of dimension $< \lambda_0$ out of N^b.
299. (V13.14) Lemma 22.5: given a neighbourhood U of the minimum set from which it pushes maps, and indices $\geq \lambda_0$ off the minimum, the minimum set pushes maps of dimension $< \lambda_0$ out of every compact sublevel set.
300. (V13.15) In the model of broken geodesics, a neighbourhood retraction of Ω^d gives a neighbourhood of the minimum that deforms into it.
301. (V13.16) Theorem 22.1 holds for an open and closed set Ω' of paths when the index bound holds for the geodesics in Ω'.

302. (V14.1) $\exp \operatorname{diag}(ia_1, \dots, ia_n) = -I$ iff every a_j is an odd multiple of π.
303. (V14.2) For $A = \operatorname{diag}(i\pi k_j)$, $\operatorname{tr} AA^* = \pi^2 \sum k_j^2$ and $\operatorname{tr} A = i\pi \sum k_j$.
304. (V14.3) For odd k_j, $\sum k_j^2 \geq n$ with equality iff all $k_j = \pm 1$; with $\sum k_j = 0$ and $n = 2m$, then m are $+1$ and m are -1.
305. (V14.4) A skew-Hermitian A has $A^2 = -\pi^2 I$ iff $A = i\pi(2P - I)$ for a Hermitian projection P, and then $\operatorname{tr} A = 0$ iff $\operatorname{tr} P = n/2$.
306. (V14.5) For diagonal $A = \operatorname{diag}(i\pi k_j)$, $\frac{1}{4}[[A, W], A]$ multiplies the (j, l) entry of W by $\frac{\pi^2}{4}(k_j - k_l)^2$.
307. (V14.6) E_{jl}, E'_{jl} are skew-Hermitian traceless eigenvectors with eigenvalue $\frac{\pi^2}{4}(k_j - k_l)^2$, diagonal matrices are in the kernel, and together they span the skew-Hermitian matrices.
308. (V14.7) The conjugate parameters of an eigenvalue e are the $t \in (0, 1)$ in $\pi\mathbb{Z}/\sqrt{e}$, and for $e = (\pi d)^2$ there are $d - 1$ of them.
309. (V14.8) The index of a spectrum is the number of conjugate parameters in $(0, 1)$ counted with multiplicity, equivalently the sum over the eigenvalues of their numbers of conjugate parameters.
310. (V14.9) For odd k_j, Milnor's spectrum has index $\sum_{k_i > k_j} (k_i - k_j - 2)$.
311. (V14.10) The formula is non-negative, symmetric under $k \mapsto -k$, independent of the order, and zero when all $k_j = \pm 1$.
312. (V14.11) For odd $k_1, \dots, k_{2m}$ with sum 0, not all ± 1, the formula is at least $2m + 2$.
313. (V14.12) In $U(n + 1)$, $k = (3, 1, \dots, 1)$ gives index 0 although it is not minimal, and its $\sum k_j$ is larger than that of any minimal geodesic.
314. (V14.13) $U(n)$ and $SU(n)$ are compact submanifolds of the space of matrices and matrix groups, with Lie algebras $\mathfrak{u}(n)$ and $\mathfrak{su}(n)$.
315. (V14.14) Every geodesic from I to $-I$ in $SU(n)$ is $t \mapsto \exp(tA)$ with $A = T \operatorname{diag}(i\pi k_j)T^*$, T unitary, k_j odd, $\sum k_j = 0$.
316. (V14.15) In $SU(2m)$ the distance from I to $-I$ is $\pi\sqrt{2m}$, and $\exp(tA)$ is minimal iff $A^2 = -\pi^2 I$, iff all $k_j = \pm 1$.
317. (V14.16) The initial velocities of the minimal geodesics form a space homeomorphic to $G_m(\mathbb{C}^{2m})$, by $A \mapsto \frac{1}{2}(I - \frac{i}{\pi}A)$.
318. (V14.17) The space Ω^d of minimal paths is homotopy equivalent to that space of velocities, and homeomorphic to it after identifying paths that agree on $[0, 1]$.
319. (V14.18) Theorem 20.5 for $SU(n)$: the multiplicity of a conjugate parameter along $\exp(tA)$ is the dimension of an explicit kernel, which equals the count from Milnor's spectrum.
320. (V14.19) Every geodesic from I to $-I$ in $SU(n)$ has index $\sum_{k_i > k_j} (k_i - k_j - 2)$.
321. (V14.20) In $SU(2m)$ every geodesic from I to $-I$ not of least energy has index $\geq 2m + 2$.
322. (V14.21) $\Omega^d(SU(2m); I, -I)$ is a neighbourhood retract, so $\pi_i \Omega^d \cong \pi_{i+1} SU(2m)$ for $i \leq 2m$ (groups for $i \geq 1$).
323. (V14.22) Theorem 23.3: $\pi_i G_m(\mathbb{C}^{2m}) \cong \pi_{i+1} SU(2m)$ for $i \leq 2m$, at every point, as groups for $i \geq 1$.
324. (V14.23) An equivariant map onto a homogeneous space with a local section of the orbit map is locally trivial; restricting frames and $U(n) \to V_k(\mathbb{K}^n)$ are locally trivial.
325. (V14.24) The frame-to-span map $V_m(\mathbb{K}^n) \to G_m(\mathbb{K}^n)$ is locally trivial with fibre $U(m)$.
326. (V14.25) $V_k(\mathbb{C}^n)$ is $2(n-k)$-connected, and $U(m) \to U(m+k)$ is bijective on π_i for $i < 2m$, onto for $i \leq 2m$.
327. (V14.26) Lemma 23.4: $\pi_i G_m(\mathbb{C}^{2m}) \cong \pi_{i-1} U(m)$ for $1 \leq i \leq 2m$, as groups for $i \geq 2$.
328. (V14.27) $SU(n) \to U(n)$ is bijective on π_j for $j \geq 2$.

329. (V14.28) $\pi_{i-1}U(m) \cong \pi_{i+1}U(2m)$ for $1 \le i \le 2m$, and periodicity $\pi_{i-1}U(m) \cong \pi_{i+1}U(m)$ for $1 \le i \le 2m-2$ (groups for $i \ge 2$).
330. (V14.29) Theorem 23.5: for $j \le 2m-1$, $\pi_j U(m)$ is trivial for even j and in bijection with $\mathbb{Z}$ for odd j.
331. (V14.30) $\pi_1 S^1 \cong \mathbb{Z}$ and $\pi_{2k+1}U(m) \cong \mathbb{Z}$ for $2k+2 \le 2m$, as groups.
332. (V15.1) A complex structure is an orthogonal J with $J^2 = -I$, equivalently a skew J with $J^2 = -I$; it forces n even.
333. (V15.2) If $J^2 = -I$ then $\exp(\theta J) = \cos\theta\, I + \sin\theta\, J$, so $\exp(\pi J) = -I$.
334. (V15.3) For odd positive a_j, Milnor's spectrum for $O(2m)$ has index $\sum_{i<j}(a_i + a_j - 2) + \sum_{a_i > a_j}(a_i - a_j - 2)$, which is 0 when all $a_j = 1$ and at least $2m-2$ otherwise.
335. (V15.4) $\Omega_k(n)$ is the set of complex structures anticommuting with $J_1, \dots, J_{k-1}$; these sets decrease with k and are compact.
336. (V15.5) For $P, J' \in \Omega_k(n)$, the reflection $PJ'^{-1}P = -PJ'P$ lies in $\Omega_k(n)$.
337. (V15.6) For skew A anticommuting with J, $J\exp A$ is a complex structure, and it anticommutes with every J_α that anticommutes with J and commutes with A.
338. (V15.7) $J' \in \Omega_{k+1}(n)$ iff $J' = J_k A$ with A a complex structure anticommuting with J_k and commuting with $J_1, \dots, J_{k-1}$; then $J_k \exp(tA) = \cos t\, J_k + \sin t\, J'$ stays in $\Omega_k(n)$ and passes through J' at $t = \pi/2$.
339. (V15.8) $P \oplus P'$ lies in $\Omega_k(n+n')$ for the structures $J_\alpha \oplus J'_\alpha$ whenever $P \in \Omega_k(n)$ and $P' \in \Omega_k(n')$.
340. (V15.9) Explicit matrices give 1, 3, 7, 8 anticommuting complex structures on $\mathbb{R}^2$, $\mathbb{R}^4$, $\mathbb{R}^8$, $\mathbb{R}^{16}$, and 8 on every $\mathbb{R}^{16r}$.
341. (V15.10) The products $J_1J_2J_3$, $J_1J_4J_5$, $J_2J_4J_6$, $J_1J_6J_7$, J_3J_4, J_7J_8 and $J_1 \cdots J_7$ have exactly the squares and commutation properties Milnor uses on pp. 139–143.
342. (V15.11) $J' \mapsto J_a J_b J'$ is a bijection of $\Omega_{k+1}(n)$ onto the symmetric involutions commuting with J_a, J_b and anticommuting with the other J_c, $c \le k$.
343. (V15.12) With eight structures, $V_1 \supset W \supset X \supset X_1$ halve the dimension at each step, and in $\mathbb{R}^{16r}$ they have dimensions $8r$, $4r$, $2r$, r.
344. (V15.13) A minimal space is a non-zero space with no proper non-zero subspace closed under $J_1, \dots, J_k$; m_k is Milnor's table.
345. (V15.14) For $k \le 8$, m_k divides the dimension of every space with k Clifford maps, and is the least positive dimension carrying k anticommuting complex structures.
346. (V15.15) k Clifford maps make a space a module over the Clifford algebra C_k of dimension 2^k, so a minimal space has dimension at most 2^k.
347. (V15.16) For $k \le 8$ a space with k Clifford maps is minimal iff its dimension is m_k.
348. (V15.17) For $k \le 8$, $k \not\equiv 3 \pmod 4$, any two minimal spaces are isomorphic by a linear isomorphism commuting with the J_α.
349. (V15.18) For anticommuting complex structures, $J_1u, \dots, J_ku$ are orthonormal and orthogonal to a unit vector u.
350. (V15.19) Eigenvalues greater than π^2 each contribute at least one to the index; so $s-1$ such eigenvalues give index at least $s-1$.
351. (V15.20) For odd a_h with some $|a_h| \ge 3$ and sum c, the pairs $a_h > a_j$ alone give $4\lambda \ge r - |c|$.
352. (V15.21) If $x\exp A \in N$ and $A \in T_{x\exp A}$ for $A \in T_x$, with $\dim T_x = \dim N$, then the geodesics of N are exactly the curves $t \mapsto x\exp(tA)$, $A \in T_x$, and N is complete.
353. (V15.22) Assertions 1 and 2: for $x \in \Omega_k(n)$ and $A \in \mathcal{T}_x$, $x\exp A \in \Omega_k(n)$; conversely, if C is small and $x\exp C \in \Omega_k(n)$, then $C \in \mathcal{T}_x$.
354. (V15.23) The set of points of $\Omega_k(n)$ where $\mathcal{T}$ has a given dimension is open and closed in

$\Omega_k(n)$, compact, a submanifold of $M_n(\mathbb{R})$ with tangent spaces $x\mathcal{T}_x$, and its geodesics are the curves $x\exp(tA)$, $A \in \mathcal{T}_x$.

355. (V15.24) A skew V with $\exp V = -I$ has $\|V\|^2 \geq \pi^2 n$, with equality only if $V^2 = -\pi^2 I$.
356. (V15.25) If some $A \in \mathcal{T}_p$ has $A^2 = -I$, the distance from p to $-p$ in the stratum of p is $\pi\sqrt{n}$, and the paths of least energy are, on $[0,1]$, exactly the curves $p\exp(\pi tA)$ with $A \in \mathcal{T}_p$, $A^2 = -I$.
357. (V15.26) These A, for $p = J_k$, correspond homeomorphically to the points $J' = J_k A$ of $\Omega_{k+1}(n)$, the geodesic of J' being $\cos(\pi t)J_k + \sin(\pi t)J'$; and $\Omega_{k+1}(n)$ is homotopy equivalent to the space of minimal paths from J_k to $-J_k$ (homeomorphic after identifying paths that agree on $[0,1]$).
358. (V15.27) The minimal paths from p to $-p$ form a retract of an open neighbourhood in the path space.
359. (V15.28) For skew X commuting with p and $J_1, \ldots, J_{k-1}$, $X\gamma - \gamma X$ is a Jacobi field along $\gamma(t) = p\exp(tV)$; spaces of such X commuting with $\exp(\tau V)$ but not with V bound the multiplicity of $\gamma(\tau)$, and so the index, from below.
360. (V15.29) $\mathbb{R}^n$ splits orthogonally into subspaces minimal under $A, J, J_1, \ldots, J_{k-1}$; on each, A^2 is a scalar $-c$, with $c \geq 1$ if $\exp(\pi A) = -I$; each is a minimal space for $k+1$ structures; and for $k \leq 7$, $k \not\equiv 2 \pmod 4$, any two are related by an isomorphism as in Milnor's p. 145.
361. (V15.30) For $k \leq 7$, $k \not\equiv 2 \pmod 4$, a geodesic from J_k to $-J_k$ in $\Omega_k(n)$ that is not of least energy has index $\geq n/m_{k+1} - 1$, and $\geq n-2$ in $O(n)$ ($k = 0$).
362. (V15.31) For $k \leq 7$, $k \not\equiv 2 \pmod 4$ and $i \leq n/m_{k+1} - 3$, the map $J' \mapsto (\cos(\pi t)J_k + \sin(\pi t)J')$ is bijective on π_i at every point, and $\pi_i\Omega_{k+1}(n) \cong \pi_{i+1}\Omega_k(n)$.
363. (V15.32) Theorem 24.3: for $i \leq n-4$, $\Omega_1(n) \to \Omega(O(n); I, -I)$ is bijective on π_i, and $\pi_i\Omega_1(n) \cong \pi_{i+1}O(n)$.
364. (V15.33) For a continuous F of modulus 1, the set of paths along which the angle of F ends at a given value is open and closed in the path space.
365. (V15.34) $\det_{\mathbb{C}}$, defined without a complex basis, is multiplicative, has modulus 1 on orthogonal matrices commuting with $\mathbf{i}$, and $\det_{\mathbb{C}}\exp(tV) = e^{-\sqrt{-1}\,t\,\mathrm{tr}(V\mathbf{i})/2}$.
366. (V15.35) $\mathbf{i} = J_1 \cdots J_{k-1}$ has the properties Milnor uses for $k = 2, 6$; $f = \det_{\mathbb{C}}(J_k^{\mathsf{T}}\,\cdot\,)$ has modulus 1 on $\Omega_k(n)$; and $\Omega^{(c)}$, the paths whose f-angle ends at $\pi m_{k+1}c/2$, is open and closed and contains the geodesic $J_k\exp(\pi tA)$ exactly when $-\mathrm{tr}(A\mathbf{i}) = m_{k+1}c$.
367. (V15.36) For $k = 2, 6$ every path from J_k to $-J_k$ lies in exactly one $\Omega^{(c)}$, c an integer.
368. (V15.37) For $k \equiv 2 \pmod 4$, on each minimal piece $A = a_h\mathbf{i}$ with a_h an odd integer, and $\mathrm{tr}(A\mathbf{i}) = -m_{k+1}\sum a_h$.
369. (V15.38) For $k = 2, 6$, a geodesic in $\Omega^{(c)}$ that is not of least energy has index $\geq \lfloor (n/m_{k+1} - |c|)/4 \rfloor$.
370. (V15.39) For $k = 2, 6$ and $j + 3 \leq \lfloor (n/m_{k+1} - |c|)/4 \rfloor$, at a point J' whose semicircle lies in $\Omega^{(c)}$, the map of item 31 is bijective on π_{j+1}, and $\pi_{j+1}\Omega_{k+1}(n) \cong \pi_{j+2}\Omega_k(n)$.
371. (V15.40) For $k = 2, 6$ and $|c| \leq n/m_{k+1} - 8$, the path components of $\Omega^{(c)}$ correspond bijectively to those of the part of $\Omega_{k+1}(n)$ mapped into it.
372. (V15.41) On $\mathbb{R}^{16r}$ with the explicit structures, $J \mapsto J \oplus J'_k$ maps $\Omega_k(16r)$ into $\Omega_k(16r+16)$ compatibly with the maps of item 31, and for $k = 2, 6$ without changing c.
373. (V15.42) $\Omega_8(16r)$ is homeomorphic to $O(r)$, the base point J_8 corresponding to I.
374. (V15.43) For $r \geq 2i + 7$: $\pi_i\Omega_8(16r) \cong \pi_{i+8}O(16r)$ and $\pi_i O(r) \cong \pi_{i+8}O(16r)$, as sets.
375. (V15.44) $\pi_i O(n)$ does not depend on $n \geq i + 2$, and $\pi_i U(n)$ does not depend on $n > i/2$, up to bijection.

376. (V15.45) Theorem 24.7 in the stable range: $\pi_i O(n) \cong \pi_{i+8} O(n')$, as sets, for $n \geq i+2$ and $n' \geq i+10$.
377. (V15.46) $\pi_i U(n) \cong \pi_{i+2} U(n')$, as sets, for $i < 2n$ and $i+2 < 2n'$.
378. (VA.1) A homotopy equivalence is a map with a two-sided homotopy inverse; a map with a left and a right homotopy inverse is one.
379. (VA.2) The telescope is $X_\Sigma = \bigcup_n X_n \times [n, n+1] \subseteq X \times \mathbb{R}$, and X is the homotopy direct limit of $\{X_i\}$ when $(x, \tau) \mapsto x$ is a homotopy equivalence; raising τ stays in X_Σ.
380. (VA.3) If the X_n are closed, a map on X_Σ (or $P \times X_\Sigma$) continuous on each strip is continuous.
381. (VA.4) If f_Σ is a homotopy equivalence between homotopy direct limits, so is f.
382. (VA.5) $X_n \to X_\Sigma^n$, $x \mapsto (x, n)$, is a homotopy equivalence, with no closedness.
383. (VA.6) A function $f \geq 0$ with $f \geq n+1$ off X_n makes X a homotopy direct limit, and X_Σ is always the homotopy direct limit of its truncations.
384. (VA.7) Example 1 holds for paracompact Hausdorff X; sublevel sets $\{E \leq a_i\}$ of a continuous E with $a_i \to \infty$ need no paracompactness.
385. (VA.8) Example 3: $[0, 1]$ is not the homotopy direct limit of $\{0\} \cup [\frac{1}{n+1}, 1]$.
386. (VA.9) For closed X_n, Milnor's maps h_1 are homotopy equivalences with inverse g.
387. (VA.10) Case 1: for closed X_n and $\varphi_n \simeq \mathrm{id}$, φ_Σ is a homotopy equivalence; the third line of h_u is $h^{n+1}_{(4t-3)u}$.
388. (VA.11) Case 2: if the Y_n are closed and every f_n is a homotopy equivalence, f_Σ is a homotopy equivalence; the X_n need not be closed.
389. (VA.12) Theorem A by Milnor's proof, with the Y_i closed.
390. (VA.13) The mapping telescope of $K_0 \to K_1 \to \cdots$ is the homotopy direct limit of its closed truncations, each homotopy equivalent to the corresponding K_n.
391. (VA.14) The mapping telescope of the inclusions $X_n \subseteq X_{n+1}$ is homotopy equivalent to X_Σ, and to X when X is the homotopy direct limit.
392. (VA.15) Theorem A holds exactly as Milnor states it, with no closedness hypothesis.
393. (VA.16) A homotopy direct limit of the X_n is homotopy equivalent to the mapping telescope of any sequence K_n with compatible homotopy equivalences $X_n \to K_n$.
394. (VA.17) The Corollary, with the mapping telescope of the K_n in place of a CW-complex.
395. (VA.18) A countable cell complex is the mapping telescope of a tower of cell attachments.
396. (VA.19) A space exhausted by sublevel sets homotopy equivalent to the compatible stages of a cell tower has the homotopy type of the countable cell complex.